CompTIA®

A+® Complete

Study Guide

Volume 2: Core 2 Exam 220-1202

Sixth Edition

Quentin Docter

Jon Buhagiar

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For the fam.
—Quentin Docter

For my wife and son.
—Jon Buhagiar

Acknowledgments

As we were putting together this book, I was reminded of the proverb that begins "It takes a village . . ." That beginning definitely holds true for creating a book of this scope and size. From beginning to end, scores of dedicated professionals have focused on delivering the best book possible to you, the readers.

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—Quentin Docter

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—Jon Buhagiar

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Introduction

Welcome to the *CompTIA® A+® Complete Study Guide Volume 2: Core 2 Exam 220-1202 Sixth Edition*. This is the sixth edition of our best-selling study guide for the A+ certification sponsored by CompTIA (Computing Technology Industry Association). Thank you for choosing us to help you on your journey toward certification!

This book is written at an intermediate technical level; we assume that you already know how to *use* a personal computer and its basic peripherals, such as USB devices and printers, but we also recognize that you may be learning how to *service* some of that computer equipment for the first time. The exams cover basic computer service topics as well as more advanced issues, and they cover topics that anyone already working as a technician should be familiar with. The exams are designed to test you on these topics in order to certify that you have enough knowledge to fix and upgrade some of the most widely used types of personal computers and operating systems.

In addition to the information in the chapters, we've included a lot of extra material to help your study prep. At the end of each chapter is a list of exam essentials to know as well as 20 review questions to give you a taste of what it's like to take the exams. In addition, there are eight bonus exams of at least 75 questions each. Finally, there are flashcards designed to help your recall. Before you dive into those, though, we recommend you take the assessment test at the end of this introduction to gauge your current knowledge level.

Don't just study the questions and answers—the questions on the actual exams will be different from the practice ones included with this book. The exams are designed to test your knowledge of a concept or objective, so use this book to learn the objective behind the question. That said, we're confident that if you can do well on our quizzes, you will be well equipped to take the real exam.

This book covers more than just the exams, however. We believe in providing our students with a foundation of IT knowledge that will prepare them for real jobs, not just to pass a test. After all, life is not a multiple-choice test with the answers clearly laid out in front of you!

For experienced IT professionals, you can use the book to fill in the gaps in your current computer service knowledge. You may find, as many PC technicians have, that being well versed in all the technical aspects of hardware and operating systems is not enough to provide a satisfactory level of support—you must also have customer-relations skills, understand safety concepts, and be familiar with change management and environmental impacts and controls. We include helpful hints in all of these areas.

What Is CompTIA A+ Certification?

The A+ certification program was developed by CompTIA to provide an industrywide means of certifying the competency of computer service technicians. The A+ certification is granted to those who have attained the level of knowledge and troubleshooting skills that are needed

to provide capable support in the field of personal computers. It is similar to other certifications in the computer industry, such as the Cisco Certified Support Technician (CCST) program and the Microsoft certification programs. The theory behind these certifications is that if you need to have service performed on any of their products, you would sooner call a technician who has been certified in one of the appropriate certification programs than just call the first "expert" in the phone book.

The A+ certification program was created to offer a wide-ranging certification, in the sense that it is intended to certify competence with personal computers and mobile devices from many different makers/vendors. You must pass two tests to become A+ certified:

- The Core 1 220-1201 exam, which covers basic computer concepts, PC hardware, basic networking, mobile devices, and hardware troubleshooting
- The Core 2 220-1202 exam, which covers operating systems, security, software troubleshooting, and operational procedures

You don't have to take the 220-1201 and the 220-1202 exams at the same time. However, the A+ certification is not awarded until you've passed both tests.

Why Become A+ Certified?

There are several good reasons to get your A+ certification. The CompTIA Candidate's Information packet lists five major benefits:

- It demonstrates proof of professional achievement.
- It increases your marketability.
- It provides greater opportunity for advancement in your field.
- It is increasingly a requirement for some types of advanced training.
- It raises customer confidence in you and your company's services.

Provides Proof of Professional Achievement The A+ certification is a status symbol in the computer service industry. Organizations that include members of the computer service industry recognize the benefits of A+ certification and push for their members to become certified. And more people every day are putting the "A+ Certified Technician" emblem on their business cards.

Increases Your Marketability A+ certification makes individuals more marketable to potential employers. A+ certified employees also may receive a higher base salary because employers won't have to spend as much money on vendor-specific training.

Provides Opportunity for Advancement Most raises and advancements are based on performance. A+ certified employees work faster and more efficiently and are thus more productive. The more productive employees are, the more money they make for their company. And, of course, the more money they make for the company, the more valuable they are to the company. So, if an employee is A+ certified, their chances of being promoted are greater.

Fulfills Training Requirements Most major computer hardware vendors recognize A+ certification. Some of these vendors apply A+ certification toward prerequisites in their own respective certification programs, which has the side benefit of reducing training costs for employers.

Raises Customer Confidence As the A+ Certified Technician moniker becomes better known among computer owners, more of them will realize that the A+ technician is more qualified to work on their computer equipment than a noncertified technician.

How to Become A+ Certified

A+ certification is available to anyone who passes two exams. You don't have to work for any particular company. It's not a secret society. It is, however, an elite group. To become A+ certified, you must do two things:

- Pass the Core 1 220-1201 exam.
- Pass the Core 2 220-1202 exam.

The exams can be taken at any Pearson VUE testing center. If you pass both exams, you will get a certificate in the mail from CompTIA saying that you have passed, and you will also receive a lapel pin and business card.

To register for the tests, go to `www.pearsonvue.com/comptia`. You'll be asked for your name, Social Security number (an optional number may be assigned if you don't wish to provide your Social Security number), mailing address, phone number, employer, when and where you want to take the test, and your credit card number. (Payment arrangements must be made at the time of registration.)

Tips for Taking the A+ Exam

Here are some general tips for taking your exam successfully:

- Bring two forms of ID with you. One must be a photo ID, such as a driver's license. The other can be a major credit card or a passport. Both forms must include a signature.
- Arrive early at the exam center so that you can relax and review your study materials, particularly tables and lists of exam-related information. When you enter the testing room, you will need to leave everything outside; you won't be able to bring any materials into the testing area.
- Read the questions carefully. Don't be tempted to jump to an early conclusion. Make sure that you know exactly what each question is asking.
- Don't leave any unanswered questions. Unanswered questions are scored against you. Some questions will have multiple correct responses. When a question has more than one correct answer, a message at the bottom of the screen will prompt you to, for example, choose two. Be sure to read the messages displayed to know how many correct answers you must choose.

- When answering multiple-choice questions that you're not sure about, use a process of elimination to get rid of the obviously incorrect answers first. Doing so will improve your odds if you need to make an educated guess.
- For the hands-on questions where you need to complete a task, double-check that you have finished all parts of it, including checking all tabs available to you if they give you an interface to click on. Also, understand that these questions will take longer than multiple-choice ones. Spend the time needed on them; you will be able to make up time on the shorter multiple-choice questions.
- On form-based tests (nonadaptive), because the hard questions will take the most time, save them for last. You can move forward and backward through the exam.
- For the latest pricing on the exams and updates to the registration procedures, visit CompTIA's website at `www.comptia.org`.

Who Should Read This Book?

If you are one of the many people who want to pass the A+ exams, and pass them confidently, then you should buy this book and use it to study for the exams.

This book was written to prepare you for the challenges of the real IT world, not just to pass the A+ exams. This study guide will do that by describing in detail the concepts on which you'll be tested.

What Does This Book Cover?

CompTIA A+ Complete Study Guide: Exam 220-1201 and Exam 220-1202 covers everything you need to know to pass the CompTIA A+ exams.

Volume 2 of the book starts at Chapter 1 and concludes with Chapter 10. It covers all the topics on which you will be tested for Exam 220-1202:

Chapter 1: Operating System Basics Chapter 1 begins by covering the fundamentals of understanding various operating systems. The chapter then shifts to focus on supporting applications' installations. The chapter concludes by introducing Windows 10/11 editions, features, the Windows interface, and file management.

Chapter 2: Windows Configuration Chapter 2 explores Control Panel applets, Microsoft Management Console snap-ins, and Windows settings used to configure and customize the Windows operating system. The chapter also covers the Windows Registry and disk management.

Chapter 3: Windows Administration Chapter 3 guides you through the installation and upgrade of the Windows operating system. The chapter then explores the various command-line tools used to administer Windows. Finally, the chapter covers the administration and support of networking that is common with Windows.

Chapter 4: Working with macOS and Linux Chapter 4 covers common features and tools found in both the macOS and Linux operating systems. The chapter concludes by covering command-line tools used on Linux operating systems.

Chapter 5: Security Concepts Just when you think this book couldn't get any better, we toss in a chapter devoted to security. This chapter introduces key security concepts you need to know for the exam, including information on physical security, social engineering, wireless networks, best practices, and data destruction methods.

Chapter 6: Securing Operating Systems This chapter follows up on Chapter 5, which introduced security concepts, by providing specifics on how to secure Microsoft and mobile operating systems, prevent and remove malware, and configure network security on wired and wireless networks.

Chapter 7: Troubleshooting Operating Systems and Security Troubleshooting is a major part of almost every technician's job. In Chapter 7, we talk about troubleshooting common operating system issues as well as security challenges you may encounter.

Chapter 8: Scripting, Remote Access, and Artificial Intelligence Network administrators and technicians need to know how to create and modify scripts for automation. They also need remote access to systems in order to use or troubleshoot them. The chapter shows you technologies used to do just that. In addition, this chapter includes a wealth of knowledge on how artificial intelligence (AI) operates, as well as considerations for the adoption of AI.

Chapter 9: Safety and Environmental Concerns This chapter discusses the impacts computers can have on people as well as the environment. You learn how to avoid damaging a computer as well as avoid having it damage you, how to perform backups and disaster recovery, and how to properly dispose of computer components that are no longer needed.

Chapter 10: Documentation and Professionalism This chapter covers the "softer" side of working with computers. Specifically, you learn to practice proper communication, show professionalism, and implement best practices associated with documentation and change management.

Volume 1 of the book covers all the topics on which you will be tested for Exam 220-1201.

What's Included in the Book

We've included several learning tools throughout the book:

Assessment Tests At the end of this introduction is an assessment test that you can use to check your readiness for the exams. Take this before you start reading the book; it will help you determine the areas on which you might need to brush up. The answers

to the assessment test questions appear on a separate page after the last question of the test. Each answer includes an explanation and the chapter the material appears in.

Objective Map and Opening List of Objectives In this introduction, we have included a detailed exam objective map that shows you where each exam objective is covered. In addition, each chapter opens with a list of the exam objectives it covers. Use these resources to see exactly where each exam topic is covered.

Exam Essentials Each chapter, just after the summary, includes a number of "exam essentials." These are the key topics that you should focus on when preparing for the exam.

Chapter Review Questions To test your knowledge as you progress through the book, each chapter ends with review questions. As you finish each chapter, answer the review questions and then check your answers—the correct answers and explanations are in Appendix A. You can go back to reread the section that deals with each question you got wrong to ensure that you answer correctly the next time that you're tested on the material.

Performance-Based Questions The A+ exams may ask you to perform tasks to correctly answer a question. This may mean dragging and dropping items into the correct order or clicking the correct area of an image or screenshot. While it's hard to simulate those in a paper book, the end of each chapter features a performance-based question. These questions will have you think through a topic presented in the chapter, and then perform a task. The correct answers are found in Appendix B.

Interactive Online Learning Environment and Test Bank

The interactive online learning environment that accompanies *CompTIA A+ Complete Study Guide: Exam 220-1201 and Exam 220-1202* provides a test bank with study tools to help you prepare for the certification exams and increase your chances of passing them the first time! The test bank includes the following elements:

Sample Tests All the questions and answers in this book are provided, including the assessment tests, which you'll find at the end of this introduction, and the review questions at the end of each chapter. In addition, there are eight practice exams. Use these questions to test your knowledge of the study guide material. The online test bank runs on multiple devices.

Flashcards Four sets of questions are provided in digital flashcard format (a question followed by a single correct answer). You can use the flashcards to reinforce your learning and provide last-minute test prep before the exam.

Glossary The key terms from this book and their definitions are available as a fully searchable PDF.

eBooks Read eBook versions of this Study Guide in multiple formats on your favorite device.

Audio Instruction Over 2 hours of recorded instruction from the authors to help you study for the certification exam with confidence.

Go to www.wiley.com/go/sybextestprep to register and gain access to this interactive online learning environment and test bank with study tools.

Like all exams, the CompTIA A+ certification is updated periodically and may eventually be retired or replaced. At some point after CompTIA is no longer offering this exam, the old editions of our books and online tools will be retired. If you have purchased this book after the exam was retired, or are attempting to register in the Sybex online learning environment after the exam was retired, please know that we make no guarantees that this exam's online Sybex tools will be available once the exam is no longer available.

How to Use This Book

If you want a solid foundation for preparing for the A+ exams, this is the book for you. We've spent countless hours putting together this book with the intention of helping you prepare for the exams.

This book is loaded with valuable information, and you will get the most out of your study time if you understand how we put the book together. Here's a list that describes how to approach studying:

1. Take the assessment tests immediately following this introduction. It's okay if you don't know any of the answers—that's what this book is for. Carefully read over the explanations for any questions you get wrong, and make note of the chapters where that material is covered.
2. Study each chapter carefully, making sure you fully understand the information and the exam objectives listed at the beginning of the chapter. Again, pay extra-close attention to any chapter that includes material covered in questions you missed on the assessment test.
3. Read over the Summary and Exam Essentials. These will highlight the sections from the chapter you need to be familiar with before sitting for the exam.
4. Answer all the review questions at the end of each chapter. Specifically note any questions that confuse you, and study the corresponding sections of the book again. Don't just skim these questions. Make sure that you understand each answer completely.
5. Go over the electronic flashcards. These help you to prepare for the latest A+ exams, and they're really great study tools.
6. Take the practice exams.

Performance-Based Questions

CompTIA includes performance-based questions on the A+ exams. These are not the traditional multiple-choice questions with which you're probably familiar. These questions require the candidate to know how to perform a specific task or series of tasks. The candidate will be presented with a scenario and will be asked to complete a task. They will be taken to a simulated environment where they will have to perform a series of steps and will be graded on how well they complete the task.

The Sybex test engine does not include performance-based questions. However, at the end of each chapter, we have included a section called "Performance-Based Question," which is designed to measure how well you understood the chapter's topics. Some simply ask you to complete a task for which there is only one correct response. Others are more subjective, with multiple ways to complete the task. We provide the most logical or practical solution in Appendix B. Note that these questions may cover topic areas not covered in the actual A+ performance-based questions. However, we feel that being able to think logically is a great way to learn.

The CompTIA A+ Exam Objectives

The A+ exams consist of the Core 1 220-1201 exam and the Core 2 220-1202 exam. The following are the detailed exam objectives for Core 2 220-1202.

Exam objectives are subject to change at any time without prior notice and at CompTIA's sole discretion. Please visit the A+ Certification page of CompTIA's website (`https://comptia.org/certifications/a`) for the most current listing of exam objectives.

A+ Certification Exam Objectives: 220-1202

The following table lists the domains measured by this examination and the extent to which they are represented on the exam.

Domain	Percentage of exam
1.0 Operating Systems	28%
2.0 Security	28%
3.0 Software Troubleshooting	23%
4.0 Operational Procedures	21%
Total	100%

Objective Map

The following table lists where you can find the objectives covered in Volume 2 of this book.

Objective	Chapter(s)
1.0 Operating Systems	
1.1 Explain common operating system (OS) types and their purposes.	1, 2

- Workstation systems (OSs)
  - Windows
  - Linux
  - macOS
  - Chrome OS
- Mobile OSs
  - iPadOS
  - iOS
  - Android
- Various filesystem types
  - New Technology File System(NTFS)
  - Resilient File System (ReFS)
  - File Allocation Table 32 (FAT32)
  - Fourth extended filesystem (ext4)
  - Extended filesystem (XFS)
  - Apple File System (APFS)
  - Extensible File Allocation Table (exFAT)
- Vendor life-cycle limitations
  - End-of-life (EOL)
  - Update limitations
- Compatibility concerns between operating systems

Objective	Chapter(s)
1.2 Given a scenario, perform OS installations and upgrades in a diverse environment.	1, 2, 3

- Boot methods
  - Universal Serial Bus (USB)
  - Network
  - Solid-state/flash drives
  - Internet-based
  - External/hot-swappable drive
  - Internal hard drive (partition)
  - Multiboot
- Types of installations
  - Clean install
  - Upgrade
  - Image deployment
  - Remote network installation
  - Zero-touch deployment
  - Recovery partition
  - Repair installation
  - Other considerations
    - Third-party drivers
- Partitioning
  - GUID [globally unique identifier] partition Table (GPT)
  - Master boot record (MBR)
- Drive format
- Upgrade considerations
  - Backup files and user preferences
  - Application and driver support/backward compatibility
  - Hardware compatibility
- Feature updates
  - Product life cycle

Objective	Chapter(s)
1.3 Compare and contrast basic features of Microsoft Windows editions.	1

- Windows 10 editions
  - Home
  - Pro
  - Pro for Workstations
  - Enterprise
- Windows 11 editions
  - Home
  - Pro
  - Enterprise
- N versions
- Feature differences
  - Domain vs. workgroup
  - Desktop styles/user interface
  - Availability of Remote Desktop Protocol (RDP)
  - Random-access memory (RAM) support limitations
  - BitLocker
  - gpedit.msc
- Upgrade paths
  - In-place upgrade
  - Clean install
- Hardware requirements
  - Trusted Platform Module (TPM)
  - Unified Extensible Firmware Interface (UEFI)

Objective	Chapter(s)

1.4 Given a scenario, use Microsoft Windows operating system features and tools.
▪ Task Manager
  ▪ Services
  ▪ Startup
  ▪ Performance
  ▪ Processes
  ▪ Users
▪ Microsoft Management Console (MMC) snap-in
  ▪ Event Viewer (`eventvwr.msc`)
  ▪ Disk Management (`diskmgmt.msc`)
  ▪ Task Scheduler (`taskschd.msc`)
  ▪ Device Manager (`devmgmt.msc`)
  ▪ Certificate Manager (`certmgr.msc`)
  ▪ Local User and Groups (`lusrmgr.msc`)
  ▪ Performance Monitor (`perfmon.msc`)
  ▪ Group Policy Editor (`gpedit.msc`)
▪ Additional tools
  ▪ System Information (`msinfo32. exe`)
  ▪ Resource Monitor (`resmon.exe`)
  ▪ System Configuration (`msconfig. exe`)
  ▪ Disk Cleanup (`cleanmgr.exe`)
  ▪ Disk Defragment (`dfrgui.exe`)
  ▪ Registry Editor (`regedit.exe`)
	2

1.5 Given a scenario, use the appropriate Microsoft command-line tools.
▪ Navigation
  ▪ cd
  ▪ dir
▪ Network
  ▪ ipconfig
  ▪ ping
	3

Objective	Chapter(s)
▪ netstat ▪ nslookup ▪ net use ▪ tracert ▪ pathping ▪ Disk management ▪ chkdsk ▪ format ▪ diskpart ▪ File management ▪ md ▪ rmdir ▪ robocopy ▪ Informational ▪ hostname ▪ net user ▪ winver ▪ whoami ▪ [command name] /? ▪ OS management ▪ gpupdate ▪ gpresult ▪ sfc	
1.6 Given a scenario, configure Microsoft Windows settings. ▪ Internet Options ▪ Devices and Printers ▪ Program and Features ▪ Network and Sharing Center ▪ System ▪ Windows Defender Firewall	2

Objective	Chapter(s)

- Mail
- Sound
- User Accounts
- Device Manager
- Indexing Options
- Administrative Tools
- File Explorer Options
  - View hidden files
  - Hide extensions
  - General options
  - View options
- Power Options
  - Hibernate
  - Power plans
  - Sleep/suspend
  - Standby
  - Choose what closing the lid does
  - Turn on fast startup
  - USB selective suspend
- Ease of Access
- Time and Language
- Update and Security
- Personalization
- Apps
- Privacy
- System
- Devices
- Network and Internet
- Gaming
- Accounts

Objective	Chapter(s)
1.7 Given a scenario, configure Microsoft Windows networking features on a client/desktop.	3

▪ Domain joined vs. workgroup
  ▪ Shared resources
  ▪ Printers
  ▪ File servers
  ▪ Mapped drives
▪ Local OS firewall settings
  ▪ Application restrictions and exceptions
  ▪ Configuration
▪ Client network configuration
  ▪ Internet Protocol (IP) addressing scheme
  ▪ Domain Name System (DNS) settings
  ▪ Subnet mask
  ▪ Gateway
  ▪ Static vs. dynamic
▪ Establish network connections
  ▪ Virtual private network (VPN)
  ▪ Wireless
  ▪ Wired
  ▪ Wireless wide area network (WWAN)/cellular network
▪ Proxy settings
▪ Public network vs. private network
▪ File Explorer navigation–network paths
▪ Metered connections and limitations	

1.8 Explain common features and tools of the macOS/desktop operating system.	4

▪ Installation and uninstallation of applications
  ▪ File types
  ▪ .dmg
  ▪ .pkg
  ▪ .app	

Objective	Chapter(s)
App Store	
Uninstallation process	
▪ System folders	
▪ /Applications	
▪ /Users	
▪ /Library	
▪ /System	
▪ /Users/Library	
▪ Apple ID and corporate restrictions	
▪ Best practices	
▪ Backups	
▪ Antivirus	
▪ Updates/patches	
▪ Rapid Security Response (RSR)	
▪ System Preferences	
▪ Displays	
▪ Networks	
▪ Printers	
▪ Scanners	
▪ Privacy	
▪ Accessibility	
▪ Time Machine	
▪ Features	
▪ Multiple desktops	
▪ Mission Control	
▪ Keychain	
▪ Spotlight	
▪ iCloud	
▪ iMessage	
▪ FaceTime	
▪ Drive	

Objective	Chapter(s)
▪ Gestures ▪ Finder ▪ Dock ▪ Continuity ▪ Disk Utility ▪ FileVault ▪ Terminal ▪ Force Quit	
1.9 Identify common features and tools of the Linux client/desktop operating system. ▪ File management ▪ ls ▪ pwd ▪ mv ▪ cp ▪ rm ▪ chmod ▪ chown ▪ grep ▪ find ▪ Filesystem management ▪ fsck ▪ mount ▪ Administrative ▪ su ▪ sudo ▪ Package management ▪ apt ▪ dnf ▪ Network ▪ ip ▪ ping ▪ curl ▪ dig ▪ traceroute	4

Objective	Chapter(s)
▪ Informational	
▪ man	
▪ cat	
▪ top	
▪ ps	
▪ du	
▪ df	
▪ Text editors	
▪ nano	
▪ Common configuration files	
▪ /etc/passwd	
▪ /etc/shadow	
▪ /etc/hosts	
▪ /etc/fstab	
▪ /etc/resolv.conf	
▪ OS components	
▪ systemd	
▪ kernel	
▪ bootloader	
▪ Root account	
1.10 Given a scenario, install applications according to requirements.	1
▪ System requirements for applications	
▪ 32-bit vs. 64-bit dependent application requirements	
▪ Dedicated vs. integrated graphics card	
▪ Video random-access memory (VRAM) requirements	
▪ RAM requirements	
▪ Central processing unit (CPU) requirements	
▪ External hardware tokens	
▪ Storage requirements	

Objective	Chapter(s)
▪ Application to OS compatibility	
▪ Distribution methods	
▪ Physical media vs. mountable ISO file	
▪ Downloadable package	
▪ Image deployment	
▪ Impact considerations for new applications	
▪ Device	
▪ Network	
▪ Operation	
▪ Business	
1.11 Given a scenario, install and configure cloud-based productivity tools.	1
▪ Email systems	
▪ Storage	
▪ Sync/folder settings	
▪ Collaboration tools	
▪ Spreadsheets	
▪ Videoconferencing	
▪ Presentation tools	
▪ Word processing tools	
▪ Instant messaging	
▪ Identity synchronization	
▪ Licensing assignment	
2.0 Security	
2.1 Summarize various security measures and their purposes.	5
▪ Physical security	
▪ Bollards	
▪ Access control vestibule	
▪ Badge reader	
▪ Video surveillance	
▪ Alarm systems	
▪ Motion sensors	

Objective	Chapter(s)

- Door locks
- Equipment locks
- Security guards
- Fences

- Physical access security
  - Key fobs
  - Smart cards
  - Mobile digital key
  - Keys
  - Biometrics
    - Retina scanner
    - Fingerprint scanner
    - Palm print scanner
    - Facial recognition technology (FRT)
    - Voice recognition technology
  - Lighting
  - Magnetometers
- Logical security
  - Principle of least privilege
  - Zero Trust model
  - Access control lists (ACLs)
  - Multifactor authentication (MFA)
    - Email
    - Hardware token
    - Authenticator application
    - Short Message Service (SMS)
    - Voice call
    - Time-based one-time pass word (TOTP)
    - One-time password/passcode (OTP)
  - Security Assertions Markup Language (SAML)
  - Single sign-on (SSO)
  - Just-in-time access

Objective	Chapter(s)
▪ Privileged access management (PAM) ▪ Mobile device management (MDM) ▪ Data loss prevention (DLP) ▪ Identity access management (IAM) ▪ Directory services	
2.2 Given a scenario, configure and apply basic Microsoft Windows OS security settings. ▪ Defender Antivirus ▪ Activate/deactivate ▪ Update definitions ▪ Firewall ▪ Activate/deactivate ▪ Port security ▪ Application security ▪ User and groups ▪ Local vs. Microsoft account ▪ Standard account ▪ Administrator ▪ Guest user ▪ Power user ▪ Log-in OS options ▪ Username and password ▪ Personal identification number (PIN) ▪ Fingerprint ▪ Facial recognition ▪ SSO ▪ Passwordless/Windows Hello ▪ NTFS vs. share permissions ▪ File and folder attributes ▪ Inheritance ▪ Run as administrator vs. standard user ▪ User Account Control (UAC)	5, 6

Objective	Chapter(s)
▪ BitLocker	
▪ BitLocker-To-Go	
▪ Encrypting File System (EFS)	
▪ Active Directory	
▪ Joining domain	
▪ Assigning log-in script	
▪ Moving objects within organizational units	
▪ Assigning home folders	
▪ Applying Group Policy	
▪ Selecting security groups	
▪ Configuring folder redirection	
2.3 Compare and contrast wireless security protocols and authentication methods.	6
▪ Protocols and encryption	
▪ Wi-Fi Protected Access 2 (WPA2)	
▪ WPA3	
▪ Temporal Key Integrity Protocol (TKIP)	
▪ Advanced Encryption Standard (AES)	
▪ Authentication	
▪ Remote Authentication Dial-in User Service (RADIUS)	
▪ Terminal Access Controller Access-control System (TACACS+)	
▪ Kerberos	
▪ Multifactor	
2.4 Summarize types of malware and tools/methods for detection, removal, and prevention.	5
▪ Malware	
▪ Trojan	
▪ Rootkit	
▪ Virus	
▪ Spyware	
▪ Ransomware	

Objective	Chapter(s)
▪ Keylogger ▪ Boot sector virus ▪ Cryptominer ▪ Stalkerware ▪ Fileless ▪ Adware ▪ Potentially unwanted program (PUP) ▪ Tools and methods ▪ Recovery console ▪ Endpoint detection and response (EDR) ▪ Managed detection and response (MDR) ▪ Extended detection and response (XDR) ▪ Antivirus ▪ Antimalware ▪ Email security gateway ▪ Software firewalls ▪ User education regarding common threats ▪ Antiphishing training ▪ OS reinstallation	
2.5 Compare and contrast common social engineering attacks, threats, and vulnerabilities. ▪ Social engineering ▪ Phishing ▪ Vishing ▪ Smishing ▪ QR code phishing ▪ Spear phishing ▪ Whaling ▪ Shoulder surfing ▪ Tailgating ▪ Impersonation ▪ Dumpster diving	5

Objective	Chapter(s)

▪ Threats
  ▪ Denial of service (DoS)
  ▪ Distributed denial of service (DDoS)
  ▪ Evil twin
  ▪ Zero-day attack
  ▪ Spoofing
  ▪ On-path attack
  ▪ Brute-force attack
  ▪ Dictionary attack
  ▪ Insider threat
  ▪ Structured Query Language (SQL) injection
  ▪ Cross-site scripting (XSS)
  ▪ Business email compromise (BEC)
  ▪ Supply chain/pipeline attack
▪ Vulnerabilities
  ▪ Non-compliant systems
  ▪ Unpatched systems
  ▪ Unprotected systems (missing antivirus/missing firewall)
  ▪ EOL
  ▪ Bring your own device (BYOD)	

2.6 Given a scenario, implement procedures for basic small office/home office (SOHO) malware removal.
▪ 1. Investigate and verify malware symptoms.
▪ 2. Quarantine infected system.
▪ 3. Disable System Restore in Windows Home.
▪ 4. Remediate infected systems.
  a. Update antimalware software.
  b. Scan and removal techniques (e.g., safe mode, preinstallation environment)
  c. Reimage/reinstall.
▪ 5. Schedule scans and run updates.
▪ 6. Enable System Restore and create a restore point in Windows Home.
▪ 7. Educate the end user.
	7

Objective	Chapter(s)
2.7 Given a scenario, apply workstation security options and hardening techniques.	5

▪ Data-at-rest encryption
▪ Password considerations
  ▪ Length
  ▪ Character types
  ▪ Uniqueness
  ▪ Complexity
  ▪ Expiration
▪ Basic input/output system (BIOS)/Unified Extensible Firmware Interface (UEFI) passwords
▪ End-user best practices
  ▪ Use screensaver locks
  ▪ Log off when not in use
  ▪ Secure/protect critical hardware (e.g., laptops)
  ▪ Secure personally identifiable information (PII) and passwords
  ▪ Use password managers
▪ Account management
  ▪ Restrict user permissions
  ▪ Restrict log-in times
  ▪ Disable guest account
  ▪ Use failed attempts lockout
  ▪ Use timeout/screen lock
  ▪ Apply account expiration dates
▪ Change default administrator's user account/password
▪ Disable AutoRun
▪ Disable unused services	

2.8 Given a scenario, apply common methods for securing mobile devices.	6

▪ Hardening techniques
  ▪ Device encryption
  ▪ Screen locks
  ▪ Facial recognition	

Objective	Chapter(s)
▪ PIN codes ▪ Fingerprint ▪ Pattern ▪ Swipe ▪ Configuration profiles ▪ Patch management ▪ OS updates ▪ Application updates ▪ Endpoint security software ▪ Antivirus ▪ Antimalware ▪ Content filtering ▪ Locator applications ▪ Remote wipes ▪ Remote backup applications ▪ Failed log-in attempts restrictions ▪ Policies and procedures ▪ MDM ▪ BYOD vs. corporate-owned devices ▪ Profile security requirements	
2.9 Compare and contrast common data destruction and disposal methods. ▪ Physical destruction of hard drives ▪ Drilling ▪ Shredding ▪ Degaussing ▪ Incineration ▪ Recycling or repurposing best practices ▪ Erasing/wiping ▪ Low-level formatting ▪ Standard formatting ▪ Outsourcing concepts ▪ Third-party vendor ▪ Certification of destruction/recycling ▪ Regulatory and environmental requirements	5

Objective	Chapter(s)
2.10 Given a scenario, apply security settings on SOHO wireless and wired networks.	6

▪ Router settings
  ▪ Change default passwords
  ▪ IP filtering
  ▪ Firmware updates
  ▪ Content filtering
  ▪ Physical placement/secure locations
  ▪ Universal Plug and Play (UPnP)
  ▪ Screened subnet
  ▪ Configure secure management access
▪ Wireless specific
  ▪ Changing the service set identifier (SSID)
  ▪ Disabling SSID broadcast
  ▪ Encryption settings
  ▪ Configuring guest access
▪ Firewall settings
  ▪ Disabling unused ports
  ▪ Port forwarding/mapping	

2.11 Given a scenario, configure relevant security settings in a browser.	6

▪ Browser download/installation
  ▪ Trusted sources
  ▪ Hashing
  ▪ Untrusted sources
▪ Browser patching
▪ Extensions and plug-ins
  ▪ Trusted sources
  ▪ Untrusted sources	

Objective	Chapter(s)
▪ Password managers	
▪ Secure connections/sites–valid certificates	
▪ Settings	
▪ Pop-up blocker	
▪ Clearing browsing data	
▪ Clearing cache	
▪ Private-browsing mode	
▪ Sign-in/browser data synchronization	
▪ Ad blockers	
▪ Proxy	
▪ Secure DNS	
▪ Browser feature management	
▪ Enable/disable	
▪ Plug-ins	
▪ Extensions	
▪ Features	
3.0 Software Troubleshooting	
3.1 Given a scenario, troubleshoot common Windows OS issues.	7
▪ Blue screen of death (BSOD)	
▪ Degraded performance	
▪ Boot issues	
▪ Frequent shutdowns	
▪ Services not starting	
▪ Applications crashing	
▪ Low memory warnings	
▪ USB controller resource warnings	
▪ System instability	
▪ No OS found	
▪ Slow profile load	
▪ Time drift	

Objective	Chapter(s)
3.2 Given a scenario, troubleshoot common mobile OS and application issues.	7

▪ Application fails to launch
▪ Application fails to close/crashes
▪ Application fails to update
▪ Application fails to install
▪ Slow to respond
▪ OS fails to update
▪ Battery life issues
▪ Random reboots
▪ Connectivity issues
  ▪ Bluetooth
  ▪ Wi-Fi
  ▪ Near-field communication (NFC)
▪ Screen does not autorotate	

3.3 Given a scenario, troubleshoot common mobile OS and application security issues.	7

▪ Security concerns
  ▪ Application source/unofficial application stores
  ▪ Developer mode
  ▪ Root access/jailbreak
  ▪ Unauthorized/malicious application
  ▪ Application spoofing
▪ Common symptoms
  ▪ High network traffic
  ▪ Degraded response time
  ▪ Data-usage limit notification
  ▪ Limited Internet connectivity
  ▪ No Internet connectivity
  ▪ High number of ads
  ▪ Fake security warnings
  ▪ Unexpected application behavior
  ▪ Leaked personal files/data	

Objective	Chapter(s)
3.4 Given a scenario, troubleshoot common personal computer (PC) security issues.	7

- Common symptoms
  - Unable to access the network
  - Desktop alerts
  - False alerts regarding antivirus protection
  - Altered system or personal files
    - Missing/renamed files
    - Inability to access files
  - Unwanted notifications within the OS
  - OS updates failures
- Browser-related symptoms
  - Random/frequent pop-ups
  - Certificate warnings
  - Redirection
  - Degraded browser performance

4.0 Operational Procedures	
4.1 Given a scenario, implement best practices associated with documentation and support systems information management.	10

- Ticketing systems
  - User information
  - Device information
  - Description of issues
  - Categories
  - Severity
  - Escalation levels
  - Clear, concise written communication
    - Issue description
    - Progress notes
    - Issue resolution

Objective	Chapter(s)
▪ Asset management ▪ Inventory lists ▪ Configuration management database (CMDB) ▪ Asset tags and IDs ▪ Procurement life cycle ▪ Warranty and licensing ▪ Assigned users ▪ Types of documents ▪ Incident reports ▪ Standard operating procedures (SOPs) ▪ Software package custom installation procedure ▪ New user/onboarding setup checklist ▪ User off-boarding checklist ▪ Service-level agreements (SLAs) ▪ Internal ▪ External/third-party Knowledge base/articles	
4.2 Given a scenario, apply change management procedures. ▪ Documented business processes ▪ Rollback plan ▪ Backup plan ▪ Sandbox testing ▪ Responsible staff members ▪ Change management ▪ Request forms ▪ Purpose of the change ▪ Scope of the change ▪ Change type ▪ Standard change	10

Objective	Chapter(s)
▪ Normal change ▪ Emergency change ▪ Date and time of change ▪ Change freeze ▪ Maintenance windows ▪ Affected systems/impact ▪ Risk analysis ▪ Risk level ▪ Change board approvals ▪ Implementation ▪ Peer review ▪ End-user acceptance	
4.3 Given a scenario, implement workstation backup and recovery methods. ▪ Backup ▪ Full ▪ Incremental ▪ Differential ▪ Synthetic full ▪ Recovery ▪ In-place/overwrite ▪ Alternative location ▪ Backup testing ▪ Frequency ▪ Backup rotation schemes ▪ On-site vs. off-site ▪ Grandfather-father-son (GFS) ▪ 3-2-1 backup rule	10

Objective	Chapter(s)
4.4 Given a scenario, use common safety procedures.	9
▪ Electrostatic discharge (ESD) straps	
▪ ESD mats	
▪ Electrical safety	
▪ Equipment grounding	
▪ Proper component handling and storage	
▪ Cable management	
▪ Antistatic bags	
▪ Compliance with government regulations	
▪ Personal safety	
▪ Disconnect power before repairing PC	
▪ Lifting techniques	
▪ Fire safety	
▪ Safety goggles	
▪ Air filter mask	
4.5 Summarize environmental impacts and local environment controls.	9
▪ Material safety data sheet (MSDS) documentation for handling and disposal	
▪ Proper battery disposal	
▪ Proper toner disposal	
▪ Proper disposal of other devices and assets	
▪ Temperature, humidity-level awareness, and proper ventilation	
▪ Location/equipment placement	
▪ Dust cleanup	
▪ Compressed air/vacuums	
▪ Power surges, brownouts, and blackouts	
▪ Uninterruptible power supply (UPS)	
▪ Surge suppressor	

Objective	Chapter(s)
4.6 Explain the importance of prohibited content/activity and privacy, licensing, and policy concepts.	9

▪ Incident response
  ▪ Chain of custody
  ▪ Informing management/law enforcement as necessary
  ▪ Copy of drive (data integrity and preservation)
  ▪ Incident documentation
  ▪ Order of volatility
▪ Licensing/digital rights management (DRM)/end-user license agreement (EULA)
  ▪ Valid licenses
  ▪ Perpetual license agreement
  ▪ Personal-use license vs. corporate-use license
  ▪ Open-source license
▪ Non-disclosure agreement (NDA)/mutual non-disclosure agreement (MNDA)
  ▪ Regulated data
  ▪ Credit card payment information
  ▪ Personal government-issued information
  ▪ PII
  ▪ Healthcare data
  ▪ Data retention requirements
▪ Acceptable use policy (AUP)
▪ Regulatory and business compliance requirements
  ▪ Splash screens	

4.7 Given a scenario, use proper communication techniques and professionalism.	10

▪ Present a professional appearance and wear appropriate attire.
  ▪ Match the required attire of the given environment.
  ▪ Formal
  ▪ Business casual	

Objective	Chapter(s)

- Use proper language and avoid jargon, acronyms, and slang, when applicable.
- Maintain a positive attitude/project confidence.
- Actively listen and avoid interrupting the customer.
- Be culturally sensitive.
  - Use appropriate professional titles and designations, when applicable.
- Be on time (if late, contact the customer).
- Avoid distractions.
  - Personal calls
  - Texting/social media sites
  - Personal interruptions
- Appropriately deal with difficult customers or situations.
  - Do not argue with customer and/or be defensive.
  - Avoid dismissing customer issues.
  - Avoid being judgmental.
  - Clarify customer statements (i.e., ask open-ended questions to narrow the scope of the issue, restate the issue, or question to verify understanding).
  - Use discretion and professionalism when discussing experiences/encounters.
- Set and meet expectations/timeline and communicate status with the customer.
  - Offer repair/replacement options, as needed.
  - Provide proper documentation on the services provided.
  - Follow up with customer/user at a later date to verify satisfaction.
- Appropriately handle customers' confidential and private materials.
  - Located on a computer, desktop, printer, etc.

Objective	Chapter(s)
4.8 Explain the basics of scripting.	8

▪ Script file types
  ▪ .bat
  ▪ .ps1
  ▪ .vbs
  ▪ .sh
  ▪ .js
  ▪ .py
▪ Use cases for scripting
  ▪ Basic automation
  ▪ Restarting machines
  ▪ Remapping network drives
  ▪ Installation of applications
  ▪ Automated backups
  ▪ Gathering of information/data
  ▪ Initiating updates
▪ Other considerations when using scripts
  ▪ Unintentionally introducing malware
  ▪ Inadvertently changing system settings
  ▪ Browser or system crashes due to mishandling of resources	

4.9 Given a scenario, use remote access technologies.	8

▪ Methods/tools
  ▪ RDP
  ▪ VPN
  ▪ Virtual network computer (VNC)
  ▪ Secure Shell (SSH)
  ▪ Remote monitoring and management (RMM)
  ▪ Simple Protocol for Independent Computing Environments (SPICE)
  ▪ Windows Remote Management (WinRM)
  ▪ Third-party tools
  ▪ Screen-sharing software	

Objective	Chapter(s)
▪ Videoconferencing software	
▪ File transfer software	
▪ Desktop management software	
▪ Security considerations of each access method	
4.10 Explain basic concepts related to artificial intelligence (AI).	8
▪ Application integration	
▪ Policy	
▪ Appropriate use	
▪ Plagiarism	
▪ Limitations	
▪ Bias	
▪ Hallucinations	
▪ Accuracy	
▪ Private vs. public	
▪ Data security	
▪ Data source	
▪ Data privacy	

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Assessment Test for Exam 220-1202

1. Which will accomplish a specific task for the user?
   - **A.** Driver
   - **B.** Application
   - **C.** Operating system
   - **D.** Filesystem
2. Which operating system is a workstation that operates solely in the cloud?
   - **A.** Linux
   - **B.** Android
   - **C.** Windows 11
   - **D.** ChromeOS
3. Which term defines the precompiled instructions that are programmed and allows an operating system and its applications to operate?
   - **A.** Shell
   - **B.** Source
   - **C.** GUI
   - **D.** Multithreading
4. Which term best describes the Android operating system?
   - **A.** Server
   - **B.** Workstation
   - **C.** Mobile
   - **D.** Cloud-based
5. Which Windows Update branch will allow you to install preview releases of updates for Windows 11?
   - **A.** Semi-Annual Channel
   - **B.** Semi-Annual Channel (Targeted)
   - **C.** Long-Term Servicing Channel
   - **D.** Insider Program
6. Which tool will allow you to ready the operating system for imaging?
   - **A.** Microsoft Deployment Toolkit
   - **B.** Windows Assessment and Deployment Kit
   - **C.** `sysprep.exe`
   - **D.** Windows Imaging

7. Which element of the boot process holds the information that instructs the operating system to load from a specific partition?
   - **A.** `winload.efi`
   - **B.** `bootmgr`
   - **C.** `winresume.exe`
   - **D.** BCD

8. You need to configure a static IP address for Windows 11. Which of the following will allow you to perform this task?
   - **A.** The Network & Sharing Center
   - **B.** Windows Defender Firewall
   - **C.** The Network & Internet Settings screen
   - **D.** The VPN settings screen

9. Which filesystem performs on-the-fly defragmentation?
   - **A.** FAT
   - **B.** NTFS
   - **C.** ext4
   - **D.** FAT32

10. Which place can you find updates for Apple operating systems?
    - **A.** iTunes
    - **B.** App Store
    - **C.** Keychain
    - **D.** Mission Control

11. Which macOS feature is similar to Windows File Explorer?
    - **A.** Keychain
    - **B.** iCloud
    - **C.** Spotlight
    - **D.** Finder

12. You want to find all files in the `/home` directory and subdirectories that contain the word `password`. Which command should you use?
    - **A.** `find /home -name "password"`
    - **B.** `grep -r "password" /home`
    - **C.** `ls /home | grep "password"`
    - **D.** `grep "password" /home`

13. Your system log files report an ongoing attempt to gain access to a single account. This attempt has been unsuccessful to this point. What type of attack are you most likely experiencing?
    - **A.** Password-guessing attack
    - **B.** Rootkit attack
    - **C.** Worm attack
    - **D.** TCP/IP hijacking

14. One of the vice presidents of the company calls a meeting with the information technology department after a recent trip to competitors' sites. She reports that many of the companies she visited granted access to their buildings only after fingerprint scans, and she wants similar technology employed at this company. Of the following, which technology relies on a physical attribute of the user for authentication?
    - **A.** Smartcard
    - **B.** Biometrics
    - **C.** Geo-fencing
    - **D.** Tokens

15. What type of malware is able to conceal itself from many parts of the operating system and will obtain/retain elevated privileges?
    - **A.** Worm
    - **B.** Trojan
    - **C.** Rootkit
    - **D.** Botnet

16. Which component of physical security addresses outer-level access control?
    - **A.** Fences
    - **B.** Access control vestibule
    - **C.** Multifactor authentication
    - **D.** Strong passwords

17. Which type of device can detect weapons on a person entering a facility?
    - **A.** Biometrics
    - **B.** Magnetometer
    - **C.** Motion sensor
    - **D.** Badge reader

18. A user wants to ensure that all contacts from their mobile device are copied onto their computer before migrating to a new device. Which of the following processes can accomplish this?
    - **A.** Mirroring
    - **B.** Synchronization
    - **C.** Calling each contact
    - **D.** Attaching the contacts to an email

19. Which feature of Windows Defender Firewall allows you to create custom rules for applications?

A. Real-time protection

B. Cloud-delivered protection

C. Advanced Security MMC

D. Tamper protection

20. A technician is configuring a new Windows computer for a home office. Which of the following steps should the technician take to secure the workstation?

A. Rename default accounts.

B. Configure Single Sign-On.

C. Disable Windows Update.

D. Disable Action Center pop-ups.

21. You need to protect files on the desktop operating system with encryption, but find out you do not have a TPM. What can you use to protect the files?

A. BitLocker

B. Encrypted File System

C. BitLocker To Go

D. Full drive encryption

22. Which is a benefit of setup of a device with a Microsoft account?

A. Access to the online store

B. Automatic synchronization of OneDrive

C. Active Directory authentication

D. Offline authentication

23. Which critical system generally requires third-party tools to troubleshoot performance problems?

A. RAM

B. CPU

C. Graphics

D. Network

24. What is the last step that should be performed when removing malware?

A. Investigate and verify symptoms.

B. Enable System Protection.

C. Educate the end user.

D. Schedule scans and run updates.

25. Your mobile device is suffering from an intermittent wireless connection. What is recommended to alleviate signal drops?

A. Shortening the SSID

B. Using the 5 GHz band

C. Reducing power on the WAP

D. Using the 2.4 GHz band

26. You are receiving USB controller resource warning messages. What should you try to resolve the issue?

A. Move USB devices around on the USB ports.

B. Manually allot more endpoints.

C. Upgrade drivers for the USB devices.

D. Manually increase the output amperage.

27. Why is time drift a big problem on virtual machines?

A. Lack of configuration for the NTP server

B. Availability of the NTP server

C. Shared physical RTC

D. Emulated RTC

28. Which tool can be used to restart a failed service?

A. `msconfig.exe`

B. WinRE

C. Computer Management MMC

D. Resource Monitor

29. Which is a safety risk from overheating a lithium-ion battery?

A. Reduced voltage

B. Shock

C. Explosion

D. Shutdown

30. A user is experiencing frequent shutdowns on their Windows computer. What should you check first to identify the cause?

A. Event Viewer

B. Device Manager

C. Task Manager

D. Windows Update

31. Which network protocol and port does RDP operate on?

A. TCP port 3389

B. TCP port 22

C. TCP port 5900

D. TCP port 443

32. Which line would be used to comment Windows Batch script code?

A. `//comment`

B. `'comment`

C. `REM comment`

D. `# comment`

33. Which command will launch the Remote Desktop Connection utility?

A. `msra.exe`

B. `mstsc.exe`

C. `quickassist.exe`

D. `ssh.exe`

34. A client computer connects to the main office and is configured with an IP address from the main office, on the client computer. What is being described?

A. Site-to-site VPN

B. Remote Desktop connection

C. SSH connection

D. Host-to-site VPN

35. Zoom and Teams are classic examples of what?

A. Screen-sharing software

B. Videoconferencing software

C. File transfer software

D. Desktop management software

36. Which of the following is a hallmark of human intelligence that AI does not possess?

A. Pattern recognition

B. Emotional experience

C. Data processing

D. Task automation

37. Which of the following are good measures to take to help prevent ESD? (Choose two.)

A. Decrease the humidity.

B. Tie long hair back.

C. Take your shoes off.

D. Perform self-grounding.

38. Which of the following screwdrivers has a splined head?

- **A.** Flat-tipped
- **B.** Phillips
- **C.** Axial
- **D.** Torx

39. What type of software licensing agreement usually provides free trial software, with the expectation that you will pay for it if you decide to keep it?

- **A.** Freeware
- **B.** Shareware
- **C.** Open source
- **D.** Single user

40. How do antistatic mats and wrist straps drain the potential ESD?

- **A.** The use of magnets
- **B.** Through a resistor
- **C.** High-voltage probe
- **D.** Direct ground

41. When cleaning dust out of computer equipment, you should always use which of the following?

- **A.** Multimeter
- **B.** Flashlight
- **C.** Air filter mask
- **D.** Mirror

42. Which is the most likely source of interference to wireless communications?

- **A.** Magnets
- **B.** ESD
- **C.** Surges
- **D.** Microwave ovens

43. While working on a user's system, you discover a sticky note attached to the bottom of the keyboard that has their username and password written on it. The user is not around, and you need to verify that the network connection is working. What should you do?

- **A.** Log in, verify access, and log out.
- **B.** Log in and stay logged in when you are finished.
- **C.** Text the user.
- **D.** Log in and change the user's password.

44. While installing a new network card, you accidentally broke a component off of the card. What should you do?

- **A.** Explain that the card was broken out of the box.
- **B.** Install the card anyway.

C. Inform the customer of the situation.

D. Leave everything as is until you can locate a new card.

45. What should you do when you cannot meet the timeline you proposed for resolving an issue?

A. Avoid informing the customer.

B. Communicate the delay and provide a new estimate.

C. Blame external factors.

D. Close the ticket prematurely.

46. Which ticket entry method is the easiest for end users to submit their own tickets?

A. Email entry

B. Portal entry

C. Manual entry

D. Application entry

47. Which method is used to test a change in an isolated environment?

A. Primary plan

B. Backout plan

C. Sandbox testing

D. Technical evaluation

48. You need to change the computer's name and join it to a domain, which tool will allow you to perform these actions?

A. Device Manager

B. User Accounts

C. System Properties

D. Credential Manager

49. Which tab in the Task Manager will allow you to see processes separated by each user on the system?

A. Processes

B. Performance

C. App History

D. Users

50. Which RAID system requires three or more disks to provide fault tolerance?

A. Mirroring

B. Striping

C. RAID-1

D. Striping with Parity

Answers to Assessment Test 220-1202

1. B. An application is written to perform a specific task for the user of the operating system, such as word processing or data analysis. Drivers allow the operating system to communicate with the hardware. The operating system and the filesystem assist in running applications. See Volume 2 Chapter 1 for more information.

2. D. The Google ChromeOS is a workstation that operates solely in the cloud, because all the data is stored in the cloud. Cloud-based operating systems require an Internet connection to access information. Android and Windows 11 use cloud storage, but they are not considered cloud-based operating systems. Linux is strictly a workstation operating system. See Volume 2 Chapter 1 for more information.

3. B. The source, also known as the source code, is the code to be compiled, and this allows the operating system and applications to execute. The shell is a component of the operating system that allows execution of applications. The GUI is a graphical extension of the shell to allow the execution of applications. Multithreading is a term used to describe how an application can create multiple simultaneous requests to the processor. See Volume 2 Chapter 1 for more information.

4. C. The Android operating system is a mobile operating system often used with tablets and smartphones. A server is an operating system that is optimized to serve information. A workstation is an operating system that is optimized to execute and display the information retrieved from a server operating system. A cloud-based operating system is an operating system that requires network connectivity to access applications. See Volume 2 Chapter 1 for more information.

5. D. The Insider Program allows for the installation of brand-new features before they are publicly released. The Semi-Annual Channel is normally delayed by three to four months. The Semi-Annual Channel (Targeted) branch will install updates as they are released to the general public. The Long-Term Servicing Channel never installs new features during the life of the version of Windows. See Volume 2 Chapter 3 for more information.

6. C. The `sysprep.exe` utility will allow you to ready the operating system for imaging, by resetting specific information, such as the computer name. The Microsoft Deployment Toolkit can assist in creating the steps, but it calls on the `sysprep` tool. The Windows Assessment and Deployment Kit allows you to customize the Windows operating system for imaging, but it does not ready the operating system for imaging. Windows Imaging (WIM) is a file format to contain the image. See Volume 2 Chapter 3 for more information.

7. D. The Boot Configuration Data (BCD) holds the information that instructs the Windows Boot Manager (`bootmgr`) to load the operating system from a specific partition. `winload.efi` loads the operating system kernel. `bootmgr` is the initial bootstrap program that reads the BCD. `winresume.exe` is used when resuming a previous session that has been suspended. See Volume 2 Chapter 3 for more information.

8. C. By clicking the Start menu, then Settings, and choosing Network & Internet you can configure the properties of the network adapter to setup a static IP address. The Network & Sharing Center is used to view the network connection firewall profile and connection status. The Windows Defender Firewall is used to protect the operating system from malicious network connectivity. The VPN settings screen allows you to configure a virtual private network (VPN) connection. See Volume 2 Chapter 3 for more information.

9. C. The ext4 filesystem does not suffer from fragmentation, because it performs on-the-fly defragmentation. FAT is a 16-bit filesystem that suffers from fragmentation. NTFS is a journaled filesystem that suffers from fragmentation. FAT32 is a 32-bit filesystem that suffers from fragmentation. See Volume 2 Chapter 4 for more information.

10. B. The App Store is where you can find updates for the Apple operating system. iTunes is used to purchase and download music, videos and other content. The Keychain is used to store credentials on behalf of the user. Mission Control is used to view all of the currently running applications. See Volume 2 Chapter 4 for more information.

11. D. The macOS Finder is the equivalent to the Windows File Explorer. The Keychain is a password management system for the operating system. iCloud is a cloud-based storage and backup service. Spotlight helps you find applications, documents and other files. See Volume 2 Chapter 4 for more information.

12. B. The `grep -r` command searches recursively for the word `password` in all files within the `/home` directory. The `find /home -name "password"` command will not recursively search the `/home` directory. The `ls /home | grep "password"` command will only return files with password in the name. The `grep "password" /home` command will not recursively search directories. See Volume 2 Chapter 4 for more information.

13. A. A password-guessing attack occurs when a user account is repeatedly attacked using a variety of passwords. A rootkit attack would not be immediately seen and not shown as an ongoing effort. A worm attack would not be visible as an ongoing attempt to gain access. TCP/IP hijacking is a form of an on-path (formerly man-in-the-middle) attack. See Volume 2 Chapter 5 for more information.

14. B. Biometrics relies on a physical characteristic of the user to verify identity. Biometric devices typically use either a hand pattern or a retinal scan to accomplish this. Smartcards contain a private certificate key and are protected with a passphrase. Geo-fencing uses GPS coordinates to require that authentication only happens within those parameters. Tokens are rotating numerical keys that you must physically have with you. See Volume 2 Chapter 5 for more information.

15. C. Rootkits are software programs that have the ability to hide certain things from the operating system. A worm is malware that replicates itself and infects other system. Trojans are programs that enter a system or network under the guise of another program. A botnet is a group of infected computers that can be remotely controled via a command and control server. See Volume 2 Chapter 5 for more information.

16. A. Fences are intended to delay or deter entrance into a facility's outer-level of access. Access control vestibules are used for mid-layer access control to prevent tailgating. Multifactor authentication (MFA) is used for mid- and inner-layer access control. Strong passwords are used for mid- and inner-layer access control. See Volume 2 Chapter 5 for more information.

17. B. A magnetometer, also known as a metal detector, can detect weapons on a person entering a facility. Biometrics are used to authenticate someone based on their physical traits. A motion sensor is used to detect motion. A badge reader is used to scan security ID badges. See Volume 2 Chapter 5 for more information.

18. B. Synchronization can copy all contacts, programs, email messages, pictures, music, and videos between a mobile device and a computer. Mirroring is a term reserved for when devices replicate themselves to similar devices. Calling the contacts won't achieve the desired result, and emailing contacts is inefficient and does not immediately result in placing the contacts on the computer. See Volume 2 Chapter 6 for more information.

19. C. The Advanced Security MMC allows you to create custom rules for applications in Windows Defender Firewall. Real-time protection provides real-time antivirus and antimalware protection. Cloud-delivered protection allows for the automatic submission of malware samples. Tamper protection notifies you if Windows Defender Antivirus is turned off or tampered with. See Volume 2 Chapter 6 for more information.

20. A. Renaming the default accounts on the new Windows computer is the easiest way to secure the operating system. Configuring Single Sign-On should only be performed if a resource is required that only uses SSO as authentication. Disabling Windows Update will do the opposite of securing the operating system. Disabling Action Center pop-ups will also do the opposite of securing the operating system. See Volume 2 Chapter 6 for more information.

21. B. Encrypted File System (EFS) will encrypt and protect files on an NTFS filesystem without the need of the TPM. BitLocker requires a TPM and therefore will not work. BitLocker To Go is used with removable media. Full drive encryption is used with mobile devices. See Volume 2 Chapter 6 for more information.

22. B. When you set up a device with a Microsoft account, you automatically synchronize files in OneDrive, along with other known folders. Access to the online store can be achieved by launching the Microsoft Store and logging in; it does not require the user to log into the operating system with the Microsoft account. Active Directory authentication is applicable if the workstation is joined to an Active Directory domain. Offline authentication is the default method if you are not signing in with a Microsoft account. See Volume 2 Chapter 6 for more information.

23. C. Graphics cards usually require third-party tools to diagnosis performance problems. RAM problems can be diagnosed inside the operating system from Task Manager. CPU problems can be diagnosed inside the operating system from Task Manager. Network problems can be diagnosed inside the operating system from Task Manager. See Volume 2 Chapter 7 for more information.

24. C. The last step in the malware removal process is the education of the end user to prevent future occurrences of infection. Investigate and verify malware symptoms is the first step in the malware removal process. Enabling System protection is one of the last steps, but it is not the last step. Scheduling scans and running updates should be performed after you have remediated the malware. See Volume 2 Chapter 7 for more information.

25. B. By using the 5 GHz wireless band you can limit the amount of interference from external devices such as microwave ovens and Bluetooth devices. Shortening the SSID will have no effect on intermittent signal drops. Reducing power on the WAP will have no effect on intermittent signal drops. Using the 2.4 GHz band will probably make signal drops worse. See Volume 2 Chapter 7 for more information.

26. A. Moving USB devices around on the USB ports is the first step in remediating USB issues related to resources. USB 2.0 devices should be consolidated to USB 2.0 ports and the same should be done for USB 3.0 devices. You cannot manually allot endpoints for USB devices. Upgrading the drivers for the USB devices will not solve power issues or endpoint issues; drivers for the USB controller should be updated. You cannot manually increase the output amperage for USB devices. See Volume 2 Chapter 7 for more information.

27. D. Normally a dedicated chip called the real time clock (RTC) keeps time for physical machines. However, with virtual machines their RTC is emulated and susceptible to time drift. By default, Windows 10/11 are configured for `time.windows.com` as an NTP. The availability of the NTP server of `time.windows.com` is not any different than physical machines. When deploying virtual machines, the physical RTC is not shared; each VM gets an emulated RTC. See Volume 2 Chapter 7 for more information.

28. C. The Computer Management MMC contains the Services add-in that can be used to restart and configure failed services. Although services can be viewed with the `msconfig .exe` tool, they cannot be restarted with the tool, only enabled and disabled on startup. The Windows Recovery Environment (WinRE) is used to troubleshoot and repair problems offline. The Resource Monitor cannot be used to restart services. See Volume 2 Chapter 7 for more information.

29. C. The risk of thermal runaway is attributed from overheating a lithium-ion battery. When thermal runaway happens, you run the risk of explosion of the battery. Overheating a li-ion battery can cause reduced voltage, but that is not a safety risk. Overheating a li-ion battery will not cause a shock. Overheating of the li-ion battery will cause a shutdown, but that is not a safety risk. See Volume 2 Chapter 7 for more information.

30. A. Event Viewer logs can help identify the cause of frequent shutdowns by showing entries related to Kernel-Boot or Kernel-General. Device Manager manages hardware devices. Task Manager monitors tasks and performance. Windows Update manages updates for the operating system. See Volume 2 Chapter 7 for more information.

31. A. The Remote Desktop Protocol operates on TCP port 3389. The SSH protocol operates on TCP port 22. The VNC protocol operates on TCP port 5900, and HTTPS operates on TCP port 443. See Volume 2 Chapter 8 for more information.

32. C. The line of `REM comment` is used to comment Windows Batch script code. The line of `//comment` is used to comment JavaScript code. The line of `'comment` is used to comment VBScript code. The line of `# comment` is used to comment Bash script code and PowerShell code. See Volume 2 Chapter 8 for more information.

33. B. The command `mstsc.exe` will launch the Windows Remote Desktop Connection utility. From this utility you can remotely connect to a server or other workstation. The command `msra.exe` will launch the Microsoft Remote Assistance utility to allow a trusted helper to remote in to help. The command `quickassist.exe` will launch the Quick Assist remote assistance utility to allow an assistant to remote in to help. The command `ssh.exe` will launch the Secure Shell client that allows you to connect to a Linux/UNIX server or networking equipment. See Volume 2 Chapter 8 for more information.

34. D. The scenario being described is a host-to-site connection. A site-to-site VPN connection will be built between two sites, such as the main site and a branch site. A Remote Desktop connection or SSH connection will not configure a site IP address on the client computer. See Volume 2 Chapter 8 for more information.

35. B. Zoom and Teams are classic examples of videoconferencing software, because they allow multiple attendees and have a form of attendee management. Microsoft Quick Assist is an example of screen-sharing software. Dropbox is an example of file transfer software. Splashtop and TeamViewer are examples of desktop management software. See Volume 2 Chapter 8 for more information.

36. B. Emotional experience is a hallmark of human intelligence that AI does not possess. AI can recognize patterns. AI is excellent at processing large volumes of data. AI can automate tasks. See Volume 2 Chapter 8 for more information.

37. B, D. Long hair or neckties can very easily hold a static charge and damage computer parts. In addition, if you don't have an ESD strap or mat, you should perform self-grounding before you touch sensitive computer equipment. Lower humidity actually increases the risk of ESD. Taking your shoes off will not limit ESD. See Volume 2 Chapter 9 for more information.

38. D. A Torx screwdriver has a splined head used for greater gripping of the screw. A flat-tipped screwdriver as the name suggests has a flat tip. A Phillips screwdriver has a cross used for better gripping over a normal flat-tip screwdriver. An axial screwdriver is not a type of screwdriver. See Volume 2 Chapter 9 for more information.

39. B. Shareware often provides a free trial of a program, with the expectation that you will pay for it if you keep it. Freeware and open source applications are generally totally free, although the developers may ask for donations. Single user software is licensed for one user. See Volume 2 Chapter 9 for more information.

40. B. Typically, a 1 mega-ohm resistor is used to slowly drain or discharge the potential electrostatic discharge (ESD). Magnets are used to hold screws onto screwdrivers. High-voltage probes are used to drain high voltage stored up in electrical circuits and not ESD. Direct grounds are never used because they have the potential for discharge and possible electrocution. See Volume 2 Chapter 9 for more information.

41. C. An air filter mask will protect you from inhaling dust particles. A multimeter is used to measure voltage. Although useful, a flashlight and a mirror are not needed for cleaning dust out of computer equipment. See Volume 2 Chapter 9 for more information.

42. D. Microwave ovens operate in the 2.45 GHz range and can directly affect wireless communications. Magnets can generate a magnetic field, but they will not disrupt wireless communications. Electrostatic discharge (ESD) can kill components, but a discard from ESD in the same space as wireless will not cause interruptions. Electrical surges can affect the equipment running wireless, but it will not affect wireless communications. See Volume 2 Chapter 9 for more information.

43. C. You should text the user and let them know that you need to verify their network connection. You should not log in as another person, with their username and password. You should not log in and stay logged in when you are finished. You should definitely not log in and change the user's password. See Volume 2 Chapter 10 for more information.

44. C. You should be honest and inform the customer of the situation. Explaining the card was broken out of the box is untruthful and deceptive. Installing the card anyway will cause new problems. Leaving everything as is until you can locate a new card, does not help the customer since you've created a new dilemma. See Volume 2 Chapter 10 for more information.

45. B. Communicating the delay and providing a new estimate will help maintain transparency and manage customer expectations. Avoiding informing the customer will not meet the customer's expectations for communicating the timeline. Blaming external factors is deceitful and not professional. Closing the ticket prematurely is an unprofessional behavior because the customer will not have a status on the resolution. See Volume 2 Chapter 10 for more information.

46. A. Email entry is the easiest for end users because it allows them to email the ticketing system and use a method of communication they are familiar with. Portal entry requires the end user to log into the portal, then enter fields in a form to submit a ticket. Manual ticket entry is not done by the end user; it is completed by help desk staff. Application entry is an automated ticket that submits when an application has an issue; this is independent from what the user needs solved. See Volume 2 Chapter 10 for more information.

47. C. Sandbox testing is used to test a change before placing it into production. The process from the sandbox testing will become the primary plan used when the change in production is green-lighted. The backout, also known as the rollback plan is created in the event the primary plan fails in production. The technical evaluation is a technical evaluation of the change, along with a draft of the primary plan. See Volume 2 Chapter 10 for more information.

48. C. The System Properties applet (SYSDM.CPL) allows you to change the computer name and join the system to a domain. The Device Manager is used to manage hardware resources. The User Accounts applet is used to manage user accounts. Credential Manager is used to manage stored credentials. See Volume 2 Chapter 2 for more information.

49. D. The Users tab in Task Manager will allow you to see processes sorted by each user. The Processes tab will show you processes sorted by forefront and background applications. The Performance tab will show you the overall performance of the system with graphical charts. The App History tab will display historical information about processes, such as CPU time. See Volume 2 Chapter 2 for more information.

50. D. Striping with parity is also known as RAID-5, it requires three or more disks and provides fault tolerance. Mirroring is also known as RAID-1 and only requires two disks. Striping is also known as RAID-0 and provides no fault tolerance. See Volume 2 Chapter 2 for more information.

CompTIA A+ Core 2 Exam (220-1202)

VOLUME 2

Chapter 1

Operating System Basics

THE FOLLOWING COMPTIA A+ 220-1202 EXAM OBJECTIVES ARE COVERED IN THIS CHAPTER:

- ✓ **1.0 Operating Systems**
- ✓ **1.1 Explain common operating system (OS) types and their purposes.**
  - Workstation systems (OSs)
    - Windows
    - Linux
    - macOS
    - Chrome OS
  - Mobile OSs
    - iPadOS
    - iOS
    - Android
  - Vendor life-cycle limitations
    - End-of-life (EOL)
    - Update limitations
  - Compatibility concerns between operating systems
- ✓ **1.2 Given a scenario, perform OS installations and upgrades in a diverse environment.**
  - Upgrade considerations
    - Backup files and user preferences
    - Application and driver support/backward compatibility
    - Hardware compatibility
  - Feature updates
    - Product life cycle

✓ **1.3 Compare and contrast basic features of Microsoft Windows editions.**

- Windows 10 editions
  - Home
  - Pro
  - Pro for Workstations
  - Enterprise
- Windows 11 editions
  - Home
  - Pro
  - Enterprise
- N versions
- Feature differences
  - Domain vs. workgroup
  - Desktop styles/user interface
  - Availability of Remote Desktop Protocol (RDP)
  - Random-access memory (RAM) support limitations
  - BitLocker
  - gpedit.msc
- Upgrade paths
  - In-place upgrade
  - Clean install
- Hardware requirements
  - Trusted Platform Module (TPM)
  - Unified Extensible Firmware Interface (UEFI)

✓ **1.10 Given a scenario, install applications according to requirements.**

- System requirements for applications
  - 32-bit vs. 64-bit dependent application requirements
  - Dedicated vs. integrated graphics card

- Video random-access memory (VRAM) requirements
- RAM requirements
- Central processing unit (CPU) requirements
- External hardware tokens
- Storage requirements
- Application to OS compatibility

- Distribution methods
  - Physical media vs. mountable ISO file
  - Downloadable package
  - Image deployment
- Impact considerations for new applications
  - Device
  - Network
  - Operation
  - Business

✓ **1.11 Given a scenario, install and configure cloud-based productivity tools.**

- Email systems
- Storage
  - Sync/folder settings
- Collaboration tools
  - Spreadsheets
  - Videoconferencing
  - Presentation tools
  - Word processing tools
  - Instant messaging
- Identity synchronization
- Licensing assignment

The first half of this book focused mainly on the hardware and physical elements of the computing environment. You learned about physical components, or hardware, of personal computers, laptops, and mobile devices, as well as networking, printers, and troubleshooting procedures. That completes the coverage of the topics on the 220-1201 exam. This chapter marks a departure from that.

In this chapter—and several to come—the focus is on operating systems (OSs). To be specific, the majority of information will be on the Microsoft Windows operating systems, which you must know well for the 220-1202 certification exam. However, the 220-1202 exam also requires basic knowledge of the macOS and Linux operating systems.

Understanding Operating Systems

Computers are pretty much useless without software. A piece of hardware might just as well be used as a paperweight or doorstop if you don't have an easy way to interface with it. Software provides that way. While there are many types of software, or programs, the most important one you'll ever deal with is the operating system. Operating systems have many different, complex functions, but two of them jump out as being critical: interfacing with the hardware and providing a platform on which other applications can run.

Here are three major distinctions of software about which you should be aware:

Operating System The *operating system (OS)* provides a consistent environment for other software to execute commands. The OS provides users with an interface with the computer so that they can send commands (input) and receive feedback or results (output). To do this, the OS must communicate with the computer hardware to perform the following tasks, as illustrated in Figure 1.1:

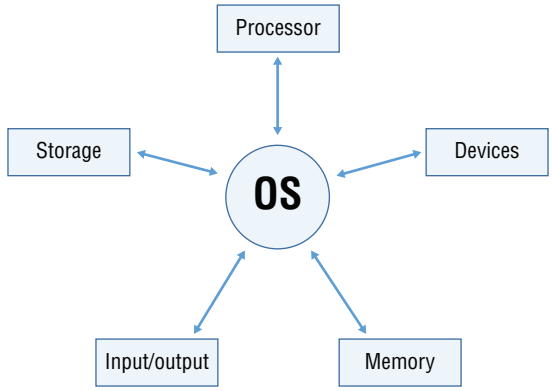
- Disk storage and file management
- Device access
- Memory management
- Output formatting

Once the OS has organized these basic resources, users can give the computer instructions through input devices (such as a keyboard or a mouse). Some of these commands are built into the OS, whereas others are issued through the use of applications. The OS becomes the center through which the system hardware, other software, and the user communicate; the rest of the components of the system work together through the OS, which coordinates their communication.

Application Used to accomplish a particular task, an *application* is software that is written to supplement the commands available to a particular OS. Each application is specifically compiled (configured) for the OS on which it will run. For this reason, the application relies on the OS to do many of its basic tasks. Examples of applications include complex programs, such as Microsoft Word and Microsoft Edge, as well as simple programs, such as a command-line utility. Whether they are complex or simple, when accessing devices and memory, the programs can simply request that the OS do it for them. This arrangement saves substantially on programming overhead because much of the executable code is *shared*—that is, it is written into the operating system and can therefore be used by multiple applications running on that OS.

Drivers *Drivers* are extremely specific software written for the purpose of instructing a particular OS on how to access a piece of hardware. Each modem or printer has unique features and configuration settings, and the driver allows the OS to understand how the hardware works and what it is able to do.

FIGURE 1.1 The operating system interacts with resources.



You can learn about drivers in detail in Volume 1 Chapter 2, "Expansion Cards, Storage Devices, and Power Supplies."

In the following sections, we'll look at some terms and concepts central to all operating systems. Then we'll move into specific discussions of Microsoft Windows operating systems.

Are These the Only Versions of Windows?

In the workplace, it is likely that you will encounter some different versions of the Windows operating systems. Although older versions of Windows are no longer supported by Microsoft, these operating systems can still be found in certain instances. These older operating systems are typically used to support legacy systems, such as kiosks, medical equipment,

and many other applications. Windows 11 has rapidly become the mainstream operating system for both the workplace and home over the past several years, since its release in 2021. Microsoft has scheduled the end of support for Windows 10 in late 2025, as of the writing of this book.

Know that this version of the exam focuses only on the Windows 10/11 operating systems mentioned here, as well as macOS, Linux (both detailed in Volume 2 Chapter 4, "Working with macOS and Linux"), ChromeOS, iOS, and iPadOS. We highly recommend that you become familiar with all of the operating systems your job requires, but these are the ones that you will need to know for the A+ certification exam.

Exam Objective 1.1, "Explain common operating system (OS) types and their purposes," requires you to know the workstation and mobile operating systems that you might encounter as a technician. Be sure to know the differences between each of the operating systems.

Operating System Terms and Concepts

Before we get too far into our discussion of PC workstation operating systems, it will be useful to define a few key terms. The following are some terms that you will come across as you study this chapter and work in the computer industry:

Version A *version* is a particular revision of a piece of software, normally described by a number that tells you how new the product is in relation to other versions of the product.

Source Code The *source code* is the actual code that defines how a piece of software works. Computer operating systems can be *open source*, meaning that the OS can be examined and modified by anyone, or they can be *closed source*, meaning that only an owner or developer can modify or examine the code. As an example, Microsoft and macOS are closed source operating systems and *Linux* is an open source operating system.

A word often used interchangeably with *closed source* is *proprietary*.

Shell A *shell* is a program that runs on top of the OS and allows the user to issue commands through a set of menus or another interface (which may or may not be graphical). Shells make an OS easier to use by changing the user interface. File Explorer is the name of the shell that has been used with the Windows operating system since Windows 95.

Graphical User Interface A *graphical user interface*, or *GUI*, is a method by which a person communicates with a computer using graphical images, icons, and methods other than text. GUIs allow a user to use a mouse, touchpad, or another mechanism (in addition to the keyboard) to interact with the computer to issue commands.

Network A *network* is any group of computers that have a communication link between them. Networks allow computers to share information and resources quickly and securely.

Cooperative Multitasking *Cooperative multitasking* is a multitasking method that depends on the application itself to be responsible for using the processor and then freeing it for access by other applications. This is the way very early versions of Windows managed multiple applications. If any application locked up while using the processor, the application was unable to free the processor to do other tasks and the entire system locked, usually forcing a reboot.

Preemptive Multitasking *Preemptive multitasking* is a multitasking method in which the OS allots each application a certain amount of processor time and then forcibly takes back control and gives another application or task access to the processor. This means that if an application crashes, the OS takes control of the processor away from the locked application and passes it on to the next application, which should be unaffected. Although unstable programs still lock, only the locked application will stall, not the entire system. Preemptive multitasking is what is used today in modern operating systems.

Multithreading *Multithreading* is the ability of a single application to have multiple requests in to the processor at one time. This results in faster application performance because it allows a program to do many things at once.

32-Bit A *32-bit operating system* is one that can not only run on a 32-bit processor but can utilize the capabilities of the processor fully. While this may sound simple, the truth of the matter is that it took many years after the 32-bit processor became available before operating systems (which were 16-bit at the time) were able to utilize their features. You cannot execute 64-bit software on 32-bit operating systems; the instructions sets are functionally different.

64-Bit A *64-bit operating system* is one that is written to utilize the instructions possible with 64-bit processors. Originally, these were more common with servers than desktops, but they have now become ubiquitous in the market with both Intel and AMD processors. As mentioned earlier, you cannot mix 64-bit software with 32-bit hardware (but you can run most 32-bit software on 64-bit hardware).

Exam Objective 1.10 specifically calls out 32-bit vs. 64-bit dependent application requirements.

x86 The term *x86* is commonly used to refer to operating systems intended to run on the Intel processor. Intel initially identified its 32-bit processors with numbers ending in 86 prior to switching to the Pentium line.

x64 The term *x64* is commonly used to denote operating systems that can run on 64-bit processors. This is also commonly referred to as *AMD64*, because AMD defined the 64-bit instruction set used today.

ARM The term *ARM* is commonly used to denote operating systems that can run on Advanced RISC Machine (ARM) processors. ARM operating systems use a reduced instruction set computing (RISC) instruction set commonly found on tablet and phone processors. ARMv8-A processors can support 64-bit operating systems, so you will find 32-bit and 64-bit operating systems.

Random Access Memory The term *random access memory (RAM)* is used to describe the amount of memory installed and accessible for an operating system. The operating system uses RAM to hold its operating system code, as well as the programs you open on the operating system. Each program you install will have a minimum RAM requirement and recommended RAM requirement. Most computer systems on the market today are sold with at least 4 gigabytes (GB) of RAM. PC manufacturers have identified that newer programs are requiring more and more RAM to operate properly, so upgrades of 8 GB of RAM are common and have become the new normal.

A 32-bit operating system has the limitation of addressing only 4 GB of RAM. Most new computer systems on the market today come with a preinstalled 64-bit operating system. The 4 GB limitation will present itself if you are re-installing a computer system that has more than 4 GB of RAM and use a 32-bit version of the operating system. You will find out quickly that you just downgraded the computer.

Operating system vendors implemented a Physical Address Extension (PAE) workaround for the 4 GB limit. For the purposes of the exam, however, you should be aware that 32-bit operating systems have a hard limit of addressing 4 GB of RAM.

Operating Systems

When we think of an operating system, Windows or macOS is probably the first that comes to mind. These operating systems comprise only one category of operating system. We use other operating systems, such as Android and Apple iOS, and don't even realize how much they have become part of our daily life. The CompTIA exam recognized these various operating systems and integrated them into the 220-1202 exam.

Operating System Categories

An operating system category defines the use and function of both the operating system and the hardware. All operating systems fit into one of four different broad categories: *server*, *workstation*, *mobile*, or *cloud-based* operating systems.

Server Server operating systems allow for the sharing of information and applications by workstation operating systems. The operating system is optimized for background processes used for access by clients, as opposed to the foreground processes, such as the GUI.

Workstation Workstation operating systems allow end users to access information and applications on server operating systems as well as independently run applications locally. The operating system is optimized for foreground processes, such as the GUI. Background sharing is usually limited to a specific number of users; workstation versions of Microsoft Windows are limited to 10 concurrent users.

Mobile Mobile operating systems are found on mobile devices such as phones and tablets. The mobile operating system is generally optimized for touch-based devices where one program is used at a time.

Cloud-Based Cloud-based operating systems are a new breed of operating system that has emerged from the mobile computing era. Cloud-based operating systems are also considered stateless operating systems, since personal data is not primarily stored on the device but in the cloud. Mobile operating systems can be considered cloud-based operating systems. So this category takes on a dual role of defining a category and describing where the data is stored.

Common Operating Systems

Now that you understand the various categories of operating systems you may encounter, we will explore the common operating systems you will find in your everyday work as an A+ technician.

Microsoft Windows The Microsoft Windows operating system is one of the most popular operating systems adopted by business and personal users. Windows 11, the current version, is considered a workstation operating system.

Apple Macintosh OS *macOS* is the current version of the operating system for Apple Macintosh laptop and computer hardware platforms. Ironically, macOS is the original operating system name from back in the 1980s. Between 1999 and 2015, the macOS was called Mac OS X. The macOS operating system is considered a workstation operating system.

Linux Linux was created as a free, open source operating system in 1991 by Linus Torvalds. Linux was created to mimic the design of the Minix (mini-UNIX) operating system, which was a minimal version of UNIX that was originally used for academic purposes. Today, the Linux kernel (core of the OS) remains a free, open source platform upon which many different Linux distributions have been created, such as Ubuntu, Red Hat, Fedora, and Debian, just to name a few. The Linux operating system is considered both a workstation operating system and a server operating system, depending on its installation and configuration.

Microsoft Windows Tablet In 2012, with the release of the tile-based Windows 8 operating system, Microsoft released the Surface tablet. Today the Surface tablet runs the Windows 11 operating system and uses an ARM-based processor. It competes on the market with other *Ultrabooks* (subnotebook computers), such as the iPad Pro. Newer versions of the Surface tablet even have a neural processing unit (NPU) for artificial

intelligence (AI) processing. The operating system itself is a full version of Windows 11, and it's considered a mobile operating system.

Android Google created the *Android* operating system as a tile-based mobile operating system. Android was originally released in 2008 with HTC phones; the release of Android tablets followed in 2011. Android-based phones and tablets have become quite popular in the mobile operating system market. The popularity of Android is attributed to the application ecosystem that the Google Play store creates. The Google Play store enables users to purchase, download, and install applications, books, music, and video for the Android operating system.

iOS Apple created the iOS operating system as a tile-based mobile operating system. Originally released on the first iPhone in 2007, the operating system was known as the iPhone Operating System (iOS). The release of the iPod Touch followed a few months later, and it also furnished the iPhone Operating System. In 2010, the first iPad was released, and it furnished the iPhone Operating System as well. Today, the operating system's name has been shortened to iOS; the current version, as of this writing, is iOS version 18. Apple iOS is a popular mobile operating system in the marketplace today. Its popularity is attributed to Apple's strong support for the accompanying hardware. The Apple App Store also creates a strong ecosystem for downloading, installing, and purchasing applications, books, music, and video for the iOS operating system.

iPadOS *iPadOS* version 13.1 was released in the fall of 2019 for the Apple iPad. iPadOS is a variant of the popular Apple iOS with better multitasking features and a better home screen. This allows the Apple iPad to behave more like a light-weight laptop than a tablet. The Apple iPad Pro can be expanded with a keyboard, and has built-in AI called Apple Intelligence. This makes it a real competitor in the Ultrabook market.

ChromeOS *ChromeOS*, formerly styled as Chrome OS, is an operating system that is designed around the Chrome web browser by Google. ChromeOS was released in 2011 on the hardware platform known as the Chromebook, which is considered one of the first Ultrabooks. ChromeOS is a true cloud-based operating system, since all its data is stored in the Google Drive cloud-based storage system. With the initial release of the Chromebook, workstation-type devices called Chromeboxes had also been released to market. Applications, which are really just web plug-ins, can be installed from the Chrome Web Store. The applications are written in HTML, HTML5, CSS, and JavaScript, which allow cross-platform support between ChromeOS and the Chrome web browser installed on a PC or Mac.

Stateless Operating Systems

The lines have blurred in the past few years with the rapid adoption of the cloud for both enterprise and consumer installations. The workstation-based operating systems, such as Windows 11 and macOS, have embraced the cloud and to some extent have become

stateless devices like mobile-based operating systems. Other devices like ChromeOS are strictly cloud-based devices and contain no local filesystem, yet act as workstations for cloud-based applications.

In the enterprise, an operating system can be set up with all the applications a user needs to do their job, along with access to all their files. The installation of applications and the configuration of the applications is managed by mobile device management (MDM) software, such as Intune. Then, if the device breaks, is stolen, or is upgraded to new hardware, a new device can be provisioned in the MDM software and everything gets set up just like their previous device.

Consumer installations have also benefited from the adoption of the cloud. When a consumer purchases a new device to replace their old device, all they have to do is sign into the cloud service and their files and applications will automatically be transferred over and installed. The adoption of the cloud and the new behavior now makes traditional operating systems act more like stateless mobile-based operating systems.

Exam Objective 1.2, "Given a scenario, perform OS installations and upgrades in a diverse environment." Be sure to understand the operating system life cycle in the following section.

Operating System Life Cycle

An operating system life cycle begins when an operating system is introduced and ends when the operating system is no longer supported. As a computer technician, you should pay close attention to an operating system's life cycle, because the *end-of-life (EOL)* date for an OS means that it will no longer receive updates.

When an operating system is considered end-of-life, newer features will not be added to the OS in the future. More importantly, security updates will no longer be offered, which will put your operating system and information at risk of compromise.

The network administrator should pay close attention to the dates an operating system is considered end-of-life. If the organization has a support contract for problems, the support of the operating system at the end-of-life could no longer be honored. Of course, the lack of security updates for a corporate network is more concerning, since the stability of an organization can be compromised.

It has been common practice for network administrators to skip versions of every other operating system. This is mainly attributed to the *long-term support* for the current operating system and the amount of work required to upgrade an entire organization's operating systems. In addition to these factors, many hardware vendors support the current operating system plus the last release. Administrators can purchase a new laptop or PC and request the prior operating system. However, once the next operating system is released, the support for the old operating system will force an upgrade.

When Windows Vista was released, Windows XP was the dominant corporate operating system. Windows Vista was not highly adopted, but when Windows 7 was released, many administrators planned on the highly anticipated upgrade to Windows 7. We've seen the same trend with Windows 10; many administrators completely skipped rolling out Windows 8 entirely. Windows 11 is the current operating system and there is no way to tell what will happen in the future. Microsoft has changed its naming convention for Windows and released platform upgrades as feature releases. This has made it easier for administrators to adopt the new Windows version.

The following common terminology is used by operating system developers in relation to the life cycle of an OS:

Alpha An operating system in alpha is a first draft of an operating system. Operating systems that are available in alpha are considered previews. It is common to find these in open source communities, such as Linux operating systems, but it is uncommon to find a corporate-based operating system in alpha.

Beta An operating system in beta is a second draft of an operating system. This stage of an operating system means most of the bugs have been discovered and are avidly being patched. This stage is typically used to discover missing features and new bugs. Some operating system developers require an opt-in to run a beta operating system, and some require invites to a beta program that mandates participation.

Release Candidate An operating system that is deemed a release candidate is in the final stage before it is released to market. During the release candidate stage, administrators are urged to install and use the operating system as though it were the final release. The benefit is that the administrator can get a previewed look at the operating system while helping to polish the final product, also called the *release to market (RTM)*. An operating system in this stage is generally stable and has relatively few bugs.

Current A current operating system, as the term states, is the current release of an operating system. Many operating system developers are on a one- to two-year release cycle for the future version of the current operating system.

Service Pack Although the term *service pack* is generally associated with updates, it is also used to describe a milestone in the life cycle of an operating system. When a service pack is released for an operating system, it adds major features as well as patches for both security and functionality. Microsoft has replaced this term with the introduction of Windows 8; it is now just called an update or Windows Update. However, many other operating systems still use "service pack."

Mainstream Support During the *mainstream support* of an operating system, all *hotfixes*, *security updates*, *feature updates*, and general support for the operating system is supported. An operating system developer will usually have a predetermined mainstream support date when the operating system is released. When the mainstream support ends

for an operating system, it is either no longer supported or enters into an extended support period.

Extended Support The *extended support* period, sometimes called the *long-term support (LTS)* period, for an operating system is the final state of an OS's life cycle. For example, during this time, hotfixes and features will no longer be supported, but critical security updates will be supported with Microsoft operating systems. Additional contract purchases may be required if hotfixes are required during this period. An administrator should have all systems upgraded to the current operating system by the end of the extended support date.

Microsoft has announced end of support date of October 14, 2025 for Windows 10 Home and Pro. The end of support for Windows 10 (21H2) LTSC is January 12, 2027. You can read about other versions of Windows 10 at `https://learn.microsoft.com/en-us/windows/release-health/release-information`.

Exam Objective 1.3, "Compare and contrast basic features of Microsoft Windows editions," requires you to know the hardware requirements for Windows 10/11. Be sure to know the various hardware requirements.

Minimum System Requirements

In the upcoming chapters, we'll explore how to install and upgrade each of the operating systems that you need to know for the exam. However, the hardware requirements of the operating system that you are thinking of installing can prevent you from even considering these options. Before you can begin to install an OS, you must consider several items. You must perform the following tasks before you even start to think about undertaking the installation:

1. Determine hardware compatibility and *minimum requirements*.
2. Determine installation options.
3. Determine the installation method.
4. Prepare the computer for installation.

Let's begin our discussion by talking about hardware compatibility issues and the requirements for installing the various versions of Windows.

Determining Hardware Compatibility and Minimum Requirements

Before you can begin to install any version of an operating system, you must determine whether it supports the hardware that you will be using. That is, will the version of the operating system have problems running any of the device drivers for the hardware that you have?

To answer this question, operating system vendors have developed several versions of *hardware compatibility lists (HCLs)*. An HCL is a list of all the hardware that works with the operating system. Microsoft published HCLs for many prior versions of Windows. Since the release of Windows 10, the HCL has disappeared completely and it is now the responsibility of the vendor to certify the compatibility of the hardware with Microsoft. However, many other operating system vendors still have a recommended HCL.

The point is, before installing an operating system, you should check all your computer's components against an HCL or the manufacturer of the component to ensure that they are compatible with the version of operating system you plan to install. If a product is not on the list, that does not mean it will not work; it merely means it has not been tested. The list represents tested software and hardware that vendors have stated are compatible, but it is by no means all-inclusive.

In addition to general compatibility, it is important that your computer have enough "oomph" to run the version of an operating system that you plan to install. For that matter, it is important for your computer to have enough resources to run any software that you plan to use. Toward that end, Microsoft (as well as other software publishers) releases a list of both minimum and recommended hardware specifications that you should follow when installing Windows.

"Minimum specifications" are the absolute minimum requirements for hardware that your system should meet in order to install and run the OS version you have chosen. "Recommended hardware specifications" are what you should have in your system in order to realize usable performance. Always try to have the recommended hardware (or better) in your system. If you don't, you may have to upgrade your hardware before you upgrade your operating system if you're running anything beyond a minimal environment. Table 1.1 lists the minimum hardware specifications for Windows 10 and Windows 11. Note that in addition to these minimum requirements, the hardware chosen must be compatible with the selected version of Windows. Also be aware that additional hardware may be required if certain features are installed (for example, a fingerprint reader is required to use biometric logins). Windows 11 is only available in a 64-bit version and requires a Trusted Platform Module 2.0 and UEFI for added security of the operating system and its applications.

TABLE 1.1 Windows 10 and Windows 11 minimum system requirements

Operating System	Windows 10	Windows 10	Windows 11
Architecture	32-bit	64-bit	64-bit
Processor	1 GHz or faster processor or System on a Chip (SoC)	1 GHz or faster processor or System on a Chip (SoC)	1 GHz or faster processor with 2 or more cores or System on a Chip (SoC)
Memory	1 GB	2 GB	4 GB
Free hard disk space	16 GB	32 GB	64 GB

Operating System	Windows 10	Windows 10	Windows 11
Graphics card	Microsoft DirectX 9 or later graphics device with WDDM 1.0 driver	Microsoft DirectX 9 or later graphics device with WDDM 1.0 driver	Microsoft DirectX 12 or later graphics device with WDDM 2.0 driver
Display	800 × 600	800 × 600	High-definition 720p
Additional hardware	N/A	N/A	UEFI & TPM 2.0 required

If there is one thing to be learned from Table 1.1, it is that Microsoft is nothing if not optimistic. For your own sanity, though, we strongly suggest that you always take the minimum requirements with a grain of salt. They are, after all, *minimum* requirements. Even the recommended requirements should be considered minimum requirements. The bottom line is to make sure that you have a good margin between your system's performance and the minimum requirements listed. Always run Windows on more powerful hardware rather than less!

Other hardware—sound cards, network cards, modems, video cards, and so on—may or may not work with Windows. If the device is fairly recent, you can be relatively certain that it was built to work with the newest version of Windows. If it is older, however, you may need to find out who made the hardware and check their website to see if there are drivers available for the version of Windows that you are installing.

There's one more thing to consider when evaluating installation methods. Some methods work only if you're performing a clean installation, not an upgrade.

Understanding Applications

An operating system without any applications installed on it isn't very useful. Even back when Windows 1.0 was released 30 years ago, it came with applications, such as Minesweeper, a file manager, and a simple word processor. In this section, we will explore installing applications for Windows 10/11. However, these concepts can be applied to any operating system.

Exam Objective 1.10, "Given a scenario, install applications according to requirements," requires you to know the various considerations for installing applications. You should understand application requirements, distribution methods, and impact considerations for new applications.

System Requirements for Applications

Before installing an application, you must collect information on the requirements of the application to make sure that your system can satisfy the requirements. If your system cannot satisfy the requirements for the application, you may have to upgrade the system. You can find application requirements on the vendor's website. For example, if you need to install the Microsoft Power BI Desktop application, you would search Microsoft's site to find the system requirements for this particular application. The documentation would reveal several different requirements, such as the following:

Memory The most common requirement that needs to be met before installing an application is the memory requirement. Memory is often displayed in the form of total random access memory (RAM) for the operating system.

CPU The requirement for the *central processing unit (CPU)* required for an application is often displayed in the form of total CPU processing power of the system. It is often expressed in the form of GHz, and other requirements can include the number of physical CPUs, the number of cores, and even the model of processor.

Architecture The architecture of the CPU and the operating system installed needs to match the application requirements for 32-bit or 64-bit applications. The architecture must meet or exceed the requirements of the application. For example, if the application needs a 64-bit architecture, then the CPU and operating system need to support 64-bit operations. However, if the application requires a 32-bit architecture, then a 32-bit or 64-bit CPU and operating system can accommodate the requirement.

Storage The requirement for the storage is often displayed in GB or TB, depending on the application. The storage requirement is often the free space required to install the application. However, some software packages that require the storage of user data also display a separate storage requirement for the data.

Prerequisites A common prerequisite to install an application might be a specific version of the .NET Framework. However, some applications have specific prerequisite requirements, such as a Microsoft SQL Server for data storage.

Display When data is to be displayed in a specific format, the application designer has formatted the application to a minimum display area. This requirement is often in the form of pixels, such as 1024 × 768. Although pixels are the most common display requirement, another type of requirement is a graphics processing unit (GPU). A specific GPU may be required if the application will process video or graphics.

Video Random Access Memory (VRAM) VRAM requirements are typically related to applications that are graphic intensive, such as gaming. VRAM is typically much faster than conventional RAM, because it must be accessed faster to sustain the higher bandwidth requirements for the graphics applications being used.

Built-in Video Cards vs. Dedicated Some video cards that are built into the motherboard of the computer share VRAM with the RAM of the system. These video cards are much slower because they depend on the conventional RAM. These video cards also consume valuable RAM that is typically dedicated to applications. The RAM on the motherboard is divided between the CPU and video card and can be configured through the Basic Input/Output System/Unified Extensible Firmware Interface (BIOS/UEFI) interface. Typically, you can disable these built-in videos cards when you install a dedicated video card. Dedicated video cards are always preferable if you are going to use video. The built-in video card is typically an added bonus many vendors include, and typically they are never used.

Licensing When installing an application, the last step is usually licensing the application. The license can come in several different forms, such as a node-locked license tied to a MAC address or drive serial number. A network-based license server allows for concurrent use by several people at once. External hardware tokens license an application with the use of a USB dongle or hardware device that allows the program to run.

Licensing is covered in detail in Volume 2 Chapter 9, "Safety and Environmental Concerns."

Other In addition to the common requirements, other requirements must be met before an application can be installed. The operating system requirement is one such requirement. Every application will have a minimum operating system on which it can be installed and supported. Other requirements can be for specific hardware, licensing, or drivers.

Additional Application Requirement Considerations

When an application's requirements are published, they are the bare minimum requirements, not the optimal recommended specifications. The requirements are a conservative estimate so that a potential customer is not scared away with expensive hardware upgrades. When extremely low requirements are advertised, we always recommend calling a presales support person. You should describe how the application will be used and seek their recommendations for optimizing the application's performance. You will often find that they are double or triple the advertised bare minimum requirements.

Storage is probably the biggest consideration when evaluating the installation of an application that will store user data. It is extremely difficult to gauge how much storage should be set aside for the application for user data. If too much storage is allocated, the precious commodity might go unused for the life of the application. If too little is allocated, you may have to upgrade the storage—or, worse, you could fill the entire drive up and be forced to do it in a panic.

The most common architectures for CPUs are 32-bit and 64-bit. They are noted as x86 for 32-bit and x64 or AMD64 for 64-bit. The 64-bit CPU extension was adopted from the AMD version of their 64-bit CPU. When installing an operating system, the operating system

requirements must be met or exceeded for the CPU architecture type. If you are installing a 64-bit operating system, then you must have at least a 64-bit CPU. If you are installing a 32-bit operating system, you can install it on a 32-bit CPU or 64-bit CPU. However, if you have a 64-bit CPU it always makes sense to install a 64-bit operating system, since it gives you maximum flexibility and you typically can't upgrade to a different architecture.

Operating system compatibility is a consideration for older applications that are still required by organizations. Application compatibility has been incorporated into the Windows operating system since Windows XP. It allows for an application to behave as though the operating system was an older operating system. For example, over the years, Windows has become more restrictive of local permissions on the filesystem. If an application that was written for an older operating system expects to write to a Registry key, application compatibility will allow it to think it's directly writing to the location.

Application Installation

Once the requirements are met for an application and the considerations are investigated, you are ready to install the application. You must now consider how the application will be installed; the number of machines on which the application will be installed will factor into this consideration. This section discusses several different ways that an application can be installed.

Physical Media

If the installation is a one-off installation, then a CD/DVD drive might be your best option. In recent years, applications have even been shipped on Blu-ray media. If you must install the application on several different PCs, then this method may not be the preferred installation method. When installing from optical media, even the fastest optical drive is slow compared to other methods, such as USB flash drives.

ISO Mountable

Although optical discs have been around since the mid-1990s and have been the most popular method of installing applications and operating systems, the optical disc is quickly becoming a relic. When you use a virtualization product such as Hyper-V or VMware Workstation, the optical disc is just too clumsy and slow to use. Mountable ISO images of the physical media has become the new norm. When you need to install an operating system or application, you simply download the media from the vendor, mount the ISO, and install it as if you had a virtual optical drive. ISO sizes will vary from 500 MB to 9.6 GB with normal CD and DVD formats, but Blu-ray discs can be up to 45 GB.

USB Drive

Applications are outgrowing optical media such as CD-ROM and DVD-ROM, so USB drives have become popular. USB drives are faster and bigger than optical media. If a handful of computers require the application, then this might be a better option. However, the disadvantage is that simultaneous installations are limited to the number of flash drives you have

with the application loaded. Another common problem with USB drive installations is that the USB drives are lost from time to time or inadvertently overwritten. For this reason, many application vendors lock their drives so they can't be overwritten and repurposed.

Downloadable Media

Optical and USB media are slowly being replaced, by online market places like Steam for games, and Amazon for software, just to name a few. These market places allow for a downloadable package that can be installed. Some of these market places will even install the application for you from a digital library. These ecosystems are very common with video games and every major video game vendor has their own platform for software delivery.

The software box still has a place in retail, because it satisfies a consumer's need for a tangible product when walking out of the store. However, many software titles sold in a box, will have a digital download code inside.

Image Deployment

Typically, when you have a group of computers that require the same software packages to be installed, it's faster to deploy a software image. A software image is nothing more than a digital image of an operating system with all the software packages installed. Once everything is installed, the image needs to be prepared for deployment with a tool called sysprep. You can then use a tool like Deployment Image Servicing and Management (DISM) to create a digital image of the hard drive. The deployment of the image can be done a number of ways, such as USB, Windows Deployment Services (WDS), Microsoft Deployment Toolkit (MDT), and even DISM. A complete chapter could be dedicated to deployment techniques.

Network Installation

When you need to install an application on many different PCs, a network installation should be your first choice. The application is typically uploaded to a file share by the administrator, and then the file share is set to read-only access for the user who is performing the installation. Depending on the speed of the network, this could be the fastest method to install an application on several different PCs simultaneously.

There are several different methods for deploying an application over the network. Each method depends on the number and location of the computers.

User-Initiated Installation

These installations are used when the administrator of the PC will start the installation manually. This method is preferable when the administrator is expected to answer specific questions during the installation, such as where to install the application.

The installation of any application generally requires that the user be an administrator of the operating system or have elevated privileges to install an application. If user-initiated installation is chosen as the method for deploying an application, you must be sure that the user account has the appropriate permissions to install the application.

Automated Installation

Two different types of automated installations can be employed by the administrator: push installations and pull installations. Either installation type is used when the conformity of the installation is required.

Automated installation products, such as Microsoft Endpoint Configuration Manager (MECM), can deploy applications to multiple PCs with a push installation. Microsoft MECM, is considered the Swiss Army knife of installation and reporting services. Each client in the network will have the MECM agent installed prior to the push installation of the application. The agent will then be responsible for reporting on the current operating system. This allows you to collect information for determining if you can satisfy the requirements of the application. The agent can then be used as the contact point for the push installation of the application.

Group Policy can also be used to automate the installation of applications. The Group Policy method is a pull-based installation method, where the client will pull the application from the network share. This method contains no agent, so reporting is not available for client resources, installation requirements, or installation status. The benefit is that it requires very little infrastructure if you have Microsoft Active Directory (AD) installed already.

Regardless of the installation type, push- or pull-based, most automated installation methods do not require the user logged in to be an administrator. MECM can be configured to use a system account that has elevated privileges in the target operating system to install the application. Group Policy pull-based installations can also be configured to use a system account that has elevated privileges to install the application.

Push installations are installations in which the administrator can push an application out to clients. Pull installations are installations in which the client pulls, or queries, for the application to be installed. Group Policy application deployment is a pull type of installation, since the client is responsible for pulling the application and installing it. MECM is a push type installation method. For more information about MECM, visit:

`https://learn.microsoft.com/en-us/mem/configmgr/core/understand/configuration-manager-faq`.

Security Considerations

When an application is installed in the operating system, the overall security of the operating system can potentially be compromised. Of course, it is not the intent of the application to weaken the security of the operating system, but vulnerabilities in the application can exist. Application developers realize that their code is not perfect and that vulnerabilities can exist, so many applications now include self-updating capabilities.

Applications weakening the security of the local device should not be your only concern. Applications can actually weaken the security of the entire network, especially if they are not updated frequently. Applications that operate over the network with client-server

functionality pose the largest security risks, since they allow for remote exploitation of vulnerabilities. One measure that can be employed with network applications that are not updated frequently is to firewall the services. There are many other active measures, which we will discuss in further detail in Volume 2 Chapter 6, "Securing Operating Systems."

Other Considerations for New Applications

Once the requirements of the application are satisfied, the security of the application is evaluated, and an installation method is chosen; several other considerations should be reviewed before installing a new application. These considerations must be evaluated to ensure that the installation of the application does not impose unintended problems on the organization and network.

Impact to Device Although the impact to the device should be minimal if the application requirements are satisfied, the impact should still be considered. For example, if you install an application that constantly logs, you could find that the hard drive will fill up quickly and other applications could run more slowly due to the drive activity. Although this is an extreme example, other impacts may be more subtle and manifest over a longer period of time.

Impact to Network The impact to the network should also be considered. The bandwidth the application needs to operate, and consequently uses, should be evaluated. Although an application at first glance might not appear to use a lot of bandwidth, as more demand is placed on the application, bandwidth will increase. For example, if you installed a network video recorder (NVR), a few cameras might not use too much network. However, as more and more cameras are added, bandwidth usage can sharply increase. If the bandwidth traverses a wide area network (WAN), the application could quickly use all the available bandwidth.

Impact to Operations Depending on the type of application being installed, the operations of the operating system, or the information system the operating system serves, can be negatively impacted. Therefore, the impact to operations should always be considered before installing an application. For example, if you were to install an Office package on a point-of-sale (POS) system, you could negatively affect the ability of the system to process sales.

Impact to Business Some applications can have an impact on the business or organization itself. Therefore, the impact on day-to-day business should be evaluated. Examples of applications that could affect business are human resource (HR), customer relationship management (CRM), and sales systems, just to name a few. If these applications were to be changed, they could impact the organization.

Cloud-Based Productivity Tools

The cloud was a buzzword 10 years ago and slowly gained traction, as administrators held tightly onto their on-premises servers with skepticism. Since the COVID 19 pandemic of 2020, the cloud is no longer a buzzword. During the pandemic every administrator who had not embraced the cloud found themselves rapidly deploying cloud-based applications.

Today, every system put in place should be considered for the cloud—not because we expect another pandemic, but because the work force has forever changed! Pandora's box is open, and organizations have found out that people are actually productive when they work from home (WFH). Working from home also benefits the employee and affords a greater degree of life-balance. No more worrying that the kids will get off the bus and come home to an empty house, since you are working from your home office. Many jobs today are considered hybrid, meaning the employee works from home a few days a week and also comes into the office a few days a week. Some jobs are completely remote!

All of this means that at some point your organization needs to embrace cloud-based productivity tools. The following are the most common productivity tools you may need to support. We will focus primarily on Microsoft 365 cloud-based tools in this section, mainly because they are the most common. However, keep in mind that they are not the only tools out there in the marketplace.

Exam Objective 1.11, "Given a scenario, install and configure cloud-based productivity tools," requires you to know the various cloud-based productivity tools in the following section.

Cloud-Based vs. On-Premises

The great debate of cloud-based versus on-premises productivity software centers on flexibility, cost, and control. The flexibility of on-premises systems is only as good as your Internet connection and how you have deployed the system. If you've deployed the email server to a single server, it is possible that demand from the outside of the organization could impact users inside the organization. This could be normal usage or a threat actor impacting the email system. Alternately, if you have a poor Internet connection, users outside could have a poor experience while internal users are fine.

Cost is another consideration in the debate between cloud-based and on-premises systems. Typically, when you set up an on-premises server there is a one-time cost for the infrastructure, such as the server and the software. Keeping in mind that, although your organization might use virtualization, there is a still a shared cost for the CPU, RAM, and disk space. After the initial costs, there might be a support contract or licensing that is yearly.

When we purchase a cloud-based service, we are renting the infrastructure. Typically cloud-based services are charged per-user/per-month, with a discount for committing to a full year and paying up front. The positive is you don't need to purchase servers or tap into precious virtualization resources. Also, most cloud-based systems include basic support costs. The negative is that you need to pay a larger amount each year you use the service.

The last consideration is control. When we use a cloud-based service we give up control to the vendor; this could be good or bad. If you need complete control over the version of the system and how the system is configured, then cloud-based might be a bad decision. However, updates and new versions means value for the customer. Another big benefit is security; when applications become cloud-based, you no longer need to worry about patching the software. The vendor will typically have a continuous patch cycle and offer release information on new

features in scheduled releases. Organization can realize value from cloud-based systems, when they simply don't have the expertise on hand to maintain the systems.

Cloud-Based vs. Traditional Software

Cloud-based software is typically delivered via a standard web browser and requires very little configuration when deployed in this fashion. The storage and processing are all performed on the remote servers—only the user interface is rendered on the client. The only processing on the client is the sorting or presentation of data.

Traditional software typically needs to be installed on the operating system, and the install is specific to the application. Storage and processing are typically all done on the local operating system. In some cases, you can point the application to a centralized server for the storage of information.

Cloud-based software that is only used via a web browser can be cumbersome. Therefore, today's cloud apps allow a traditional install of the application that points to the cloud for storage. The cloud application might also allow a web interface. This is the case with the Microsoft 365 suite of products; you can install the Microsoft 365 applications or you can use them through the `Office.com` web portal. Keep in mind that the applications installed on the operating systems typically have more advanced features then their web-based browser counterpart.

This marriage of web-based cloud app and traditional installation gives you tremendous flexibility. You can utilize a Google Chromebook to do day-to-day tasks via the web browser, then a Windows computer for specific tasks that require the full version. Since the storage and format of the document are the same, you can literally pick right back up where you left off, regardless of which platform you are on. This also means you can use the Microsoft 365 applications on computers where you might not have administrative rights to install software and just need access to a web browser.

Cloud-Enabled Applications

Connectivity is a consideration when using cloud-based applications. You need access to an Internet connection to use them. This is where a traditional installation or a cloud-enabled application is a benefit. In the following sections, we will explore two cloud-enabled applications that are installed locally on your computer. They both allow offline use and then communicate with the cloud when Internet is restored.

Exchange Online and Outlook

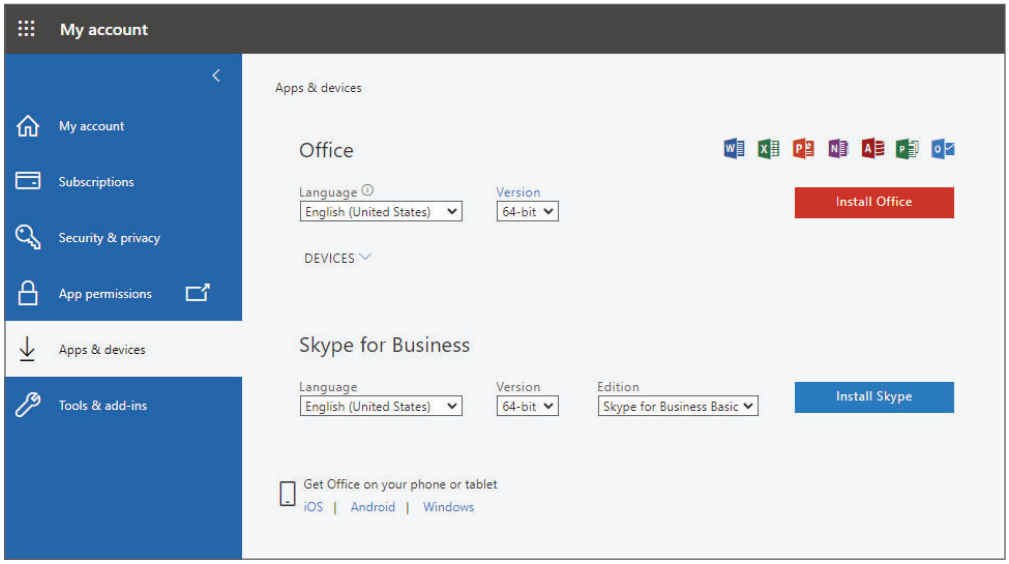
Microsoft Exchange has been the foundation of the Microsoft email and collaboration services since its initial release in 1993. The on-premises Exchange Server offered organizations a robust solution for email, calendaring, contact management, and overall communications. However, with the rise of cloud computing, this led to the launch of Exchange Online in 2008 as part of Microsoft's Business Productivity Online Suite (BPOS) and later integrated into Office 365 (now Microsoft 365). Unlike traditional on-premises Exchange Server, Exchange Online operates entirely in the cloud. This eliminates the need for complex local

infrastructure while providing seamless updates, scalability, and advanced features like AI-driven insights, enhanced security, and compliance tools.

Microsoft Outlook was introduced in 1997 as part of the Office 97 suite. It quickly became the keystone of personal and professional email. Initially designed as a desktop application, Outlook provided users with a unified platform for email, calendars, tasks, and contact management, primarily connecting to on-premises servers like Microsoft Exchange. Over time, Outlook evolved with features such as integration with mobile devices, advanced search, and shared calendaring, just to name a few. Today, paired with Exchange Online, Outlook has transcended its desktop roots to offer a robust cloud-enabled application.

Just like its initial debut with the install of Office 97, today it is installed as an integral part of the Microsoft 365 Apps suite. The Microsoft 365 Apps suite is even preinstalled in Windows 11 by default; all you need to do is sign in with your Microsoft Account credentials. If you need the installation of Microsoft 365 Apps, you can visit `www.microsoft365.com` and click Install And More, then select Install Microsoft 365 Apps. This will take you to the familiar dialog shown in Figure 1.2. The executable that downloads will install the latest Microsoft 365 Apps (Office) applications of Word, Excel, PowerPoint, OneNote, Access, Publisher, and of course Outlook.

FIGURE 1.2 Microsoft 365 Apps (Office) download



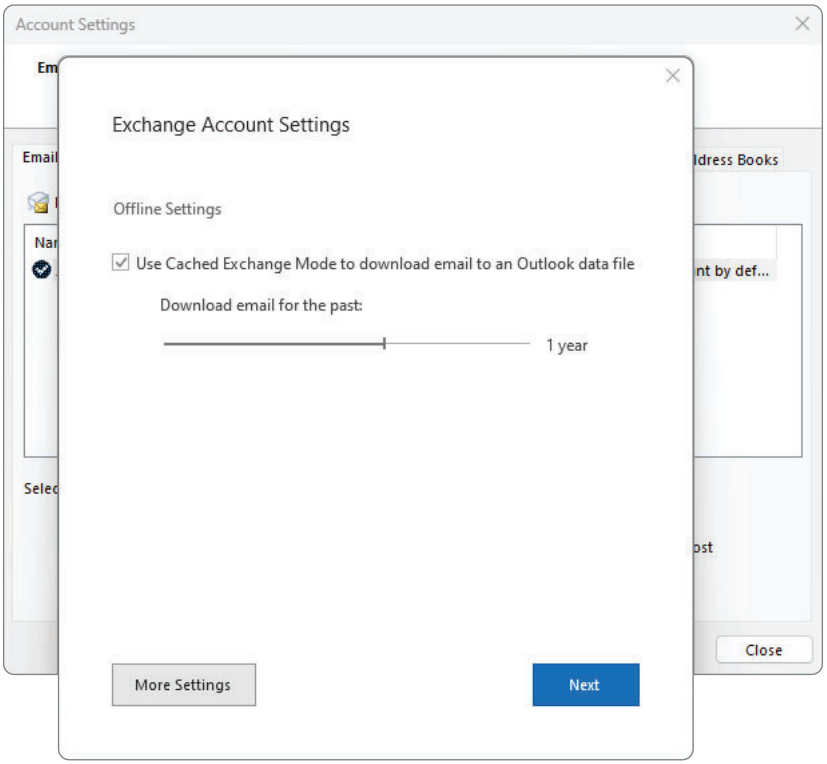
When you first launch Outlook, it will ask you to sign in. After entering your email address, it will check DNS for an autodiscover record that resolves to the Exchange Online service. Outlook will then set up your profile pointed to the Exchange Online service.

Outlook will be configured in Cached Exchange Mode by default if you use an Exchange Server or Microsoft 365 account. Cached Exchange Mode allows you to work offline by downloading the last 12 months of email to the computer. When you search, it will search this first 12 months, then ask you if you want to expand the search to online. When you create new emails and don't have an Internet connection, the emails will sit in the outgoing mailbox until you connect to the Internet and launch Outlook again.

You can turn Cached Exchange Mode on or off manually, or even vary the length of time email is cached with the following procedure:

1. In the Outlook client, click File ➢ Account Settings ➢ Account Settings.
2. Click Exchange or Microsoft 365, and then click Change.
3. Under Offline Settings, select Use Cached Exchange Mode to turn it on or off, as shown in Figure 1.3.

FIGURE 1.3 Cached Exchange Mode



4. Move the slider to lower or increase the length of time emails are cached.
5. Click Next, then Done to save the changes.

The autodiscover record is created by the administrator when the Exchange Online server is initially configured for an organization. If you are using a personal Microsoft 365 account, `http://autodiscover.outlook.com` points to Microsoft's internal `Outlook.com` infrastructure.

Microsoft OneDrive

Microsoft OneDrive was launched in 2007 as Windows Live SkyDrive. It began as a simple online storage service for users to save and share files. Over time, it evolved into a cloud storage platform and was rebranded as Microsoft OneDrive in 2014.

OneDrive is integrated tightly with Microsoft 365 and in Windows 11, OneDrive is an integral part of the File Explorer interface. A dedicated OneDrive section appears in the navigation pane, allowing users to access their cloud-stored files alongside local ones. Additionally, integration with Windows 11's notification system ensures users are alerted about file activity, such as completed syncs, sharing updates, or storage limit warnings.

OneDrive provides users with effortless access to their files from any device, along with advanced collaboration tools and automatic backup capabilities. A standout feature of OneDrive is its synchronization functionality, which enables users to keep their files updated across devices and the cloud.

The Files On-Demand feature is also enabled by default, displaying cloud files as placeholders until opened, which conserves local disk space. OneDrive also includes built-in sharing options, encryption, and integration with tools like Word, Excel, and Teams, making it a vital part of both personal and professional file management.

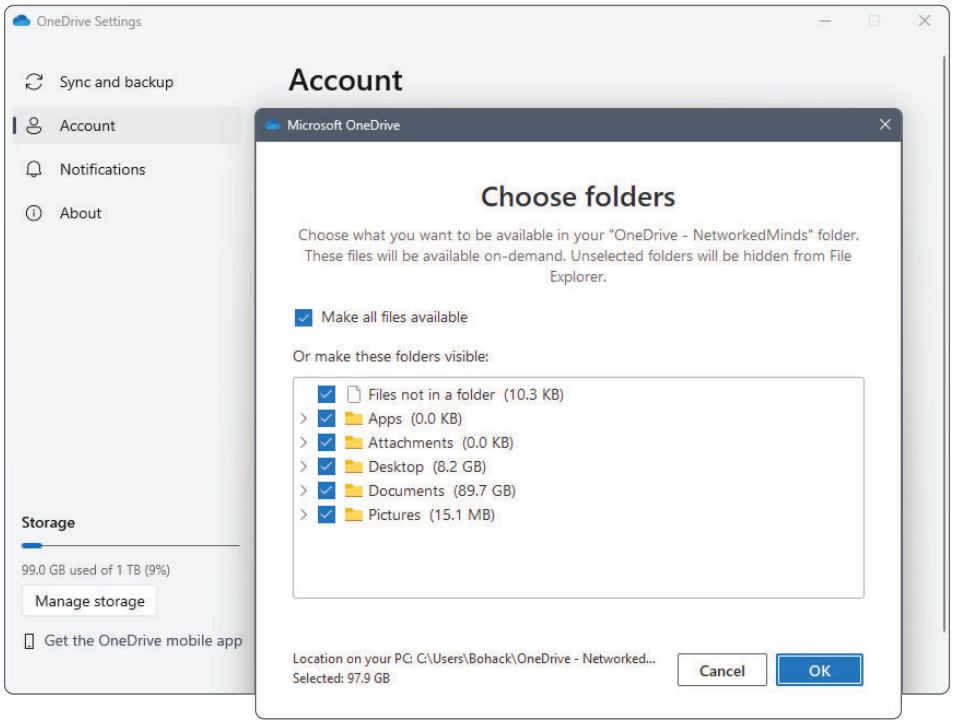
During installation or setup, the user signs in with their Microsoft or organization account to link their OneDrive storage. By default, key system folders like `Desktop`, `Documents`, and `Pictures` are offered for automatic backup and sync to OneDrive, ensuring critical files are protected without additional configuration. Through the OneDrive app, users can select specific folders to sync, ensuring that their essential files are always current across devices.

To change the current folders syncing to OneDrive, use the following procedure:

1. Click the OneDrive cloud icon in the Taskbar.
2. Select the gear icon in the upper-right corner.
3. Choose Settings from the drop-down menu.
4. Choose Account from the left menu.

The Choose Folders dialog will pop up, where you can choose which files are synced with OneDrive, as shown in Figure 1.4.

FIGURE 1.4 OneDrive Choose Folders



Microsoft OneDrive is not the only cloud-enabled storage sync tool available. Google Drive is another popular option, and if you are on a Mac, then iCloud is the preferred method. There are also third-party solutions like Carbonite Backup, DropBox, and Backblaze—these are just a few of the many different vendor offerings.

Collaboration Tools

Cloud-enabled collaboration tools have revolutionized the way teams work together, offering seamless integration and real-time collaboration across various platforms. The Microsoft 365 suite, which includes tools like Microsoft Teams, SharePoint, and OneDrive, provides a comprehensive solution for businesses to communicate, share files, and collaborate on projects from anywhere in the world. These tools are designed to enhance productivity and streamline workflows, making it easier for teams to stay connected and work efficiently.

Google Workspace, formerly known as G Suite, offers a robust set of applications such as Google Drive, Google Docs, and Google Meet. These tools enable users to create, edit, and share documents in real time, facilitating smooth collaboration and communication.

Both Microsoft 365 and Google Workspace are equipped with advanced security features, ensuring that data is protected while enabling teams to collaborate effectively in a cloud-based environment. Microsoft 365 Productivity Apps are typically bundled with Word, Excel, PowerPoint, Outlook, Access, Teams, and of course OneDrive to back up and store all your documents. Google Workspace Apps are typically bundled with Docs, Sheets, Slides, Gmail, Meet, Calendar, Chat, and of course Google Drive to back up and store everything. Here are some of the apps you might encounter:

Word Processing Tools Word processing applications such as Microsoft Word, and Google Docs offer robust platforms for creating, editing, and sharing documents. These tools support real-time collaboration, enabling multiple users to work on the same document simultaneously. With features like Track Changes, comments, and various formatting options, users can efficiently produce and refine their written content. The cloud integration ensures that documents are automatically saved and accessible from any device.

Spreadsheets Cloud-enabled spreadsheet applications like Microsoft Excel and Google Sheets have transformed data management and analysis. These tools allow multiple users to collaborate in real time, making it easy to update and share data instantly. With features such as pivot tables, charts, and various formulas, users can efficiently organize and interpret large data sets. These functionalities enable users to perform detailed data analysis, create visual representations of data, and generate insightful reports. The ability to use complex formulas and functions further enhances the analytical capabilities of these tools, making them indispensable for data-driven decision-making. The cloud integration ensures that all changes are saved automatically and accessible from any device, enhancing productivity and collaboration.

Videoconferencing Videoconferencing tools such as Microsoft Teams and Google Meet have become essential for remote communication and collaboration. These platforms offer high-quality video and audio calls, screen sharing, and virtual meeting rooms, enabling teams to connect and collaborate from anywhere in the world. With features like chat, file sharing, and integration with other productivity tools, videoconferencing applications facilitate seamless communication and efficient teamwork. Microsoft Teams and Google Meet are the two dominant videoconferencing solutions in enterprises today. However, there are many other third-party products, such as Zoom, GoTo Meeting, and Cisco Webex, just to name a few popular apps; there are many others.

Presentation Tools Cloud-enabled presentation tools like Microsoft PowerPoint and Google Slides provide powerful platforms for creating and sharing engaging presentations. These tools support real-time collaboration, allowing multiple users to work on the same presentation simultaneously. With a wide range of templates, design options, and multimedia integration, users can create visually appealing and informative presentations. The multimedia integration capabilities of these tools allow users to embed videos, images, and audio clips, enhancing the overall impact of their presentations. Additionally, features like animations and transitions can be used to make the content

more engaging and interactive. The cloud storage ensures that presentations are always accessible and up-to-date.

Instant Messaging Instant messaging tools like Microsoft Teams and Google Chat provide quick and efficient communication channels for teams. These platforms support real-time text messaging, file sharing, and integration with other productivity tools, making it easy for team members to stay connected and collaborate. With features like group chats, direct messages, and notifications, instant messaging applications enhance communication and streamline workflows. There are many other instant messaging cloud-enabled apps, but these are the two most popular in enterprises today.

Hybrid Identity

Active Directory (AD) is a directory service developed by Microsoft for Microsoft Windows domain networks. It provides a centralized system for authentication, authorization, and user management, and these are just a few capabilities. AD stores information about objects on the network and makes this information easy for administrators and users to find and use. AD is typically used for on-premises networks in which you need to authenticate to a domain controller.

Entra ID, formerly known as Azure Active Directory (Azure AD), is Microsoft's cloud-based identity and access management (IAM) service. Entra ID also allows for authentication, authorization, and user management, in addition to many other functions. Some of the security features Entra ID supports are single sign-on (SSO), multifactor authentication (MFA), and conditional access policies to enhance security and streamline user access, and that's just scratching the surface. Entra ID is considered an identity provider (IdP), and it integrates with any cloud application that supports SSO. Entra ID is also required if you are using any Microsoft 365 platform apps, such as Exchange Online or OneDrive, just to name a few.

Both AD and Entra ID serve a similar function and support identities differently, depending if you are authenticating on a domain-joined computer or signing into a cloud-enabled application. Typically, when an organization decides to expand to the Microsoft 365 cloud platform, it is necessary to support a hybrid identity. You can achieve a hybrid identity two main ways; with Entra ID Connect Sync or Entra ID Cloud Sync. However, the basic idea is synchronizing AD identities to Entra ID, and both tools achieve this. Therefore, when a domain user is created, a cloud-based identity will also be created in Entra ID to support cloud-based services.

Entra ID Connect Sync performs a one-way synchronization of users, groups, and computers in AD to Entra ID. It can be installed on only one server, and it's typically installed on the domain controller. It requires dedicated credentials and synchronizes every 30 minutes. Entra ID Connect Sync has been around since the introduction of Azure.

A newer approach is installing and configuring Entra Cloud Sync. Entra ID Cloud Sync is an agent-based installation that is installed on your domain controller. It does not require a service account and can be installed on more than one domain controller. It syncs every 2 minutes, and it can even sync cloud-based groups back to the AD domain. Entra ID Cloud Sync is configured and controlled through the cloud, as opposed to Entra ID Connect Sync,

which is administered on the server it is installed on. Obviously Entra ID Cloud Sync has a ton of features compared to the previous version and should be your preference.

Both Entra ID Connect Sync and Entra ID Cloud Sync both support password writeback. This feature enables you to change your password on the `Office.com` platform and the password will write back to the AD domain.

Cloud Licensing

When a user, also known as an identity, is created in the cloud, it is created because we want to allow the user access to cloud-based services. Cloud licensing for the Microsoft 365 platform provides a flexible and scalable way for organizations to manage user access to various Microsoft services. With cloud licensing, administrators can assign licenses to individual user accounts or groups of users, enabling them to use applications such as Microsoft Teams, Exchange Online, and OneDrive, just to name a few.

Microsoft offers licensing to organizations in the form of tiers for the Microsoft 365 platform. Each tier will add additional functionality to Exchange, security, compliance, and Entra ID, among others. Each tier is also priced incrementally based on the features, and they are charged per user/per month. The two major commercial tiers for Microsoft 365 are E3 and E5. Academics institutions have three tiers—A1, A3, and A5—with the lowest tier typically used for students.

Along with the Microsoft 365 platform, you can purchase additional licensing for other Office products. There are many other add-ons, such as telephony and larger storage for Exchange, among other add-ons.

To associate a license, an administrator would log into the Microsoft 365 admin center, navigate to the Users section, select the user account, and then assign the appropriate license from the available options. Additionally, an administrator can assign a group to a license; this is called group-based licensing. The benefit is that when a user is added to the specific group, they automatically are assigned a license.

Introduction to Windows

Microsoft released Windows 10 on July 29, 2015 and offered a free upgrade from Windows 7, Windows 8, and Windows 8.1. The free upgrade was to be limited to the first year of the Windows 10 release and expired on July 29, 2016. However, it's been reported that Microsoft still allows the upgrade and activation of Windows 10 for free, but your luck may vary. For these reasons, Windows 10 was rapidly adopted. Microsoft released Windows 11 on October 5, 2021, and it too is offered as a free upgrade for any Windows 10 device that meets the minimum requirements. Windows 10 is scheduled for end-of-life October 14, 2025, and Windows 11 will become the current version.

Exam Objective 1.3, "Compare and contrast basic features of Microsoft Windows editions," requires you to know the difference between the Windows 10 and 11 versions. Be sure to understand the feature differences.

Windows Editions

Windows 11 has had a total of 10 editions released. Only two of the 10 editions were made available in the retail channel: Windows 11 Home and Windows 11 Pro. Microsoft has made it easy for the retail consumer to pick the right edition of Windows 11, narrowing down retail choices to the two editions. Technically, there is a third retail edition called Windows 11 Pro for Workstations, which is preinstalled on high-end hardware for *high-performance computing (HPC)* requirements and allows up to 4 CPUs and 6 TB of RAM.

The editions available for Microsoft volume licensing options are Windows 11 Enterprise, Windows 11 Enterprise LTSC (Long-Term Servicing Channel), Windows 11 Education, and Windows 11 Pro Education. Windows 11 volume license editions include features such as AppLocker, BranchCache, and DirectAccess, just to name a few. Windows 11 Enterprise LTSC is an edition that is released every 2–3 years and is supported for 10 years after its initial release. Windows 11 LTSC receives normal Windows updates for security but does not receive feature upgrades. The *Microsoft Store* and bundled apps are also omitted from the Windows 11 Enterprise LTSC edition. Both Windows 11 Enterprise and Windows 11 Education editions have the same features over and above Windows 11 Pro and Windows 11 Home editions. Windows 11 Education editions are made available only to academic institutions and K–12 schools.

Although there are 10 different editions of Windows, the 220-1202 exam focuses on the following four Windows 10 editions: Home, Pro, Pro for Workstations, and Enterprise. In addition, the 220-1202 exam focuses on the three editions of Windows 11: Home, Pro, and Enterprise. Although Windows 11 Pro for Workstations is not an exam objective, we included it for completeness.

Windows 10 Pro and Windows 11 Home This edition is sold as a retail product, and it's preinstalled by the hardware vendor. When the operating system is preinstalled, it is referred to as an original equipment manufacturer (OEM) operating system. Windows 10/11 Home also lacks the key feature of joining a domain that can only be found in Windows 10/11 Pro or Enterprise.

Windows 10 Pro and Windows 11 Pro (Professional) This is the premier retail edition of the Windows 10/11 operating system. It is the most common OEM operating system because it allows users to join corporate domains and becomes value-added for the hardware vendor.

Windows 10 Pro for Workstations and Windows 11 Pro for Workstations This edition is used primarily for high-end computer workstations that require more than two CPUs and that need to address more than 2 TB of memory. Windows 10/11 Pro for Workstations supports up to four CPUs and 6 TB of memory. In addition, it supports Remote Direct Memory Access and Non-Volatile Dual In-line Memory Modules (NVDIMM).

Windows 10 Enterprise and Windows 11 Enterprise This edition supports all features of Windows 10/11 Pro and can be found in mid- to large-sized organizations. It also

supports features generally found in mid- to large-sized organizations. Windows 10/11 Enterprise can be purchased with a volume license subscription from Microsoft or an E5 Microsoft 365 license.

More information on Windows 11 editions can be found here:

`www.microsoft.com/en-us/windows/compare-windows-11-home-vs-pro-versions#tabs1-2`

There are 32-bit and 64-bit versions available for each of the Windows 10 editions listed except Windows 10 Pro for Workstations, since it is specifically used for high-performance computing. With the introduction of Windows 11, Microsoft only offers Windows 11 in a 64-bit version.

N Version

The Windows N version is a special edition of the Windows operating system designed to comply with European Union (EU) regulations. Unlike standard Windows editions, Windows N versions do not include built-in multimedia features such as Windows Media Player, Groove Music, Movies & TV, Voice Recorder, or Skype. This decision was made to provide users with the freedom to choose and install their preferred third-party media applications, leveling the playing field for other software developers. The absence of these features means that users may need to install the Media Feature Pack to restore multimedia functionality. This edition is particularly useful for those who prefer a more customizable operating system or have specific requirements for their multimedia software.

For Exam Objective 1.2, "Given a scenario, perform OS installations and upgrades in a diverse environment," be sure to understand upgrade considerations for various edition upgrades and operating system upgrades covered in the following section.

Upgrade Paths

If you want to perform an upgrade of Windows 11 instead of a clean installation, there are a few rules. When you upgrade from Windows 10 Home to Windows 11 Pro, you'll need to provide the new activation key for the higher edition. You cannot upgrade to Windows 10 to a lower version. Although you can upgrade the edition when performing an operating system upgrade, it is recommended that you use like-to-like editions of Windows. A like-to-like edition is an edition with similar functionality, such as Windows 10 Home to Windows 11 Home, or Windows 10 Pro to Windows 11 Pro.

Upgrading from Windows 10 to Windows 11 requires meeting several key prerequisites to ensure a smooth transition. First and foremost, your device must have a compatible 64-bit processor with at least 1 GHz clock speed and two or more cores. Additionally, a minimum of 4 GB of RAM and 64 GB of storage is essential. The system firmware must support Unified Extensible Firmware Interface (UEFI) and the ability for Secure Boot. The motherboard must have a Trusted Platform Module (TPM) version 2.0 enabled for enhanced security. A DirectX 12-compatible graphics card with a Windows Display Driver Model 2.0 (WDDM) driver is also necessary. Ensuring these requirements are met will help facilitate a seamless upgrade to Windows 11, providing access to its advanced features and improved performance.

If you want to retain settings and applications, an in-place upgrade is required in which you upgrade the current operating system to the desired version.

Another major restriction is that you cannot switch architecture during an in-place upgrade. If you have a 32-bit installation of Windows 10, then you will need to perform a clean install of Windows 11. Windows 11 supports only the 64-bit architecture, not the 32-bit architecture. You can switch architectures only when performing a clean installation, in which the old operating system is overwritten. This means that you must back up and restore settings, files, and applications.

It is always recommended to have a backup of your installation before you begin upgrading. It is equally important to check the application compatibility before upgrading. You should always check with the vendor of the application prior to upgrading to make sure they support Windows 11. An upgrade of the application may be required, or a completely new version of the application may be needed.

In addition to checking application compatibility prior to upgrading, you should check the hardware compatibility. Most of the time, a simple upgraded driver is required; however, sometimes you will find that the hardware is not compatible and that new hardware is required. In rare instances, the PC hardware is just not compatible and a new PC is required. This is usually the case with laptops, since components are not easily upgradable or upgrades are impossible. So do your homework prior to upgrading and know what to expect after the upgrade process is complete.

Windows Features

Every Windows operating system edition has a set of features that are bundled into the edition being purchased. The difference between the four main editions of the operating system—Home, Pro, Pro for Workstations, and Enterprise—is how the edition is purchased and its accompanying features. Windows 10/11 Home and Windows 10/11 Pro are both available through retail channels and can be purchased off the shelf. Windows 10/11 Pro for Workstations is typically preinstalled on high-performance workstations. Windows 10/11 Enterprise edition is unavailable without a volume license agreement from Microsoft or E5 Microsoft 365 subscription. Table 1.2 compares the features of the four main editions of Windows 10/11.

TABLE 1.2 Windows 10/11 features and editions

Edition	Maximum RAM supported	Maximum physical CPUs supported (multiple cores)	Notes
Home	128 GB	2	Lacks support for Remote Desktop (client only), BitLocker, Windows To Go, Hyper-V, joining to a domain, and participating in Group Policy. This edition is strictly for consumer use.
Pro	2 TB	2	Can join a Windows server domain; includes Remote Desktop Server, BitLocker, Windows To Go, Hyper-V, and participating in Group Policy.
Pro for Workstations	6 TB	4	Includes all features of the Pro edition, with support for 6 TB of RAM and up to 4 physical CPUs.
Enterprise	6 TB	2	Includes BitLocker, support for domain joining and Group Policy, DirectAccess, AppLocker, and BranchCache. This edition is available only through a volume license subscription.

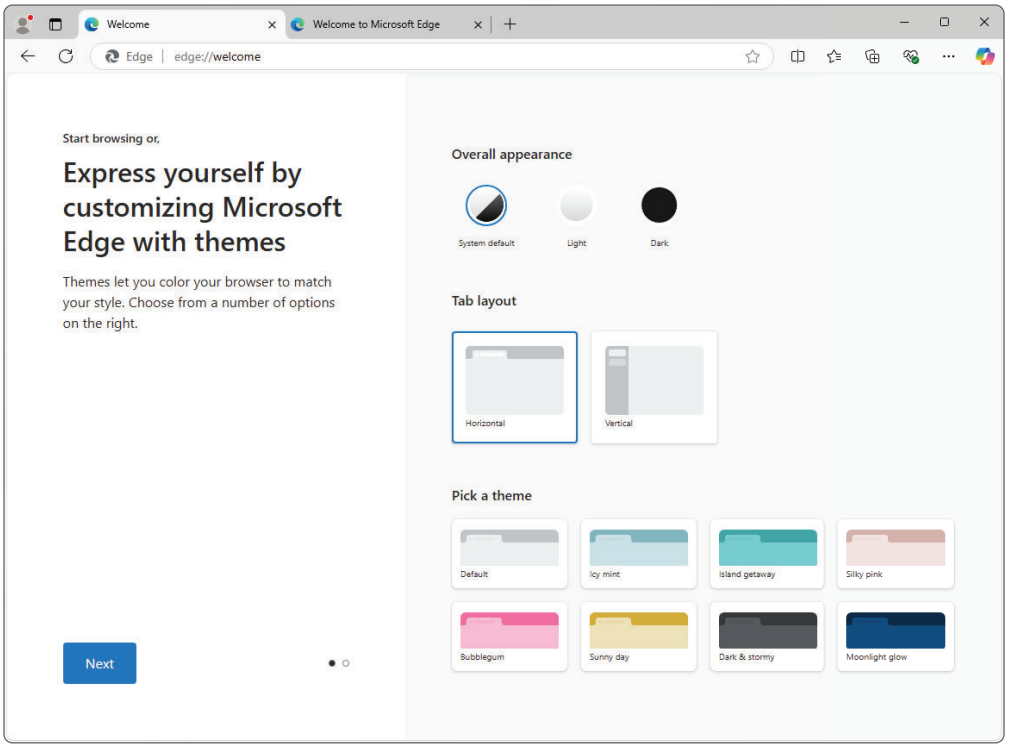
The following is a list of features introduced and associated with the Windows 10/11 operating system that you should know for the exam, along with a brief description of each:

Domain vs. Workgroup Windows 10/11 Pro edition offers a significant advantage for businesses through its support for domain joining. Domain joining is a feature that allows computers to be part of a centralized network managed by Active Directory. This setup provides enhanced security, centralized management, and streamlined access to network resources, making it ideal for mid to larger organizations. In contrast, a workgroup is a simpler network model where each computer is managed independently, suitable for smaller networks or home environments. While workgroups offer ease of setup and maintenance, they lack the robust security and administrative controls available in a domain environment. Domain joining is not a feature in Windows 10/11 Home edition.

Remote Desktop Protocol Windows 10/11 Pro edition includes the Remote Desktop Protocol (RDP), a powerful feature that allows administrators to remotely access and control the operating system. RDP provides a secure and seamless connection, enabling administrators to run applications, manage files, and perform tasks as if they were physically present at the computer. For example, a network administrator can use RDP to troubleshoot issues on a workstation located in a different office. The Windows 10/11 Home edition includes the Remote Desktop client, but the operating system does not permit remote access connections inbound.

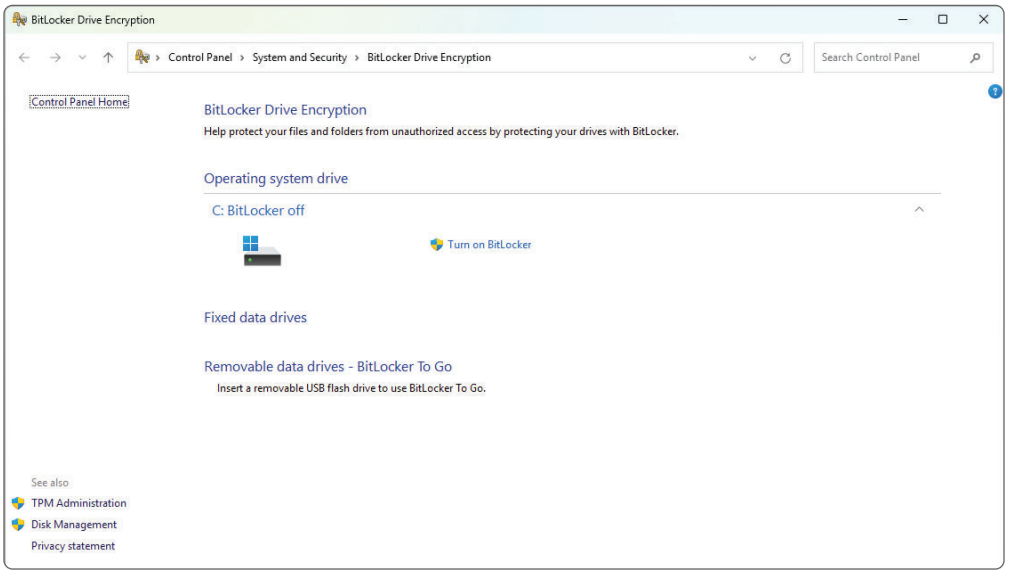
Microsoft Edge The Microsoft Edge web browser was developed as a lightweight web browser, as shown in Figure 1.5. Microsoft Edge is the successor to Internet Explorer and has already replaced Internet Explorer as the default web browser on Windows 10/11. The original version of Edge was released on January 15, 2020, and it was redesigned based upon the Chromium source code. Microsoft Edge does not support ActiveX and browser helper objects. Internet Explorer 11 reached end-of-life (EOL) on June 15, 2022. Internet Explorer 11 is no longer installed in Windows 11. You can still use the IE Mode in Microsoft Edge to provide backward compatibility. The Microsoft Edge browser can be seen in Figure 1.5.

FIGURE 1.5 The Microsoft Edge web browser



BitLocker BitLocker allows you to use drive encryption to protect files, including those needed for startup and login, as shown in Figure 1.6. This feature is available in all editions of Windows 10/11, except for Windows 10/11 Home. For removable drives, BitLocker To Go provides the same encryption technology to help prevent unauthorized access to the files stored on them. BitLocker is covered in Volume 2 Chapter 6.

FIGURE 1.6 The BitLocker Control Panel applet



Group Policy Group Policy is a powerful feature in Windows 11 that allows administrators to manage and configure operating system settings for the local computer. When used with an Active Directory domain, it provides centralized control over user and computer configurations, enhancing security and efficiency over multiple computers. The `gpedit.msc` tool, also known as the Group Policy Editor, is the interface used to create, modify, and apply these policies. This tool is available in all Windows 10/11 editions, except for the Windows 10/11 Home edition.

Desktop Styles Windows offers a variety of desktop styles that allow you to personalize your experience. From the sleek and modern look of Windows 11 to the classic feel of Windows 10, each version provides unique customization options. Users can choose from a range of themes that change the background, window colors, and sounds. You can even download new themes from the Microsoft Store. Additionally, Windows 11 introduces three different modes: Dark, Light, and Custom, each with its own color customization options for the Start menu, Taskbar, and Action Center, as shown in Figure 1.7.

Microsoft Defender Windows 10/11 includes an antivirus, antimalware utility that is built into the operating system. You can always provide your own antivirus or antimalware, but Microsoft Defender allows for protection from the time that Windows is installed. In addition to antivirus/antimalware, Microsoft Defender can protect against ransomware, as shown in Figure 1.8.

FIGURE 1.7 Windows Theme settings

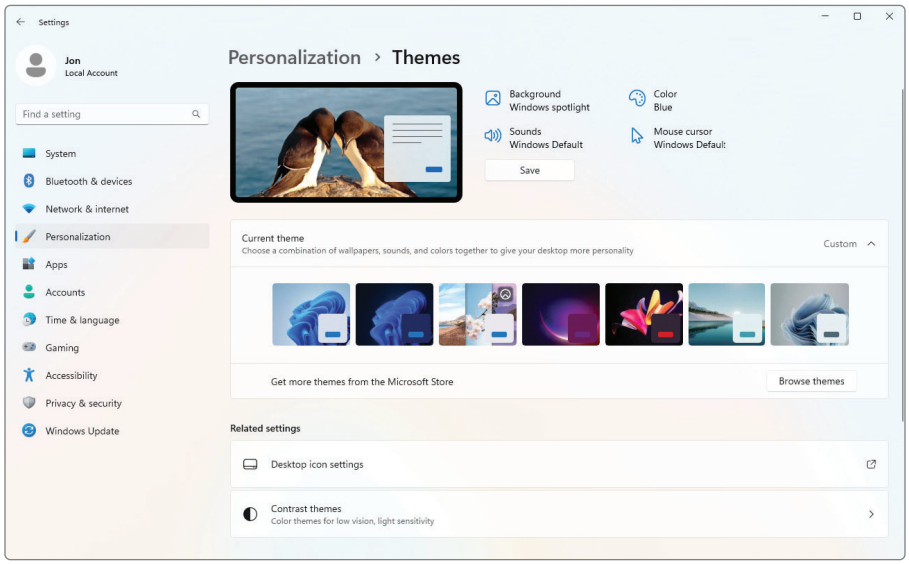
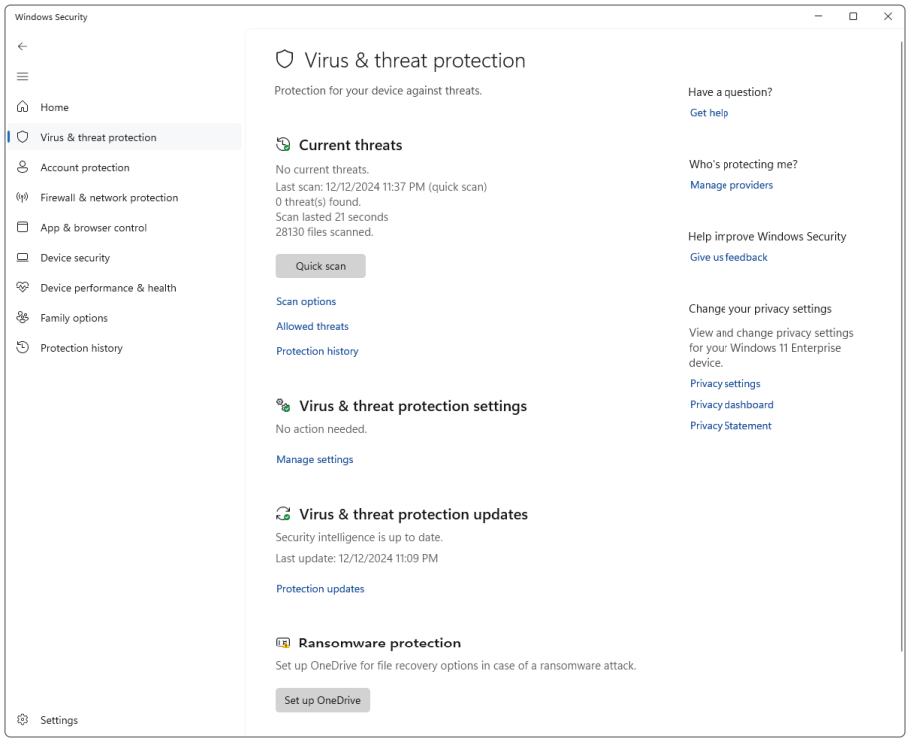


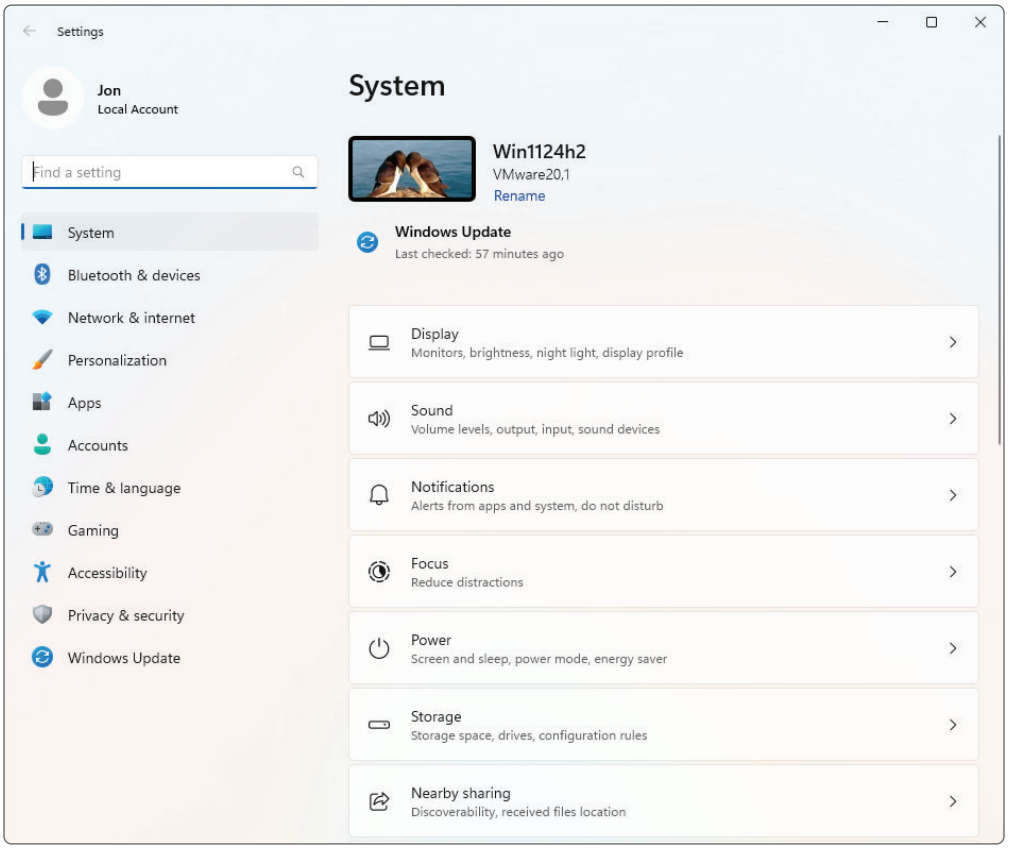
FIGURE 1.8 Microsoft Defender



Microsoft Defender is covered in detail in Volume 2 Chapter 6.

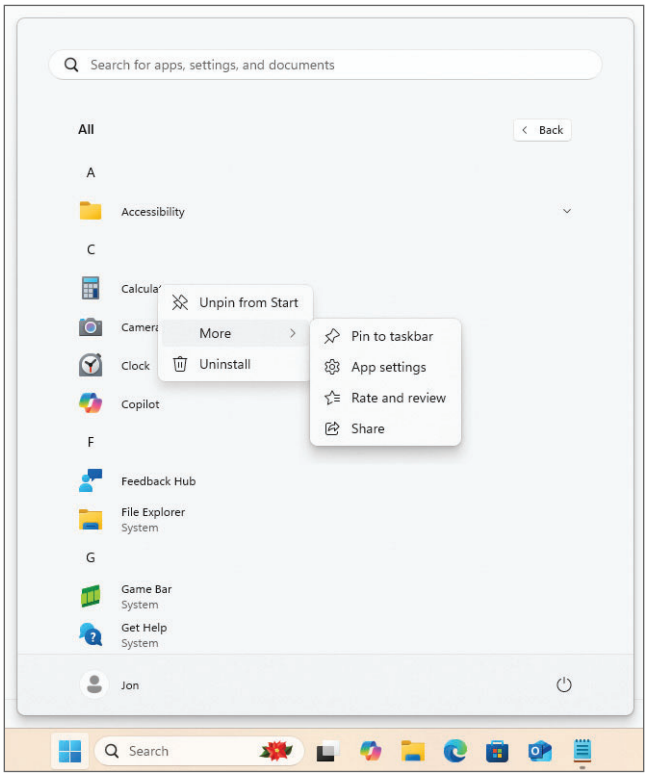
Settings App The Settings app is similar to Control Panel in functionality, because you can control many aspects of the operating system. However, any new feature being added to the Windows 10/11 operating system is added to the Settings app in lieu of the creation of new Control Panel applets. At this point, it looks like Control Panel will become a legacy component in future versions of Windows. You can access the Settings app by clicking the Start Menu and selecting the gear on the left-hand side. You can also right-click the Start menu and select Settings from the context menu. The Settings app is shown in Figure 1.9.

FIGURE 1.9 The Settings app in Windows 11



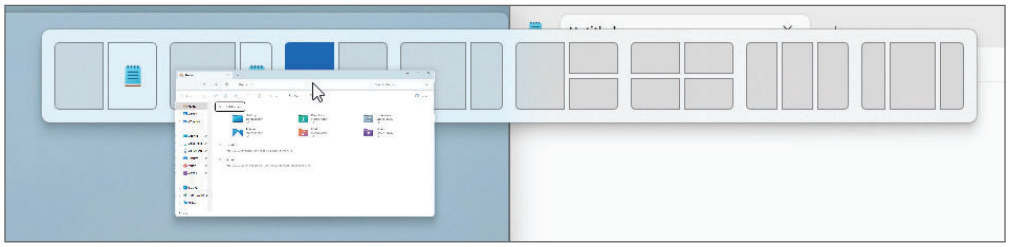
Pinning to the Taskbar The context menu for each application allows you to choose whether you want to pin (add) it to the Start menu, pin it to the Taskbar, or remove it if it is already there. You can also uninstall an application with the context menu, as shown in Figure 1.10.

FIGURE 1.10 An application context menu



Snap Assist Windows Snap was first introduced with Windows 7. Snap allows for side-by-side automatic windowing of windows on the screen. All you have to do is move a window over to a side, and the window will resize to half of the screen. Windows 10 introduced Snap Assist, which allows for up to four quadrants of the screen to automatically be adjusted using snapping. When a window is snapped over to a side, a preview in the blank area will show a preview of windows so that you can choose the next window to snap. Windows 11 added the functionality of dragging the window to the top drop-down bar to automatically place the window in a division, after the first one has been snapped in place, as shown in Figure 1.11.

FIGURE 1.11 Windows Snap Assist



Windows Hello Windows Hello was first introduced as a new feature of Windows 10 that allows the user to sign into Windows with biometrics. Windows Hello uses the built-in camera on the laptop or PC so that the device can be unlocked with facial recognition. On devices with a fingerprint reader, you can also use your fingerprint to sign into Windows.

Windows features are added with every version/update of the Windows operating system. You can view information about upcoming features that Microsoft is working on with the Windows Insider program. For more information, visit `https://insider.windows.com/en-us`.

The Windows Interface

If you've worked with older versions of Windows (such as Windows 7), you'll notice that it looks similar to the current Windows interface. While there are some differences, most of the basic tasks are accomplished in an almost identical fashion on everything from a Windows 95 workstation computer on up. Also, although the tools that are used often vary between the different OSs, the way that you use those tools remains remarkably consistent across platforms.

We will begin with an overview of the common elements of the Windows GUI. We will then look at some tasks that are similar across Windows operating systems. You are encouraged to follow along by exploring each of the elements as they are discussed.

As you follow along, you may notice that there are numerous icons and options that we do not mention. Quite honestly, there are too many to cover, and they're beyond the scope of this chapter. For now, simply ignore them or browse through them on your own, and then return to the text.

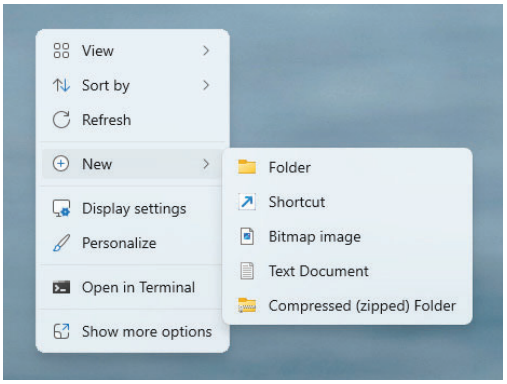
The Desktop

The *Desktop* is the virtual desk on which all your other programs and utilities run. By default, it contains the Start menu, the Taskbar, and a number of icons. The Desktop can contain additional elements, such as shortcuts or links to web page content. Because the

Desktop is the foundation on which everything else sits, the way that the Desktop is configured can have a major effect on how the GUI looks and how convenient it is for users. When you click the lower-left corner of the Desktop, the Start menu appears. (Right-clicking the Windows icon in Windows 8.1 and above displays a set of operating system functions.)

You can change the background patterns, screen saver, color scheme, and size of elements on the Desktop by right-clicking any area of the Desktop that doesn't contain an icon. The menu that appears, similar to the one shown for Windows 11 in Figure 1.12, allows you to do several things, such as create new Desktop items, change how your icons are arranged, or personalize your desktop.

FIGURE 1.12 The Windows 11 Desktop context menu



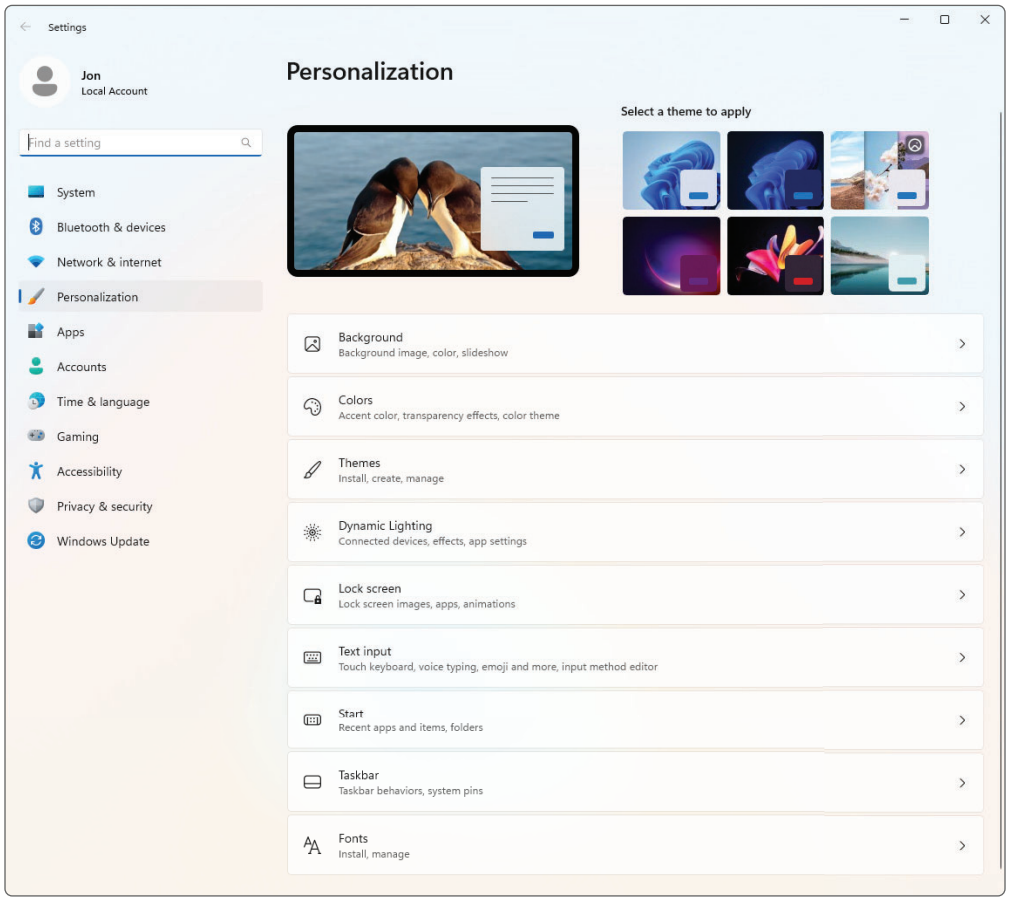
The Three Clicks in Windows

When it comes to interacting with a mouse in Windows, there are three possibilities:

- **Primary Mouse Click**—A single click (typically the left mouse button) is used to select an object or place a cursor.
- **Double-Click**—Two primary mouse clicks in quick succession are used to open a program through an icon or for other application-specific functions.
- **Secondary Mouse Click (or Alternate Click)**—Most mice have two buttons. Clicking once on the secondary button (usually the one on the right, although that can be modified) is interpreted differently from a left mouse click. Generally, this click displays a context-sensitive menu in Windows from which you can perform tasks or view object properties.

When you right-click the Desktop in Windows 11 and choose Personalize, you will see the Personalization Settings screen, as shown in Figure 1.13.

FIGURE 1.13 The Windows 11 Personalization setting screen



With the rapid adoption of Windows 11, this book will cover the most current version of Windows 11 (24H2), since the CompTIA objectives focus on Windows 11. However, every prior operating system Microsoft has produced has similar settings for personalization. We will cover the main ones in the Display Settings window for Windows 11.

Background The Background section is used to select a picture to display on the Desktop. In addition to setting a picture, you can select a group of pictures to display as a slideshow or display a static color.

Colors The Color section allows you to select a color scheme for the Start menu or to change the color or size of other Desktop elements.

Themes The Themes section allows you to select a theme to customize the look and feel of your machine quickly. Selecting a theme sets several items at once, such as a picture to display on the Desktop, the look of icons, the sounds to use, and so on. All these options can also be selected individually through the other personalization tabs.

Dynamic Lighting Dynamic lighting in Windows 11 settings allows users to manage RGB lighting for their devices directly from the operating system, eliminating the need for third-party applications.

Lock Screen Windows 8 and above allows you to easily change the lock screen wallpaper as well as the applications that are displayed on the lock screen.

Text Input In Windows 11, Text Input settings offer advanced options to enhance your typing experience. You can enable features such as autocorrect misspelled words, highlight misspelled words, and show text suggestions as you type. These settings help improve typing accuracy and speed, making it easier to compose messages and documents. Additionally, you can customize the touch keyboard settings to suit your preferences and enhance your overall typing experience.

Screen Saver Although the screen saver is not in the Personalization menu, it's worth covering. The screen saver can be configured to automatically display on your screen if your computer has been inactive for a certain period of time. Originally used to prevent burn-in of monitors, screen savers are now generally used for entertainment or to password-protect users' desktops. You can access the screen saver options in Windows 11 by clicking Lock Screen under Personalization and then clicking Screen Saver Settings. The Screen Saver Control Panel tab also gives you access to other power settings.

Start The Start section allows you to set the options for the Start menu, such as to show the most used apps or to show recently added apps, just to name a few.

Taskbar The Taskbar section allows you to change the behavior of the Taskbar, such as automatically hiding it or using smaller buttons, just to name a few.

Fonts The Fonts section allows you to review any font installed on the operating system. These fonts are used by applications that allow changing of fonts, such as a Microsoft Office document. In addition, you can uninstall fonts and install new fonts from the Microsoft Store.

Windows 10/11 has replaced many of the Control Panel applets with the Settings app. If you cannot find a setting from a prior operating system in the Control Panel Desktop app, you can find the setting in the new Settings app.

In Exercise 1.1, you will see how to change a screen saver.

EXERCISE 1.1

Changing a Screen Saver in Windows

1. Right-click the Desktop.
2. Choose Personalize from the context menu.
3. Click Lock Screen on the left side.
4. Scroll to the bottom and select Screen Saver Settings.
5. Choose a screen saver from the drop-down menu.
6. Click Preview to view how the screen saver will look.
7. Click OK to exit the Settings window.

The Taskbar

The *Taskbar* (see Figure 1.14) is another standard component of the Windows interface. Note that although the colors and feel of the Desktop components, including the Taskbar, have changed throughout the operating system versions, the components themselves are the same. In versions prior to Windows 11, the Taskbar contains two major items: the Start menu and the notification area, previously called the *system tray* (systray). The Start menu in Windows 11 is now left-center of the Taskbar and is easily identifiable: it is a button that has the Windows logo. The system tray is located on the right side of the Taskbar and contains a clock and date by default, speaker for sound volume, network connection status and OneDrive status, but other Windows utilities (for example, screen savers or antivirus utilities) may put their icons there to indicate that they are running and to provide the user with a quick way to access their features. On the far left of the Taskbar, you will find an icon that changes between, weather, sports, or headlines. If you click on the area, a news slide-out panel will fill the left side of the screen.

FIGURE 1.14 The Windows 11 Taskbar

Windows also uses the middle-right area of the Taskbar. When you open a new window or program, it gets a button on the Taskbar with an icon that represents the window or program. To bring that window or program to the front (or to maximize it if it was minimized), click its button on the Taskbar. As the middle-right area of the Taskbar fills with buttons, the buttons become smaller so that they can all be displayed.

Windows 11 allow you to pin commonly used programs to the Taskbar. The icon will appear on the Taskbar, and when the program is running, a short line will appear under the icon. You can pin any running task by right-clicking the icon in the Taskbar and selecting Pin To Taskbar. You can just as easily remove pinned icons by right-clicking them and selecting Unpin From Taskbar.

You can make the Taskbar automatically hide itself when it isn't being used (thus freeing that space for use by the Desktop or other windows). You can also make the Taskbar look like previous versions of Windows by selecting the alignment. In Exercise 1.2, we will show you how to do this.

EXERCISE 1.2

Modifying the Taskbar

1. Right-click the Taskbar.
2. Choose Taskbar Settings to bring up the Settings app for the Personalization options.
3. Click Taskbar Behaviors.
4. Select the option Automatically Hide The Taskbar. The Taskbar will automatically hide when the mode is turned on.
5. Select Taskbar Alignment and change it from Center to Left.
6. Move the mouse pointer to the bottom of the screen. The Taskbar will pop up and be available for normal use.

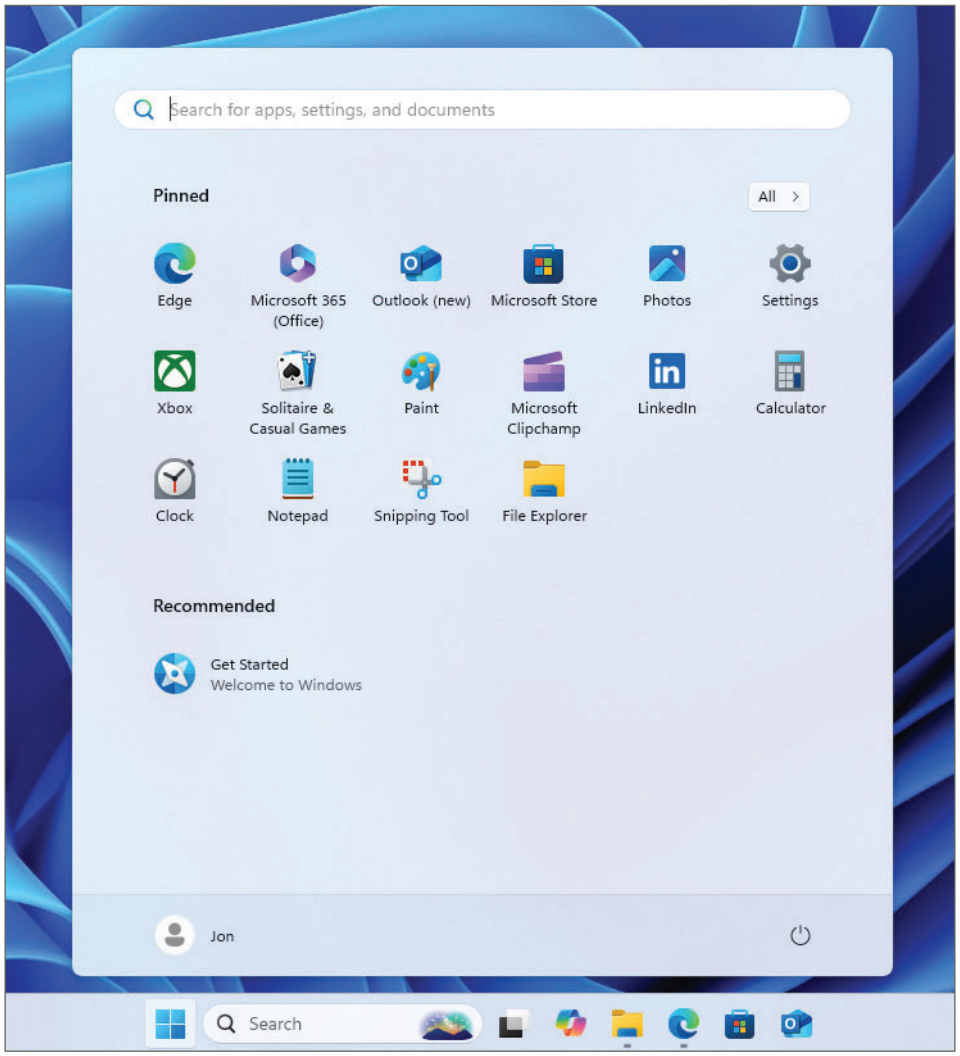
The Start Menu

Back when Microsoft officially introduced Windows 95, it bought the rights to use the Rolling Stones' song "Start Me Up" in its advertisements and at the introduction party. Microsoft chose that particular song because the Start menu was the central point of focus in the new Windows interface, as it was in all subsequent versions.

To display the Start menu, you can press the Windows key on your keyboard at any time. You can also click the Windows logo button in the Taskbar in Windows 11. You'll see a Start menu similar to the one shown in Figure 1.15 for Windows 11.

When Windows 8 was released, the Start menu was replaced with a Start screen. There was considerable backlash from the user community over the removal of the Start menu in Windows 8. This major change in the Start menu was partially responsible for Windows 8's slow adoption. A number of third-party vendors developed alternatives and marketed them to fill this demand.

FIGURE 1.15 Windows 11 Start menu



From the Start menu, you can select any pinned application. If you can't find your application in the pinned area, you can select All from the upper right of the menu. This will allow you to scroll down alphabetically to find your application. You can also use the search box to locate the application. Once you locate the application, if it is frequently used it is best to pin it to the Start menu. To pin an icon, move the mouse pointer over the icon and right-click. The context menu will appear, allowing you to pin or uninstall an application. You can also select More to open another context menu that allows you to Pin To The Taskbar, select the app settings, Rate and Review, or Share.

To check which OS you are using, press the Windows key and R key at the same time, type **winver.exe**, and then press Enter. The About Windows Application dialog will display the version and build number of the operating system.

Let's explore the principal features of the Windows 11 Start menu.

The Run Command

It is possible to run commands and utilities from the Start menu search box or from the Run dialog. To access the Run dialog box in Windows 11, press the Windows key and the R key at the same time. To execute a particular program, type its name in the Open field. If you don't know the exact path, you can browse to find the file by clicking the Browse button. Once you have typed the executable name, click OK to run the program.

To open a command prompt, type **command** in the Run box, and then click OK. You might need to run this as Administrator if you want to change system settings. From the Start menu, type **cmd** or **command** in the search field and then press Ctrl+Shift+Enter to run as Administrator.

Applications can easily be started from the Run window. You often will find it faster to open programs this way than to search for their icons in the Start menu maze. In Exercise 1.3, you will see how to start a program from the Run window.

EXERCISE 1.3

Starting a Program from the Run Window

1. Press the Windows key and the R key at the same time.
2. In the Open field, type **notepad**.
3. Click OK. Notepad will open in a new window.

If the program that you want to start has been run from the Run window before, you can find it on the Open field's drop-down list. Click the down arrow to display the list, and then select the program that you want by clicking its name and then OK.

Shut Down Command

Windows operating systems are very complex. At any one time, many files are open in memory. If you accidentally hit the power switch and turn off the computer while these files are open, there is a good chance that they will be corrupted. For this reason, Microsoft has added the Shut Down command under the Start menu. The command appears as an icon of a power button without a label. When you select this option, Windows presents you with several choices. The submenu will display Lock, Sleep, Shut Down, and Restart.

If the operating system requires security updates to install, there will be an orange dot in the lower-right corner of the power button icon. Additionally, the submenu will display Update And Shut Down and Update And Restart. If there is an optional feature update, the dot will be blue to grab your attention.

Recommended

The Recommended section of the Start menu displays recommended applications. It also shows frequently used applications based on your usage patterns.

Icons

Icons are shortcuts that allow a user to open a program or a utility without knowing where that program is located or how it needs to be configured. Icons consist of the following major elements:

- Icon label
- Icon graphic
- Program location or path

The label and graphic of the icon typically tell the user the name of the program and give a visual hint about what that program does. The icon for the Notepad program, for instance, is labeled Notepad and its graphic is a notepad. By right-clicking an icon once, you make it the active icon and a drop-down menu appears. One of the selections is Properties. Clicking Properties brings up the icon's attributes (see Figure 1.16), and it is the only way to see exactly which program an icon is configured to start and where the program's executable is located. You can also specify whether to run the program in a normal window or maximized or minimized.

Additional functionality has been added to an icon's properties to allow for backward compatibility with older versions of Windows (known as *compatibility mode*). To configure this, click the Compatibility tab and specify the version of Windows for which you want to configure compatibility. Note that you cannot configure compatibility if the program is part of the version of Windows that you are using. Figure 1.17 shows the compatibility settings available for an older program.

This feature is helpful if you own programs that used to work in older versions of Windows but no longer run under the current Windows version. In addition, you can specify different display settings that might be required by older programs.

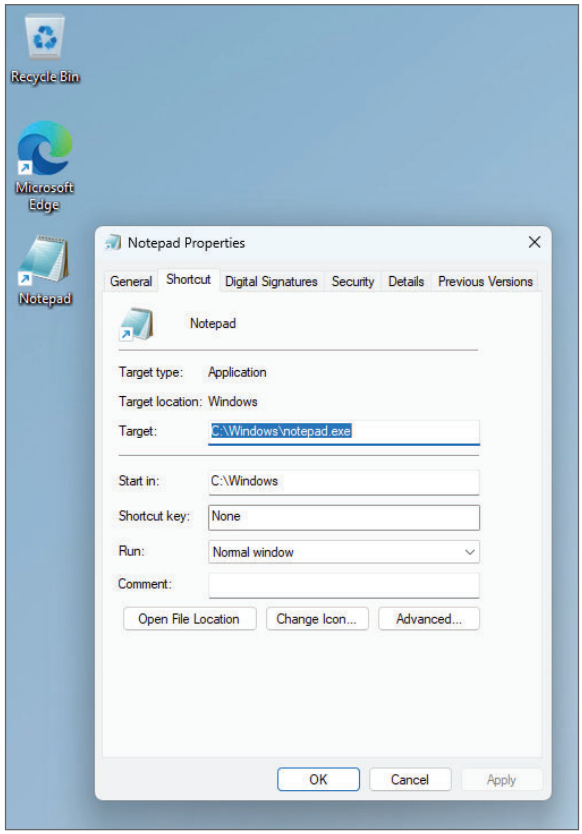
Standard Desktop Icons

In addition to the options in your Start menu, a number of icons are placed directly on the Desktop in Windows. The Recycle Bin icon is one of these icons. In the latest version of Windows 11 (24H2), the Microsoft Edge icon can also be found on the Desktop. In older versions of Windows, the Computer icon could also be found on the desktop.

The Recycle Bin All files, directories, and programs in Windows are represented by icons. These icons are generally referred to as *objects*. When you want to remove

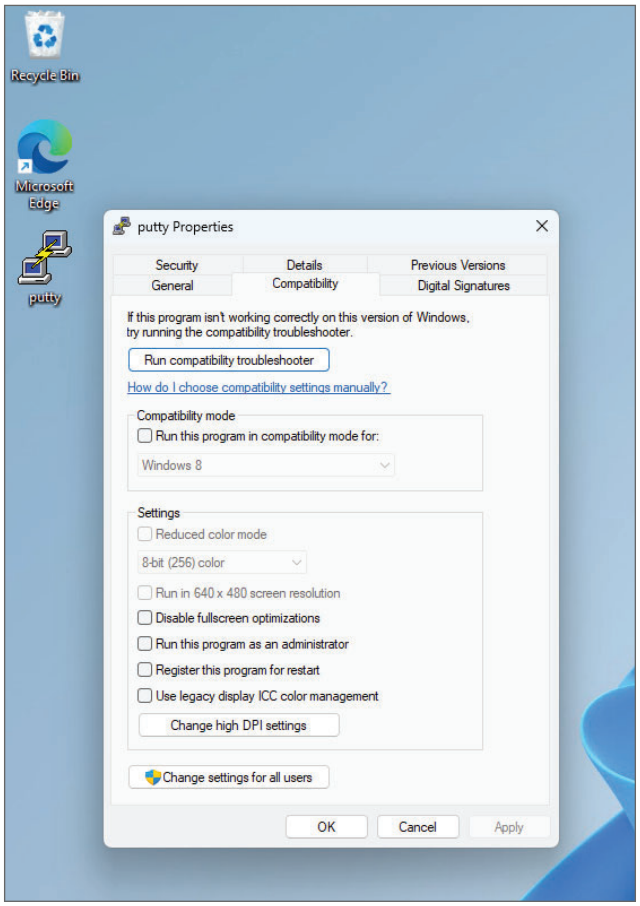
an object from Windows, you do so by deleting it. Deleting doesn't just remove the object, though; it also removes the ability of the system to access the information or application that the object represents. Therefore, Windows includes a special folder where all deleted files are placed: the Recycle Bin. The Recycle Bin holds the files until it is emptied or until you fill it. It gives users the opportunity to recover files that they delete accidentally. By right-clicking the Recycle Bin icon, you can see how much disk space is allocated. Some larger files that cannot fit in the Recycle Bin will be erased after a warning.

FIGURE 1.16 The Properties window of an application



You can retrieve a file that you have deleted by opening the Recycle Bin and then dragging the file from the Recycle Bin to where you want to restore it. Alternatively, you can right-click a file and select Restore. The file will be restored to the location from which it was deleted.

FIGURE 1.17 The Compatibility settings possible with an older program



In versions of Windows that this exam tests you on, the "deleted" files are stored in a folder called `\$Recycle.Bin`. This was not the case in all previous versions of Windows.

To erase files permanently, you need to empty the Recycle Bin, thereby deleting any items in it and freeing the hard drive space they took up. If you want to delete only specific files, you can select them in the Recycle Bin, right-click, and choose Delete. You can also permanently erase files (bypassing the Recycle Bin) by holding down the Shift key as you delete them (by dragging the file and dropping it in the Recycle Bin, pressing

the Del key, or clicking Delete on the file's context menu). If the Recycle Bin has files in it, its icon looks like a full trash can; when there are no files in it, it looks like an empty trash can.

Microsoft Edge If the eyes are the windows to the world, then Microsoft Edge is the window to the Internet (no pun intended). Because the Internet is such an integral part of a user experience on an operating system, it is placed on the desktop by default. Microsoft Edge is Microsoft's preferred browser of choice and it's based off the Chromium project.

This PC Icon If you double-click the This PC icon, it displays a list of all the disk drives installed in your computer. In addition to displaying disk drives, it displays a list of other devices attached to the computer, such as scanners, cameras, and mobile devices. The disk devices are sorted into categories such as Hard Disk Drives, Devices with Removable Storage, Scanners And Cameras, and so on.

You can delve deeper into each disk drive or device by double-clicking its icon. The contents are displayed in the same window.

In addition to allowing you access to your computer's files, the This PC icon on the Desktop lets you view your machine's configuration and hardware, also called the About page.

The This PC icon is not displayed on the Desktop by default. In Windows 11, you can add this icon by navigating to Settings ➪ Personalization ➪ Themes and clicking Desktop Icon Settings. You can then select the icons you want to display on the Desktop.

What's in a Window?

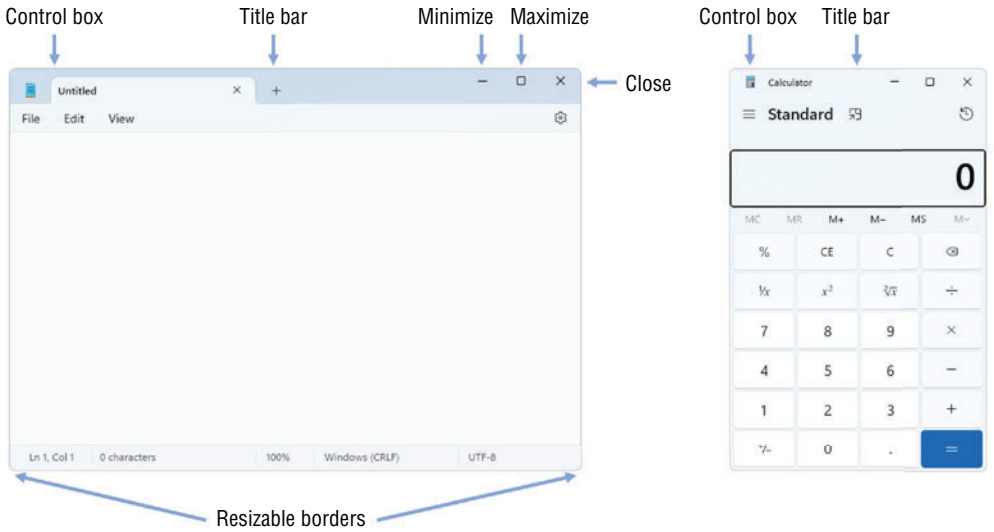
We have now looked at the nature of the Desktop, the Taskbar, the Start menu, and icons. Each of these items was created for the primary purpose of making access to user applications easier. These applications are, in turn, used and managed through the use of *windows*—the rectangular application environments for which the Windows family of operating systems is named. We will now examine how windows work and what they are made of.

A *program window* is a rectangular area created on the screen when an application is opened within Windows. This window can have a number of different forms, but most windows include at least a few basic elements.

Elements of a Window

Several basic elements are present in a standard window. Figure 1.18 shows the control box, title bar, Minimize/Maximize button, Close button, and resizable border in the text editor Notepad (`notepad.exe`) and Calculator (`calc.exe`) that has all the basic window elements—and little else.

FIGURE 1.18 The basic elements of a window, as seen in Notepad and Calculator



The basic window elements are as follows:

Control Box Located in the upper-left corner of the window, the control box is used to control the state of the application. It can be used to maximize, minimize, and close the application. Clicking it once brings into view a selection menu. Double-clicking it closes the window and shuts down the application. Some actions might differ from application to application, such as a double-click might maximize a window or the application.

Minimize and Maximize/Restore Buttons Used to change the state of the window on the Desktop. They are discussed in the section "States of a Window" later in this chapter.

Close Button Used to easily end a program and return any resources that it was using to the system. It essentially does the same thing as double-clicking the control box but with one fewer click.

Title Bar The area between the control box and the Minimize button. It states the name of the program and, in some cases, gives information about the particular document being accessed by that program. The color of the title bar indicates whether the window is the active window. Clicking and holding on it is an easy way to move the window on the screen.

Menu Bar Used to present useful commands within an application in an easily accessible format. Clicking one of the menu choices displays a list of related options from which you may choose.

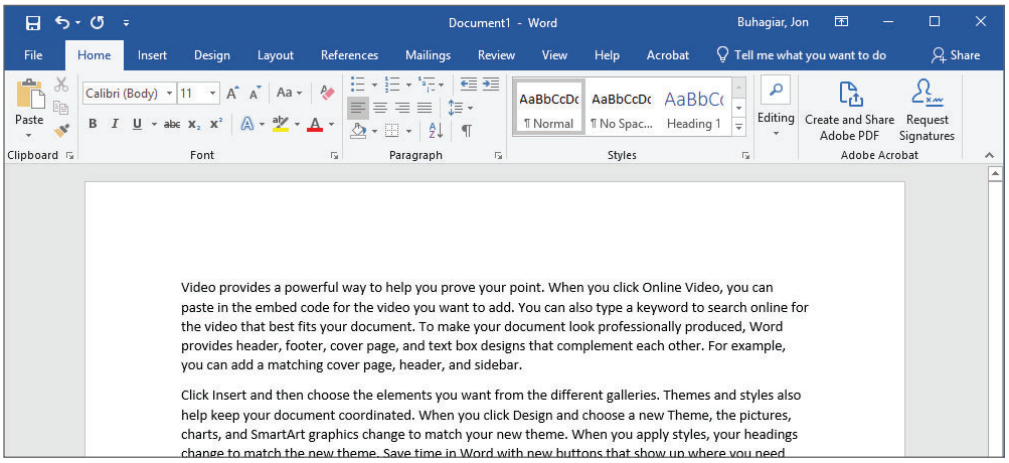
Active Window The window that is currently being used. It has two attributes. First, any keystrokes that are entered are directed into the active window by default. Second, any other windows that overlap the active window are pushed behind it.

Border A thin line that surrounds the window in its restored down state and allows it to be resized.

Not every element is found in every window, because application programmers can choose to eliminate or modify each item. Still, in most cases, they will be consistent, with the rest of the window filled in with menus, toolbars, a workspace, or other application-specific elements. For instance, Microsoft Word, the program with which this book was written, adds a Ribbon control. It also has a menu bar, a number of optional toolbars, scroll bars at the right and bottom of the window, and a status bar at the very bottom. Application windows can become quite cluttered.

Notepad and Calculator are very simple Windows programs. Notepad has a single menu bar and the basic elements shown in Figure 1.18. Calculator has a simple menu that you can open by clicking the 3 lines in the upper-left of the application. Figure 1.19 shows a Microsoft Word window. Both Word and Notepad are used to create and edit documents, but Word is far more configurable and powerful and therefore has many more optional components available within its window.

FIGURE 1.19 A window with many more components, as seen in Microsoft Word



A quick way to access Help is to press the F1 key. In operating systems before Windows 10, a help and support application would launch. In Windows 10/11, the F1 key launches the web browser.

States of a Window

There is more to the Windows interface than the specific parts of a window. Windows also are movable, stackable, and resizable, and they can be hidden behind other windows (often unintentionally).

When an application window has been launched, it exists in one of three states:

Maximized A maximized window takes up all the available space on the screen. When it is in front of other programs, it is the only thing visible—even the Desktop is hidden. It takes up the entire space of the Desktop, and the middle button in the upper-right corner displays two rectangles rather than one. The sides of the window no longer have borders. The window is flush with the edges of the screen. Maximizing a window provides the maximum workspace possible for that window's application, and the user can actively access the window. In general, maximized mode is the preferred window size for most word processing, graphics creation, and other types of user applications.

Restored A restored window can be used interactively, and it is identical in function to a maximized window, with the simple difference that it does not necessarily take up the entire screen. Restored windows can be very small, or they can take up as much space as maximized windows. Generally, how large the restored window becomes is the user's choice. Restored windows display a Maximize button (the middle button in the upper-right corner) with a single rectangle in it; this is used to maximize the window. Restored windows have a border.

Minimized Minimized program windows are represented by nothing but an icon and title on the Taskbar, and they are not usable until they have been either maximized or restored. The difference between a minimized program and a closed program is that a minimized program is out of the way, but it is still taking up resources and is therefore ready to use if you need it. It also leaves the content of the window in the same place where it was when you minimized it.

When one program is open and you need to open another (or maybe you need to stop playing a game because your boss has entered the room), you have two choices. First, you can close the program currently in use and simply choose to reopen it later. If you do this, however, the contents of the window (your current game, for example) will be lost, and you will have to start over. Once the program has been closed, you can move on to open the second program.

The second option is to minimize the active window. Minimizing the game window, for example, removes the open window from the screen and leaves the program open but displays nothing more than an icon and title on the Taskbar. Later, you can restore the window to its previous size and finish the game in progress.

Keep in mind that applications in the background are still running. Therefore, if you minimize your game, you might return to find that you've been eaten by whatever monster you were running from in the game. A program running while minimized can be a good thing, however, if you're running a useful utility, such as a long search or an Internet download.

File Management

File management is the process by which a computer stores data and retrieves it from storage. Although some of the file-management interfaces across Windows may have a different look and feel, the process of managing files is similar across the board.

Files and Folders

In order for a program to run, it must be able to read information off the hard disk and write information back to the hard disk. To be able to organize and access information—especially in larger new systems that may have thousands of files—it is necessary to have a structure and an ordering process.

Windows provides this process by allowing you to create *directories*, also known as *folders*, in which to organize files. Windows also regulates the way that files are named and the properties of files. The filename for each file created in Windows has to follow certain rules, and any program that accesses files through Windows must also comply with these rules:

- Each file has a filename of up to 255 characters.
- Certain characters, such as a question mark (`?`) and slash (\ or /), are reserved for other uses and cannot be used in the filename. Periods are used to separate the filename from the extension, and the backslash is used to separate the directories in a path from the filename.
- A filename extension (generally three or four characters) can be added to identify the file's type.
- Filenames are not case sensitive. (You can create files with names that use both upper- and lowercase letters, but to identify the file within the filesystem, it is not necessary to adhere to the capitalization in the filename.) Thus, you cannot have a file named `working.txt` and another called `WORKING.TXT` in the same folder. To Windows, these filenames are identical, and you can't have two files with the same filename in the same folder. We'll get into more detail on this topic a little later.

In Windows 3.*x* and DOS, filenames were limited to eight characters and a three-character extension, separated by a period—known as the 8.3 file-naming convention. Windows 95 introduced long filenames, which allowed the 255-character filename convention.

The Windows filesystem is arranged like a filing cabinet. In a filing cabinet, paper is placed into folders, which are internal dividers, which are in a drawer of the filing cabinet. In the Windows filesystem, individual files are placed in subdirectories that are inside directories, which are stored on different disks or different partitions.

Windows also protects against duplicate filenames; no two files on the system can have exactly the same name and path. A *path* indicates the location of the file on the disk; it is composed of the letter of the logical drive on which the file is located and, if the file is located in a folder or subfolder, the names of those directories. For instance, if a file named `pagefile.sys` is located in the root of the `C:` drive—meaning it is not within a folder—the path to the

file is `C:\pagefile.sys`. As another example, if a file called `notepad.exe` is located under Windows under the root of `C:`, then the path to this file is `C:\Windows\notepad.exe`.

The *root folder* of any drive is the place where the hierarchy of folders for that drive begins. On a `C:` drive, for instance, `C:\` is the root folder.

Common filename extensions that you may encounter include `.exe` (for executable files, aka applications), `.dll` (for dynamic linked library files), `.sys` (for system files), `.log` (for log files), `.drv` (for driver files), and `.txt` (for text files). Note that DLL files contain additional functions and commands that applications can use and share. In addition, specific filename extensions are used for the documents created with each application. For example, the filenames for documents created in Microsoft Word have a `.doc` or `.docx` extension. You'll also encounter extensions such as `.mpg` (for video files), `.mp3` (for music files), `.png` and `.tif` (for graphics files), `.htm` and `.html` (for web pages), and so on. Being familiar with different filename extensions is helpful in working with the Windows filesystem.

Capabilities of File Explorer

Although it is technically possible to use the command-line utilities provided within the command prompt to manage your files, this generally is not the most efficient way to accomplish most tasks. The ability to use drag-and-drop techniques and other graphical tools to manage the filesystem makes the process far simpler. The *File Explorer* is a utility that allows you to accomplish a number of important file-related tasks from a single graphical interface.

Here are some of the tasks you can accomplish using File Explorer:

- View files and directories
- Open programs or data files
- Create directories and files
- Copy objects to other locations
- Move objects to other locations
- Delete or rename objects
- Change file attributes

You can access many of these functions by right-clicking a file or folder and selecting the appropriate option, such as Copy or Delete, from the context menu.

The File Explorer utility used in Windows 8/8.1 and Windows 10/11 was previously named Windows Explorer. Although the functionality remains almost identical since the introduction of Windows 95, the rebranding was just one of many changes when Windows 8 was released.

Navigating and Using File Explorer

Using File Explorer is simple. A few basic instructions are all you need to start working with it. First, the Explorer interface has a number of parts, each of which serves a specific purpose. The top area of File Explorer is dominated by a set of menus and toolbars that give you easy access to common commands. The main section of the window is divided into two panes: the left pane displays the drives and folders available, which is called the navigation pane, and the right pane displays the contents of the currently selected drive or folder, which is called the results pane. In recent versions of Windows, the Navigation pane is turned off. The following list describes some common actions in File Explorer:

Expanding a Folder You can click a folder in the left pane to expand it (show its subfolders in the left pane) and display its contents in the right pane. Clicking the right-pointing arrow (>) to the left of a folder expands the folder without changing the display in the right pane, as shown in Figure 1.20.

Collapsing a Folder Clicking the down arrow next to a folder collapses it.

Selecting a File If you click the file in the right pane, Windows highlights the file by marking it with a darker color.

Selecting Multiple Files The Ctrl or Shift key allow you to select multiple files at once. Holding down Ctrl while clicking individual files selects each new file while leaving the currently selected file(s) selected as well. Holding down Shift while selecting two files selects both of them and all the files in between.

Opening a File Double-clicking a file in the right pane opens the program if the file is an application; if it is a data file, it will open using the application for which the filename extension is configured.

Changing the View Type Windows has several different view types: Extra Large Icons, Medium Icons, Large Icons, Small Icons, List, Details, Content, and Tiles. The choices vary a bit between the Windows versions. You can move between these views by clicking the View menu and selecting the view that you prefer. In Windows 10, the View menu is a ribbon that will drop down when clicked. The Navigation pane can be turned on from the View tab.

Finding Specific Files You access this option by using the Search button or bar. You can search for files based on their name, file size, file type, and other attributes.

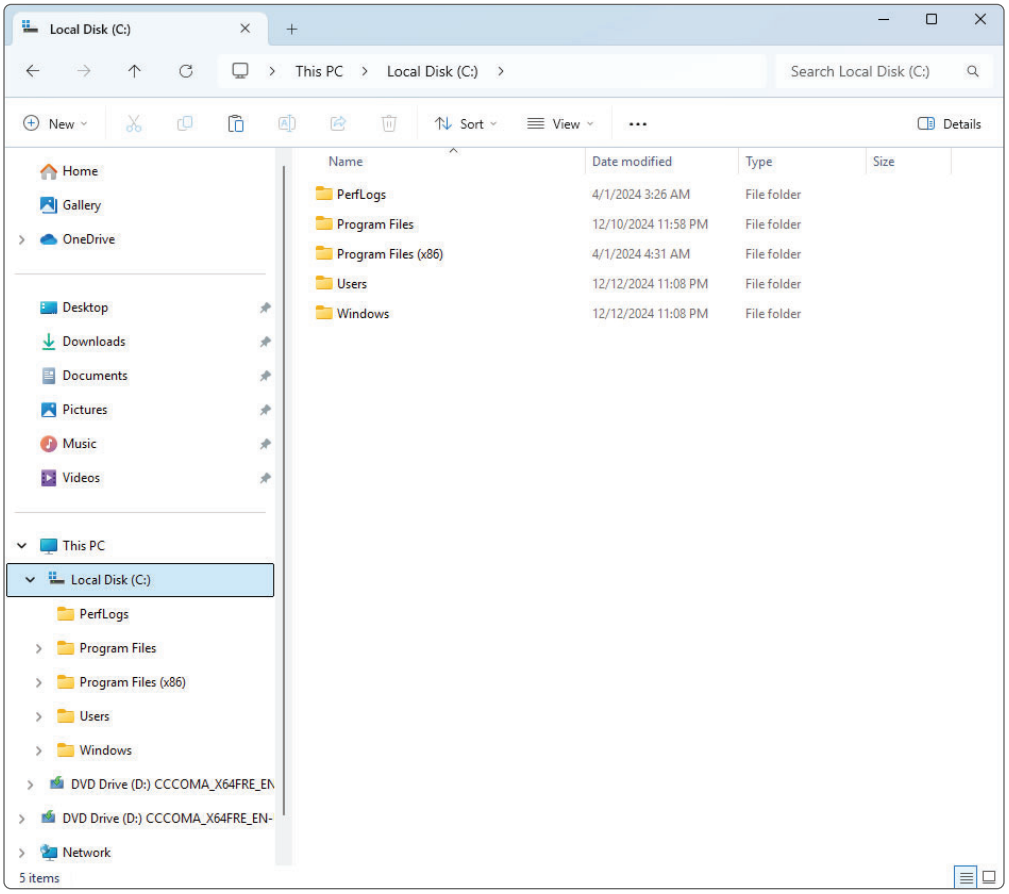
When you're searching, you can also use wildcards. *Wildcards* are characters that act as placeholders for a character or set of characters, allowing, for instance, a search for all files with a `.txt` filename extension. To perform such a search, you'd type an asterisk (**`*`**) as a stand-in for the filename: **`*.txt`**. An asterisk takes the place of any number of characters in a search. A question mark (?) takes the place of a single number or letter. For example, `autoex??.bat` would return the file `autoexec.bat` as part of its results.

Creating New Objects To create a new file, folder, or other object, navigate to the location where you want to create the object, and then right-click in the right pane (without selecting a file or folder). In the menu that appears, select New and then choose the object that you want to create.

Deleting Objects Select the object and press the Del key on the keyboard, or right-click the object and select Delete from the menu that appears.

The simplicity of deleting in Windows makes it likely that you or one of the people you support will delete or misplace a file or a number of files that are still needed. In such a case, the Recycle Bin (mentioned earlier) is a lifesaver.

FIGURE 1.20 Windows 11 File Explorer



Besides simplifying most file-management commands as shown here, File Explorer allows you to complete a number of disk-management tasks easily. For example, you can format and label removable media, which is discussed further in Volume 2 Chapter 3, "Windows Administration."

Preparing for the Exam

Future chapters will delve further into operating systems and the tools, utilities, and features available with each. There is also additional coverage, as applicable, in the chapters on troubleshooting. For purposes of exam study, Table 1.3 offers a complete list of the features for each of the Windows operating systems that you need to know for the exam. It also details which chapter(s) in this book has more coverage of that particular topic.

TABLE 1.3 Windows features

Feature	Purpose	More information
BitLocker	Encrypts drives; available in each OS but not in every edition	Volume 2 Chapter 6
Domain access vs. workgroup	Shared security for computers and users	Volume 2 Chapter 3
Desktop styles/user interface	Customization of the desktop and the user interface	Volume 2 Chapter 2
Remote Desktop Protocol (RDP)	Allows users and administrators to connect remotely to obtain a desktop session	Volume 2 Chapter 8
Group Policy	A mechanism inside of Active Directory that allows for the management of user and computers	Volume 2 Chapters 2, 5

Summary

In this chapter, you learned about the basic operating systems, application installation, and the Windows 10/11 features. Additionally, we covered the basics of the Windows structure and window management. Because Windows is a graphical system, the key to success in learning to use it is to explore the system to find out what it can do. You will then be better prepared to decipher later what a user has done.

First, we explored the various operating systems you may encounter, along with their characteristics, such as their life cycles, categories, and minimum requirements.

Next, we discussed applications as they apply to the operating system and underlying architecture, as well as the various factors that should be evaluated before installing the application.

Finally, we introduced Windows 10/11 and its various editions. We then covered some basic Windows management concepts, including file management, as well as the folder structure. We also discussed using approved hardware and updating Windows.

With the basic knowledge gained in this chapter, you are now ready to learn how to interact with the most commonly used tools, the subject of the following chapter.

Exam Essentials

Know the RAM limitations between operating system architectures. 32-bit operating systems have a RAM limitation of 4 GB, whereas 64-bit operating systems can address over 4 GB of RAM.

Understand an operating system's life cycle. When development is started on an operating system, it is considered alpha or beta code. Release candidates precede the release to market (RTM) of an operating system. The end of mainstream support generally means only security updates will continue to be supported during the extended support period.

Know the various operating system types and the common operating systems. The four basic types of operating systems are server, workstation, mobile, and cloud-based. Windows, macOS, and Linux can be used as server, workstation, and mobile operating systems, whereas Google Chrome is primarily a cloud-based operating system.

Know the various editions of Windows 10/11 and their limitations. The four most common Windows 10/11 editions are Windows 10/11 Home, Pro, Pro for Workstations, and Enterprise. Windows 10/11 Home does not allow joining of domains and has many premium features. Windows 10 Pro is the most flexible edition because it allows for domain joining. Windows 10/11 Pro for Workstations allows for up to 4 CPUs and 6 TB of RAM. Windows 10/11 Enterprise is a volume licensed edition of Windows 10/11 that offers premium features.

Know the various requirements for applications. The system requirements for an application can be architecture, such as 32-bit versus 64-bit, dedicated graphics cards versus integrated, VRAM and RAM requirements, CPU requirements, the use of external licensing dongles or hardware tokens, storage requirements, and application compatibility.

Know the various cloud applications you will commonly support. You should be familiar with Exchange Online and Outlook terminology and basic setup. You should also know what each of the Microsoft 365 applications are used for, such as Word, Excel, PowerPoint, and Teams, as well as their Google counterparts.

Understand how to manage files in Windows. Nearly all file management—including moving, copying, renaming, and deleting files and changing file attributes—is accomplished through Windows File Explorer.

Know where files are located. The various versions of Windows that you need to know for this exam store files in multiple locations. You should be able to identify the location of those files mentioned in this chapter and be able to identify items such as where the Recycle Bin files are on each Windows operating system.

Review Questions

The answers to the chapter review questions can be found in Appendix A.

1. Which of the following can you type in the Start menu in Windows to open a command prompt? (Choose two.)
   - **A.** `run`
   - **B.** `cmd`
   - **C.** `command`
   - **D.** `open`
2. Which part of the operating system can be described as extremely specific software written for the purpose of instructing the OS on how to access a piece of hardware?
   - **A.** Source code
   - **B.** Application
   - **C.** Kernel
   - **D.** Driver
3. Which application will sync your files to the cloud?
   - **A.** Excel
   - **B.** OneDrive
   - **C.** Exchange
   - **D.** Outlook
4. Which of the following installation methods would you use if you wanted to install a program on a virtual machine? (Choose the best method.)
   - **A.** ISO
   - **B.** USB
   - **C.** Optical disk
   - **D.** ZIP files
5. The Windows File Explorer program can be used to do which of the following? (Choose two.)
   - **A.** Browse the Internet
   - **B.** Copy and move files
   - **C.** Change file attributes
   - **D.** Create backup jobs
6. What is the maximum allowable length for the name of a file in Windows?
   - **A.** 8 characters
   - **B.** 32 characters

C. 64 characters

D. 255 characters

7. Which of the following is a program that runs on top of the OS and allows the user to issue commands through a set of menus or some other graphical interface?

A. Taskbar

B. Shell

C. Desktop

D. Source

8. If a program doesn't have a shortcut on the Desktop or in the All Apps submenu, you can start it by which of the following methods? (Choose the best answer.)

A. Using the Shut Down command

B. Typing the program name in the Start box

C. Using the `run` command and typing in the name of the program

D. Typing **`cmd`** in the Start box followed by the program name

9. Which operating system feature offers the ability for a single application to have multiple requests into the processor at one time?

A. Multiuser mode

B. Dystopia

C. Preemption

D. Multithreading

10. In Windows, a deleted file can be retrieved using which of the following?

A. My Computer icon

B. Recycle Bin

C. Control Panel

D. Settings app

11. To turn off a Windows 11 operating system gracefully, you should do which of the following?

A. Run the turnoff command at a command prompt.

B. Turn off the switch and unplug the machine.

C. Press Ctrl+Alt+Del.

D. Select Start, then click the power icon, and then choose Shut Down.

12. Which is required for a 64-bit installation of Windows 11?

A. TPM 2.0

B. 32 GB Storage

C. Microsoft DirectX 9 graphics device

D. 4 GB RAM

13. What is the minimum amount of free hard drive space required for the installation of Windows 11 64-bit?

A. 1.5 GB

B. 15 GB

C. 32 GB

D. 64 GB

14. What is the minimum required memory for a 64-bit installation of Windows 11?

A. 512 MB

B. 1 GB

C. 2 GB

D. 4 GB

15. In Windows, a quick way to access Help is to press which keyboard key?

A. F12

B. The Windows key on the keyboard

C. F1

D. Alt

16. Which key(s) will open a run box?

A. Windows key+R

B. Ctrl+R

C. Windows key

D. F1

17. Which of the following is located on the rightmost portion of the Taskbar?

A. Start menu

B. Pinned apps

C. Notification area

D. Shutdown options

18. In addition to right-clicking the Desktop, how else can you access the Display Properties settings?

A. By clicking the Settings gear in the Start menu

B. By clicking the System icon under Control Panel

C. By pressing Ctrl+Alt+Esc

D. By pressing Ctrl+Alt+Tab

19. Which of the following allows you to navigate file/folder structures in Windows?

A. Start menu

B. File Explorer

C. KDE

D. GNOME

20. What is the minimum required processor speed for a 64-bit installation of Windows 11?

A. 2.2 GHz

B. 3.3 GHz

C. 1.0 GHz

D. 2.3 GHz

Performance-Based Question

You will encounter performance-based questions on the A+ exams. The questions on the exam require you to perform a specific task, and you will be graded on whether or not you were able to complete the task. The following requires you to think creatively in order to measure how well you understand this chapter's topics. You may or may not see similar questions on the actual A+ exams. To see how your answer compares to the authors', refer to Appendix B.

Your organization is planning to upgrade its operating system to Windows 11. Currently all of the laptops used in your organization use Windows 10 Pro. The organization eventually wants to roll out the Windows 11 feature of BranchCache. How should you proceed to accommodate the upgrade and the future feature needs?

Chapter 2

Windows Configuration

THE FOLLOWING COMPTIA A+ 220-1202 EXAM OBJECTIVES ARE COVERED IN THIS CHAPTER:

- ✓ **1.0 Operating Systems**
- ✓ **1.1 Explain common operating system (OS) types and their purposes.**
  - Various filesystem types
    - New Technology File System (NTFS)
    - Resilient File System (ReFS)
    - File Allocation Table 32 (FAT32)
    - Fourth extended filesystem (ext4)
    - Extended filesystem (XFS)
    - Apple File System (APFS)
    - Extensible File Allocation Table (exFAT)
- ✓ **1.2 Given a scenario, perform OS installations and upgrades in a diverse environment.**
  - Partitioning
    - GUID [globally unique identifier] Partition Table (GPT)
    - Master boot record (MBR)
  - Drive format
- ✓ **1.4 Given a scenario, use Microsoft Windows operating system features and tools.**
  - Task Manager
    - Services
    - Startup
    - Performance

- Processes
- Users

- Microsoft Management Console (MMC) snap-in
  - Event Viewer (`eventvwr.msc`)
  - Disk Management (`diskmgmt.msc`)
  - Task Scheduler (`taskschd.msc`)
  - Device Manager (`devmgmt.msc`)
  - Certificate Manager (`certmgr.msc`)
  - Local Users and Groups (`lusrmgr.msc`)
  - Performance Monitor (`perfmon.msc`)
  - Group Policy Editor (`gpedit.msc`)
- Additional tools
  - System Information (`msinfo32.exe`)
  - Resource Monitor (`resmon.exe`)
  - System Configuration (`msconfig.exe`)
  - Disk Cleanup (`cleanmgr.exe`)
  - Disk Defragment (`dfrgui.exe`)
  - Registry Editor (`regedit.exe`)

✓ **1.6 Given a scenario, configure Microsoft Windows settings.**

- Internet Options
- Devices and Printers
- Programs and Features
- Network and Sharing Center
- System
- Windows Defender Firewall
- Mail
- Sound
- User Accounts
- Device Manager

- Indexing Options
- Administrative Tools
- File Explorer Options
  - View hidden files
  - Hide extensions
  - General options
  - View options
- Power Options
  - Hibernate
  - Power plans
  - Sleep/suspend
  - Standby
  - Choose what closing the lid does
  - Turn on fast startup
  - USB selective suspend
- Ease of Access
- Time and Language
- Update and Security
- Personalization
- Apps
- Privacy
- System
- Devices
- Network and Internet
- Gaming
- Accounts

The previous chapter introduced the basic components of the Windows operating systems. This chapter builds on that and focuses on the configuration of the Windows 11 operating system.

Interacting with Operating Systems

In the following sections, we will look at the Microsoft GUI from the ground up. In Volume 2 Chapter 1, "Operating System Basics," we took a detailed look at its key components, and we will build on that with an exploration of basic tasks for Windows 11. Although the exam objectives are focused on Windows 11, most of these configuration concepts can be applied to Windows 10, and even to old Windows operating systems in many cases.

Microsoft has included a number of tools with each iteration of Windows to simplify system configuration. Although some tools have specific purposes and are used only on rare occasions, you will come to rely on a number of tools and access them on a regular basis. It is this latter set that we will examine in the following sections.

Exam Objective 1.4, "Given a scenario, use Microsoft Windows operating system features and tools," requires you to know the elements of the Task Manager, various Microsoft Management Console (MMC) snap-ins, and additional tools. Be sure to know how to use the Task Manager, and know what each of the MMC snap-ins and additional tools provide.

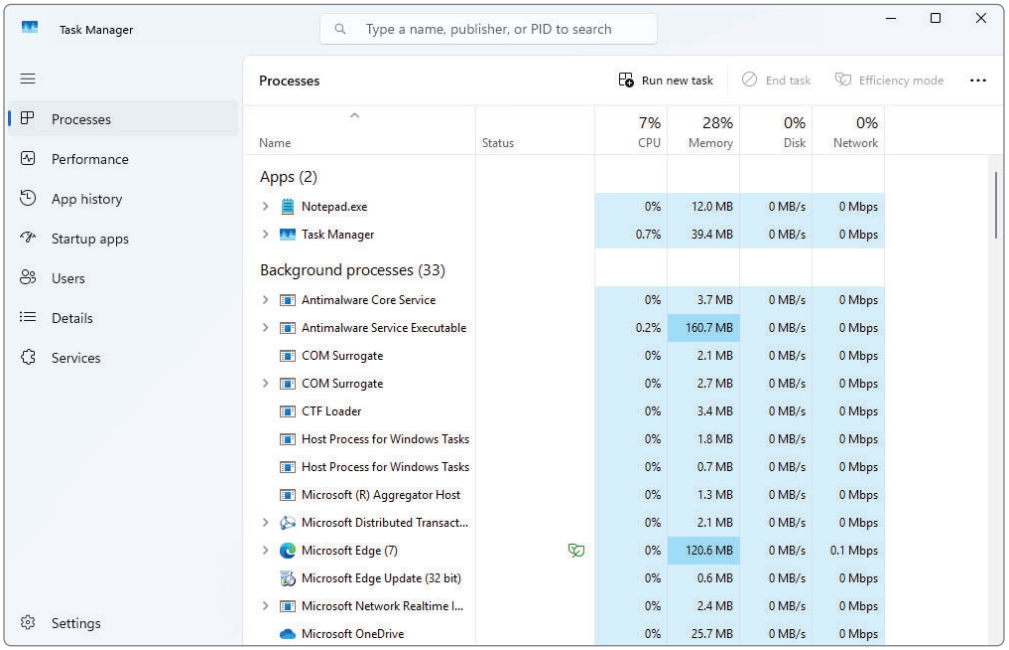
Task Manager

Task Manager lets you shut down nonresponsive applications selectively in all Windows versions. In current versions of Windows, it can do so much more, allowing you to see which processes and applications are using the most system resources, view network usage, see connected users, and so on. To display Task Manager, press Ctrl+Alt+Delete and click the Task Manager button or option. You can also right-click an empty spot in the Taskbar and choose Task Manager from the context menu.

To get to Task Manager directly in any of the Windows versions that include it, you can press Ctrl+Shift+Esc.

Depending on the Windows version, Task Manager has various tabs or views. Figure 2.1 shows the common default display in Windows 11, but other versions vary from the seven tabs shown here: Processes, Performance, App History, Startup Apps, Users, Details, and Services. In the following figure the side menu is expanded by clicking the three lines in the upper-left of the Task Manager. By default the menu is collapsed and only the icons are displayed.

FIGURE 2.1 The default Task Manager in Windows



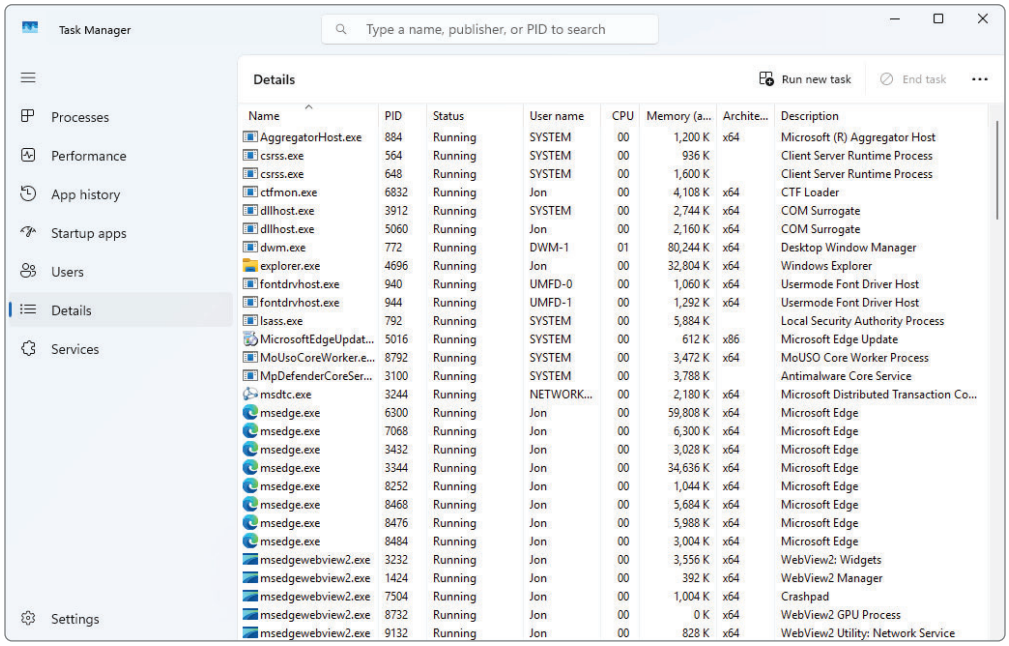
Let's look at these tabs in more detail:

Processes The Processes tab lets you see which tasks are open on the machine. You also see the status of each task, which can be either Running or Not Responding. If a task/application has stopped responding (that is, it's hung), you can right-click the task in the list and click End Task. Doing so closes the program, and you can try to open it again. Often, although certainly not always, if an application hangs, you have to reboot the computer to prevent the same thing from happening again shortly after you restart the application. You can also use the Processes tab to switch to a different task or create new tasks. In Windows 11, many of the elements that appear in Processes are copied between the Processes and Details tabs; an App History tab was added that, as the name implies, displays usage settings for applications and the currently logged-in user account.

Details In Windows 11 the Details tab, shown in Figure 2.2, gives you more detail than the Processes tab. The Details tab can display a multitude of metrics and details, such as

Program ID (PID), Status, User Name, and UAC Virtualization Status, and these are just the defaults. You can select the columns you want to see by right-clicking on the header, clicking Select Columns, then choosing the columns you want to inspect, as shown in Figure 2.3.

FIGURE 2.2 The Details tab of Task Manager



To end a process, right-click it in the list and click End Task. Be careful with this choice, because ending some processes can cause Windows to shut down. If you don't know what a particular process does, you can look for it in any search engine and find a number of sites that explain it.

You can also change the priority of a process in Task Manager's Details tab by right-clicking the name of the process and choosing Set Priority. The six priorities, from lowest to highest, are as follows:

Low For applications that need to complete sometime but that you don't want interfering with other applications. On a numerical scale from 0 to 31, this equates to a base priority of 4.

Below Normal For applications that don't need to drop all the way down to Low. This equates to a base priority of 6.

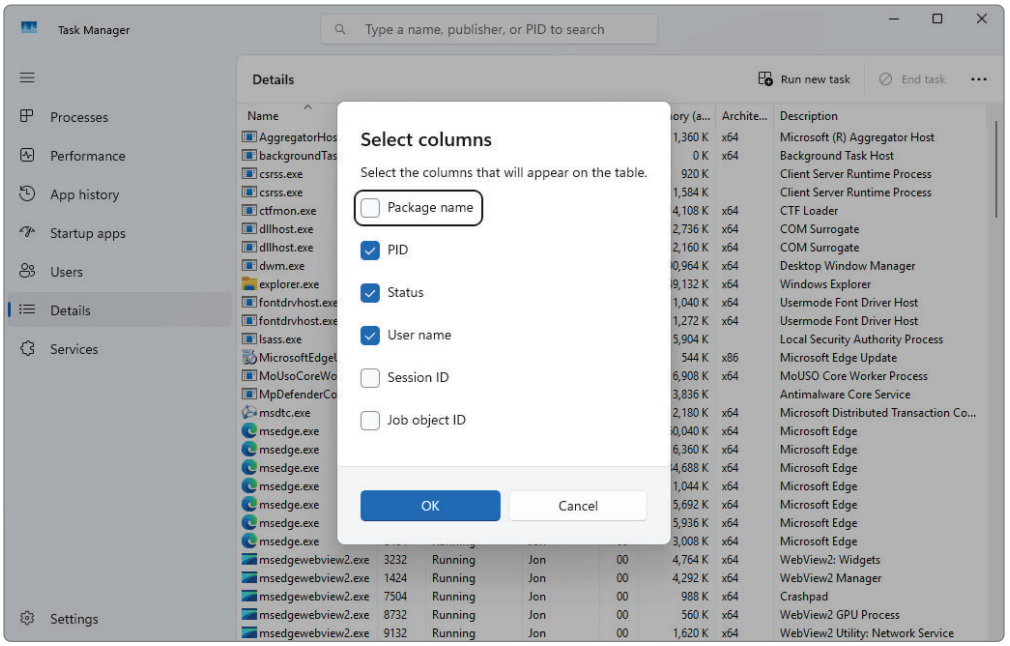
Normal The default priority for most applications. This equates to a base priority of 8.

Above Normal For applications that don't need to boost all the way to High. This equates to a base priority of 10.

High For applications that must complete soon, when you don't want other applications to interfere with the applications' performance. This equates to a base priority of 13.

Realtime For applications that must have the processor's attention to handle time-critical tasks. Applications can be run at this priority only by a member of the Administrators group. This equates to a base priority of 24.

FIGURE 2.3 The various columns for the Details tab



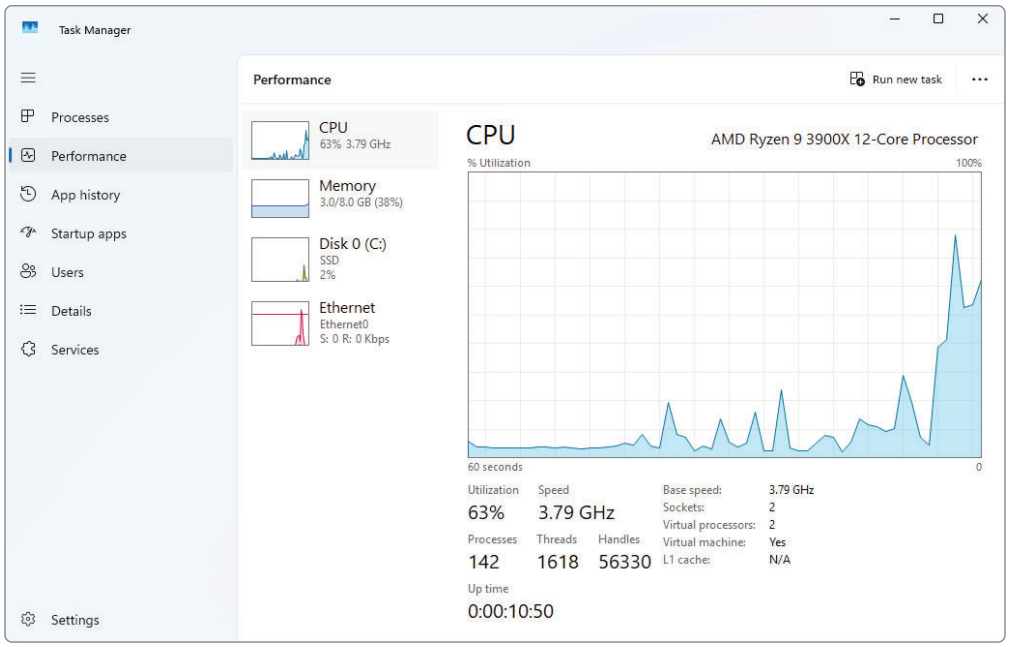
If you decide to change the priority of an application, you'll be warned that doing so may make it unstable. You can generally ignore this option when changing the priority to Low, Below Normal, Above Normal, or High, but you should heed this warning when changing applications to the Realtime priority. Realtime means that the processor gives precedence to this process over all others—over security processes, over spooling, over everything—and this is sure to make the system unstable.

Task Manager changes the priority only for that instance of the running application. The next time the process is started, priorities revert back to that of the base (typically, Normal).

Services The Services tab lists the name of each running service as well as the process ID associated with it and its description, status, and group. A button labeled Open Services appears on the bottom of this screen. Clicking it will open the *Microsoft Management Console (MMC)* snap-in for Services, where you can configure each service. Within Task Manager, right-clicking a service will open a context menu listing six choices: Start, Stop, Restart, Open Services (which opens the Service MMC), Search Online, and Go To Details (which takes you to the Processes or Details tab).

Performance The Performance tab displays a variety of information, including overall CPU usage percentage, a graphical display of CPU usage history, the number of processes, and a graphical display of physical memory. This tab also provides you with additional memory-related information, such as physical and kernel memory usage as well as the total number of handles, threads, and processes. Total, limit, and peak commit-charge information also appears, as shown in Figure 2.4.

FIGURE 2.4 The Performance tab of Task Manager



Some of the items are beyond the scope of this book, but it's good to know that you can use the Performance tab to keep track of system performance. Note that the number of processes, CPU usage percentage, and commit-charge information always appear at the bottom of the Task Manager window, regardless of which tab you have currently selected.

Users The Users tab provides you with information about the users logged into the local machine. You'll see the username, status, CPU usage, memory usage, disk usage, and network usage. You can right-click the name of any connected user to perform a variety of functions, including viewing active processes, sending the user a message, signing off the user, switching accounts, and initiating a remote-control session to the user's machine, if they are connected remotely.

Startup The Startup tab lists the names of services configured to begin at startup as well as the publisher, status, and startup impact. From this tab, you can select any service listed and choose to disable it. This functionality was provided by the `msconfig` command in prior versions of Windows, such as Windows 7.

Use Task Manager whenever the system seems bogged down by an unresponsive application. You will work with Task Manager in Exercise 2.1.

EXERCISE 2.1

Working with Task Manager

1. Select Start, type **`notepad`**, and then press Enter.
2. Right-click an empty portion of the Taskbar and select Task Manager.
3. Find Notepad on the Processes tab, right-click, and select Go To Details.
4. Right-click on `notepad.exe` on the Details tab that opened and select Set Priority, then Above Normal.
5. Confirm the change by clicking Change Priority in the dialog box that opens to confirm your change.
6. Right-click on the column header and choose Select Columns from the context menu.
7. Scroll through the list and select the Base Priority option, then click OK.
8. Observe the status of `notepad.exe`.
9. Right-click `notepad.exe` and select End Task.
10. Confirm the ending of the task by clicking End Process in the dialog box that displays.

Microsoft Management Console

Microsoft created the Microsoft Management Console (MMC) interface as a frontend in which you can run administrative and configuration tools. Many administrators don't even know that applications they use regularly run within an MMC. In the following sections, we will cover many of the different MMC snap-ins you will use in your day-to-day

administration and configuration of the operating system. You can start the MMC by pressing Windows key+R to open Run, typing **mmc.exe** in the box, and then clicking OK. Once the MMC is started, you can create a custom MMC, adding the snap-ins discussed next. Click File ➢ Add Or Remove Snap-ins, and then select which snap-ins to add. For more information about the MMC, visit `https://learn.microsoft.com/en-us/troubleshoot/windows-server/system-management-components/what-is-microsoft-management-console`.

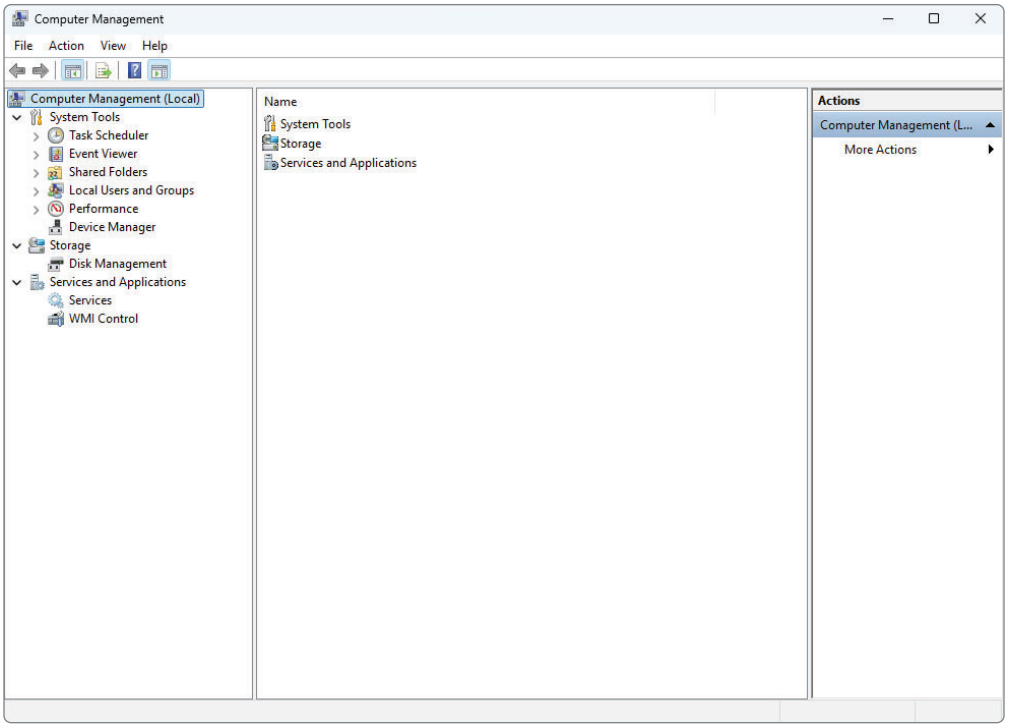
Computer Management

Windows includes a snap-in to manage computer settings: the Computer Management console. The Computer Management console can manage more than just the installed hardware devices; it can manage all the services running on a computer in addition to Device Manager, which functions almost identically to the one that has existed since Windows 9*x*. It also contains Event Viewer, which shows any system errors and events as well as methods to configure the software components of all the computer's hardware.

To access the Computer Management console, right-click the Start menu and select Computer Management from the context menu. You can also press Windows key+R and type **compmgmt.msc** to start the console.

After you are in Computer Management, you will see all the tools available, as shown in Figure 2.5. This is one power-packed snap-in and includes the following system tools:

FIGURE 2.5 Computer Management



Task Scheduler Allows for the scheduling of tasks using a simple command, script, or external program. Many of the operating system's functions are tasks such as checking for updates.

Event Viewer A link to the tool that allows you to view application error logs, security audit records, and system errors. We cover Event Viewer in detail shortly.

Shared Folders Allows you to manage all your computer's shared folders. You can also view active remote sessions and open files from this console.

Local Users and Groups Allows you to create and manage local user and group accounts.

Performance Shows you how your system hardware is performing and alerts you if system performance goes under a threshold that you set.

Device Manager Allows you to manage hardware devices and perform tasks such as updating drivers, disabling devices, and uninstalling devices. Device Manager is the first place to visit if a device is physically installed but not working.

Storage Allows you to perform basic disk management on the operating system.

Services and Applications Allows you to control services in the operating system.

Event Viewer

Event Viewer (`eventvwr.msc`) is an MMC snap-in that shows a lot of detailed information about what is running on your operating system. You can start it in several different ways—for example, by clicking Start, typing **Event Viewer**, and selecting the Event Viewer app, or by right-clicking the Start button and selecting Event Viewer from the context menu. In addition, you can add it as a snap-in inside the MMC or press Windows key+R, type **`eventvwr.msc`**, and press Enter.

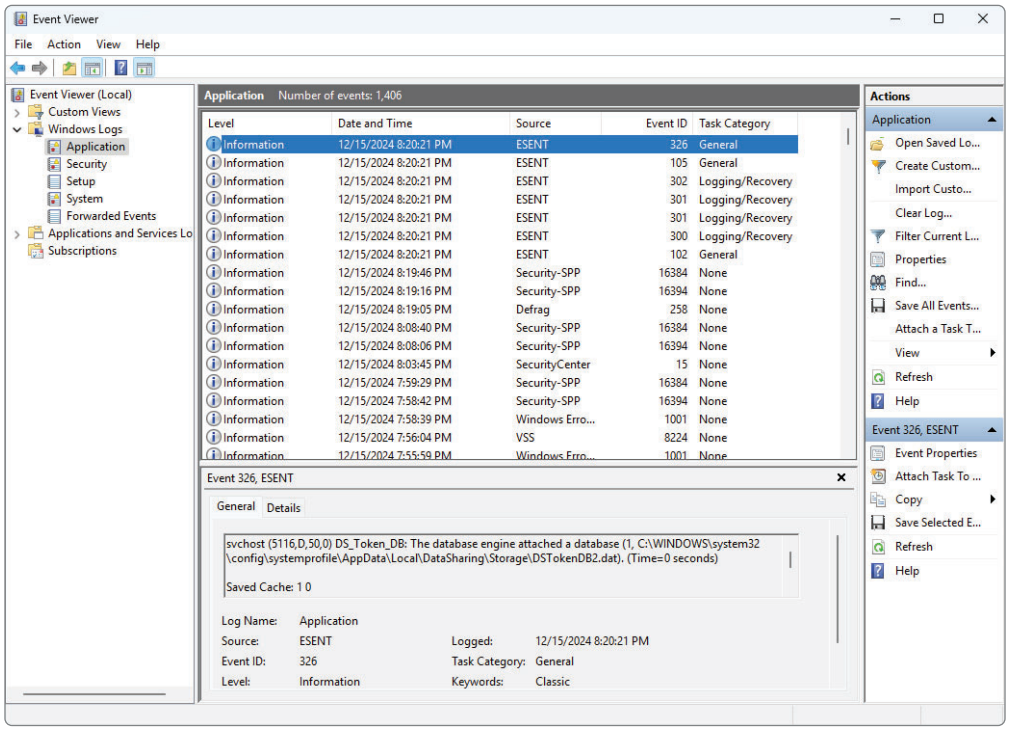
Event Viewer should be the first place you look when you are trying to solve a problem whose solution is not evident. The system and applications will often create an entry in Event Viewer that can be used to verify operation or diagnosis problems, as shown in Figure 2.6.

Table 2.1 highlights the three main event logs that you should be concerned with for the exam. Each feature in Windows has the ability to store its events in specific log files, in application and service logs.

TABLE 2.1 Event Viewer logs

Event Log	Description
Application	Events generated by applications installed on the operating system
Security	Events generated by the Security Reference Monitor in the Executive kernel
System	Events generated by the operating system

FIGURE 2.6 Event Viewer



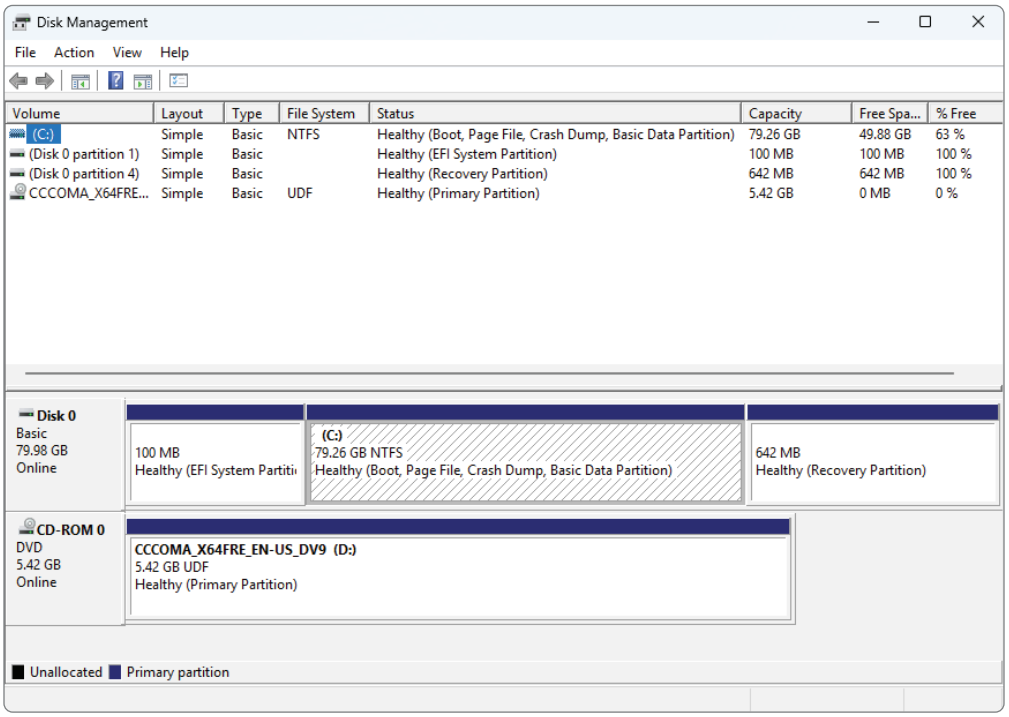
Although you might think that all the security-related information is in the Security log, you're only half right. The Security log is used by the Security Reference Monitor inside the *Executive kernel*. It is responsible for reporting object audit attempts. Examples of object audit attempts include file access, group membership, and password changes.

Most of the useful security-related information will be in the application and system logs. Using these logs, you can see errors and warnings that will alert you to potential security-related problems. When you suspect an issue with the operating system or an application that interacts with the operating system, you should check these logs for clues. The event log won't tell you exactly what is wrong and how to fix it, but it will tell you if there is an issue that needs to be investigated.

Disk Management

The Disk Management (`diskmgmt.msc`) snap-in is used to view disk information, such as volumes configured on the physical disk and the *filesystems* that are formatted on the volume. Disk Management, shown in Figure 2.7, isn't used just to view information; you can also use it to partition volumes on a new or existing disk, format filesystems, and mount volumes to drive letters. These are just a few configuration tasks; we will cover Disk Management later in this chapter.

FIGURE 2.7 Disk Management



Task Scheduler

Task Scheduler (`taskschd.msc`) is accessible beneath the Computer Management snap-in or inside Windows Tools from the Start menu under All Apps. This snap-in will allow you to configure an application to run automatically or at any regular interval (see Figure 2.8). A number of terms are used to describe the options for configuring tasks: *action* (what the task actually does), *condition* (an optional requirement that must be met before a task runs), *setting* (any property that affects the behavior of a task), and *trigger* (the required condition for the task to run).

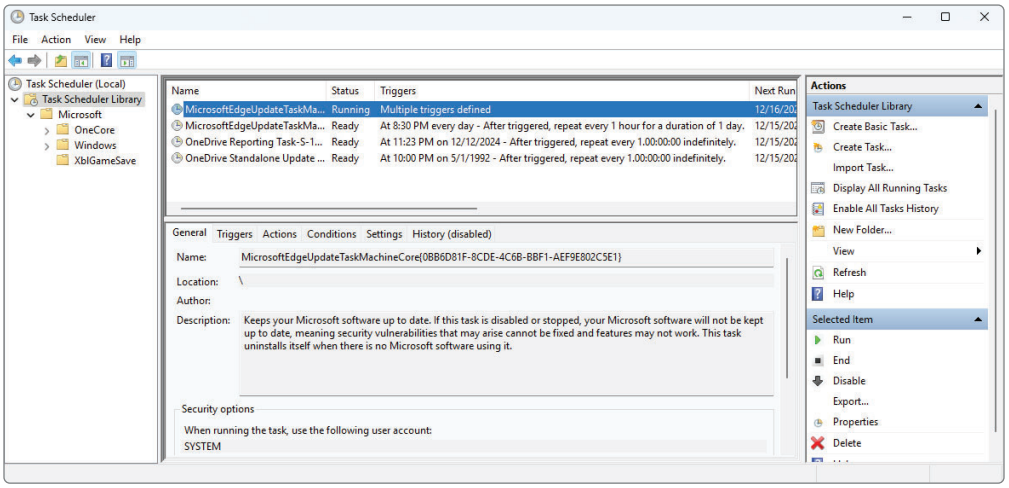
For example, you could configure a report to run automatically (action) every Tuesday (trigger) when the system has been idle for 10 minutes (condition), and only when requested (setting).

Device Manager

Device Manager (`devmgmt.msc`) is an indispensable tool for the management of peripherals and components attached to the computer. You can view all the devices in the system, as shown in Figure 2.9. Device Manager has been around since Windows 95, and it hasn't changed all that much. Because computer systems have evolved and all newer computer

systems are plug-and-play capable, this tool is usually only used when you are in doubt that a component is working or you're troubleshooting a problem, or when you want to check the driver. Device Manager was used in the past a lot more since most peripherals had to be manually configured.

FIGURE 2.8 Task Scheduler



Device Manager allows you to manually update the driver for a device, roll back a driver to a prior version, uninstall a device, and disable a device. Some peripherals, such as the network card, allow you to configure if you want the device to wake the computer. Some peripherals, such as a disk drive or USB device, can be configured to fall asleep if they are not used for a specified period of time.

There are a number of states that a device inside of Device Manager can be in. The following are the possible states:

Working In a working state the device will not have any symbol or icon overlayed and the device will function as normal.

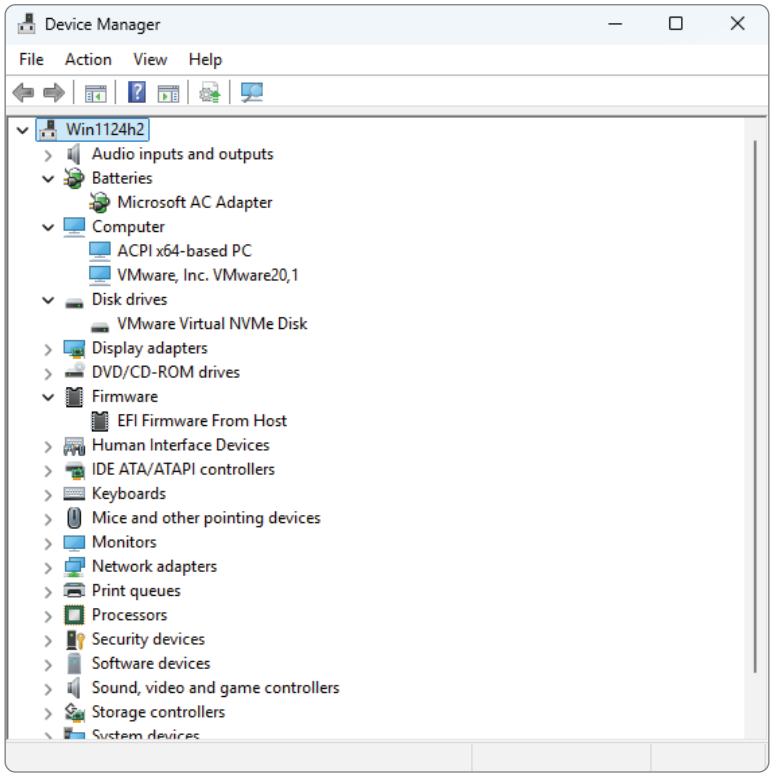
Disabled In a disabled state the device will not function. The symbol or icon overlayed will be a down arrow in a circle.

Problem In a problem state the device can function, but there is an underlying issue. The symbol or icon overlayed on the device will be a yellow exclamation mark in either a circle or triangle.

Driver Issue In a missing driver issue state the device is functioning with a compatible driver and lacks a device-specific driver. The symbol or icon overlayed will be a question mark.

Manual In a manual state the device can function as normal. The symbol or icon overlayed will be in the form of a lowercase i. This signifies that the Automatic Settings option is not selected for the device.

FIGURE 2.9 Device Manager



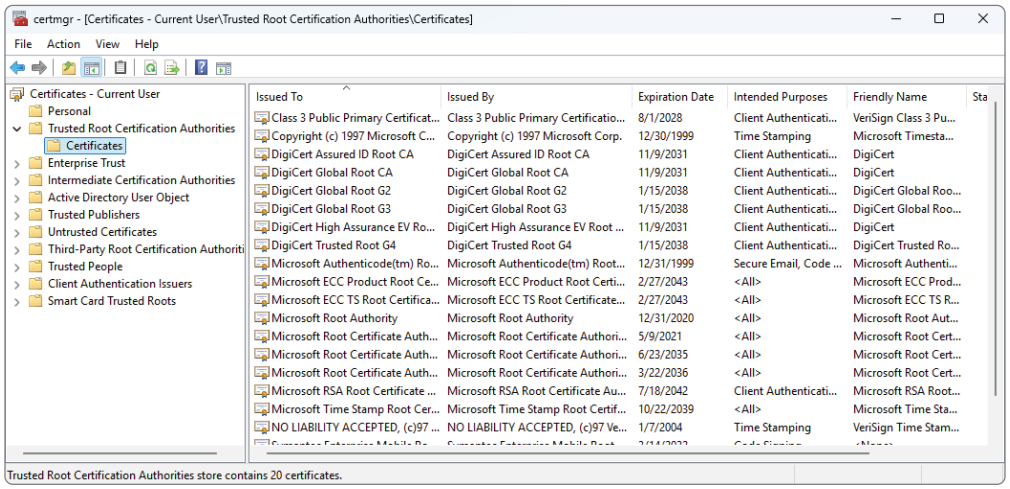
Certificate Manager

Certificate Manager (`certmgr.msc`) is used to view and manage certificates used by the web browser and the operating system, as shown in Figure 2.10. Certificates are used for a number of reasons; the most common are encryption and digital signatures that provide trust.

Certificate Manager allows you to manage certificates for your user account, a service account, or the computer account. When you choose to manage your user account, the certificates managed can be used only by your account. So, the certificates are only relevant while you are logged in and using an application that requires a certificate from the certificate store. When you choose to manage certificates for a particular service account, the certificate is only relevant to that specific service, such as a virtual private network (VPN) service. When you choose to manage certificates for the computer account, these

certificates are relevant only for the operating system, even if someone is not logged in. This configuration mode is commonly used when configuring a certificate for the Internet Information Services (IIS) web server which is integrated into most editions of Windows Server.

FIGURE 2.10 Certificate Manager



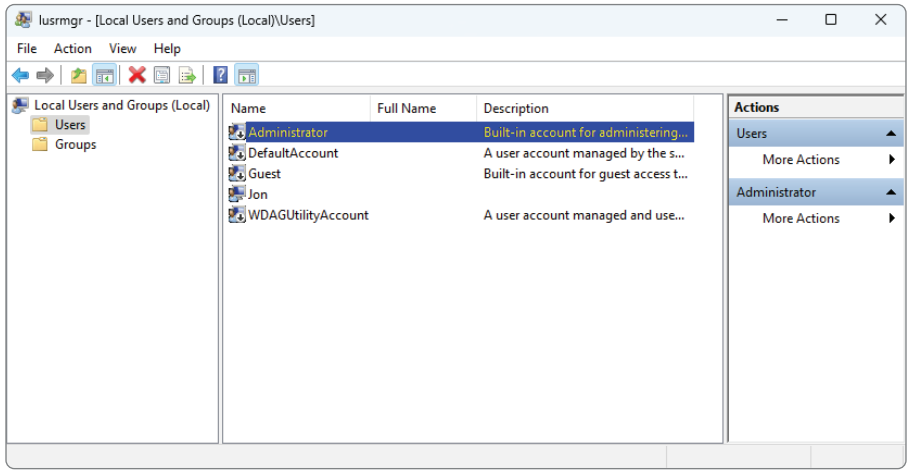
Local Users and Groups

The Local Users and Groups (`lusrmgr.msc`) MMC snap-in allows for granular control over local user accounts and groups for the Windows operating system. You can access the Local Users and Groups MMC by right-clicking the Start menu and choosing Computer Management. You can also press the Windows key+R, type **`lusrmgr.msc`**, and press Enter. When the tool launches, if you click Users you will see several built-in user accounts, all of which are disabled (depicted with the down arrow), as shown in Figure 2.11. The only active account on a brand-new Windows 11 operating system is the first account setup.

The MMC extension is divided into the following two parts:

- The Users section allows for the viewing, creation, and modification of individual users. You can also access disabled users, such as the built-in Administrator account and Guest account.
- The Groups section allows for the viewing, creation, and modification of groups. The User Accounts Control Panel applet only allows assignment of the administrator group with a radio button–type dialog box. However, from this MMC extension, you can assign many different groups. Each group assigns more permissions to a user.

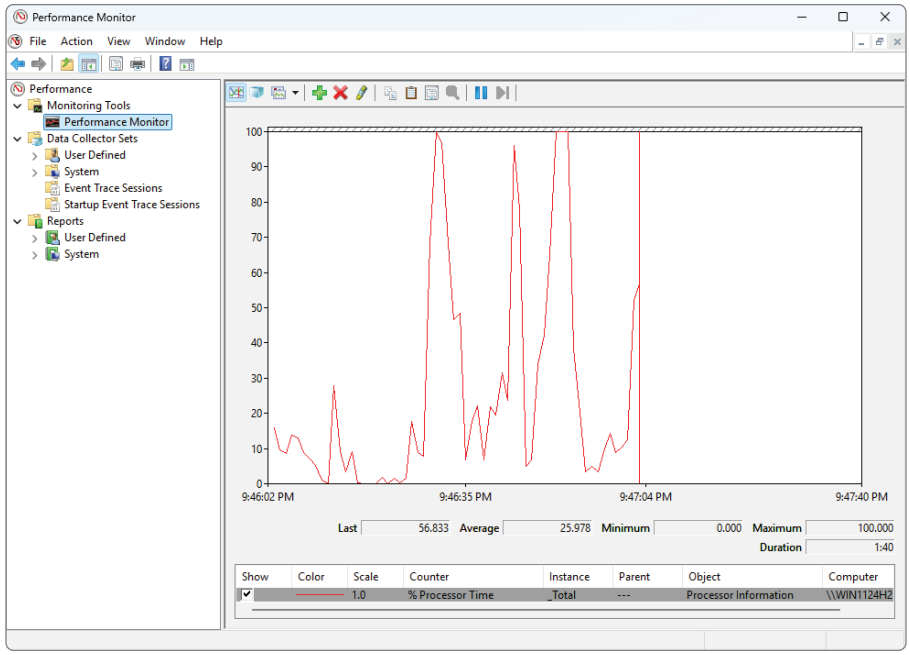
FIGURE 2.11 Local Users and Groups



Performance Monitor

Performance Monitor (`perfmon.msc`) varies a bit in different versions of Windows, but it has the same purpose throughout: to display performance counters, as shown in Figure 2.12. The tool collects counter information and then sends that information to a console or event log.

FIGURE 2.12 Performance Monitor



Performance Monitor's objects and counters are very specific; you can use Performance Monitor as a general troubleshooting tool as well as a security troubleshooting tool. For instance, you can see where resources are being used and where the activity is coming from. In Exercise 2.2, you see how to work with Performance Monitor.

EXERCISE 2.2

Working with Performance Monitor

1. Select Start, type **`perfmon`**, and then press Enter.

   Make sure that you select Performance Monitor, as opposed to System Monitor or another default that may come to the forefront.

2. Choose the Performance Monitor section under Monitoring Tools.
3. Click the plus sign (+) or right-click in the graphical display area and select Add Counters.
4. Expand the Processor section, and then select the %Processor Time object.
5. Click Add and then click OK.
6. Open Windows File Explorer, click the C: drive, type `*` into the search box, and then press Enter.
7. Quickly change to Performance Monitor and watch the impact of this search on the processor.

   This action is time consuming and therefore will help you notice the changes that take place in Performance Monitor.

8. Run the same operation again. This time, however, change your view within Performance Monitor to the histogram bar by clicking the button directly to the left of the plus sign.
9. Run the same operation again, changing your view within Performance Monitor to Report.
10. Exit Performance Monitor.

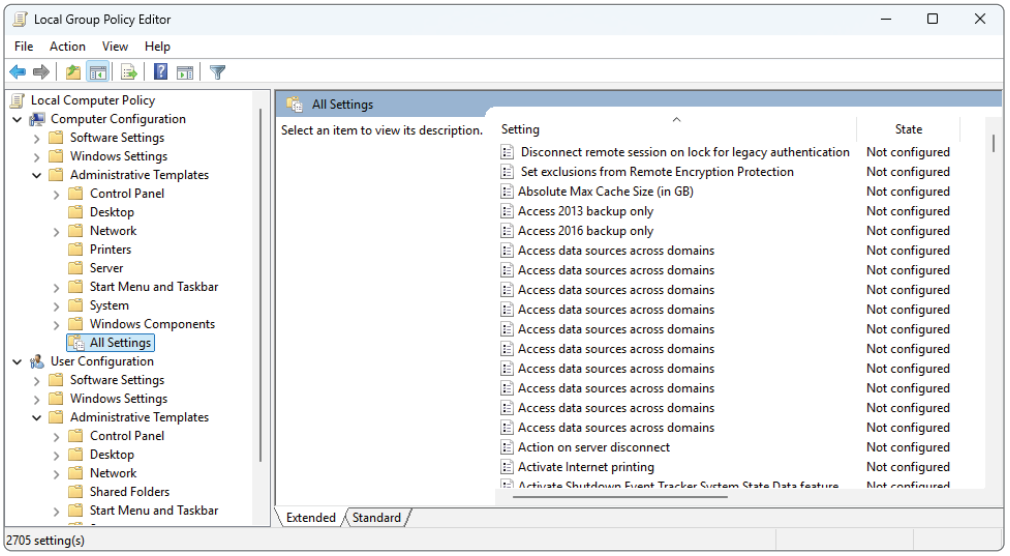
Group Policy Editor

The Group Policy Editor (`gpedit.msc`) tool allows you to edit the local Group Policy for the operating system. *Group Policy* is a mechanism that allows an administrator to set various settings to customize the operating system. These settings can restrict the operating

system in a number of ways, such as removing the Settings tab from a built-in application. Group Policy can also control aspects of security for the operating system. Group Policy also contains a mechanism to enforce these settings by reapplying the settings periodically in the event they are changed.

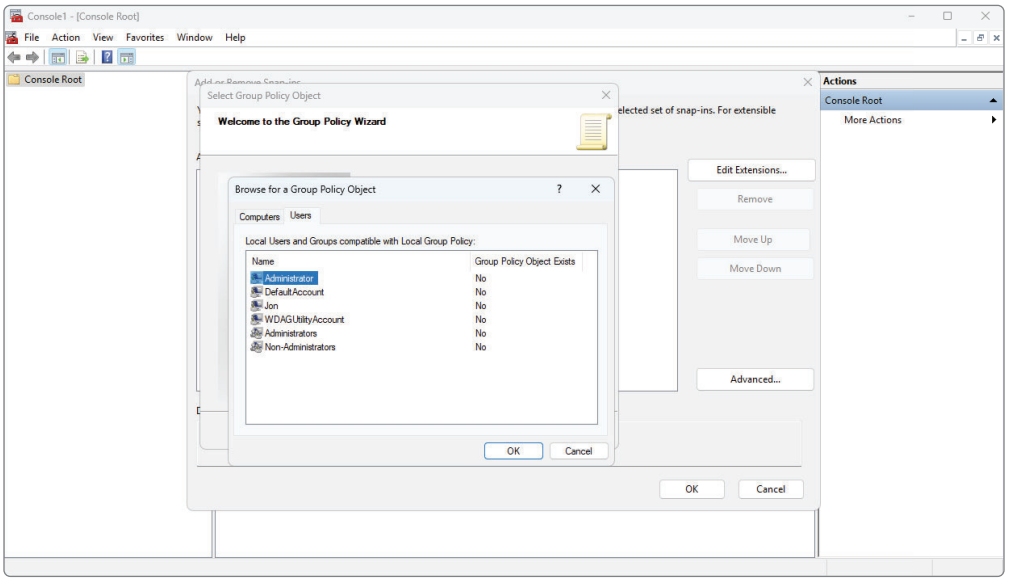
The Group Policy Editor divides all settings between computer settings and user settings. The computer settings affect the operating system and how it behaves. The user settings affect the user logged onto the operating system. If you open the Group Policy Editor by typing **`gpedit.msc`** in the Run dialog box and pressing Enter, the tool will open and display a tree of settings, as shown in Figure 2.13.

FIGURE 2.13 Group Policy Editor



There is another way to open the Group Policy Editor that allows you to edit the Group Policy for a specific user, administrators, or all non-administrators. You must first start the MMC tool by typing **`mmc.exe`** in the Run dialog box and then pressing Enter. Then, add the snap-in Group Policy Object Editor. After it is selected, another dialog box will pop up where you can browse to select a specific user or group or a special group of users, as shown in Figure 2.14.

FIGURE 2.14 Group Policy Editor browse dialog box



Additional Tools

When the Microsoft Management Console (MMC) was first introduced with Windows 2000, it was to be a single pane of glass for monitoring and configuration of the Windows operating system. Almost 25 years later, we still use a mixture of tools outside the MMC to monitor and configure Windows. Next, we discuss some of these additional utilities and tools.

System Configuration Tool

The System Configuration (`msconfig.exe`) tool allows you to configure how Windows 11 starts up, as well as launching additional tools. The tabs of the System Configuration tool differ a bit based on the Windows version you are running. The main tabs are General, Boot, Services, Startup, and Tools. Figure 2.15 shows the General tab for Windows 11. From here, you can configure the startup options.

Figure 2.16 shows the Boot tab for Windows 11. Note that from here, you can configure the next boot to be a safe boot, and you can turn on the boot information so that you can see drivers as they load—which is quite useful when a system keeps hanging during boot. Although the option is titled Safe Boot, this is classically referred to as booting into Safe mode.

FIGURE 2.15 System Configuration General tab

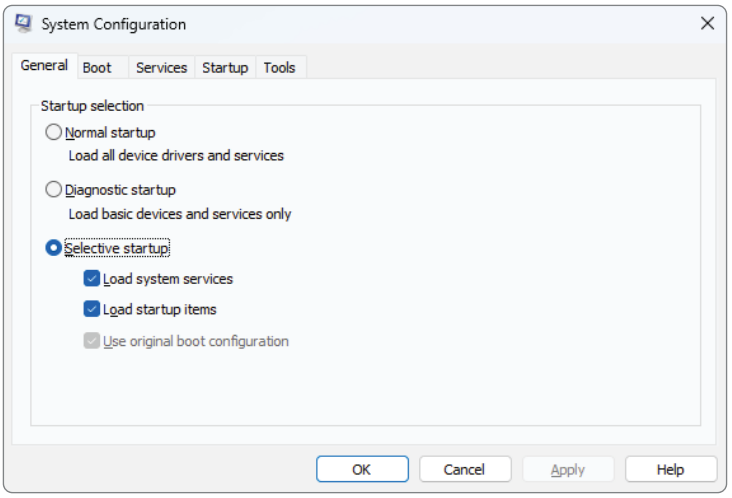


FIGURE 2.16 System Configuration Boot tab

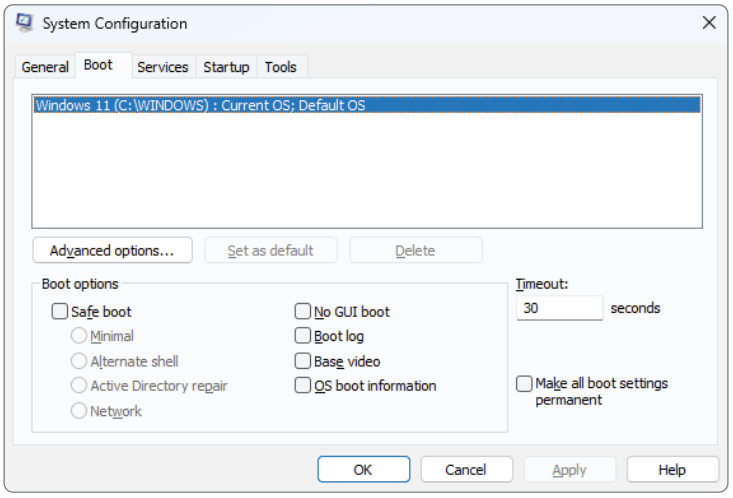


Figure 2.17 shows the Services tab for Windows 11. On this tab, you can view the services installed on the system and their current status (running or stopped). You can also enable or disable services.

In Windows 7 and earlier, the Startup tab allowed you to configure applications that start up when any user logs in. In Windows 10/11, the Startup tab redirects you to the Startup tab in Task Manager, where these tasks can be performed, as shown in Figure 2.18.

FIGURE 2.17 System Configuration Services tab

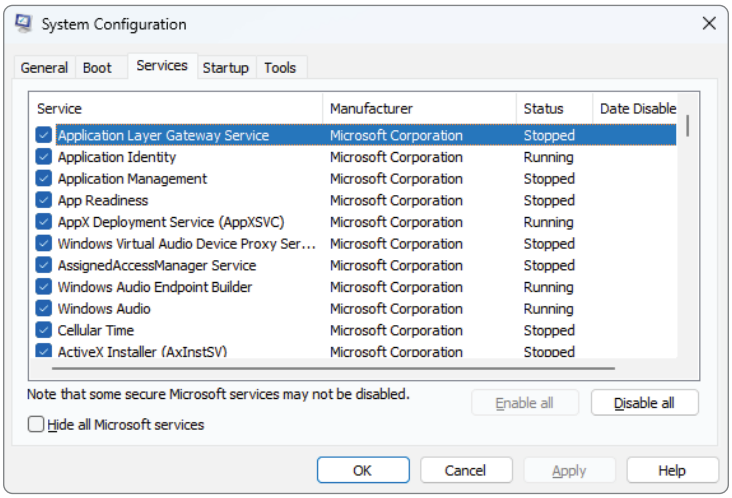


FIGURE 2.18 System Configuration Startup tab and Task Manager Startup tab

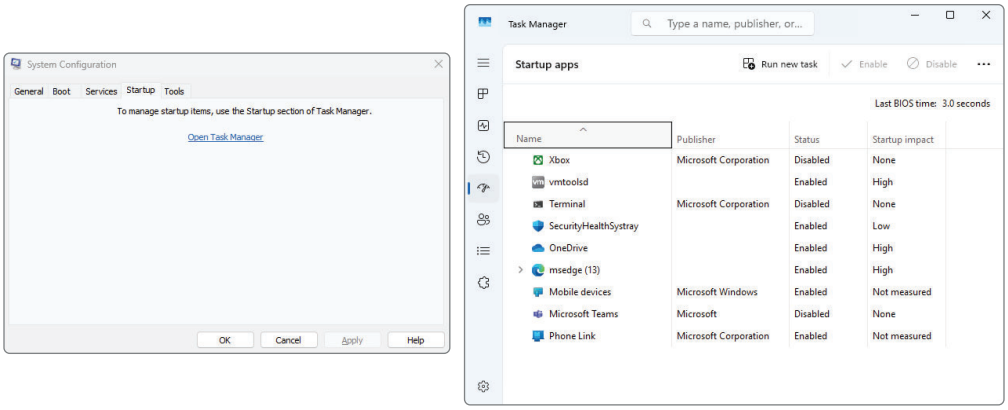


Figure 2.19 shows the Tools tab for Windows 11. On this tab, you can launch a number of administrative tools to configure various Windows features.

Keep in mind that the tabs differ slightly based on the operating system version. We walked through the CompTIA objectives related to this tool in this discussion.

System Information Tool

The System Information (`msinfo32.exe`) tool displays a fairly thorough list of settings on the machine (see Figure 2.20). You cannot change any values here, but you can search, export, and save the output. Several command-line options can be used when starting `msinfo32`; Table 2.2 summarizes them.

FIGURE 2.19 System Configuration Tools tab

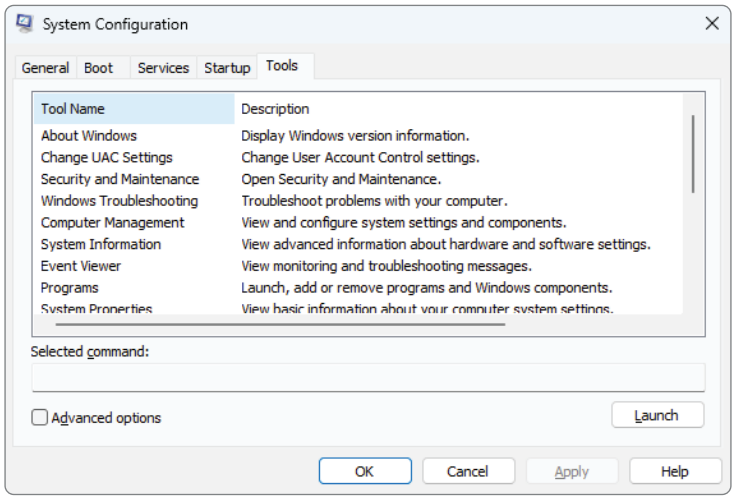


FIGURE 2.20 The Msinfo32 interface shows configuration values for the system.

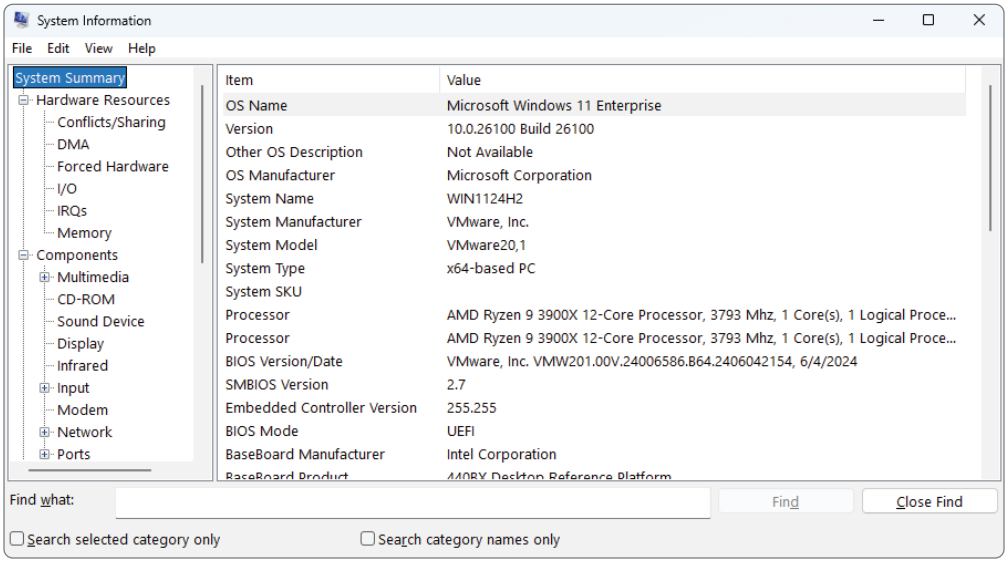


TABLE 2.2 `msinfo32` command-line options

Option	Function
`/computer`	Allows you to specify a remote computer on which to run the utility
`/nfo`	Creates a file and saves it with an `.nfo` extension
`/report`	Creates a file and saves it with a `.txt` extension

The use of command-line options will be covered in Volume 2 Chapter 3, "Windows Administration," when you learn about installing and upgrading to Windows 11.

Resource Monitor

Resource Monitor (`resmon.exe`) is used to identify resource utilization of CPU, disk, network, and memory on Windows. The utility was originally introduced in Windows Vista and has been included with all new releases of Windows. The utility can be launched a few different ways. The first is by entering the command **`resmon.exe`** in the Run dialog box. Another way to open the Resource Monitor is from inside Task Manager. The upper-right of the Performance tab is three dots, if you click on them and then select the option Resource Monitor; clicking it will launch Resource Monitor in a new window. Figure 2.21 shows Resource Monitor and the four tabs for each resource.

The CPU tab allows you to identify the process with the highest amount of CPU utilization on the operating system. Each column can be sorted; it is as simple as clicking the CPU column to sort the column and find the process with the highest CPU utilization. When you do find the process, you can right-click the entry and choose to end the process, suspend the process, and even search the process online.

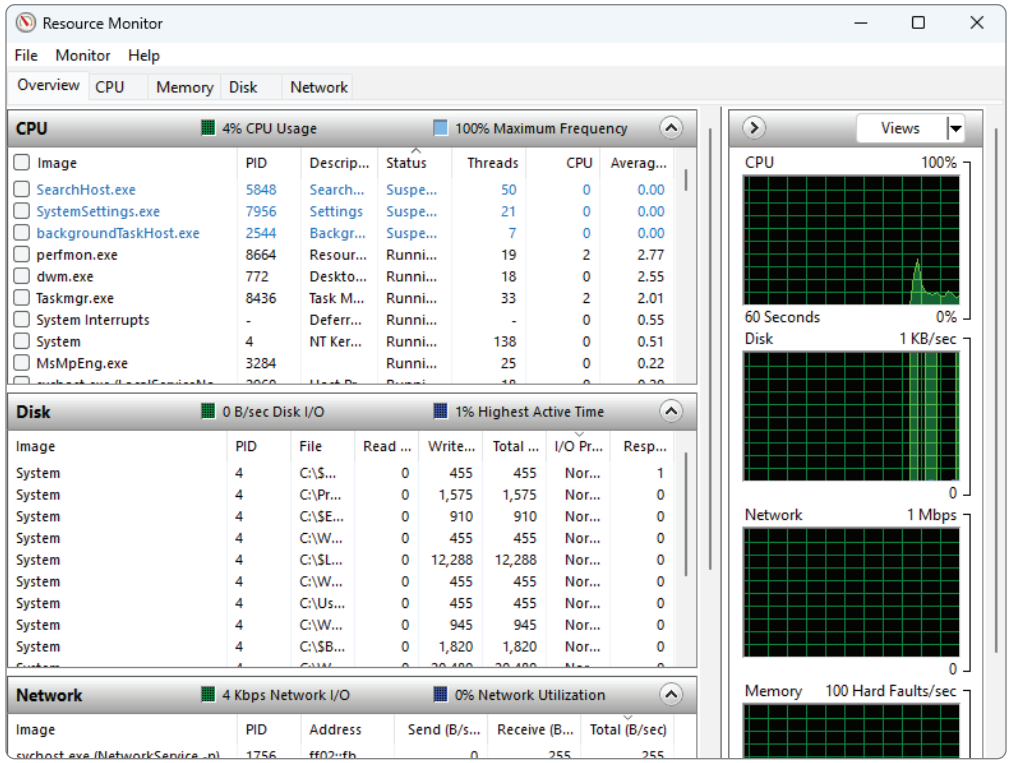
The Memory tab displays detailed memory usage of the processes running on the operating system, as shown in Figure 2.22. This tab can help you identify memory in use, memory reserved by hardware, and memory available by the operating system and processes.

The Disk tab helps you identify a process that is overusing the hard drive with a high amount of read requests, write requests, or overall usage. The Disk tab will also allow you to identify the I/O priority of processes and their response time. This tab is extremely useful when you suspect that a process is slowing down the system.

The Network tab displays all the processes that are currently utilizing the network. The processes can be sorted by send, received, and total bytes per second. The Network tab does a lot more than just displaying activity; it also shows the destination addresses for each

process. This is valuable information if you suspect name resolution is a problem with the remote application. Opening the TCP Connections drop-down, you can view the active TCP connections on the operating system, along with packet loss and latency. Normally this active view of network traffic can only be performed with a packet capture tool or another third-party tool. The Network tab is not just useful for information on outgoing applications, it can also display processes listening on TCP and UDP ports. The Resource Monitor tool will also display *firewall* status for the processes listening on the operating system.

FIGURE 2.21 Resource Monitor



Disk Cleanup

Storage space is finite in a computer, and it is inevitable that it will be consumed over time. It can be consumed by a number of files, such as the operating system, downloaded programs, temporary files, updates, deleted files, and of course your data files. The Disk Cleanup (`cleanmgr.exe`) tool can clean up operating system files to free up space without affecting your data files. Disk Cleanup, shown in Figure 2.23, can be launched by entering the command **`cleanmgr.exe`** in the Run dialog box, or by searching for Disk Cleanup on the Start menu.

FIGURE 2.22 Resource Monitor Memory tab

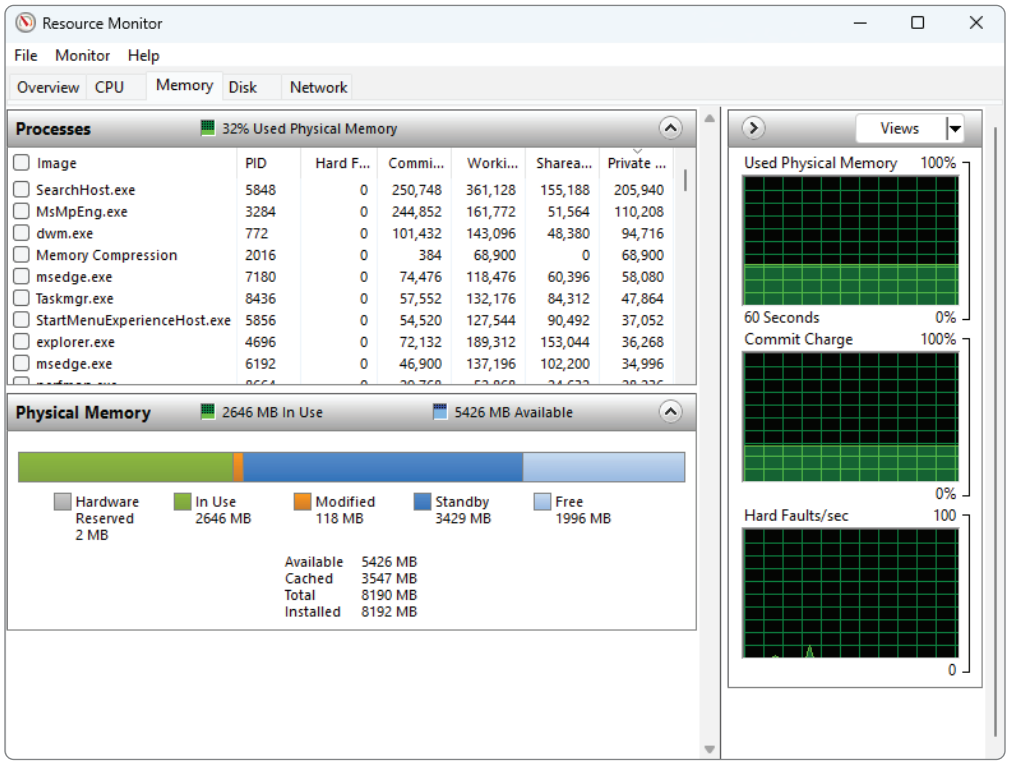
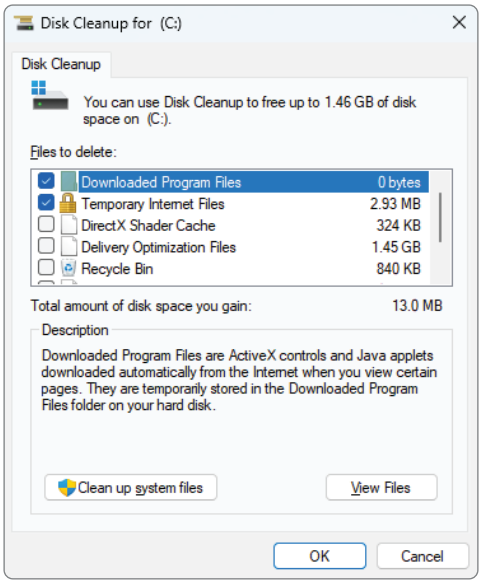


FIGURE 2.23 Disk Cleanup



Once Disk Cleanup is launched, you can select the various files you want to clean up on the system drive. The categories you can delete are Downloaded Program Files, Temporary Internet Files, DirectX Shader Cache, Delivery Optimization Files, Recycle Bin, Temporary Files, Thumbnails, Windows Update Files, Windows Defender Antivirus, and countless other files. For some categories, you can view the files that you can potentially delete; two such categories are Temporary Internet Files and Downloaded Program Files. However, there are other categories in which you can view the individual files. After selecting the categories you want to delete, you can purge them from the disk, thus freeing up space.

Objective 1.6, "Given a scenario, configure Microsoft Windows settings," requires you to know various Control Panel settings and their controls.

Control Panel

Although, for the most part, Windows is functional from the time it is installed, Microsoft realized that if someone were going to use computers regularly, they would probably want to be able to customize their environment so that it would be better suited to their needs—or at least more fun to use. As a result, the Windows environment has a large number of utilities that are intended to give you control over the look and feel of the operating system.

This is, of course, an excellent idea. It is also a bit more freedom than some less-than-cautious users seem to be capable of handling. You will undoubtedly serve a number of customers who call you to restore their configuration after botched attempts at changing one setting or another.

More than likely, you will also have to reinstall Windows yourself a few times because of accidents that occur while you are studying or testing the system's limits. This is actually a good thing, because no competent computer technician can say that they have never had to reinstall because of an error. You can't really know how to fix Windows until you are experienced at breaking it. So, it is extremely important to experiment and find out what can be changed in the Windows environment, what results from those changes, and how to undo any unwanted results. To this end, we will examine the most common configuration utility in Windows: Control Panel, as shown in Table 2.3, which describes some popular Control Panel applets. Also, not all applets are available in all versions and many of them have gradually been move by Microsoft to the Settings.

TABLE 2.3 Selected Windows Control Panel applets

Applet name	Function
Device Manager	Adds and configures new hardware.
Programs and Features	Changes, adds, or deletes software.
Administrative Tools	Performs administrative tasks on Windows 10. In Windows 11 it has been replaced with Windows Tools.

TABLE 2.3 Selected Windows Control Panel applets *(continued)*

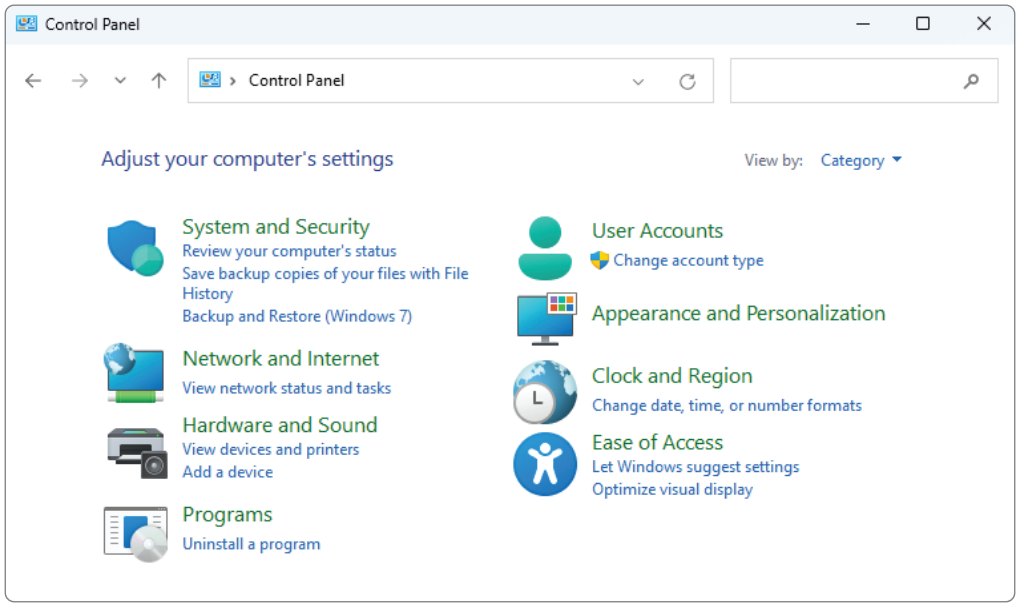
Applet name	Function
File Explorer Options	Configures the look and feel of how folders are displayed in Windows File Explorer.
Internet Options	Sets a number of Internet connectivity options.
Network and Sharing Center	Sets options for connecting to other computers.
Power Options	Configures different power schemes to adjust power consumption.
Devices and Printers	Configures printer settings and print defaults.
System	Allows you to view and configure various system elements (discussed in more detail later in this chapter).
Windows Defender Firewall	Configures basic firewall exemptions.
Mail (Microsoft Outlook)	Configures the Outlook mail profile (only appears if Outlook is installed).
Sound	Configures audio.
User Accounts	Configures local accounts on the operating system.
Indexing Options	Configures the folders to index for search capabilities.
Ease of Access Center	Allows for accessibility options for users.

In the current version of Windows, when you first open Control Panel, it appears in Category view, as shown in Figure 2.24. Control Panel programs have been organized into various categories, and this view provides you with the categories from which you can choose. When you choose a category and pick a task, the appropriate Control Panel program opens. Or, you can select one of the Control Panel programs that is part of the category.

You can change this view to Small Icons or Large Icons in Windows 11, which displays all the Control Panel programs in a list, as in older versions of Windows. The items are organized by category and not by large icons in Windows by default. Therefore, we *strongly*

suggest that administrators change to the large icon view. To do so, select Large Icons in the View By drop-down box in the right corner of Control Panel. Throughout this chapter, when we refer to accessing Control Panel programs, we will assume that you have changed the view to the Large Icons view.

FIGURE 2.24 The Windows Control Panel in Category view

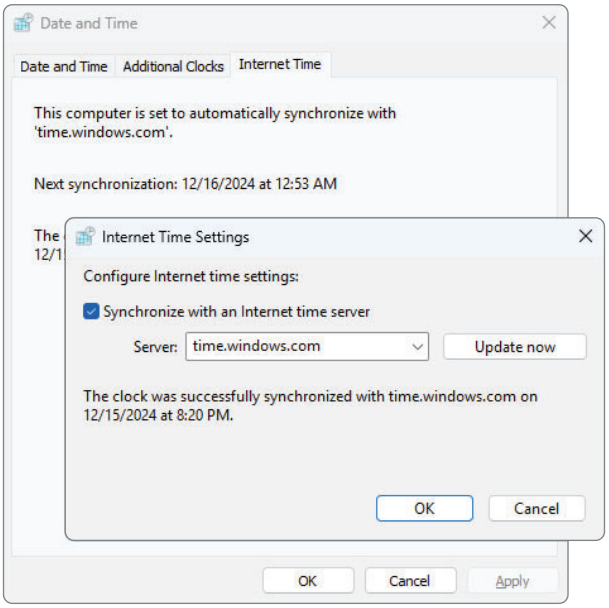


For a quick look at how the Control Panel programs work, in Exercise 2.3, you'll examine some of the settings in the Date and Time applet (`timedate.cpl`).

Filenames appearing in parentheses, such as `timedate.cpl` in the preceding paragraph, can be used to start the utility without needing to go through the Control Panel dialog boxes.

The Date and Time applet is used to configure the system time, date, and time zone settings, which can be important for files that require accurate time stamps (or for users who don't wear a watch). Because Date and Time is a simple program, it's a perfect example to use. Current versions of Windows have an Internet Time Settings tab, which enables you to synchronize time on the computer with an Internet time server (the options in Windows 11 are shown in Figure 2.25). By default, the Internet time server is set to `time.windows.com`.

FIGURE 2.25 Windows Date and Time/Internet Time settings



EXERCISE 2.3

Changing the Time Zone

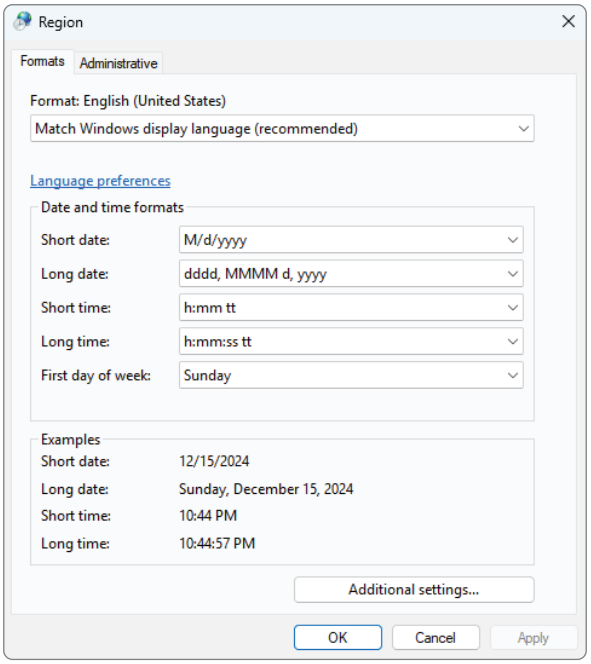
1. Click Start, type **Control Panel** or **`timedate.cpl`**, and then select the appropriate result from the search.
2. From Control Panel, double-click the Date and Time icon. (By default, the programs are listed alphabetically.)
3. Click the Change Time Zone button, and from the drop-down menu, select (GMT –03:30) Newfoundland.
4. Hop a plane to Newfoundland, secure in the knowledge that you will know what time it is once you get there.
5. If you skipped step 4, change the time zone back to where it should be before closing the window.

Microsoft is slowly moving away from Control Panel in favor of the Settings app. Many of the same settings appear in both Control Panel and the Settings app. Some settings can only be set in the Settings app, such as Windows Update settings, whereas others, such as Storage Spaces, can only be set in the classic Control Panel.

Regional and Language Options

You can configure regional settings through the Control Panel applet Region in Windows 11. Using this applet (`intl.cpl`), you can choose which format is used for numbers (see Figure 2.26), your geographic location, and the language to be used for non-Unicode programs. In Windows 11, the Settings app is also used to set the layout of the keyboard you are using as well as set the regional settings previously mentioned. We will explore the Settings app further on in this chapter.

FIGURE 2.26 Windows Region Control Panel applet



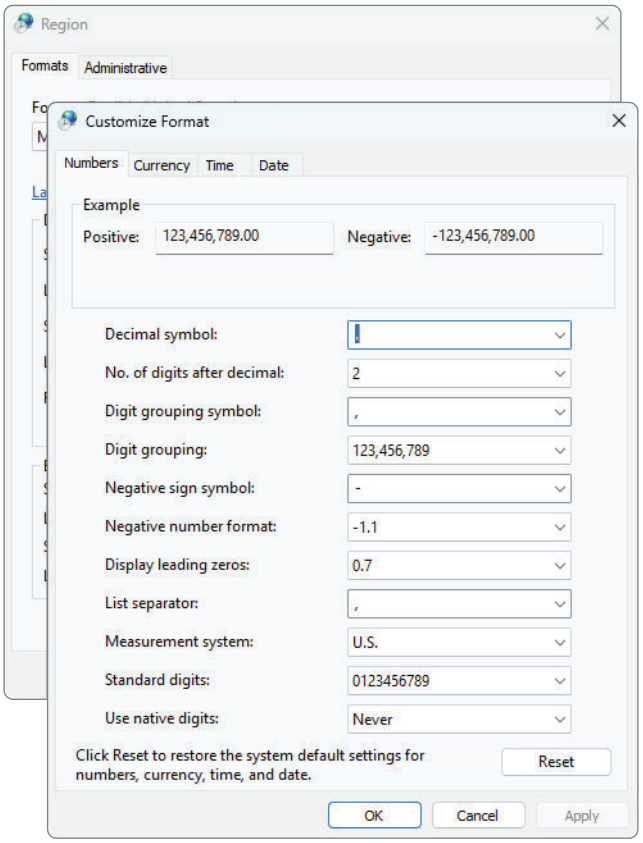
The ability to support so many languages is provided through the use of the *Unicode* standard. In Unicode, and the Unicode Character Set (UCS), each character has a 16-bit value. This allows the same character to be interpreted/represented by 65,536 different entities.

If you click Additional Settings, you can go beyond the date and time formats and also configure number and currency, as shown in Figure 2.27.

Internet Options Applet

The Internet Options applet (`inetcpl.cpl`) brings up the Internet Properties dialog box, as shown in Figure 2.28. The tabs include General, Security, Privacy, Content, Connections, Programs, and Advanced. Use this applet to configure the browser environment for any program running on Windows that makes use of Internet Explorer settings, such as user proxy settings or Internet security zone setting, just to name a few.

FIGURE 2.27 Windows Additional Settings Region Control Panel applet



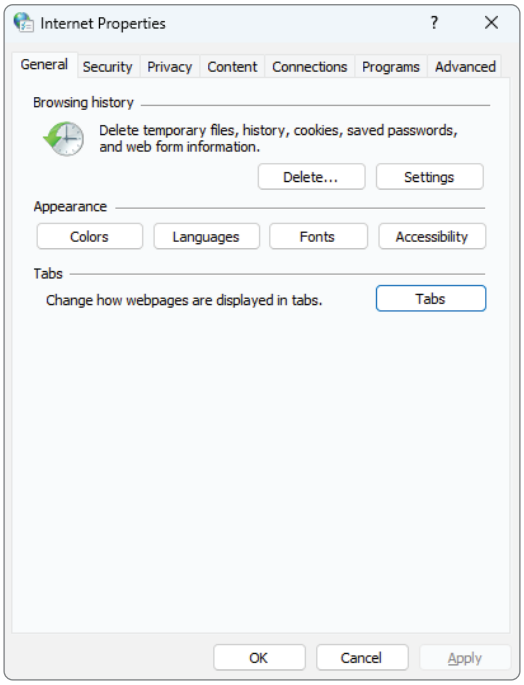
It's important to note that in Windows 10, Internet Explorer exists alongside the Microsoft Edge browser. Microsoft has incorporated an Internet Explorer mode for Microsoft Edge; this will allow a smoother transition to Edge. The sunset date (EOL) for Internet Explorer was June 15, 2022, and Windows 11 does not include Internet Explorer. The settings in this Control Panel applet can also be set in the Settings app.

File Explorer Options Applet

The File Explorer Options applet will open to the General tab, as shown in Figure 2.29. Using this tab, you can change the default opening pane for File Explorer. The Home option, the default, displays quick access shortcuts, favorites, and recent files. The This PC option displays the common folders found under This PC, along with devices and drives.

In addition, you can control how folders open, such as opening folders in the same window or opening folders in their own new windows. You can change how items are opened, such as single-click or double-click. This tab also lets you change privacy settings, such as showing recently used file folders in the Home view. By default, Windows 11 will be helpful by showing these recently used files and folders, but you may want to shut that behavior off. After doing so, you should clear the File Explorer history by clicking the Clear button in the General tab of the Folder Options applet.

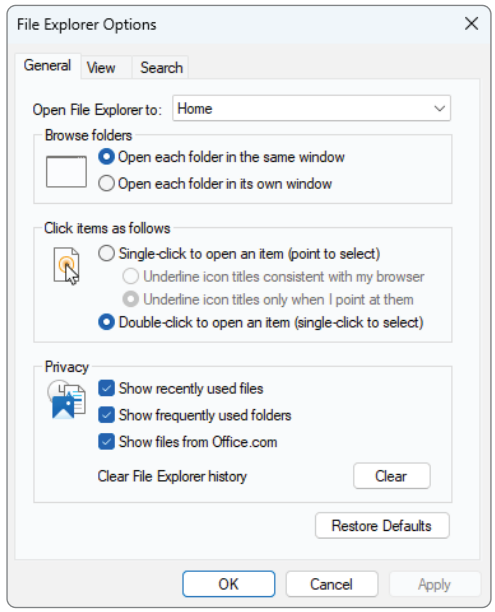
FIGURE 2.28 Internet Options Control Panel applet



The View tab in the File Explorer Options applet allows you to change how files and folders are viewed in File Explorer. There are a number of settings on this tab that will allow you to change the way File Explorer is viewed. The settings range from always showing menus to showing all folders in the Navigation pane. One of the first settings that is usually changed is Hide Extensions For Known File Types, because seeing the extensions is really handy.

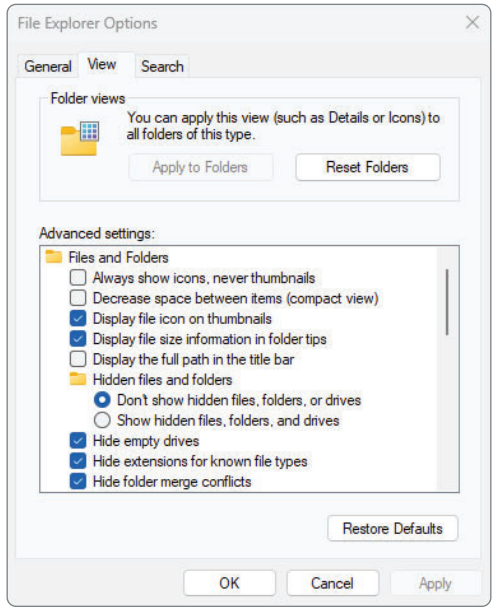
In later versions of Windows 10/11, the ability to see extensions is the default. This was changed because many malicious files can hide their extensions in an attempt to be executed or opened.

FIGURE 2.29 File Explorer General Options



Some of the more important files that you will need to work on are hidden by default as a security precaution. To make certain folders or files visible, you need to change the display properties of Windows File Explorer, as shown in Figure 2.30. You learn how to do this in Exercise 2.4.

FIGURE 2.30 File Explorer View Options



EXERCISE 2.4

Showing Hidden Files and Folders

1. Open Windows File Explorer on a Windows system.
2. Browse to the root of the C: drive and note the entries that are there.
3. If it is available, choose View, then Show, and select the Hidden Items option so that they are now visible.
4. Note the entries that now appear but that did not previously.
5. Open the File Explorer Options applet from Control Panel and select the View tab, if needed.
6. Deselect Hide Protected Operating System Files (Recommended).
7. Deselect Hide File Extensions For Known File Types.
8. Click OK.

   You will now be able to see the Windows system files discussed in the following sections. For security reasons, you should set these settings back to the defaults after you've read this chapter.

The Search tab in the File Explorer Options applet controls the Search and Index feature in Windows 11. You can turn off using the search feature and change the behavior of non-indexed locations.

System Applet

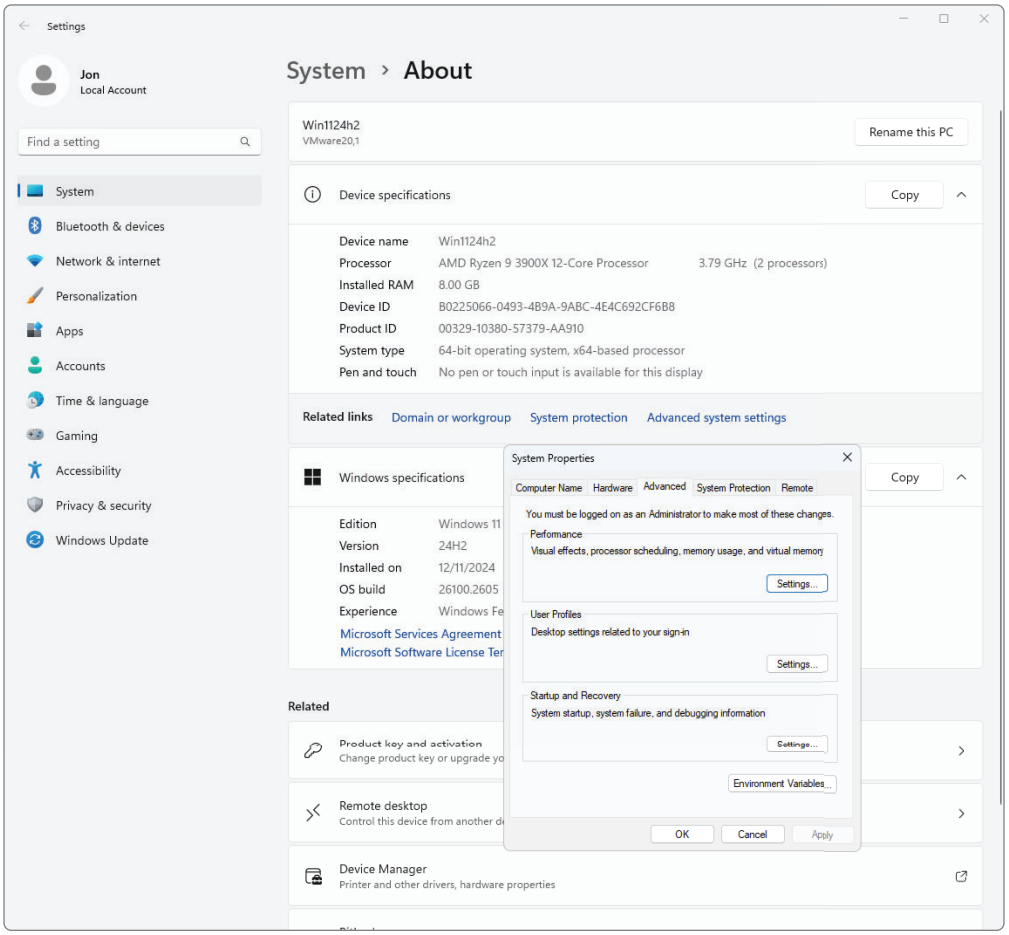
The System applet in Control Panel is one of the most important applets, and technically it's not an applet. In most recent versions of Windows 11, the System applet in Control Panel will open the Settings app to the About screen. From within the Settings app panel, you can make a large number of configuration changes to a Windows machine. If you click Advanced System Settings, the classic System Properties (`sysdm.cpl`) will open. (See Figure 2.31 for the Windows 11 classic System Properties applet.) You can perform a number of functions in this applet, which can include some of the following options:

- Network Identification
- Device Manager
- Hardware
- User Profiles
- Environment Variables
- Startup and Recovery
- Performance

- System Protection
- Remote
- Computer Name
- Advanced

In the following sections, we will look more closely at the functionality of the tabs.

FIGURE 2.31 Windows System Properties Control Panel applet



Computer Name

This tab is used to define whether the machine is in a workgroup or a *domain* environment. We talk more about networking in Volume 2 Chapter 3, but in general terms, here's the difference between a workgroup and a domain:

Workgroup Loosely associated computers, each of which is its own security authority, that share a common *workgroup* name. Often used for 10 or fewer computers/workstations that do not require a server computer.

Domain A group of computers that are tightly connected or associated and that share a common domain name. Has a single authority (called a *domain controller*) that manages security for all the computers. Often used for 10 or more computers/workstations that connect to one or more server computers.

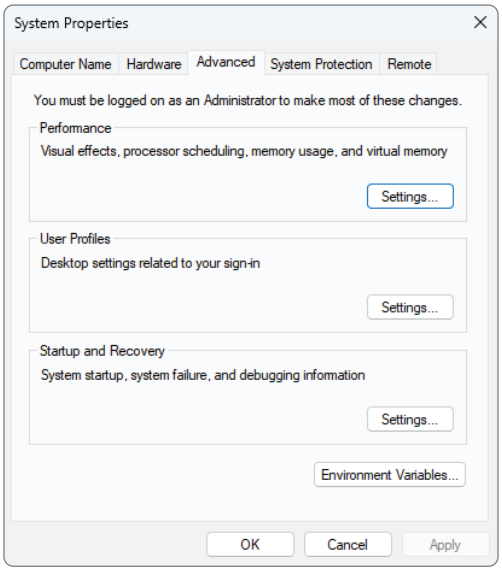
Hardware

This tab includes a number of tools that enable you to change how the hardware on your machine is used. The most useful is the ability to open the Device Manager directly from this tab. The other setting on this tab is how Windows behaves when you plug in devices. By default, Windows will automatically download *drivers*, *apps*, and custom icons from the devices plugged in. When you purchase a hardware device, odds are that it's been in the box for a while. By the time it gets made, packaged, stored, delivered to the store, stored again at the retailer, and then purchased by you, it's entirely likely that the company that made the device has updated the driver—even possibly a few times if there have been a lot of reported problems.

Advanced

The Advanced tab has several subheadings, each of which can be configured separately, as shown in Figure 2.32. The following options are among those on this tab.

FIGURE 2.32 Windows System Properties Advanced tab



Performance

Although it is hidden in the backwaters of Windows' system configuration settings, the Performance option holds some important settings that you may need to configure on a system. To access it, on the Advanced tab, click Settings in the Performance area.

In the Performance window, you can specify the size of your virtual memory and how the system handles the allocation of processor time. In Windows, you also use Performance to configure visual effects for the GUI.

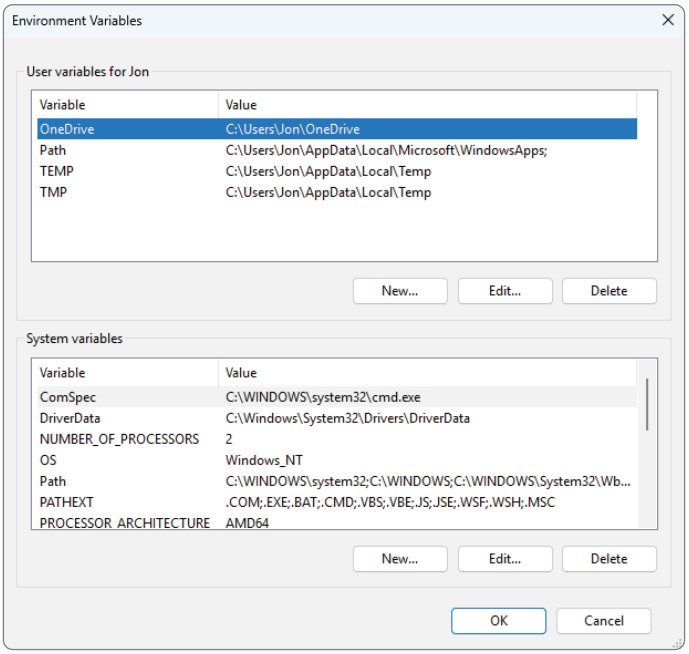
How resources are allocated to the processor is normally not something that you will need to modify. It is set by default to optimize the system for foreground applications, making the system most responsive to the user who is running programs. This is generally best, but it means that any applications (databases, network services, and so on) that are run by the system are given less time by the system.

If the Windows machine will be working primarily as a network server, you may want to change the Performance option to Background Services. You can find this in the Performance Options on the Advanced tab. Otherwise, leave it as is.

Environment Variables

There are two types of *environment variables* (as shown in Figure 2.33), and you can access either one by clicking the Environment Variables button in the System Properties window.

FIGURE 2.33 Windows environment variables



User Variables Specify settings for an individual user and do not affect others who log into the machine.

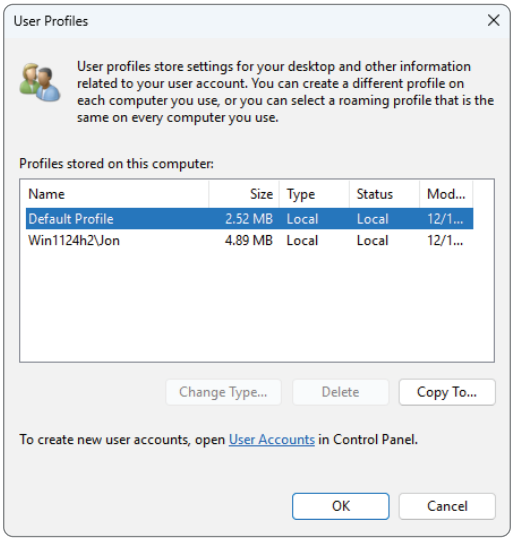
System Variables Set for all users on the machine. System variables are used to provide information needed by the system when running applications or performing system tasks.

System and user variables were extremely important in the early days of DOS and Windows. Their importance has lessened with recent Windows versions, but this is the interface where TEMP variables, the location of the OS, and other important settings for Windows can be found.

User Profiles

In Windows, every user is automatically given a *user profile* when they log into the workstation. This profile contains information about their settings and preferences. Although it does not happen often, occasionally a user profile becomes corrupted or needs to be destroyed. Alternatively, if a particular profile is set up appropriately, you can copy it so that it is available for other users. To do either of these tasks, use the User Profiles settings to select the user profile with which you wish to work. You will be given three options, as shown in Figure 2.34.

FIGURE 2.34 Windows User Profiles Settings



Change Type The Change Type option enables you to configure a profile as local (the default) or roaming. If a user works at two machines, each machine will use a different profile. Updates to one machine will not be reflected on the other. If you have a network,

roaming profiles can be configured to allow a user to have a single profile anywhere on the network. Further discussion of this topic is beyond the scope of this book.

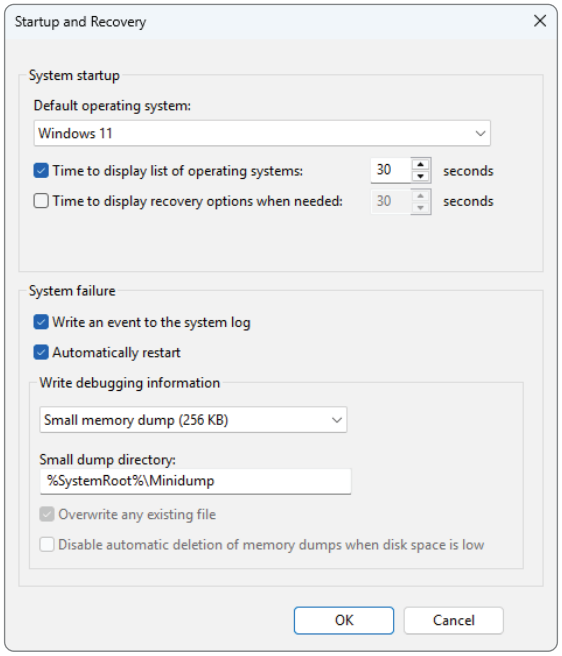
Delete The Delete option removes the user's profile entirely. When that user logs in again, they will be given a fresh profile taken from the system default. Any settings that they have added will be lost, as will any profile-related problems that they have caused.

Copy To The Copy To option copies a profile from one user to another. Often the source profile is a template set up to provide a standard configuration.

Startup And Recovery

The Windows Startup And Recovery options, shown in Figure 2.35, are relatively straightforward. They involve two areas: what to do during system startup and what to do in case of unexpected system shutdown.

FIGURE 2.35 Windows Startup And Recovery options



System Startup The System Startup option defaults to the Windows OS you installed, but you can change this default behavior if you like. Unless you are *dual-booting*, only one option is available; however, if you have another OS installed, you can change the Windows boot manager to load that as the default. You can also reduce the time the menu (time to display list of operating systems or recovery options) is displayed or remove the menu entirely.

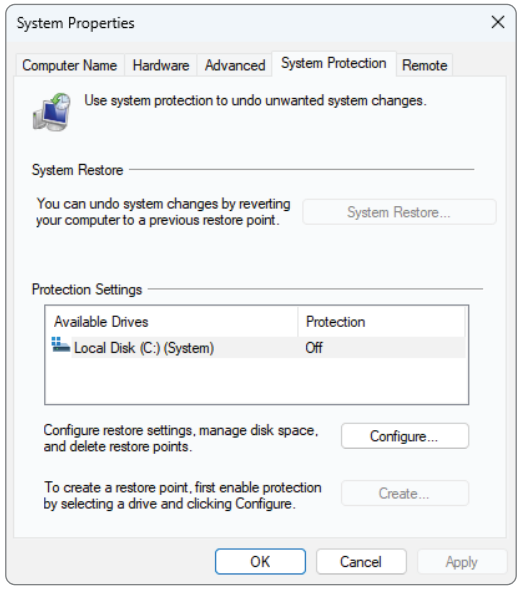
If you choose to disable the menu completely on a dual-boot system, you will find that doing so may cause you annoyance in the future when you want to boot into a different OS but no longer have a choice to do so. Thus, you should always let the boot menu appear for at least 2–5 seconds if you are dual-booting.

System Failure A number of options are available in the Startup And Recovery dialog box for use in case of problems. These include writing an event about the problem, sending an alert to the network, and saving information about the problem to disk. These options come into play only in case of a major system problem.

System Protection

The System Protection tab lets you disable/enable and configure the System Restore feature, as shown in Figure 2.36. When System Restore is enabled on one or more drives, the operating system monitors the changes that you make on your drives. From time to time, it creates what is called a *restore point*. Then, if you have a system crash, it can restore your data back to the restore point. You can turn on System Restore for all drives on your system or for individual drives. Note that turning off System Restore on the system drive (the drive on which the OS is installed) automatically turns it off on all drives.

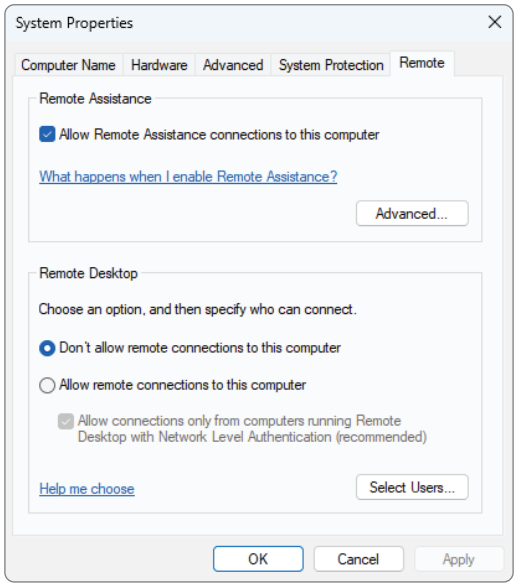
FIGURE 2.36 Windows System Protection Options



Remote

The Remote tab lets you enable or disable Remote Assistance and Remote Desktop, as shown in Figure 2.37. Remote Assistance permits people to access the system in response to requests issued by the local user using the Windows Remote Assistance tool. Remote Desktop permits people to log into the system at any time using the Remote Desktop Connection tool. This can help an administrator or other support person troubleshoot problems with the machine from a remote location.

FIGURE 2.37 Windows Remote options



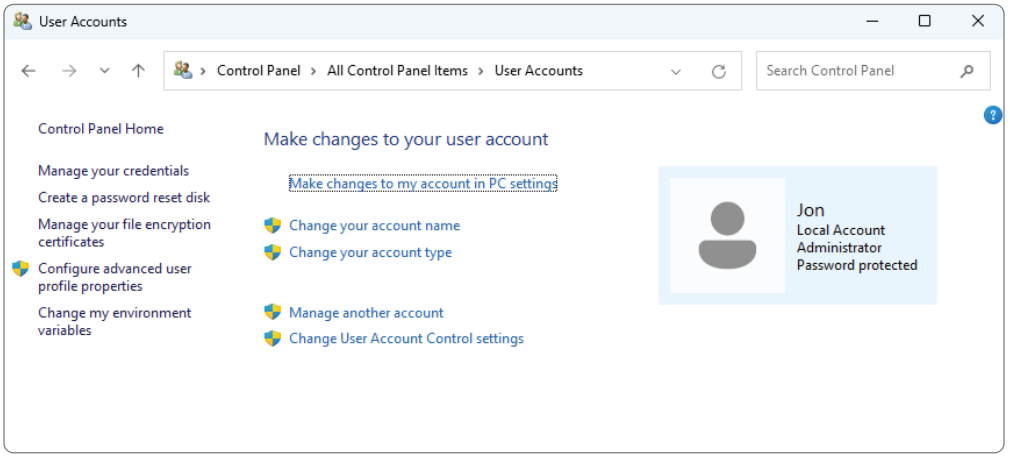
Remote Assistance is enabled by default. It is handled at two levels. Having just Remote Assistance turned on allows the person connecting to view the computer's screen. To let that person take over the computer and be able to control the keyboard and mouse, click Advanced, and then, in the Remote Control section, click Allow This Computer To Be Controlled Remotely. You can also configure Remote Desktop here.

User Accounts Applet

The User Accounts applet allows you to view and create accounts for the Windows operating system. You can change the account name that appears on the Welcome and Start screen. You can also change the account type by selecting the Standard or Administrator radio button. In Windows 11, the Settings app for Accounts allows you to change the picture that is displayed along with your username.

In addition to the management of user accounts, the User Accounts applet allows you to change the *User Account Control (UAC)* settings for the operating system, as shown in Figure 2.38.

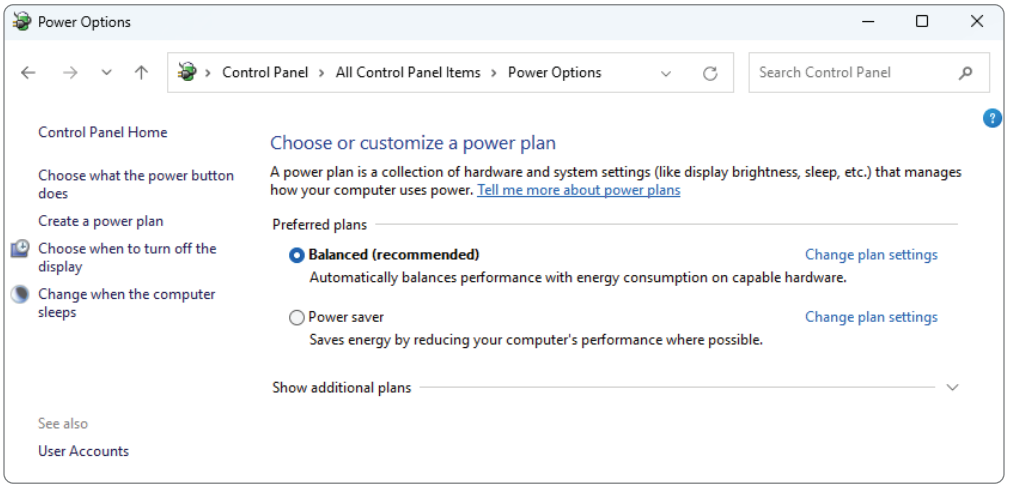
FIGURE 2.38 Windows User Accounts applet



Power Options Applet

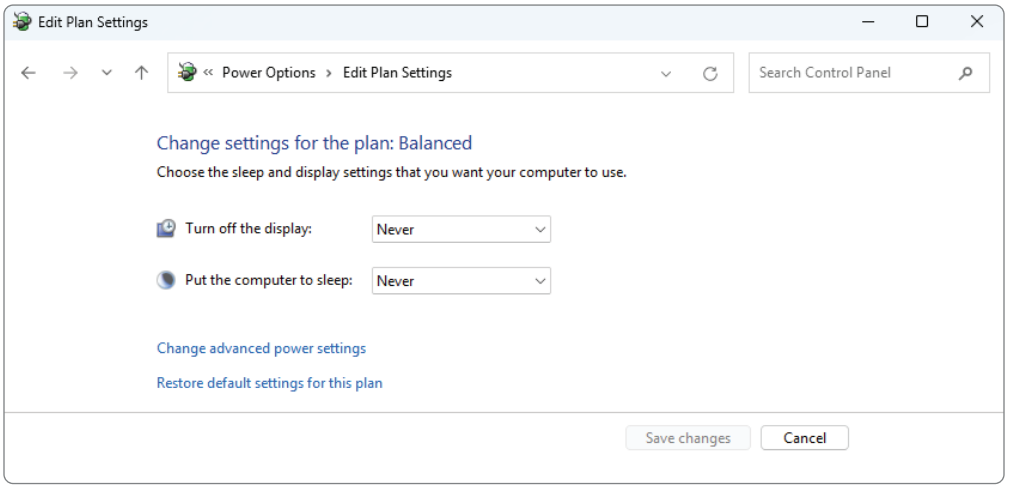
The Power Options applet (`powercfg.cpl`) allows you to choose a power plan of Balanced and Power Saver, as shown in Figure 2.39. In addition to the Balanced and Power Saver plans, a system might contain CPU-specific power plans, High Performance, or Ultimate Performance, depending on the type of computer. Each power plan dictates when devices—namely, the display device and the computer—will turn off or be put to sleep.

FIGURE 2.39 Windows Power Options applet



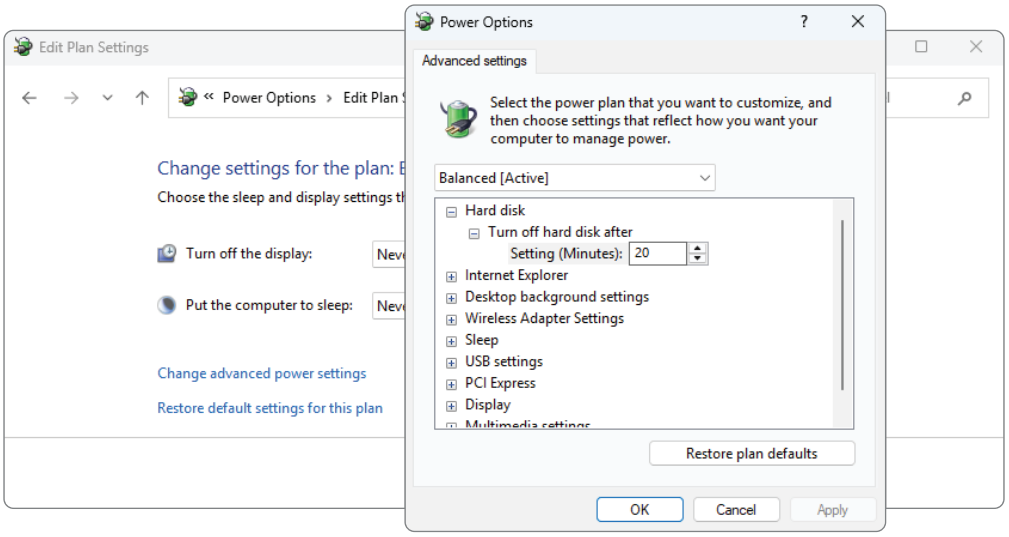
When you click Change Plan Settings, you can change how fast the display is turned off and how fast the computer is put to sleep, as shown in Figure 2.40.

FIGURE 2.40 Windows Edit Plan Settings



Clicking Change Advanced Power Settings allows you to configure a number of settings based on power, as shown in Figure 2.41. These settings include specifying when the hard drive turns off, turning off the wireless adapter, specifying Internet options for JavaScript Timer Frequency, and determining the system cooling policy. The applet allows you to tweak your power policy, and you can always restore the plan defaults.

FIGURE 2.41 Windows Advanced Power Options settings



Power Management

The power plan configured in Windows will interface with the Advanced Configuration and Power Interface (ACPI). The ACPI must be supported by the system BIOS/UEFI in order to work properly. However, most computer hardware made in the last decade will support the ACPI. The ACPI on the computer hardware provides the operating system with the necessary methods for controlling the hardware. This is in contrast to Advanced Power Management (APM), which gave only a limited amount of power to the operating system and let the BIOS do all the real work. Because of this, it is not uncommon to find legacy systems that can support APM but not ACPI.

There are four main states of power management common in most operating systems:

Hibernate This state saves all the contents of memory to the hard drive, preserves all data and applications exactly where they are, and allows the computer to power off completely. When the system comes out of hibernation, it returns to its previous state by restoring the contents of memory from the hard drive. This power state consumes the least amount of power and the system will appear to be completely off in this state. This power state is usually only available on laptops and not desktops.

Sleep/Suspend In some operating systems, *Sleep* is used interchangeably with *Hibernate*, but that should not be the case. In Windows, Sleep puts the system in a low-power state, whereas Hibernate turns the system off—or at least it appears to be off.

Standby The *standby* state might appear to be identical to the sleep mode, as it restores the system quickly. However, it is not really a power state. Standby mode typically reduces power and turns the screen off.

Shutdown/Turn Off As the name implies, this shuts down everything and then turns off the system. You can shut down the system and have it stay off, or you can shut down the system and have it power back on (Restart). The latter option is often used when updates need to be applied.

Advanced Power Savings

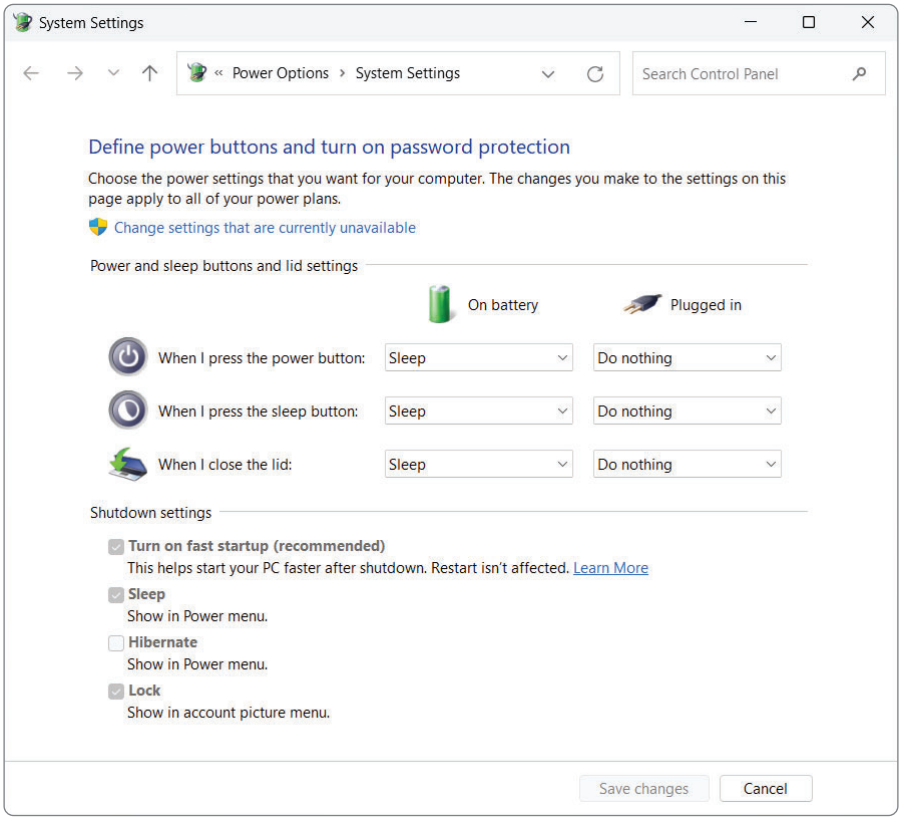
If you are interested in saving power with a system that is not accessed often, one option is to employ Wake on LAN (WoL). Wake on LAN is an Ethernet standard implemented via a network interface card that allows a "sleeping" machine to awaken when it receives a wakeup signal. Wake on LAN cards have more problems than standard network cards. In our opinion, this is because they're always on. In some cases, you'll be unable to get the card working again unless you unplug the PC's power supply and reset the card.

Choose What Closing The Lid Does

Another power-saving feature is to choose what happens when you close the lid of your laptop. This option is only available on laptop devices that have a lid closure sensor. The option can be found on the left-hand side of the Power Options Control Panel applet, and it's labeled Choose What Closing The Lid Does. When the dialog box opens, you will see

drop-down menus for pressing the power button or sleep button and closing the lid, as shown in Figure 2.42. Depending on whether the laptop is plugged in or on battery power, you can choose a different option.

FIGURE 2.42 Windows Define Power Buttons And Turn On Password Protection dialog box



Fast Startup

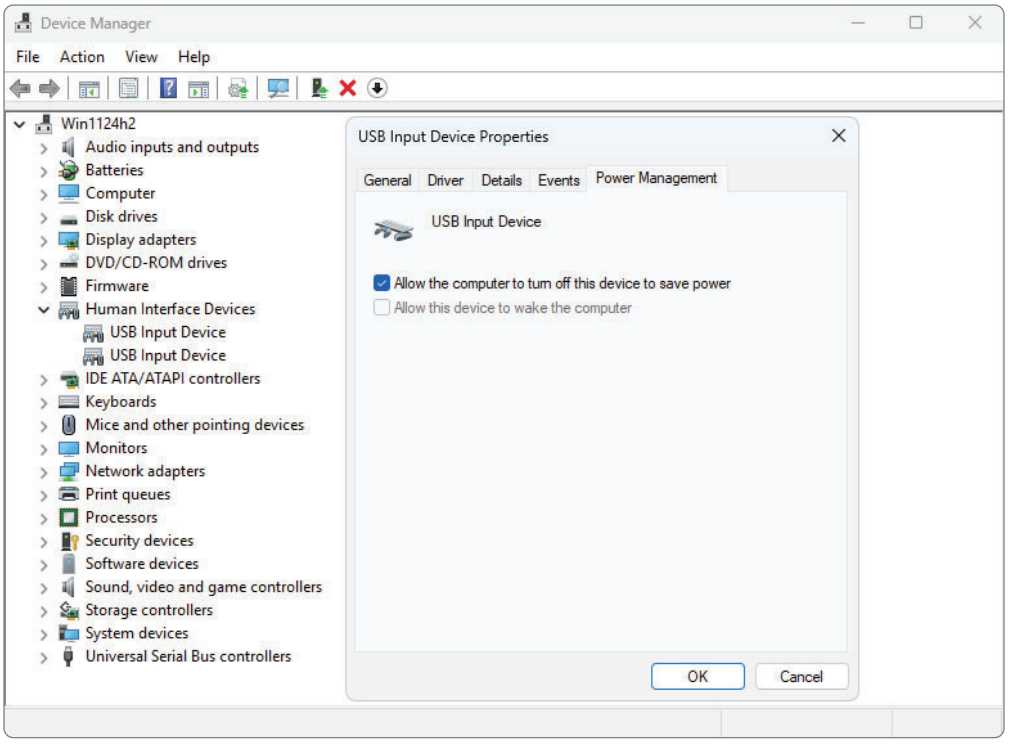
Windows Fast Startup is another advanced feature that was originally introduced with Window 8 as Fast Boot. The feature allows the system to hibernate during shutdown so that the system will appear to start up more quickly. The Fast Startup feature will attempt to know when a cold boot is required, such as when installing a program. If a cold boot is required, you have two options. The first option is to turn Fast Startup off in the Power Options System Settings, as shown in Figure 2.42. You can also use the `shutdown /s /t 0` command; the `/s` switch will shut down the system and the `/t 0` switch will do it immediately. The `shutdown` command also allows for shutdown of remote computers with the command `shutdown /m \\computername /s /t 0`.

USB Selective Suspend

The Universal Serial Bus (USB) selective suspend feature will allow the USB hub on the motherboard to suspend power to a device via the USB port. This is a handy feature to save battery power on a laptop if an external hard drive, a mouse, or some other device that requires power is not being used. However, this feature can also be problematic if you have a communication device connected via USB. You can turn this feature off on the Power Management tab of the device inside Device Manager, as shown in Figure 2.43.

There is a lot more going on than just powering off a USB port with the USB selective suspend feature. The operating system communicates to the device via the device driver and signals the device itself to power down. The system or the device can then send a wake-up signal to restore operations. You can find more information at `https://learn.microsoft.com/en-us/windows-hardware/drivers/usbcon/usb-selective-suspend`.

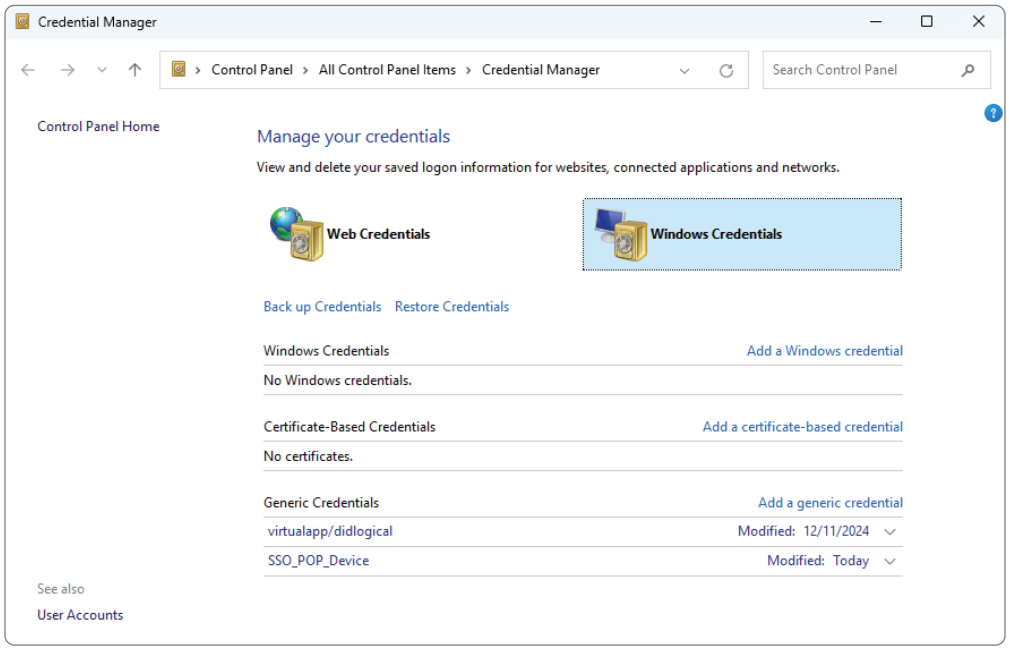
FIGURE 2.43 Device Manager, Device Properties detail, Power Management tab



Credential Manager Applet

The Credential Manager applet allows you to manage stored credentials for applications such as the web-based credentials and Microsoft Edge browsers, as well as the operating system itself. The Credential Manager service built into Microsoft operating systems stores username and password credentials in an encrypted database. The applet allows you to interact with the service to make changes to the stored credentials, as shown in Figure 2.44.

FIGURE 2.44 Windows Credential Manager applet

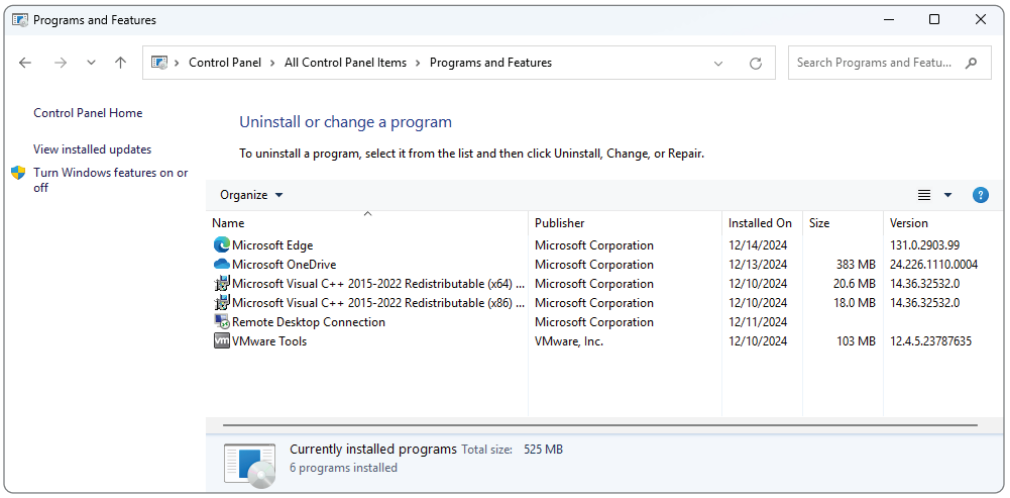


Programs and Features Applet

The Programs and Features applet (`appwiz.cpl`) allows you to view and uninstall desktop applications that are installed in Windows, as shown in Figure 2.45. You can also see the installed updates by clicking View Installed Updates on the left side of the applet.

The Programs and Features applet also allows you to install and remove features in Windows. When you click Turn Windows Features On Or Off on the left side, a dialog box will appear with a list of OS features that can be selected or deselected. Features are different from desktop applications because they are an integral part of the operating system. Examples include Hyper-V, Windows Subsystem for Linux, and Windows PowerShell 2.0, just to name a few.

FIGURE 2.45 Windows Programs and Features applet



Devices and Printers Applet

In Windows 11 the Devices and Printers Control Panel applet will now open the Bluetooth & Devices Settings screen. This Settings app will allow you to manage external devices such as external hard drives, printers, webcams, and any other external devices. From the Settings screen, you can discover and add new devices that are connected to your computer or network. Bluetooth & Devices is shown in Figure 2.46.

Sound Applet

The Sound applet (`mmsys.cpl`) allows you to view and change the default playback and recording device for sound on the system. By right-clicking the device and choosing Properties, you can view and modify the properties of a playback or recording device. The changes will vary by the device and the vendor of the device—common options are playback and recording levels and enhancements to the playback and recording levels. In addition, the Sound applet enables you to change the operating system's sound scheme, allowing you to change the various sounds, as shown in Figure 2.47.

Troubleshooting Applet

In Windows 11 the Troubleshooting applet will open into the Troubleshooting Settings screen. The Troubleshooter Settings screen does exactly what its name says it does—it allows troubleshooting of Programs, Hardware and Sound, Network and Internet, and System and Security. The Troubleshooter screen can be seen in Figure 2.48. Troubleshooting has one notable feature: application compatibility troubleshooting. When this feature identifies any application that is problematic, the Troubleshooting wizard will appear and Windows will try to troubleshoot the application compatibility.

FIGURE 2.46 Windows Bluetooth & Devices settings

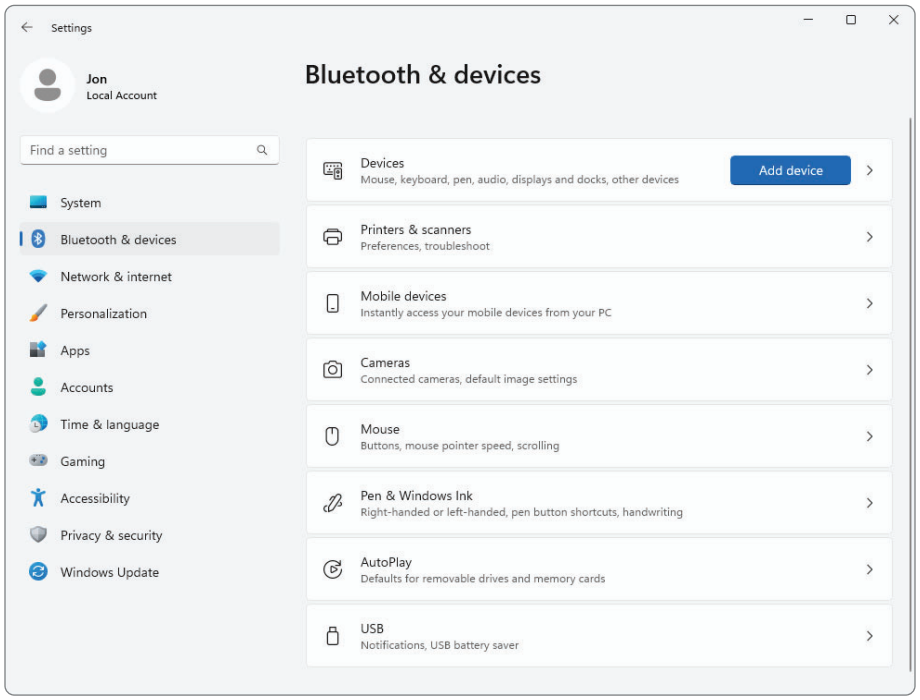


FIGURE 2.47 Windows Sound applet

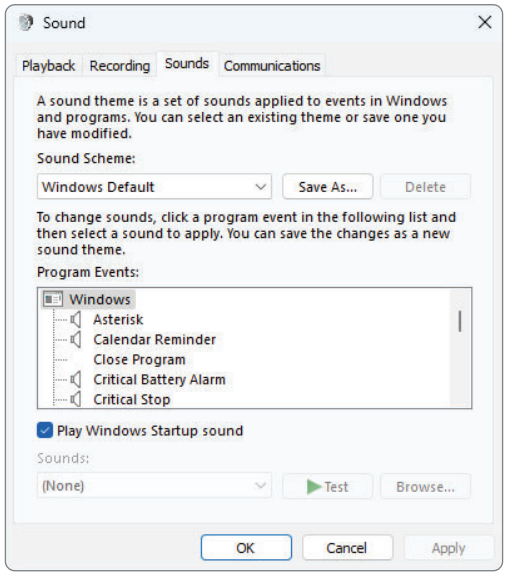
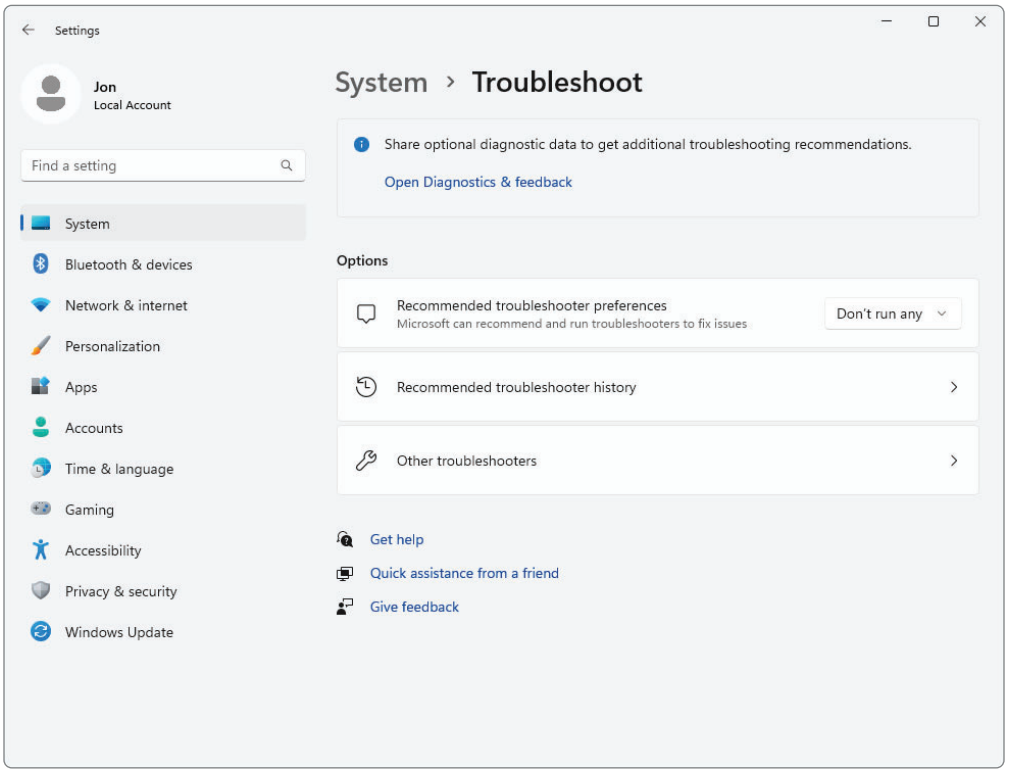


FIGURE 2.48 Windows Troubleshooting Settings app



Network and Sharing Center Applet

The Network and Sharing Center applet allows you to view and change the active network connections for the operating system, as shown in Figure 2.49. The applet displays the current network profile that the computer has been placed into by the Windows Firewall service. From the main page of the applet, you can click Change Adapter Settings to see the classic network adapter view, which allows the adapters to be configured manually. From the main page of the applet, you can also click Change Advanced Sharing Settings on the left side. This allows you to change network discovery options for the network profile chosen as well as turn on or off file and printer sharing globally for the operating system.

Device Manager Applet

As previously mentioned, the Device Manager applet (`hdwwiz.cpl`) was first introduced in Windows 95 and has hardly changed since its introduction over 20 years ago. If the applet looks familiar, it is the same as the MMC Device Manager (`devmgmt.msc`). Device Manager

allows you to view and change hardware devices on the operating system, as shown in Figure 2.50. This applet allows the administrator to load and update third-party drivers as well as drivers in the Windows Catalog. Device Manager is usually the first place you should check if a hardware device does not function after installation.

FIGURE 2.49 Windows Network and Sharing Center applet

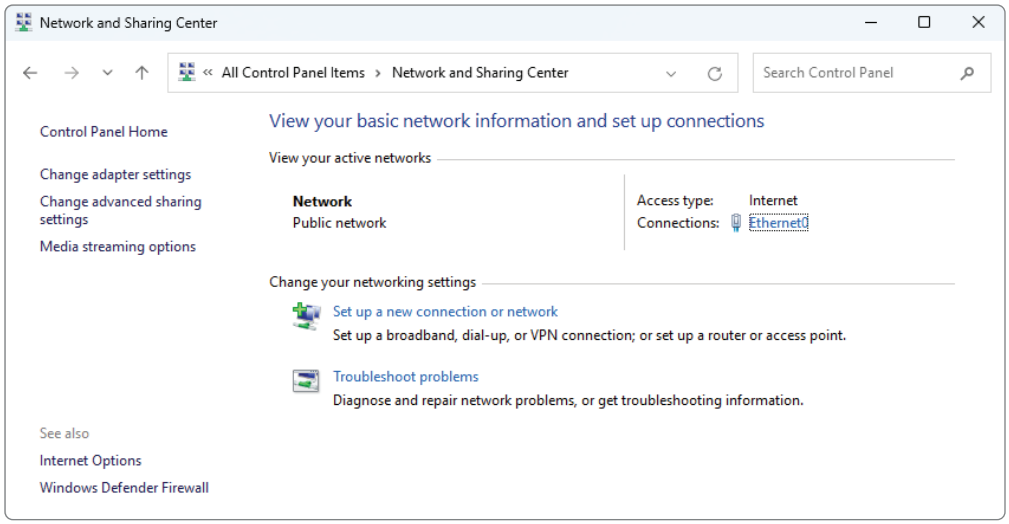
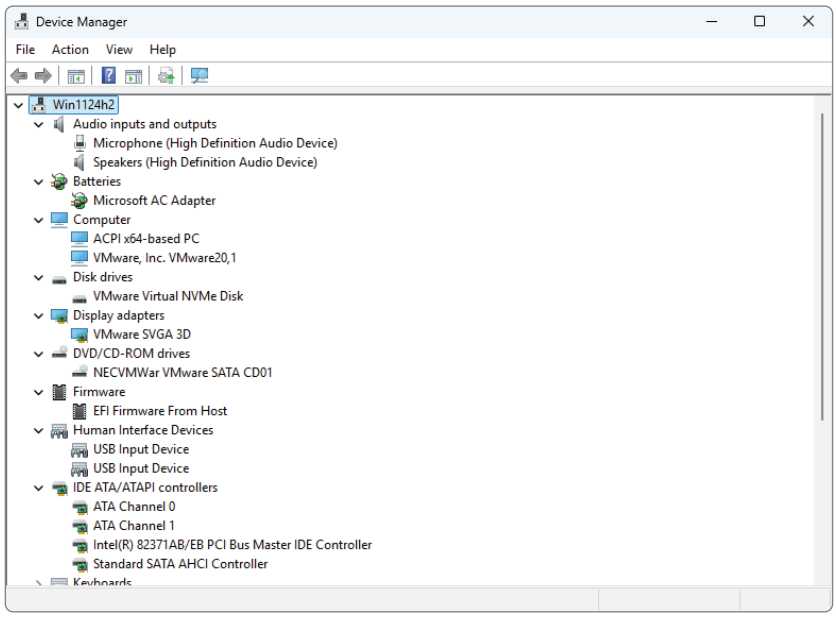


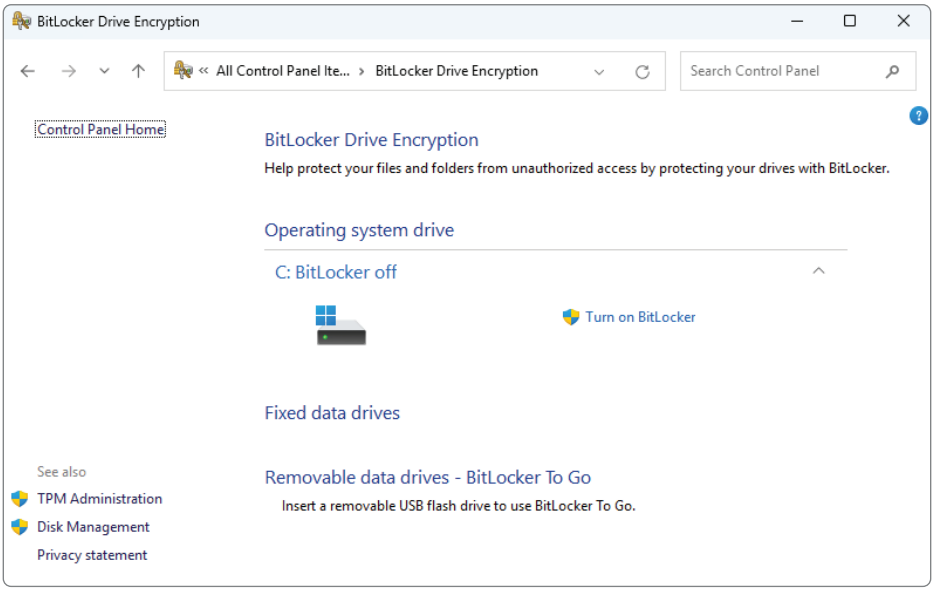
FIGURE 2.50 Windows Device Manager



BitLocker Applet

BitLocker drive encryption is a low-level full-disk encryption feature that can be controlled from the BitLocker Drive Encryption applet, as shown in Figure 2.51. The BitLocker Drive Encryption applet also allows you to turn on BitLocker drive encryption on removable media devices.

FIGURE 2.51 Windows BitLocker Drive Encryption applet



Windows Defender Firewall

The Windows Firewall (`firewall.cpl`), which in Windows 11 is named Windows Defender Firewall, is used to block access from the network (be it internal or the Internet). While *host-based firewalls* are not as secure as other types of firewalls, this was a great move in the right direction. It first appeared in Windows XP Service Pack 2 but was released as a polished product in Windows Vista.

Figure 2.52 shows the opening screen of Windows Defender Firewall in Windows 11. Windows Defender Firewall is turned on by default. It also blocks incoming traffic by default.

Clicking Advanced Settings on the left opens the Windows Defender Firewall with Advanced Security MMC, as shown in Figure 2.53. This MMC allows great control of the Windows Defender Firewall controls for inbound and outbound rules. Inbound rules can be created to allow and deny network traffic inbound to applications and the operating system. Outbound rules can be created to allow and deny network traffic leaving the operating system. Only inbound firewall rules are restricted by default. When an application attempts to listen to specific network traffic, the operating system will display a dialog box asking you to confirm the network activity. This dialog box will automatically create an inbound rule

for the application's traffic. The Windows Defender Firewall with Advanced Security MMC allows you to pre-create the rule. In addition, you can create connection security rules for authenticating and encrypting traffic.

FIGURE 2.52 Windows Defender Firewall in Windows 11

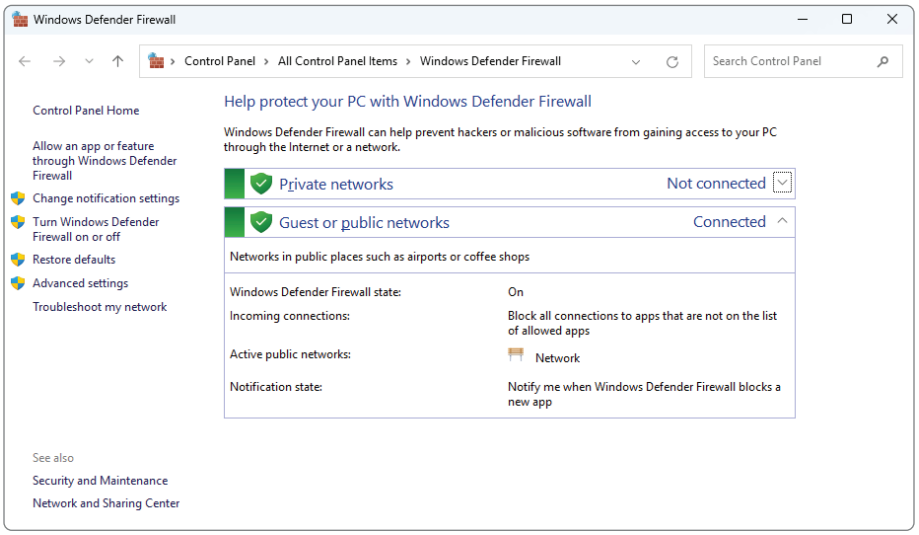
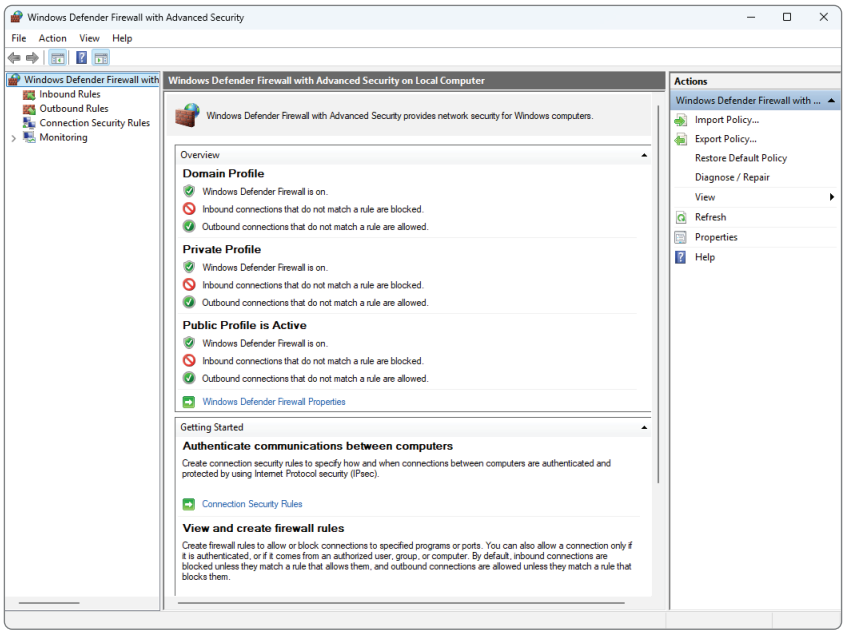


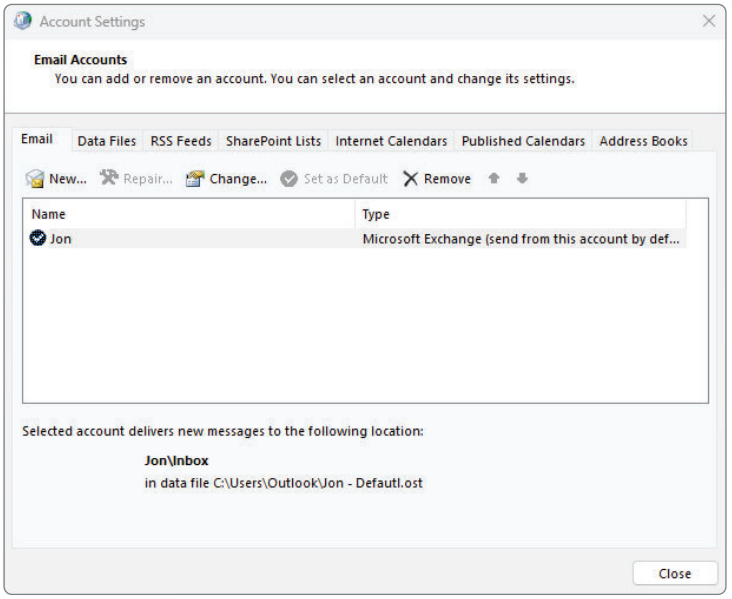
FIGURE 2.53 Windows Defender Firewall with Advanced Security in Windows



Mail Applet

When Microsoft Outlook is installed in Windows, it is configured through the Mail applet. If you do not have Microsoft Outlook installed, the icon will simply not show up in Control Panel. The Mail applet is how you configure an email account, additional data files (OST files), RSS feeds, SharePoint lists, Internet calendars, published calendars, and address books, as shown in Figure 2.54. In the main dialog box you can even set up different profiles for different accounts. If more than one profile is configured, you will be prompted when Outlook launches to specify the profile you want to use.

FIGURE 2.54 Outlook Mail applet



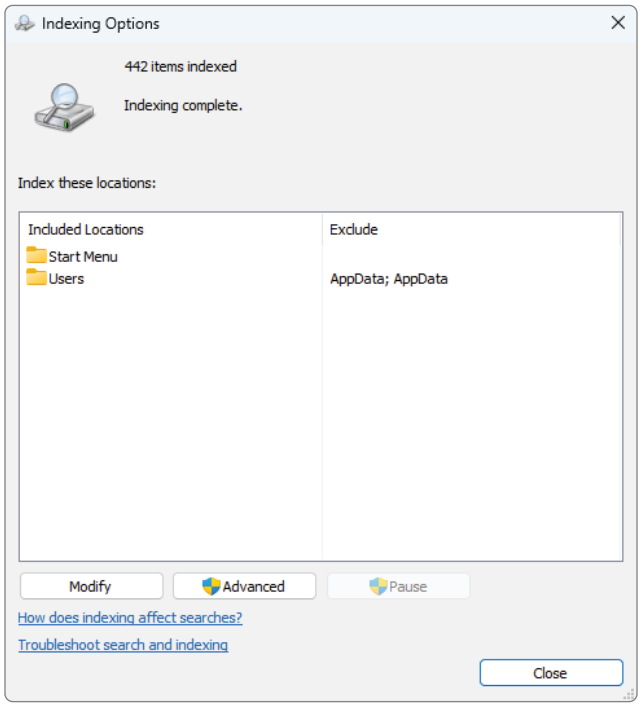
Outlook will usually configure itself automatically by asking a series of questions when it first launches. The initial configuration will be stored and can be accessed by launching the Mail applet.

Indexing Options Applet

The Indexing service was introduced as a desktop search engine with Windows NT 4.0. Today the Indexing service is an integral part of Windows 11. It's an exceptional feature that is a requirement for today's volume of data. The Indexing service will systematically index files such as Microsoft Office documents, PDFs, text files, and many other files types. When searching for a word using File Explorer in a folder that is indexed, the Indexing service is queried directly, thus returning fast results. If the folder is not indexed, then the search process grinds through each file in the folder and produces results at a much slower pace.

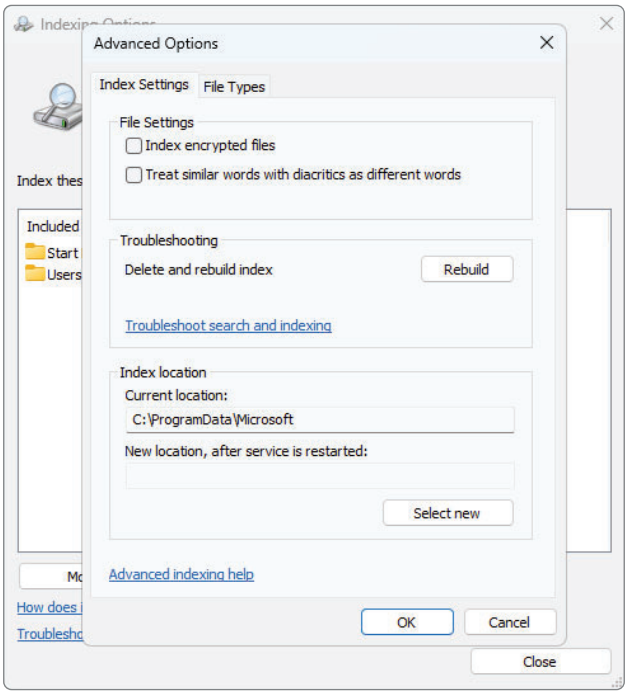
You access the Indexing Options dialog box, shown in Figure 2.55, by using the Index Options applet in Control Panel. The default locations indexed are Start Menu and the Users folder (excluding AppData files). You can add locations to be indexed if you store files outside the normal Documents or Desktop locations that are contained inside the Users folder.

FIGURE 2.55 Indexing Options dialog box



By clicking Advanced in the Indexing Options dialog box, you open the dialog box shown in Figure 2.56. There you can choose to index encrypted files and to treat words with diacritics (accents) as different words. You can also rebuild the index in an attempt to fix missing documents from your search. This dialog box also allows you to relocate the index database. The File Types tab allows you to add various file types to index. It contains a very inclusive range of file types, but by default many are set to index just metadata on the file. Important file types like DOCX and PDF are set to index the contents.

FIGURE 2.56 Advanced Options for indexing



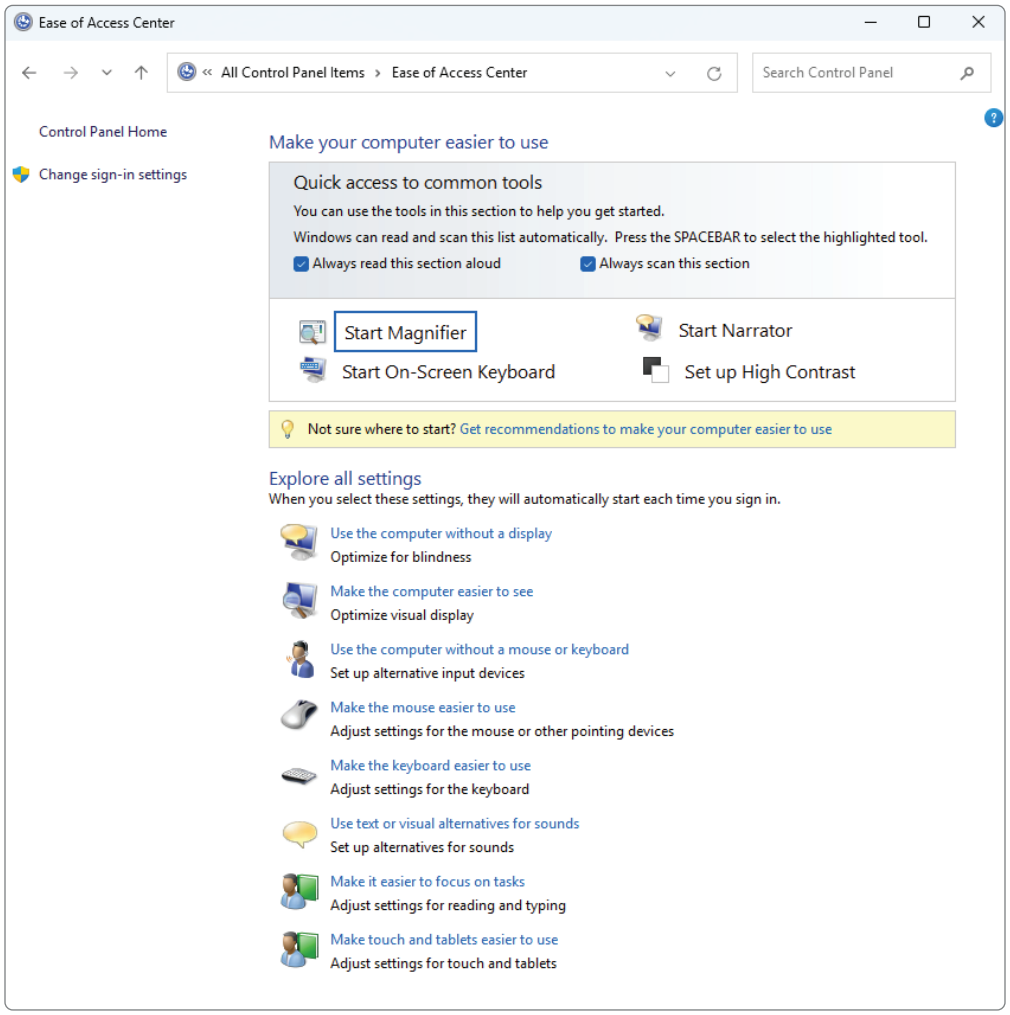
Ease of Access Center Applet

The Ease of Access Center applet contains various settings that make it easier to use Windows for motor and sensory impaired users, as shown in Figure 2.57. There is a wide range of tools, such as a magnifier, narrator, onscreen keyboard, and a high-contrast color scheme. In the many versions of Windows over the years, the accessibility tools have become expansive. Microsoft has continually added tools and settings to this applet to allow anyone with an impairment to use the operating system to its fullest.

Administrative Tools and Windows Tools Applet

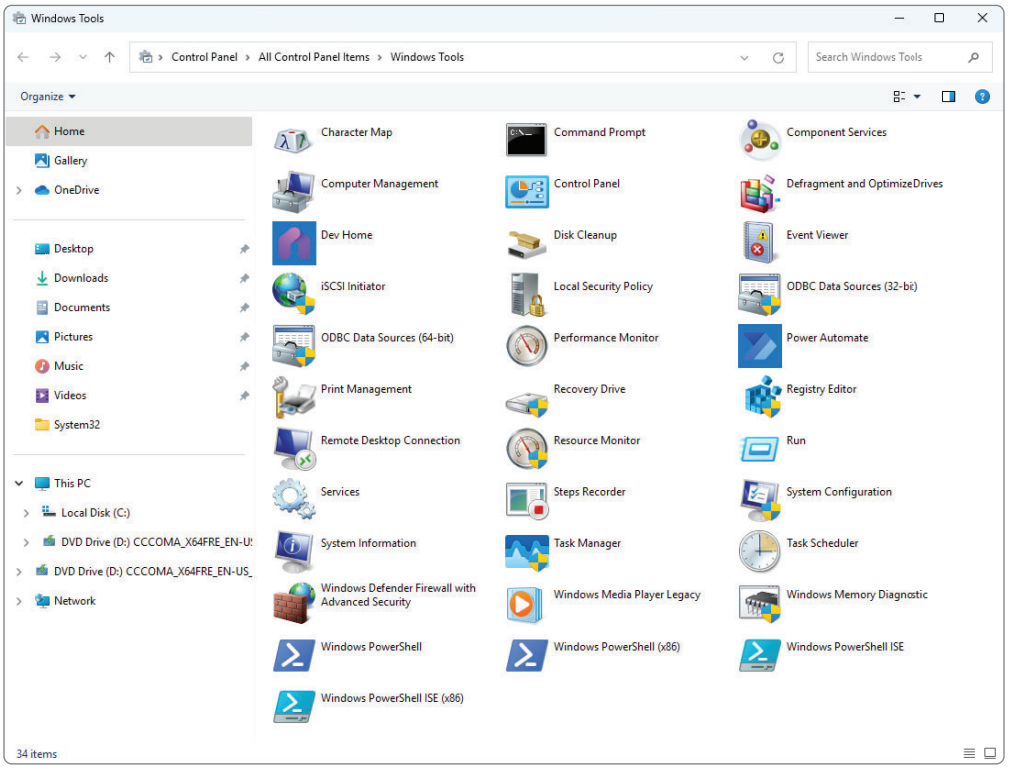
In Windows 11 the Administrative Tools applet has been replaced by the Windows Tools applet in Control Panel. Actually, it isn't really an applet at all; it is like a shortcut to various tools, as shown in Figure 2.58. These tools all have a common theme: administering the operating system. Many of these tools can be accessed in other ways, such as right-clicking the Start button or using the Start menu and expanding Windows Administrative Tools in versions prior to Windows 11 or Windows Tools in Windows 11.

FIGURE 2.57 Ease of Access Center applet



In this array of tools, you can configure component services, perform disk cleanup and defragment the disk, set up iSCSI connections, edit the local security policy, set up Open Database Connectivity (ODBC) connectors, schedule tasks, and perform memory diagnostics. These are just a few tasks that we haven't already covered.

FIGURE 2.58 Windows 11 Windows Tools applet



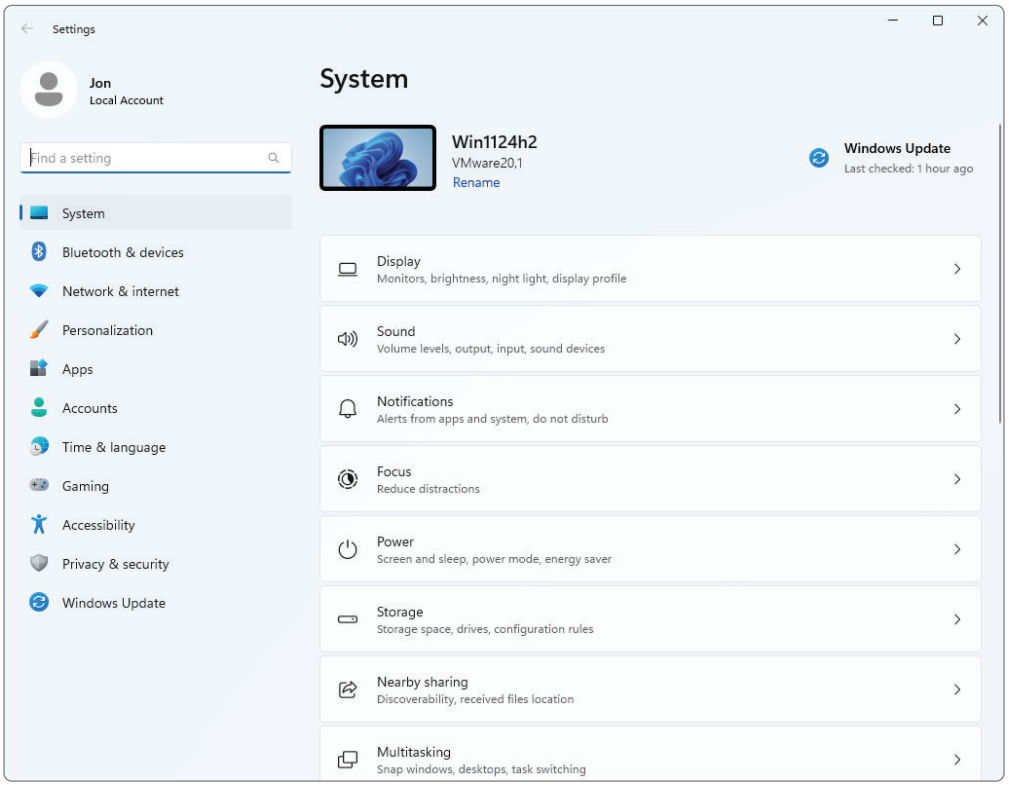
Windows Settings

The Windows Settings app first made its debut in Windows 8. It was Microsoft's attempt to make configuring Windows simpler for end users. Many of the configuration tasks formerly performed in Control Panel have been either duplicated in the Settings apps or replaced entirely. The appearance of the Settings app has created anxiety for both end users and administrators alike, because it's a change (albeit unwanted) from the Control Panel that has been around since Windows 95.

You can open the Settings app by clicking the Start menu and selecting the gear icon with the title Settings. The Settings app will open to the screen shown in Figure 2.59. Here, you can search for the setting you need, or you can choose from various categories. The search capability has been a welcomed feature, since every release of Windows introduces new settings.

Many of the settings that are covered in this section have also been covered in the Control Panel section, as per the objectives of the CompTIA 220-1202 exam. It is good to know both ways to access settings, since many of the settings in Control Panel have not been moved over to the Settings app. Likewise, many of the settings in the Settings app can only be found in the app, because they are entirely new settings.

FIGURE 2.59 Windows 11 Settings app



Time & Language

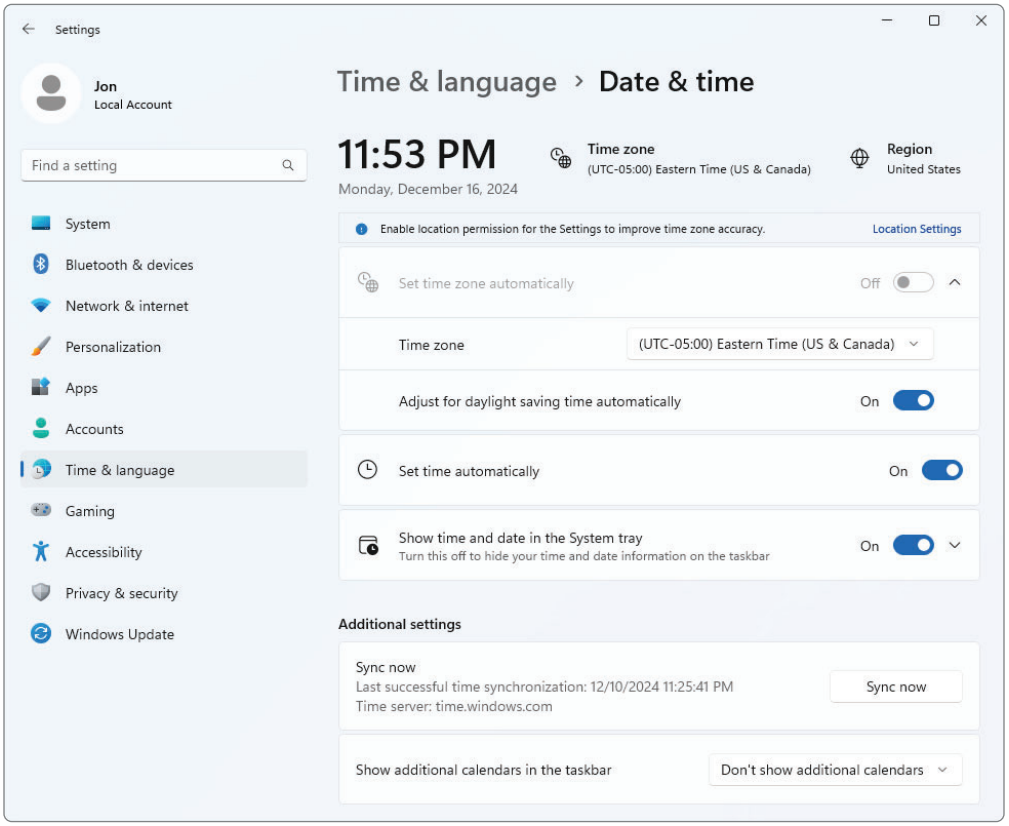
The Time & Language screen in the Settings app allows you to change settings related to the date and time, language and region, keyboard preferences, and speech in Windows 11. Click Time & Language to change anything that you can change in Control Panel. The date and time can be changed, as shown in Figure 2.60.

The Set Time Automatically switch is set by default and synchronizes with Network Time Protocol (NTP) server of `time.windows.com`. You can choose to have Windows adjust the time zone automatically, or you can manually change the time zone. This dialog box also allows for customized calendars in the Taskbar.

In the upper-right corner, you can select Region or you can open Time & Language and select Language & Region. Changing the Country Or Region setting allows Windows to deliver content relevant to the area where you reside. This dialog box also allows you to

change the way values such as money, time, and date are formatted. If you select Windows Display Language, you can set the Windows display language as well as the preferred language, as shown in Figure 2.61. In addition, you can set the keyboard language and speech language preferences from the Time & Language menu.

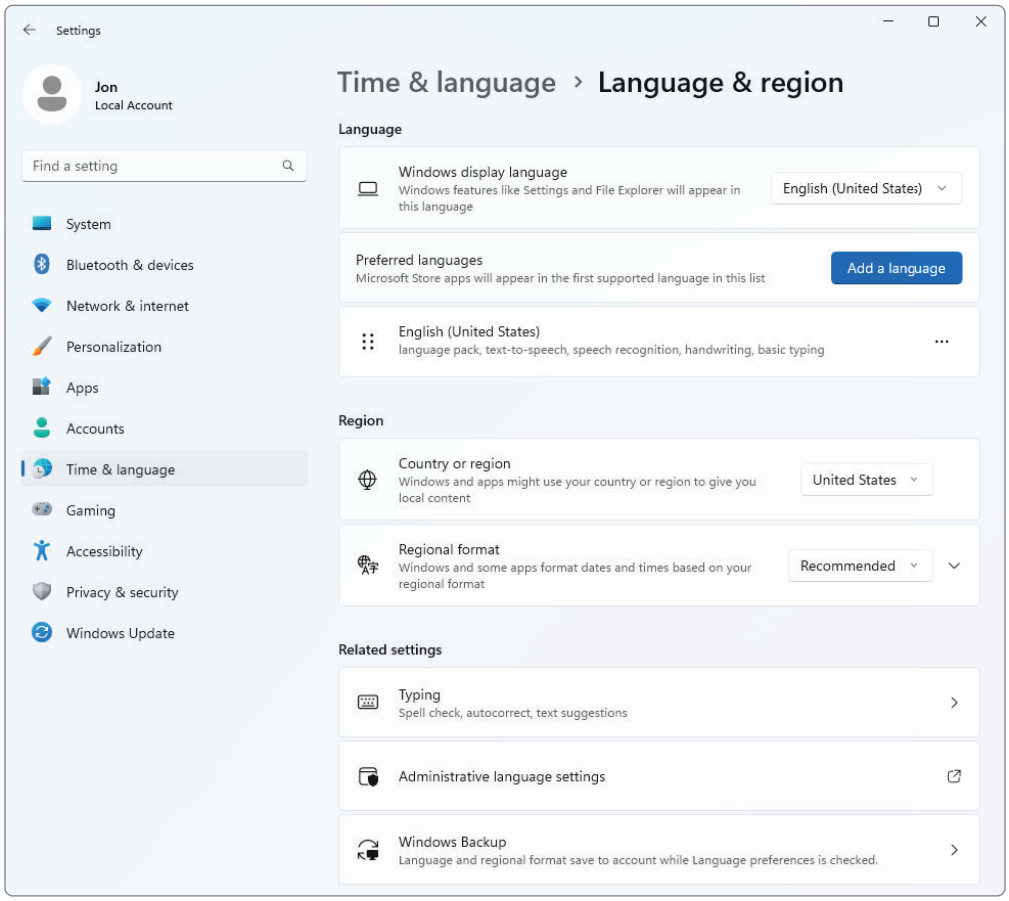
FIGURE 2.60 Date & Time settings



Windows Update

Update and Security from Windows 10 has been replaced by Windows Update in Windows 11. The Windows Update screen is where you can access all settings related to Windows Updates and their delivery, as shown in Figure 2.62.

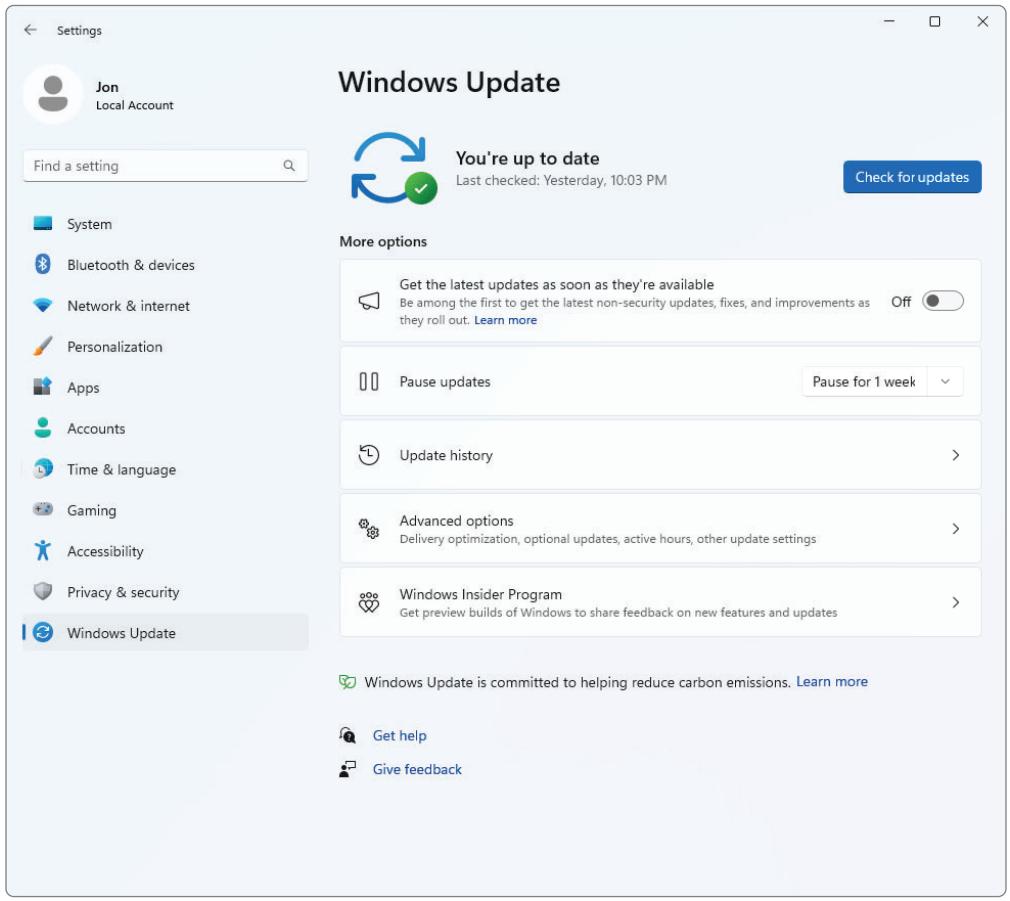
FIGURE 2.61 Language settings



From this initial screen, you can check for Windows Updates, as well as control downloads and installation, view the optional updates, pause updates, view update history, and set advanced options. Advanced options allow you to specify whether you want to receive updates for other Microsoft products, download updates over metered connections, change your active hours, determine how soon Windows is restarted after updates are applied, update notifications, and whether to pause updates until a certain date.

If you click Delivery Optimization on the Advanced Options screen for the Windows Update settings, you can change where Windows receives updates. By default, Windows will attempt to download updates from other PCs on the local network in order to conserve bandwidth. If the updates are not available, then Windows will download the updates from the Windows Update service.

FIGURE 2.62 Windows Update Settings



Windows Security

The Windows Security screen can be accessed via the Privacy & Security screen and selecting Windows Security. This will allow you to change a number of security-related items, as shown in Figure 2.63. From this screen, you can view Virus & Threat Protection, Account Protection, Firewall & Network Protection, App & Browser Control, Device Security, Device Performance & Health, and Family Options. The Windows Security screen is a one-stop shop for everything security-related for the operating system. You can click on each of the items and it will open in the respective settings for the item.

If you select Open Windows Security, as shown in Figure 2.63 a new standalone Windows Security screen will open that contains many more items that can be controlled. The standalone Windows Security settings screen can be seen in Figure 2.64.

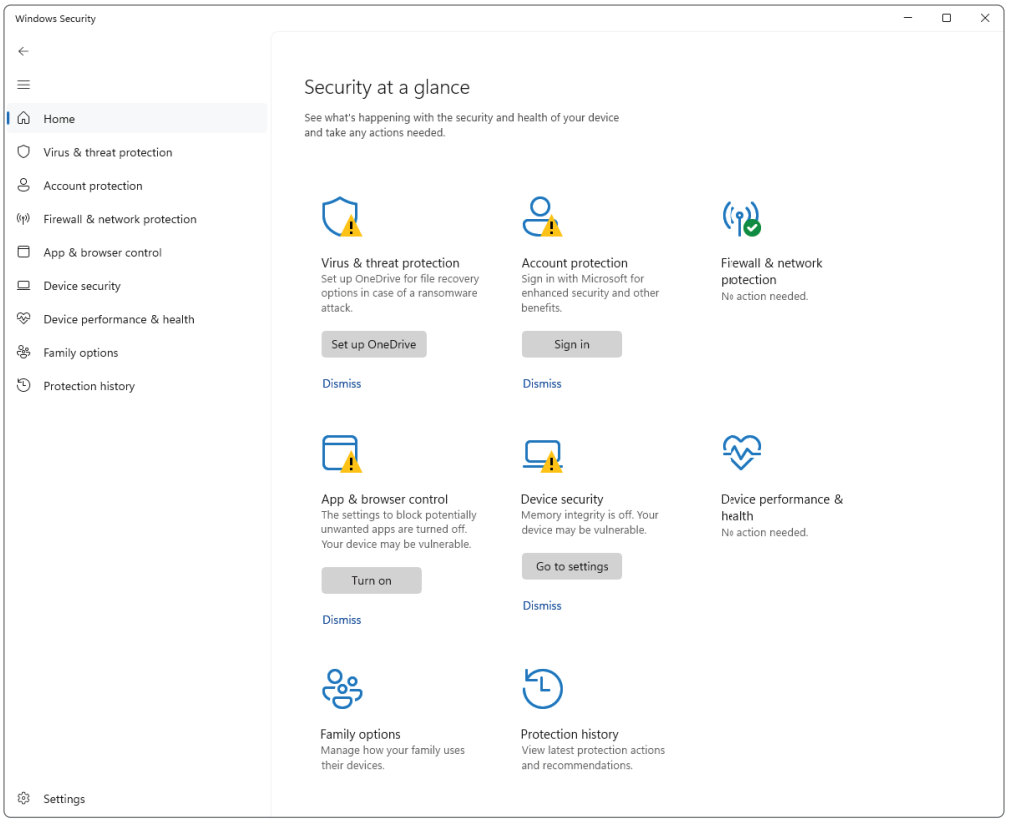
FIGURE 2.63 Windows Security settings

Virus & Threat Protection The Virus & Threat Protection section provides you with the ability to initiate scans of the operating system, manage Virus & Threat protection settings, check for updates related to Virus & Threat protection, and set up Ransomware Protection, as shown in Figure 2.65.

Account Protection In this section you can sign into a Microsoft Account that allows for seamless synchronization of security settings and updates across multiple devices, ensuring consistent and up-to-date protection. Hello provides you with the ability to sign in with biometrics. Windows Dynamic Lock is a security feature that automatically locks your computer when a paired Bluetooth device, such as your smartphone,

moves out of range. This helps prevent unauthorized access if you step away from your PC and forget to lock it manually. These options can be found seen in Figure 2.66.

FIGURE 2.64 Stand-alone Windows Security settings

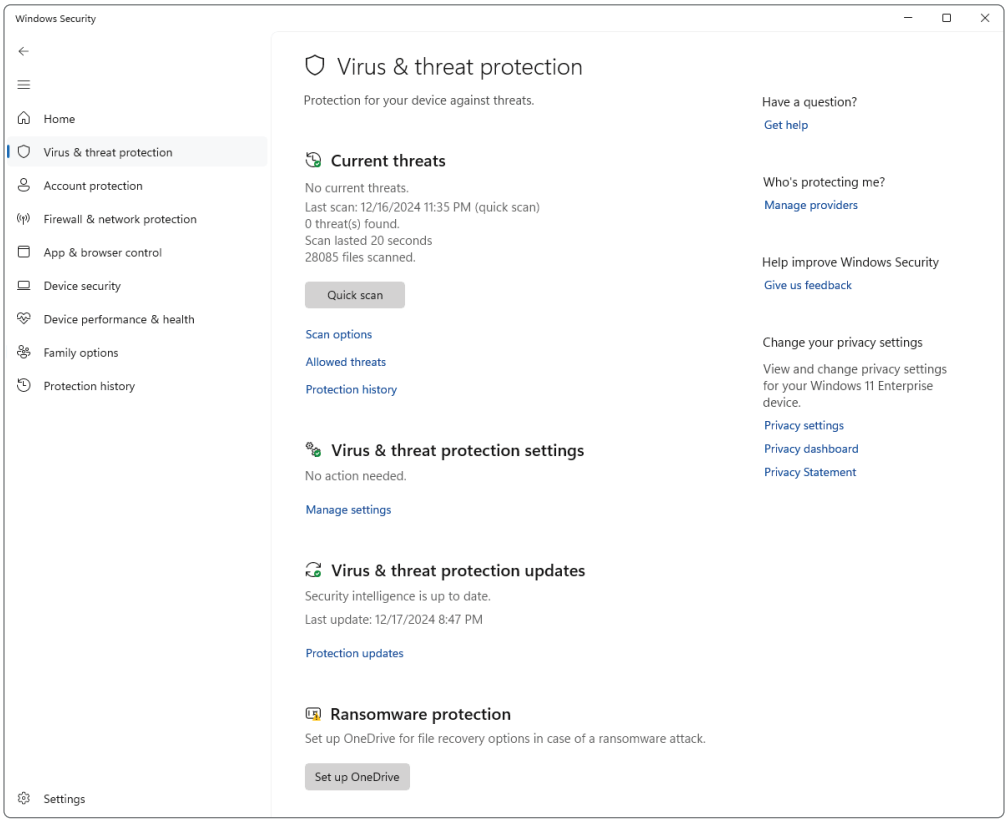


Firewall & Network Protection The Firewall & Network Protection screen in the Settings app allows you to view and manage your active firewall setting, as well as allowing apps through the firewall, as shown in Figure 2.67.

App & Browser Control The App & Browser Control screen protects you from untrusted applications and malicious websites. Smart App Control is available on clean installs of Windows 11; it starts in evaluation mode to see if you are a good candidate for the setting to be turned on. Once it is turned on, it will block programs that act

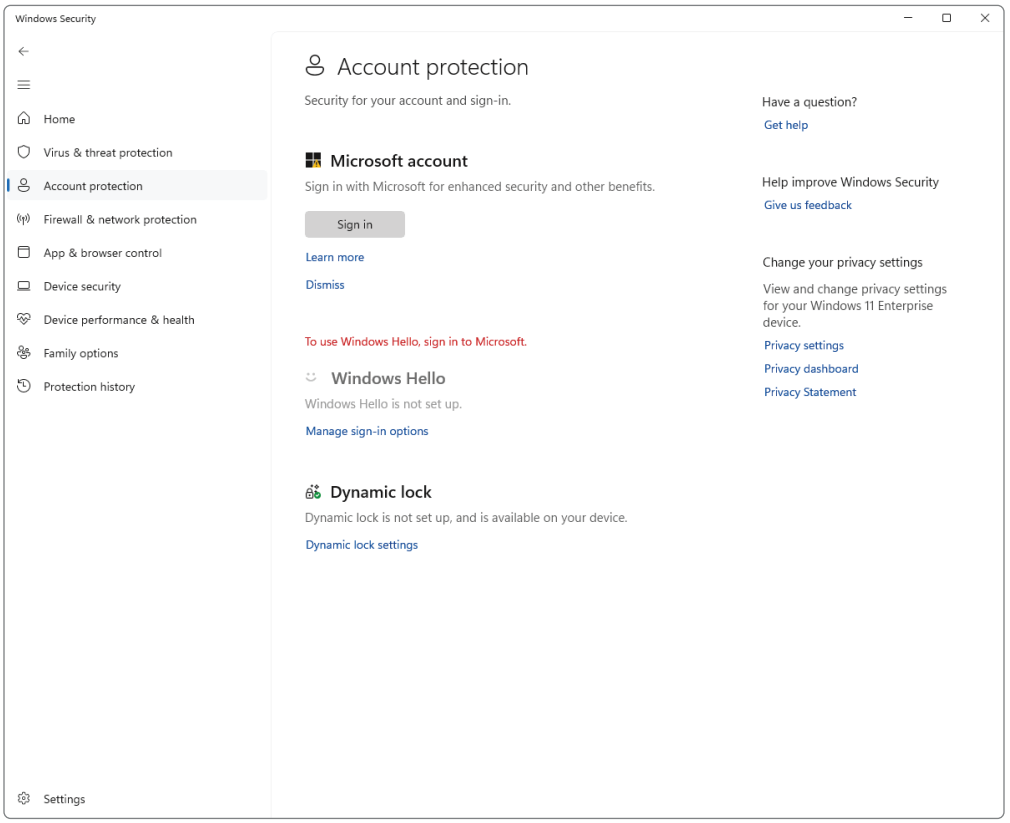
malicious or evasive. This setting is not recommended for developers or administrators, as some of the tweaks to the operating system might be blocked or perceived as malicious. Reputation-Based Protection will protect you from potentially unwanted application (PUA) by monitoring a program's reputation. In addition, you can control SmartScreen for Edge, which protects you from phishing sites and malicious downloads. Exploit Protection protects you from specific attacks and works in combination with Microsoft Defender subscriptions.

FIGURE 2.65 Virus & Threat Protection settings



Device Security The Device Security screen provides you with three options of Core Isolation, Security Processor, and Data Encryption. Core Isolation allows you to control memory integrity, which can be used to identify malicious software. Security Processor provides details on the Trusted Platform Module (TPM) specification. Data Encryption is a link that opens the BitLocker Control Panel applet.

FIGURE 2.66 Account Protection settings

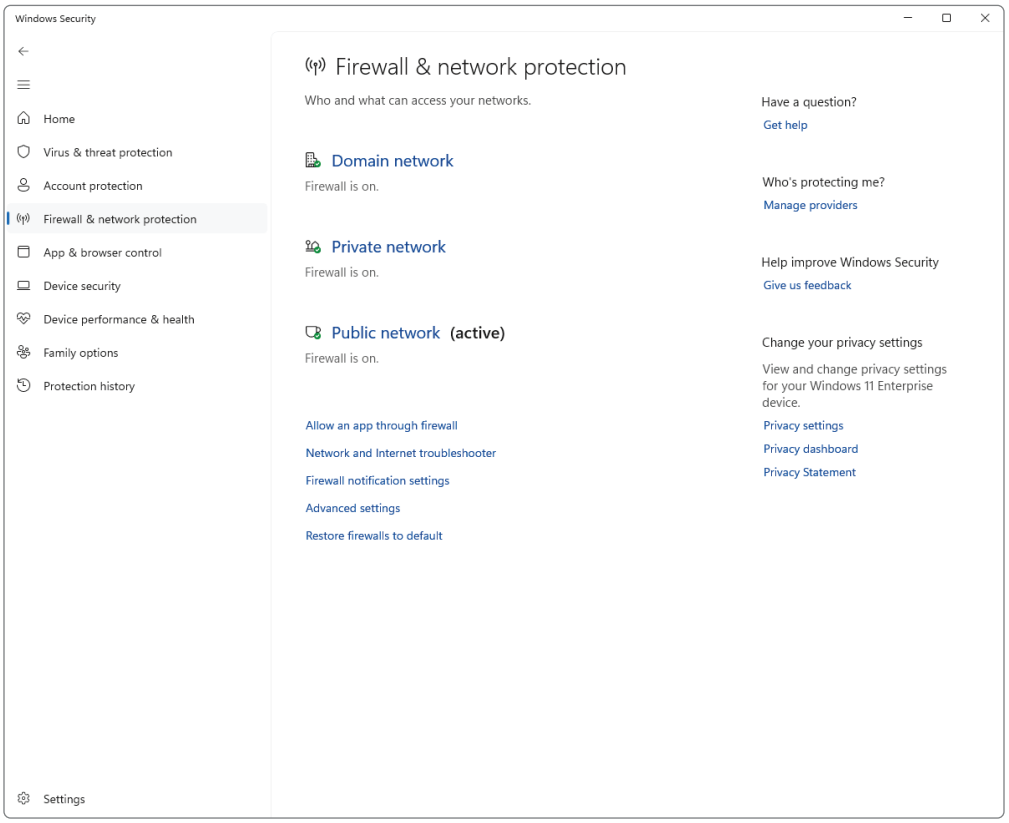


Device Performance & Health This screen does not allow any controls of settings—it is simply a display of a health report for storage, apps and software, and the Windows time service, as shown in Figure 2.68. The screen also explains what Fresh Start is, and it provides a link to online content that offers details about the Device Performance & Health screen.

Family Options This screen does not allow any control of settings—it simply explains some of the parental controls you can employ. The screen also supplies a link to view family settings online. The online portal allows you to set up the controls for your family, such as content filters, screen time, spending, activity reporting, email, and calendar, just to name a few. The controls in the web page are outside the scope of the CompTIA exam, but we included this topic for completeness.

Protection History This screen does not allow any control of settings. It does provide the most recent security-related events, as shown in Figure 2.69.

FIGURE 2.67 Firewall & Network Protection settings



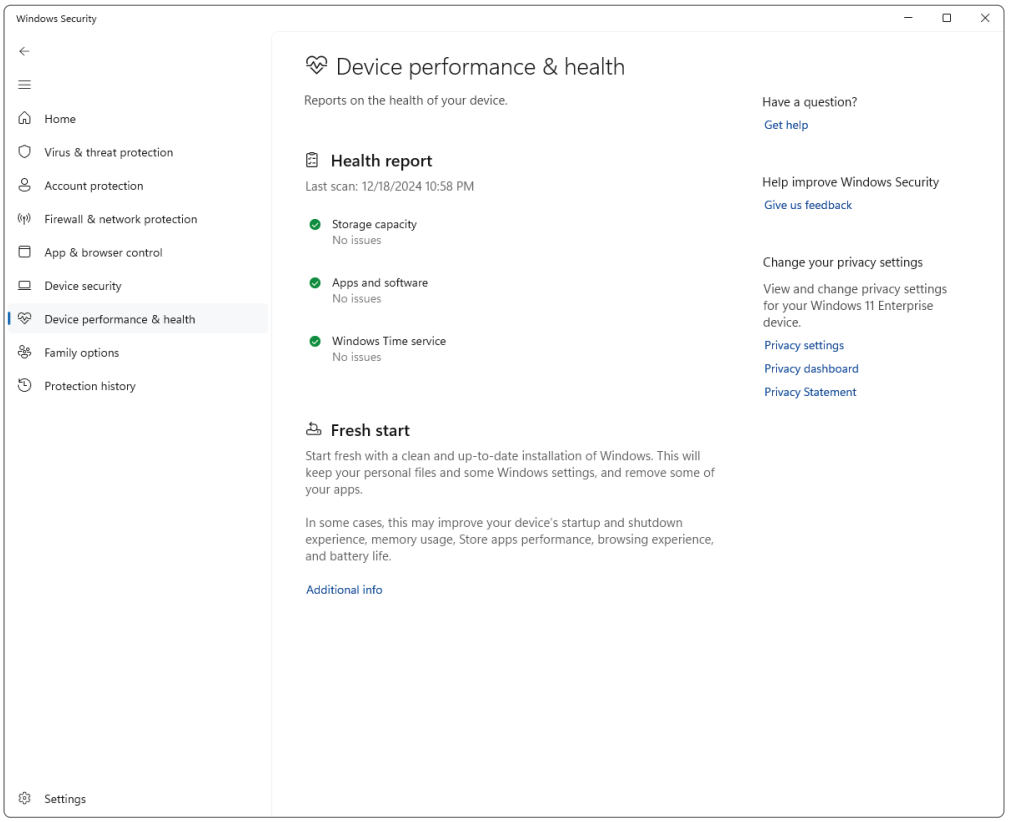
Personalization

The Personalization Settings screen allows you to personalize the entire operating system. The Personalization screen consolidated a number of other Control Panel applets when Windows 10 was first introduced. The new functionality introduced in Windows 10 was carried forward to Windows 11, shown in Figure 2.70.

Background Obviously, this section allows you to configure the background, formerly known as the wallpaper. You can choose a standard image, or choose to use your own photo. You can even choose Windows Spotlight for your background. The Windows Spotlight will download pictures from Bing. In addition, you can choose contrast themes for low vision or light sensitivity.

Colors The Colors section allows you to change the colors for the Windows controls and application controls. You can choose the transparency effects for Windows and the accent colors. An option to choose contrast themes for low vision or light sensitivity also exists.

FIGURE 2.68 Device Performance & Health

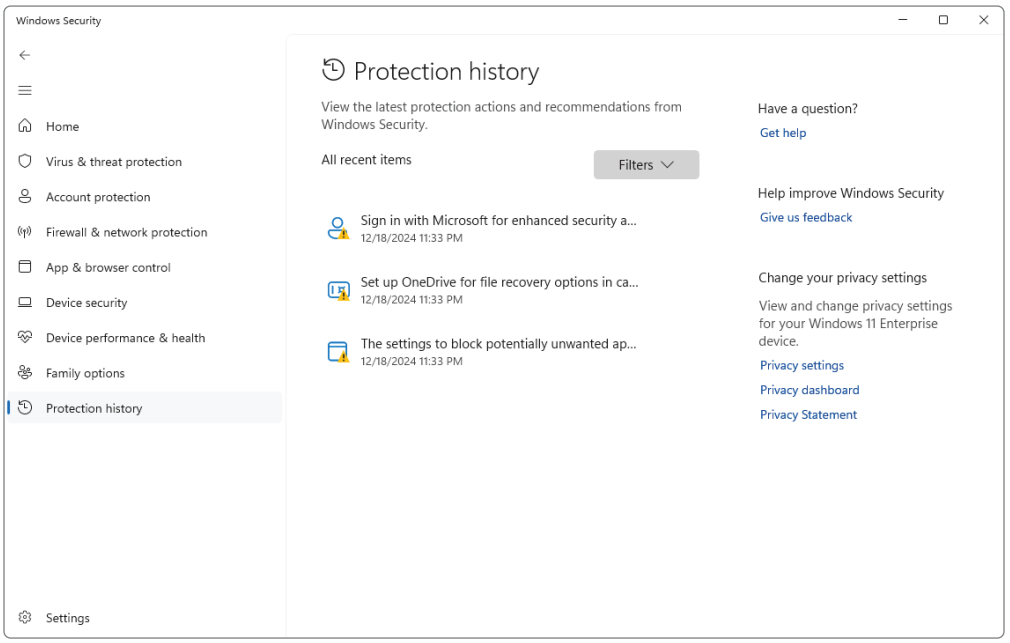


Themes As with prior versions of Windows, the theme can be changed using the Themes section. Changing the theme will change the background, colors, sounds, and mouse cursor. You can even download more themes from the Microsoft Store. The option to choose contrast themes for low vision or light sensitivity also exists.

Dynamic Lighting Dynamic Lighting allows you to change the RGB colors of connected devices so that third-party software is not needed. This section allows you to control a myriad of devices that have RGB LEDs.

Lock Screen Lock Screen allows you to configure how the lock screen looks and what is displayed on the screen when it is locked. By default, the lock screen is set to Windows Spotlight. The Spotlight feature downloads pictures from Bing and displays them on the lock screen. You can also configure which applications will display their status on the lock screen. This section also allows you to control the screen timeout settings related to power and the screen saver timeout.

FIGURE 2.69 Protection History



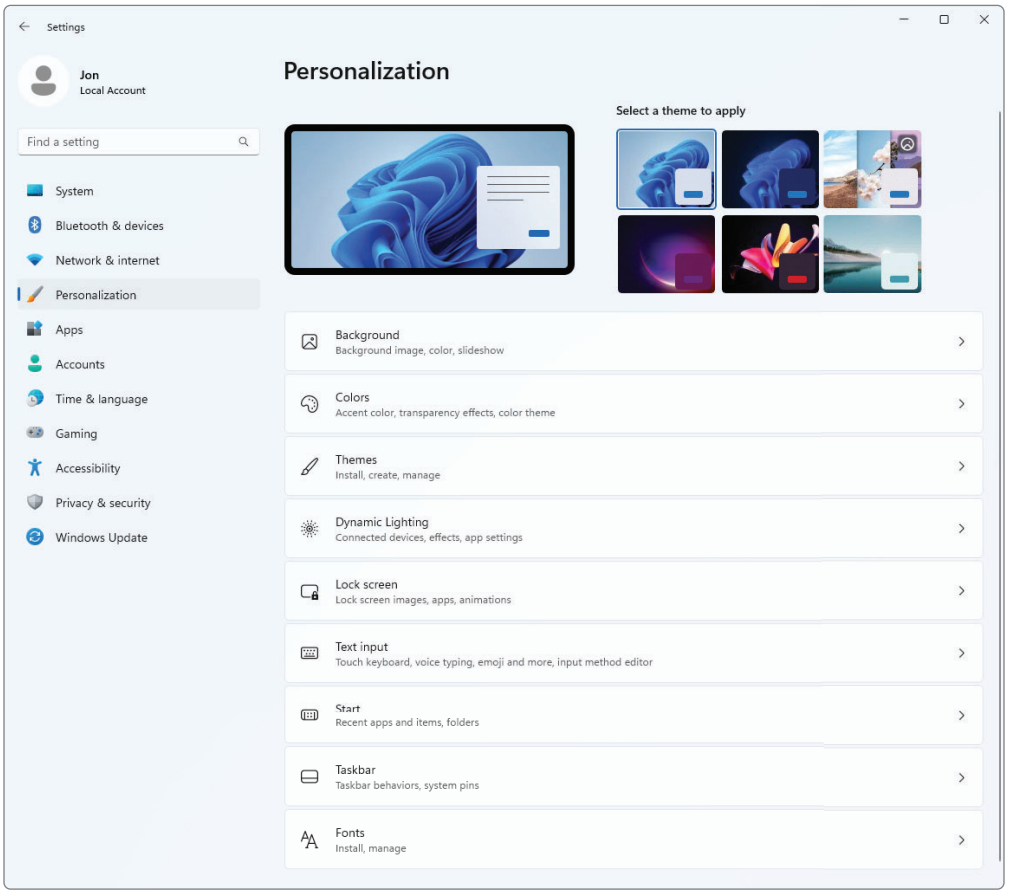
Text Input This section allows you to customize the text inputs prompts, the touch keyboard look, the voice typing prompt, and emojis. You can apply preinstalled themes or create a customized theme.

Start The Start section allows you to personalize the Start menu. You can change a number of settings, such as displaying the app list in the Start menu, showing recently added apps, and showing the suggestions, just to name a few settings.

Taskbar The Taskbar section allows you to change a number of settings. You can turn on or off Taskbar items, such as Search, Task View, and widgets. In this section, you can also control system tray icons for the pen menu and touch keyboard. By changing Taskbar Behaviors, you can automatically hide the Taskbar, and change the justification of the Taskbar from center to left. There are several other settings, such as combining Taskbar buttons into a single icon, flashing icons, and several other settings.

Fonts The Fonts section allows you to view all the installed fonts on Windows, as well as install new fonts by dragging and dropping them. The Fonts section also contains a link to open the Microsoft Store so you can download additional fonts.

FIGURE 2.70 Windows 11 Personalization settings

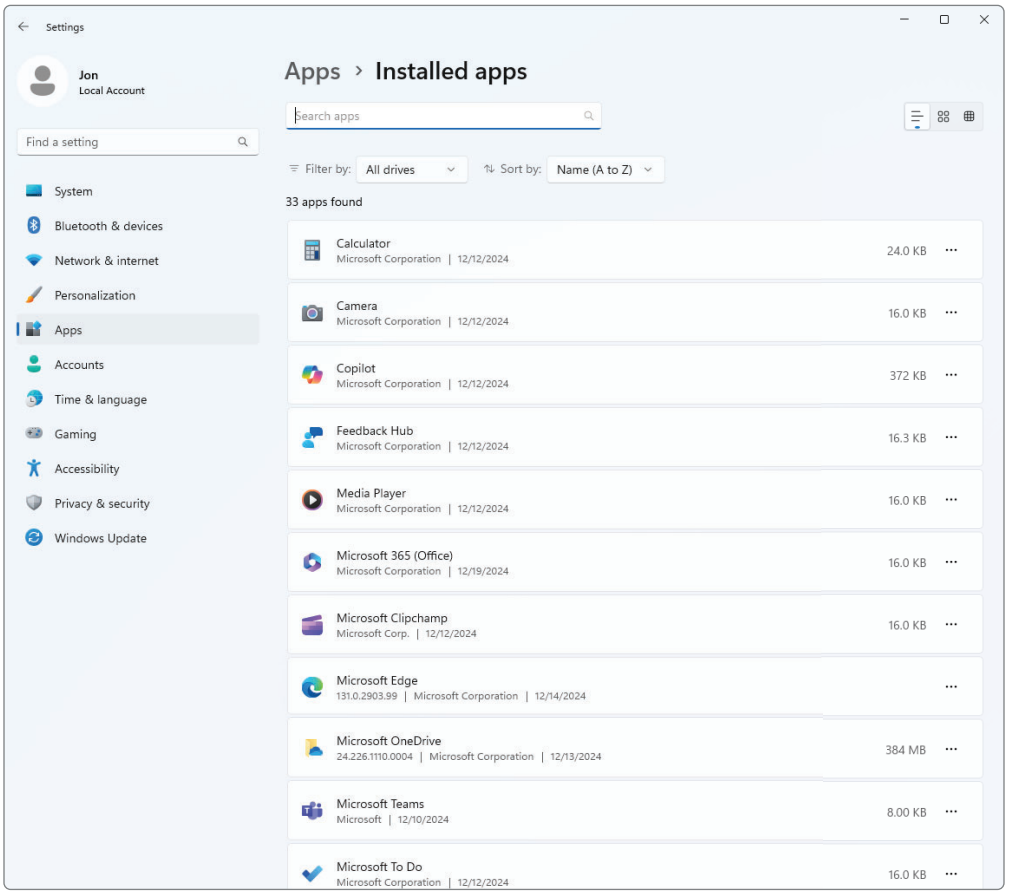


Apps

The Apps Settings screen controls every aspect about application on the operating system. This screen will eventually replace the Program and Features Control Panel applet, since it performs many of the same functions.

Installed Apps The Installed Apps section displays all the applications installed on the operating system, as shown in Figure 2.71. You can also uninstall apps by clicking the three dots on the right-hand side of the application section and selecting Uninstall. By clicking Advanced Options, you can choose to reset the application, repair the application, uninstall the application, and terminate the application. When you click the three dots on some applications, you can only choose to modify the application.

FIGURE 2.71 Installed Apps settings



Advanced App Settings The Advanced App Settings will allow you to change the source of your applications. The default is Anywhere, so you can continue to install downloaded applications. You can also choose the Microsoft Store Only option, or Microsoft Store Preferred if there is a comparable application. The Advanced App Settings also allow you to choose your sharing level for applications across your other devices. In addition, you can uninstall updates through the Advanced App Settings.

Default Apps In the Default Apps section, you can select the default app for email, maps, music, photos, videos, and websites. You also have the option to set the default app by the file type of the file or the network protocol being used. This is also known as the link type, which can be changed by selecting Choose Defaults By Link Type.

Offline Maps Offline Maps allows you to download maps, as well as update the already downloaded maps. You can select the storage location for offline maps, as well as control map downloads over a metered connection.

Apps for Websites Use the Apps For Websites section to associate an app with a website. Doing so causes the app to "spring into action" when a website is visited, making it a seamless experience.

Video Playback The Video Playback section allows you to tweak how a video will look when played back on Windows. You can choose to play video at a lower resolution to save network bandwidth. Other settings include high dynamic range (HDR) settings and a shortcut to System ➢ Display And System ➢ Power.

Startup Use the Startup section to specify which applications start in the background. By adjusting unwanted application starting when you log in, you can speed up your logon experience.

Privacy

The Windows Privacy Settings are located on the Privacy & Security Settings screen. The Privacy Settings are alongside of the Windows Security settings we previously covered. The Privacy Settings allow you to control all your privacy concerns with the Windows operating system, as shown in Figure 2.72.

Find My Device This section allows you to sign into a Microsoft account so that you can track the device if it is lost or stolen. This section is mainly focused at mobile devices.

General Here you can control your advertising ID, how websites make content decisions for you, the tracking of app launches, and suggested content settings.

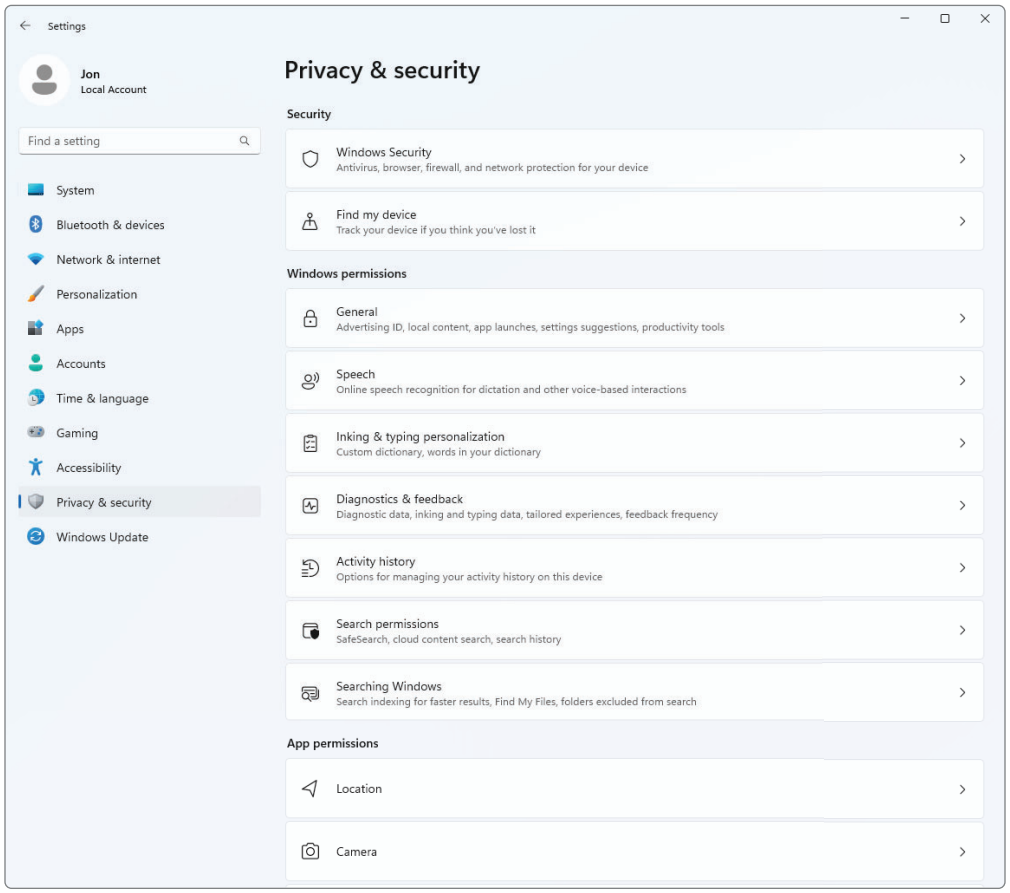
Speech The Speech section allows you to control how your voice is used for speech recognition with Microsoft's online speech recognition technology.

Inking & Typing Personalization The Inking & Typing Personalization section allows you to control whether your handwriting is used to build a personal dictionary of words.

Diagnostic & Feedback In Diagnostic & Feedback, you control how your personal information is used to provide diagnostic and feedback statistics to Microsoft. You can improve your inking and typing, and allow serving of tailored tips and ads. You can also view diagnostic data and delete the personalized data, such as the custom dictionary.

Activity History The Activity History section lets you control whether your activity is stored and tracked on the device. The history is not limited to recent documents—Windows can also track previous website visits. This allows you to continue where you left off.

FIGURE 2.72 Privacy settings



In prior versions of Windows, when you pressed the Windows key+Tab, Windows would show you all your previous documents that you had opened. This feature was called Windows Timeline. This feature was retired from later versions of Windows 11. Pressing the Windows key+Tab will still show you open windows.

Search Permissions The Search Permissions section allows you to control content moderations. You can filter out explicit content from web searches using the Strict setting, or turn it off completely; by default, it is set to Moderate. There are several other controls that allow you to search your personal, work, or school account for documents.

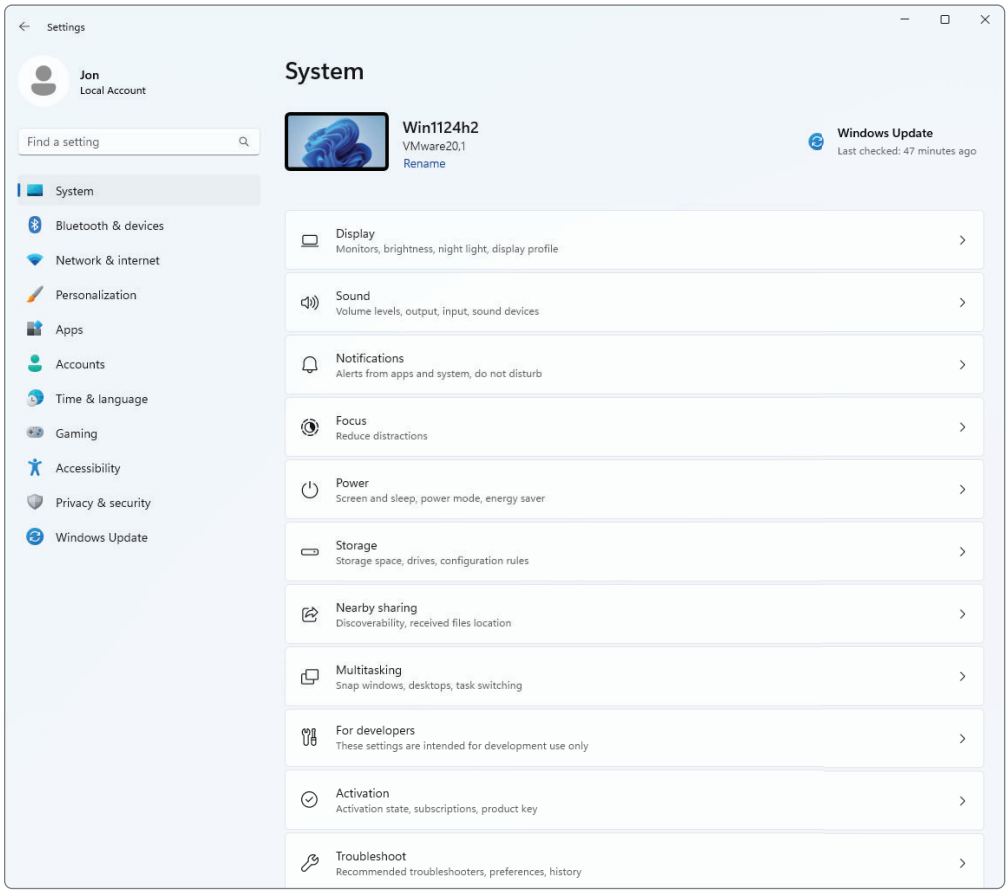
Searching Windows The Searching Windows section allows you to control how documents are searched on your computer. You can exclude folders from being included in searches. If you click the Advanced Indexing Options section, the Indexing Options Control Panel applet will open.

App Permissions In addition to the aforementioned Windows permission settings, you can view and control App permissions. This is the most powerful feature of the Privacy section, and it mirrors the controls found on mobile operating systems today. The various permissions that can be viewed and controlled are as follows: Location, Camera, Microphone, Voice Activation, Notifications, Account Info, Contacts, Calendar, Phone Calls, Call History, Email, Tasks, Messaging, Radios, Other Devices, Background Apps, App Diagnostics, Automatic File Downloads, Documents, Pictures, Videos, and File System. This is quite an exhaustive list, and you can review and control each of these permissions for a particular application.

System

The System section allows you to change a multitude of settings that pertain to the operating system, as shown in Figure 2.73. Just about every aspect of the operating system can be controlled.

FIGURE 2.73 System settings



Display In the Display section, you can arrange your monitors if you have more than one. You can also change how the additional monitors operate, such as extending or duplicating your desktop. You can also turn on the feature called Night Light, which restricts the blue light the display normally emits. The Display section also allows you to tune the Windows high dynamic range (HDR) of colors on your display. The most important settings are probably the display resolution and Scale And Layout settings, which allow you to get the most out of your display.

Sound In the Sound section, you select your output and input devices. You can also click the Troubleshoot button to help you identify sound issues. The Sound section is similar to the Sounds Control Panel applet, because it allows you to change the sound devices and control volume levels.

Notifications The Notifications section allows you to change the way the operating system notifications behave. You can control all operating system notifications, change lock screen notifications, control reminder and incoming VoIP calls on the lock screen, and specify whether notifications play sounds, among other settings.

Focus The Focus section allows you to control which notifications come to your attention and when they notify you; you can, for example, choose to suppress notifications when you are playing a video game.

Power The Power section is identical to the Power Control Panel applet. Here you can change when the screen turns off and when the operating system enters sleep mode. You can even change the power mode, similar to the Power Control Panel applet. You can also review recommendations to conserve energy.

Storage The Storage section provides a graphical overview of space used on the local disk. Clicking Installs Apps displays all the applications you can uninstall on the operating system. Temporary Files displays all the various temporary files on the operating system (you can then choose to remove them). This option is similar to the disk cleanup feature. Another feature called Storage Sense can be turned on, which automatically frees up space on the local disk by removing unneeded files.

Nearby Sharing The Nearby Sharing section allows you to control how your device is discovered by other devices. By default, the Nearby Sharing feature is turned off. You can turn the feature on for only your devices or for everyone. In addition, you can choose where to save received files, and you can change the discoverable name of your device.

Multitasking Use the Multitasking section to control Snap Assist, which is how an application or window snaps into a corner of the screen. You can also change the way the Alt + Tab keys display applications. In addition, you can configure how virtual desktops are used in Windows 11.

For Developers There are several controls in the For Developers section that only a developer can truly appreciate and only a developer should configure. Several of the settings in this section can make your PC insecure, as they share data on the local network

and to other applications. There are some notable features that even a network admin could appreciate, such as the File Explorer options. You can adjust whether hidden files or file extensions are shown. Set these settings at your own risk. The rule of thumb is unless you need it, don't set it.

Activation The Activation section allows you to see the current state of the operating system's activation. You can also change the product key in this section.

Troubleshooting This section allows you to control the behavior of the troubleshooter. By default, it is set to Don't Run Any. This setting can be changed to automatically run or ask you before running. In addition, if you click the Other Troubleshooters menu item, you will open another screen where you can run a multitude of troubleshooters.

Recovery The Recovery setting screen allows you to fix problems without resetting your PC by directing you to the Troubleshooters section. You can fix problems with your PC using Windows Update and by reinstalling your operating system. The most popular options are Reset This PC and Advanced Startup; both of these are covered Volume 2 Chapter 5, "Security Concepts."

Projecting To This PC The Projecting To This PC section allows you to control how other devices project their displays to Windows 11. The protocol used is called Miracast, which is a technology that allows screen sharing across devices.

Remote Desktop Use the Remote Desktop section to enable and disable the Remote Desktop feature, which allows you to connect remotely to a PC. We will cover these options in Volume 2 Chapter 8, "Scripting, Remote Access, and Artificial Intelligence."

Clipboard The Clipboard section allows you to control how the Clipboard operates. You can turn on features like Clipboard History, which enables you to have multiple items in your Clipboard. You can even sync Clipboards across multiple devices.

System Components The System Components section allows you to control the various components of the operating system, such as the Game Bar, Get Help, and Microsoft Store, just to name a few. You have similar options in resetting, repairing, and terminating the process as you do with installed apps.

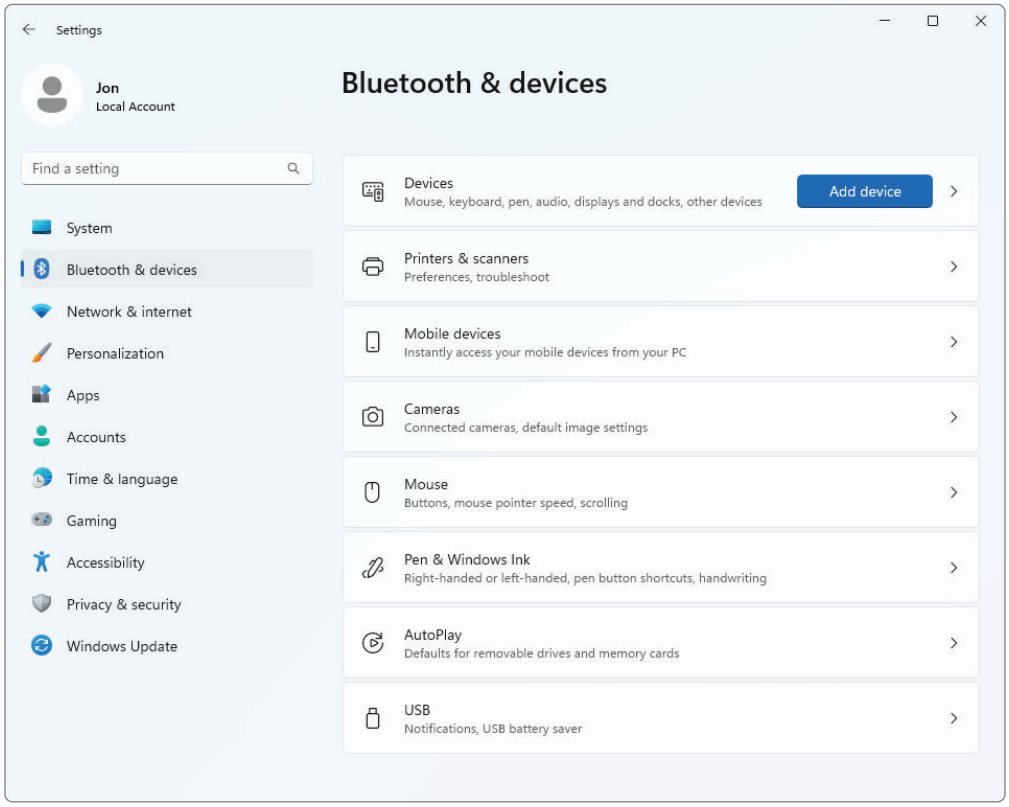
Optional Features This section allows you to remove features, such as VBScript, Notepad, Print Management, and several others that are built in. In addition to removing, you can also install features like Hyper-V, .NET Framework, legacy components, and many others.

About The last section, About, allows you to view information about the PC and rename it, if you want. In addition, you can join an Active Directory domain or change workgroups. We will cover domains and workgroups in Volume 2 Chapter 3. There are several other options that will take you to other settings screens or Control Panel applets. These options include System Protection, Advanced System Settings, Product Key and Activation, Remote Desktop, Device Manager, and BitLocker.

Bluetooth & Devices

The Bluetooth & Devices Settings screen allows you to view, control, and configure all devices connected to the PC. If you remember back in the Control Panel section of this chapter, this is the settings screen that the applet is directed back to when you click the Devices And Printers Control Panel applet. The settings screen is shown in Figure 2.74.

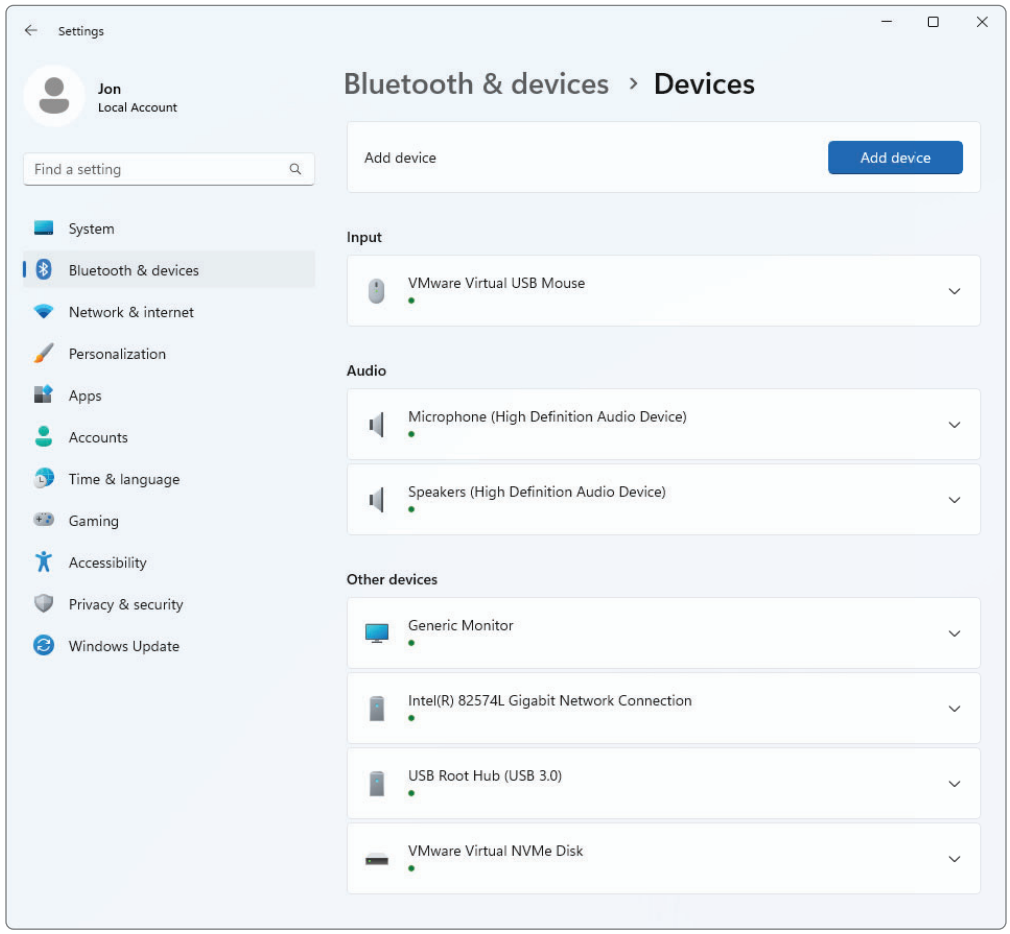
FIGURE 2.74 Bluetooth & Devices settings



Devices The Devices section allows you to view and configure devices that are directly connected to the system, connected via Bluetooth, connected via another wireless technology, or wired directly, as shown in Figure 2.75. On this screen, you can add a Bluetooth, wireless display or dock, or any other device with a pairing process.

Printers & Scanners The Printers & Scanners section allows you to view and configure all of the installed printers and imaging devices on the operating system. You can allow Windows to manage the default printer, and Windows will select the most recently used printer as your default. You also have the option to add additional printers or scanners.

FIGURE 2.75 Devices settings



Mobile Devices This section allows you to add mobile devices that can be managed by the operating system. You can then configure the Phone Link options so you can receive calls on your PC. This feature can be set up for Android or iPhone.

Cameras This section allows you to search for network cameras on your network so that they can be used by the operating system. In addition, this is where you will find your webcam settings.

Mouse The Mouse section lets you change how the mouse behaves on the operating system. You can change settings like which button on the mouse is your primary button, your cursor speed, how the scroll wheel advances, and other mouse-related settings.

Pen & Windows Ink The Pen & Windows Ink section contains settings related to handwriting, such as the font you want to use when converting handwriting.

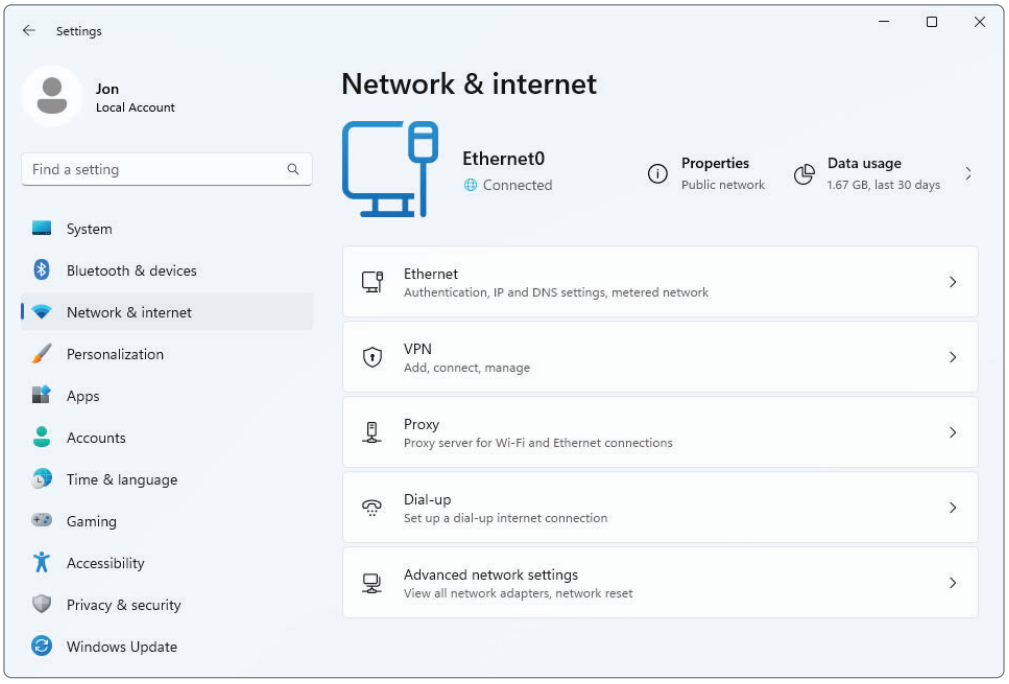
AutoPlay The AutoPlay section allows you to control how AutoPlay works for media and devices connected to Windows. You can choose a default action for removable drives or memory cards inserted.

USB The USB section allows you to control notifications if a USB device is not working correctly or has issues.

Network & Internet

Use the Network & Internet section to view and control network settings for Windows as well as Internet settings. The opening screen is the status section, and it displays the current network status for the Network connection, as shown in Figure 2.76. On this screen you can click Properties and then change various properties of the network connection, such as the public or private firewall posture, metered connection setting, and IP addressing. A newer feature lets you view the data usage for each application, which makes it easy to find an application that uses a lot of bandwidth.

FIGURE 2.76 Network & Internet settings



Ethernet or Wi-Fi The Ethernet or Wi-Fi section will change depending on your connection to the network. This section works as described earlier, mainly because the status section link redirects to this menu item.

VPN Use the VPN (virtual private network) section to view and configure settings for VPN connections. You can add a VPN connection or change advanced options, such as allowing VPN over metered networks or allowing VPN connections while roaming if a cellular modem is being used.

Proxy The Proxy Settings screen is where you can configure a proxy for the currently logged-on user. Here you set the proxy that Microsoft Edge will use. Other applications that use the common Microsoft web controls will also use the proxy server. By default, Automatically Detect Settings is enabled, but you can elect to use a setup script instead. It is a common task to manually set a proxy server, which will be in the form of an IP address or *fully qualified domain name (FQDN)*, along with the port. You also specify a bypass list of addresses that is allowed to bypass the proxy server.

Dial-Up Although it's unlikely you have a dial-up connection, the Network & Internet settings screen includes a section for configuring dial-up connections.

Advanced Network Settings The Advanced Network Settings screen allows you to configure a number of advanced settings for the network adapters. Use this section to configure advanced sharing options, such as network discovery and file and printer sharing. You can also reset the network connection and open the Firewall and Internet Options from this section.

Gaming

The Gaming settings screen was originally introduced with Windows 8 to create a seamless interface between the Xbox platform and the PC. Today, it has evolved into a very rich feature, and here you control how gaming is performed on Windows 11. The Gaming settings screen is shown in Figure 2.77.

Game Bar In this section, you can control whether the game bar is active during a game and how it launches. You can change any of the shortcut keys related to the Xbox Game Bar.

The most important key sequence is Windows key+ G, which launches the Xbox Game Bar, as shown in Figure 2.78. However, if you have an Xbox controller, the Xbox key will launch the game bar.

Captures The Captures section allows you to configure where screenshots and recorded captures are saved. You can manage all aspects of the capture in this section, such as recording length, recording audio, audio quality, microphone and system volume levels, recorded frames per second, and overall video quality.

Game Mode Use the Game Mode section to control the game mode, which turns off Windows updates so they don't interrupt gameplay. You can also adjust the quality of gameplay to deliver the best frame rate, and manually change the Graphics settings for performance of either desktop apps or Microsoft Store apps.

FIGURE 2.77 Gaming settings

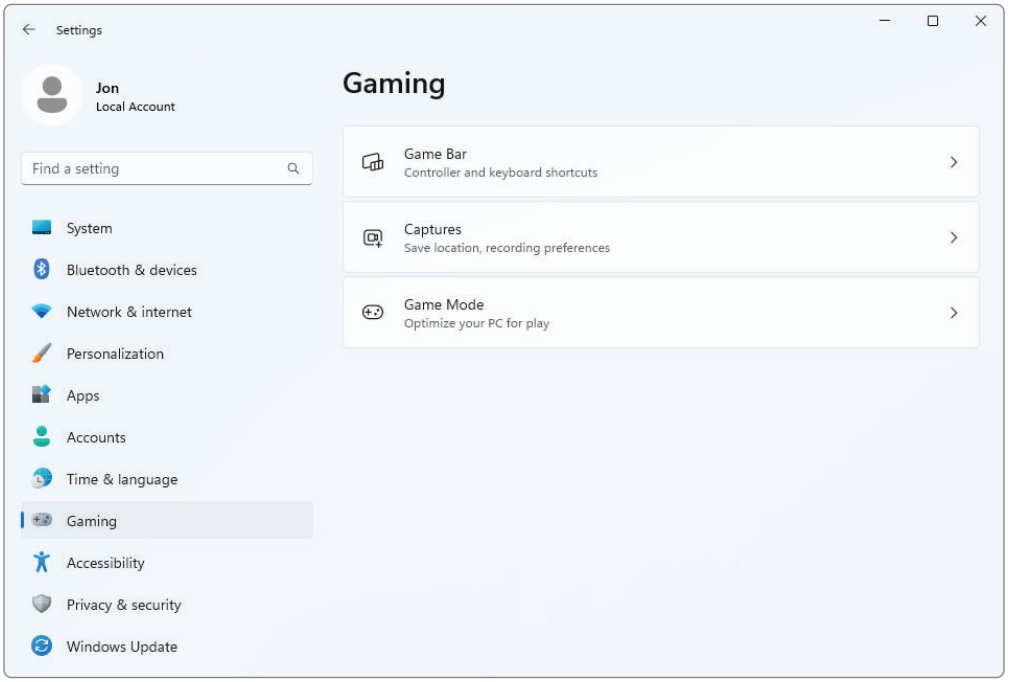
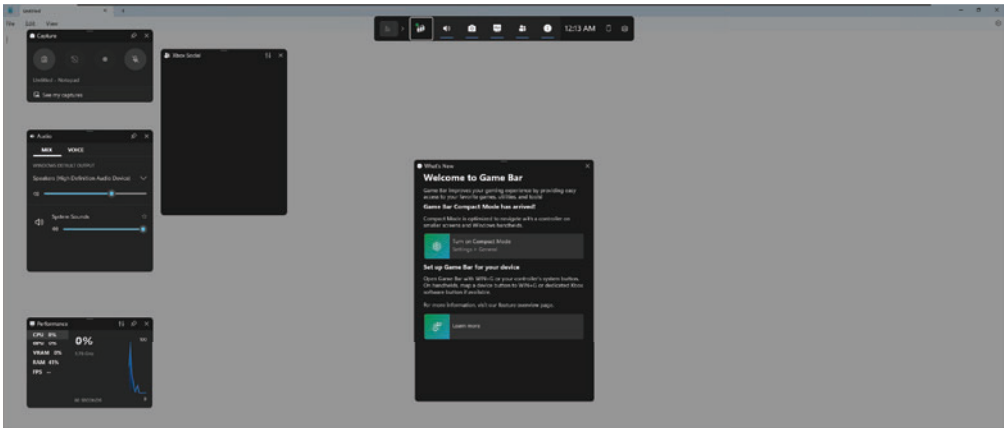


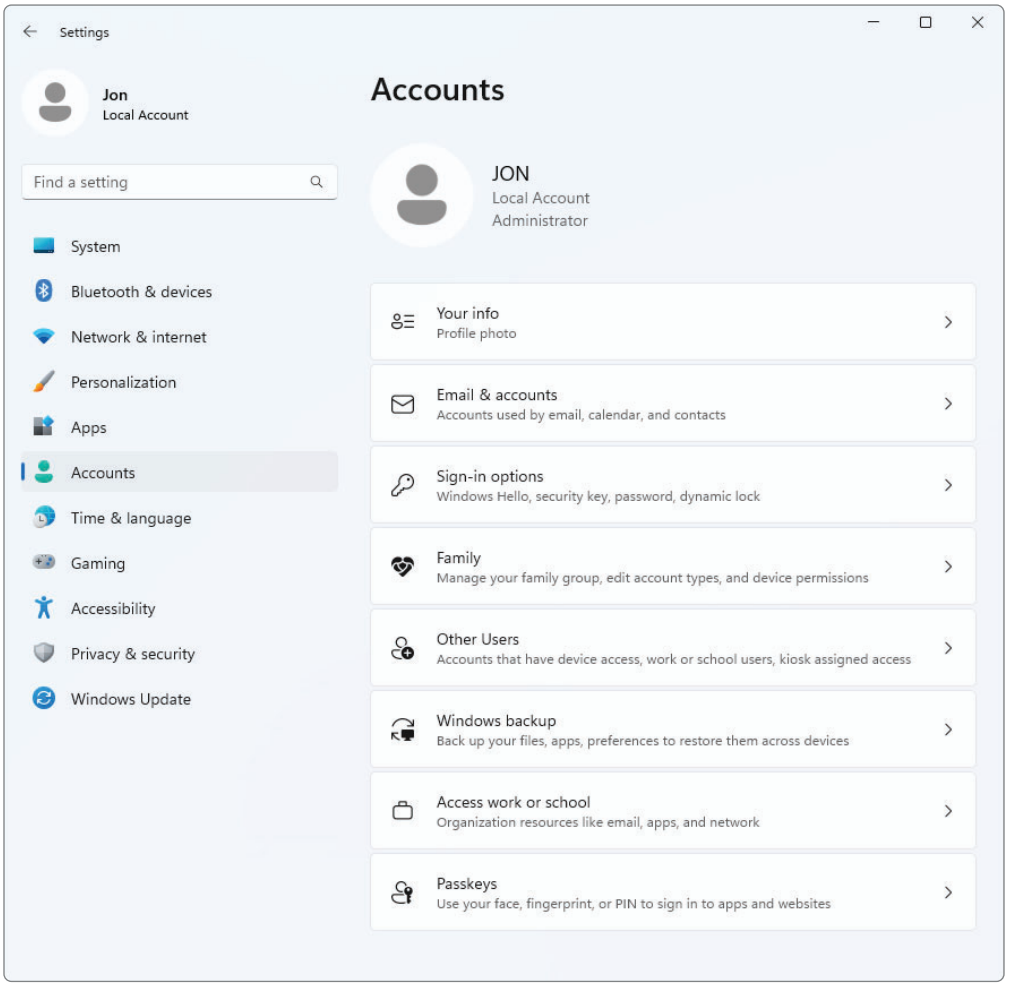
FIGURE 2.78 Windows 10 Xbox Game Bar



Accounts

The Accounts section allows you to view and configure all the settings for your user account, as well as other accounts on the operating system. The setting screen is shown in Figure 2.79.

FIGURE 2.79 Accounts settings



Your Info The default screen is the Your Info screen, and it will display all of the information about your account, such as name, email address, and account type. You also have the option of managing your Microsoft account online.

Email & Accounts The Email & Accounts section enables you to add an email account that is used for email, calendar, and contact information. You can also change accounts used by other apps, such as the Microsoft Store app, that require a login.

Sign-in Options Use Sign-in Options to change the way you log into Windows. The Windows Hello feature is configured on this screen. Windows Hello allows you to substitute your face, fingerprint, PIN, security key, or picture password for your actual password. The Hello feature works by storing your real credentials, such as your username and password, in Credential Manager. Credential Manager is then locked with this process. When you attempt to log in with a picture of your face, Windows Hello will unlock the credentials stored in Credential Manager and pass the actual username and password to the operating system. Dynamic Lock is another feature that can be configured in this section. Dynamic Lock will dynamically lock your computer when you walk away with a device that is paired to the laptop, such as a mobile device.

Family This screen does not allow any control of settings; it simply explains some of the parental controls you can employ. The screen also supplies a link to view family settings online. This is the same screen that can be found in the Windows Security settings screen under Family Options.

Other Users This screen allows you to set up other user accounts for the operating system. You can also set up the operating system to act as a kiosk. Uses might be a digital sign, interactive display, or something else where you don't need someone logged into the desktop. You launch a wizard that will create a local account, and you can then choose a kiosk app. When the kiosk mode is enabled, the operating system will boot up and automatically log in as the local user created, and the configured app with run.

Windows Backup The Backup section allows you to back up files to OneDrive. You can remember the apps on the current operating system. This allows you to restore both the apps and the data when you log into a new computer. This feature requires a Microsoft account and enough storage to back up your files.

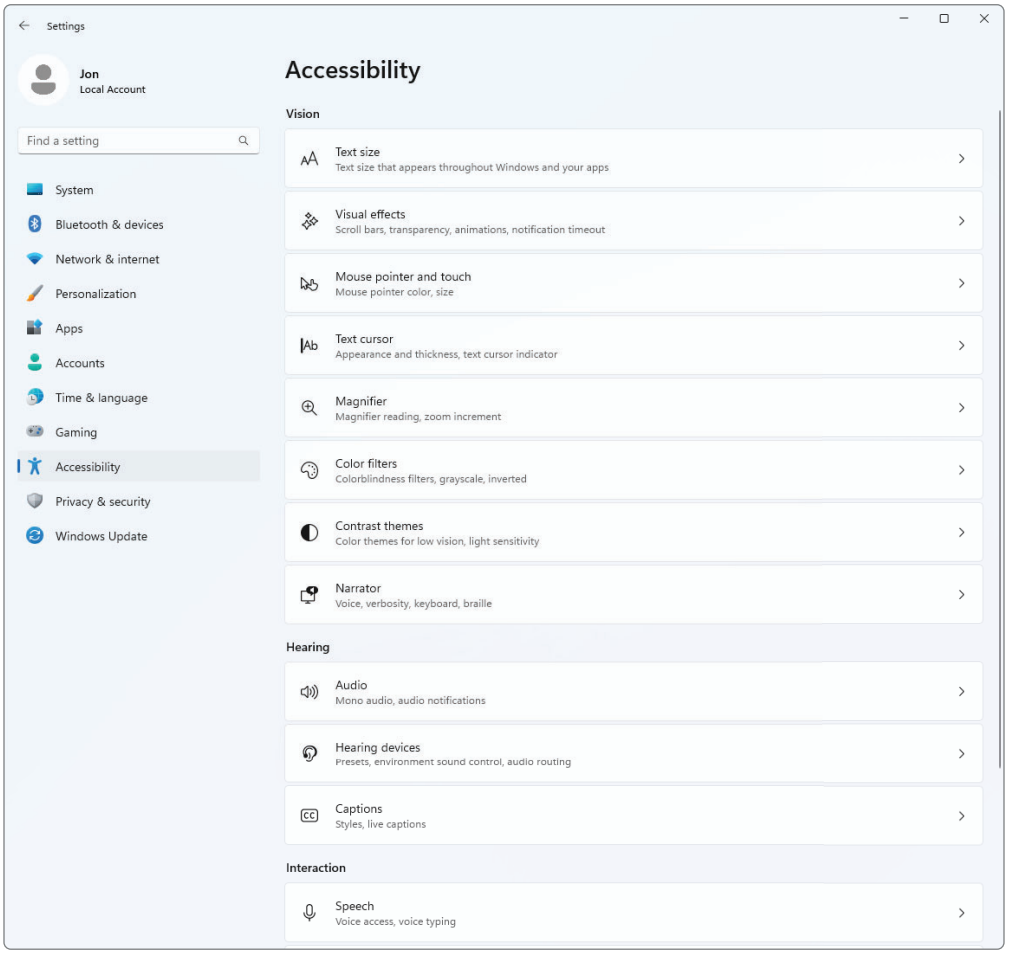
Access Work Or School The Access Work Or School section is used to connect the operating system with a corporate or school account. These accounts usually contain mobile device management (MDM) settings. This passes some or all of the control of the operating system to the organization responsible for the account. The enrollment of the operating system into an MDM system can be performed with a provisioning package to help simplify the enrollment process. You can also export management log files for analysis if something is not functioning properly with the MDM control. You can set up an account for test taking, which locks down the operating system when it is logged into.

Passkeys The Passkeys section provides you with a way to manage the various passkeys for websites and applications. You can then use biometrics or a PIN through Windows Hello to unlock and give the application the key.

Accessibility

The Accessibility settings screen is almost identical to the Ease of Access Control Panel applet. In the future, we will probably see the Ease of Access Control Panel applet converged into the Accessibility Settings screen, like many other applets. The Accessibility Settings screen is divided into three categories—Vision, Hearing, and Interactions, as shown in Figure 2.80.

FIGURE 2.80 Accessibility settings



Text Size The Text Size screen allows you to adjust the size of the text displayed on the screen. This feature is particularly useful for individuals with visual impairments, as it enables them to increase the readability of text across the operating system and

applications. By modifying the text size, users can ensure that all textual content, including menus, dialog boxes, and system messages, is easily legible.

Visual Effects The Visual Effects setting provides options to customize the appearance and behavior of visual elements in Windows 11. You can enable or disable animations, transparency effects, and other visual enhancements to improve system performance or reduce visual clutter. This setting is beneficial for users who experience motion sensitivity or who prefer a more streamlined interface.

Mouse Pointer and Touch This setting allows you to customize the appearance and behavior of the mouse pointer and touch input. You can change the pointer size, color, and speed, as well as enable features like pointer trails and touch feedback. These adjustments help individuals with motor impairments or visual difficulties to interact more effectively with their devices.

Text Cursor The Text Cursor setting enables you to modify the appearance and behavior of the text cursor, also known as the caret. You can change the cursor's thickness, color, and blink rate to make it more visible and easier to locate within text fields. This feature is particularly useful for individuals with visual impairments or those who have difficulty tracking the cursor's movement.

Magnifier The Magnifier is a screen magnification tool that allows you to enlarge portions of the screen for better visibility. It offers various modes, including full-screen, lens, and docked views, to accommodate different user preferences. The Magnifier also provides options for adjusting the zoom level, following the mouse pointer, and inverting colors to enhance contrast.

Color Filter The Color Filter setting enables you to apply color filters to the screen to improve visibility and differentiate between colors. This feature is particularly helpful for individuals with color blindness or other visual impairments. Users can choose from various filter options, such as grayscale, inverted, and specific color filters, to suit their needs.

Contrast Themes Contrast Themes are high-contrast color schemes designed to make text and interface elements more distinguishable. These themes use a limited color palette with high contrast between foreground and background elements, making it easier for users with low vision or color blindness to navigate and interact with the system.

Narrator The Narrator is a screen reader built into Windows 11 that provides spoken feedback for onscreen content. It reads aloud text, buttons, and other interface elements, allowing users with visual impairments to navigate and interact with their devices. The Narrator offers various customization options, including voice selection, verbosity levels, and keyboard shortcuts.

Audio The Audio settings allows you to adjust sound-related accessibility features, such as mono audio, audio alerts, and sound schemes. These options help individuals with hearing impairments or auditory processing difficulties to better perceive and respond to audio cues within the operating system and applications.

Hearing Devices This setting provides support for connecting and configuring hearing devices, such as hearing aids and cochlear implants, to Windows 11. You can pair their devices via Bluetooth or other compatible methods and adjust settings to optimize audio output for their specific hearing needs.

Captions The Captions setting allows you to customize the appearance of closed captions and subtitles displayed in videos and other media. You can change the font, color, size, and background of captions to improve readability and ensure that the text is easily distinguishable from the video content.

Speech The Speech setting enables user to control their devices using voice commands and dictation. This feature is particularly useful for individuals with mobility impairments or those who prefer hands-free interaction. Users can configure speech recognition settings, train the system to recognize their voice, and customize voice command preferences.

Keyboard The Keyboard setting provides options to customize keyboard behavior and accessibility features. You can enable features like Sticky Keys, Filter Keys, and Toggle Keys to assist with typing and keyboard navigation. These settings are beneficial for individuals with motor impairments or those who have difficulty using a standard keyboard.

Mouse The Mouse setting allows you to adjust the behavior and appearance of the mouse pointer. You can change the pointer speed, enable pointer precision, and configure additional mouse buttons. These adjustments help individuals with motor impairments or those who require a more customized pointing experience.

Eye Control Eye Control is an accessibility feature that enables users to control their devices using eye-tracking technology. This setting allows individuals with severe mobility impairments to navigate the operating system, type text, and interact with applications using only their eye movements. Users can calibrate the eye-tracking device and customize settings to optimize the experience.

The Windows Registry

Windows configuration information is stored in a special configuration database known as the *Registry*. This centralized database contains environmental settings for various Windows programs. It also contains registration information that details which types of filename extensions are associated with which applications. So, when you double-click a file in Windows File Explorer, the associated application runs and opens the file that you double-clicked.

Exam Objective 1.4, "Given a scenario, use Microsoft Windows operating system features and tools," requires you to know tools such as Registry Editor (`regedit.exe`).

The Registry was introduced with Windows 95. Most operating systems up until Windows 95 were configured through text files, which could be edited with almost any text editor. However, the Registry database is contained in a special binary file that can be edited only with the Registry Editor provided with Windows. The Registry Editor can be accessed a few different ways. You can launch it from the icon named Registry Editor in Control Panel ➢ Windows Tools ➢ Registry Editor. You can also launch it from the Start Menu ➢ All ➢ Windows Tools ➢ Registry Editor. The fastest way is to open the Run box with Windows key+R and type **`regedit.exe`**.

Current versions of Windows have what appear to be two applications that can be used to edit the Registry: `regedit.exe` and `regedt32.exe` (with no i). In reality, however, `regedt32.exe` opens `regedit`. In Windows 11, they are the same application. In prior Windows versions they work similarly, but each had slightly different options for navigation and browsing.

The Registry is broken down into a series of separate areas called *hives*. The keys in each hive are divided into two basic sections—user settings and computer settings. In Windows, a number of files are created corresponding to each of the different hives. The names of most of these files do not have extensions, and their names are `SYSTEM`, `SOFTWARE`, `SECURITY`, `SAM`, and `DEFAULT`. One additional file, whose name does have an extension, is `NTUSER.DAT`.

The basic hives of the Registry are as follows:

`HKEY_CLASSES_ROOT` Includes information about which filename extensions map to particular applications.

`HKEY_CURRENT_USER` Holds all configuration information specific to a particular user, such as their desktop settings and history information.

`HKEY_LOCAL_MACHINE` Includes nearly all configuration information about the actual computer hardware and software.

`HKEY_USERS` Includes information about all users who have logged into the system. The `HKEY_CURRENT_USER` hive is actually a subkey of this hive.

`HKEY_CURRENT_CONFIG` Provides quick access to a number of commonly needed keys that are otherwise buried deep in the `HKEY_LOCAL_MACHINE` structure.

Modifying a Registry Entry

If you need to modify the Registry, you can modify the values in the database or create new entries or keys. You will find the options for adding a new element to the Registry on the

Edit menu. To edit an existing value, double-click the entry and modify it as needed. You need administrator-level access to modify the Registry.

Windows uses the Registry extensively to store all kinds of information. Indeed, the Registry holds most, if not all, of the configuration information for Windows. Modifying the Registry in Windows is a potentially dangerous task. Control Panel and other configuration tools are provided so that you have graphical tools for modifying system settings. Directly modifying the Registry can have unforeseen—and unpleasant—results. You should modify the Registry only when told to do so by an extremely trustworthy source or if you are absolutely certain that you have the knowledge to do so without wreaking havoc.

Restoring the Registry

Windows stores Registry information in several files on the hard drive. With Windows 10 later versions, you need to restore the files from a backup for the `systemroot\repair` directory by using the Windows Backup program. Repairing the Registry from a backup overwrote the Registry files in `systemroot\system32\config`.

The Last Known Good Configuration option no longer exists at boot time in Windows 10/11. If the operating system suffers two consecutive failed attempts to boot into Windows, the Windows Recovery Environment (WinRE) will automatically boot. From WinRE, the operating system will try to automatically fix common issues. However, the only way to recover a blue screen on boot is to choose Reset This PC in the WinRE and select Keep My Files. More information on WinRE can be found here:

```
https://learn.microsoft.com/en-us/troubleshoot/
windows-server/performance/
use-winre-to-troubleshoot-startup-issue
```

Real World Scenario

Beware of Editing the Registry

Just in case it hasn't sunk in yet: Be careful editing the Registry! There is no Undo button, nor do you have the safety net of choosing not to save your edits before you close. Once you make the change, it's made—for better or for worse.

There have been countless examples throughout our careers of people going in to edit the Registry without really knowing what they were doing. In many cases, making small

changes to the Registry, without having a viable backup, means having to reinstall Windows. At the very least, this is inconvenient.

Windows can help in this regard if you are in a networked environment with Windows-based servers. You can create system policies that prevent users from performing certain tasks, and the most important task to restrict is running Registry editors.

Remember that a system restore will restore the Registry to the state it was in when a restore point was saved. As a very last-resort option for system recovery, Windows uses the Windows Recovery Environment (WinRE) to do a complete PC reset. It is your goal to make sure that you never need to use this.

Disk Management

Where there are files, there are disks. That is, all the files and programs that we've talked about so far reside on disks. Disks are physical storage devices, and they also need to be managed. There are several aspects to disk management. One is concerned with getting disks ready to be able to store files and programs; another deals with backing up your data; and yet another involves checking the health of disks and optimizing their performance. We'll look at these aspects in more detail.

Exam Objective 1.1, "Explain common operating system (OS) types and their purposes," requires you to know the various filesystem types and disk partitioning. Be sure to know the differences between each of the filesystem types.

Getting Disks Ready to Store Files and Programs

In order for a hard disk to be able to hold files and programs, it has to be partitioned and formatted. *Partitioning* is the process of creating logical divisions on a hard drive. A hard drive can have one or more partitions. *Formatting* is the process of creating and configuring a file allocation table (FAT) and creating the root directory. The *New Technology Filesystem (NTFS)* is available with all the versions of Windows you need to know about for the exam, but others are also recognized and supported. The file table for NTFS is called the *Master File Table (MFT)*.

The following is a list of the major filesystems that are on the CompTIA objectives. They are split into three categories of Linux, Apple, and Windows.

Linux Filesystems

Linux filesystems are an essential component of the Linux operating system, providing a structured way to store, organize, and manage data on disk drives or partitions. They define how

data is stored, accessed, and retrieved, ensuring efficient data management and integrity. Linux supports a wide variety of filesystems, each with its own unique features and use cases. The following list is from the CompTIA objectives, but there are far more filesystems than these.

Extended File Allocation Table Created by Microsoft, *Extended File Allocation Table (exFAT)* is a proprietary filesystem of choice for flash drives where NTFS cannot be used (because of overhead) and FAT32 is not acceptable (due to filesystem limitations). Although exFAT is a proprietary filesystem, it is being adopted by third-party device vendors to overcome the filesystem limitations of FAT32. It is ideal for SD cards that hold a lot of information, and it is supported in all current versions of Windows. exFAT is the default filesystem on USB flash drives over 32 GB in size.

Network File System Network File System (NFS) was created by Sun Microsystems a number of decades ago and is widely used in UNIX and Linux environments. Starting with Windows 7, Microsoft includes support for NFS with Windows (but it is not always installed). Windows versions prior to Windows 7 (Vista) can install Windows Services for UNIX (SFU) to gain this support.

Extended File System The Extended File System (XFS) is a high-performance 64-bit journaling filesystem designed to handle large files and high-throughput environments. It divides the storage space into smaller sections, which helps in managing data more efficiently and allows for faster data access. XFS also has features that reduce data fragmentation, meaning it keeps files stored in a more organized way. It ensures data integrity by keeping track of changes, which helps in quick recovery if something goes wrong. Windows does not natively support the XFS filesystem.

Fourth Extended File System The name might sound the same as the Extended File System (XFS), but the ext filesystem is not derived from the XFS filesystem; each of these filesystems is unique. The ext3 (Third Extended File System) became the default filesystem for many distributions of Linux due to its journaling capabilities. Enhancements were added to it to create *ext4* (Fourth Extended File System), which is used with Android and other operating systems. It is possible to use ext4 with Windows, but doing so requires making some sacrifices in features for the sake of compatibility.

Swap Partition Swap partitions, which are found only within Linux and UNIX installations, are the equivalent of the page file (`pagefile.sys`) in Windows, except that they are their own partition type. They are used for *virtual memory* when the physical memory is exhausted.

Apple Filesystems

Apple filesystems are a fundamental part of the macOS, iOS, iPadOS, watchOS, and tvOS operating systems, providing a robust framework for organizing, storing, and managing data on various storage devices. The following are two of the main filesystems you may encounter on Apple devices.

Apple File System *Apple File System (APFS)* is a proprietary filesystem that was developed by Apple to replace HFS and HFS+. The filesystem was mainly developed to introduce new features with *solid-state storage drives (SSDs)*, such as compression, encryption, and snapshots, just to name a few. APFS also fixed some core issues with the HFS+ filesystem, such as the number of files and data integrity. The filesystem is the default filesystem for macOS Sierra 10.12.4, iOS 10.3, and tvOS 10.2 and later. Windows operating systems cannot read drives formatted in APFS.

Windows Filesystems

Windows filesystems are integral to the Windows operating system, providing a framework for organizing, storing, and managing data on various storage devices. The following are the most common files systems you may see as a technician.

File Allocation Table *FAT* is the acronym for the file table used to keep track of where files are within the filesystem. It's also the name given to this type of filesystem, introduced in 1981. The filesystems for many OSs have been built on the design of FAT but without its limitations. A FAT filesystem uses the *8.3 naming convention* (eight letters for the name, a period, and then a three-letter file identifier). This later became known as *FAT16* (to differentiate it from FAT32) because it used a 16-bit binary number to hold cluster-numbering information. Because of that number, the largest FAT disk partition that could be created was approximately 2 GB.

FAT32 The *FAT32* filesystem was introduced along with Windows 95 OEM Service Release 2. As disk sizes grew, so did the need to be able to format a partition larger than 2 GB. FAT32 was based more on VFAT than on FAT16. It allowed for 32-bit cluster addressing, which, in turn, provided for a maximum partition size of 2 terabytes (2,048 GB). It also included smaller cluster sizes to avoid wasted space. FAT32 support is included in current Windows versions.

New Technology Filesystem Introduced along with Windows NT, *New Technology Filesystem (NTFS)* is available with all current versions of Windows. NTFS is a much more advanced filesystem in almost every way than all versions of the FAT filesystem. It includes such features as individual file security encryption and compression, disk quotas, and RAID support, as well as support for extremely large file and partition sizes and disk transaction monitoring. It is the filesystem of choice for higher-performance computing.

Resilient File System The Resilient File System (ReFS) is a modern filesystem developed by Microsoft, designed to maximize data availability, scalability, and integrity. ReFS was developed by Microsoft to keep your data safe and reliable. It uses special checks to detect and fix any data corruption, ensuring your files stay intact. It can handle large amounts of data and supports long filenames, making it great for big projects. Additionally, ReFS regularly scans the storage to find and fix any hidden issues, keeping your system running smoothly.

Compact Disc File System Although not a filesystem that can be used on a hard drive, *Compact Disc File System (CDFS)* is the filesystem of choice for CD media. It has been used with 32-bit Windows versions since Windows 95. A CD mounted with the CDFS driver appears as a collection.

When you're installing Windows 11, the installer defaults to NTFS. However, you can use the partitioning tool to format the partition with exFAT. It's really unnecessary, since NTFS is a much better filesystem with respect to stability, avoiding corruption, and security. When you are formatting a storage drive, you can format it with FAT or NTFS. Storage devices formatted in FAT can be read by other operating systems, but when they are formatted in NTFS the operating system must explicitly support NTFS.

To format a partition from the command line, use the `format` command, which is available with all versions of Windows. You can run `format` from a command prompt or by right-clicking a drive in Windows File Explorer and selecting Format. However, when you install Windows, it performs the process of partitioning and formatting for you if a partitioned and formatted drive does not already exist.

You can usually choose between a *quick format* and a *full format*:

- With a quick format, a new file table is created on the hard disk, but files are not fully overwritten or erased from the disk.
- A quick format is much faster since the normal format fully erases any and all existing data on the hard disk (a time-consuming process).

Be extremely careful with the `format` command! When you format a drive, treasured data can be lost. Before using the `format` command, make sure that you back up the data you want to save.

Disk Management MMC

In Windows, you can manage hard drives through the Disk Management console. To access Disk Management in Windows 11, right-click the Start menu and then click Disk Management. Alternatively, you could start the Disk Management MMC by typing **`diskmgmt.msc`** in the Run dialog box (Windows key+R).

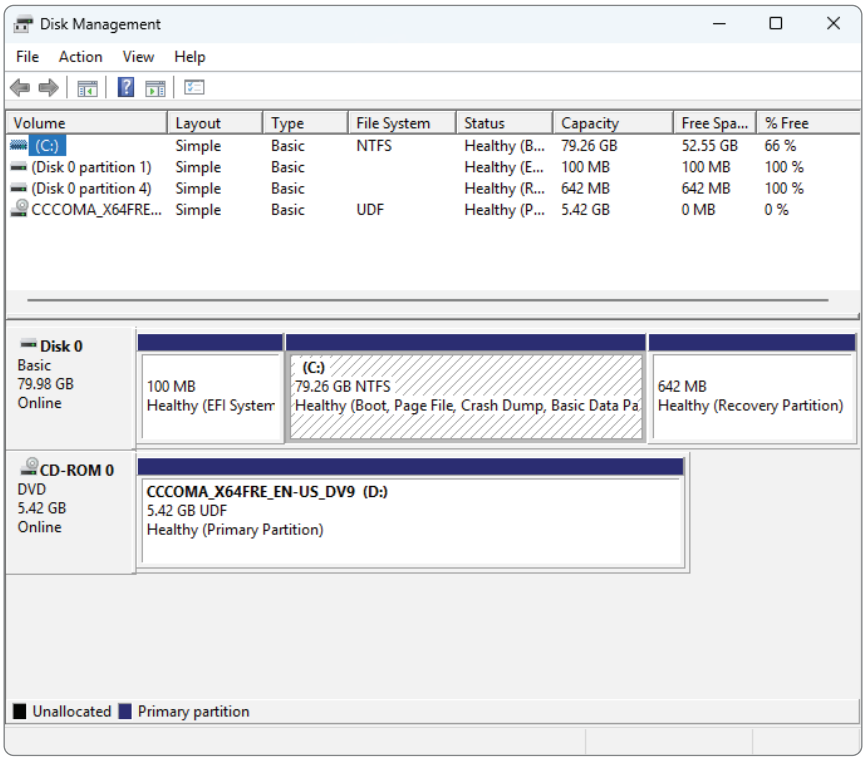
The Disk Management screen lets you view a lot of information about all the drives installed in your system, including CD/DVD-ROM drives (see Figure 2.81).

Given a scenario, there are many instances in which an administrator would turn to Disk Management during the course of trying to find the right storage solution. The Disk Management administrative console allows you to review all the logical partitions (volumes) configured and physical drives (disks) that are connected to the computer.

The logical volume is displayed in the upper portion of the Disk Management console. This view allows you to see at a glance all the configured volumes on the operating system.

Details about the volume letter or type, the layout type, the type of disk (basic vs. *dynamic*), filesystem, status, capacity, and free space can be seen in this logical view.

FIGURE 2.81 Disk Management MMC



The physical layout, displayed below the logical view, allows you to see the layout of the physical disks in a graphical format. Using both views, you can see the layout of a physical disk and its corresponding logical layout. The physical layout allows you to view the status of the drive, the size of the drive, and its health, in addition to the layout of volumes on the physical disk.

Windows supports three types of partition styles: Basic, GPT, and Dynamic. Partition styles are also known as partition schemes; the terms are used to describe the underlying structure of the partitioning of the physical disk.

Basic Disks *Master Boot Record (MBR)* is the standard BIOS partition table used and predates GPT. MBR disk storage can have a maximum of four primary partitions or three primary partitions and one extended partition containing multiple logical partitions, for a maximum of 26 logical partitions. The partition from which the operating

system boots must be designated as active and must be a primary partition. Only one partition on a disk can be marked active.

GPT Disks *GUID [globally unique identifier] Partition Table (GPT)* is the current partition table specification used with *Unified Extensible Firmware Interface (UEFI)* BIOS. GPT partition tables allocate 64-bit logical block addressing for larger partitions. If you need to create partitions 2 TB or larger, it is required. A GPT disk can contain up to 128 primary partitions. The large number of primary partitions supported eliminates the need to create extended/logical partitions.

Dynamic Disks *Dynamic disks* are proprietary to Windows operating systems. Dynamic disks can contain up to 2,000 volumes. All the dynamic disks installed in the operating system contain an identical 1 MB database (partition table) that describes all the other disks installed. The true benefit of this 1 MB database is moving fault-tolerant disks to another Windows operating system. The operating system will automatically recognize that the disks work together, and the *Logical Disk Management (LDM)* will configure itself according to the layout in the 1 MB database.

The terms *partition* and *volume* are interchangeable; they both mean the same thing. They are logical subdivisions of a physical disk. Dynamic disks use the terminology of volumes, and MBR and GPT use the terminology of partitions. Sometimes partitions will be referred to as volumes, such as when you use the `diskpart` command-line utility.

Right-clicking any volume opens a context menu that allows you to change the drive letter or paths, format, extend, shrink, delete, or add a mirror. Right-clicking any drive opens a context menu that allows you to create a new spanned, striped, mirrored, or RAID-5 volume. You can also convert to a dynamic or GPT disk.

Let's discuss the features and functions in further detail:

Initializing Disks The phrase *initializing a disk* means different things depending on if it is a hardware initialization or, in this case, a Windows initialization. When you initialize a disk in Windows, you are allowing the LDM to recognize the disk. When you initialize a disk in hardware disk systems, you are erasing the partition table and even the data. To initialize a disk, right-click the drive and select Initialize. This needs to be performed only once.

Mounting Partitions Once a partition (volume) is created, you need to make it available to the operating system for use. You have two basic options. The first option is to assign the volume a drive letter. The second option is to mount the volume in an empty NTFS folder. So you can mount an entire volume in a specific folder. You can perform this task by right-clicking a partition and selecting Change Drive Letter And Paths.

Extending Partitions When you install an operating system on a disk, you will more than likely use the entire disk for the installation of the operating system. However, if you ever fill it all up and buy a bigger drive, when you clone it over to the new larger disk, you will find that the partition is the same size. This is where extending a partition comes in handy. It is also very common for virtualized operating systems, since adding space to a virtual disk can be done on the fly. As long as there is unallocated space on the disk, you can right-click the partition (volume) and choose Extend Volume.

Shrinking Partitions Shrinking a partition, also known as splitting a partition, is not a common task, but it is done from time to time. It is usually performed when an installation is not using a fraction of the physical disk storage and another partition is required. You can perform this action by right-clicking the partition (volume) and choosing Shrink Volume.

Extending partitions and shrinking partitions require that you perform the task on the last partition (volume) created on the drive. This is not always the case. These functions will not always work from within the operating system. An open source tool called SystemRescueCD contains a tool called GParted. This tool allows you to reconfigure the disk layout and move partitions around, as well as shrink and extend partitions.

Adding Drives When you add a disk to the system, you normally have to power the computer down to attach the disk. However, servers have the capability to add drives on the fly, called *hot swap disks*. When a disk is added to the operating system, the operating system needs to know it. You can make the operating system take inventory of its disks by selecting Rescan Disks from the Action menu in the Disk Management console.

Adding Arrays Arrays of disks are multiple disks working together for a specific purpose. These are also known as a *Redundant Array of Independent* (*or Inexpensive*) *Disks* (*RAID*). They are used for fault tolerance so that in the event of a disk failure, the system will continue to function. They can also be used to add performance to a system by using multiple disks. Striped disks, also known as *RAID-0*, require two or more disks and are used for performance; they do not offer any fault tolerance. A mirror, also known as *RAID-1*, is the most basic type of fault tolerance; it creates an identical mirror of the information on another disk. Striped disks with parity, also known as *RAID-5*, offer performance and fault tolerance and require three or more disks. Windows is unique in that it allows volumes to participate in array groups, rather than requiring the entire disk. A combination of mirroring (*RAID-1*) and striping (*RAID-0*) can be used together as *RAID-10*. *RAID-10* offers both fault tolerance of a mirrored disk and the performance of striping across the mirrors. *RAID-10* requires a minimum of four disks to operate.

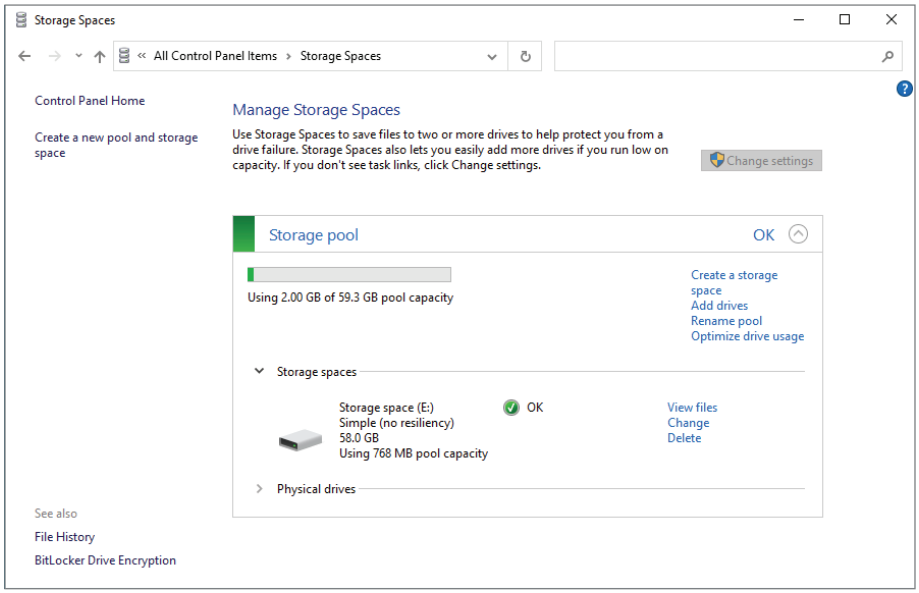
Windows can create RAID groups of 0, 1, and 5, but RAID 10 is not directly supported. There are various tricks to making a RAID 10 volume, such as creating two mirrors and then striping across the mirrors. However, this adds a level of complexity, and we recommend that if you need RAID 10 you purchase a hardware solution such as a dedicated RAID card. Volume 1 Chapter 2, "Expansion Cards, Storage Devices, and Power Supplies," provided more information about the hardware implementations of RAID.

Storage Spaces This Windows feature was initially introduced in Windows Server 2012 and Windows 8. It allows for a group of drives to be placed into a pool of storage that can be configured for fault tolerance. The benefit is that more disks can be added later to extend a storage pool dynamically. The one caveat is that the hard drives can either be managed by the Disk Management MMC or Storage Spaces, but not both.

The *Storage Spaces* feature lets you manage a variety of hard drives. A unique feature is its capability to create a storage pool using two or more external (USB-attached) hard drives. Storage Spaces is not exclusive to external hard drives; internal hard drives can be used as well.

In Storage Spaces, you can create a two-way mirror, a three-way mirror, or parity resiliency of your data. A two-way mirror is identical in functionality to a *RAID-1* mirror. A three-way mirror is similar to a RAID-1 mirror in that it duplicates the data on two other drives; unlike RAID-1, however, you can lose two hard drives and retain data. A parity resiliency type is identical in functionality to a *RAID-5* (striping with parity). You can also use Storage Spaces without resiliency, which is called a *simple volume*, as shown in Figure 2.82.

FIGURE 2.82 Storage Spaces



Checking the Health of Hard Disks and Optimizing Their Performance

Exam Objective 1.4, "Given a scenario, use Microsoft Windows operating system features and tools," requires you to know tools such as Disk Defragment (`dfrgui.exe`).

As time goes on, it's important to check the health of Windows computers' hard disks and to optimize their performance. Windows provides you with several tools to do so, some of which we've already mentioned in this chapter. One important tool is Disk Defragmenter, which has existed in almost all versions of Windows.

When files are written to a hard drive, they're not always written contiguously, or with all the data in a single location. Files are stored on the disk in numbered blocks, similar to PO boxes. When they are written, they are written to free blocks. As a result, file data is spread out over the disk, and the time it takes to retrieve files from the disk increases. Defragmenting a disk involves analyzing the disk and then consolidating fragmented files and folders so that they occupy a contiguous space (consecutive blocks). This increases performance during file retrieval, since the hard drive arm in mechanical hard disks needs to travel less to retrieve the blocks. Defragmentation of the filesystem is not required on solid-state drives (SSDs), since the fragmentation of data blocks does not slow down the retrieval of the memory locations.

To access Disk Defragmenter, follow these steps:

1. Click Start ➢ All ➢ Windows Tools ➢ Defragment And Optimize Disk. Or you can click the Start menu and type **defragment**, then select it.
2. In the list of drives, select the drive that you want to defragment, and then click Analyze. When the analysis is finished, Disk Defragmenter tells you how much the drive is fragmented and whether defragmentation is recommended.
3. If defragmentation is recommended, click Defragment.

   Be aware that for large disks with a lot of fragmented files, this process can take quite some time to finish.

In Windows, you can also access Disk Defragmenter through the properties of any partition listed in Disk Management (or, even easier, any disk root under This PC in Windows File Explorer). Select the Tools tab, and then click Defragment.

Summary

In this chapter, you learned about some of the tools that can be used with Windows. We covered basic Windows management concepts, including managing disks, using filesystems, and understanding directory structure. Keeping your computer healthy will save you a lot of stress.

With the basic knowledge gained in this chapter, you are now ready to learn how to interact with the most popular operating systems in use today. These topics are covered in the next four chapters.

Exam Essentials

Know the various filesystems and the differences between them. The most commonly used filesystem on Windows hard drives is NTFS. FAT32 is older and perhaps a bit quicker for smaller hard drives and USB thumb drives, but NTFS adds a plethora of important features, including security and auditing. exFAT is a filesystem Microsoft adopted to address large filesystems that FAT32 had problems addressing. The Extended filesystem (XFS) is a high-performance 64-bit journaling file system designed to handle large files and high-throughput environments and used with Linux. The Resilient File System (ReFS) is a Windows filesystem designed to maximize data availability, scalability, and integrity. Additionally, you should know that the ext3, ext4, and APFS filesystems are not compatible with Windows without third-party software.

Understand the various partitioning for Windows installs. The GUID Partition Table (GPT) is a modern specification used with UEFI BIOS, allowing for larger partitions and supporting up to 128 primary partitions. The Master Boot Record (MBR) is the traditional BIOS partition table, allowing for a maximum of four primary partitions or three primary partitions and one extended partition. Formatting a drive involves creating a file allocation table and configuring the root directory, with NTFS being the most commonly used filesystem.

Know which Windows Control Panel utility to use for a given scenario. A number of Control Panel applets allow you to change Windows' functionality. With the introduction of Windows 10, some of the functionality is split between the Settings app and the classic Control Panel applets; this is carried forward with Windows 11. You should be proficient with both for a given scenario in which changes to the operating system need to be made.

Know which Windows Settings section is used for a given scenario. The Settings app is slowly replacing Control Panel as the main method of configuring Windows. You should know the major categories of the Settings app and the various settings each of them contains.

Know the main administrative tools. You should know the primary graphical tools for troubleshooting Windows and configuring the operating system. These include the disk management tools, Windows Tools, Device Manager, Task Manager, System Information, System Restore, Task Scheduler, as well as all other tools covered in this chapter. Know the various disk management concepts.

Know the concepts related to disk management. You should know the various concepts related to disk management to initialize and mount a drive to assign a drive letter; the concepts of extending, splitting, and shrinking partitions (volumes); fault tolerance arrays; and Storage Spaces. GPT should be used with UEFI/BIOS, addressing 2 TB and larger drives, or if more than 26 logical volumes are required.

Review Questions

The answers to the chapter review questions can be found in Appendix A.

1. Which MMC snap-in allows you to start a program based on a condition?
   - **A.** Programs and Features
   - **B.** Task Scheduler
   - **C.** Disk Management
   - **D.** Group Policy Editor

2. You need to select a format for a flash drive that is compatible with other operating systems. Which of the following is the filesystem should you choose?
   - **A.** NTFS
   - **B.** EFS
   - **C.** FAT32
   - **D.** CDFS

3. Which utility can be used to schedule a remote shutdown?
   - **A.** `taskmgr`
   - **B.** `kill`
   - **C.** `shutdown`
   - **D.** `netstat`

4. Which event log should be checked if you suspect that the computer has rebooted unexpectedly?
   - **A.** Application
   - **B.** Boot
   - **C.** Security
   - **D.** System

5. Which of the following filesystems is a proprietary filesystem created by Microsoft for use with large flash drives?
   - **A.** GPT
   - **B.** NFS
   - **C.** ext3
   - **D.** exFAT

6. Which section in the Settings app lets you configure backups?
   - **A.** System
   - **B.** Devices
   - **C.** Apps
   - **D.** Accounts

7. Which of the following filesystems is a proprietary filesystem created by Apple that offers SSD support?
   - **A.** HFS
   - **B.** ext4
   - **C.** APFS
   - **D.** exFAT

8. Which Control Panel applet would you use to join the operating system to a domain?
   - **A.** System
   - **B.** User Accounts
   - **C.** Windows Tools
   - **D.** Programs And Features

9. What can you do if a program is not responding to any commands and appears to be locked up?
   - **A.** Open the System Control Panel applet and choose Performance to see what process is causing the problem.
   - **B.** Add more memory.
   - **C.** Press Ctrl+Alt+Delete to reboot the computer.
   - **D.** Open Task Manager, select the appropriate task, and then click End Task.

10. Which command will start the Event Viewer snap-in?
    - **A.** `eventviewer.exe`
    - **B.** `eventvwr.msc`
    - **C.** `lusrmgr.msc`
    - **D.** `devmgmt.msc`

11. You need to configure an iSCSI connection; where will you find iSCSI settings?
    - **A.** Storage Spaces applet
    - **B.** Disk Management snap-in
    - **C.** Windows Tools
    - **D.** Device Manager snap-in

12. Which type of resources do you configure in Device Manager?
    - **A.** Hardware
    - **B.** Files and folders
    - **C.** Applications
    - **D.** Memory

13. What should be periodically done with mechanical hard drives to speed up performance when launching and saving files?

A. Free up space

B. Empty Recycle Bin

C. Defragment

D. Trim

14. You have a number of files with a `.des` extension and you want to be able to search each file's metadata; which applet should this be configured in?

A. Indexing Options applet

B. File Explorer Options applet

C. Windows Tools applet

D. Ease of Access Center applet

15. Which of the following partitions is specifically the partition from which the operating system boots?

A. Primary partition

B. Extended partition

C. Dynamic partition

D. Logical partition

16. Which of the following Registry hives contains information about the computer's hardware?

A. `HKEY_CURRENT_MACHINE`

B. `HKEY_LOCAL_MACHINE`

C. `HKEY_MACHINE`

D. `HKEY_RESOURCES`

17. Which of the following replaces the MBR in 64-bit versions of the Windows operating system?

A. LILO

B. NFS

C. GPT

D. GRUB

18. Which of the following utilities will rearrange the files on your hard disk to occupy contiguous chunks of space?

A. Disk Defragmenter

B. Windows File Explorer

C. Scandisk

D. Windows Backup

19. Which power mode will write the contents of the RAM to the disk?

A. Hibernate

B. Standby

C. Sleep

D. Suspend

20. Which partition style allows up to 2,000 volumes to be partitioned on a physical disk?

A. GPT

B. MBR

C. Dynamic Disk

D. Basic Disk

Performance-Based Question

You will encounter performance-based questions on the A+ exams. The questions on the exam require you to perform a specific task, and you will be graded on whether or not you were able to complete the task. The following requires you to think creatively in order to measure how well you understand this chapter's topics. You may or may not see similar questions on the actual A+ exams. To see how your answers compare to the authors', refer to Appendix B.

You are working at a company that has standardized on Windows 10 workstations for all. The phone rings, and it is your supervisor. He tells you that his workstation is running incredibly slowly, almost to the point where is it is unusable. When you ask what he is running, he reports that he has exited out of everything but the operating system. You suspect there are background processes tying up the CPU and memory. Which utility can you have him use to look for such culprits?

Chapter 3

Windows Administration

THE FOLLOWING COMPTIA A+ 220-1202 EXAM OBJECTIVES ARE COVERED IN THIS CHAPTER:

✓ **1.0 Operating Systems**

✓ **1.2 Given a scenario, perform OS installations and upgrades in a diverse environment.**

- Boot methods
  - Universal Serial Bus (USB)
  - Network
  - Solid-state/flash drives
  - Internet-based
  - External/hot-swappable drive
  - Internal hard drive (partition)
  - Multiboot
- Types of installations
  - Clean install
  - Upgrade
  - Image deployment
  - Remote network installation
  - Zero-touch deployment
  - Recovery partition
  - Repair installation
  - Other considerations
    - Third-party drivers

✓ **1.5 Given a scenario, use the appropriate Microsoft command-line tools.**

- Navigation
  - cd
  - dir
- Network
  - ipconfig
  - ping
  - netstat
  - nslookup
  - net use
  - tracert
  - pathping
- Disk Management
  - chkdsk
  - format
  - diskpart
- File Management
  - md
  - rmdir
  - robocopy
- Informational
  - hostname
  - net user
  - winver
  - whoami
  - [command name] /?
- OS Management
  - gpupdate
  - gpresult
  - sfc

✓ **1.7 Given a scenario, configure Microsoft Windows networking features on a client/desktop.**

- Domain joined vs. workgroup
  - Shared resources
  - Printers
  - File servers
  - Mapped drives
- Local OS firewall settings
  - Application restrictions and exceptions
  - Configuration
- Client network configuration
  - Internet Protocol (IP) addressing scheme
  - Domain Name System (DNS) settings
  - Subnet mask
  - Gateway
  - Static vs. dynamic
- Establish network connections
  - Virtual private network (VPN)
  - Wireless
  - Wired
  - Wireless wide area network (WWAN)/cellular network
- Proxy settings
- Public network vs. private network
- File Explorer navigation—network paths
- Metered connections and limitations

The previous chapter introduced the basic components of Windows operating systems and discussed the various tools used to configure Windows. This chapter builds on the previous chapters and focuses on Windows administration. In this chapter you will learn how to install and upgrade Windows 11 and learn various command-line tools. We'll also explore advanced concepts, such as Microsoft networking and design. All the content is generic to the Windows operating systems you'll be tested on during the 220–1202 exam. The content can be applied to any of the previous Windows operating systems.

Installing and Upgrading Windows

Exam Objective 1.2, "Given a scenario, perform OS installations and upgrades in a diverse environment," requires you to know boot methods and the various types of installations. Be sure to know the various boot methods and the various types of installations for Windows 11.

Windows 11, released on October 5th, 2021, was the successor to Windows 10. The user interface (UI) looked familiar, and it was a well-needed upgrade. Microsoft also made the upgrade notification so irritating that many were forced to upgrade just to escape the notifications.

Regardless of the motivation to upgrade to Windows 11, its adoption was successful because of its meager system requirements. The hardware requirements for Windows 11 are similar to prior versions of Windows. However, starting with Windows 11, requirements have been scaled up for performance, as shown in Table 3.1. The meager hardware requirements were one of the driving factors for many to just hit the upgrade button.

TABLE 3.1 Windows 10/11 system requirements

Component	Windows 10 requirements	Windows 11 requirements
Processor	1 GHz or faster	1 GHz or faster (2 or more cores) 64-bit or System on a Chip (SoC)
RAM	1 GB (32-bit) or 2 GB (64-bit)	4 GB (64-bit)
Available Hard drive space	16 GB (32-bit) or 32 GB (64-bit)	64 GB (64-bit)

Component	Windows 10 requirements	Windows 11 requirements
Graphics card	DirectX 9 with WDDM 1.0 or higher driver	DirectX 12 with WDDM 2.0 driver
Display	800 x 600	High definition 720p
Additional hardware	N/A	UEFI & TPM 2.0 required

If you are planning to install Windows 11 on a computer, it must meet the specifications in Table 3.1, which are pretty easy to meet or exceed. However, sometimes you'll find that your organization's computers are older than you think. Windows 11 requires a realistic amount of computing power, RAM, and storage. In addition to these requirements, a UEFI firmware that is Secure Boot capable and a TPM 2.0 is required by Windows 11. To learn more about Windows 11 system requirements, visit `https://learn.microsoft.com/en-us/windows-hardware/design/minimum/minimum-hardware-requirements-overview`.

Checking Hardware Compatibility

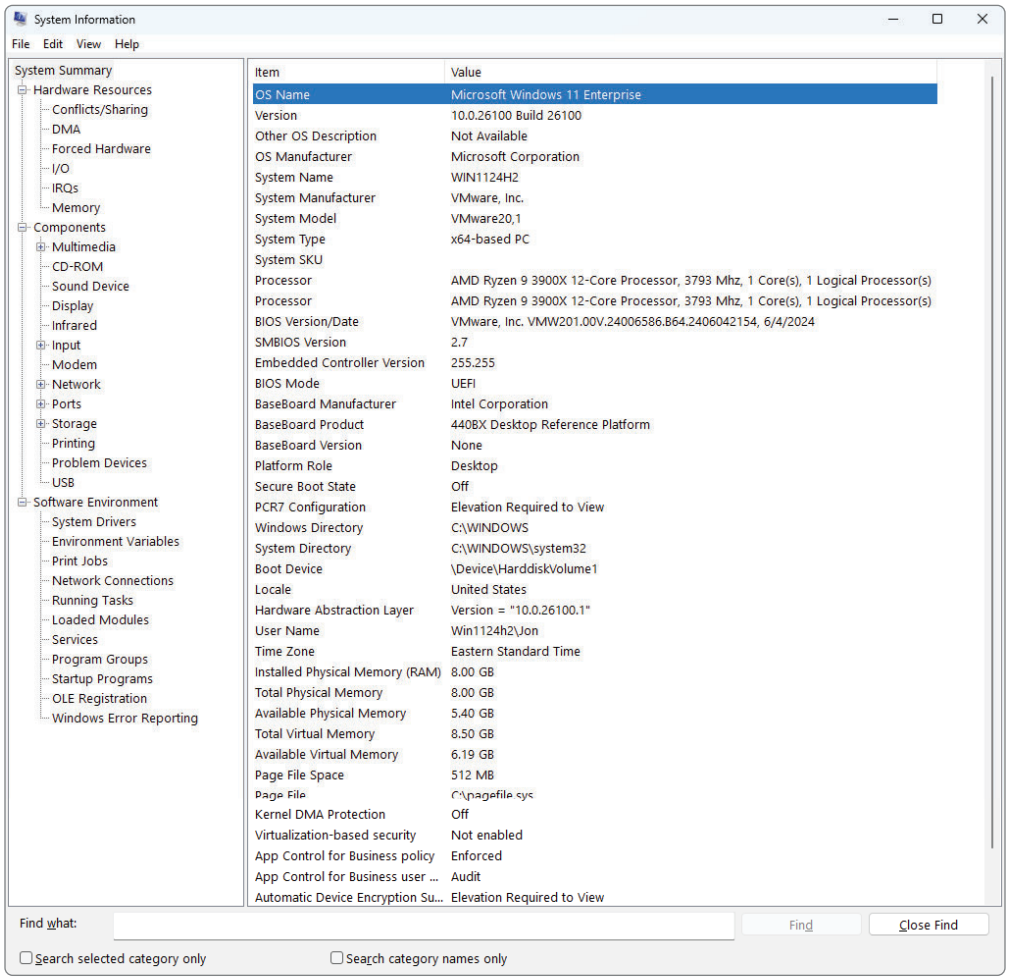
In the past, Microsoft published a hardware compatibility list (HCL) for Windows XP and prior Windows operating systems. Microsoft went away from a published list of vendors in a static document and moved to an online version, which listed not only compatible hardware but compatible applications as well. You can access the Windows Compatible Products List at `https://partner.microsoft.com/en-us/dashboard/hardware/search/cpl`. However, the best way to check is to verify with the manufacturer that drivers exist and that the hardware is compatible with Windows 11.

Sometimes you need to deploy Windows 11 to a group of computers. Several tools can be used to collect information about the hardware of the current operating system. The easiest tool to use is the System Information utility. To access this utility, you simply log in and select System Information from the Start menu. You can use System Information to investigate the processor, RAM, hard drive, and video card, as well as specifics about each of the peripherals connected (see Figure 3.1). You can also access System Information by pressing Windows key+R, typing **`msinfo32.exe`**, and clicking OK.

You can also use System Information remotely. Select View and then choose Remote Computer from the drop-down menu and select the remote computer. You can also access this utility from the command line by typing **`msinfo32.exe`**. The `msinfo32.exe` command allows remote collection of information to a flat text file using the following command:

```
msinfo32.exe /computer computername /report c:\report.txt
```

FIGURE 3.1 System Information



The `msinfo32.exe` command can also be used to output the information to its native format using the following command:

```
msinfo32.exe /computer computername /nfo c:\report.nfo
```

However, this NFO file will need to be opened with the System Information utility, which might be difficult if there are more than 10 or so machines.

An alternative to the System Information (`msinfo32.exe`) utility is the Microsoft Assessment and Planning Toolkit (MAP), which can be downloaded from the Microsoft Download Center (`www.microsoft.com/downloads`). The MAP Toolkit allows for the automated inventory collection of hardware and software from the current operating systems. The MAP Toolkit can then produce a report from the collection of data and provide the administrator with the readiness of current hardware for Windows 11. The MAP Toolkit

requires the installation of a SQL database, which is included in the installation of the MAP Toolkit. The toolkit requires a dual-core 1.5 GHz processor, 2 GB of RAM, and a minimum of 1 GB of free hard drive space.

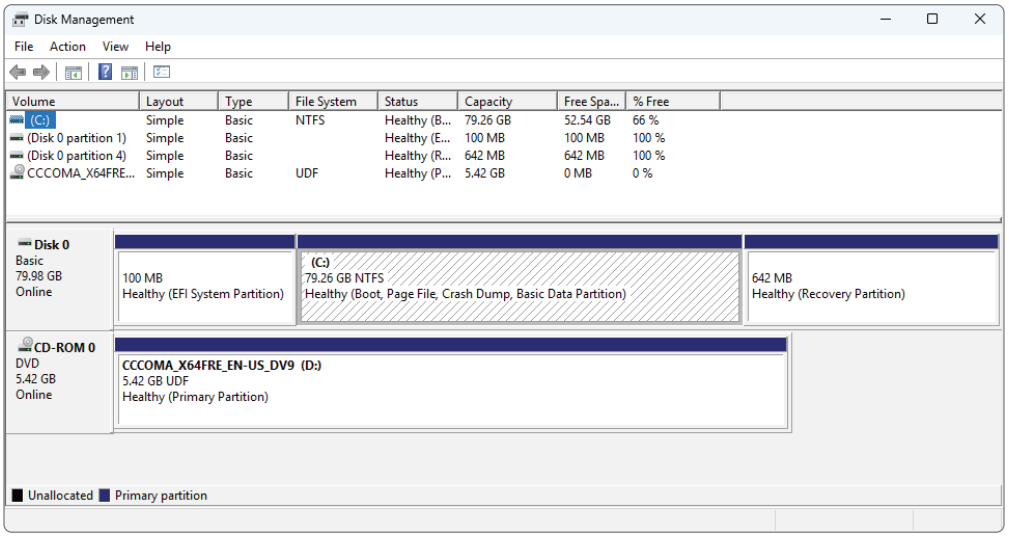
Windows Installation Options

Windows 11 can be installed as an upgrade or as a clean installation. When you choose Custom on Windows 11 23H2 or older, you can decide whether or not to format the hard disk. If you choose not to format the hard disk, the old operating system is placed in a folder called `WINDOWS.OLD`. When you choose to format the hard disk, it will erase your files, programs, and settings. You can also upgrade Windows 11 inside the existing operating system; we will cover how later in this chapter.

When installing Windows 10, you have the option to install it on *Basic Input/Output System (BIOS)*-based hardware or *Unified Extensible Firmware Interface (UEFI)*-based hardware. When installing Windows 11, you must install it on UEFI-based hardware. The hardware must support the newer standard of UEFI to install it in this fashion. UEFI hardware provides a feature called *Secure Boot*. Secure Boot operates by checking the signatures of the hardware, including the UEFI drivers (also called option ROMs), EFI applications, and, finally, the operating system. If the signatures are verified, the operating system is then given control of the boot process and the hardware. Windows 11 also requires that the hardware be Secure Boot–capable and that a TPM 2.0 module is installed.

For a UEFI installation, the partitioning of the drive will be laid out like Figure 3.2. The Recovery partition holds a bootable copy of the Windows Recovery Environment (WinRE) and is roughly 640 MB in size. The *EFI System Partition (ESP)* is a System partition used to hold the *Boot Configuration Data (BCD)* for the booting of the boot partition containing the *Windows kernel*.

FIGURE 3.2 Windows default disk layout



The Windows Recovery Environment (WinRE) is a bootable environment used by Windows for recovery of the operating system. WinRE is automatically booted when the operating system does not start properly after two consecutive boots. WinRE can also be booted into by restarting the operating system and holding down the Shift key. However, this process assumes that you can successfully boot the operating system.

The Installation Process

The elements of the installation of Windows 11 are almost identical to the installation of Windows 10 and prior versions of Windows. For that matter, it is similar to most operating systems, such as macOS or Linux. There are several common elements during the setup process that must be addressed, such as *locale* and where to install the operating system.

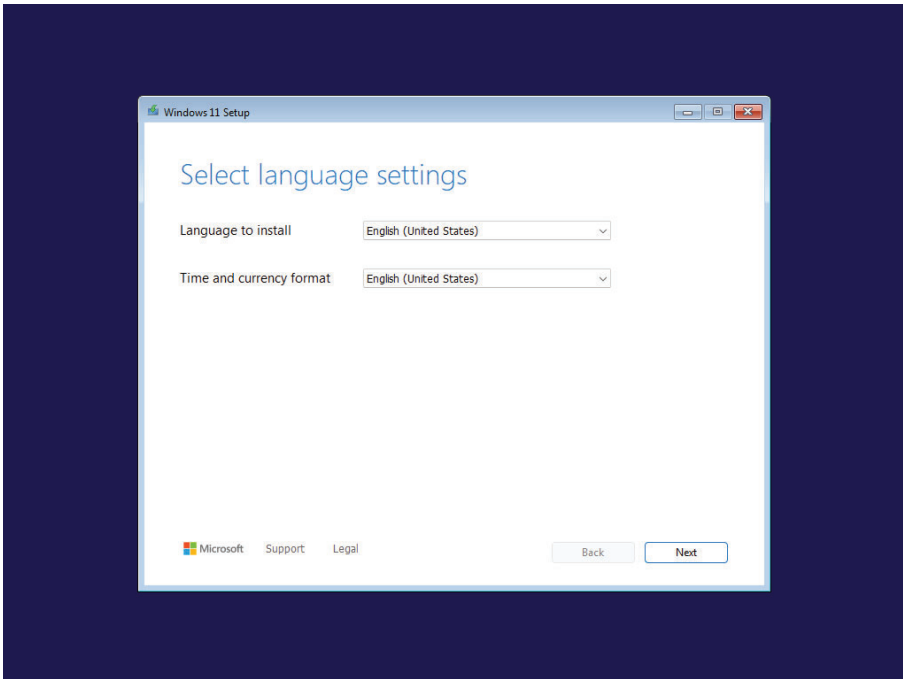
The installation of Windows 11 can be performed from a Windows 11 installation DVD-ROM. However, optical media such as DVD-ROM is rarely used because most laptops and tower computers no long include optical drives. Universal Serial Bus (USB) installation media is a preferred method with most computer vendors. Installation media can be created with the Windows 11 Media Creation Tool, which can be downloaded from `www.microsoft.com/en-us/software-download/windows11`.

Windows XP was the last operating system to support CD-ROM installation media. Starting with Windows Vista, the size of every Windows release has grown and required DVD media for the installation. In addition to using the media creation tool to create USB media, you can download ISO files. The files can then be burned to DVD or used with a virtual machine by mounting the ISO file.

The version of Windows 11 as of the writing of this book is 24H2. Prior to Windows 11 24H2, the installation media would provide the means to install or upgrade when the media was booted. As of Windows 11 24H2, if the media is booted you only have the option to perform a clean installation of Windows 11. You can still upgrade with the installation media, but you must start the upgrade from within the currently installed operating system. We will cover the upgrade process later in this chapter. The following example explains a clean installation of Windows 11 with the latest Windows 11 24H2 media:

1. Start by booting the computer and directing it to boot from the physical media. Each computer will have a different way of entering the one-time boot menu, such as pressing the F12 key or the Esc key, or sometimes by initiating it from the BIOS/UEFI.
2. After the media is selected and detected, as the media boots you may see the message "Press any key to boot from CD or DVD." This prevents the operating system from automatically booting into the installer. Once it boots, you will see the initial setup screen shown in Figure 3.3.

FIGURE 3.3 Select Language Settings screen



3. Select the language for the installation, as well as the time and currency format; these are often referred to as the *locale*. Click Next to continue.
4. On the next screen you will need to select the keyboard or input method language, as shown in Figure 3.4. Click Next to continue.
5. The next screen presents you with three options, as shown in Figure 3.5:
   - The first option is the most obvious: to install the operating system by selecting Install Windows 11.
   - The second option is to enter into recovery mode by selecting Repair Your PC.
   - The third option is less obvious: to exit the installation by pressing the X in the upper-right corner of the dialog box.

   Before you can click the Next button, you must check the option "I agree everything will be deleted including files, apps, and settings." Once you confirm by selecting the option, you can click Next.

FIGURE 3.4 Select Keyboard Settings screen

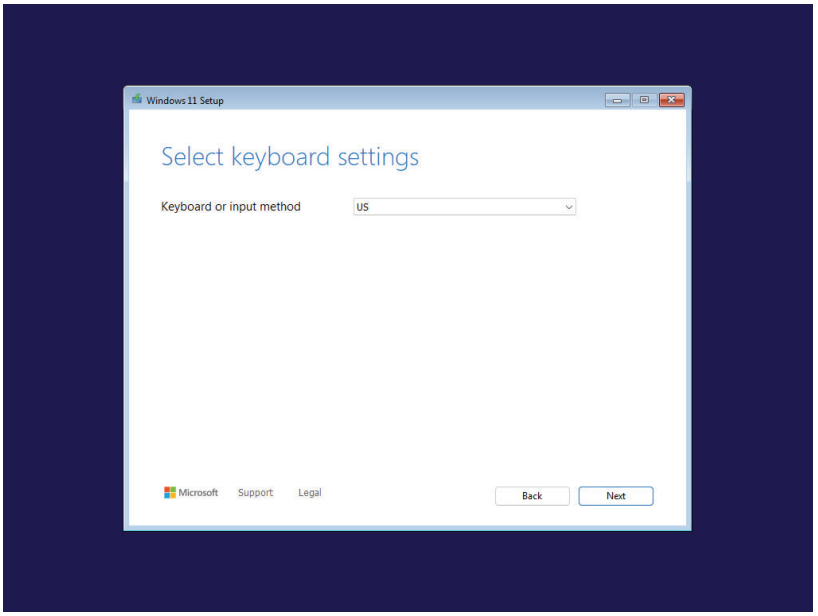
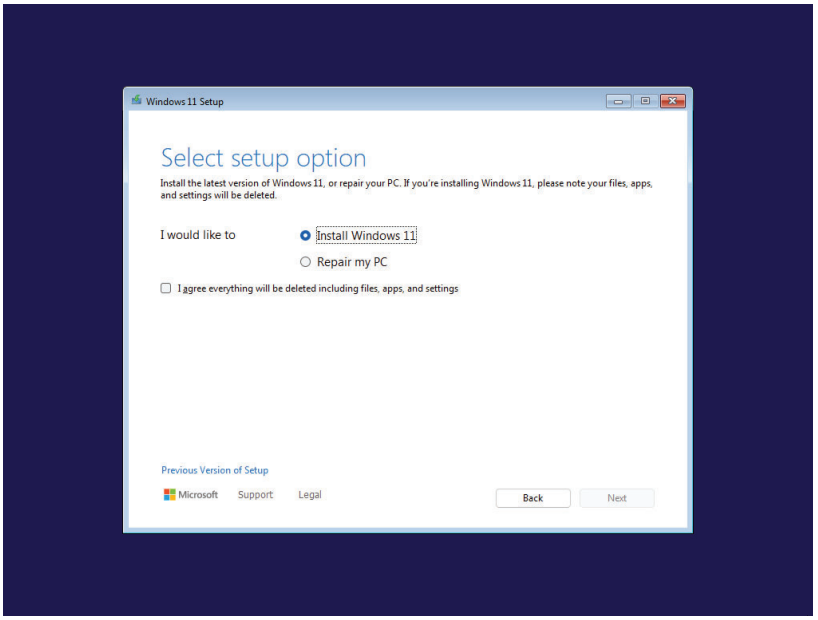
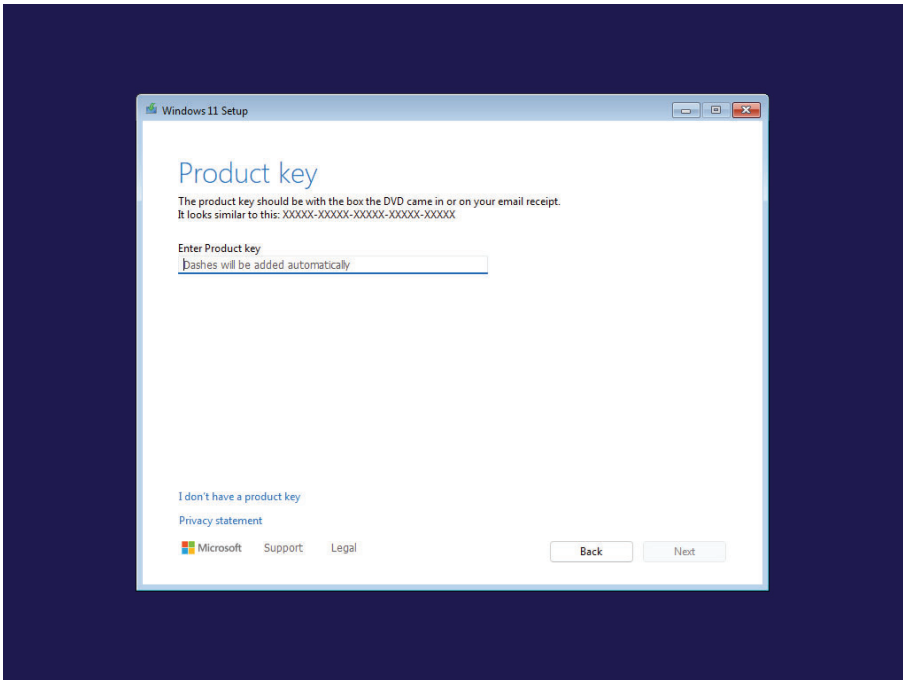


FIGURE 3.5 Select Setup Option screen



6. You will be prompted to enter the product key for the installation, as shown in Figure 3.6. You can skip the product key entry by selecting "I don't have a product key." This will allow you to bypass the product key screen and continue with the install. You will need to activate the operating system after it is installed.

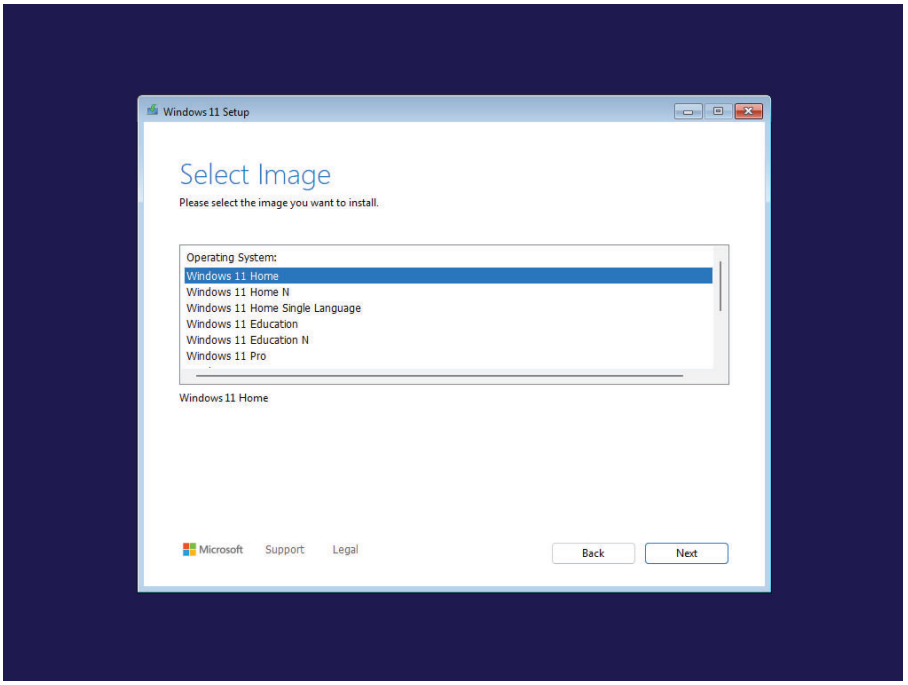
FIGURE 3.6 Product Key screen



You can install Windows 11 without the product key. However, you will need to activate Windows 11 with a valid 25-digit product key to allow personal customization, such as wallpaper and Start menu settings. Windows 11 installs into *reduced functionality mode (RFM)*. If you don't enter a product key, RFM will eventually nag you by displaying an Activate Windows message in the lower-right corner of the Desktop.

7. On the next screen, you will be asked which version of the operating system you want to install, as shown in Figure 3.7. Select to install only the operating system for which you have a valid license key for activation, since you cannot change editions later without a complete reinstallation of the operating system.

FIGURE 3.7 Windows edition selection



8. After selecting the edition, you will be prompted with the *end-user license agreement (EULA)*, as shown in Figure 3.8. To continue, read the license terms and if you agree, click Accept.

In previous versions of Windows 10 and prior to Windows 11 24H2, the next screen would prompt for the type of installation. The two options were Custom: Install Windows Only (Advanced) or Upgrade: Install Windows And Keep Files, Settings, And Applications. If you chose Custom, you could perform a clean installation of Windows, meaning that the drive would be wiped clean. Always remember that *custom* means a clean installation with Windows 11 23H2 and prior.

9. The next screen requires you to specify where you want to install the new operating system. You can delete, create, and extend partitions. If a partition exists and you are committed to a clean installation, then deleting the existing partitions is the most common task. In rare instances where no drives show up, you may need to install custom drivers specified by the vendor. You can perform that task from this screen as well by selecting Load Driver, as shown in Figure 3.9. Select the drive for the installation of Windows 11, and then click Next.

FIGURE 3.8 Windows end-user license agreement

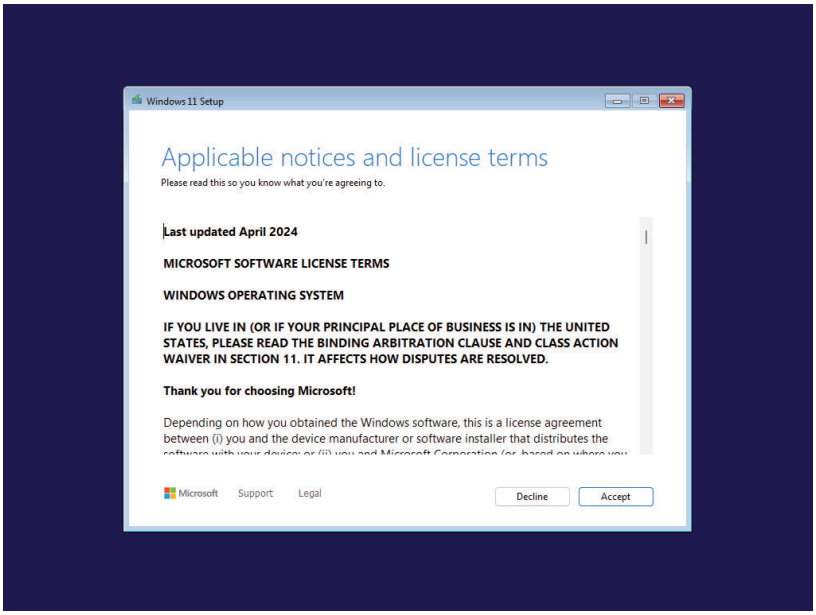
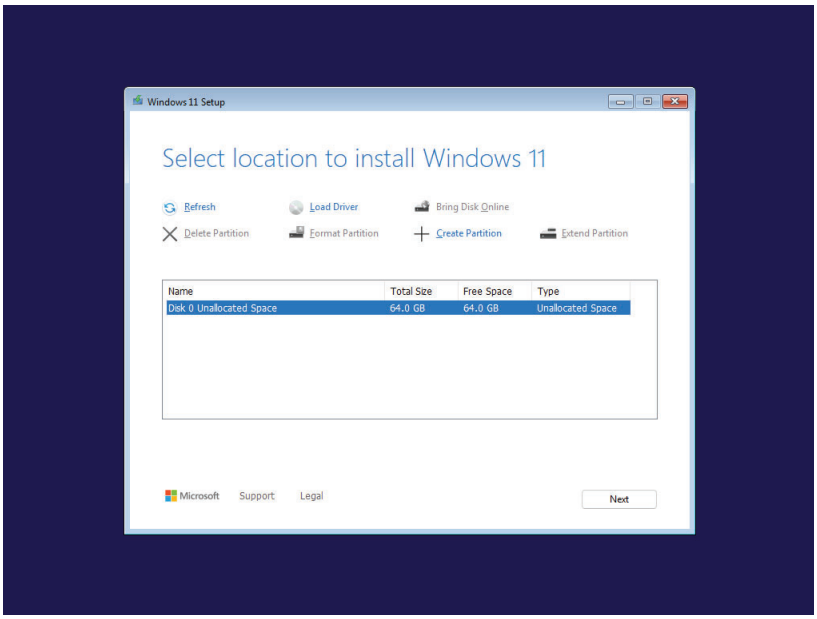


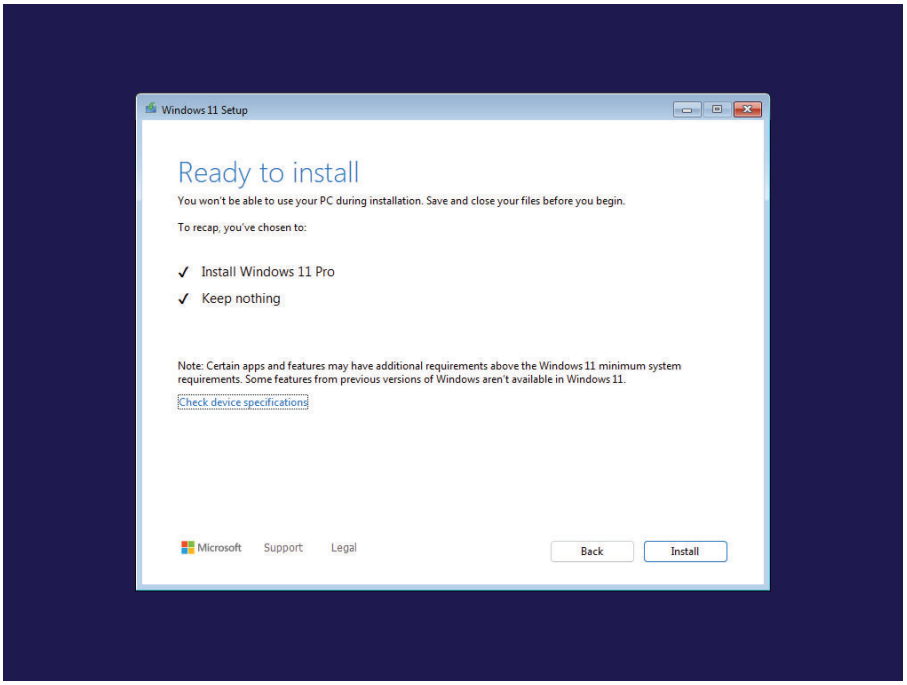
FIGURE 3.9 Windows installation partitioning



The changes made in the partition utility are immediate.

10. You will be prompted with a Ready To Install confirmation screen. This screen will confirm that if you proceed you will install Windows 11 with the selected edition. The screen will also confirm that you will keep nothing, meaning it is a clean installation, as shown in Figure 3.10.

FIGURE 3.10 Ready To Install screen

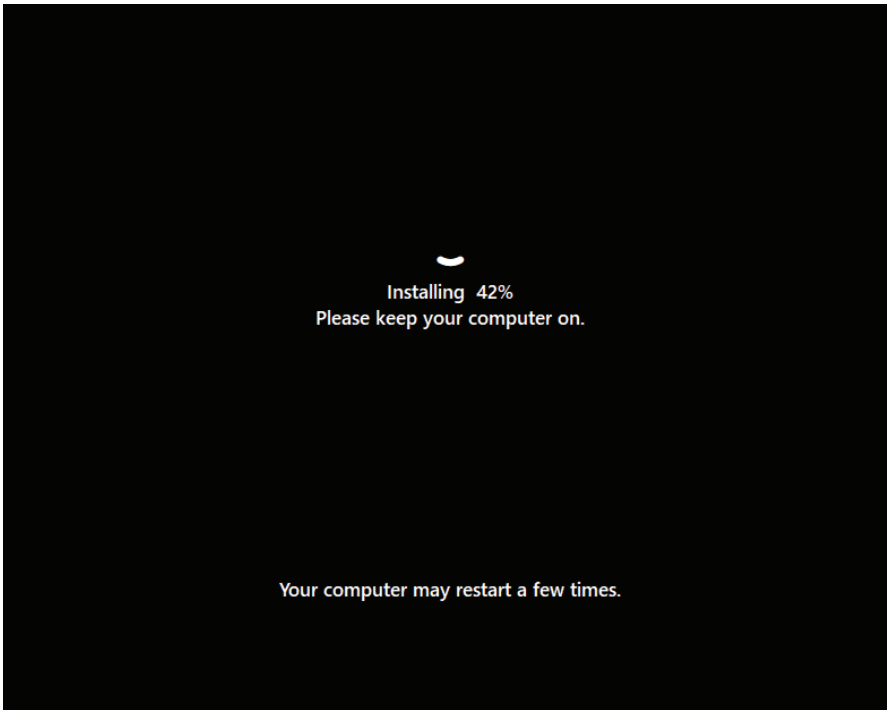


11. This step is where the filesystem is formatted, the boot files are copied, and the operating system's files are applied to the disk. You will only see a progress counter, as shown in Figure 3.11.

    When this process completes, the files for the operating system are on the hard drive but the operating system is not generalized to your computer; that happens in the next boot. After the computer reboots, it will detect hardware and run through what is called a *generalize pass*. The screen will show an installation progress percentage, as shown in Figure 3.12. During this process the various device drivers are being detected. This is the point where the operating system is adjusting itself to the computer hardware.

FIGURE 3.11 Windows installation progress

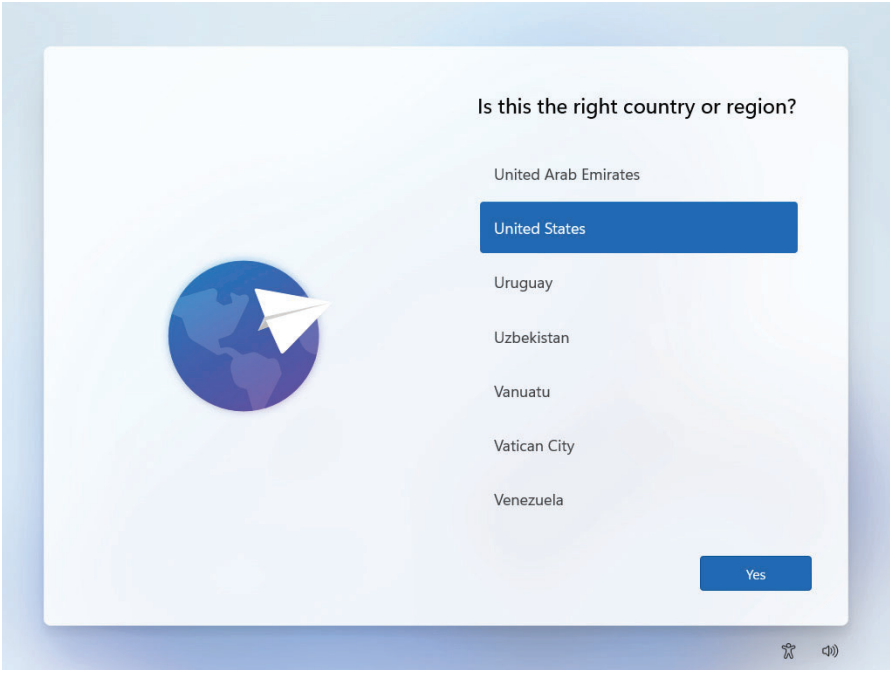
FIGURE 3.12 Windows Installing screen



Once the drivers are detected, the operating system will reboot again. During this boot, the drivers detected in the prior stage will be instantiated and the *specialize pass* will begin. In this pass, the locale (region) of the operating system will be chosen, as well as the keyboard layout. You can even choose a second keyboard layout, if you're switching out a keyboard.

12. Confirm the region by clicking Yes, as shown in Figure 3.13.

FIGURE 3.13 Operating system locale (region) setting



13. Confirm the keyboard layout by clicking Yes, as shown in Figure 3.14. It will also ask you if you want to add a second keyboard layout; just click Skip.

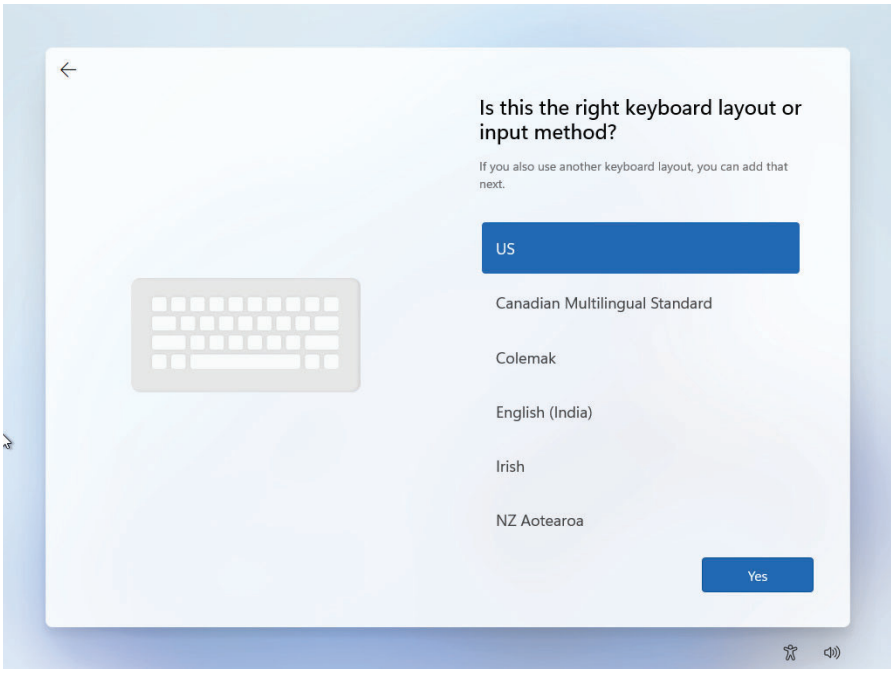
    The Windows installation will display a message that it is checking for updates after the selection of the locale, keyboard layout, and additional keyboard layout question. The operating system will do some background work. The screen will assure you that something is going on in the background, and the message will change periodically. Then the operating system will reboot again.

During the check for updates, the operating system will make a query to Microsoft Intune to check if the operating system's unique hash is contained in its global catalog. If the hash is in the global catalog, then

the respective organization will be queried for the Autopilot configuration. Windows Autopilot is a suite of cloud-based technologies designed to simplify the deployment and management of new Windows devices. This process eliminates the need for custom images and extensive setup, reducing the time and infrastructure required for device management. For more detailed information on Autopilot please refer to:

`https://learn.microsoft.com/en-us/autopilot/overview`

FIGURE 3.14 Operating system keyboard layout setting



14. When Windows 11 boots up, you will be given an opportunity to name your device during the setup, as shown in Figure 3.15. This is different from older operating systems, such as Windows 10, where a random name was created for you.

    The operating system will seem to reboot after you confirm the name of the system. You will then see a Just A Moment screen while the operating system boots, as shown in Figure 3.16.

15. The next set of screens will help you set up the first user account as well as assign the administrator of the operating system.

FIGURE 3.15 Operating system name

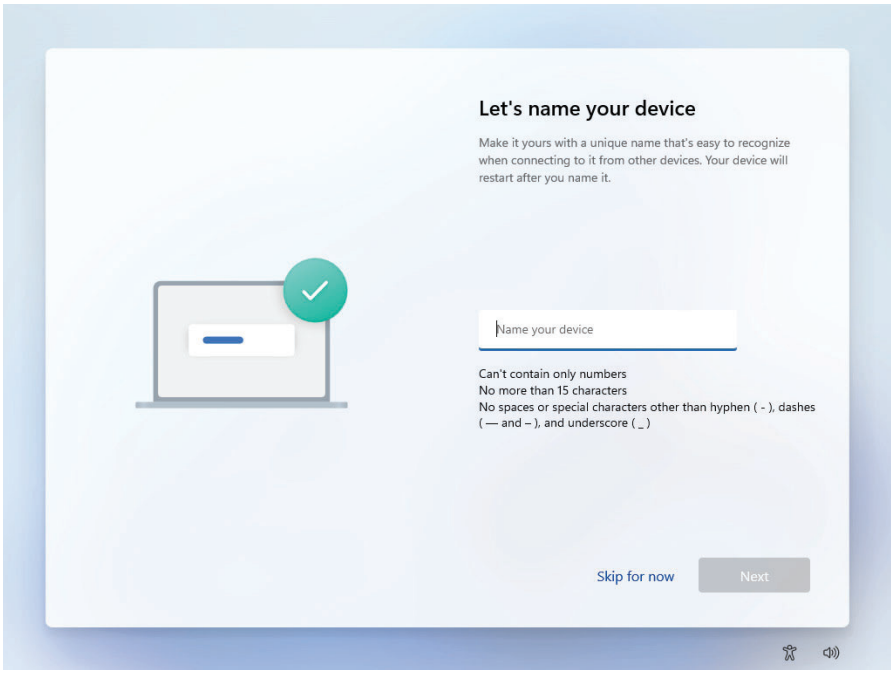
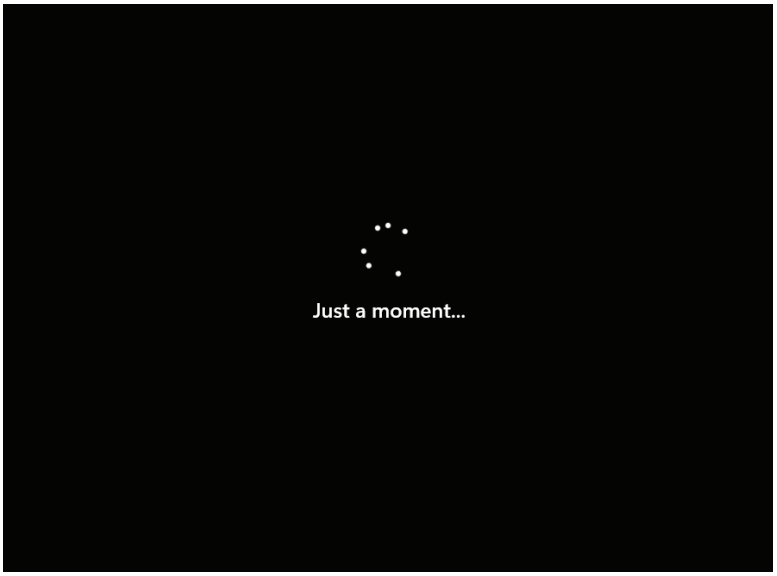
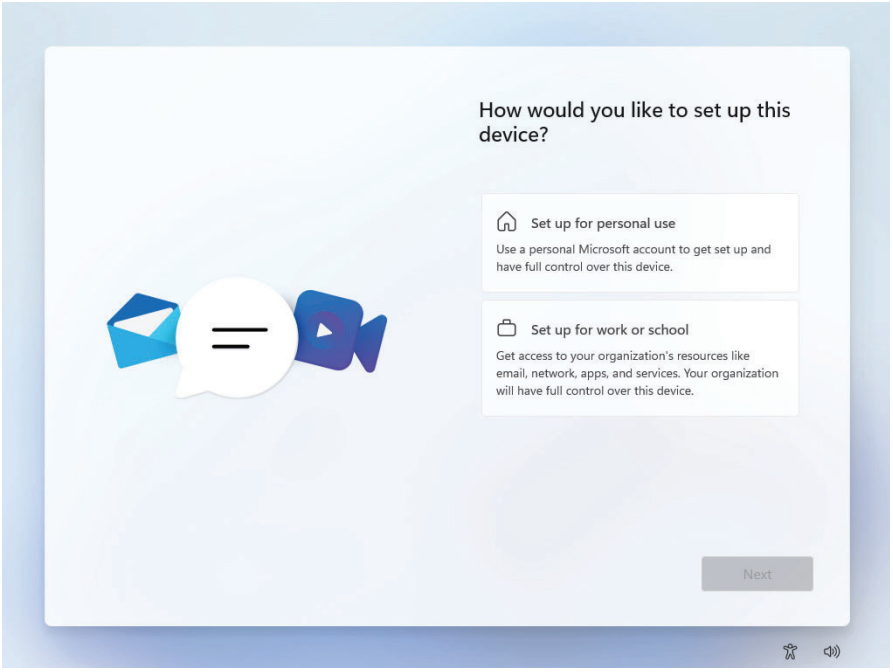


FIGURE 3.16 Just A Moment screen



This is considered the OOBE pass, or the *out-of-box experience pass*, where the first account is set up for the operating system. You have the choice of logging in with a Microsoft account by choosing Set Up For Personal Use. This is great if you have the account combined with a Microsoft 365 subscription, since the licensing of the Office products are all streamlined when performing a login of this nature. However, if you want to join the computer to an organization's Intune mobile device management (MDM) service, you can select Set Up For Work Or School, as shown in Figure 3.17.

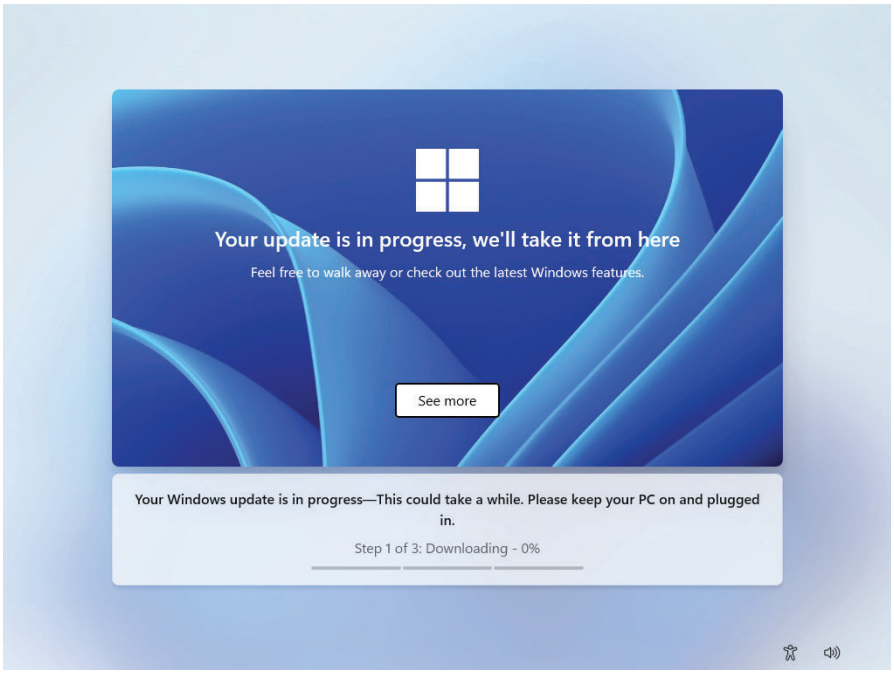
FIGURE 3.17 Windows account options



If you click Set Up For Personal Use, Windows 11 will start downloading Windows Updates and installing the updates, as shown in Figure 3.18. This ensures that your operating system is up to date when you first log in. Once updates are installed, you will be asked to sign in with a Microsoft account so that your apps, files, and services will sync to the device.

If you choose Set Up For Work Or School, you'll be asked to sign in with a work account or school account that is attached to an Intune service. This will give control of the device over to the organization, where it will be controlled by the MDM policies in Intune. Windows updates will not automatically install like the Set Up For Personal Use option, since the update policy will be decided for you by your organization.

FIGURE 3.18 Windows Updates installation



By choosing the Set Up For Work Or School option, you can choose to join the operating system to a domain. This will allow you to create a local admin account on the operating system.

We will install using a Microsoft account, so select the Set Up For Personal Use option, and click Next.

16. When the operating system is done installing Windows Updates, you will be presented with the Let's Add Your Microsoft Account dialog box, as shown in Figure 3.19. Enter your Microsoft account and password in the screens that follow, and select Next.
17. On the following screen you'll be asked to set a personal identification number (PIN). This allows you to easily log into the operating system. Click Create Pin. On the following screen create a new PIN, then type it again to confirm it, as shown in Figure 3.20, and click OK.

All new sign-in requests to the operating system can be performed with your PIN.

FIGURE 3.19 Microsoft account options for personal use

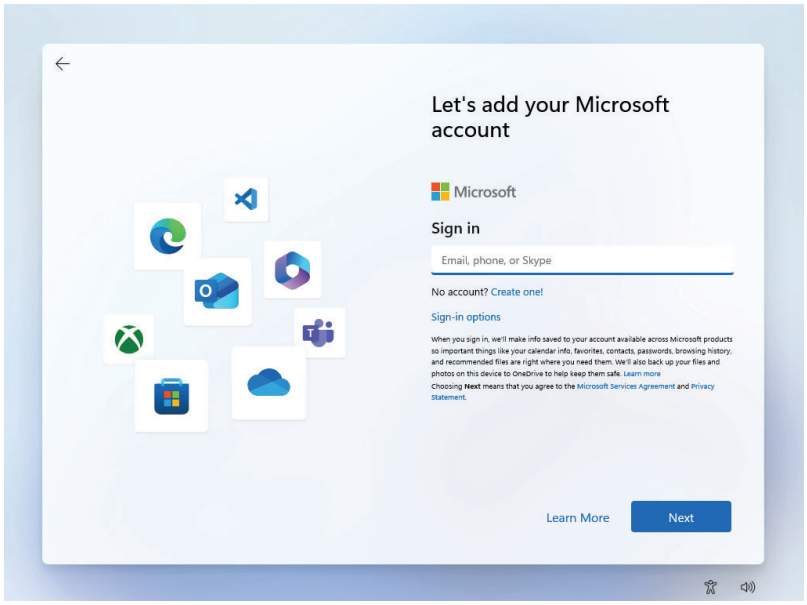
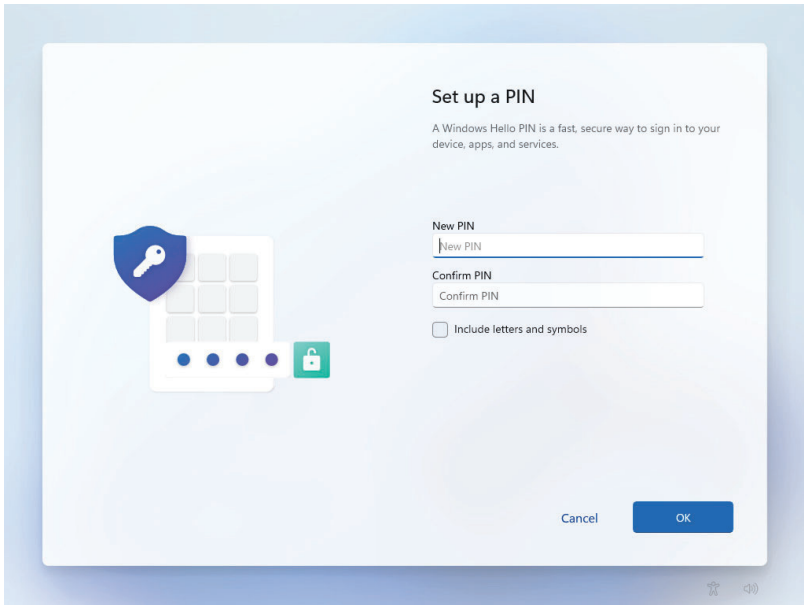
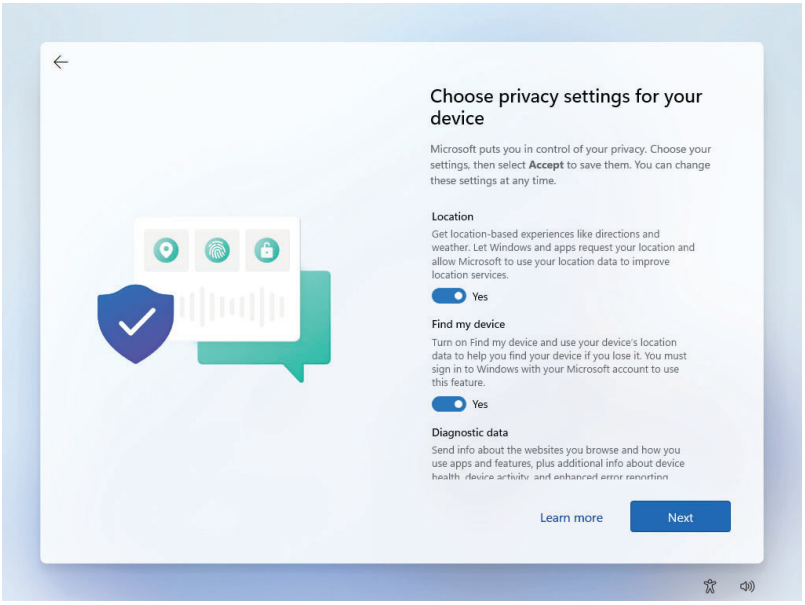


FIGURE 3.20 Set Up A PIN



18. 18.The next screen asks you to Choose Privacy Settings For Your Device. This screen is known as *telemetry* options, as shown in Figure 3.21. Choose the level of privacy on the device as well as features like Location, Find My Device, Diagnostic Data, and other features that transmit to Microsoft's cloud. Once you have made your selection, click Accept to continue.

FIGURE 3.21 Choose Privacy Settings For Your device



19. If you have previously logged into another operating system with the Microsoft login, you will be presented with a Welcome Back screen, as shown in Figure 3.22. You can choose to Restore From This PC, or you can select More Options, then select Set Up As A New PC.

    There are a number of advertisements to connect your phone, Microsoft 365 offers, OneDrive cloud storage offers, as well as informational advertisements. You can choose to click Decline to process or Continue to learn more. In addition, there are ads for Game Pass that you can choose to Skip For Now or Join. These ads will rotate as new versions of Windows are released, but the Microsoft 365, OneDrive, and Game Pass subscriptions will most likely remain.

20. Windows will display the Getting Things Ready For You screen while it configures the operating system. You will then be prompted with the Let's Customize Your Experience screen, as shown in Figure 3.23. You can choose how you will use the operating system, and a tailored experience will be provided. For example, you can choose development, gaming, entertainment, just to name a few. Choosing these will deliver ads, content, and tips to the operating system. You can always switch at a later time by selecting Start menu ➢ Settings ➢ Personalization ➢ Device Usage. You can choose Skip to continue.

FIGURE 3.22 Restore From This PC screen

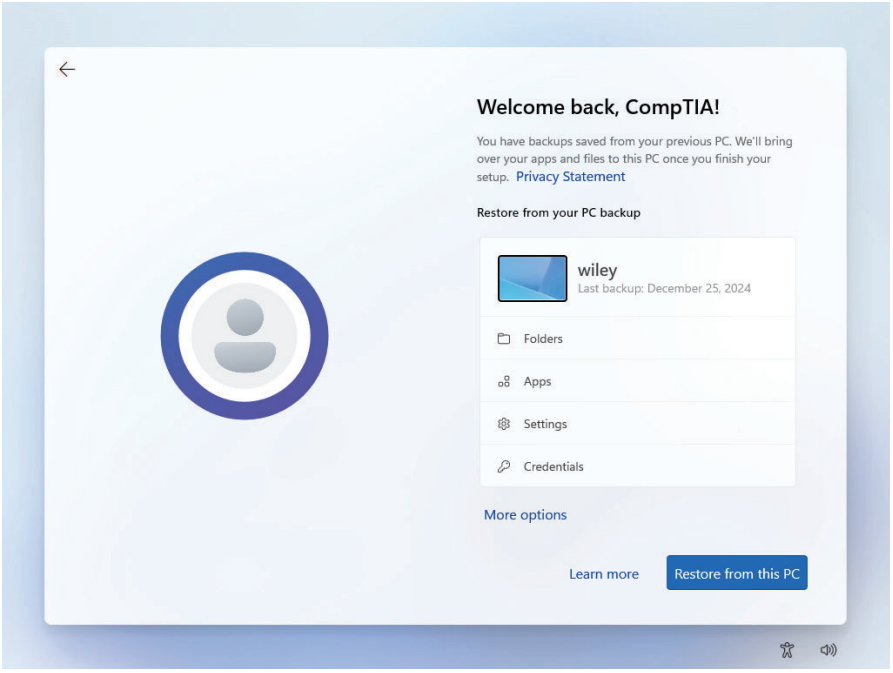
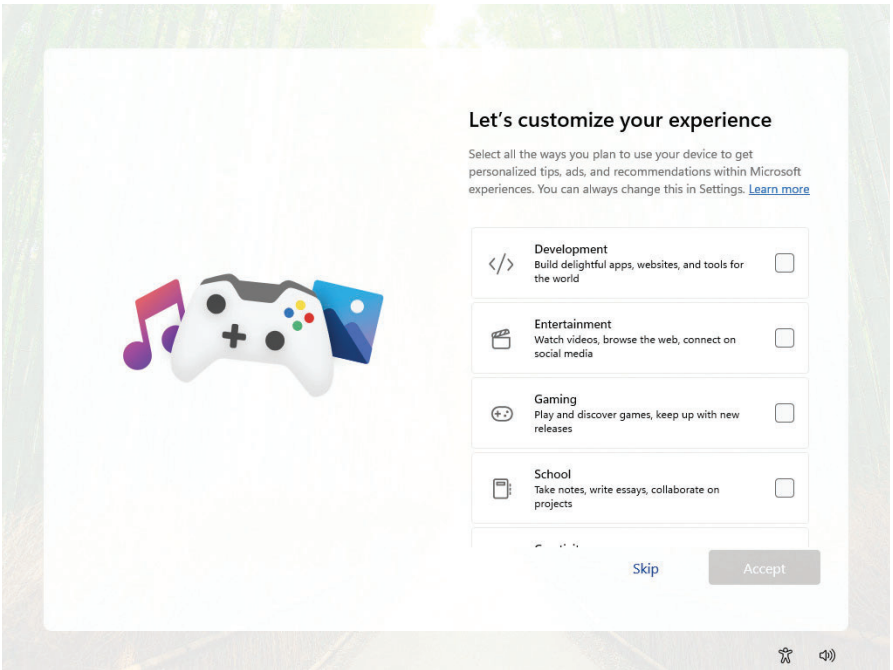
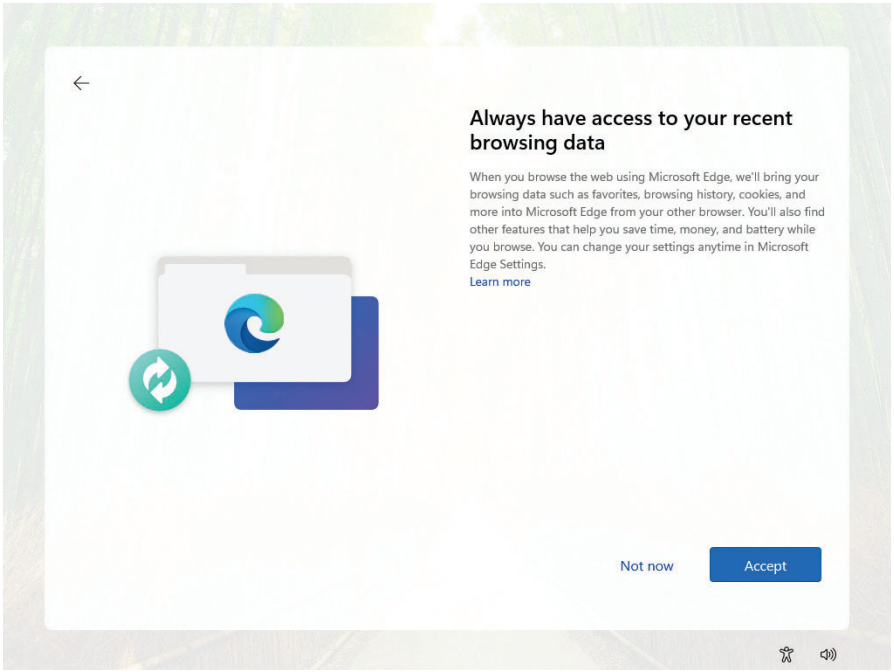


FIGURE 3.23 Let's Customize Your Experience



21. The next screen will ask you to opt into synchronizing your browsing data. The Always Have Access To Your Recent Browsing Data screen will allow you to opt into Microsoft Edge synchronization, as shown in Figure 3.24. Click Accept to continue.

FIGURE 3.24 Browsing data synchronization opt-in



22. The operating system will briefly state Finishing Up. Then you will be logged into the operating system. You will be greeted with a web page in Edge welcoming you to Windows, as shown in Figure 3.25.

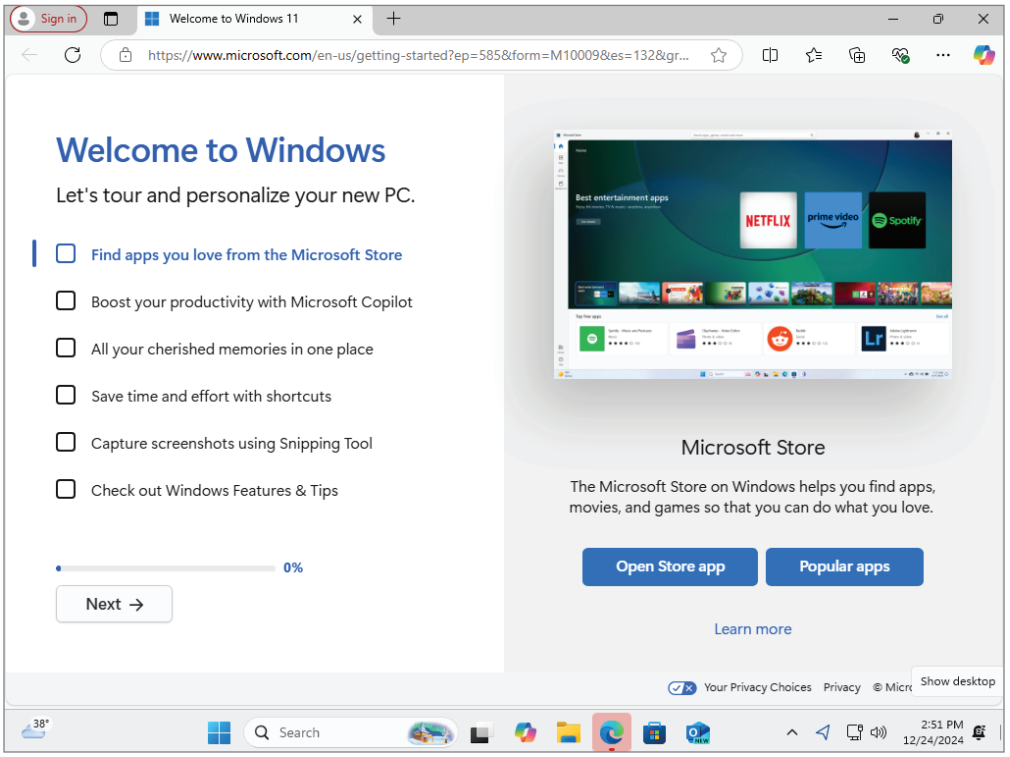
Post-Installation Setup

After installation of the operating system, you have a few post-installation tasks. In this section we will cover all of the post-installation tasks that you should complete.

Time Zone Adjustment

At this point, you are pretty much done. The next screen you will see is the Windows 11 Desktop. However, throughout the entire setup, you were never asked for your time zone. The time zone is automatically calculated based on your IP address. If it is incorrect, then the following procedure will manually adjust it.

FIGURE 3.25 Welcome To Windows



To manually adjust the time, perform the following steps:

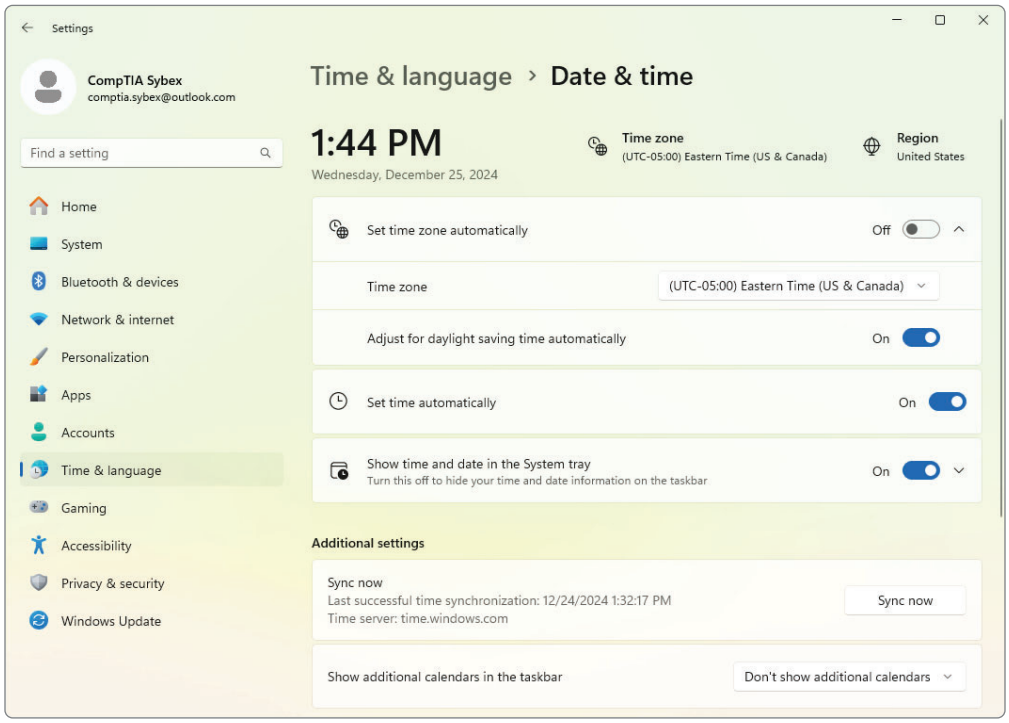
1. Right-click the clock in the lower-right corner of the Desktop.
2. Select Adjust Date And Time.

   This will open the Date & Time screen, as shown in Figure 3.26.
3. Turn off Set Time Automatically if you want to change the time. If you want to change the time zone, turn off Set Time Zone Automatically if it is turned on.

Sign In To OneDrive

One of the key features of using a Microsoft account is the ability to back up your files with OneDrive. Before you can use this feature, you will need to log in. You can begin this process by clicking the gray cloud icon with the line through it, located in the system tray, as shown in Figure 3.27.

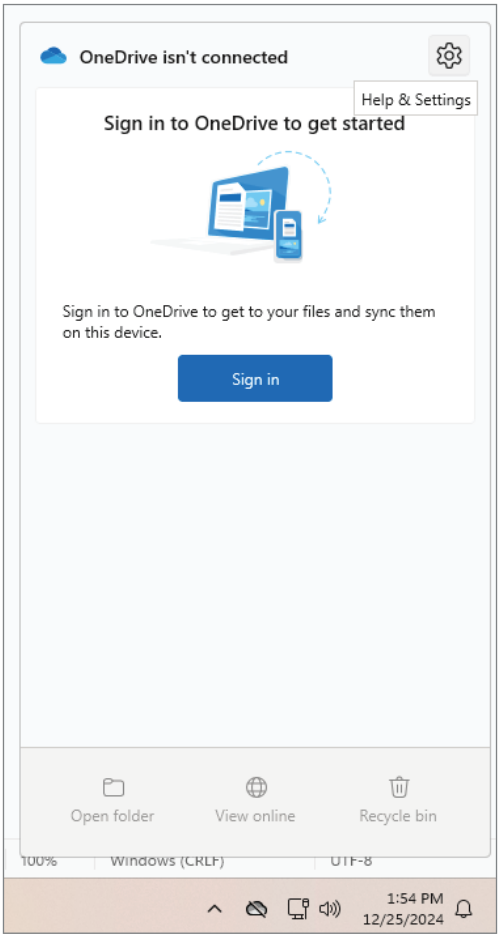
FIGURE 3.26 Adjusting Time Zone



1. Begin by clicking Sign In.
2. In the Set Up OneDrive dialog box, enter your email address that was set up with a Microsoft account. Then click Sign In.
3. Acknowledge the Privacy assurance screen by clicking Next.
4. Choose to send optional diagnostic usage data, then click Accept.
5. Confirm or change your OneDrive folder location, then click Next.
6. Confirm which folders OneDrive will back up, as shown in Figure 3.28. Then click Start Backup.
7. Choose to Upgrade your account for more storage, or select Not Now.
8. You can click Next through the tutorial, there are a few screens you will be presented with.

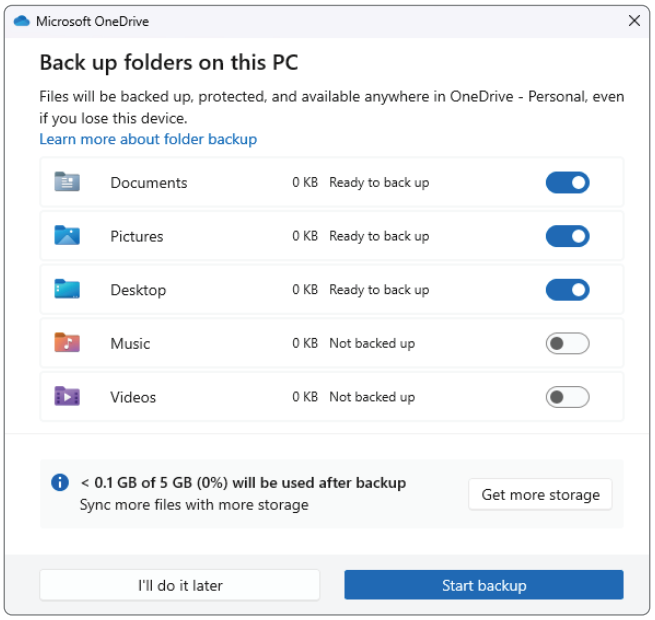
9. Click Later on the offer to Get The Mobile App.
10. Click Open My OneDrive Folder to exit the sign-in process.

FIGURE 3.27 Sign In To OneDrive



You will now notice that the line through the cloud has disappeared. Keep in mind that you will be limited to the free 5 GBs of space all Microsoft accounts are allotted. If you need more storage, you will need to purchase a Microsoft 365 account or OneDrive storage. More information can be found at `www.microsoft.com/en-us/microsoft-365/onedrive/compare-onedrive-plans`.

FIGURE 3.28 Back Up Folders On This PC



Sign In To Edge

When you launch Edge for the first time you will be asked to sign in. This allows you to sync favorites and browsing data. You can begin the process by clicking Sign In To Sync Data in the upper-left corner of the Edge browser, as shown in Figure 3.29.

If you have already signed in to OneDrive, the operating system will detect your sign-in and sign you into Edge automatically.

The Upgrade Process

Now let's take a look at the in-place upgrade process for a Windows 10 operating system to Windows 11. It is recommended that you start by performing all of the current Windows Updates first. During Windows Update, you may see an ad to upgrade to Windows 11, as shown in Figure 3.30. You can begin the upgrade from this ad by selecting Download And Install.

The tried and true method of upgrading Windows is by inserting the Windows 11 media or connecting to a network share that contains the media and launching `setup.exe`. This will require answering a User Account Control (UAC) prompt. When the setup process starts, it will give you the option to change how Windows Setup downloads updates, as shown in Figure 3.31, or you can just click Next at this point. The default is to proceed with the download of Windows Updates for the installation.

FIGURE 3.29 Signing in to Edge

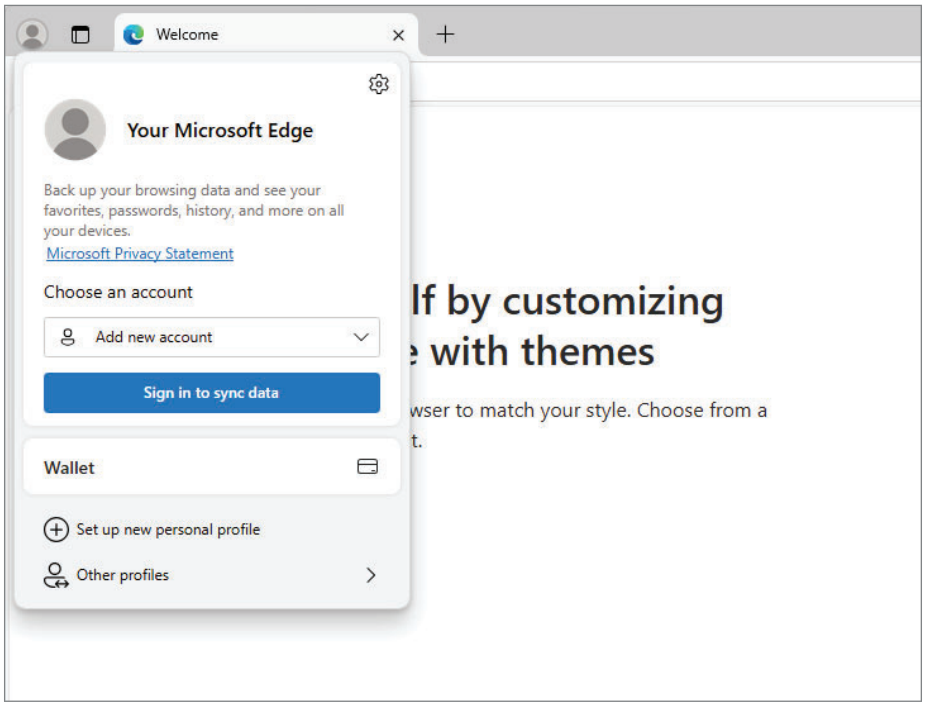


FIGURE 3.30 Download And Install Windows 11

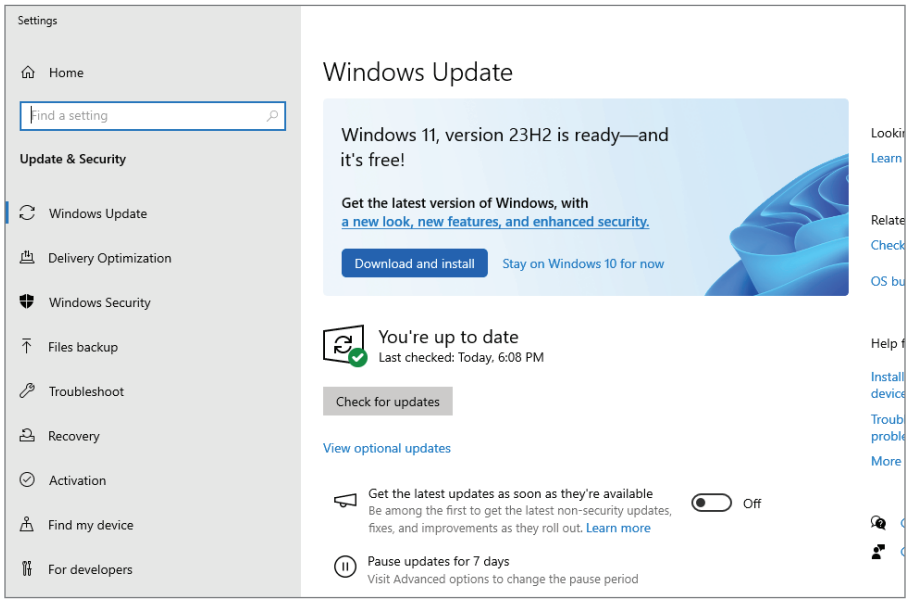
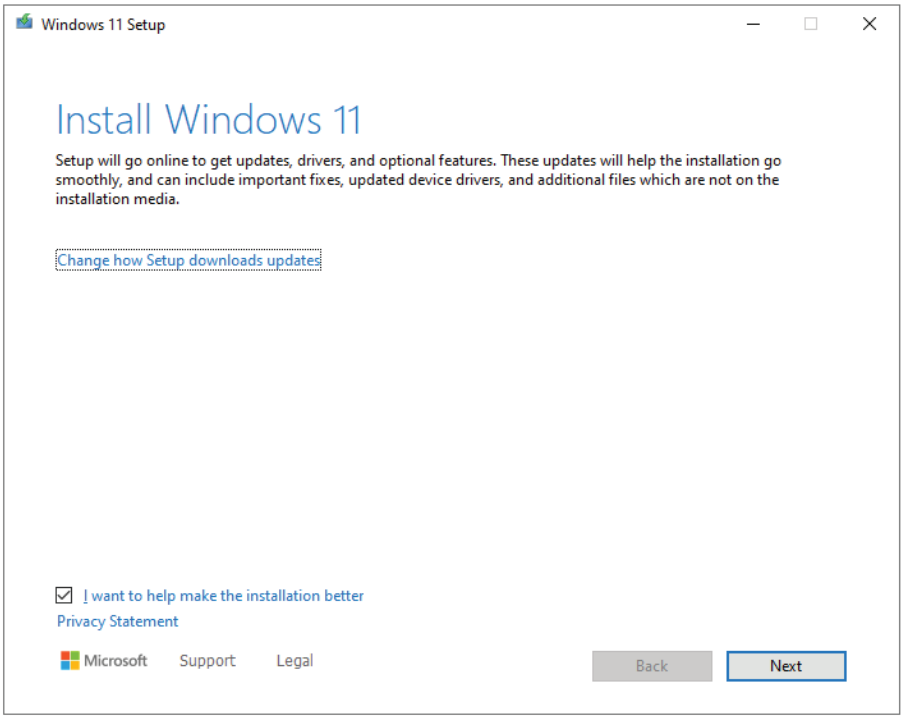


FIGURE 3.31 Install Windows 11 screen



Before proceeding, you must accept the end-user license agreement (EULA), also known as the license terms, as shown in Figure 3.32.

The installer will check and download updates necessary for the installation before continuing, as shown in Figure 3.33. This will ensure that you are secure and the upgrade process is smooth and without complication.

The next screen is the Ready To Install screen, shown in Figure 3.34. This confirms the edition detected and also confirms that personal files and apps will be kept. You have the option to change what is kept during the upgrade process by clicking Change What To Keep. You can choose one of these options: Keep Personal Files And Apps, Keep Personal Files Only, or Nothing.

The upgrade process will begin, and you will see the familiar progress percentage in the upper-left corner of the screen, as shown in Figure 3.35. The computer will reboot, and Setup will continue with a different progress screen, as shown in Figure 3.36. The computer will reboot several times during this process. As mentioned earlier, device drivers will be detected, and a reboot is required for the drivers to be properly loaded and instantiated.

FIGURE 3.32 Windows 11 end-user license agreement

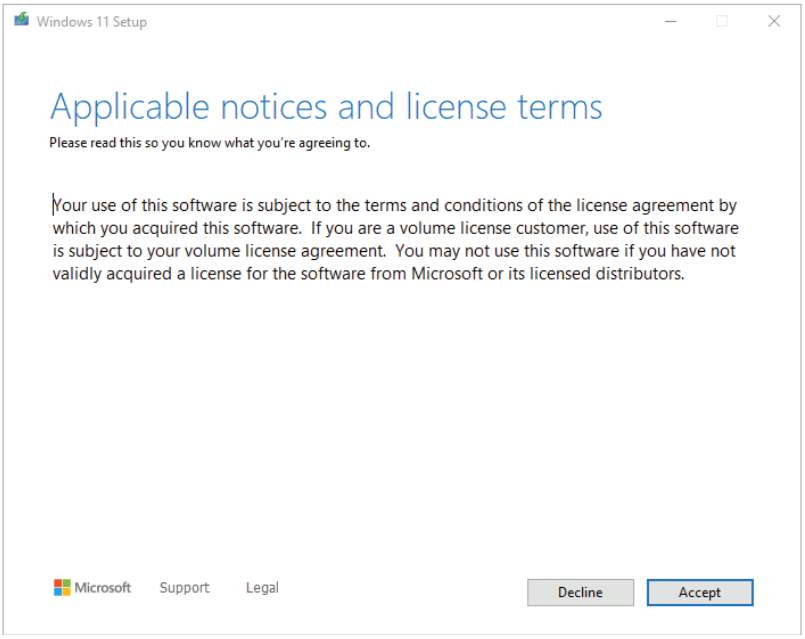


FIGURE 3.33 Windows 11 updates check

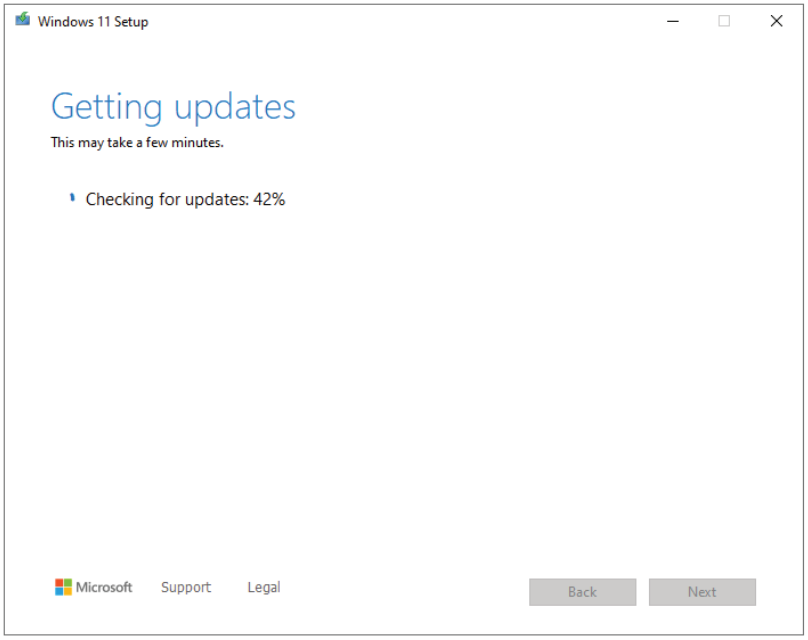


FIGURE 3.34 Windows 11 Ready To Install screen

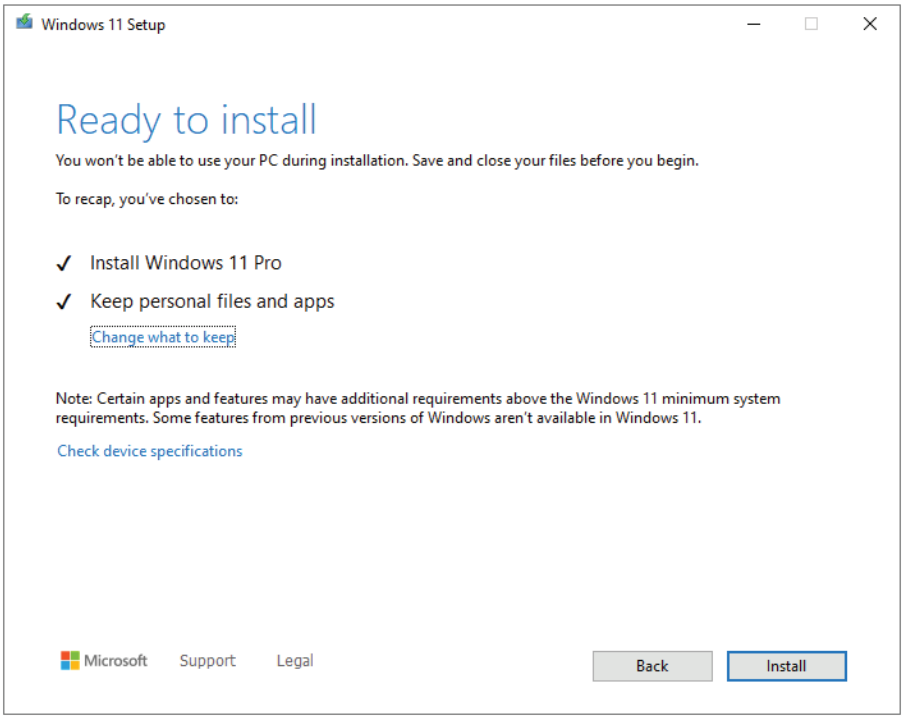
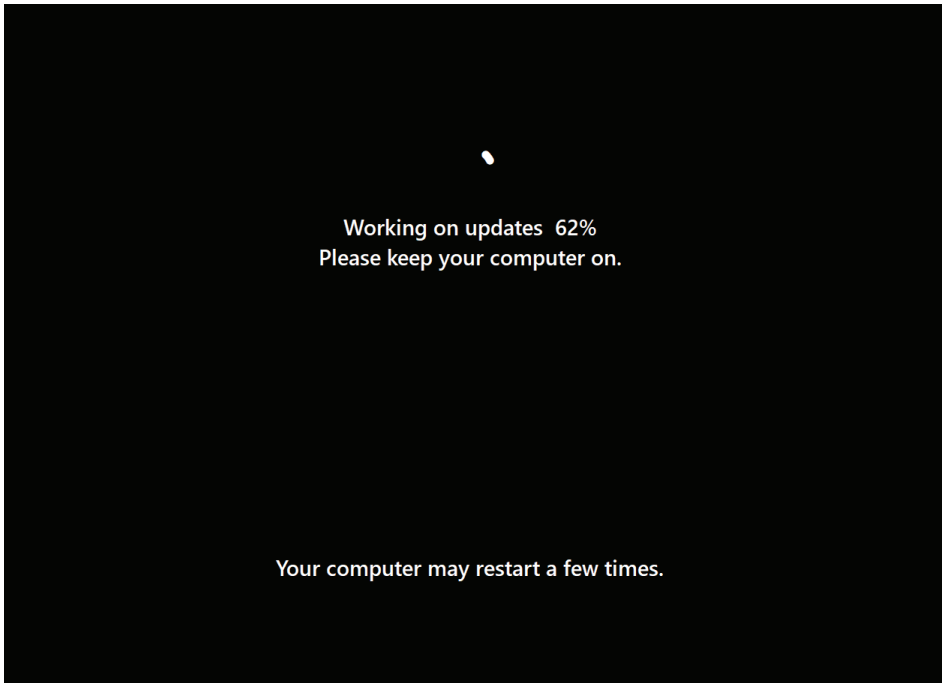


FIGURE 3.35 Installing Windows 11 upgrade percentage

FIGURE 3.36 Windows 11 upgrade percentage after reboot



The Windows 11 upgrade process has omitted the possibility of choosing the wrong edition of Windows 11 by automatically detecting the currently installed edition. The Windows 11 upgrade process has also omitted the telemetry settings screen, inheriting the settings from the original install.

You have 10 days to revert to the prior versions of Windows after the upgrade. The previous operating system files are placed in the `C:\WINDOWS.OLD` folder. To roll back, click Start menu ➢ Settings ➢ System ➢ Recovery ➢ Go Back, and you can revert back to Windows 10.

Repair Installation

A *repair installation* is used when you want to reinstall the operating system without losing personal data files, application settings, or applications you've installed. The installation is similar to an upgrade, as described in the previous section, except that Windows 11 will detect that it is installed already. You will be presented with the default option of Keep Personal Files And Apps. However, you can choose one of these options: Keep Personal Files And Apps, Keep Personal Files Only, or Nothing. The setup process will then reinstall the OS without affecting your personal files, applications, and their corresponding settings. It will, however, reinstall the operating system files, so it is considered a *repair* installation.

Another option for reinstalling Windows 11 is to reset the operating system with the Reset This PC option. This option is used to reset the operating system back to its original state. It provides another way to fix the operating system when it looks to be corrupted. This method should be used as a last resort. It does allow you the choice to keep personal files or completely erase them along with the operating system. On Windows 11, you can reset the operating system with the Reset This PC option by clicking Start menu ➢ Settings ➢ System ➢ Recovery, and then choosing Reset This PC.

Recovery Partition

Some vendors will supply a recovery partition that contains the original image for the OS the system came installed with. In the event the operating system is corrupted, the recovery partition can be used to reimage the device back to the factory image. When vendors started to include the recovery partition, the image could be restored by booting the computer into the recovery partition, and a third-party utility would be used to reimage the device. However, with Windows 11 the use of third-party utilities is no longer needed—Windows has a provision for locally installing the image via WinRE.

Side-by-Side Upgrading Windows 10/11

Side-by-side upgrades require additional hardware, since the original operating system will not be modified during the upgrade. The new hardware will be the hardware that is installed from scratch with Windows 11. The user settings and data will then be migrated over from the previous operating system to the new one.

Side-by-side upgrades are the best way to upgrade when you need to upgrade a system but the current system is in use. When dealing with one system, side-by-side upgrades are great because you don't have to worry about backups of the OS. However, applications must be reinstalled, so this approach does have its disadvantages. When multiple systems require upgrading, a *rolling upgrade* can be performed. A rolling upgrade is a variation on the side-by-side upgrade that allows the decommissioned equipment to be used to upgrade the next user, and this creates a continuous upgrade cycle. In other words, as you complete the upgrade of one user, their old system becomes the next user's new system, and the process continues until you get to the last user.

There are a few ways of migrating the user data from the old device to the new device, depending on how your deployment is configured. If the devices are using OneDrive to back up users' data folders, then users can simply sync their data onto their new devices. This method will work for data files and Microsoft Store apps, but OneDrive will not back up and restore Win32 applications.

If you are not using OneDrive, then you can migrate user data with Windows 11 using the Microsoft Windows User State Migration Tool (USMT). USMT allows you to migrate user file settings related to the applications, Desktop configuration, and accounts from one computer to another computer. The migration can be performed via a network connection or a hard drive.

USMT is compatible with Windows 10 and Windows 11, but it is currently available only as part of the Windows 11 Assessment and Deployment Kit (ADK). Although USMT is not part of the CompTIA objectives for the 220-1202 exam, it is important to know that a tool like USMT exists, because the migration of user information is a component of an upgrade.

Additional information about USMT can be found here:

`https://learn.microsoft.com/en-us/windows/deployment/usmt/usmt-overview`

Using USMT requires an investment of time. If you are performing only a few side-by-side upgrades, you can copy the user's profile directory manually. Then, when the new computer is ready, you can copy over the various folders, such as videos, pictures, and documents.

Image Deployment

Upgrading operating systems takes time and complex steps to complete, such as running Windows Upgrade, installing updates, upgrading the OS, and installing programs. These steps can take a tremendous amount of time, but if you have fewer than a dozen devices to upgrade, then you could manually upgrade the devices. However, if you have more than a dozen, there are alternatives such as image deployment.

The image deployment process is simple in concept. It starts with a base operating system on a reference computer utilizing real hardware or a virtual machine. The software is then installed on the operating system, an answer file is created using the Windows 11 Assessment and Deployment Kit (ADK), and the Sysprep tool is run on the operating system. The last step is to use a tool like Windows Deployment Services (WDS) to create an image of the operating system. Once the image is captured, it can be deployed to the other computers using the Windows Preinstallation Environment (WinPE).

Zero-Touch Deployment

Zero-touch deployments represent a modern approach to device management, automating the setup and configuration of new devices with minimal manual intervention. Two popular tools for zero-touch deployments are Microsoft Deployment Toolkit (MDT) and Windows Autopilot.

On-Premises Deployments

MDT is an on-premises solution that provides extensive customization and control over the deployment process, making it ideal for organizations with specific and complex deployment needs. Zero-touch deployments using MDT are designed to automate the deployment of Windows operating systems with minimal manual intervention. This process is often integrated with Microsoft Endpoint Configuration Manager (MECM) to enhance its capabilities. In a zero-touch deployment, MDT and MECM work together

to create a seamless deployment experience. The process begins with the creation of a deployment network share inside MDT, where all necessary files, applications, and drivers are stored. These resources are then packaged into a task sequence, which is a series of steps that automate the installation and configuration of the operating system on target devices.

One of the key components of zero-touch deployments is the use of task sequences. These sequences are highly customizable and can include steps such as partitioning the hard drive, applying the operating system image, installing applications, and configuring settings. The task sequence is executed automatically when the target device boots up, typically via a Preboot Execution Environment (PXE) boot or a bootable media. This automation ensures that each device is configured consistently according to the organization's standards, reducing the risk of human error and speeding up the deployment process.

Zero-touch deployments with MDT also support advanced features such as user state migration and application deployment. User State Migration Tool (USMT) can be integrated into the task sequence to capture and restore user data and settings, ensuring a smooth transition for end users. Additionally, MDT can deploy applications as part of the task sequence, either by installing them directly from the deployment share or by downloading them from a network location. This flexibility allows IT administrators to tailor the deployment process to meet the specific needs of their organization, making zero-touch deployments a powerful and efficient solution for managing large-scale Windows deployments.

Cloud-Based Deployments

Zero-touch deployments of Windows 11 leverage cloud-based technologies to streamline the setup and configuration of new devices, minimizing the need for IT intervention. This approach is primarily facilitated by Windows Autopilot, a feature within Microsoft Intune, which allows organizations to preconfigure devices before they even reach the end user. When a user receives their device, they simply power it on and connect to the Internet, and the device automatically downloads and applies the necessary configurations, applications, and policies. This ensures a consistent and secure setup for all devices.

One of the key benefits of zero-touch deployments is the enhanced user experience. Employees can start using their new devices almost immediately, without waiting for manual setup or configuration. This is particularly advantageous for remote or hybrid work environments, where employees may not have easy access to IT support. Additionally, zero-touch deployments help maintain security and compliance by ensuring that all devices are configured according to the organization's policies from the day they are setup. This reduces the risk of misconfigurations and potential security vulnerabilities.

Zero-touch deployments are highly scalable, making them ideal for organizations of all sizes. Whether deploying a handful of devices or thousands, the process remains efficient and manageable. IT administrators can create and manage deployment profiles, monitor the status of deployments, and make adjustments as needed, all from a

centralized console. This scalability, combined with the automation and consistency provided by Windows Autopilot, makes zero-touch deployments a powerful tool for modern IT management.

Deployment of the Windows operating system can be dedicated to a book itself. The CompTIA exam requires basic knowledge of zero-touch deployments. If you want to learn more about AutoPilot, visit `https://learn.microsoft.com/en-us/autopilot/overview`. To learn more about the MDT you can visit `https://learn.microsoft.com/en-us/mem/configmgr/mdt`. Information about MECM can be found at `https://learn.microsoft.com/en-us/mem/configmgr`.

Upgrading Editions of Windows

Windows 11 has the ability to upgrade at any time from one edition of the operating system to a higher one (e.g., from Windows 11 Home to Windows 11 Pro). This can easily be accomplished by entering the appropriate activation product key. You can access the activation screen by choosing Start menu ➢ System ➢ Activation and selecting Change from the Change Product Key menu. You can only upgrade editions; downgrading of retail editions is not supported. However, downgrading of volume license editions, such as Windows 11 Enterprise to Windows 11 Pro, can be achieved. Downgrading of Windows 11 Education to Windows 10/11 Pro can also be done. Downgrading of volume license editions is not formally supported by Microsoft but can be accomplished.

Security and Feature Updates

The Windows platform has always had security updates and feature updates. In older versions of Windows such as XP, the features were subtle additions to the operating system. However, with the introduction of Windows 10 entire releases have been dedicated to major feature releases, such as the Creators update and the Anniversary update. Since then, the features have not been so dramatic, but Windows continues to get feature updates with every major/minor update.

Windows versions typically change twice a year (semi-annually) and have done so since the introduction of Windows 10. They just haven't been obvious, because they were downloaded as a Windows Update. The original version of Windows 10 is 1507, which was released in July 2015. You may have noticed the pattern. The version is a *date code*, consisting of the last two digits of the year (15) and the two-digit month (07). So, it's simple to calculate when the last major update was released, and what is currently installed. With the October 2020 release of Windows 10, Microsoft deviated from this naming convention, using H1 for the first half and H2 for the second half of the year. For example, version 21H1 was released in the first half of 2021. Windows 11 also follows the same date code with its initial release in October 2021; its date code is 21H2.

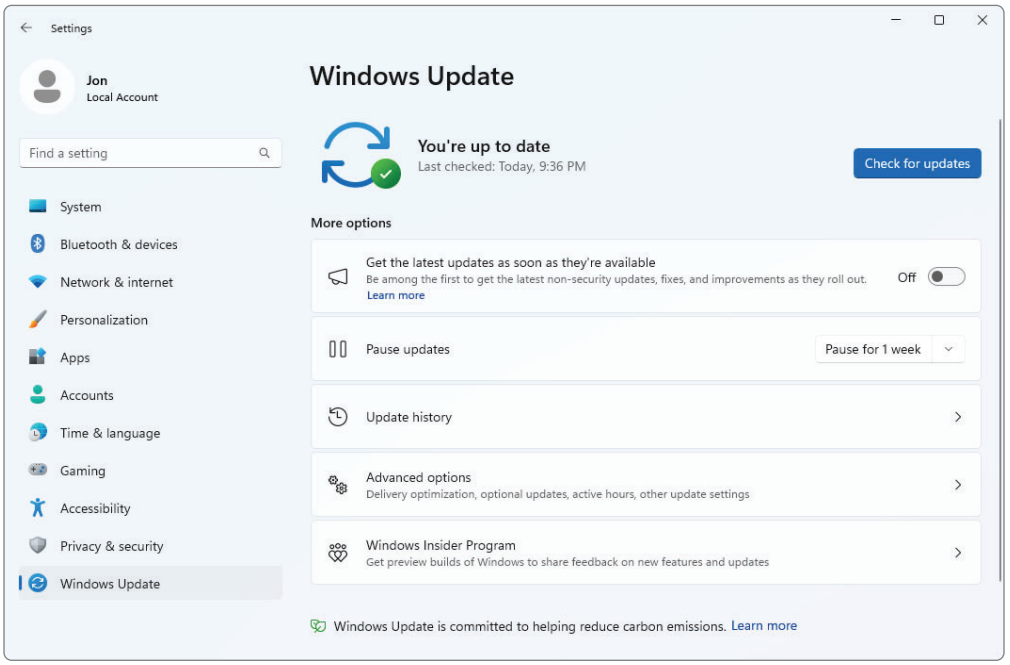
Versions are updated twice a year (semi-annually), usually in spring and fall. They are often referred to as the Windows 11 Spring Update or Windows 11 Fall Update, respectively. In the past few years Microsoft has only released an H2 for Windows 11, which equates to one major update released per year, in the fall. As of this writing, the current Windows 10 version is 22H2, and the Windows 11 version is 24H2. They were both used in the development of this book. Using `winver.exe`, you can see the actual version of the operating system, as shown in Figure 3.37.

FIGURE 3.37 Discovering your version of Windows 11

The life cycle of a Windows product has an end date for support, generally a year to two years after the product has been released. Once the end date is met, Microsoft will no longer support that version of Windows. The retirement date of Windows is generally 10 to 12 years, depending on the operating system's popularity. When the product reaches its retirement date, security updates are no longer furnished for the operating system. Therefore, you should be planning OS upgrades on a consistent schedule.

Probably the biggest change to Windows 11 is the way updates are delivered to the operating system. The end user no longer has the option to choose if they want to set up Windows Updates; they are mandatory and unavoidable. You can pause updates for 1 week to 5 weeks (35 days), but inevitably updates will be installed. To pause Windows Updates, click the Start menu ➢ Settings ➢ Windows Update ➢ Pause Updates. Then you can choose to pause updates for up to 5 weeks. Figure 3.38 shows the options to pause Windows Updates.

FIGURE 3.38 Pause options for Windows Updates



Windows 11 has three different build branches of code (updates) that will be installed on a regular basis with automatic updates. The branches are as follows:

General Availability Channel This is the default servicing channel for Windows 11 devices, with the exception of Windows 11 Enterprise LTSC (Long-Term Servicing Channel).

Insider Program The Insider Program allows you to get the latest product updates from Microsoft before they are available for the targeted release. You must enroll your organization first, and then you can enroll your device to get the latest updates from the Windows Insider Program build channel.

Long-Term Servicing Channel Only users of the Windows 11 Enterprise LTSC Edition can select this channel for updates. The operating system will receive monthly updates via Windows Updates, but no new features will be added over the lifetime of the product.

Table 3.2 details the list of the various Windows 11 editions and the options for changing their service update channels.

TABLE 3.2 Windows servicing channel options

Windows 11 Edition	General Availability Channel	Insider Program	Long-Term Servicing Channel
Home	Yes	Yes	No
Pro	Yes	Yes	No
Enterprise	Yes	Yes	No
Enterprise LTSC	No	No	Yes
Education	Yes	Yes	No

With the introduction of Windows 10 and the initial Windows Update servicing channel options, Microsoft has introduced the concept of Windows as a service (WaaS). A deeper dive on Windows servicing channels can be viewed here:

`https://learn.microsoft.com/en-us/windows/deployment/update/waas-overview`

Installation/Upgrade Boot Options

You can begin the installation or upgrade process by booting from a number of sources. There are three sources in particular with which you should be familiar: USB flash drive, CD-ROM/DVD ISO image, and network boot (PXE). The one method most commonly used for an attended installation is the USB flash drive boot. Another common installation method is booting from ISO image. This installation method is used with virtual machines, since the DVD-ROM is in the form of an ISO image and can be stored on a network filesystem.

You can boot a PC over the network (rather than from an ISO image, USB, or hard disk) with *Windows Preinstallation Environment (WinPE)*, which is a stub operating system that creates a *Preboot Execution Environment (PXE)*. A stub operating system is characterized as a scaled-down version of the primary operating system. A stub operating system usually has basic functionality to perform a basic task, such as the installation of the Windows OS. When Windows 11 is installed with this method, it is considered a *remote network installation*.

You can install or upgrade Windows with traditional installation media, such as DVD ISO image, USB flash drives, and network boot (PXE), but they all tend to be a bit slow. An alternate method of installation and upgrade of Windows is from an external/hot-swappable drive, such as a USB hard drive. You can even use an external hot-swappable drive, such as an eSATA. However, the top speed on eSATA is 6 Gbps, compared to the top speed of USB 3.1 at 10 Gbps and USB 4 at 80 Gbps!

There is one other final option for installing Windows that is often used for reinstallation of the operating system by the original equipment manufacturer (OEM). That option is an internal hidden partition that contains the installation media. The OEM will often include this partition on the hard drive to allow the user to easily reinstall the operating system in the event of problems. In recent years, OEMs have moved away from an internal hard drive partition in favor of the Reset Your PC option in Windows.

Installing or booting Windows 11 can be accomplished through several versatile methods, each catering to different needs and scenarios. Whether you're setting up a new system, upgrading an existing one, or managing multiple devices in an organization, understanding these methods can streamline the process and ensure a smooth installation. From traditional USB drives to advanced network booting and Internet-based installations, each method offers unique advantages. This chapter will walk you through the technical details of each approach, helping you choose the best method for your specific requirements. Let's explore the various ways to get Windows 11 up and running on your device.

Windows 11 Boot Process

The Windows 11 boot process is a sequence of steps that the operating system follows to initialize and prepare the system for use. This process begins when the computer is powered on and continues until the Windows Desktop is fully loaded and ready for user interaction. Understanding these steps can help diagnose and troubleshoot boot-related issues. The following is a typical boot process for Windows 11:

1. Power-On Self-Test (POST)

   When the computer is powered on, the UEFI firmware performs a Power-On Self-Test (POST) to check the hardware components. This step ensures that all critical hardware, such as the CPU, memory, and storage devices, are functioning correctly.

2. UEFI Initialization

   After a successful POST, the UEFI firmware looks for the boot loader in the EFI System Partition (ESP). The firmware executes the Windows Boot Manager (`bootmgfw.efi`) found in this partition.

3. Boot Configuration Data (BCD) Reading

   The Windows Boot Manager reads the Boot Configuration Data (BCD) to determine which operating system to load. The BCD contains essential information about the operating system's location and boot parameters.

4. Loading the Windows Boot Loader

   The Boot Manager loads the Windows Boot Loader (`winload.efi`), which is responsible for initializing the Windows kernel. This step sets the stage for the operating system to take control.

5. Kernel Initialization

   The Windows Boot Loader initializes the kernel (`ntoskrnl.exe`), the core component of the operating system. The kernel then initializes the hardware abstraction layer (HAL), essential drivers, and the system registry.

6. Session Manager Subsystem (`smss.exe`)

   Control is handed over to the Session Manager Subsystem (`smss.exe`), which starts the user session. The Session Manager initializes system processes, loads necessary drivers and services, and sets up the virtual memory paging file.

7. Winlogon Process (`winlogon.exe`)

   The Winlogon process is started, handling user logins and launching the Windows shell. This process is crucial for managing user authentication and starting the graphical user interface.

8. Windows Shell Initialization

   The Windows shell (`explorer.exe`) loads the Desktop environment, including the Taskbar, Start menu, and Desktop icons. This step prepares the system for user interaction.

9. User Interaction

   At this point, the system is fully booted, and the user can begin interacting with Windows 11. The Desktop environment is ready, and all necessary background services are running.

Each of these steps is critical for a successful boot. Any failure along the way can result in boot errors or a system that fails to start properly. The installation process follows a similar boot process, the Windows Preinstallation Environment (WinPE) is a mini–Windows operating system used to install Windows 11 and prior versions of Windows. Understanding this process can help in diagnosing issues and ensuring a smooth boot experience.

Universal Serial Bus (USB)

Installing Windows 11 via a USB drive is a widely used and efficient method. To begin, you'll need a USB flash drive with at least 8 GB of storage. Download the Windows 11 ISO file from the official Microsoft website. Use a tool like Rufus to create a bootable USB drive by selecting the ISO file and the USB drive as the target. Ensure the partition scheme is set to GPT for UEFI systems or MBR for BIOS systems. Once the USB drive is ready, insert it into your computer, restart the system, and enter the BIOS/UEFI settings (usually by pressing F2, F12, Delete, or Esc during startup). Set the USB drive as the primary boot device. Save the changes and exit the BIOS/UEFI. Your computer will boot from the USB drive, and you can follow the onscreen instructions to complete the installation process.

Network

Network booting, or PXE (Preboot Execution Environment), allows you to install Windows 11 over a network connection, which is particularly useful for deploying the OS across multiple machines. PXE booting requires the installation and configuration of a DHCP server to provide network boot options and a TFTP server to host the Windows 11 installation files. On the client machines, enable PXE boot in the BIOS/UEFI settings. When the client

computers start, they will receive the network boot information from the DHCP server, connect to the TFTP server, and download the necessary boot files. The Windows Deployment Services (WDS) can be used to manage and automate this process, making it easier to deploy Windows 11 across an organization.

Solid-State/Flash Drives

Installing Windows 11 from a solid-state drive (SSD) or flash drive offers faster read/write speeds compared to traditional hard drives. To prepare the SSD or flash drive, format it using FAT32 and ensure it contains the Windows 11 installation files. You can use tools like DiskPart to clean and format the drive. Connect the SSD or flash drive to your computer, restart the system, and access the BIOS/UEFI settings. Set the SSD or flash drive as the primary boot device. Save the changes and exit the BIOS/UEFI. Your computer will boot from the SSD or flash drive, and you can follow the installation prompts to complete the process, benefiting from quicker data transfer rates and reduced installation time.

Internet-Based

Internet-based installation of Windows 11 leverages cloud services to download and install the operating system directly from the Internet. This method is ideal for devices with limited local storage or for users who prefer not to use physical media. To perform an Internet-based installation, connect your device to a stable internet connection. Access the Windows 11 download page `www.microsoft.com/en-us/software-download/windows11` and download the Windows Installation Assistant. Run the Windows 11 Installation Assistant, which will guide you through the process of downloading and installing Windows 11. This method ensures you have the latest updates and drivers during the installation process.

External/Hot-Swappable Drive

Using an external or hot-swappable drive to install Windows 11 provides flexibility and ease of use. These drives can be connected and disconnected without shutting down the system. To install Windows 11, connect the external drive to your computer and ensure it contains the installation files. Format the drive using NTFS if necessary. Access the BIOS/UEFI settings and set the external drive as the primary boot device. Save the changes and exit the BIOS/UEFI. Restart your computer, and it will boot from the external drive. Follow the on-screen instructions to complete the installation. This method is particularly useful for systems that require frequent OS changes or testing, such as a development or testing lab.

Internal Hard Drive (Partition)

Booting and installing Windows setup from an internal hard drive partition allows you to reimage the computer rapidly. This is commonly known as a recovery partition, as previously discussed. To create your own recovery partition, you would use unpartitioned space on the drive to create a primary partition and set it to be an active partition. Setting it to active would allow it to be bootable in the BIOS. Next you will format the space with NTFS,

then copy the Windows 11 setup files to the new partition. The `bcdedit.exe` tool can then be used to register the new partition as a boot option.

This method was commonly used in prior operating systems, because Windows lacked the tools to recover the operating system. However, starting with Windows 10, a recovery partition is created, and this is no longer needed. You can still perform this type of install, but the built-in functionality works the same.

Multiboot

Multibooting allows you to install multiple operating systems on a single computer, giving you the option to choose which OS to boot into at startup. To set up a multiboot system with Windows 11, first ensure you have separate partitions for each operating system. Use disk management tools to create or resize partitions as needed. Install the first OS as usual, then install Windows 11 on a different partition. During the installation, the Windows boot manager will be configured to allow you to select the desired OS each time you start your computer. This setup is ideal for users who need to run different operating systems for various tasks, such as development, testing, or personal use.

Multiboot systems were a mainstream option 20 years ago to boot into two separate operating systems on one computer. However, with the introduction of virtualization, such as Hyper-V, multiboot operating systems have lost their utility. Starting with Windows Vista SP2 released in 2009, Hyper-V was a downloadable feature.

Considerations

With every great plan there are unplanned consequences; we call them *considerations*. When installing Windows 11 or upgrading to Windows 11, there are several considerations surrounding the applications you need to use and the hardware Windows 11 is being installed on.

When you upgrade an operating system, you have the potential for data loss. Therefore, it is always advisable to back up files and user preferences before starting an upgrade. Depending on the user you are upgrading in your organization, if files are lost it could cause catastrophic loss of sales, payments, and most importantly, time. Whenever possible, perform a full-drive backup, because settings may be in a spot you wouldn't normally back up. If a full-drive backup cannot be performed, then perhaps replace the drive with another one. Label the original drive *removed* and recycle it after you confirm that everything is acceptable with the end user.

The applications must be supported on Windows 11. You may think that if it runs, then it's supported, but you'd be wrong. Just because a program runs on Windows 11 doesn't mean that it was meant to be run on the operating system. It is up to the discretion of the software vendor if they will support you on the latest operating system. This is often discovered in the event you need help from them. So always check if the application is supported on the latest OS before upgrading. In many cases you'll find that software vendors require the latest version of their product purchased or installed for it to be supported on the latest

OS. Applications are also backward compatible with older operating systems, because not everyone will be on the latest and greatest operating system.

The hardware that Windows 11 is running on is another consideration. Many motherboards and peripherals need third-party drivers. These drivers must be supported on Windows 11 or they may not function correctly. Older network interface cards (NICs) are notorious for not being supported on Windows 11. So, always check the hardware vendor's website before upgrading to Windows 11.

Command-Line Tools

Although the exam focuses on the Windows operating systems, it tests a great number of concepts that carry over from the Microsoft Disk Operating System (MS-DOS). MS-DOS was never meant to be extremely user friendly. Its roots are in Control Program for Microcomputers (CP/M), which, in turn, has its roots in UNIX. Both of these older OSs are command line–based, and so is MS-DOS. In other words, they all use long strings of commands typed in at the computer keyboard to perform operations. Some people prefer this type of interaction with the computer, including many folks with technical backgrounds. Although Windows has left the full command-line interface behind, it still contains a bit of DOS, and you get to it through the command prompt.

Although you can't tell by looking at it, the Windows command prompt is actually a Windows program that is intentionally *designed* to have the look and feel of a DOS command line. Because it is, despite its appearance, a Windows program, the command prompt provides all the stability and configurability you expect from Windows. You can access a command prompt by running `cmd.exe`.

A number of diagnostic utilities are often run at the command prompt. They can be broken into two categories: networking and operating system. The utilities associated with networking appear in other chapters, but the focus here is on the utilities associated with the operating system.

Exam Objective 1.5, "Given a scenario, use the appropriate Microsoft command-line tools," requires you to know various command-line tools. The OS command-line tools that you are expected to know for the exam are `cd`, `dir`, `ipconfig`, `ping`, `netstat`, `nslookup`, `net use`, `tracert`, `pathping`, `chkdsk`, `format`, `diskpart`, `md`, `rmdir`, `robocopy`, `hostname`, `net user`, `winver`, `whoami`, `[command name] /?`, `gpupdate`, `gpresult`, and `sfc`.

In the following sections we will cover a number of commands you will see on the exam, along with the commands available with standard privileges, as opposed to those with administrative privileges.

Navigation and File Management

The power of the command line comes from the level of detail that can be attained with simple commands. There are several commands that allow you to extract a high level of detail. However, the command line is an unforgiving user interface that requires specific commands, whereas with the GUI you just need to know how to point and click, but it lacks the high level of detail. In the following section we will cover basic navigation at the command line and basic file management commands.

dir Command

The `dir` command is used to display a list of the files and folders/subdirectories within a directory. When you use it without any parameters, `dir` will show you not only that information but also the volume label and serial number, along with the amount of free space, in bytes, remaining on the disk.

Wildcards can be used with the command to list all files that begin with a certain letter or end with certain letters. An example is typing **`dir *.txt`** to list all the text files in a directory. A plethora of parameters are available that can be used to customize the results or the display. Table 3.3 lists some of the most common switches available for `dir`.

TABLE 3.3 Common `dir` switches

Switch	Purpose
`/a`	Allows you to specify the attributes of files you are seeking (hidden, system, and so on)
`/o`	Allows you to specify a different display order (alphabetic is the default)
`/l`	Returns the results unsorted and in lowercase format
`/s`	Recursively searches through subdirectories as well as the current directory
`/t`	Sorts the files according to time order
`/p`	Displays the results one page/screen at a time
`/q`	Shows file ownership

cd, md, rd Commands

The `cd`, `md`, and `rmdir` commands are used to change (or display), make, and remove directories, respectively. The commands `cd`, `md`, and `rd` are shorthand versions of the `chdir`, `mkdir`, and `rmdir` commands, respectively. Table 3.4 lists their usage and switches.

TABLE 3.4 cd/md/rmdir usage and switches

Command	Purpose
`cd [path]`	Changes to the specified directory.
`cd /d [drive:][path]`	Changes to the specified directory on the drive.
`cd ..`	Changes to the directory that is up one level.
`cd \`	Changes to the root directory of the drive.
`md [drive:][path]`	Makes a directory in the specified path. If you don't specify a path, the directory will be created in your current directory.
`rmdir [drive:][path]`	Removes (deletes) the specified directory.
`rmdir /s [drive:][path]`	Removes all directories and files in the specified directory, including the specified directory itself.
`rmdir /q [drive:][path]`	Quiet mode. You won't be asked whether you're sure you want to delete the specified directory when you use `/s`.

Drive Navigation

So far, you've seen the basics of looking at directories with the `dir` command, changing directories, making directories, and removing them. However, up to this point we have assumed you are on the same partition. The `cd` command will change directories within a drive letter, such as the C: drive, but it will not change drive letters unless you supply the `/d` switch. To change drives without using the `dir` command, just enter the drive letter and append a semicolon to it. For example, if you want to change to the D: drive, enter **`d:`** at the command prompt. You can then use the `cd` command followed by the drive letter, and if you want to change back, enter **`c:`** at the command prompt.

Now that you've learned how to navigate the command prompt to look at files, let's use that knowledge in Exercise 3.1.

EXERCISE 3.1

Command-Line Directory Management

1. Open a command prompt by selecting the Start menu, typing **cmd** in the search box, right-clicking the result Command Prompt (System), and selecting Run As Administrator. Alternately you can press the Windows key+R, type **cmd**, then press Ctrl+Shift+Enter to run as administrator.
2. Change to the root of your C: drive by typing **`cd /d C:\`** and pressing Enter.

EXERCISE 3.1 *(continued)*

Note: If you are already in C:, all you have to do is type **`cd \`** and press Enter.

3. Create a directory called C16 by typing **`md C16`** and pressing Enter.
4. Change to the C16 directory by typing **`cd C16`** and pressing Enter.
5. Create several layers of subdirectories at once. Type **`md A1\B2\C3\D4`** and press Enter. Notice that these commands create each of the directories that you specified. You now have a directory structure that looks like this: `C:\C16\A1\B2\C3\D4`.
6. Change back to your root directory by typing **`cd \`**.
7. Attempt to delete the C16 directory by typing **`rd C16`** and pressing Enter.

   Windows won't let you delete the directory because the directory is not empty. This is a safety measure. Now let's really delete it.
8. Delete the C16 directory and all of its subdirectories by typing **`rmdir /s C16`** and pressing Enter.
9. You will be asked whether you're sure that you want to delete the directory. If you are, type **Y** and press Enter.
10. To close the command prompt window, type **`exit`**.

    Note that if you had used the `/q` option in addition to `/s`, your system wouldn't have asked whether you were sure; it would have just deleted the directories.

copy Command

The `copy` command does what it says: It makes a copy of a file in a second location. (To copy a file and then remove it from its original location, use the `move` command.) Here's the syntax for `copy`:

```
copy [filename] [destination]
```

It's pretty straightforward. There are several switches for `copy`, but in practice they are rarely used. The three most commonly used switches are `/a`, which indicates an ASCII text file; `/v`, which verifies that the files are written correctly after the copy; and `/y`, which suppresses the prompt asking whether you're sure that you want to overwrite files if they exist in the destination directory.

The `copy` command cannot be used to copy directories. Use `xcopy` for that function.

One useful tip is to use wildcards. For example, at the command prompt, the asterisk (*) is a wildcard that means *everything*. So, you could type **copy *.exe** to copy all files that have an .exe filename extension, or you could type **copy *.*** to copy all files in your current directory. The other popular wild card is the question mark (?), which does not mean everything but instead means one thing. For example, copy ABC?.exe would copy only .exe files with four-letter names, of which the first three letters are ABC.

xcopy Command

If you are comfortable with the copy command, learning xcopy shouldn't pose too many problems. It's basically an extension of copy with one notable exception—it's designed to copy folders as well as files. The syntax is as follows:

```
xcopy [source] [destination][switches]
```

There are 26 xcopy switches. Some commonly used ones are listed in Table 3.5.

TABLE 3.5 xcopy switches

Switch	Purpose
/a	Copies only files that have the Archive attribute set and does not clear the attribute (useful for making a quick backup of files while not disrupting a normal backup routine)
/e	Copies directories and subdirectories, including empty directories
/f	Displays full source and destination filenames when copying
/g	Allows copying of encrypted files to a destination that does not support encryption
/h	Copies hidden and system files as well
/k	Copies attributes (by default, xcopy resets the Read-Only attribute)
/o	Copies file ownership and ACL information (NTFS permissions)
/r	Overwrites read-only files
/s	Copies directories and subdirectories but not empty directories
/u	Copies only files that already exist in the destination
/v	Verifies the size of each new file

Perhaps the most important switch is `/o`. If you use `xcopy` to copy files from one location to another, the filesystem creates new versions of the files in the new location without changing the old files. In NTFS, when a new file is created, it inherits permissions from its new parent directory. This could cause problems if you copy files. (Users who didn't have access to the file before might have access now.) If you want to retain the original permissions, use `xcopy /o`.

robocopy Command

`robocopy.exe`, short for Robust File Copy, is a powerful command-line utility included with recent versions of Windows designed to handle complex file-copying tasks. You can use `robocopy` to mirror entire directory trees, while preserving file attributes, file ownership, and NTFS permissions. Its extensive set of switches, such as `/mir` for mirroring directories and `/sec` for copying security settings, makes it an indispensable tool for system administrators and advanced users.

An excellent resource on how to use `robocopy` can be found at `https://learn.microsoft.com/en-us/windows-server/administration/windows-commands/robocopy`. The `robocopy` utility is the Swiss army knife of file and folder copy utilities. Let's look at some examples.

The `/mir` switch mirrors the ownership, attributes, and permissions from one directory to another. The following command-line command will mirror the contents of `C:\Source` to `D:\Destination`:

```
robocopy C:\Source D:\Destination /mir
```

The `/sec` switch ensures that files are copied along with their NTFS permissions, ensuring that the security settings are preserved. The following command-line command will copy all files from `C:\Source` to `D:\Destination` along with their NTFS security:

```
robocopy C:\Source D:\Destination /sec
```

The `/mov` switch moves files and deletes the old files from the source after copying. The following command-line command moves files from the source to the destination, deleting them from the source once the copy is complete:

```
robocopy C:\Source D:\Destination /mov
```

The `/xd` switch can exclude directories from being copied. The following command-line command will copy all files from the source directory to the destination directory. The command will also exclude the `$Recycle.bin` and `System Volume Information` directories.

```
robocopy C:\Source D:\Destination /
xd $Recycle.Bin," System Volume Information"
```

The `/log:` switch allows you to specify a log file where `robocopy` will write its output, making it easier to review the results of the copy operation. The following command-line command will copy files from `C:\Source` to `D:\Destination` and log the output to `C:\Logs\robocopy.log`:

```
robocopy C:\Source D:\Destination /log:C:\Logs\robocopy.log
```

Network Connectivity Tools

Windows is a network operating system, which means that the operating system and its principal user rely on the network for connectivity to information. This is where the command line becomes really useful to the administrator of the PC. The command line will return a large amount of data that is normally not suited for a graphical user interface (GUI). The following is a short list of commands that can help you diagnose network connectivity issues from the command line.

ipconfig Command

The `ipconfig` command is a network administrator's best friend—it assists in the diagnosis of network problems with the operating system. The `ipconfig` command without any switches displays basic information, such as the IP address, subnet mask, default gateway, and DNS suffix. The command `ipconfig /all` lists adapters and each one's assigned IP address, subnet mask, default gateway, DNS suffix, DNS server(s), DHCP server, and MAC address, just to name the most important elements. Viewing these assignments can help you diagnose the current network status of the connection.

In addition to verifying the status of a network connection's assignments, you can release and renew DHCP-assigned IP addresses. The `/release` switch releases the IP address, and the `/renew` switch renews the lease of an IP address.

The `ipconfig` command also allows you to view the local DNS cache with the `/displaydns` switch. You can flush the local DNS cache with the `/flushdns` switch. These switches come in handy when a DNS entry has changed, and you want to immediately flush the cache and verify any cached entries.

ping Command

Next to the `ipconfig` command, the `ping` command is the runner-up as the network administrator's best friend. The `ping` command allows you to verify network connectivity via *Internet Control Message Protocol (ICMP)* packets. A common troubleshooting step used by network administrators is to ping the default gateway. If it returns a ping, then the network connectivity problem is probably beyond that device, or your subnet mask is incorrect. However, after a successful ping, you can verify that your computer has basic connectivity to it. An example of a successful ping is:

```
C:\Users\Sybex>ping 172.16.1.1

Pinging 172.16.1.1 with 32 bytes of data:
Reply from 172.16.1.1: bytes=32 time<1ms TTL=64
Reply from 172.16.1.1: bytes=32 time<1ms TTL=64
Reply from 172.16.1.1: bytes=32 time<1ms TTL=64
Reply from 172.16.1.1: bytes=32 time<1ms TTL=64

Ping statistics for 172.16.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\Sybex>
```

tracert Command

The `tracert` command allows the network administrator to verify the path a network packet travels to its destination. As the diagnostic packet passes through the internetwork, each router responds with a response time. This enables you to locate a fault in an internetwork. Here's an example:

```
C:\Users\Sybex>tracert 8.8.8.8

Tracing route to google-public-dns-a.google.com      [8.8.8.8]
over a maximum of 30 hops:
  1    <1 ms    <1 ms    <1 ms  pfsense.Sybex.local  [172.16.1.1]
  2    13 ms    12 ms    17 ms  96.120.62.213
  3    15 ms    15 ms    12 ms  te04012.comcast.net  [68.86.101.141]
  4    13 ms    19 ms    12 ms  162.151.152.153
  5    12 ms    13 ms    20 ms  96.108.91.78
  6    22 ms    14 ms    20 ms  96.108.91.121
  7    21 ms    24 ms    20 ms  be-7016-cr02.comcast.net  [68.86.91.25]
  8    20 ms    20 ms    26 ms  be-10130-pe04.comcast.net  [68.86.82.214]
  9    20 ms    20 ms    21 ms  as040-2-c.comcast.net  [75.149.229.86]
 10    22 ms    21 ms    20 ms  108.170.240.97
 11    20 ms    23 ms    21 ms  108.170.226.85
 12    20 ms    22 ms    18 ms  google-public-dns-a.google.com  [8.8.8.8]

Trace complete.

C:\Users\Sybex>
```

pathping Command

`pathping`, another command-line tool, combines the benefits of `tracert` and `ping`. The tool can be used to diagnose packet loss (or suspected packet loss) to a destination website. It is invaluable to network administrators to help prove to their ISP that packet loss is a problem on their network.

The tool will first trace the entire path to a destination IP address or DNS host. Then, each of the hops will be tested with ICMP for packet loss and round-trip time. It easily identifies router hops that are causing the delay or packet loss. The following is an example of a `pathping` to a provider's DNS server:

```
C:\Users\Sybex>pathping 75.75.75.75

Tracing route to cdns01.comcast.net [75.75.75.75]
```

```
over a maximum of 30 hops:
 0 Wiley.sybex.local [172.16.1.101]
 1 pfSense.sybex.local [172.16.1.1]
 2 96.120.62.213
 3 te-0-5-0-12-sur02.pittsburgh.pa.pitt.comcast.net [69.139.166.77]
 4 be-11-ar01.mckeesport.pa.pitt.comcast.net [68.86.147.109]
 5 be-7016-cr02.ashburn.va.ibone.comcast.net [68.86.91.25]
 6 ae-4-ar01.capitolhghts.md.bad.comcast.net [68.86.90.58]
 7 ur13-d.manassascc.va.bad.comcast.net [68.85.61.242]
 8 dns-sw01.manassascc.va.bad.comcast.net [69.139.214.162]
 9 cdns01.comcast.net [75.75.75.75]

Computing statistics for 225 seconds…
      Source to Here  This Node/Link
Hop RTT  Lost/Sent = Pct Lost/Sent = Pct Address
 0                        Wiley.sybex.local [172.16.1.101]
                0/ 100 = 0%  |
 1  0ms   0/ 100 = 0%   0/ 100 = 0% pfSense.sybex.local [172.16.1.1]
                0/ 100 = 0%  |
 2  14ms   0/ 100 = 0%   0/ 100 = 0% 96.120.62.213
                0/ 100 = 0%  |
 3  15ms   0/ 100 = 0%   0/ 100 = 0% te-0-5-0-12-sur02.pittsburgh.pa.pitt
.comcast.net [69.139.166.77]
                0/ 100 = 0%  |
 4  15ms   0/ 100 = 0%   0/ 100 = 0% be-11-ar01.mckeesport.pa.pitt.comcast.net
[68.86.147.109]
                0/ 100 = 0%  |
 5  23ms   0/ 100 = 0%   0/ 100 = 0% be-7016-cr02.ashburn.va.ibone.comcast.net
[68.86.91.25]
                0/ 100 = 0%  |
 6  23ms   0/ 100 = 0%   0/ 100 = 0% ae-4-ar01.capitolhghts.md.bad.comcast.net
[68.86.90.58]
                0/ 100 = 0%  |
 7  25ms   0/ 100 = 0%   0/ 100 = 0% ur13-d.manassascc.va.bad.comcast.net
[68.85.61.242]
                0/ 100 = 0%  |
 8  24ms   0/ 100 = 0%   0/ 100 = 0% dns-sw01.manassascc.va.bad.comcast.net
[69.139.214.162]
                0/ 100 = 0%  |
 9  23ms   0/ 100 = 0%   0/ 100 = 0% cdns01.comcast.net [75.75.75.75]

Trace complete.

C:\Users\Sybex>
```

netstat Command

The `netstat` command allows you to view listening and established network connections for the operating system. Several switches can be used with the `netstat` command. One of the most useful is the `-b` switch, which displays the name of the application and its current established connections. Adding the `-a` switch displays all the listening connections in addition to the established connections. A basic example follows:

```
C:\Users\Sybex>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    127.0.0.1:49750        view-localhost:50912   ESTABLISHED
  TCP    127.0.0.1:50912        view-localhost:49751   ESTABLISHED
  TCP    172.16.1.181:49208     104.20.60.241:https    ESTABLISHED
  TCP    172.16.1.181:49599     172.67.181.149:https   ESTABLISHED
  TCP    172.16.1.181:49600     52.167.17.97:https     TIME_WAIT
  TCP    172.16.1.181:49602     20.50.80.210:https     ESTABLISHED
  TCP    172.16.1.181:49603     a104-75-163-105:http   TIME_WAIT
  TCP    172.16.1.181:56759     151.101.1.140:https    ESTABLISHED
  TCP    172.16.1.181:64151     iad23s96-in-f10:https  CLOSE_WAIT
  TCP    172.16.1.181:64152     iad66s01-in-f13:https  CLOSE_WAIT
  TCP    172.16.1.181:64154     iad23s96-in-f10:https  CLOSE_WAIT

C:\Users\Sybex>
```

nslookup Command

DNS is one of the most important network services that an operating system and user relies on for resolution of `www.sybex.com` to an IP address. Without DNS, we just couldn't remember the millions of IP addresses; it would be like trying to remember the phone number of every person you've ever met or are going to meet.

When DNS problems arise, the `nslookup` command allows you to verify that DNS is working correctly and that the correct results are being returned. The simplest way to use DNS is to use an inline query, such as **`nslookup www.sybex.com`**. This will return the IP address associated with the *fully qualified domain name (FQDN)* of `www.sybex.com`. The `nslookup` command can also be used in the interactive mode by typing **`nslookup`** and pressing Enter. This mode allows you to query more than the associated IP address, depending on the type of DNS record you are trying to diagnose. By default, the record looked up with the `nslookup` command is the A or CNAME DNS records. These records are the most commonly looked-up DNS records for diagnosing connectivity issues. By specifying the `-type` argument, you can change the default record queried. The following is an example of retrieving the IP address for the FQDN of `www.sybex.com`, as well as the use of the `-type` argument:

```
C:\Users\Sybex>nslookup www.sybex.com
Server:  pfsense.wiley.local
```

```
Address:  172.16.1.1

Non-authoritative answer:
Name:    www.sybex.com
Address:  63.97.118.67

C:\Users\Sybex>nslookup -type=mx sybex.com
Server: pfSense.wiley.local
Address: 172.16.1.1

Non-authoritative answer:
sybex.com    MX preference = 20, mail exchanger = cluster1a.us.messagelabs.com
sybex.com    MX preference = 10, mail exchanger = cluster1.us.messagelabs.com

C:\Users\Sybex>
```

net Command

The `net` command can be used with several different subcommands. Most of the subcommands have been deprecated with the last few releases of Windows.

`net use` The `net use` subcommand is still widely used by administrators to map drive letters to network shares. The syntax for mapping a drive of Z: to the network location of `\\server\share` is as follows:

```
net use Z: \\server\share
```

`net user` `net user` is another subcommand widely used by administrators. This command allows an administrator to list all the local accounts on a Windows installation by entering **`net user`**. It can also be used to list local accounts or domain accounts by supplying some arguments to the command. If you wanted to create a local account of `usertwo` with a password of *Passw0rd*, you would use the following syntax:

```
net user usertwo Passw0rd /add
```

OS Management Tools

There are several other command-line tools just as useful as the connectivity tools we've discussed, but they are used for various other purposes. The following tools are part of the objectives for the 220-1202 exam, and they are commonly used by Windows administrators.

gpupdate Command

Active Directory refreshes local and Active Directory–based policies every 90 minutes in what is called a *background refresh cycle*. When the background refresh happens, policies are reapplied, forcing the settings that the administrator has configured in the Group Policy settings.

The gpupdate command is used to update Group Policy settings. It refreshes, or changes, both local and Active Directory–based policies and replaces some of the functionality that previously existed with the secedit command.

The gpupdate command can force the refresh cycle immediately with the /force switch. In addition, you can target the computer or the user, which is particularly useful when trying to diagnose a problem with Group Policy Objects (GPOs). The following is an example of forcing a refresh for the computer GPO settings:

```
C:\Users\bohack>gpupdate /force /target:computer
Updating policy...

Computer Policy update has completed successfully.

C:\Users\bohack>
```

gpresult Command

The gpresult command is used to show the Resultant Set of Policy (RSoP) report/values for a remote user and computer. Bear in mind that configuration settings occur at any number of places; they are set for a computer, a user, a local workstation, the domain, and so on. Often one of the big unknowns is which set of configuration settings takes precedence and which is overridden. With gpresult, it is possible to ascertain which settings apply.

A number of switches can be used in conjunction with the gpresult command. The most useful switches are /r and /z. The /r switch allows you to see the RSoP summary of GPOs applied. This allows you to quickly verify if a policy is being applied. You can then use the /z switch to turn on super-verbosity, which allows the output to display the exact settings being applied.

sfc Command

The System File Checker (sfc.exe) is a command line–based utility that checks and verifies the versions of system files on your computer. If system files are corrupted, sfc will replace the corrupted files with correct versions.

The syntax for the sfc command is as follows:

```
sfc [switch]
```

Table 3.6 lists the switches available for sfc.

TABLE 3.6 sfc switches

Switch	Purpose
/scanfile	Scans a file that you specify and fixes problems if they are found
/scannow	Immediately scans all protected system files and replaces corrupted files with cached copies

Switch	Purpose
`/verifyonly`	Scans protected system files and does not make any repairs or changes
`/verifyfile`	Identifies the integrity of the file specified and makes any repairs or changes
`/offbootdir`	Repairs an offline boot directory
`/offwindir`	Repairs an offline Windows directory

To run `sfc`, you must be logged in as an administrator or have administrative privileges. If the System File Checker discovers a corrupted system file, it will automatically overwrite the file by using a copy held in another directory. The most recent Windows versions store the files in a large number of discrete folders beneath `C:\WINDOWS\WINSXS` (where they are protected by the system and only TrustedInstaller is allowed direct access to them—the cache is not rebuildable). TrustedInstaller is a service in Windows 11 that enables the installation, removal, and modification of system components.

The `C:\WINDOWS\SYSTEM32` directory is where many of the Windows system files reside.

If you attempt to run `sfc` from a standard command prompt, you will be told that you must be an administrator running a console session in order to continue. Rather than opening a standard command prompt, on Windows 11 open the Start menu, type **`cmd`**, and then press Ctrl+Shift+Enter. The UAC will prompt you to continue, and then you can run `sfc` without a problem.

shutdown Command

The `shutdown.exe` utility can be used to schedule a shutdown (complete or a restart) locally or remotely. You can even have the computer enter into a hibernation power state. In addition, shutdowns of the computer can be logged with a variety of reasons, from unplanned to application maintenance. A message can also be specified and announced to users for the shutdown. The syntax of the command is as follows:

```
shutdown [/i | /l | /s | /sg | /r | /g | /a | /p | /h | /e | /o] [/hybrid]
[/soft] [/fw]  [/m \\computer][/t xxx][/d [p|u:]xx:yy [/c "comment"]]
```

There are a lot of different switches, as you can see from the previous usage syntax. Table 3.7 lists the most important switches for the `shutdown` command.

TABLE 3.7 shutdown switches

Switch	Purpose
/s	Shut down the computer.
/sg	Shut down the computer. On the next boot, restart any registered applications.
/r	Do a full shutdown and restart the computer.
/g	Do a full shutdown and restart the computer. After the system is rebooted, restart any registered applications.
/a	Abort a system shutdown.
/h	Hibernate the local computer.
/o	Go to the advanced boot options menu and restart the computer. Must be used with the /r option.
/m \\computer	Specify the target computer.
/t *xxx*	Set the timeout period before shutdown to *xxx* seconds. The valid range is 0–315360000 (10 years), with a default of 30. If the timeout period is greater than 0, the /f parameter is implied.

Disk Commands

There are several commands for the administration of drives and folders. In this section, we will cover the common command-line tools that you will use in day-to-day administration of Windows. All of these tools can be accessed in the GUI as well, but the advantage to using the command line is avoiding clicks and potential mistakes.

format Command

The `format` command is used to wipe data off disks and prepare them for new use. Before a hard disk can be formatted, it must have partitions created on it. (Partitioning was done in the DOS days with the `fdisk` command, but that command does not exist in current versions of Windows, having been replaced by `diskpart`.) The syntax for `format` is as follows:

```
format [volume] [switches]
```

The *volume* parameter describes the drive letter (e.g., `D:`), mount point, or volume name. Table 3.8 lists some common `format` switches.

TABLE 3.8 `format` switches

Switch	Purpose
`/fs:[filesystem]`	Specifies the type of filesystem to use (FAT, FAT32, or NTFS)
`/v:[label]`	Specifies the new volume label
`/q`	Executes a quick format

There are other options as well—to specify allocation sizes, the number of sectors per track, and the number of tracks per disk size. However, we don't recommend that you use these unless you have a very specific need. The defaults are just fine.

Thus, if you wanted to format your D: drive as NTFS, with a name of HDD2, you would type the following:

```
<?b Start?>format D: /fs:ntfs /v:HDD2<?b End?>
```

Before you format any drive, be sure that you have backed it up (or are prepared to lose whatever is on it).

diskpart Command

The `diskpart.exe` utility shows the partitions and lets you manage them on the computer's hard drives. You can perform the same functions in the `diskpart` utility as you can perform in the GUI, which is discussed later in this chapter. Because of the enormous power that `diskpart` holds, membership in the Administrators local group (or equivalent) is required to run `diskpart`.

chkdsk Command

You can use the Windows `chkdsk.exe` utility to create and display status reports for the hard disk. `chkdsk` can also correct filesystem problems (such as cross-linked files) and scan for and attempt to repair disk errors. You can manually start `chkdsk` by right-clicking the problem disk and selecting Properties. This will bring up the Properties dialog box for that disk, which shows the current status of the selected disk drive.

By clicking the Tools tab at the top of the dialog box and then clicking the Check button in the Error-checking section, you can start `chkdsk`. Exercise 3.2 walks you through starting `chkdsk` in the GUI, and Exercise 3.3 does the same from the command line.

EXERCISE 3.2

Running `chkdsk` within Windows

1. Open Windows File Explorer by holding down the Windows key and pressing E.
2. Right-click C: and choose Properties.
3. Click the Tools tab, and then click the Check button.
4. Choose Scan Drive.
5. Once the drive is scanned for errors, you can click Close.

   In operating systems prior to Windows 10, you can choose options such as Automatically Fix Filesystem Errors and/or Scan For And Attempt Recovery Of Bad Sectors.

EXERCISE 3.3

Running `chkdsk` at the Command Line

1. Open an administrative command prompt in Windows 10/11 by clicking the Start menu and typing **`cmd`**, and then press Ctrl+Shift+Enter.

   You can also right-click on the result and run `cmd` with administrator privileges from the context menu.

2. Type **`chkdsk /f`** and press Enter.

   The system will now scan for, and fix, filesystem errors.

Informational Tools

Command-line tools like `winver` and `hostname` are essential utilities for quickly accessing system information in Windows environments. These tools, while simple, are invaluable for IT professionals and users who need to efficiently gather system details without navigating through multiple settings menus. The following are common informational tools you need to know for the 220-1202 exam.

`hostname` Command

The `hostname` command allows the administrator to keep their sanity. The command returns the hostname of the computer that you have the command prompt open on. It can get pretty confusing for the administrator when they jump from one computer to another and remain in the command line. So by typing the command **`hostname`**, you can positively identify the system you are about to execute a command on. The following is an example of the command's use:

```
C:\Users\Sybex>hostname
```

```
Wiley-023432

C:\Users\Sybex>
```

winver Command

`winver.exe` is not a command-line tool but a useful tool to glean information about the operating system. The command will display a GUI dialog box. The command's sibling at the command line is `ver.exe`. This command has its roots all the way back to DOS, and it's still active in the latest version of Windows 11. Using the command will return the version of Windows, as shown in the following:

```
C:\Users\NetworkedMinds>ver.exe

Microsoft Windows [Version 10.0.19043.1237]

C:\Users\NetworkedMinds>
```

whoami Command

The `whoami` command in Windows is a simple yet powerful tool used to display the current user's username. When executed in the command prompt, it returns the domain and username of the account currently logged into the system. The following are some of the possible switches that can be used with the `whoami` command.

To simply find out the username of the account currently logged in, you can use the `whoami` command. This is particularly useful when you need to confirm which user is active on the system, especially in environments with multiple user accounts. The following command will display the current user's domain and username:

```
C:\Users\NetworkedMinds>whoami

DOMAIN\username
```

If you need to check what privileges the current user has, the `whoami /priv` command is very handy. This can help you understand what actions the user is authorized to perform, which is crucial for troubleshooting permission issues. The following command lists the privileges assigned to the current user, which can be useful for understanding what actions the user can perform:

```
C:\Users\NetworkedMinds>whoami /priv

PRIVILEGE_NAME                DESCRIPTION                                    STATE
============================= ============================================== ========
SeShutdownPrivilege           Shut down the system                           Disabled
SeChangeNotifyPrivilege       Bypass traverse checking                       Enabled
SeUndockPrivilege             Remove computer from docking station           Disabled
```

To see all the groups the current user belongs to, you can use the `whoami /groups` command. This is useful for verifying the user's roles and permissions within the system, as group memberships often determine access rights. The following command shows all the groups the current user is a member of, providing insight into the user's permissions and roles:

```
C:\Users\NetworkedMinds>whoami /groups

GROUP_INFORMATION
==========================
*Group Name                Attributes
========================== ========================================
DOMAIN\Domain Users        Mandatory group, Enabled by default, Enabled group
BUILTIN\Administrators     Mandatory group, Enabled by default, Enabled group
```

When you need to find the Security Identifier (SID) for the current user, the `whoami /user` command is the way to go. The SID is a unique value used to identify the user account, which can be important for various administrative tasks. The following command returns the Security Identifier (SID) for the current user, which is a unique value used to identify the user account:

```
whoami /user

USER INFORMATION
==========================
User Name                SID
==========================
DOMAIN\username           S-1-5-21-3623811015-3361044348-30300820-1013
```

help and */?* Commands

The `help` command does what it says: It gives you help. If you just type **`help`** and press Enter, your computer gives you a list of system commands that you can type. To get more information, type the name of a command that you want to learn about after typing **`help`**. For example, if you type **`help rd`** and press Enter, you will get information about the `rd` command.

You can get the same help information by typing `/?` after the command.

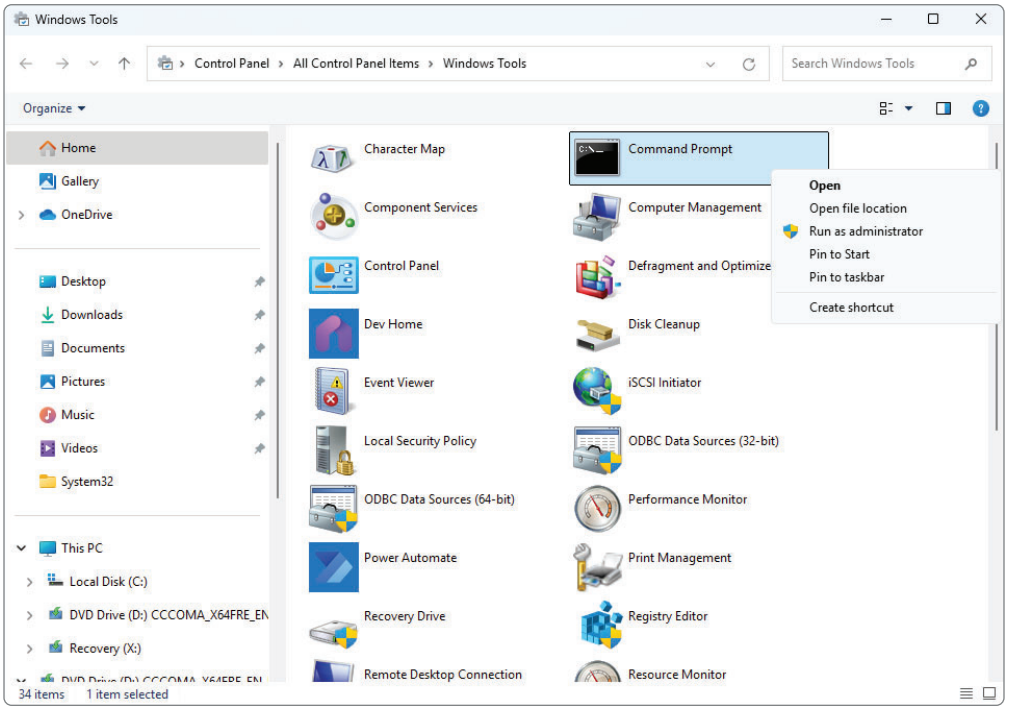
The `/?` switch is slightly faster and provides more information than the `help` command. The `help` command provides information for system commands only. (It does not include network commands.) For example, if you type **`help ipconfig`** at a command prompt, you get no useful information (except to try `/?`); however, typing **`ipconfig /?`** provides the help file for the `ipconfig` command.

Commands with Standard Privileges vs. Administrative Privileges

By default, any user can open a command prompt and begin typing the names of command-line commands. Certain commands, however, can be dangerous when run and, as a safety precaution, require administrative privileges. The `sfc` command was mentioned earlier, for example, as requiring administrative privileges.

With Windows 11, rather than opening a standard command prompt, choose Start menu ➢ All ➢ Windows Tools, right-click Command Prompt, and then choose Run As Administrator. The UAC will prompt you to continue, and then you can run `sfc` without a problem, as shown in Figure 3.39.

FIGURE 3.39 Opening a command prompt with admin privileges in Windows 11



In most cases, if you try to run a utility that requires administrative privileges and you are not currently in a console session that has the privileges, an error message will notify you of this.

Networking in Windows

CompTIA expects you to know a number of topics related to networking and Windows. This section covers the various scenarios in which you will deploy Windows in a network. First, we'll cover small office, home office (SOHO) deployments, and then we'll scale it out to an enterprise network.

Exam Objective 1.7, "Given a scenario, configure Microsoft Windows networking features on a client/desktop," requires you to know various network components of Windows 11. Be sure to know the various network models and their differences, the various ways to connect to a network, access resources, and configure a client.

Networking Models

You can choose from several different networking models to facilitate Windows authentication. There is no single way to implement a network model; it all depends on the needs of the organization to share the resources. Each model has its advantages and disadvantages. This section covers the three most popular methods.

Workgroups

The *workgroup* networking model has existed in the Windows operating system since it was first introduced. Windows can function as both a client and a server simultaneously. When an operating system can function as both a client and a server, it is considered *peer-to-peer networking*. Clients can join the network and leave the network at any time.

The core of the Windows operating system is essentially the same for both a server and workstation. There are a few differences between the two—for instance, servers normally give priority to background services like file and printer sharing versus the desktop. The second and most limiting is the restriction of connecting 20 simultaneous clients to a single workstation. After 20 simultaneous connections are made, all others clients will return an Access Denied error—that is, until one of the 20 connections is closed out and the session is terminated. This takes some time on the workstation side, so if you have more than 20 clients, you should plan to deploy an actual server system running software such as Windows Server 2022 or Windows Server 2025.

Workgroups are normally used in SOHO environments or in situations that do not require the infrastructure of a dedicated server for authentication. They should be kept to a maximum of 20 clients, with the expectation that each client will maintain its own resources (files and printer sharing). Many small offices use this networking model and never need anything more.

A typical situation where a workgroup is effective is when a printer needs to be shared from a single computer. The disadvantage is that the computer must be on in order for the

clients to use the printer. Another disadvantage with workgroups is user authentication, which we will discuss in detail later in this chapter.

When you install Windows 11, by default it is joined to the workgroup named Workgroup. However, there may be instances in which you want to join another workgroup. To join another workgroup, perform the following steps:

1. Click the Start menu.
2. Select Settings.
3. Click System.
4. Click About.
5. Click the Advanced System Settings, Related Setting.
6. Select Change Settings under the heading of Computer Name, Domain, And Workgroup Settings.
7. Click Change Next To Rename This Computer or Change Its Domain Or Workgroup.
8. Change the Workgroup option to the desired workgroup.
9. Click OK.
10. Click OK.
11. Reboot the computer.

Domains

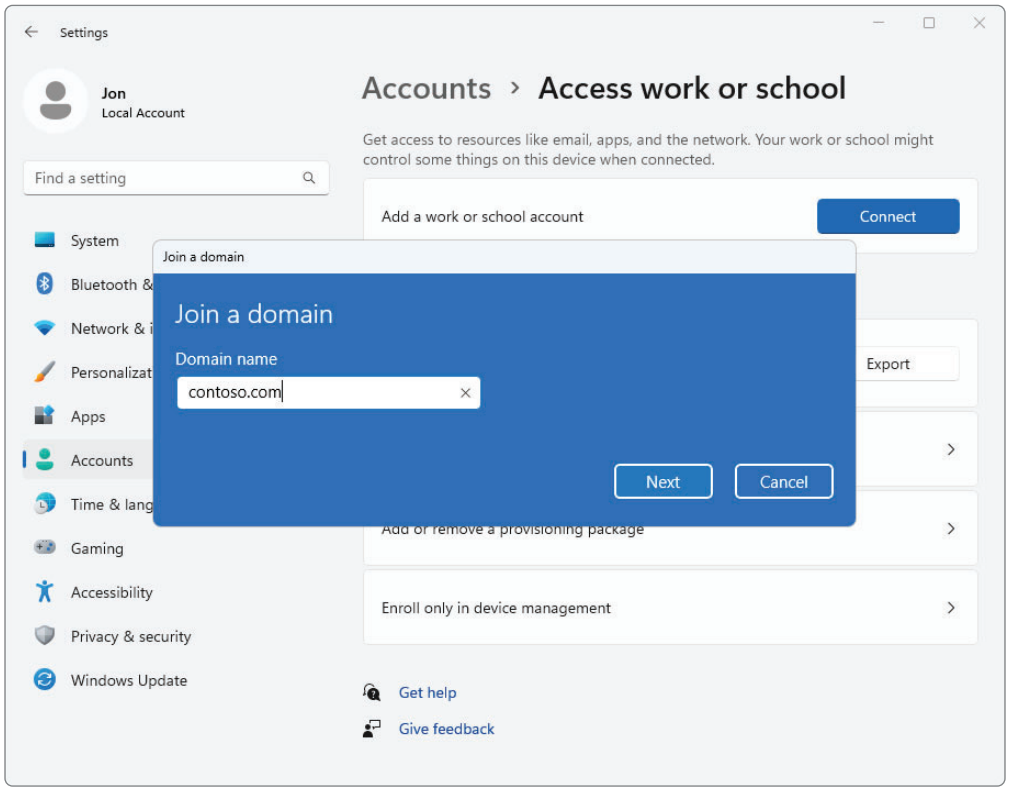
Domain functionality has existed since Microsoft Windows 3.51 (mid-1990s). Unlike the workgroup networking model, the domain networking model requires that clients be joined to a domain. Joining a domain creates a trust between the client (Windows client, for example) and the authentication server (Windows Server 2025 running Active Directory, for example). Joining a client to a domain allows users with an account in the domain to log into the client. To join a domain in Windows 11, perform the following steps:

1. From the Start menu, choose Settings ➢ Accounts ➢ Access Work Or School ➢ Connect ➢ Join This Device To A Local Active Directory Domain. You will be prompted to enter a domain name, as shown in Figure 3.40.
2. Enter your credentials on the domain when prompted, which allows the joining of the operating system, as shown in Figure 3.41.
3. Reboot the operating system for the changes to take effect.

An alternate way of joining a domain is through the System Properties dialog box. To do so:

1. Right-click the Start menu.
2. Click System.
3. Under Related Links choose Advanced System Settings.
4. Under Computer Name, Domain, And Workgroup Settings, select Change Settings.

FIGURE 3.40 Windows 11 Settings app for joining a domain

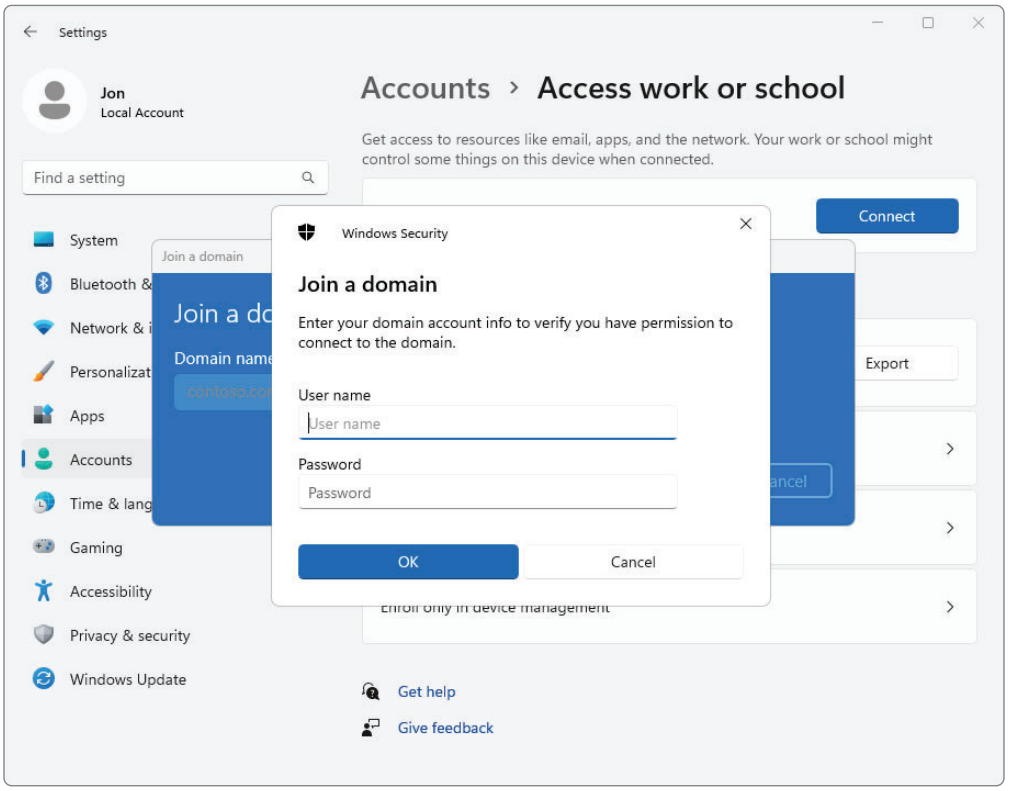


5. Click Change.
6. Select the Domain radio button under Member Of.
7. Enter the domain name and click OK, as shown in Figure 3.42.

The next dialog box will prompt you to enter your credentials on the domain that allows the joining of the operating system. You will then need to reboot the operating system for the changes to take effect. This method of joining an operating system to a domain has been supported since Windows XP.

Domains also allow for files and printers to be secured with domain credentials. The key takeaway in the benefit to domains is centralized authentication for users and computers. It is important to note that the term *domain* describes both the networking model and the friendly name of the security domain of the network.

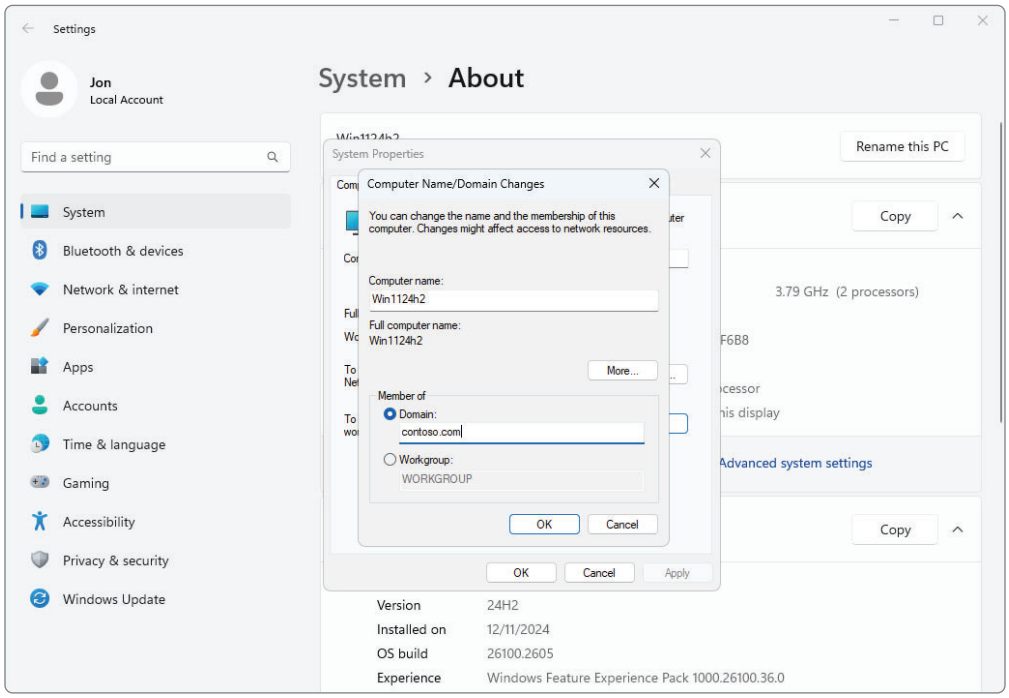
FIGURE 3.41 Windows 11 domain credentials prompt



Microsoft Active Directory is the main authentication technology used with domain controllers in a domain networking model. Active Directory contains objects such as users, computers, and printers, as well as many other types of objects, called Group Policy Objects (GPOs). Active Directory allows for the grouping of these objects logically so that they can be controlled with policies through Group Policy. Active Directory domains can scale to any number of joined clients and user credentials. Many large organizations use Active Directory for centralized user authentication, because it is very scalable.

Group Policy is a component of Active Directory as well as a component of the client-side operating system. Group Policy is used to create policies that can be applied to groups of users and a group of computers. Group Policy is considered enforced, since it reapplies policies every 90 minutes.

FIGURE 3.42 Windows domain joining



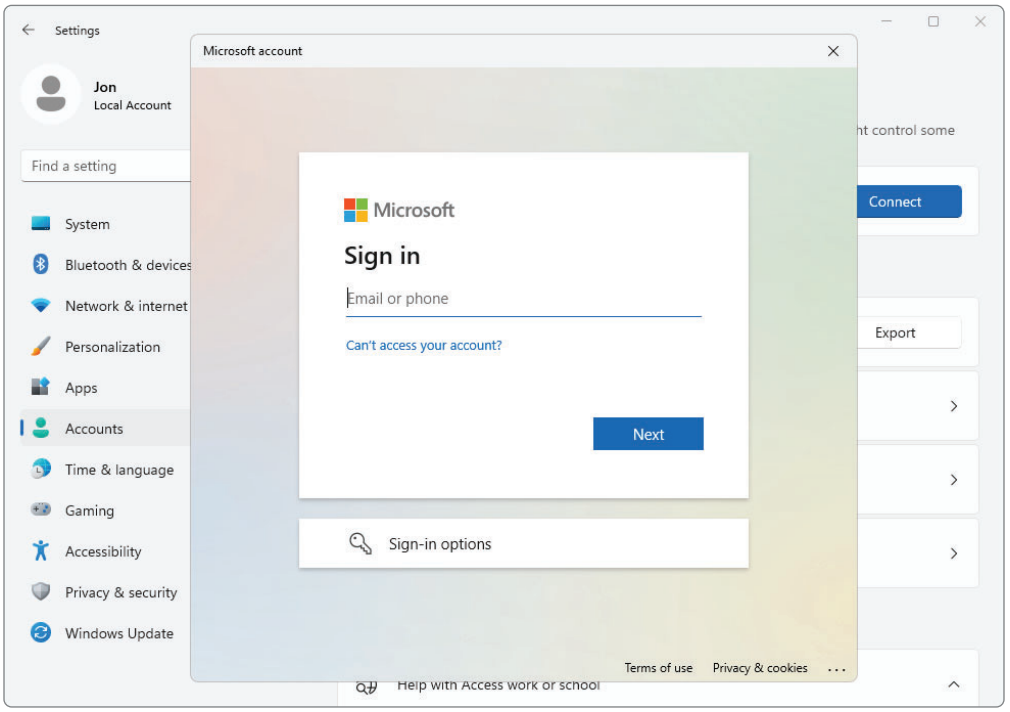
Microsoft Entra ID

Microsoft Entra ID, previously known as Azure Active Directory, is a cloud-based service that helps manage user identities and access to resources. It allows users to sign in once and access multiple applications securely. Microsoft Entra ID has features like single sign-on (SSO), multifactor authentication (MFA), and conditional access policies. These features ensure only authorized users can access important data. Entra ID also integrates easily with various applications and services, making it simpler for IT administrators to manage user permissions and keep everything secure. With the explosion of cloud-based applications, just about every application supports Microsoft Entra ID. Microsoft 365 exclusively uses Microsoft Entra ID to secure information and provide identity services to users.

To join a Windows 11 device to Microsoft Entra ID, follow these steps:

1. Click Start menu ➢ Settings ➢ Accounts ➢ Access Work Or School ➢ Connect ➢ Join This Device To Microsoft Entra ID.
2. Enter your Microsoft Entra ID email address and click Next, as shown in Figure 3.43.
3. On the next screen enter your password, and click Sign In.

FIGURE 3.43 Entra ID Sign In



4. You will need to enter your authenticator code or satisfy MFA on the next screen, then click Verify.
5. On the next screen you will verify the organization and the username that you will join with, then click Join, as shown in Figure 3.44.
6. The last screen welcomes you to the Microsoft Entra ID organization, and you can click Done.

Once authenticated, the device will be registered with Microsoft Entra ID, and any applicable policies will be applied automatically. You can verify the organization and user you are connected as, on the Access Work Or School screen, as shown in Figure 3.45. The device is now enrolled in the organization's mobile device management (MDM) solution of Intune. This enrollment allows administrators to manage the device remotely, deploy applications, and enforce security settings. The integration with MDM solutions ensures that devices remain compliant with organizational policies and can be monitored for any security issues. This is all dependent on the subscription purchased.

Microsoft Entra ID is not a CompTIA objective, but it should be. More and more organizations are adopting Microsoft Entra ID. It was included in this book for completeness. You can find more information by going to `www.microsoft.com/en-us/security/business/identity-access/microsoft-entra-id`.

FIGURE 3.44 Make Sure This Is Your Organization

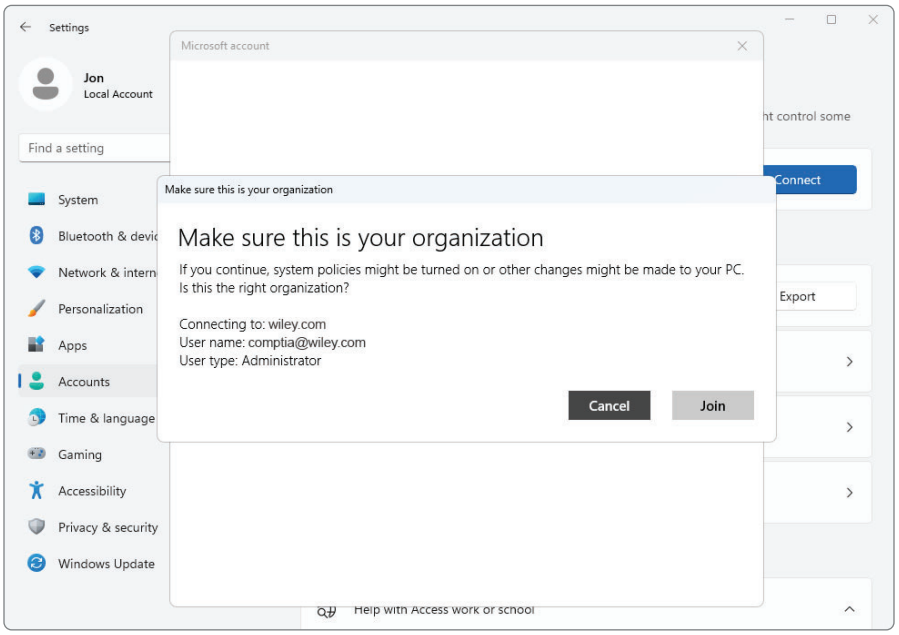
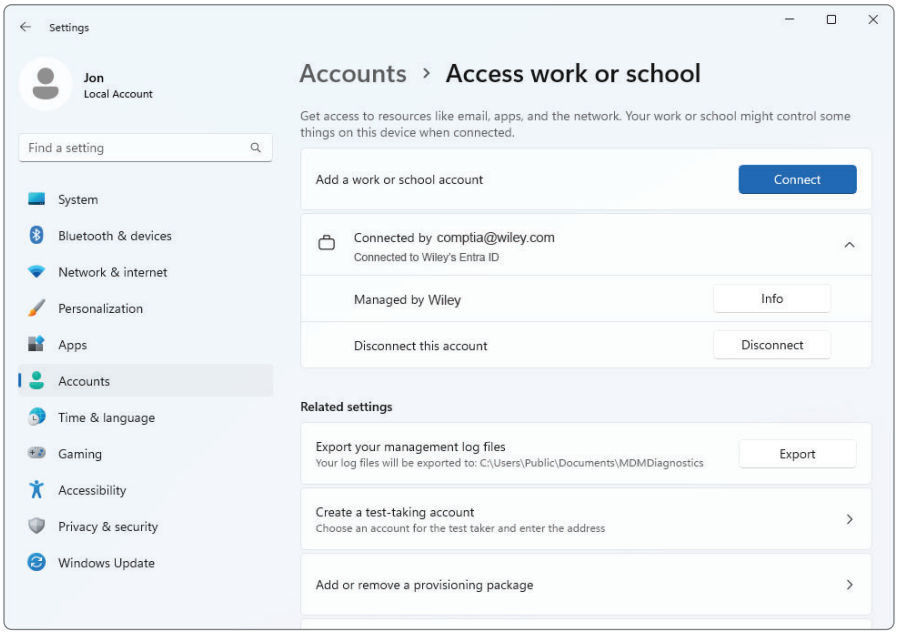


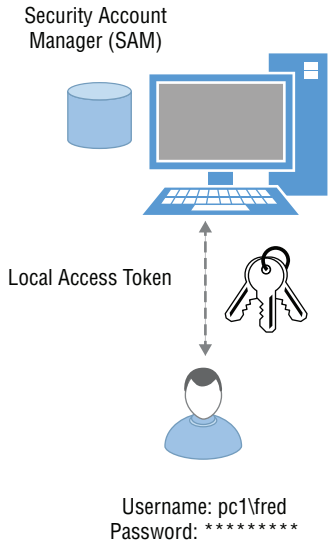
FIGURE 3.45 Access Work Or School



User Authentication

When you choose to use a workgroup networking model, all authentication is local to the operating system. Windows operating systems contain a local authentication mechanism called the *Security Account Manager (SAM)*. The SAM can be considered a local database of users and groups. All users locally authenticating to the workstation authenticate against this internal database of usernames and passwords and are granted a local access token, as shown in Figure 3.46. The local access token allows the user to access local resources secured with the user's identity.

FIGURE 3.46 Windows local authentication

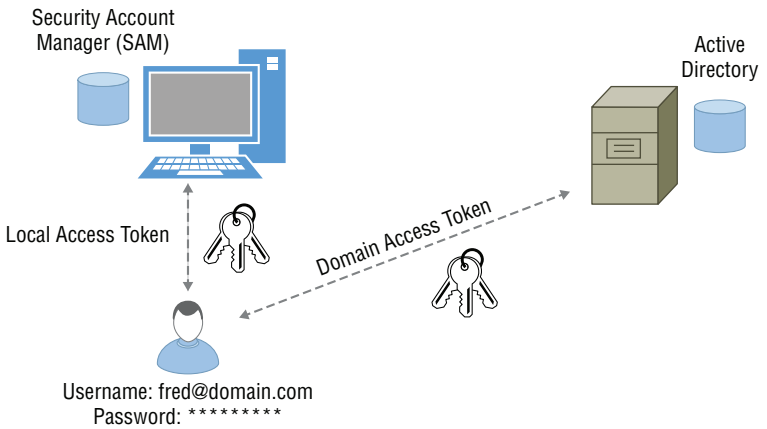


When clients are joined to a domain, a user with credentials on the domain can log into the workstation. When this happens, they are authenticating against an Active Directory *domain controller*. An Active Directory domain controller retains information about all access rights for all users and groups in the network. When a user logs into the system, AD issues the user a *globally unique identifier (GUID)*, also known as an *access token*. Applications that support Active Directory can use this access token to provide access control. When a client is joined to a domain, users can still log into the local operating system using the SAM, as if they have a local account (see Figure 3.47). However, local logins are normally restricted to administrators once a client is joined to the domain.

Active Directory simplifies the sign-on process for users and lowers the support requirements for administrators. Access can be established through groups, and it can be enforced through group memberships. Active Directory can be implemented using a Windows Server

(such as Windows Server 2022 or Windows Server 2025) computer. All users will then log into the Windows domain using their centrally created AD accounts.

FIGURE 3.47 Windows domain authentication



Microsoft Entra ID is a cloud-based alternative to hosting a traditional AD domain in a physical location. There are different benefits, such as the Microsoft 365 platform, depending on the requirements. There is also a monthly cost to maintain the licensing.

One of the big problems that larger systems must deal with is the need for users to access multiple systems or applications. This may require a user to remember multiple accounts and passwords. The purpose of a *single sign-on (SSO)* is to give users access to all the applications and systems that they need when they log in.

SSO operates on the principle that the resource trusts the authentication server. When a user logs in initially, they will authenticate against the authentication server for their organization. When the user then visits the resource, which is normally a cloud-based resource, it will prompt the authentication server to provide a claim on behalf of the user, as shown in Figure 3.48.

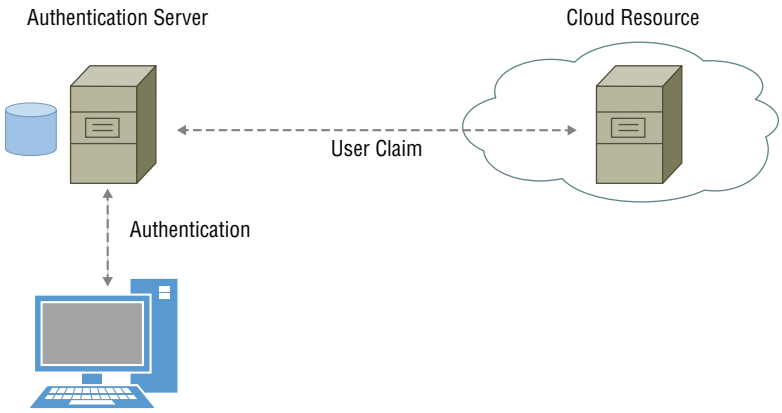
The claim normally contains basic information about the user, such as first and last name, email address, or any other attribute. At no time is the user's password sent, because they authenticated once already. Although we've oversimplified SSO in this example, it really is this simple, without the layers of encryption and complicated trust rules. As we adopt more and more cloud resources, it is becoming the number one way to provide authentication for our users because we never transmit the actual username and password.

Establishing a Network Connection

A key element to a successful network is the connection that connects the computer to the actual network. There are a number of different ways to connect to the network that we will

cover in the following section. The key takeaway is that the network system will function identically, regardless of the connection. For example, if you have a computer joined to an organization's domain, the login process will function identically with a wireless versus a wired connection.

FIGURE 3.48 Single sign-on



The type of connection you choose is based on your convenience and the requirements of the connection. As an example, wireless is extremely convenient, but it requires you to be within distance of an access point (AP), and the further away you are from the AP, the slower the connection. If your requirement is consistent high bandwidth for a voiceover IP (VoIP) call, then a wired connection is your best choice and wireless would be problematic. The following are the most common network connection methods:

Wired A wired network is most common in organizations with desktop computers. Unfortunately, a wired connection means that there is no mobility for the connection; you are literally wired to a desk.

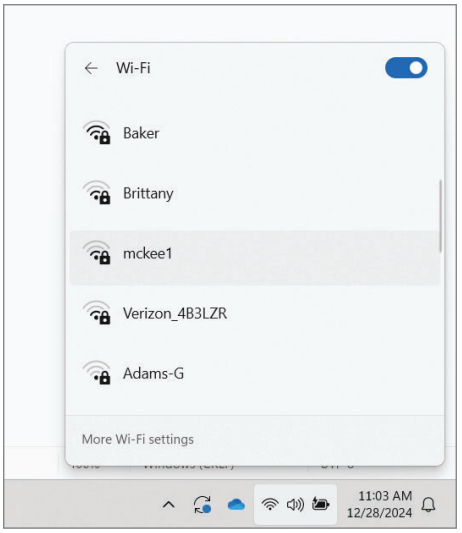
Wired connections are the most reliable and arguably easiest to diagnose when there are problems. The network link light gives you a visual indicator that you have a network connection. Windows 11 will also place a visual notification of a computer with a cable in the notification area when a wired connection is detected.

Wireless A wireless connection is found in networks where mobility is required. Wireless is often used in small office/home office (SOHO) network settings. It can also be found in small and large organizations that require workers to move around, such as a factory setting, sales workforce, or medical setting, just to name a few.

Connecting to wireless networks is not as straightforward as making a wired connection. There is rarely a visual identifier that you have a wireless connection, such as the link light

with a wired connection. Windows 11 will display the notification tray icon as a radio wave. If the radio wave is grayed out with an asterisk at the upper left, then there are wireless connections detected and you are not connected to any of them. If you click the wireless icon, all of the wireless networks available will be displayed, along with their security status, as shown in Figure 3.49.

FIGURE 3.49 Wireless connectivity



You can then choose a wireless network and select Connect. If the wireless network requires additional security, such as a preshared key (PSK) or a corporate login to a *captive portal*, the operating system will direct you. By default, wireless networks that you connect to will automatically reconnect when you are in range. When the wireless connection is established, the notification tray icon will appear as a white radio wave. As you move further away from the AP, the wireless indicator will act as a signal strength meter.

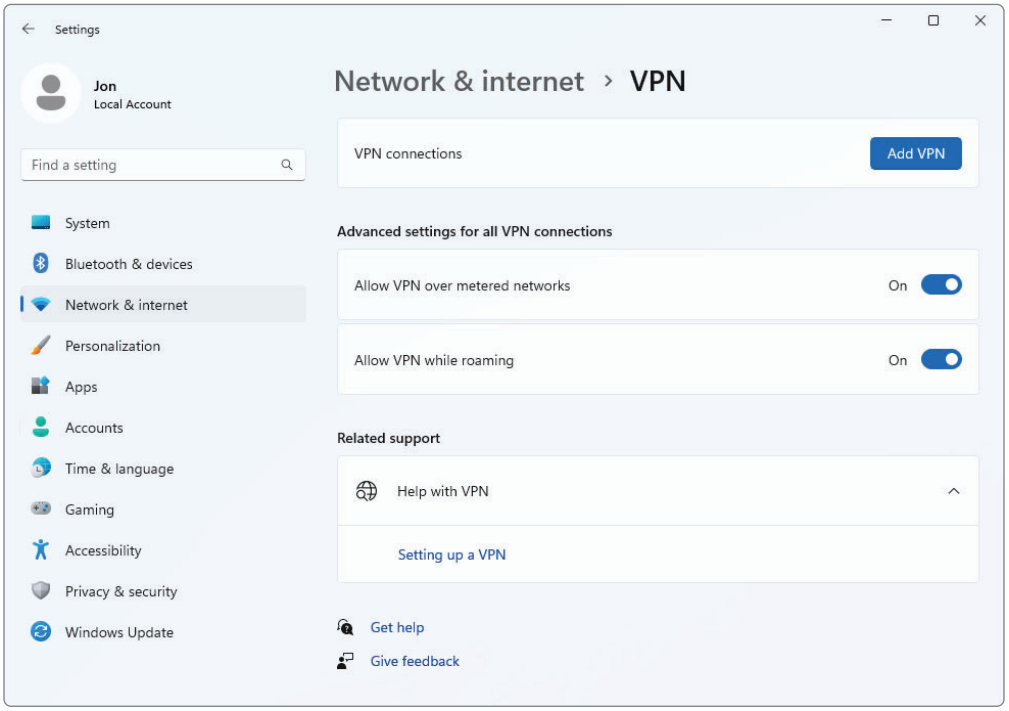
Virtual Private Network (VPN) In recent years, *virtual private networks (VPNs)* have become the rage of network privacy for browsing. These applications provide encryption to an anonymous server from which you can browse the Internet. However, the 220-1202 exam is based on traditional VPN technology that provides a secure connection between the two endpoints in an organization.

The VPN connection is an overlay network on top of an established network connection called the underlay network. So, you will need an established Internet connection before a VPN can connect. The VPN connection will provide two distinct features to the end user. The first feature it provides is an entry point to the organization's private network.

The second feature it provides is end-to-end encryption for everything transmitted over the connection between the client and the organization's private network.

Establishing a VPN connection will require information from the organization's VPN appliance or server. You will need the VPN protocol, server address, and the sign-in info. You can create a VPN connection by choosing the Start menu ➢ Settings ➢ Network & Internet ➢ VPN; from here you can add a VPN connection, as shown in Figure 3.50.

FIGURE 3.50 VPN connectivity



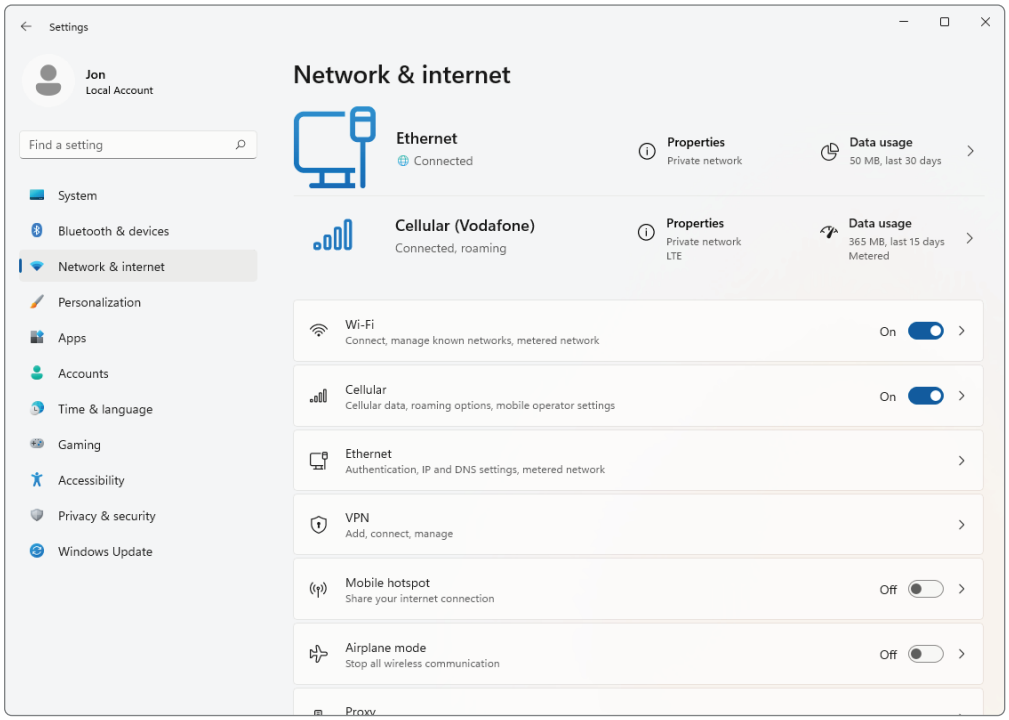
Wireless Wide Area Network (WWAN) A wireless wide area network (WWAN) connection is a connection that is created with a cellular data provider, such as Sprint, Verizon, or AT&T, just to name a few. The mobile device will require a special card called a WWAN adapter, or most mobile devices will have a built-in card. You will then need to register the connection with the cellular provider to activate the connection; this process usually requires a monthly billing for data, and the connection is often metered.

A *metered connection* is a connection in which you pay for a specific amount of data to be downloaded. Once the amount of data is reached, you generally pay for overages per gigabyte. Many features in Windows, such as Windows Updates, will work differently

depending on whether the connection is a metered connection. Windows Updates will not download over a metered connection, so the features don't use all your precious data over the metered connection. The cellular option of connectivity is not the only connection that can be classified as a metered connection; each of the methods of connectivity in this section can be set as a metered connection. It is just more common to find a cellular connection as a metered connection.

Once the connection is registered, you can connect via the wireless notification tray icon and select the cellular network you wish to connect to. You can also connect to the cellular network by choosing the Start menu ➢ Settings ➢ Network & Internet ➢ Cellular, as shown in Figure 3.51. If your device does not support a WWAN device, then the Cellular section will not appear on the Network & Internet screen. After selecting Cellular, you can then configure roaming options, as well as the preference of cellular over Wi-Fi, and you can tell Windows to treat the connection as a metered connection.

FIGURE 3.51 Windows Cellular

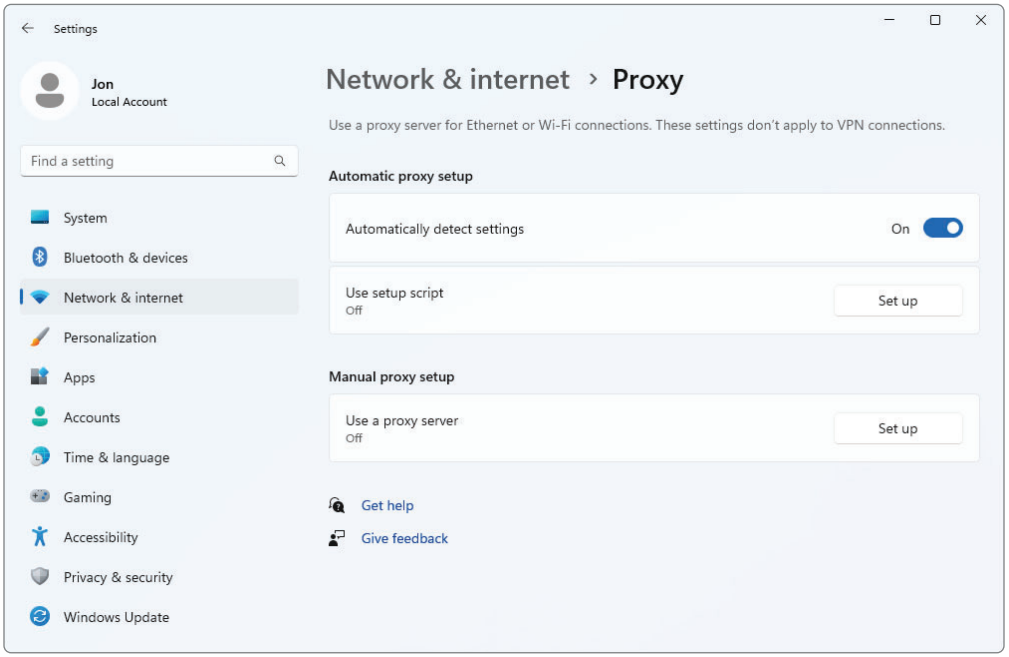


Proxy Settings In addition to establishing a connection, you may need to set a proxy server depending on your organization's policies. In many organizations, the web browser is not allowed to directly request web pages from the destination web server. An intermediary called a *proxy server* is used to request the web page on behalf of the user.

The use of a proxy server allows for caching of frequently accessed web pages, as well as the ability to filter content. The proxy is primarily for web-based traffic, such as browsing with the Edge browser or Internet Explorer. However, other applications can also elect to use the proxy server, depending on their traffic type.

To configure the proxy settings for Microsoft Edge and Internet Explorer, from the Start menu, choose Settings ➢ Network & Internet ➢ Proxy. From the proxy screen you can configure the operating system to automatically use a setup script (JavaScript) by clicking the switch for Use Setup Script and specifying the script address, as shown in Figure 3.52.

FIGURE 3.52 Windows Proxy settings



You can also specify a manual proxy setup, which is a common configuration task. You will simply click the switch Use A Proxy Server and then enter the address and port. Specific websites often require direct access and will not work with a proxy server. You can enter exceptions into the lower section and separate servers with a semicolon. You can also use wildcards if you want to exclude an entire namespace.

Accessing Resources

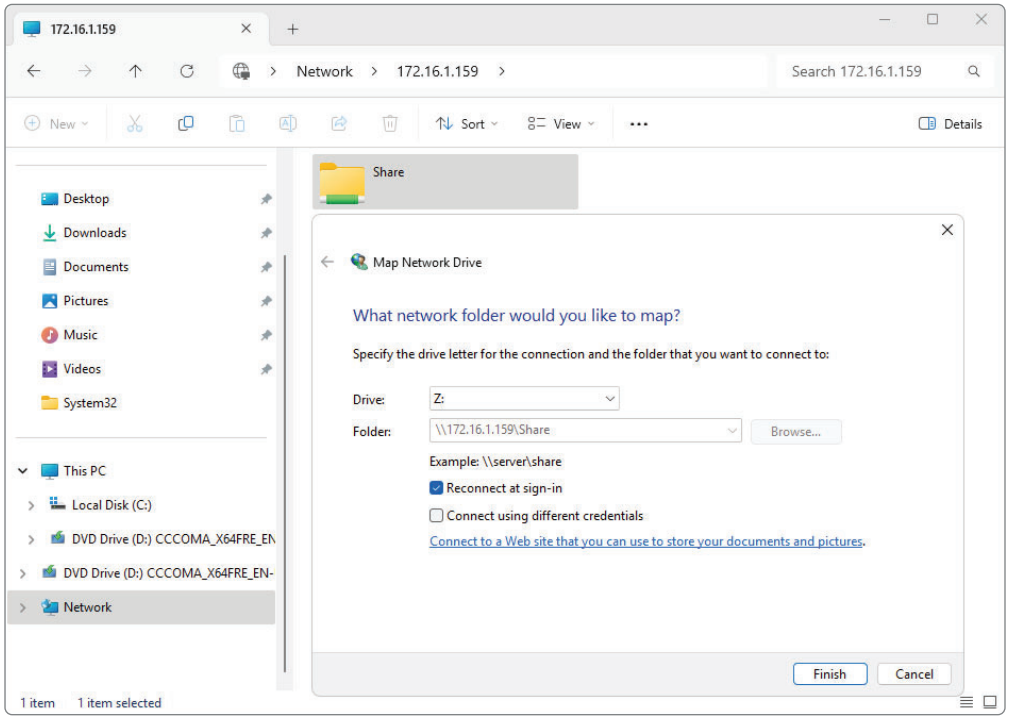
Now that you understand networking models, authentication, and how to connect to the network, let's focus on the resources that users will access. This section covers the most common types of resources.

Network Shares

A *network share* is a type of resource sharing that allows access to file and folder resources over a network from a file server. File Explorer is used to access network shares, but you can also use command-line commands to access the resources, as you will see in the following sections. As the name implies, network shares are those that exist on the network; however, they can be mapped to appear as if they are local. The `net use` command can be used to establish network connections at a command prompt, for example. If you want to connect to a shared network drive and make it your M: drive, the syntax is **`net use M: \\server\share`**. The **`\\server\share`** portion of the command is called the *Universal Naming Convention (UNC)* path. The UNC path is a standard way to describe the server and fileshare to the Windows operating system.

In addition to using the command line, you can use the GUI to map a network share. After browsing to the server by typing **`\\servername`** in the Windows File Explorer address bar, right-click the fileshare and select Map Network Drive. If you are using Windows 11, you will need to click Show More Options first. You will then be prompted with some options for mapping the network share, as shown in Figure 3.53.

FIGURE 3.53 Mapping a network drive



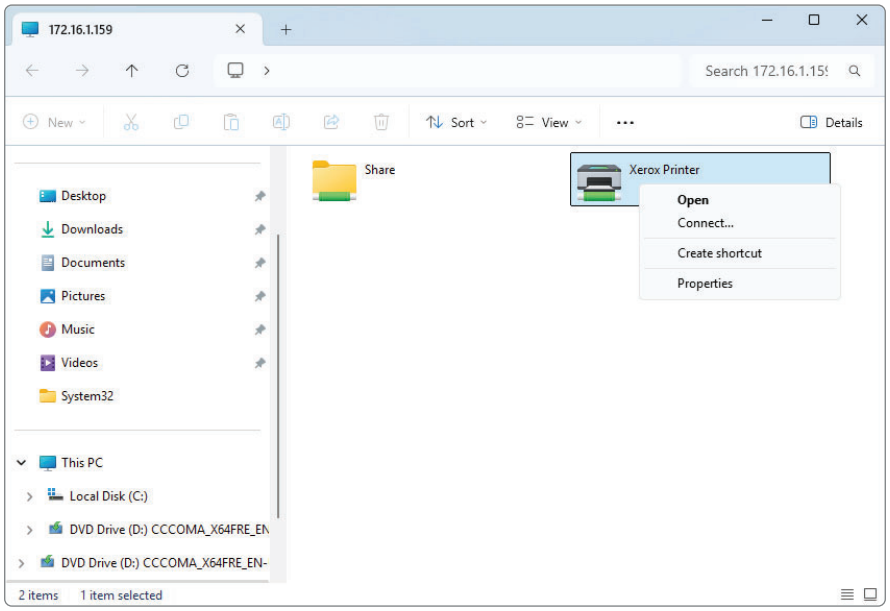
By default, the Reconnect At Sign-In option is selected. In order to ensure that the mapped drive is reconnected when the user signs in, this option must be selected.

Printer Shares

Another common type of network resource is printing. Although we often need to access files and folders, at some point we will probably need to print. You can use the same command to connect to a shared printer; the syntax of `net use lpt1: \\server\printername` will map a printer to the LPT1 device. As in the prior example, you can make the printer act as if it is locally connected to the operating system. You can also use the GUI method of connecting a printer by right-clicking the printer after browsing to the server and then selecting Connect, as shown in Figure 3.54.

Sharing printers was one of the primary reasons why networking became popular in the workplace, and it still continues to be an important reason today. Know that you can map to any printer on the network, but it still must be shared in order for you to be able to use it.

FIGURE 3.54 Connecting to printers



Administrative Shares

Administrative shares are automatically created for administrative purposes on Windows. These shares can differ slightly based on which operating system is running, but they end with a dollar sign (`$`) to make them hidden. There is one for each volume on a hard drive (`C$`, `D$`, and so forth), as well as `admin$` (the root folder—usually `C:\WINDOWS`), and `print$` (where the print drivers are located). These are created for use by administrators and usually require administrator privileges to access. It's important to note that they are hidden shares and that any shared folder created with a trailing `$` will be hidden from the users as well. Unless you know they exist, they will not be visible.

Shared Resources

Although you can access shares on file servers and printers that are shared, there are many other shared resources that can be shared and accessed. Files and printers are just the common resources that make up the file and print sharing service that Windows has had built in, arguably since the inception of MS-DOS. Application sharing is one example of a shared resource. Scanners and faxes are other examples of hardware-based shared resources. These examples only scratch the surface; the cloud is full of applications and shared resources. All these resources can be authenticated with domain-based authentication or separate locally based credentials inside the application.

Firewall Settings

Microsoft Windows has come with a preinstalled firewall since Windows XP Service Pack 2. The addition of the Windows Firewall feature was welcomed at a time in history when the Internet could be described as the *Wild West*. Today Windows Defender Firewall is an integral part of Windows. Surprisingly the interface has not changed all that much from its debut with Windows XP. The original firewall was not turned on by default, but with the introduction of Windows Vista the firewall was on by default in the inbound direction. The firewall is never configured in an outbound direction, but if needed it has the ability.

Windows Defender Firewall is scalable as a host-based firewall, because an average user can configure the firewall. However, if more complicated firewall rules need to be composed, the advanced interface allows for an administrator to intervene. The configuration of the firewall is really straightforward; you can access the basic firewall controls by choosing from the Start menu Settings ➢ Privacy & Security ➢ Open Windows Security ➢ Firewall & Network Protection. This will open the basic Windows Defender Firewall dialog, as shown in Figure 3.55. This dialog box allows you to perform basic firewall tasks, such as turning off the firewall, changing user notification, restoring defaults, and most importantly, allowing an app or feature through the firewall.

A number of applications and services are preconfigured in the firewall. For example, when you share a folder, the ports associated with filesharing are automatically enabled. Another mechanism exists to allow the firewall to easily configure itself; when a program is launched that listens to a port, an Allow or Cancel notification is sent to the user. If you click Show More,

you can select which firewall profile to apply the firewall exception, as shown in Figure 3.56. If the user selects Allow, a rule is added to the firewall for the specific application.

FIGURE 3.55 Windows Defender Firewall

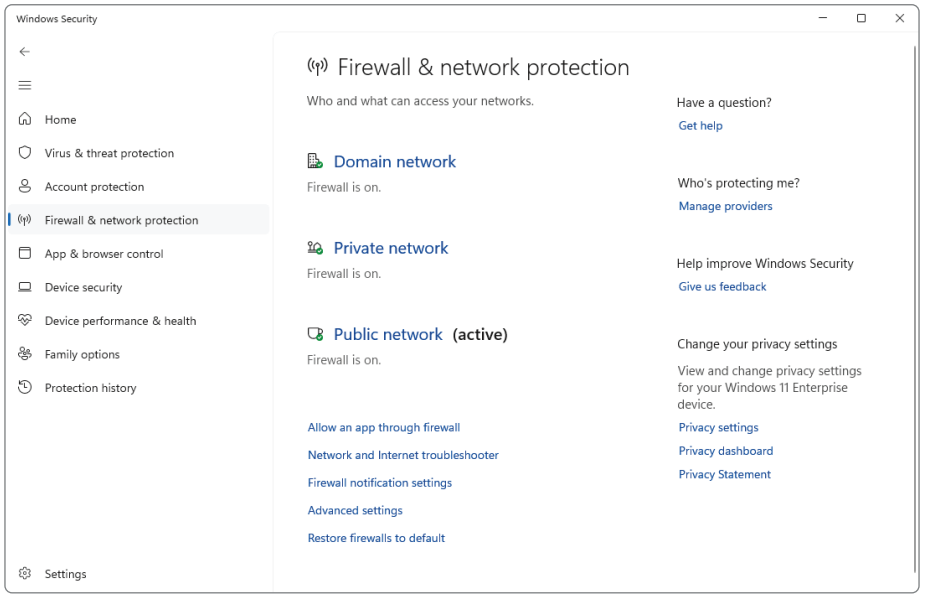
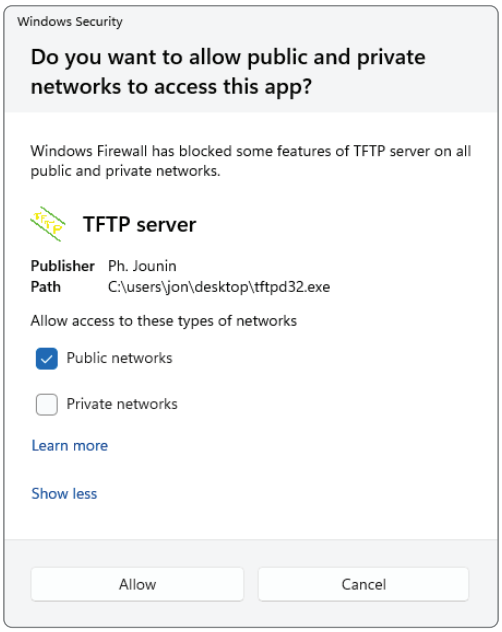


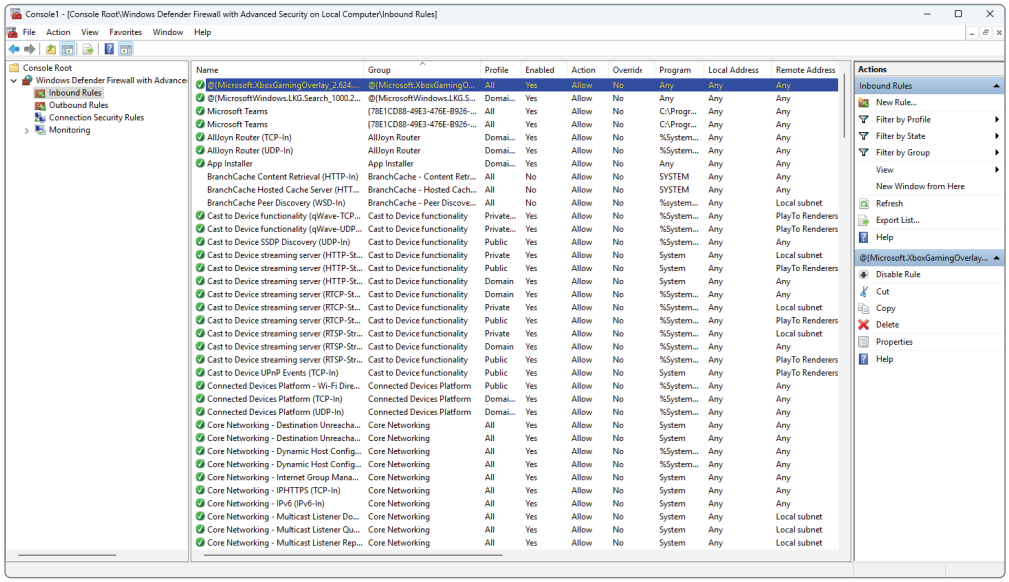
FIGURE 3.56 Windows Defender Firewall notifications



Along with the automated mechanism in which firewall rules can be added, you can specify which network profile they are active in. Network profiles are identified by the MAC address of the default gateway. The firewall will learn the internal network of your home by making a note of the MAC address of your router. If the MAC address has not been seen before by the network operating system, then the firewall will ask if the network is private or public. This way the firewall can behave differently in your home than if you were in an airport or another public setting. There are three different network profiles that firewall rules can be active for: public, private, and domain. You can control if your laptop is public or private, but if a domain controller exists and the laptop is joined to a domain, then the domain profile becomes active.

You can get a more granular view of the firewall by launching the Windows Defender Firewall with Advanced Security MMC. To launch the MMC, from the Start menu choose All ➢ Windows Tools ➢ Windows Defender Firewall With Advanced Security. The individual rules can be examined or individually configured with the MMC, as shown in Figure 3.57. You can see each of the rules along with its effective profile of public, private, domain, or all.

FIGURE 3.57 Windows Defender Firewall with Advanced Security



Adding rules manually allows for maximum granularity but comes with the price of complexity. You can add a rule based on a program, port, predefined rule, or something totally custom. For example, if you wanted to only allow an incoming port of 2233 via TCP to a specific application awaiting its request for a specific network profile, this interface will allow you to do so. If you are the administrator of a domain, you can also create rules inside a Group Policy Object (GPO) and deploy the rules out to a large group of computers.

A Group Policy Object (GPO) is a collection of Group Policy settings. GPOs allow an administrator to change the behavior and appearance of the operating system and control certain aspects of the user.

Interface Firewall Profile

Windows Defender Firewall uses network profiles to apply different security settings based on the type of network you're connected to. This ensures that your device is appropriately protected whether you're at home, at work, or on the go. The mechanism for detecting the network firewall profile to use is based off the gateway's MAC address and your prior answers of private or public. Keep in mind that a private firewall profile will allow certain resources, whereas a public firewall profile will continue to block access to certain resources. The following are the three type of firewall profiles; you can control only two of them: private and public.

Private Network Profile The Private network profile is tailored for environments where you trust the other devices on the network. This could be your home, where you have control over the network and the devices connected to it, or your workplace, where security measures are typically in place. When your device is set to a private network profile, it allows for network discovery, meaning your device can find and interact with other devices on the same network. This is useful for sharing files, using network printers, and other collaborative tasks. Essentially, the private profile assumes a higher level of trust and facilitates easier communication between devices.

Public Network Profile On the other hand, the Public network profile is designed for use in less secure environments, such as coffee shops, airports, or any public Wi-Fi network. In these settings, the risk of encountering malicious devices or individuals is higher. Therefore, the public profile disables network discovery and file sharing features to keep your device hidden from others on the same network. This minimizes the risk of unauthorized access and helps protect your personal information and data from potential threats.

Domain Network Profile The Domain network profile is used when your device is connected to a network with a domain controller, typically found in corporate environments. This profile is automatically applied when your device can authenticate to a domain controller. The domain profile allows for more relaxed security settings compared to the public profile, as it assumes that the network is managed and secured by IT professionals.

By understanding the differences between these profiles and knowing how to switch between them, you can ensure that your device is always using the appropriate security settings for your current environment. To change the profile for the active connection follow these steps:

1. From the Start menu, choose Settings ➢ Network & Internet.
2. Select the interface of either Ethernet or Wireless.
3. Set Network Profile Type with the Public Network and Private Network radio boxes.

Client Network Configuration

If you have a router or a server that supports the Dynamic Host Configuration Protocol (DHCP), the client will automatically configure itself with an IP address, subnet mask, default gateway, and the appropriate Domain Name System (DNS) servers. This is the default behavior for all devices, because it is trouble-free from a user's point of view. For example, if you turn on your laptop at home, your router will serve all the information necessary to get on the home network and the Internet. If you pick up your device and go to work, this process will also happen at your workplace to allow you to connect to servers or the Internet.

Elements of Client Configuration

Configuring a Windows client involves several critical elements that ensure network connectivity and efficient communication between devices. Understanding these elements not only helps in troubleshooting network issues but also enhances overall network performance and security. Let's look at each of the key elements in a Windows client configuration:

Internet Protocol (IP) Addressing Scheme An Internet Protocol (IP) addressing scheme is a systematic method used to assign unique addresses to devices on a network. In a Windows client configuration, IP addresses can be assigned either statically or dynamically. Static IP addressing involves manually setting a fixed IP address for a device, ensuring it remains constant over time. Dynamic IP addressing, on the other hand, uses DHCP to automatically assign IP addresses from a pool, allowing for efficient management and reuse of addresses.

IP addressing was covered in detail in Volume 1 Chapter 7, "Introduction to TCP/IP."

Domain Name System (DNS) Settings Domain Name System (DNS) settings in a Windows client configuration are essential for translating human-readable domain names into IP addresses that computers use to identify each other on the network. Proper DNS configuration ensures that when a user types a web address into their browser, the DNS server quickly resolves it to the corresponding IP address. Windows clients can be configured to use specific DNS servers, either provided by the Internet service provider (ISP) or custom servers like Google DNS or OpenDNS, to enhance speed, reliability, and security of domain name resolution.

Subnet Mask A subnet mask is a critical component in a Windows client configuration that defines the network and host portions of an IP address. It works in conjunction with the IP address to determine which part of the address identifies the network and which part identifies the individual device. For example, a common subnet mask of 255.255.255.0 indicates that the first three octets represent the network, while the last octet represents the host.

Gateway The gateway in a Windows client configuration acts as an intermediary device that routes traffic from the local network to other networks, including the Internet. It serves as the exit point for data leaving the local network and the entry point for data coming into the local network. Configuring the correct gateway address ensures

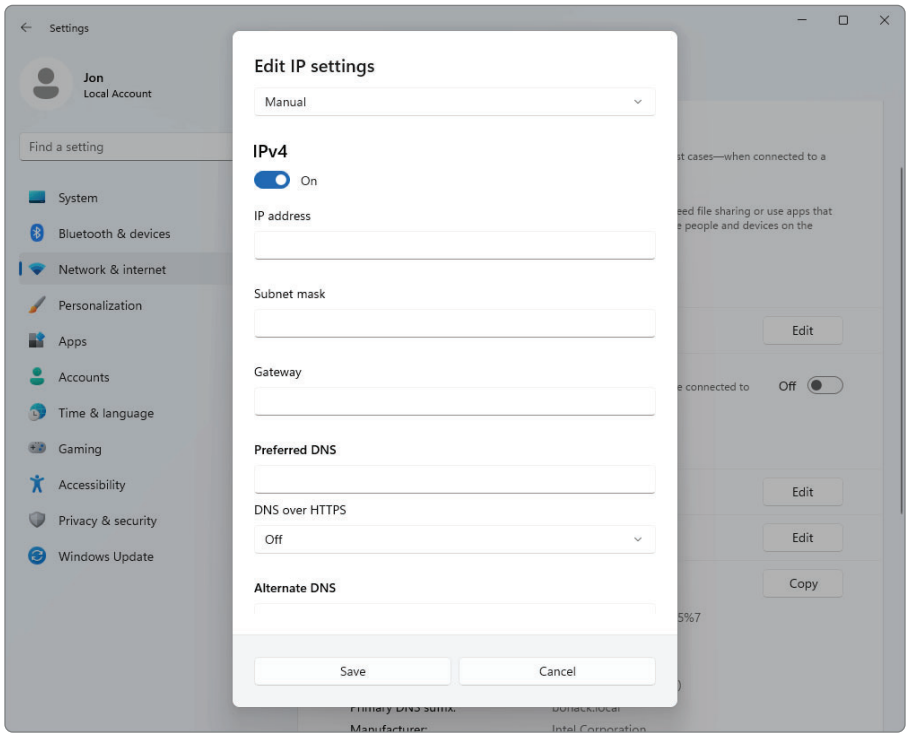
that devices on the network can communicate with external networks and access online resources. Typically, the gateway is the IP address of a router or firewall that connects the local network to the broader Internet.

Configuration of a Static IP Address

If you need to statically configure your network settings, that will require some planning and manual configuration. The first thing you will need is an IP address that is not used by another computer in the network. The subnet mask will also need to match the network you will configure the computer in. If you want to communicate outside the immediate network, you will need a default gateway, which is your router's IP address. A DNS address is also required if you want to translate simple domain names to IP addresses. There are two ways to configure the static IP address: by using the new Settings app and by using the legacy Control Panel applet. You should be familiar with both ways, as the legacy Control Panel applet offers more features, such as alternate IP address configuration.

Settings App You can configure the network addressing via the Settings app by clicking Start menu ➢ Settings ➢ Network & Internet ➢ Properties, and then clicking Edit under IP Assignment. You can then choose Manual from the drop-down menu to open two switches for IPv4 and IPv6. You can then click the switch for IPv4 and enter the IP address, subnet mask, gateway, and DNS settings, as shown in Figure 3.58.

FIGURE 3.58 Settings App network configuration

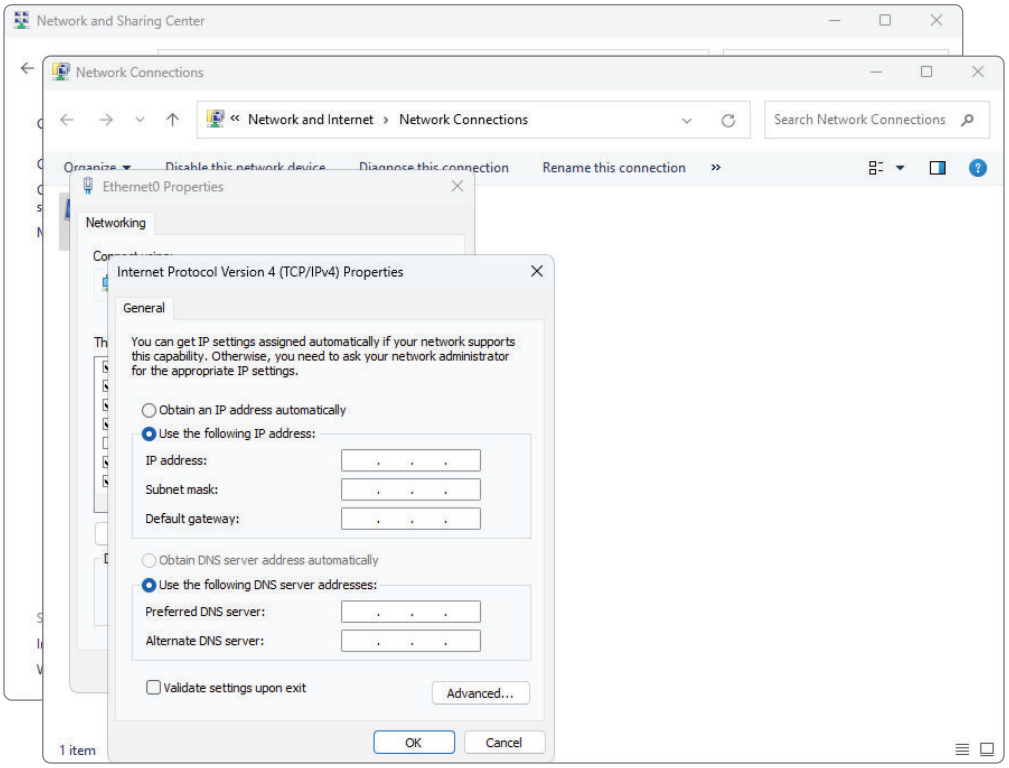


Control Panel Applet The original Control Panel applet has been around since Windows NT 4.0, and you can get there a number of ways. To open the dialog, first open Control Panel, then click Network And Sharing Center. When the dialog box opens, select Change Adapter Settings from the left side and then right-click the network adapter and select Properties. You can then double-click Internet Protocol Version 4 (TCP/IPv4). From here you can select Use The Following IP Address and manually enter the IP address, subnet mask, and gateway, as shown in Figure 3.59. As you can see, this method is not easier to get to, but it is familiar to a seasoned professional.

Configuring an Alternative IP Address in Windows

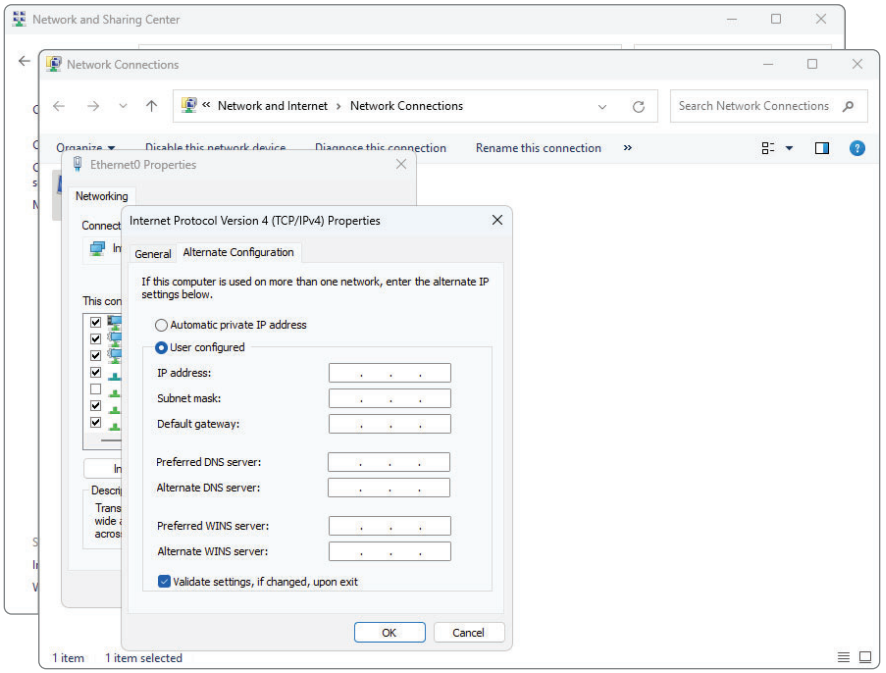
Windows 11 and prior versions of Windows also allow for the use of an alternate IP address—that is, an address configured for the system to use in the event the first choice is not available. In order for an alternate configuration to be set, the first choice has to be dynamic; the tab becomes visible only when the General configuration is set to Obtain An IP Address Automatically (as shown in Figure 3.59), and the alternate is used only if the primary address cannot be found/used, such as when the DHCP server is down. The Alternate IP address configuration is only available with the Control Panel applet. The following is a list of the elements you can configure for an alternative IP address:

FIGURE 3.59 Control Panel network configuration



IP Addressing Two radio buttons exist on the Alternate Configuration tab: Automatic Private IP Address and User Configured (see Figure 3.60). The first option is the default, meaning that the alternate address used is one in the APIPA range (`169.254.x.x`). Selecting User Configured requires you to enter a static IP address to be used in the IP Address field. The entry entered must be valid for your network in order for it to be usable.

FIGURE 3.60 Alternate Configuration tab



Subnet Mask When the User Configured radio button is selected on the Alternate Configuration tab, you must enter a value in the Subnet Mask field. This value must correspond with the subnet values in use on your network and work with the IP address you enter in the IP Address field.

DNS When the User Configured radio button is selected on the Alternate Configuration tab, you should enter values in the Preferred DNS Server and Alternate DNS Server fields. These entries are needed in order to translate domain names into IP addresses.

Gateway When the User Configured radio button is selected on the Alternate Configuration tab, you must enter a value in the Default Gateway field. This value must correspond with the subnet values and the IP address that you enter. This address identifies the router to be used to communicate outside the local network.

Summary

In this chapter, you learned the various options for the installation and upgrade of Windows. Both the installation and the upgrade process were covered in great detail, so you could see what happens in each step. We also covered the various ways to install Windows, deploy images, and recover Windows when things go wrong.

In addition, you learned some of the command-line tools that can be used to administer Windows. We covered basic Windows commands to view, create, and navigate files and folders. We then focused on commands that help you administer and diagnose the network. We also covered commands that help you manage disks and filesystems. We then explored the most important aspect of the command line: getting help.

We concluded the chapter by covering the various network models that Windows is deployed in. You learned about various authentication methods, how to access resources, and how to configure the built-in firewall to allow applications to be accessed via the network.

Exam Essentials

Know the Windows installation process and options. You should know the various options when installing Windows 11, including the various partitions created and the differences between BIOS and UEFI firmware. You should also be familiar with the Windows 11 installation process.

Know which types of installations are possible with Windows. You should know which operating systems can be upgraded to Windows 11 and which require a clean installation.

Understand upgrading to Windows. You should know that a custom installation either wipes the old system or replaces the existing system, putting the old files into `WINDOWS.OLD`. Applications then have to be reinstalled, and user data has to be migrated from the old system using tools such as USMT. An upgrade preserves the existing applications and the user data, moving them into the new operating system.

Understand what each of the command-line utilities does. Many utilities allow you to perform various functions at the command line. These include `cd`, `dir`, `ipconfig`, `ping`, `netstat`, `nslookup`, `net use`, `tracert`, `pathping`, `chkdsk`, `format`, `diskpart`, `md`, `rmdir`, `robocopy`, `hostname`, `net user`, `winver`, `whoami`, `[command name] /?`, `gpupdate`, `gpresult`, and `sfc`. The `cmd` command opens a command prompt, where you can type the rest of the commands. If you're not sure how to use a particular utility, using the `/?` switch at the end of the command will provide information about how to use it.

Understand networking in Windows. You should understand the different networking models in which a Windows operating system can participate: workgroups, domains, and Microsoft Entra ID. You should also understand the basic user authentication methods used with each of these networking models, as well as common methods for accessing networking resources.

Know the various ways to establish a connection. You should know how to connect and diagnose wired and wireless connections to a network. In addition, you should understand how a WWAN is configured and how VPN networks are used and configured.

Know the various ways to access resources. You should know how to connect to network shares and printer shares using the UNC path of the resource. In addition, you should know the other types of resources, such as administrative shares and other hardware-based resources that you may encounter.

Know the basic elements of client configuration. You should be familiar with the Internet Protocol (IP) addressing scheme, Domain Name System (DNS) settings, subnet mask, gateway, and static vs. dynamic IP addressing.

Review Questions

The answers to the chapter review questions can be found in Appendix A.

1. You are tasked with installing Windows 11 on a new computer. The computer has a 1 GHz processor, 2 GB of RAM, and 32 GB of storage. What should you do?
   - **A.** Proceed with the installation.
   - **B.** Upgrade the RAM to at least 4 GB.
   - **C.** Increase the storage to 64 GB.
   - **D.** Both B and C.
2. What is the name of the pass that detects hardware and installs drivers?
   - **A.** Generalize
   - **B.** OOBE
   - **C.** Specialization
   - **D.** WinPE
3. Which partition contains the recovery utility in the event of a problem?
   - **A.** BCD
   - **B.** System
   - **C.** ESP
   - **D.** WinRE
4. Which tool is used to create an operating system image for deployment?
   - **A.** WDS
   - **B.** MAP toolkit
   - **C.** USMT
   - **D.** `sysprep`
5. Which command will allow copying of all data to include NTFS permissions?
   - **A.** `xcopy`
   - **B.** `copy`
   - **C.** `chkdsk`
   - **D.** `robocopy`
6. Which Windows Update branch for Windows will install features as they are released to the general public?
   - **A.** Semi-Annual Channel
   - **B.** General Availability
   - **C.** Long-Term Servicing Channel
   - **D.** Insider Program

7. Which edition of Windows 11 will not allow for Insider Program branch releases?
   - **A.** Home
   - **B.** Pro
   - **C.** Education
   - **D.** Enterprise

8. You are configuring a new Windows 11 installation and need to ensure it supports Secure Boot. What type of firmware must the computer have?
   - **A.** BIOS
   - **B.** UEFI
   - **C.** Legacy BIOS
   - **D.** EFI

9. Which command is used to measure packet loss as a packet travels to a destination address?
   - **A.** `ping`
   - **B.** `nslookup`
   - **C.** `pathping`
   - **D.** `tracert`

10. Which tool allows you to report a remote computer's inventory of hardware?
    - **A.** `regedit.exe`
    - **B.** `msinfo32.exe`
    - **C.** `msconfig.exe`
    - **D.** `dxdiag.exe`

11. Which command will allow you to check a volume for corruption?
    - **A.** `diskpart`
    - **B.** `format`
    - **C.** `chkdsk`
    - **D.** `sfc`

12. Which restriction will be imposed on Windows users until they activate the operating system?
    - **A.** They won't be able to launch applications.
    - **B.** They won't be able to run Windows Updates.
    - **C.** They won't be able to change the wallpaper.
    - **D.** They won't be able to browse the Internet.

13. What is the name of the pass that configures and creates the user environment during the setup process?
    A. Generalization
    B. OOBE
    C. Specialization
    D. WinPE

14. When installing Windows, you can control telemetry data. In which step during setup is this configured?
    A. Privacy settings
    B. Locale options
    C. Partitioning options
    D. Account creation

15. You are troubleshooting a network issue and need to verify the path packets take to a destination. Which command should you use?
    A. `ping`
    B. `tracert`
    C. `nslookup`
    D. `netstat`

16. You want to upgrade from 32-bit Windows 10 Professional to a 64-bit version of Windows 11. What are your options?
    A. Upgrade directly to Windows 11 Pro 64-bit.
    B. Upgrade first to Windows 11 Pro 32-bit, and then upgrade to 64-bit.
    C. Upgrade to Windows 10 64-bit, and then upgrade to Windows 11 64-bit.
    D. You must perform a clean installation of Windows 11 Pro 64-bit.

17. Where is the boot configuration data stored on a Windows installation using EFI?
    A. EFI System Partition
    B. WinRE partition
    C. Secure Boot partition
    D. `C:\WINDOWS`

18. Which command is used to identify ports in use by applications and the operating system?
    A. `netstat`
    B. `ipconfig`
    C. `pathping`
    D. `nslookup`

19. Which type of connection is configured in the Cellular screen in the Settings app?

A. Wired

B. Wireless

C. WWAN

D. VPN

20. What is the maximum number of concurrent connections that can be made to a Windows workstation?

A. 30 connections

B. 40 connections

C. 20 connections

D. 25 connections

Performance-Based Question

You will encounter performance-based questions on the A+ exam. The questions on the exam require you to perform a specific task, and you will be graded on whether or not you were able to complete the task. The following requires you to think creatively in order to measure how well you understand this chapter's topics. You may or may not see similar questions on the actual A+ exam. To see how your answers compare to the authors', refer to Appendix B.

You need to join a Windows 11 workstation to an Active Directory domain. What are the steps you need to follow to complete the task?

Chapter

4

Working with macOS and Linux

THE FOLLOWING COMPTIA A+ 220-1202 EXAM OBJECTIVES ARE COVERED IN THIS CHAPTER:

✓ **1.0 Operating Systems**

✓ **1.8 Explain common features and tools of the macOS/ desktop operating system.**

- Installation and uninstallation of applications
  - File types
    - .dmg
    - .pkg
    - .app
  - App Store
  - Uninstallation process
- System folders
  - /Applications
  - /Users
  - /Library
  - /System
  - /Users/Library
- Apple ID and corporate restrictions
- Best practices
  - Backups
  - Antivirus
  - Updates/patches
  - Rapid Security Response (RSR)
- System Settings

- Displays
- Networks
- Printers
- Scanners
- Privacy
- Accessibility
- Time Machine

- Features
  - Multiple desktops
  - Mission Control
  - Keychain
  - Spotlight
  - iCloud
    - iMessage
    - FaceTime
    - Drive
  - Gestures
  - Finder
  - Dock
  - Continuity
- Disk Utility
- FileVault
- Terminal
- Force Quit

✓ **1.9 Identify common features and tools of the Linux client/desktop operating system.**

- File management
  - ls
  - pwd
  - mv

- cp
- rm
- chmod
- chown
- grep
- find

- Filesystem management
  - fsck
  - mount
- Administrative
  - su
  - sudo
- Package management
  - apt
  - dnf
- Network
  - ip
  - ping
  - curl
  - dig
  - traceroute
- Informational
  - man
  - cat
  - top
  - ps
  - du
  - df
- Text editors
  - nano

- Common configuration files
  - /etc/passwd
  - /etc/shadow
  - /etc/hosts
  - /etc/fstab
  - /etc/resolv.conf
- OS components
  - systemd
  - kernel
  - bootloader
- Root account

CompTIA has acknowledged that system administrators and technicians are increasingly dealing with more than just Windows on a daily basis. Therefore, they have included exam objectives based on *macOS* and *Linux*.

This chapter looks at the non-Windows operating systems from the standpoint of what you need to know to pass the exam. All the topics relevant to Exam Objectives 1.8 and 1.9 of the 220-1202 exam are covered.

macOS and Linux

In the beginning there was UNIX. UNIX System 5 (version 5) is an operating system originally created and licensed by AT&T Labs. The UNIX operating system is considered to be the root of all UNIX-based operating systems. In the mid-1970s, the University of California at Berkeley (UC Berkeley) licensed UNIX from AT&T for their computer systems and expanded on the tools shipped in the original version of UNIX. These tools became the foundation of UNIX as it is today, but UC Berkeley only licensed the operating system for specific machines. The students quickly became upset and developed and released a version of UNIX called the Berkeley Software Distribution (BSD). The term *distribution* is used today with UNIX/Linux operating systems to define the operating system and its ecosystem for application management, patching, and upgrades.

Although it began with UNIX, BSD became very popular because it was an open source license. This allowed everyone to use the operating system on any computer system they wished. In the mid-1980s, Steve Jobs created a company called NeXT and built computers that furnished the NeXTSTEP operating system. The NeXTSTEP operating system was originally built from BSD version 4.3. Unfortunately, the NeXT computer company never really took off. However, the NeXTSTEP operating system was acquired by Apple and eventually became the macOS we know today.

Linux has a very different origin story from macOS. Actually, Linux has nothing to do with the original codebase of UNIX. In the mid-1990s a Finnish student named Linus Torvalds set out to create a completely open source operating system for the world to use. Linux was the result of his efforts; it was designed from scratch, so it was completely free for anyone to use or incorporate into their own products. Today you can find a great number of Linux distributions, such as *Ubuntu*, *Debian*, Arch Linux, Gentoo, *Red Hat* . . . and the list goes on.

Although Linux has a completely different codebase from BSD and UNIX, the operating system itself functions similarly. Only the kernel and interworkings of the OS are different.

Many of the applications that were created by students on the BSD platform were ported over to Linux. Functionally, the operating systems are very similar in design and usability.

The complete history of UNIX is vast and interesting; we have covered only the big events in the UNIX timeline. Many different organizations were involved, spanning all the way back to the 1960s. Learn more about the history of UNIX here: `https://UNIX.org/what_is_UNIX/history_timeline.html`.

Applications on macOS

The applications that are available or installed on an operating system indirectly define an OS by extending functionality to the end user. As with the case of UNIX, BSD, Linux, and macOS, all of the basic command-line applications are similar in functionality, as you will learn in this chapter. These command-line applications are preinstalled with the OS distribution. When you need functionality that isn't part of the base OS, you have a few different ways to install applications.

Exam Objective 1.8, "Explain common features and tools of the macOS/desktop operating system," requires you to know about installing applications on macOS, system folders, Apple ID, best practices, system preferences, and features of macOS.

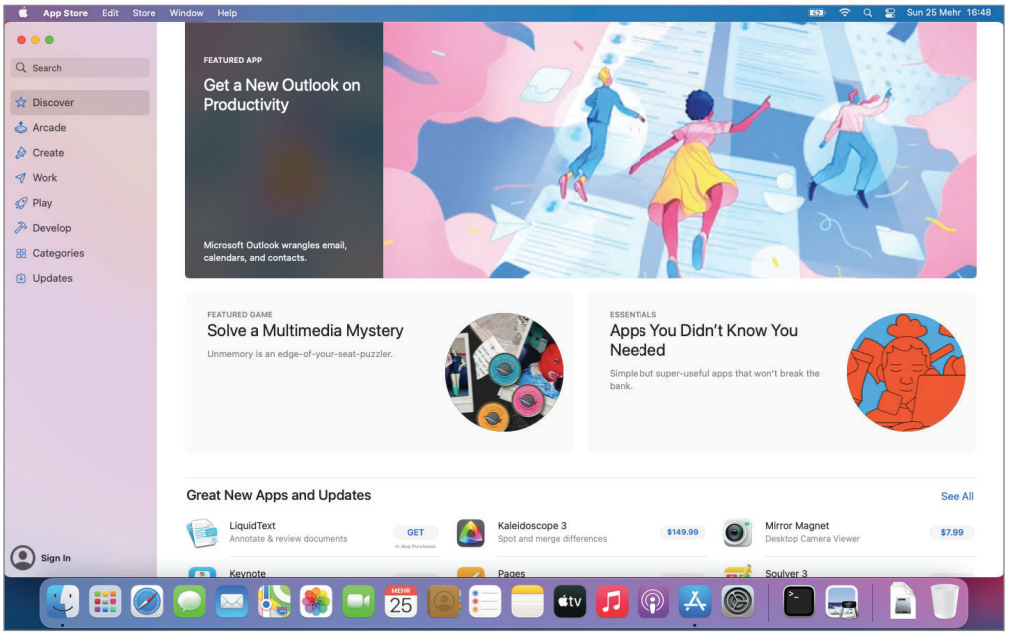
Installing Applications from the App Store

macOS contains an application ecosystem called the *App Store*, shown in Figure 4.1. From the App Store you can download and buy applications for macOS. Some applications might be free from the App Store, but they contain in-app purchases. They are called *freemium* games and applications, and they typically use this feature.

Applications installed through the App Store do not function any differently from legacy downloaded applications. Unfortunately, not all applications are available through the App Store. The developer of an application must publish it on the App Store, and not every vendor will submit their application. This is mainly due to the costs related to publishing an app to the App Store.

The benefit of using the App Store to install applications for macOS is if you purchase an app and have two devices, then you need to purchase the application only once. This is assuming you are logged in with the same Apple ID that purchased the application. Also, updates for the applications are automatically installed, compared to downloading applications from the developer directly. The Updates section in the App Store will show you all the applications containing updates.

FIGURE 4.1 macOS App Store



When you first turn on your Apple device and run through the installation, it will ask for an *Apple ID*. You really can't move on with the setup without an Apple ID. The Apple ID is your digital identity on the device, and it's what also ties your Apple Wallet to the App Store for purchases. The Apple ID will contain credit card information that can be used for purchases.

An organization may use mobile device management (MDM) software to control the installation of applications from the App Store or any software installations in general. When an organization uses MDM to restrict the installation of applications, this is considered corporate restrictions. An organization might also use the App Store as a distribution point for applications that its employees are expected to use. The organization's MDM software is associated with the employee's Apple ID, which is usually the employee's email address. These features were introduced in macOS starting with version 10.9. New features are introduced in every version of macOS.

Installing Downloadable Applications

Installing downloadable applications on macOS is not much different from the same process in Windows. The process consists of three steps: providing the installation to the operating system, mounting the installation, and then installing the applications.

The first step requires you to provide the application's installation files to the operating system. This step is normally done nowadays as a download, since we always want the latest updated applications. The process is pretty routine, as you can imagine; you navigate to a

web page and download the app. However, providing the application might also be as simple as plugging in a USB drive. macOS will automatically mount the USB drive onto the desktop and open the root folder.

The second and third step in the installation process is to launch the file you downloaded and install the application. When you launch the file, what will happen next depends on the type of file. There are a number of different file types that can be downloaded from the Internet, such as ZIP, ISO, DMG, PKG, or APP, just to name a few. The most common files for macOS applications are as follows:

APP Files The APP filename will end in `.app` and it is used to contain the application. The APP file is actually a container for various resource files and folders, and it also contains the executable for launching the application. You will rarely download an APP directly from the Internet. The APP file will often be found inside a ZIP, DMG, or ISO file or physical media.

DMG Files A DMG filename will end in `.dmg` and it stands for disk imaging file. The DMG file is directly comparable to ISO or ZIP files. The DMG file contains a file structure, and the file can be mounted as a disk on the macOS desktop. DMG files used for application installation will normally contain two files: the application itself and a shortcut to the `Applications` folder. This allows you to drag (copy) the APP file into the `Applications` folder, and doing so will install the application.

PKG Files A PKG filename will end in `.pkg` and it is an automated package installer. The PKG file is comparable to the application installation programs that are commonly found in Windows. When the PKG file is launched, it will ask the user a series of questions about the installation. PKG files are found inside DMG, ZIP, and physical media. These files are used when the installation requires customization by the end user.

Exam Objective 1.8 specifically calls out DMG, PKG, and APP file types. Be sure you are familiar with them for installation of applications in macOS.

In Exercise 4.1, you will install an application on macOS to understand the process.

EXERCISE 4.1

Installing Applications on macOS

1. Open the Safari app from the Dock.
2. When Safari opens, type **`sublimetext.com`** in the address bar and press Return.
3. When the web page loads, choose the download option of macOS.
4. A prompt will ask you to allow downloads from `www.sublimetext.com`; click Allow.
5. The download is stored in the `Downloads` folder located on the right side of the Dock. Click the `Downloads` folder and select the application `Sublime Text Build 4189.dmg`.

6. After macOS mounts the DMG as a drive and opens the folder, drag the Sublime Text icon over the Applications shortcut and release it.
7. You can then eject the Sublime Text mount on the desktop by selecting the folder and clicking Finder ➤ File ➤ Eject.
8. Press Shift+Command+A to open the `Applications` folder.

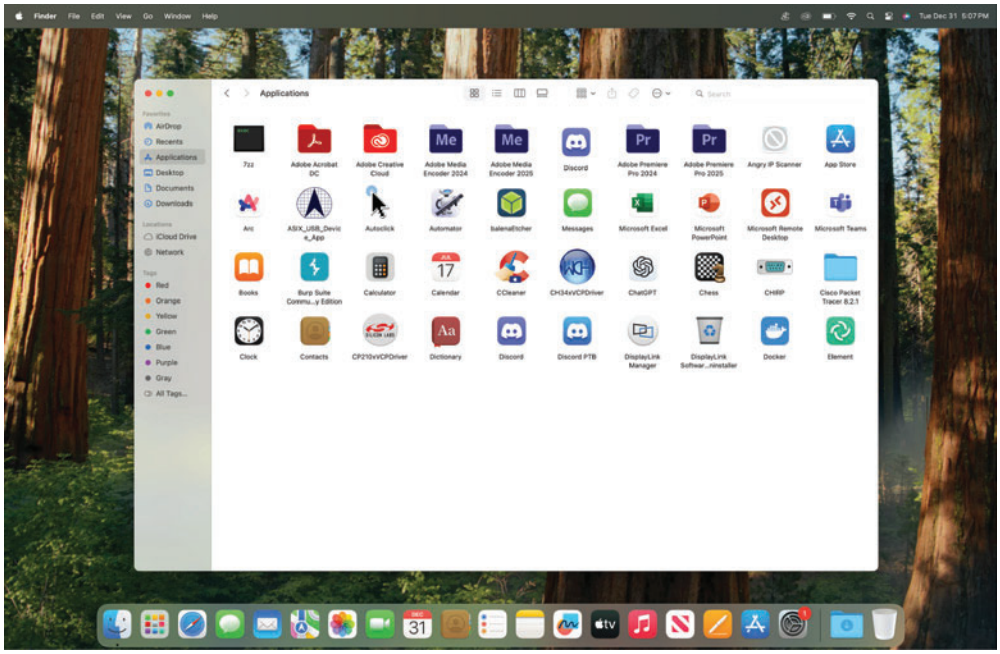
Managing Applications

The `Applications` folder contains all the applications installed on macOS, as shown in Figure 4.2. You can view and manage applications inside this folder. The two most common methods of launching the folder are using the *Finder app* and selecting Go ➤ Applications or pressing Shift+Command+A. This key sequence will also launch the `Applications` folder and requires no shuffling of applications to get to the Finder app.

From the `Applications` folder, you can delete user-installed applications. However, preinstalled applications in this folder cannot be deleted by the user, since they are technically part of the operating system. These applications are similar to the built-in applications on Windows.

To delete a user-installed application, select the folder; then on the Finder menu, select File ➤ Move To Trash. This will place the application into the Trash and remove it from the operating system. Another method is to select the application and press Command+Delete. In Exercise 4.2 you will uninstall an application to understand the process better.

FIGURE 4.2 macOS `Applications` folder



EXERCISE 4.2

Uninstalling Applications on macOS

1. Press Shift+Command+A to open the `Applications` folder.
2. Select the Sublime Text application.
3. On the Finder menu, click File ➢ Move To Trash.
4. Open the `Trash` folder from the right side of the Dock and observe that the Sublime Text application is there.

Creating Shortcuts

The last and most important topic we will cover about applications is the availability of the application. Creating shortcuts is an essential part of your overall workflow. You should be able to launch an application in as few clicks as possible. To create a shortcut, launch the `Applications` folder by pressing Shift+Command+A and drag the application to the *Dock*. The dock is covered later in this chapter in the "macOS Components" section.

macOS Components

macOS is renowned for its robust system utilities and comprehensive settings that empower users to optimize their computing experience. From essential tools like Disk Utility, which manages and repairs your storage devices, to the Accessibility settings that ensure inclusivity for all users, macOS offers a wide array of features designed to enhance productivity and usability. The System Settings provide a centralized hub for customizing your Mac, allowing you to tailor everything from privacy controls to display preferences. Additionally, features like Continuity and iCloud integration ensure a seamless experience across all your Apple devices. In the following sections, we'll delve deeper into these utilities and settings, exploring how they can help you get the most out of your macOS environment.

Interestingly, macOS shares several similarities with Windows 11, making it easier for users familiar with one operating system to transition to the other. Both operating systems offer centralized settings menus—System Settings in macOS and Settings in Windows 11—where users can manage system preferences, update software, and customize their experience. Additionally, both macOS and Windows 11 provide robust accessibility features, ensuring that users with diverse needs can effectively use their devices. The integration of cloud services, such as iCloud for macOS and OneDrive for Windows 11, further enhances the user experience by enabling seamless access to files and data across multiple devices. This familiarity in structure and functionality helps users navigate and utilize the features of both operating systems with ease.

Tools

There are a number of tools to be aware of in macOS and Linux. Most of these have counterparts in the Windows world, and we'll make comparisons where they apply. Tools are released on a daily basis, but the following are the most important for the CompTIA exam objectives and daily maintenance:

Restore/Snapshot *Snapshots* are archives of key files and settings as they exist at a moment in time. It is possible to take snapshots and restore them as needed. The macOS Time Machine feature performs a backup of your data to the external drive connected to the system. However, the external drive may not always be connected to the system, so local snapshots are also created. These local snapshots allow you to restore files using the local filesystem. To use local snapshots, start Time Machine, and if the external storage device is not connected, you'll be presented with the local snapshots. Snapshots in macOS are similar to Windows 11 System Restore points.

Image Recovery As a general rule, images are typically larger than snapshots. You can take a snapshot of a project, and that will include all the files associated with the project, whereas an image would include the project files and all files on the system at the time. Again, this is only a general rule, since images can be granular as well. Typically, however, snapshots are thought of as subsets of images.

The macOS *Disk Utility* can be used to create an image of the macOS operating system, and the image can be directed to an external storage device. Linux can use a multitude of open source tools to create an image of the operating system. The most common is the `dd` command.

Disk Maintenance Utilities A number of disk maintenance utilities are available with, or for, macOS and Linux. The most important utilities you should know are `du`, which shows how much disk space is in use; `df`, which shows how much space is free; and `fsck`, which checks and repairs disks. Although these command-line tools are available on both macOS and Linux, Disk Utility is available only on macOS. The Linux operating system can use a variety of open source disk utilities.

Disk Utility on macOS is a versatile tool designed for managing your disks and storage devices. It allows you to perform a variety of tasks such as formatting drives, creating disk images, and repairing disk errors. You can also use Disk Utility to partition your drives, and securely erase data. To launch Disk Utility, open Finder ➢ Applications ➢ Utilities ➢ Disk Utility. This powerful tool is essential for maintaining the health and efficiency of your Mac's storage system. The macOS Disk Utility is shown in Figure 4.3. This utility is comparable with the Windows Disk Management MMC.

FileVault The *FileVault* tool allows you to provide full-disk encryption to prevent unauthorized access to the information contained on the device during startup. FileVault uses XTS-AES-128 with a 256-bit key. The feature first appeared in OS X 10.3 (2003), and it is currently on its second release (FileVault 2). It can be enabled by navigating to

System Settings ➢ Security & Privacy ➢ FileVault, and from there you can click Turn On FileVault, as shown in Figure 4.4. This utility is comparable to the Windows 11 BitLocker functionality.

FIGURE 4.3 macOS Disk Utility

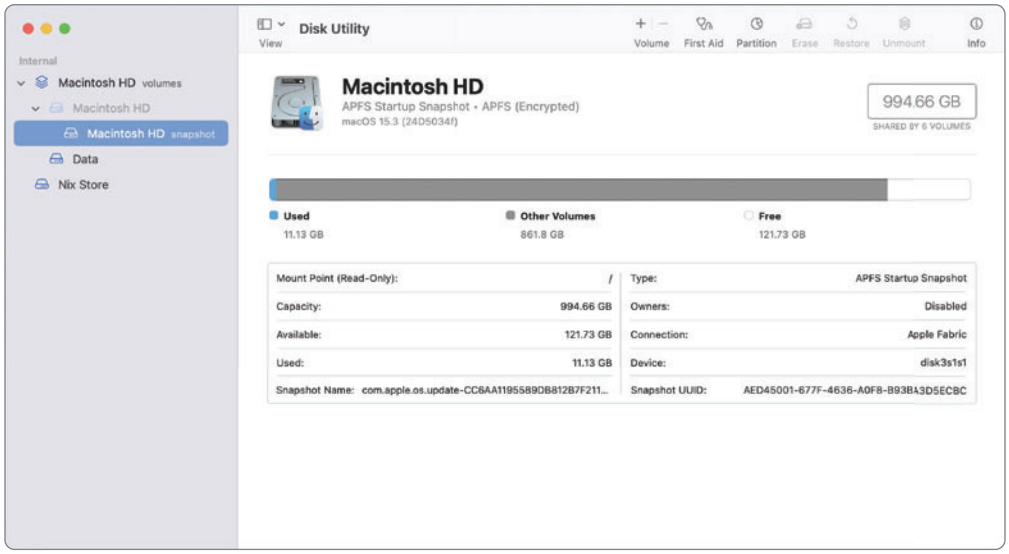
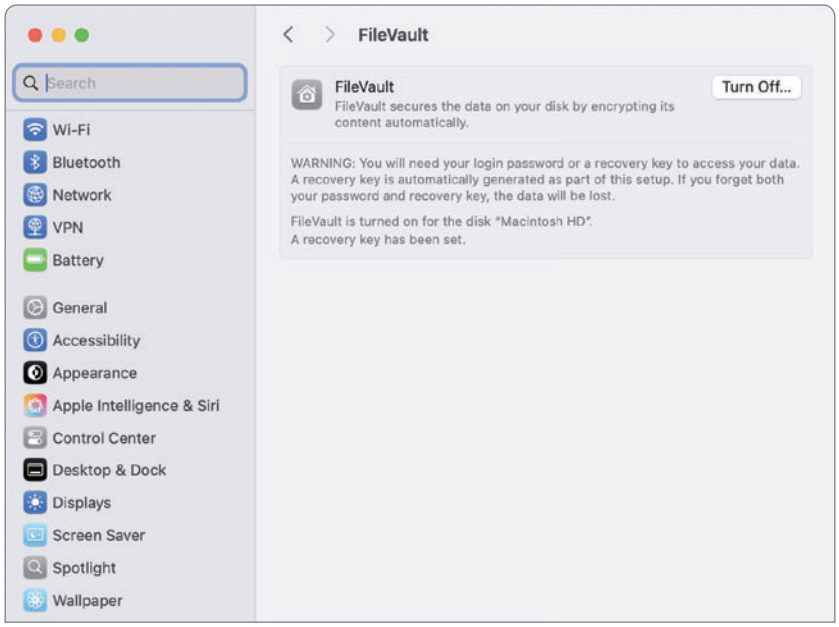


FIGURE 4.4 FileVault encryption



Shell/Terminal The command-line interface in Linux is usually reached through a Terminal session (though that need not always be the case). It is in this Terminal session that you interact with the *shell*—the interpreter between the user and operating system. The most popular shell for Linux today is *Bash (an acronym for Bourne Again Shell)*, but `csh` (C-shell), `ksh` (Korn shell), and a number of others are also in use. In the macOS/iOS world, *OpenSSH (open shell)* is often downloaded and installed. The macOS Terminal utility is accessible by going to Finder ➢ Applications ➢ Utilities ➢ Terminal. Windows 11 has also adopted the concept of a terminal for shell access, this feature is comparable to the Windows Terminal.

Screen Sharing This feature, built into macOS, enables you to share your screen with others. While your screen is shared, the user of the other computer sees what you see, and they can open, move, and close files as well as restart your system. It requires a *Virtual Network Computing (VNC)* connection client.

The Internet offers numerous versions of VNC connection clients—many are free, and some have unique features that you must pay for. An example of a free VNC connection client is TightVNC (`www.tightvnc.com/download.php`).

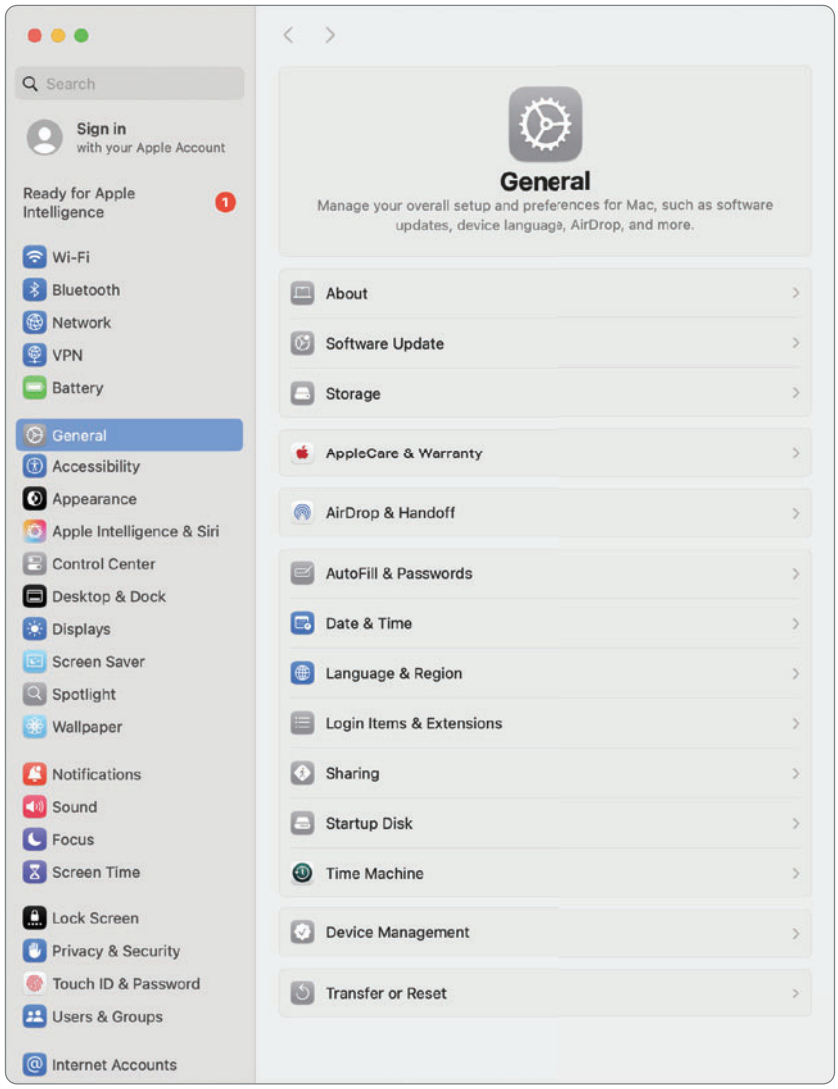
Force Quit Sometimes an application can become unresponsive. When that happens in macOS, you can use *Force Quit* to force the application to close. The Force Quit dialog box can be accessed by pressing Option+Command+Esc. This is similar to pressing Ctrl+Alt+Delete on Windows. Alternatively, you can choose Force Quit from the Apple menu in the upper-left corner of the screen. Most devices, whether running macOS or iOS, offer similar options. With an iPad, for example, if you press the Home button twice quickly, a list of apps will appear at the bottom of the screen. Swipe to get to the app in question, and then swipe up on its preview to close it.

System Settings

Like any other modern-day operating system, macOS is highly customizable. macOS has a feature called System Settings that allows you to customize the operating system, similar to the Windows 11 Settings app and Control Panel applets. In this section, we will cover the most important System Settings. After reading this section, you'll understand how to configure macOS and personalize it for your needs.

All settings are accessed through the System Settings screen, shown in Figure 4.5. The System Settings screen replaced the System Preferences screen starting in macOS Ventura 13.7.2. The System Settings screen can be accessed in a number of ways. The easiest way to launch it is from the Dock, but you can also launch it by clicking the Apple icon in the upper left of the desktop and selecting System Settings.

FIGURE 4.5 macOS General System Settings

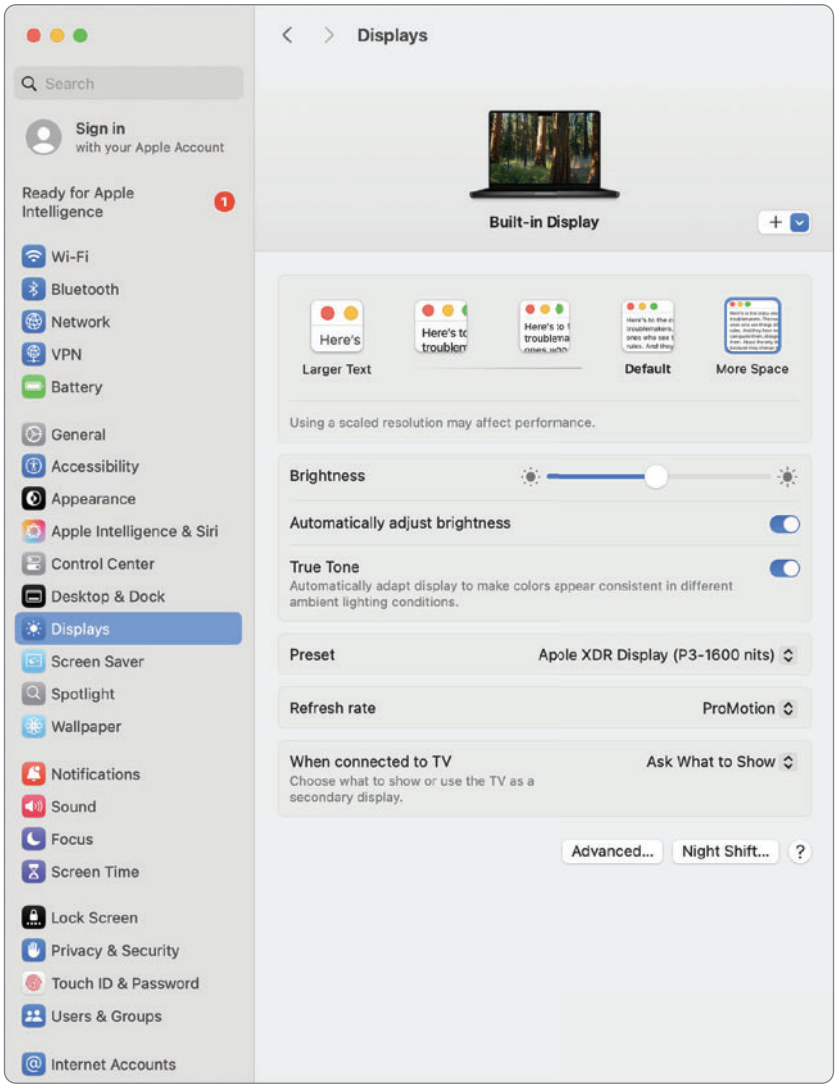


The number of icons on the System Settings screen depends on the applications installed on the operating system and if they are configurable. Let's explore the various System Settings that you need to know for the CompTIA exam:

Displays The settings on the Displays tab are primarily used to adjust display resolution and brightness, as shown in Figure 4.6. The Displays tab will also allow you to adjust the operating system if you have multiple monitors connected. You can also

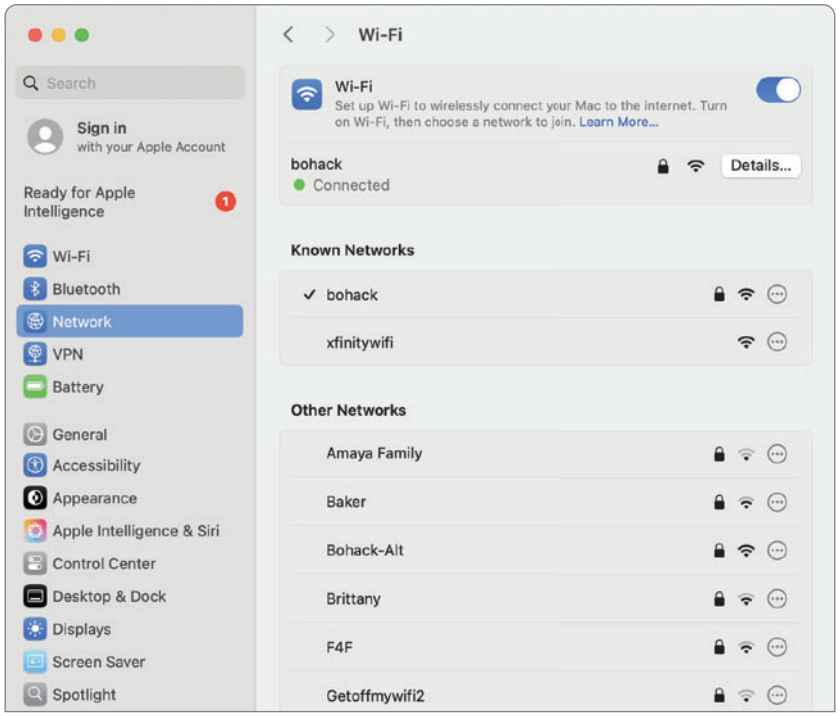
adjust the color tones the display produces by selecting the Color Profile and calibrating the display profile. This setting is generally used when editing graphic artwork is the primary function of the device. The Night Shift option is used to adjust colors at night so that you can get a better night's sleep. This feature is primarily used with laptops, since, for example, you may check the news or catch up on emails before you fall asleep. The device will automatically make the system colors warmer so that it is easier to fall asleep afterward.

FIGURE 4.6 Displays tab



Network The Network tab contains settings pertaining to the network connectivity for the device, as shown in Figure 4.7. In the figure, Wi-Fi (wireless) is the main connectivity method. If the device used a wired connection, the connectivity method would be Ethernet. You can open the Network settings by clicking the Apple icon in the upper left of the desktop and selecting System Settings and then Network.

FIGURE 4.7 Network tab

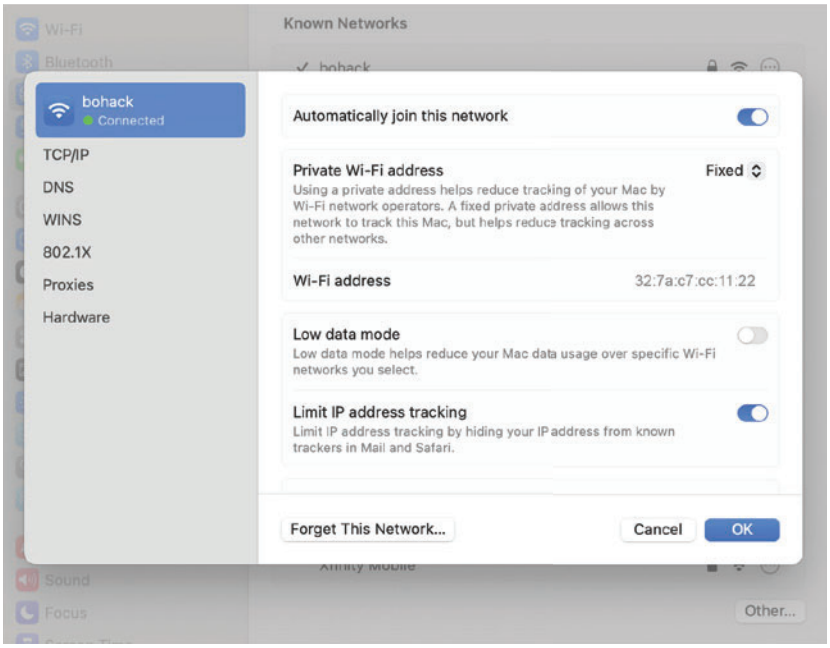


Just like Windows 11, macOS automatically detects locations, and you can change your Network setting based on these locations. For example, you could have a location of work and home. At work you might turn off Bluetooth, and at home Bluetooth might be on. This is achieved by clicking on the three dots with the down arrow under Other Services and choosing Locations ➢ Edit Locations; by default, it is set to Automatic.

The Network tab is also where you can join wireless networks and change how you join wireless networks. By default, the device will automatically join the network selected. However, there are circumstances where you would not want to automatically join the network, such as if you were directly connecting to a wireless device like a camera that broadcasts its own Service Set Identifier (SSID). If the primary wireless network was still set to automatically join, the device would keep disconnecting from the camera to join the primary network.

If you click Details, you can change your primary wireless networks and specify whether they are auto-joined, as shown in Figure 4.8. In addition to changing advanced properties for the wireless connection, you can select the TCP/IP tab and statically set the IP address. The DNS tab allows you to change the DNS servers to be queried. The WINS tab is for a deprecated Windows service, Windows Internet Name Service, that permits network browsing via broadcasts. The 802.1X tab allows you to set up 802.1X profiles for network-level security. The Proxies tab allows you to configure proxy servers for traffic on the device, as well as bypass local and select addresses. Finally, the Hardware tab allows you to set special characteristics based on the device.

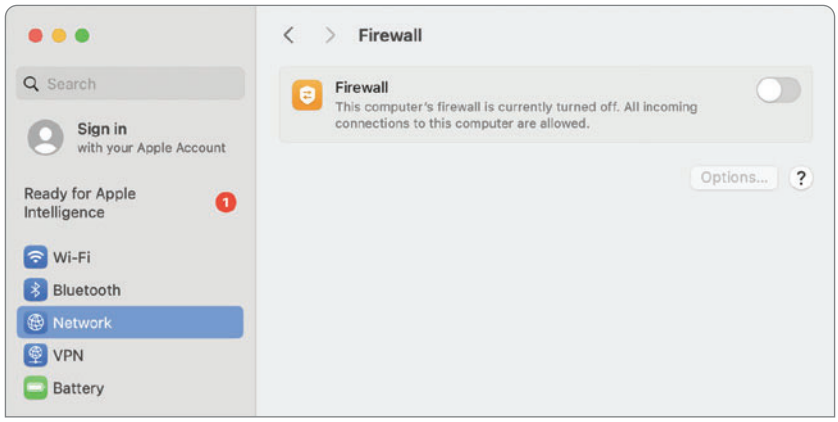
FIGURE 4.8 Details Network settings



Each device will have various methods for network connectivity. Depending on the various methods, the Network settings will differ based on the connection. The Details settings may also differ depending on the type of connectivity method for the device.

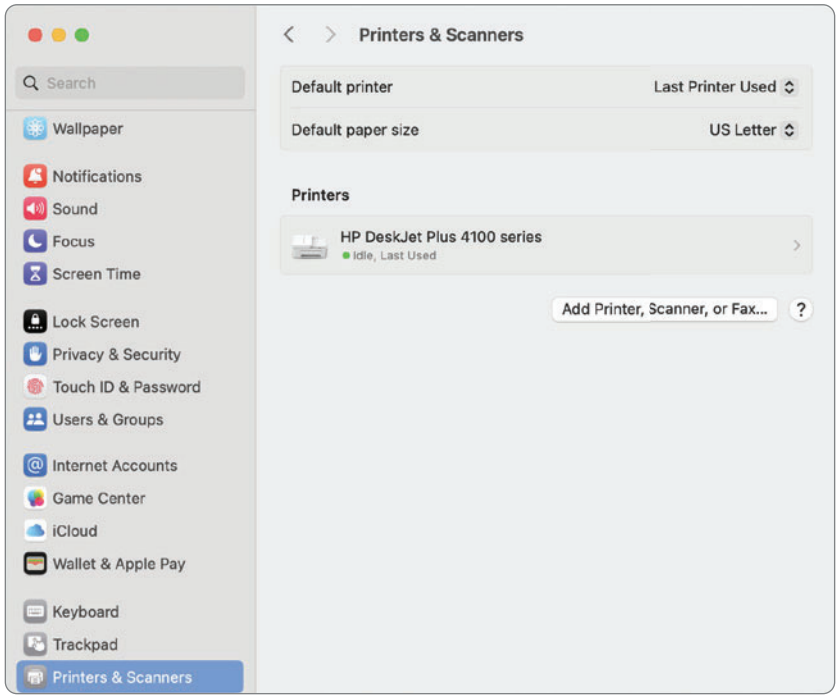
The Firewall option allows you to turn on and configure the built-in firewall for macOS. It is not turned on by default, but you can turn it on by clicking Turn On Firewall and then configure the firewall options, as shown in Figure 4.9. You can choose to block all inbound connections by default and create exceptions for only the applications you choose. By default, after turning on the firewall, the operating system allows all inbound connections to applications running on the system. You must choose to block all inbound connections and configure the exceptions.

FIGURE 4.9 Firewall settings



Printers & Scanners The Printers & Scanners tab, shown in Figure 4.10, is where you can access all settings related to printing and scanning for the device. In the figure, only an HP DeskJet Plus 4100 is connected to the device. However, there could be several different printers simultaneously connected via this System Setting. This would then allow the user to select any of the printers for their use in the desired application.

FIGURE 4.10 Printers & Scanners tab



Printers can be added and removed with this System Setting by clicking the Add Printer, Scanner, or Fax on the Printers & Scanners screen. You can then select the device and open the print queue by clicking Print Queue. This will allow you to see all the print jobs currently waiting to be printed from the local device. You can also click Options & Supplies and view the various options for the printer and check the ink or toner supply levels. In addition, you can share the printer by clicking Open Sharing Settings. You can then configure the Sharing settings and choose who can print to the printer.

Many printers today can also be purchased as multifunction copiers (MFCs), also called multifunction devices (MFDs) or multifunction printers (MFPs), which means the printer doesn't just print—it can copy and scan as well. If the device is capable of scanning, a Scan tab will be available after you select the device. Depending on the MFC device attached, the Scan tab will allow you to configure various settings. Although MFC devices are becoming common, if a stand-alone scanner was connected to the device, it would show in the Printers & Scanners settings. The device would have only one tab for scanning.

In addition to changing settings for the printers and scanners attached, you can change the default printer and the default paper size. The default printer is set to the last printer used.

Privacy & Security The Privacy & Security setting contains a number of settings that apply to the privacy of the user and the security of the device. The first setting you will notice is Location Services, as shown in Figure 4.11. Location Services on macOS provide a seamless way to enhance your user experience by allowing apps and websites to gather and use information based on your current location. You can also control the applications that request access to your data, such as Contacts, Calendars, Reminders, and Photos, just to name a few. You can add or remove the applications that need or have access to your data.

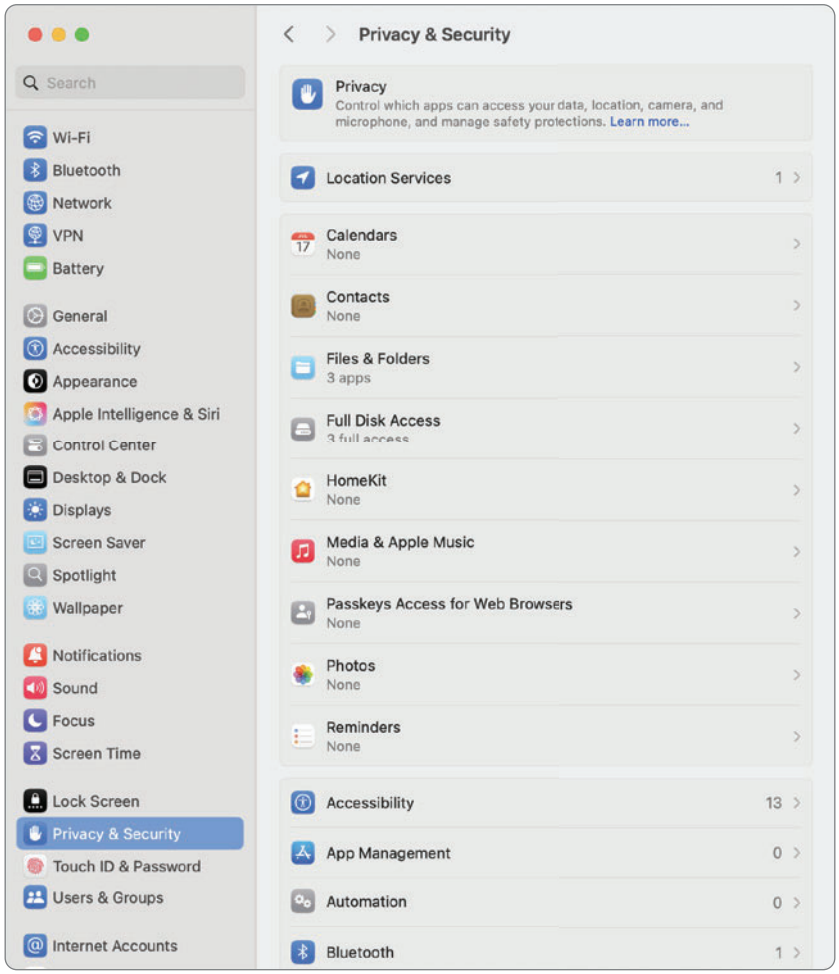
Just like in Windows 11, you can view all of the applications installed on the macOS operating system and their given permissions. Users can review and adjust permissions for each app, ensuring that only trusted applications can access their data. This granular control helps protect personal information from unauthorized access, providing peace of mind and a secure computing environment.

The Security section of the Security & Privacy screen contains some security controls. The FileVault option allows you to configure disk-level encryption to protect your files in the event the device is lost. You can turn on the FileVault feature by clicking Turn On FileVault. Once you do, you will need to unlock the disk with the user's password. If there are multiple users configured on the device, they will need to verify their passwords before the encryption is performed.

Lockdown Mode in macOS, is a feature also found in the Security section of the Privacy & Security screen. It offers an extreme level of protection for users who might be targeted by sophisticated cyberthreats. When enabled, Lockdown Mode significantly reduces the attack surface by limiting the functionality of certain apps, websites, and features. This includes blocking most message attachments, disabling complex web technologies, and restricting incoming FaceTime calls to known contacts. While this

mode enhances security, it also impacts the overall functionality and performance of the device, making it a trade-off between usability and protection. Lockdown Mode is designed for those who require the highest level of security, ensuring their sensitive data remains safeguarded against advanced digital threats.

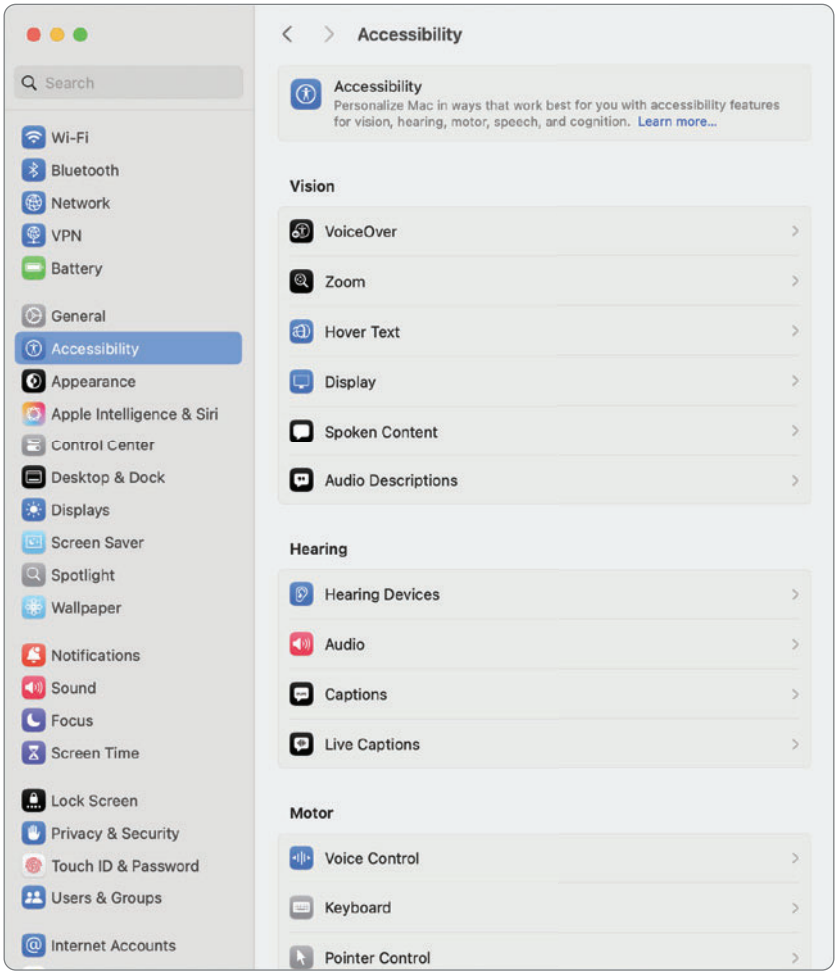
FIGURE 4.11 Privacy & Security



Accessibility The Accessibility tab allows you to customize macOS to support requirements such as vision, hearing, motor skills, or speech, as shown in Figure 4.12. In the Vision section, you can turn on VoiceOver, which will provide spoken commands as well as descriptions of items in Braille. Turn on Zoom to allow zooming with the use

of the keypad. The Display settings allow you to turn on display features for high contrast as well as reduce motion. The Spoken Content section allows you to configure the operating system to speak announcements, sections, items, and typing feedback. Use the Audio Descriptions section to turn on additional spoken content.

FIGURE 4.12 Accessibility System Settings



The Hearing section is designed to support users with hearing impairments. It offers features like mono audio, which combines stereo sound into a single channel, making it easier for users who are deaf or hard of hearing in one ear. The section also allows you to control hearing devices, as well as visually indicate when audio alerts are being

played so that the user does not miss an alert. The Captions option will display captions (subtitles) for the operating system. The Live Captions option will allow you to control how apps, like FaceTime, perform live captioning.

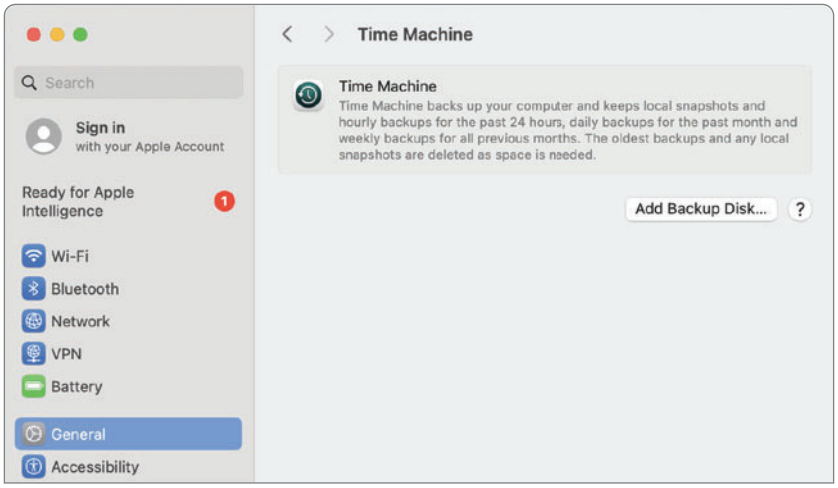
The Motor section is tailored to assist users with motor impairments. The Voice Control option provides the ability to speak to the computer to display and edit text. The Keyboard option lets you enable sticky keys and slow keys. The Pointer Control option lets you control how the trackpad operates by changing click speed. The Switch Control option allows you to specify that an adaptive device (such as a joystick) be used to control your Mac, enter text, and interact with items on your screen.

The Speech section is designed to assist users with speech impairments. It includes features like Live Speech, which allows users to use Text to Speech for applications like FaceTime. The Personal Voice option allows you to custom tailor a voice that sounds like yourself. The Vocal Shortcuts allows you to teach the operating system to recognize a custom phrase that performs an action inside the operating system.

The Siri section allows you to configure Siri to accept typed requests in lieu of spoken requests. In the Shortcut section, you can specify that a shortcut list appear when you press Option+Command+F5.

Backup/Time Machine As discussed in the previous section, backups are important. In the macOS world, *Time Machine* is a feature that can be used to make backups of various types (incremental, full, and so forth), as shown in Figure 4.13. In order to use Time Machine to back up your data, you will need an external storage device. These devices can be connected via external USB, Thunderbolt, or wirelessly with 802.11. To access Time Machine, select Apple menu ➢ System Settings ➢ General ➢ Time Machine. When an external storage device is connected for the first time and Time Machine is not configured, macOS will prompt you to set up Time Machine.

FIGURE 4.13 Time Machine settings

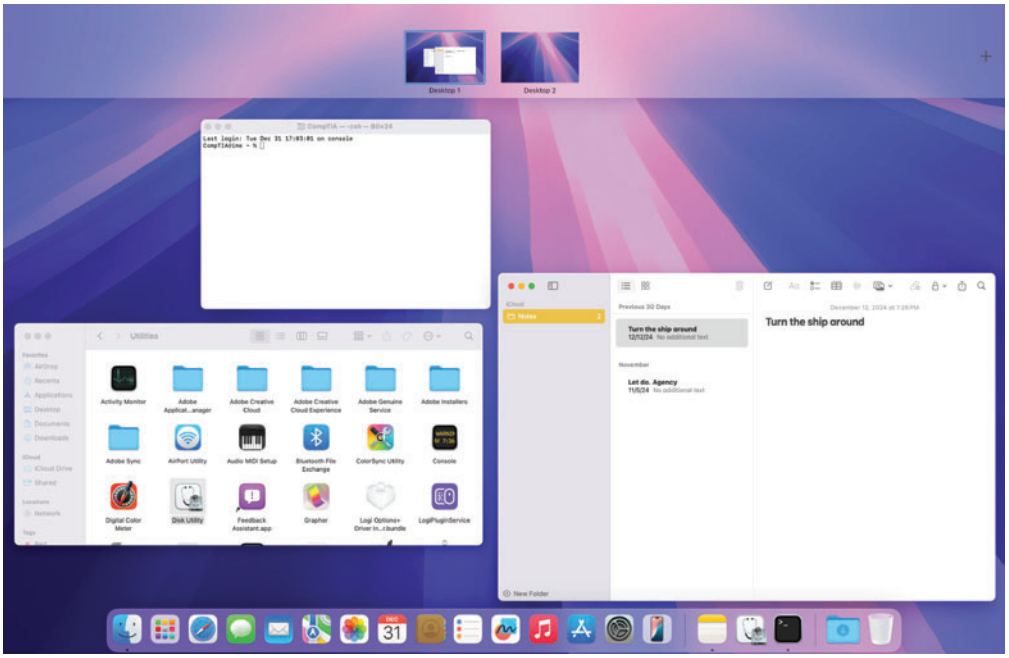


Features

Exam Objective 1.8, "Explain common features and tools of the macOS/desktop operating system," requires you to be familiar with multiple desktops, Mission Control, Keychain, Spotlight. iCloud, iMessage, FaceTime, Drive, Gestures, Finder, Dock, Continuity, Disk Utility, FileVault, Terminal, and Force Quit.

Multiple Desktops/Mission Control It is possible to run a large number of things at one time, whether those things are apps or desktops. Apple's *Mission Control* is an easy way to see what is open and switch between applications. To access Mission Control, press the Mission Control key on an Apple keyboard, click the Mission Control icon on the Dock (or Launchpad), or swipe up with three or four fingers on a trackpad. Once Mission Control is accessed, all open windows and spaces are shown—grouped by app—and you can choose between them. The Apple macOS Mission Control feature is identical in functionality to Microsoft's Task View feature. It allows you to see all the applications running on one screen so that you can choose the foreground application, as shown in Figure 4.14. By clicking the + symbol at the upper-right corner of the screen you can create a new virtual desktop.

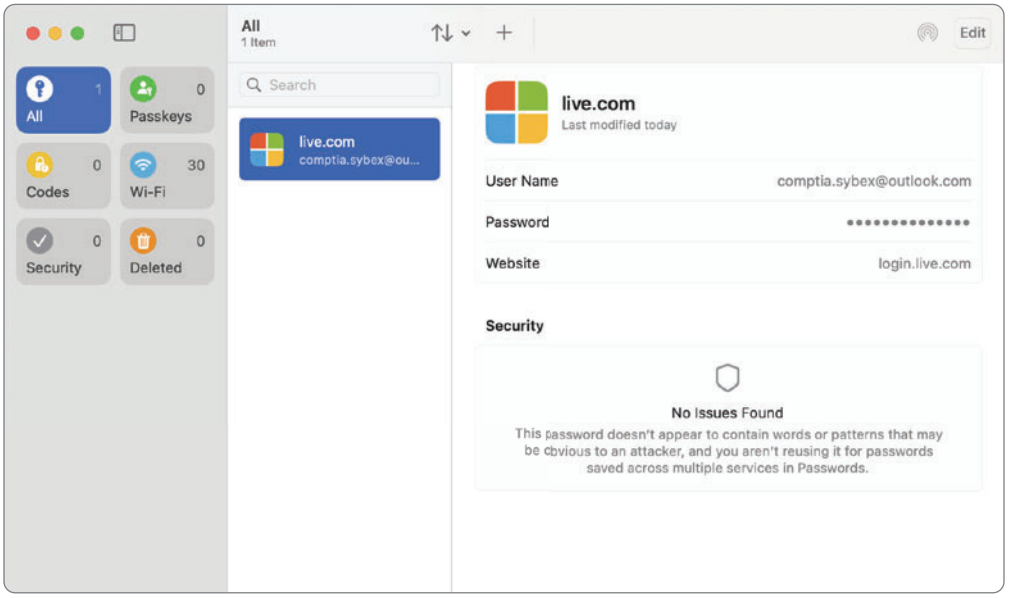
FIGURE 4.14 Viewing multiple apps with Mission Control



You can access Mission Control in a few different ways. The first is with trackpad gestures by swiping up with 3 or 4 fingers on your trackpad. The second is the key strokes of Control+Up arrow. The third way is the Mission Control (F3) button on the keyboard. The last method to initiate it is the Mission Control icon in the dock.

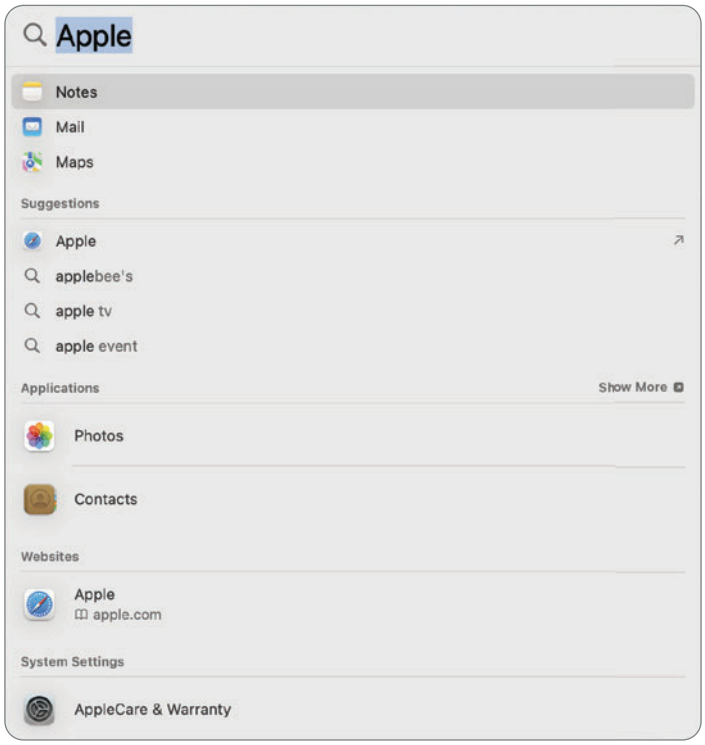
Keychain Access/Passwords *Keychain Access* is a password management system from Apple. It was originally introduced in Mac OS X, and remained in macOS for some time. However, with the introduction of macOS 15.2 (Sequoia) it has been migrated to the Passwords app. The Passwords app allows you to store passwords for websites, mail servers, Wi-Fi, and so forth, as shown in Figure 4.15. There is an iCloud variant (iCloud Passwords) that keeps such information as Safari usernames/passwords and credit card information. Values stored in the Passwords app are kept current (changes are synced) to simplify resource access. To view the current Passwords, select Applications ➢ Passwords.

FIGURE 4.15 Apple Passwords app



Spotlight The search feature within macOS is Spotlight, and a magnifying glass icon in the upper-right corner of the menu bar represents it (or you can press Command+spacebar from any app). As you type in the Spotlight utility, the results will display. As shown in Figure 4.16, we have searched for *apple*, and one of the results is the contact info for Apple Inc. Spotlight can search for documents, images, apps, and so on. Recent versions include autocomplete features as well as suggestions for additional results (usually on Wikipedia, *iTunes*, and such).

FIGURE 4.16 Apple Spotlight utility



iCloud Apple iCloud is a comprehensive cloud storage and cloud computing service that seamlessly integrates with Apple's ecosystem, providing users with a unified platform to store, sync, and access their data across all their Apple devices. iCloud ensures that your photos, documents, apps, and other important information are always up-to-date and accessible, no matter which device you're using. This service not only enhances productivity but also offers robust security features to protect your data. The iCloud settings can be seen in Figure 4.17.

One of the key components of iCloud is iCloud Drive, which allows users to store files and documents in the cloud and access them from any device. Whether you're working on a Mac, iPhone, or iPad, iCloud Drive ensures that your files are always within reach. You can organize your files into folders, share them with others, and even collaborate in real-time on documents. This makes iCloud Drive an essential tool for both personal and professional use.

iMessage is another integral part of iCloud, providing a seamless messaging experience across all Apple devices. With iMessage, your conversations are synced in real time, allowing you to start a chat on your iPhone and continue it on your Mac or iPad

without missing a beat. iCloud ensures that your messages are securely stored and backed up, so you never lose important conversations. Additionally, iMessage supports rich media, such as photos, videos, and even animated effects, making your communication more engaging.

FIGURE 4.17 iCloud configuration settings on macOS

FaceTime, Apple's video and audio calling service, is also tightly integrated with iCloud. FaceTime allows you to make high-quality video and audio calls to other Apple users, regardless of which device they are using. With iCloud, your FaceTime call history is synced across all your devices, making it easy to keep track of your communications.

Whether you're catching up with family, collaborating with colleagues, or having a virtual meeting, FaceTime provides a reliable and secure way to stay connected.

Together, these services—iCloud Drive, iMessage, and FaceTime—form an ecosystem under iCloud, enhancing the overall user experience by providing seamless integration, synchronization, and security across all Apple devices.

There can be costs associated with using iCloud. For example, the iCloud account on the macOS shown in Figure 4.17 can access 5 GB for free, but anything beyond that requires payment ($0.99/month for 50 GB, $2.99/month for 200 GB, or $9.99/month for 2 TB, as of this writing). For the latest costs visit `https://support.apple.com/en-us/108047`.

Gestures With Apple products, it is possible to scroll, tap, pinch, and swipe to interact with the macOS or other products in a way that is intended to be natural and intuitive. You can accept the default actions for these gestures, or you can configure them differently, as shown in Figure 4.18. You can access them by clicking the Apple menu ➢ System Settings ➢ Trackpad.

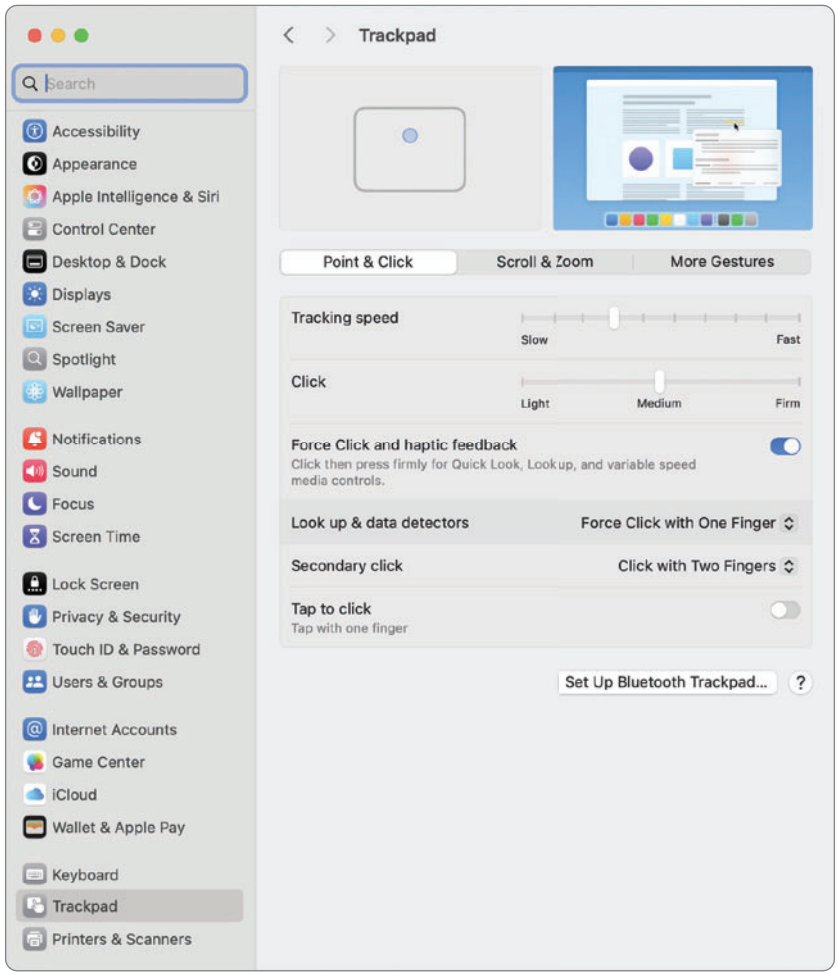
To see the basics of gestures on a macOS, visit `https://support.apple.com/en-us/102482`.

Finder To use an analogy, the Finder is to macOS as File Explorer is to Windows. It lets you browse through folders and find files, disks, apps, and so on. For example, Figure 4.19 shows an example of the Finder displaying the contents of the `Utilities` folder. You can change the view to see the entries with images, in a list view, in a column view, and so on.

Dock In macOS, a bar of icons runs along the bottom (or side, if so configured) of your screen, as shown in Figure 4.20. That set of icons, known as the *Dock*, provides easy access to key apps that come with the Mac (such as Safari, Mail, App Store, and Music) or others that you choose to add there. To see the basics of the Dock on a Mac, visit `https://support.apple.com/guide/mac-help/open-apps-from-the-dock-mh35859/mac`.

Continuity Continuity is a suite of features in macOS designed to create a seamless experience across all your Apple devices. It allows you to start a task on one device and continue it on another without missing a beat. For example, with Handoff, you can begin writing an email on your iPhone and finish it on your Mac. Universal Clipboard lets you copy text, images, or videos on one device and paste them on another. Continuity Camera enables you to take a photo or scan a document with your iPhone and instantly insert it into a document on your Mac. To use Continuity, ensure all your devices are signed in to the same Apple ID, have Bluetooth and Wi-Fi enabled, and are within close proximity to each other.

FIGURE 4.18 Settings for default gestures on macOS



There are many other features of macOS that make it a powerful operating system to use. These features, however, are the ones that CompTIA wants you to be aware of for the exam. Make sure that you know the purpose of each as you prepare.

FIGURE 4.19 Apple Finder utility

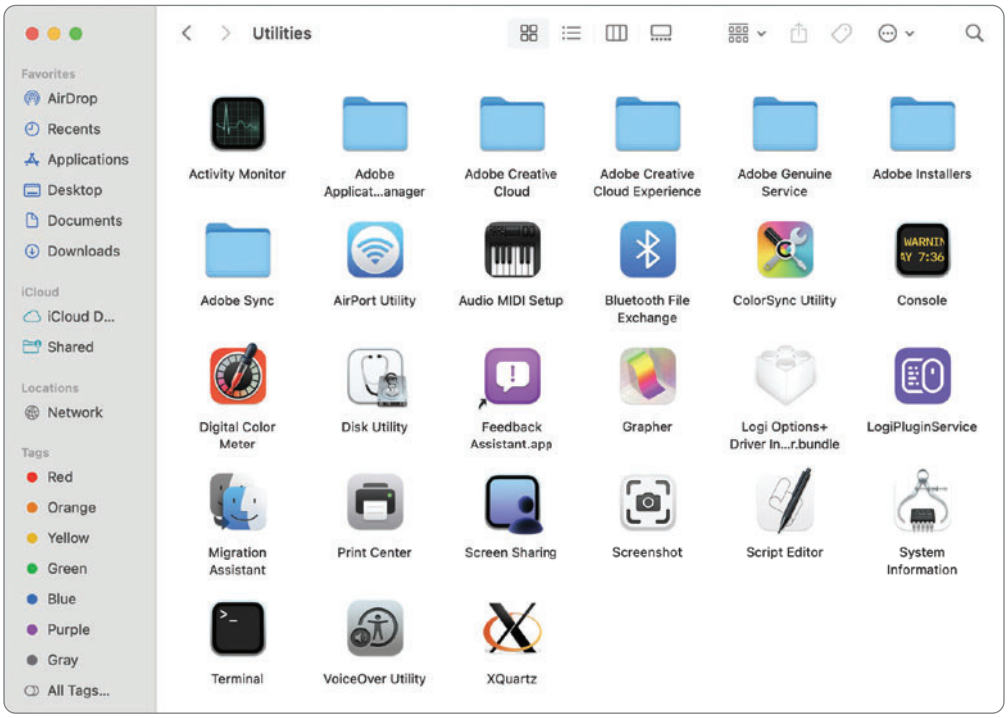
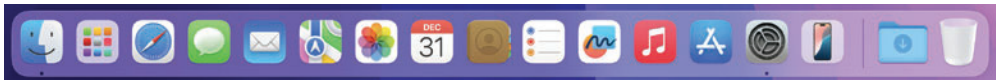


FIGURE 4.20 Apple macOS Dock



macOS System Folders

The macOS operating system organizes its filesystem into several key folders, each serving a distinct purpose to ensure the smooth operation and user experience of the operating system. Understanding these folders is crucial for both everyday users and system administrators. Each folder has a specific role, from housing system-critical files to storing user data and application resources. This organization not only helps in maintaining system stability and performance but also aids in troubleshooting and system management.

Application developers also have specific expectations for these folders. They rely on the standardized structure to ensure their applications can find and store necessary files correctly.

For instance, developers expect to place their applications in the `/Applications` folder, store system-wide resources in `/Library`, and save user-specific settings in `/Users/username/Library`. This consistency allows applications to function seamlessly across different macOS installations, providing a reliable and predictable environment for both development and usage. By adhering to these conventions, developers can create applications that integrate smoothly with the macOS ecosystem, enhancing the overall user experience.

Exam Objective 1.8, "Explain common features and tools of the macOS/desktop operating system," calls out `/Applications`, `/Users`, `/Library`, `/System`, and `/Users/Library` System folders in macOS. Be sure you know how to access and use them.

/Applications

The `/Applications` folder is a central hub for all the applications installed on a macOS system. This directory is designed to be accessible to all users, meaning that any application placed here can be used by anyone with an account on the macOS. It contains both system applications, which come pre-installed with macOS, and third-party applications, which users can install from the App Store or download from the Internet.

By default, when you install an application from the App Store or download one from the Internet, it is placed in the `/Applications` folder. This standard location makes it easy for users to find and launch their software, as they know exactly where to look for their applications. This centralized approach helps keep the system organized, as all applications are stored in one place rather than scattered across different directories.

Having a single location for applications also simplifies the process of managing and updating software. Users can easily see all installed applications in one view, making it straightforward to update, uninstall, or troubleshoot any application. For example, when a new version of an application is released, users can quickly locate the application in the `/Applications` folder and update it. Similarly, if an application needs to be removed, users can simply drag it from the `/Applications` folder to the Trash.

Moreover, this organization benefits system administrators and developers. System administrators can efficiently manage software installations across multiple user accounts, ensuring that all necessary applications are available to all users. Developers, on the other hand, can design their applications with the expectation that they will be installed in the `/Applications` folder, allowing them to streamline the installation process and ensure compatibility with macOS conventions.

/Users

The `/Users` folder is a fundamental part of the macOS filesystem, serving as the home directory for all user accounts on the system. Within this folder, each user has a dedicated subdirectory, often referred to as their home directory, which contains all their personal files,

settings, and preferences. This structure is crucial for maintaining an organized and secure environment, especially on a shared Mac.

Each user's home directory typically includes several standard folders such as `Documents`, `Downloads`, `Desktop`, and `Pictures`. These folders help users organize their files and make it easy to locate specific types of data. For example, the `Documents` folder is intended for storing text files, spreadsheets, and other document types, while the `Pictures` folder is for images and photos.

The separation of user data into individual home directories ensures that each user's information is kept private and secure. This means that one user cannot easily access another user's files, providing a layer of privacy and security. Additionally, this structure allows users to customize their settings and preferences without affecting others. For instance, each user can have their own desktop background, application settings, and browser bookmarks.

Moreover, this organization simplifies system management and troubleshooting. System administrators can manage user accounts and data more efficiently, knowing that each user's files are contained within their home directory. It also makes it easier to back up and restore user data, as each home directory can be treated as a separate unit.

/Library

The `/Library` folder contains systemwide resources and settings used by macOS and installed applications. This includes fonts, application support files, and system preferences. There are actually three `Library` folders in macOS: the system Library (`/System/Library`), the local Library (`/Library`), and the user `Library` (`/Users/username/Library`). Each serves a different scope, from systemwide resources to user-specific settings. The `/Library` folder is crucial for the proper functioning of applications and the operating system, as it stores essential files that applications rely on to run correctly.

System Library (`/System/Library`) The `System` library, located at `/System/Library`, contains the core components and resources that macOS needs to operate. This includes essential system frameworks, system extensions, and default fonts. These files are integral to the operating system's functionality and are typically protected to prevent accidental modification. Applications often rely on the frameworks and libraries found in this directory to perform system-level tasks and ensure compatibility with macOS.

Local Library (`/Library`) The local library, found at `Library`, stores resources and settings that are shared across all users on the Mac. This includes application support files, systemwide preferences, and shared libraries. For example, when an application is installed, it might place shared resources such as plug-ins, configuration files, and documentation in this directory. This ensures that these resources are available to all users, providing a consistent experience regardless of who is logged in. The local library also contains items like fonts and audio plug-ins that can be used by any application on the system.

User Library (/Users/username/Library) The user library, located at `/Users/username/Library`, is specific to each user account. This directory contains

user-specific application support files, caches, preferences, and other data. It is hidden by default to prevent accidental modification, but it can be accessed by advanced users who need to troubleshoot or customize their applications. Each user has their own `Library` folder within their home directory, ensuring that their settings and data are kept separate from other users on the system. This allows applications to store personalized settings and data, providing a tailored experience for each user.

By organizing resources and settings into these three distinct Library folders, macOS ensures that systemwide and user-specific data are properly managed and isolated. This structure helps maintain system stability, security, and performance, while also providing flexibility for both users and developers. Applications can efficiently access the resources they need, whether they are shared across the system or specific to an individual user, ensuring a seamless and reliable user experience.

/System

The /System folder is a critical directory that houses the core components of macOS. This includes the operating system kernel, essential system frameworks, and other low-level files necessary for macOS to function. This folder is typically read-only to prevent users from making changes that could destabilize the system. The `/System` folder ensures that the fundamental parts of the operating system remain intact and secure, providing a stable environment for the user.

The `/System` folder is meticulously organized to include various subdirectories that each play a vital role in the functioning of macOS. One of the key components within this folder is the operating system kernel, which is the core part of macOS responsible for managing system resources, hardware communication, and overall system stability. The kernel acts as a bridge between the hardware and software, ensuring that applications can run smoothly and efficiently. The following are some of the important functions that the `/System` folder provides:

System Frameworks Another crucial element within the `/System` folder is the system frameworks. These frameworks provide a set of standardized interfaces and libraries that developers use to build applications. By relying on these frameworks, applications can perform complex tasks such as graphics rendering, data processing, and network communication without needing to implement these functionalities from scratch. This not only simplifies the development process but also ensures that applications are compatible with the macOS environment.

System Extensions and Drivers The `/System` folder also contains system extensions and drivers that enable macOS to interact with various hardware components. These extensions are essential for the proper functioning of peripherals such as printers, scanners, and external storage devices. By keeping these files within the `/System` folder, macOS ensures that they are protected from accidental deletion or modification, which could otherwise lead to hardware malfunctions or system instability.

Default System Applications Additionally, the `/System` folder includes default system applications and utilities that are integral to the macOS experience. These applications, such as the Finder, Terminal, and Activity Monitor, provide essential functionalities that users rely on for everyday tasks. By housing these applications within the `/System` folder, macOS guarantees that they are always available and functioning correctly.

The `/System` folder is the backbone of macOS, housing essential components like the operating system kernel, system frameworks, extensions, drivers, and default applications. These elements ensure macOS runs smoothly and securely by managing system resources, facilitating hardware communication, and providing core functionalities. The folder's read-only nature protects these critical files from unintended changes, maintaining system stability and performance. This organization allows macOS to efficiently manage resources, support hardware interactions, and deliver a consistent user experience, making the `/System` folder a fundamental part of the macOS architecture.

To navigate to any of the macOS system folders, you can open the Finder, then select Go, then Go To Folder, and then type the folder name.

Best Practices

Regardless of the operating system, there are a number of best practices that an administrator should always follow. Depending on the operating system (and distribution, version, edition, and so on), it may be possible to perform operations with the utilities provided, or third-party utilities may be needed.

For Exam Objective 1.8 study, be sure you are familiar with best practices, backups, antivirus, updates/patches, and Rapid Security Response (RSR).

Scheduled Backups

Backups are duplicate copies of critical information, ideally stored in a location other than the one where the information is currently stored. Backups include both paper and computer records. Computer records are usually backed up using a backup program, backup systems, and backup procedures.

The primary starting point for disaster recovery involves keeping current backup copies of important data files, databases, applications, and paper records available for use. Your organization must develop a solid set of procedures to manage this process and to ensure that all critical information is protected.

Computer files and applications should be backed up on a regular basis. Here are some examples of critical files that should be backed up:

- Applications
- Appointment files
- Audit files
- Customer lists
- Database files
- Email correspondence
- Financial data
- Operating systems
- Prospect lists
- Transaction files
- User files
- User information
- Utilities

This list isn't all-inclusive, but it provides a place for you to start.

In most environments, the volume of information that needs to be stored is growing at a tremendous pace. Simply tracking this massive growth can create significant problems.

An unscrupulous attacker can glean as much critical information from copies as they can from the original files. Make sure that your storage facilities are secure, and it is a good idea to add physical security to the backup media as well.

You might need to restore information from backup copies for any number of reasons. Some of the more common reasons are as follows:

- Accidental deletion
- Application errors
- Natural disasters
- Physical attacks
- Server failure
- Virus/malware infection
- Workstation failure

Types of Storage Mechanisms

The information that you back up must be immediately available for use when needed. If a user loses a critical file, they won't want to wait several days while data files are sent from a remote storage facility. Several types of storage mechanisms are available for data storage:

Working Copies Working copy backups, sometimes referred to as *shadow copies*, are partial or full backups that are kept at the computer center for immediate recovery purposes. Working copies are frequently the most recent backups that have been made.

Typically, working copies are intended for immediate use. They are usually updated on a frequent basis.

Working copies aren't usually intended to serve as long-term copies. In a busy environment, they may be created every few hours.

Many filesystems used on servers include journaling. A *journaled file system (JFS)* includes a log file of all changes and transactions that have occurred within a set period of time (such as the last few hours). If a crash occurs, the operating system can check the log files to see which transactions have been committed and which transactions have not.

This technology works well, allowing unsaved data to be written after the recovery, and the system is usually successfully restored to its pre-crash condition.

On-Site Storage *On-site storage* usually refers to a location on the site of the computer center that is used to store information locally. On-site storage containers are available that allow computer cartridges, tapes, and other backup media to be stored in a reasonably protected environment in the building.

As time goes on, tape as a medium for backups is losing its popularity to other technologies, such as cloud storage.

On-site storage containers are designed and rated for fire, moisture, and pressure resistance. These containers aren't fireproof in most situations, but they are *fire-rated*. A *fireproof container* should be guaranteed to withstand damage regardless of the type of fire or temperature, whereas fire ratings specify that a container can protect the contents for a specific amount of time in a given situation.

If you choose to depend entirely on on-site storage, make sure that the containers you acquire can withstand the worst-case environmental catastrophes that could happen at your location. Make sure, as well, that they are in locations where you can easily find them after the disaster and access them (near exterior walls, on the ground floor, and so forth).

General-purpose storage safes aren't usually suitable for storing electronic media. The fire ratings used for safes generally refer to paper contents. Because paper does not catch fire until 451° Fahrenheit, electronic media are typically ruined well before paper documents are destroyed in a fire.

Off-Site Storage *Off-site storage* refers to a location away from the computer center where paper copies and backup media are kept in the event the primary site suffers a catastrophe. Off-site storage can involve something as simple as keeping a copy of backup media at a *remote office*, or it can be as complicated as a nuclear-hardened high-security storage facility. The storage facility should be bonded, insured, and inspected on a regular basis to ensure that all storage procedures are being followed.

Your determination of which storage mechanism to use should be based on the needs of your organization, the availability of storage facilities, and your budget. Most off-site storage facilities charge based on the amount of space required and the frequency of access needed to the stored information.

Although it is easy to see the need for security at any location where your files are stored, don't overlook the need for security during transportation as well.

Scheduled Disk Maintenance

When files are written to a hard drive, they're not always written contiguously or with all the data located in a single location. When discussing Windows, we talked about Disk Defragmenter, which has existed in almost all versions of Windows, and its ability to take file data that has become spread out over the disk and put it all in the same location, a process known as *defragmenting*. This process decreases the time it takes to retrieve files.

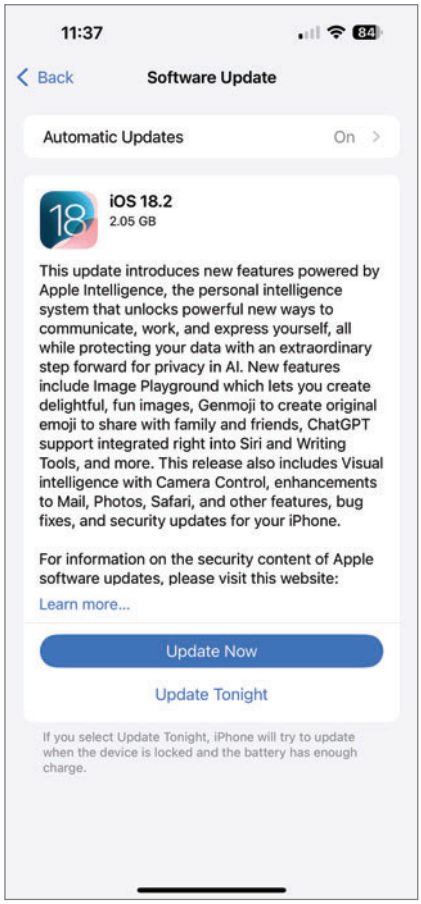
As opposed to FAT- and NTFS-based filesystems, the filesystems used on macOS and Linux rarely, if ever, need to be defragmented. The ext3 and ext4 filesystems are common to Linux, and Apple File System (APFS) is common to macOS. They all have on-the-fly defragmentation methods and implement file allocation strategies differently from their traditional Windows counterparts.

System Updates/App Store

It is important to keep the operating system current and updated. Like Windows, many other operating systems include the ability to update automatically, and almost all can look for updates and tell you when they are available. In the Apple world, the *App Store* represents a location where you can also find updates.

For example, Figure 4.21 shows that a new version of iOS is available on an iPhone. By clicking Learn More, you display the reasons the new version has been released, allowing you to read about the changes and decide whether you want to upgrade. The update notification usually includes a Learn More option and a link to Apple's release page for the iOS update.

FIGURE 4.21 Apple iOS software update



To access the Software Update area on an iPhone or iPad, choose Settings ➢ General ➢ Software Update. However, when an update is available, you will generally see a prompt on your home screen. To access the Software Update section on macOS, choose System Settings from the Apple menu, and then click Software Update. In most cases, unless a production device would be negatively impacted, you should keep systems updated with the latest releases.

Patch Management

As a general rule, updates fix a lot of issues and patches fix a few; multiple patches are rolled into updates. You can't always afford to wait for updates to be released and should install patches—particularly security-related patches—when they are released. Bear in mind that if all the security patches are not installed during the OS installation, attackers can exploit the weaknesses and gain access to information.

A number of tools are available to help with patch management, although the intentions of some are better than others. For example, rather than probe a service remotely and attempt to find a vulnerability, the Nessus vulnerability scanner (`www.tenable.com/downloads/nessus`) will query the local host to see if a patch for a given vulnerability has been applied. This type of query is far more accurate (and safer) than running a remote check. Since remote checks actually send the exploit in order to check to see if it is applicable, this can sometimes crash a service or process.

Depending on the variant of Linux you are running, *APT (Advanced Package Tool)* can be useful in getting the patches from a repository site and downloading them for installation on Debian and Ubuntu (just to name a few). The most common command used with this tool is `apt update`, which, as the name implies, gets the package for installation, as shown in Figure 4.22. The *Dandified YUM (DNF)* has replaced the *Yellowdog Updater, Modified (YUM)* tool used with *Red Hat Package Manager (RPM)-based* Linux distributions, such as Fedora and Red Hat, and works in a similar way to APT.

FIGURE 4.22 Ubuntu `apt` tool

```
login as: user
Authenticating with public key
Passphrase for key:
Welcome to Ubuntu 18.04.6 LTS (GNU/Linux 6.10.2-x86_64-linode165 x86_64)

 * Documentation:  https://help.ubuntu.com

40 updates can be applied immediately.
40 of these updates are ESM Infra security updates.
To see these additional updates run: apt list --upgradable

Last login: Sat Nov  2 14:02:45 2024 from 72.32.46.40
user@srv:~$ sudo apt update
[sudo] password for user:
Hit:1 http://mirrors.linode.com/ubuntu bionic InRelease
Get:2 http://mirrors.linode.com/ubuntu bionic-updates InRelease
Hit:3 http://mirrors.linode.com/ubuntu bionic-backports InRelease
Ign:4 http://apt-longview.linode.com bionic InRelease
Get:5 http://security.ubuntu.com/ubuntu bionic-security InRelease [102 kB]
Get:6 http://apt-longview.linode.com bionic Release
Hit:7 http://ppa.launchpad.net/certbot/certbot/ubuntu bionic InRelease
Get:8 https://esm.ubuntu.com/apps/ubuntu bionic-apps-security InRelease [7,651 B]
Get:9 https://esm.ubuntu.com/apps/ubuntu bionic-apps-updates InRelease [7,554 B]
Get:10 https://esm.ubuntu.com/infra/ubuntu bionic-infra-security InRelease [7,584 B]
Get:11 https://esm.ubuntu.com/infra/ubuntu bionic-infra-updates InRelease [7,526 B]
Reading package lists... Done
E: The repository 'http://apt-longview.linode.com bionic Release' no longer has a Release file.
N: Updating from such a repository can't be done securely, and is therefore disabled by default.
N: See apt-secure(8) manpage for repository creation and user configuration details.
user@srv:~$
```

Driver/Firmware Updates

With any operating system, it is essential to keep the drivers and *firmware* updated. Always remember to back up your configurations (such as for routers) before making any significant changes—in particular, a firmware upgrade—in order to provide a fallback in case something goes awry.

Many network devices contain firmware with which you interact during configuration. For security purposes, you must authenticate in order to make configuration changes and do so initially by using the default account(s). Make sure that the default password is changed after the installation on any network device; otherwise, you are leaving that device open for anyone recognizing the hardware to access it using the known factory password.

Updating firmware for macOS is performed via software updates. During the software update process, the firmware and the software that corresponds to it are updated. Updating firmware for the hardware installed on Linux computers will vary significantly, depending on the type of hardware. Many enterprise Linux vendors, such as Red Hat, include firmware updates in their software updates. As a rule, however, firmware is not part of the Linux software update process.

Antivirus/Antimalware Updates

At one point in time, there were so few viruses outside the Windows world that users not running Windows felt safe without protection on their systems. A significant reason for the low amount of non-Windows *malware* was that the authors of such devious programs were focusing on Windows simply because it had the lion's share of the market; they wanted to inflict as much harm as possible with their code.

As other operating systems have increased in popularity, so too have the number of malware items written for them or that can affect them. Because of this, today it is imperative to have protection on every machine. Additionally, this protection—in the form of definition files—must be kept current and up-to-date. Antivirus and antimalware definitions are released by the hour as new viruses and malware are identified. Most operating systems check daily for updates to definitions. Volume 2 Chapter 5, "Security Concepts," discusses security and antivirus/antimalware in more detail.

Rapid Security Response (RSR)

Rapid Security Response (RSR) is a feature introduced by Apple to deliver critical security updates to macOS, iOS, and iPadOS devices more swiftly than traditional software updates. These responses are designed to address significant security vulnerabilities that could be exploited by malicious actors, ensuring that devices remain protected against emerging threats.

RSRs can include updates to essential components such as the Safari web browser, the WebKit framework stack, and other critical system libraries. By focusing on these key areas, RSRs can quickly mitigate security issues without waiting for the next full software update cycle. This rapid deployment is crucial for maintaining the security integrity of Apple devices.

One of the advantages of RSRs is their ability to be applied automatically. Devices running the latest versions of the operating system will receive these updates without user intervention, although a restart may be required to complete the installation. This automation ensures that security patches are promptly applied, reducing the window of vulnerability.

RSRs are also designed to be lightweight and efficient. They require less battery power and system resources compared to full software updates, making them less disruptive to the user experience. Additionally, RSRs can be removed if necessary, allowing users to revert to the previous state if any issues arise from the update.

To verify RSRs are automatically installed, follow the following directions for your device:

iOS

1. Navigate to Settings.
2. Select General.
3. Tap Software Update.
4. Choose Automatic Updates and ensure that Security Responses & System Files is selected.

macOS

1. Open the Apple menu.
2. Select System Settings.
3. Click General in the sidebar.
4. Select Software Update on the right.
5. Click the Show Detail button next to Automatic Updates and ensure that Install Security Responses And System Files is selected.

Overall, Rapid Security Response enhances the security posture of Apple devices by providing timely and efficient updates to critical system components, ensuring that users are protected against the latest threats with minimal disruption to their daily activities.

Linux Administration

The best way to approach the following commands is to think about Microsoft Windows. That operating system offers a plethora of utilities for configuring the workstation, and just in case they don't work, or you want to go about it the hard way, you can use command-line utilities to accomplish similar tasks. The odds are good that you spend most of your time walking through graphical dialog boxes, but you're familiar enough with the command-line utilities that you can use them when you need to.

Linux is the same way. There is an overabundance of graphical utilities that can be used to configure the system, and they differ based on the distribution and the graphical interface

being used. In addition to these, command-line utilities are available in every distribution that can be used to get the job done. Those command-line utilities are what we'll focus on here.

There is only one vendor for Windows (Microsoft), but there are many vendors for Linux (Red Hat, SuSE, Ubuntu—to name just three). Also, a new version of Windows is released only every few years (Windows 7, Windows 8/8.1, Windows 10, Windows 11), but with Linux—especially because there are so many vendors—there are a lot of versions. With Ubuntu, for example, the goal is to release a new version every six months.

With all the different distributions and versions, getting to the place where you can run command-line utilities can differ a bit. In almost every implementation of Linux, you can boot into a command-line mode, and the commands entered there can then be run. Better than that, though, the easiest way to get to the command line is to open a terminal (also called a console) window. This allows you to interact with the shell, where you can type commands to your heart's content. The default shell in many Linux distributions is Bash. When you open a terminal window or log in at a text console, the Bash shell is what prompts you for commands. When you type a command, the shell executes your command.

Exam Objective 1.9, "Identify common features and tools of the Linux client/desktop operating system," requires you to know file management, filesystem management, administrative, package management, network, informational, and text editors used with Linux. In addition, you should know the various common configuration files, and OS components used with Linux.

Understanding the Syntax of Shell Commands

Because a shell interprets what you type, knowing how the shell processes the text you enter is important. All shell commands have the following general format (some commands have no options):

```
command [option1] [option2] ... [optionN]
```

On a command line, you enter a command followed by zero or more *options* (also called *arguments*). The shell uses a blank space or a tab to distinguish between the command and options. This means that you must use a space or a tab to separate the command from the options and the options from one another. If an option contains spaces, you put that option inside quotation marks. For example, to search for a name in the password file, enter the following `grep` command (`grep` is used for searching for text in files):

```
[user@localhost ~]$ grep "Jon B" /etc/passwd
```

When `grep` prints the line with the name, it looks like this:

```
filea:x:1000:100:Jon B:/home/testuser:/bin/bash
[user@localhost ~]$
```

If you create a user account with your username, type the `grep` command with your username as an argument to look for that username in the `/etc/passwd` file. In the output from the `grep` command, you can see the name of the shell (`/bin/bash`) following the last colon (:). Because the Bash shell is an executable file, it resides in the `/bin` directory; you must provide the full path to it.

The number of command-line options and their formats depend on the actual command. Typically, these options look like `-X`, where `X` is a single character. For example, you can use the `-l` option with the `ls` command. The command lists the contents of a directory, and the option provides additional details. Here is a result of typing `ls -l` in a user's home directory:

```
[user@localhost ~]$ ls -l
total 0
drwxr-xr-x 2 user users 48  2018-09-08 21:11 bin
drwx—      2 user users 320 2018-09-08 21:16 Desktop
drwx—      2 user users 80  2018-09-08 21:11 Documents
drwxr-xr-x 2 user users 80  2018-09-08 21:11 public_html
drwxr-xr-x 2 user users 464 2018-09-17 18:21 sdump
[user@localhost ~]$
```

The /etc/passwd file contains a list of the usernames and their respective attributes, such as User ID (UID), Group ID (GID), User ID info, home directory, and shell. Interestingly enough, it doesn't normally store the user's password; another file, called the shadow file (/etc/shadow), now contains a hashed version of the user's password.

If a command is too long to fit on a single line, you can press the backslash key (\) followed by Enter. Then continue typing the command on the next line. For example, type the following command (press Enter after each line):

```
[user@localhost ~]$ cat \
> /etc/passwd
root:x:0:0:Super User:/root:/bin/bash
bin:x:1:1:bin:/bin:/usr/sbin/nologin
daemon:x:2:2:daemon:/sbin:/usr/sbin/nologin
adm:x:3:4:adm:/var/adm:/usr/sbin/nologin
[Output Cut]

[user@localhost ~]$
```

The `cat` command then displays the contents of the `/etc/passwd` file.

You can concatenate (that is, string together) several shorter commands on a single line by separating the commands with semicolons (`;`). For example, the following command

changes the *current directory* to your home directory, lists the contents of that directory, and then shows the name of that directory:

```
[user@localhost ~]$ cd /dev; ls -l; pwd
total 0
crw-r--r--. 1 root root     10, 235 Jan  1 12:28 autofs
drwxr-xr-x. 2 root root         180 Jan  1 12:28 block
drwxr-xr-x. 2 root root          60 Jan  1 12:28 bsg
crw-rw----. 1 root disk     10, 234 Jan  1 12:28 btrfs-control
drwxr-xr-x. 3 root root          60 Jan  1 12:28 bus
[Output Cut]

[user@localhost dev]$
```

You can combine simple shell commands to create a more sophisticated command. For example, suppose you want to find out whether a device file named `nvme0` resides in your system's `/dev` directory because some documentation says that you need that device file for your hard drive. You can use the `ls /dev` command to get a directory listing of the `/dev` directory, and then browse through it to see whether that listing contains `nvme0`.

Unfortunately, the `/dev` directory has a great many entries, so you may find it hard to find any item that has `nvme0` in its name. You can, however, combine the `ls` command with `grep` and come up with a command line that does exactly what you want. Here's that command line:

```
[user@localhost ~]$ ls /dev | grep nvme0
nvme0
nvme0n1
nvme0n1p1
nvme0n1p2
nvme0n1p3
[user@localhost ~]$
```

The shell sends the output of the `ls` command (the directory listing) to the `grep` command, which searches for the string `nvme0`. The vertical bar (`|`) is known as a *pipe* because it acts as a conduit (think of a water pipe) between the two programs—the output of the first command is fed into the input of the second command.

Discovering and Using Linux Commands

Literally hundreds, if not thousands, of Linux commands exist within the shell and the system directories. Fortunately, CompTIA asks that you know a much smaller number than that. Table 4.1 lists common Linux commands by category.

TABLE 4.1 Essential Linux commands

Command name	Action
Managing files and directories	
`cd`	Changes the current directory.
`chmod`	Changes file permissions.
`chown`	Changes the file owner and group.
`cp`	Copies files.
`ls`	Displays the contents of a directory.
`md`	Creates a directory.
`mv`	Renames a file and moves the file from one directory to another.
`rm`	Deletes files.
`pwd`	Displays the current directory.
`find`	Searches for text in a file hierarchy.
`grep`	Searches for regular expressions in a text file.
Filesystem management	
`fsck`	Fixes the filesystem for issues or corruption.
`mount`	Utility used to mount folders and drives to the filesystem.
Administrative	
`passwd`	Changes the password.
`su`	Starts a new shell as another user. (The other user is assumed to be root when the command is invoked without any argument.)
`sudo`	Runs a command as another user (usually the root user).
`shutdown`	Shuts down Linux.

Command name	Action
Package management	
`apt`	Downloads files from a repository site, used with Debian-based Linux.
`dnf`	Downloads files from a repository site, used with Red Hat-based Linux.
Network	
`dig`	DNS query utility.
`ip`	Allows you to display and configure information related to a network interface card (NIC).
`curl`	Allows transferring data between servers using various protocols like HTTP and FTP.
`ping`	Used to test the reachability of a network host and measure the round-trip time for a packet.
`traceroute`	Used to map the path data packets take from your computer to a destination server.
Text editors	
`vi`	Starts the visual file editor, which can be used to edit files.
`nano/pico`	Starts the visual file editor, which is simple to use for editing files.
Informational	
`man`	System help for executable files.
`df`	Displays the total disk free (disk free space) for a directory.
`dd`	Copies blocks of data from one file to another (used to copy data from devices).
`du`	Allows users to analyze and report on disk usage within directories and files.
`ps`	Displays a list of currently running processes.
`kill`	Terminates a process.

TABLE 4.1 Essential Linux commands *(continued)*

Command name	Action
`cat`	Displays the contents of a file.
`top`	Displays running processes, similar to Windows Task Manager.

User Accounts

Linux is a powerful and versatile operating system widely used in various environments, from personal computers to enterprise servers. One of the fundamental principles of Linux system administration is the creation of non-root users during the initial setup. The root user, also known as the superuser, has unrestricted access to all system commands and files, making it a highly privileged account. While this level of access is necessary for certain administrative tasks, it also poses significant risks if misused or compromised.

Creating non-root users allows for the delegation of tasks and limits the potential damage that can be caused by accidental or malicious actions. Non-root users have restricted permissions, a fact that helps prevent unauthorized changes to critical system files and settings. This separation of privileges is a key aspect of the principle of least privilege concept, which aims to minimize the risk of security breaches and system failures.

For example, consider a scenario where a user accidentally runs a command that deletes important system files. If the user is operating as root, the command will execute with full privileges, potentially rendering the system inoperable. However, if the user is a non-root user, the command will fail due to insufficient permissions, protecting the system from unintended harm. This example illustrates why it is strongly discouraged to run as root for everyday tasks and highlights the importance of creating and using non-root user accounts in Linux.

Becoming Root (Superuser)

When you want to do anything that requires a high privilege level (e.g., administering your system), you have to become *root*. Normally, you log in as a regular user with your everyday username. When you need the privileges of the *superuser*, though, use the following command to become root:

```
[user@localhost ~]$ whoami
user
[user@localhost ~]$ su -
Password:
[root@localhost ~]# whoami
root
[root@localhost ~]#
```

The su command followed by a space and the minus sign (or hyphen) provides an environment similar to what the user would expect by applying the user's environment variables and initial login scripts. Once executed, the shell then prompts you for the root password. Type the password and press Enter. Your command prompt will change to a # symbol and your security context will change to the root user. The whoami command is a life saver, it helps you keep your sanity since the prompts look so familiar.

When the su command is entered without a username as an option, you will be prompted for the password of the root account, and a Bash shell will open on behalf of the root account. However, you can use the su command followed by any user account name, and as long as you supply the correct password, you will be logged in as that user.

After you've finished with whatever you want to do as root (and you have the privilege to do anything as root), type **exit** to return to your normal username.

Instead of becoming root by using the su - command, you can type **sudo** followed by the command that you want to run as root. In some distributions, such as Ubuntu, you must use the sudo command because you don't get to set up a root user when you install the operating system. If you're listed as an authorized user in the /etc/sudoers file, sudo executes the command as if you were logged in as root. Type **man sudoers** to read more about the /etc/sudoers file.

As you learn many of the commands in this chapter, you'll realize that it is similar to learning a new language. Therefore, practice makes perfect, and we urge you to practice these commands. The man command can help with nuances of Linux. The command provides a manual pages for a command or component installed on the Linux operating system. As new commands are installed the man pages are added.

Creating Accounts

Creating a new user account in Linux is a straightforward process that involves using the useradd command followed by the desired username. For example, to create a user named *CompTIA*, you would execute the following command:

```
user@server:~$ sudo useradd CompTIA
[sudo] password for user:
user@server:~$
```

After creating the account, it's essential to set a password for the new user using the passwd command:

```
user@server:~$ sudo passwd CompTIA
[sudo] password for user:
New password:
```

```
Retype new password:
passwd: password updated successfully
user@server:~$
```

The sudo command will prompt you to enter your password first, then enter and confirm the new password for the user CompTIA. Additionally, you can customize the user's home directory, shell, and other settings by using various options with the useradd command. Creating user accounts allows for better management of system resources and enhances security by assigning specific permissions and roles to different users.

Users, Groups, and Passwords

The /etc/passwd file is a crucial component of user account management in Linux. It contains essential information about each user, including the username, user ID (UID), group ID (GID), home directory, and default shell. Each line in the file represents a single user account, with fields separated by colons. Historically, this file also stored user passwords in an encrypted format. However, to enhance security, modern systems now use the /etc/shadow file for storing password information. The /etc/passwd file remains publicly readable, allowing various system utilities to access user information while keeping sensitive data secure. The following is a typical /etc/passwd file with the addition of our initial user account and the CompTIA user.

```
[user@localhost ~]$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
bin:x:1:1:bin:/bin:/usr/sbin/nologin
daemon:x:2:2:daemon:/sbin:/usr/sbin/nologin
adm:x:3:4:adm:/var/adm:/usr/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/usr/sbin/nologin
[Output Cut]

user:x:1000:1000:user:/home/user:/bin/bash
CompTIA:x:1001:1001::/home/CompTIA:/bin/bash
[user@localhost ~]$
```

In addition to the user account created in the /etc/passwd file, a group for the user is also created in the /etc/group file. You can add users to groups by using the command usermod -aG {group name} {user}. As an example, in the following we will add the CompTIA user to the root group:

```
[user@localhost ~]$ sudo usermod -aG root CompTIA
[sudo] password for user:
[user@localhost ~]$ cat /etc/group
root:x:0:CompTIA
bin:x:1:
daemon:x:2:
sys:x:3:
```

```
adm:x:4:
tty:x:5:
disk:x:6:
lp:x:7:
[Output Cut]

[user@localhost ~]$
```

The /etc/shadow file is specifically designed to securely store user password information. Unlike the /etc/passwd file, it is only accessible by the root user and contains hashed passwords along with additional fields for password aging and expiration policies. By separating password data from the more publicly accessible /etc/passwd file, Linux systems significantly improve the security of user credentials. The /etc/shadow file ensures that sensitive password information is protected from unauthorized access, reducing the risk of security breaches and enhancing overall system security. Proper management of this file is essential for maintaining a secure Linux environment.

```
[user@localhost ~]$ cat /etc/shadow
cat: /etc/shadow: Permission denied
[user@localhost ~]$ sudo cat /etc/shadow
[sudo] password for user:
root:$y$j9T$JTORk.nzkReN8x6d/4OBfUGj$RR9uljt9q0mWJP9hbtFwtEvaEynG3CBfoRQEhuq
qse1::0:99999:7:::
bin:*:19925:0:99999:7:::
daemon:*:19925:0:99999:7:::
adm:*:19925:0:99999:7:::
lp:*:19925:0:99999:7:::
[Output Cut]

user:$y$j9T$JAnsgsv9zypBiyl/bKgXOeCP$JW97KaqOym07nTMGdKGitK10JzhQYSBqk7
.UF3i0Ef9::0:99999:7:::
CompTIA:$y$j9T$e5R5vtTEfA25/HIQboCB81$TnGiEWcdTZsHpHsHhoSe3J6zi7O55e5l8
.QXxcqPWC9:20091:0:99999:7:::
[user@localhost ~]$
```

Maintaining the Operating System

The Linux operating system is a secure and functional operating system. However, it's inevitable that at some point you will need to obtain updates for security or to extend the functionality by adding packages. Thankfully, every Linux distribution has its own unique repository of security patches and additional packages. When choosing a Linux distribution, you should spend time exploring its benefits. For example, Debian packages are more stable, but the disadvantage is that the packages are older than the current version of packages.

Mint Linux is tailored to laptops and desktops, but it is not normally found on servers. These are just a few examples—the lesson is to do your homework before you decide on a Linux distribution.

Once you have chosen and installed a Linux distribution, you'll want to update the repositories and then upgrade the distribution to get the latest security patches. A *repository* is a group of packages that are available for download. The repository contains metadata, such as versions, dates, descriptions, and dependencies, about these packages. When you update the repositories, you are downloading the metadata so that you can search and begin the upgrade process.

Depending on the version, you will have one of two or three tools: the Advanced Package Tool (APT), Dandified YUM (DNF), or Yellowdog Updater, Modified (YUM). Linux distributions, such as Ubuntu, Debian, and Mint, will use the APT package management tool. In the operating system, you can update the repositories by using the `apt` or the `apt-get` command. In addition to upgrading the operating system, you can install and manage packages. In order to update the operating system, you will need to update the repositories first and then you can upgrade the operating system binaries as shown in the following:

```
user@server:~$ sudo apt update
[sudo] password for user:
Hit:1 http://archive.ubuntu.com/ubuntu bionic InRelease
Hit:2 http://archive.ubuntu.com/ubuntu bionic-updates InRelease
Hit:3 http://archive.ubuntu.com/ubuntu bionic-backports InRelease
Hit:4 http://archive.ubuntu.com/ubuntu bionic-security InRelease
Reading package lists... Done
user@server:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree
Reading state information... Done
Calculating upgrade... Done
The following packages have been kept back:
  base-files netplan.io sosreport ubuntu-advantage-tools ubuntu-server
The following packages will be upgraded:
  accountsservice apport apt apt-utils bash bcache-tools bind9-host bsdutils
[ output cut]
  unattended-upgrades update-manager-core update-notifier-common ureadahead
  util-linux uuid-runtime vim vim-common vim-runtime vim-tiny wget xkb-data
241 upgraded, 0 newly installed, 0 to remove and 5 not upgraded.
Need to get 111 MB of archives.
After this operation, 43.2 MB of additional disk space will be used.
Do you want to continue? [Y/n]
```

In the following example, you can see how a package such as `iftop` is installed using the `apt` command:

```
user@server:~$ sudo apt install iftop
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following NEW packages will be installed:
  iftop
0 upgraded, 1 newly installed, 0 to remove and 246 not upgraded.
Need to get 36.0 kB of archives.
After this operation, 91.1 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu bionic/universe amd64 iftop amd64
1.0~pre4-4 [36.0 kB]
Fetched 36.0 kB in 0s (102 kB/s)
Selecting previously unselected package iftop.
(Reading database ... 66991 files and directories currently installed.)
Preparing to unpack .../iftop_1.0~pre4-4_amd64.deb ...
Unpacking iftop (1.0~pre4-4) ...
Setting up iftop (1.0~pre4-4) ...
Processing triggers for man-db (2.8.3-2ubuntu0.1) ...
user@server:~$
```

The DNF and YUM package managers are used to update and install packages for Red Hat–based Linux distributions, such as Red Hat Enterprise Server and Fedora, among others. These tools work just like the APT tool; the first step is to update the repositories, then you can update the binaries. In the following examples, we will demonstrate both utilities:

```
[root@localhost ~]# yum update
CentOS Stream 8 - AppStream       5.6 MB/s | 16 MB 00:02
CentOS Stream 8 - BaseOS          1.6 MB/s |  6 MB 00:02
CentOS Stream 8 - Extras           35 kB/s | 15 kB 00:00
Dependencies resolved.
Nothing to do.
Complete!
[root@localhost ~]# yum upgrade
Last metadata expiration check: 0:01:40 ago on Fri 22 Oct 2021
09:52:10 PM EDT.
Dependencies resolved.
Nothing to do.
Complete!
[root@localhost ~]#
```

The yum command can also be used to install packages. In the following example, we are using yum to install the nano utility:

```
[root@localhost ~]# yum install nano
Last metadata expiration check: 0:45:41 ago on Fri 22 Oct 2021
10:51:16 PM EDT.
Dependencies resolved.
================================================================================
 Package        Architecture    Version                 Repository        Size
================================================================================
Installing:
 nano           x86_64          2.9.8-1.el8             baseos           581 k

Transaction Summary
================================================================================
Install  1 Package

Total download size: 581 k
Installed size: 2.2 M
Is this ok [y/N]:
```

The yum command has been dropped as an exam objective from the CompTIA exam. However, it is important to know it exists and functions the same as the apt and dnf commands. Therefore, it has been included for completeness. Linux is incredibly stable once it is installed and you could find yourself upgrading a system that only has the yum command available.

Newer Red Hat Linux distributions now come with the DNF package manager. You will love this tool! One simple command both updates the repositories and upgrades them. The dnf command actually looks really similar to yum in its output. You might even have both commands on the Linux distribution. Let's see the typical output of dnf updating and upgrading a Fedora installation in the following. We cut some of the output since the update/upgrade would take several pages. Anywhere there is [Output Cut] signifies we eliminated redundant output for simplicity's sake.

```
[user@localhost ~]$ sudo dnf upgrade
[sudo] password for user:
Updating and loading repositories:
 Fedora 41 openh264 (From Cisco) - x86_ 100% |  11.8 KiB/s |   4.8 KiB |  00m00s
 Fedora 41 - x86_64 - Updates           100% |   7.9 MiB/s |   7.9 MiB |  00m01s
 Fedora 41 - x86_64                     100% |  16.1 MiB/s |  36.4 MiB |  00m02s
Repositories loaded.
```

```
Package                                    Arch   Version
Reposit      Size
Upgrading:
 PackageKit                                x86_64 1.2.8-8.fc41
updates   2.9 MiB
   replacing PackageKit                    x86_64 1.2.8-7.fc41
anacond   2.9 MiB
 PackageKit-glib                           x86_64 1.2.8-8.fc41
updates 520.3 KiB
   replacing PackageKit-glib               x86_64 1.2.8-7.fc41
[Output Cut]

Installing:
 kernel                                    x86_64 6.12.6-200.fc41
updates   0.0   B
Installing dependencies:
 hiredis                                   x86_64 1.2.0-3.fc41
fedora  110.1 KiB
 kernel-core                               x86_64 6.12.6-200.fc41
updates  73.6 MiB
 kernel-modules                            x86_64 6.12.6-200.fc41
[Output Cut]

Transaction Summary:
 Installing:         9 packages
 Upgrading:        190 packages
 Replacing:        190 package

Total size of inbound packages is 518 MiB. Need to download 518 MiB.
After this operation, 182 MiB extra will be used (install 871 MiB, remove 688 MiB).
Is this ok [y/N]: y
[  1/199] kernel-0:6.12.6-200.fc41.x86_ 100% | 499.0 KiB/s | 191.1 KiB |  00m00s
[  2/199] kernel-core-0:6.12.6-200.fc41 100% |  11.0 MiB/s |  18.4 MiB |  00m02s
[  3/199] opensc-libs-0:0.26.0-1.fc41.x 100% |  10.7 MiB/s | 918.0 KiB |  00m00s
[Output Cut]

--------------------------------------------------------------------------------
[199/199] Total                         100% |  39.6 MiB/s | 518.0 MiB |  00m13s
Running transaction
[  1/391] Verify package files          100% | 153.0   B/s | 199.0   B |  00m01s
[  2/391] Prepare transaction           100% |   1.0 KiB/s | 389.0   B |  00m00s
[  3/391] Upgrading glibc-common-0:2.40 100% |  12.5 MiB/s |   1.0 MiB |  00m00s
```

```
[Output Cut]

[202/391] Erasing abrt-cli-0175:2.17.6-2.f 100% |  97.7 KiB/s | 100.0   B |  00m00s
[203/391] Erasing abrt-tui-0:2.17.6-2.f 100% |  25.9 KiB/s |  53.0   B |  00m00s
[204/391] Erasing python3-dnf-0:4.21.1- 100% |  67.1 KiB/s | 275.0   B |  00m00s
[Output Cut]

Complete!
[user@localhost ~]$
```

Let's examine how packages are added with the `dnf` command. You'll find it identical to `apt` and `yum`. In the following output, we will install the `nano` package with the command `sudo dnf install nano`.

```
[user@localhost ~]$ sudo dnf install nano
Updating and loading repositories:
Repositories loaded.
Package                 Arch    Version                 Repository        Size
Installing:
 nano                   x86_64  8.1-1.fc41              fedora         2.9 MiB

Transaction Summary:
 Installing:         1 package

Total size of inbound packages is 729 KiB. Need to download 729 KiB.
After this operation, 3 MiB extra will be used (install 3 MiB, remove 0 B).
Is this ok [y/N]: y
[1/1] nano-0:8.1-1.fc41.x86_64          100% |   3.3 MiB/s | 728.9 KiB |  00m00s
--------------------------------------------------------------------------------
[1/1] Total                             100% |   1.8 MiB/s | 728.9 KiB |  00m00s
Running transaction
[1/3] Verify package files              100% | 200.0   B/s |   1.0   B |  00m00s
[2/3] Prepare transaction               100% |  15.0   B/s |   1.0   B |  00m00s
[3/3] Installing nano-0:8.1-1.fc41.x86_ 100% |  16.4 MiB/s |   2.9 MiB |  00m00s
Complete!
[user@localhost ~]$
```

Now that you know how to update the operating system and install packages, it's inevitable that you'll run out of space on the operating system. Fortunately, you can monitor and quickly find out how much space is free on the disk by using the `df` command, otherwise known as the disk free command. By using the `df` command you can quickly see the percentage of free space, and if you supply the `-h` argument you'll get results in human-readable formats of bytes, as shown here:

```
user@server:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
```

```
udev            1.9G     0  1.9G   0% /dev
tmpfs           393M  1.5M  391M   1% /run
/dev/sda2        20G  6.0G   13G  33% /
tmpfs           2.0G     0  2.0G   0% /dev/shm
tmpfs           5.0M     0  5.0M   0% /run/lock
tmpfs           2.0G     0  2.0G   0% /sys/fs/cgroup
/dev/loop0       91M   91M     0 100% /snap/core/6350
/dev/loop1      100M  100M     0 100% /snap/core/11993
tmpfs           393M     0  393M   0% /run/user/1000
user@server:~$
```

Filesystem Management

The filesystem is a critical component of any Linux operating system. This section will cover some key aspects of filesystem management in Linux that CompTIA has identified as exam objectives for the 220-1202 exam.

mount Command

The `mount` command in Linux is a fundamental utility used to attach filesystems to the directory tree, making them accessible to the system and users. This command allows you to mount various types of filesystems, including local disk partitions, network shares, and removable media like USB drives. By specifying the device and the mount point, you can integrate additional storage into your system seamlessly. For example, to mount a USB drive located at `/dev/sdb1` to the `/mnt/usb` directory, you can use the following command:

```
[user@localhost ~]$ sudo mount -t vfat /dev/sdb1 /mnt/usb
[sudo] password for user:
[user@localhost folder]$ ls -l /mnt/usb
total 0
-rw-rw-rw-. 1 user user 44 Jan  1 13:07 pinball.exe
-rw-rw-rw-. 1 user user 34 Jan  1 12:02 file1.txt
-rw-rw-rw-. 1 user user 34 Jan  1 14:15 file2.txt
-rw-rw-rw-. 1 user user 36 Jan  1 15:05 file3.txt
[user@localhost folder]$
```

The `mount` command by itself will display all the current mount points on the operating system. This list is pretty extensive on a modern Linux operating system. So, we will use the `grep` command to only return the result for our mount of `sdb1`, as shown in the following:

```
[user@localhost ~]$ mount | grep sdb1
/dev/sdb1 on /mnt/usb type vfat (rw,nosuid,nodev,relatime,user_id=0,group_
id=0,allow_other,blksize=4096,x-gvfs-show)
[user@localhost ~]$
```

The `mount` command can also mount an ISO image to the filesystem with something called a *loop device*. A loop device is nothing more than a device driver that acts as a block storage device. `mount` typically mounts devices and not other filesystems. This command would mount an ISO file to the `/mnt/iso` folder:

```
[user@localhost folder]$ sudo mount -o loop ~/file.iso /mnt/iso
[sudo] password for user:
[user@localhost folder]$
```

The */etc/fstab* configuration file

The `/etc/fstab` file, short for *filesystems table*, complements the `mount` command by automating the mounting process. This critical configuration file defines how disk partitions, storage devices, and other filesystems are mounted and integrated into the system's directory structure at boot time. Each entry in `/etc/fstab` specifies the device, mount point, filesystem type, and various mount options, ensuring consistent and reliable access to storage resources across reboots. Here's an example of `/etc/fstab`:

```
[user@localhost folder]$ sudo cat /etc/fstab
[sudo] password for user:
# <file system> <mount point>   <type>  <options>       <dump>  <pass>
/dev/disk/by-id/dm-uuid-LVM-Oj77W3g2yUV2jWGh7e8wL7rkCD3wItJYdw4mvlia987HPTxAzB
eKjavYcK7fZKAz / ext4 defaults 0 0
/dev/disk/by-uuid/a4bdbde5-70cf-4adf-b94f-e5173310e076 /boot ext4 defaults 0 0
/swap.img       none    swap    sw      0       0
/dev/disk/by-id/usb-G-DRIVE_USB-0:0-part2 /media/Storage auto rw,nosuid,
nodev,nofail,x-gvfs-show 0 0
[user@localhost folder]$
```

In the `/etc/fstab` file, each line mounts a different portion of the filesystem. By looking at the second argument of the first line, we can see this is the mount point for `/`, also known as the root of the filesystem. The first argument explains what to mount to the root—in this case, it is a universally unique ID (UUID) used by the Logical Volume Manager (LVM) service. Looking at the third argument, we see it has the ext4 filesystem. The fourth argument contains the options to be used—in this case, we are using the default. The fifth argument specifies whether the mount point should be backed up, otherwise known as *dumped*. The sixth argument tells the OS whether during bootup the filesystem needs to have filesystem consistency checked.

Creating entries in `/etc/fstab` is a bit like baking at high altitude; it takes some time to get use to the variables. A trick is to edit your line in `/etc/fstab`, then use the command `mount -av` to remount all mount points in the `/etc/fstab`.

Now let's examine the second line in the example `/etc/fstab` file. The device UUID of `/dev/disk/by-uuid/a4bdbde5-70cf-4adf-b94f-e5173310e076` is mounted to the `/boot` folder. This is significant because the `/boot` mount point contains both the kernel and the bootloader configuration. The kernel will be a variation of `vmlinuz-*` files, and the bootloader configuration in the following example is the GRand Unified Bootloader (GRUB)

bootloader v2. However, some Linux distributions might also use the LInux LOader (LILO) bootloader.

```
[user@localhost ~]$ ls /boot/
config-6.11.4-301.fc41.x86_64
config-6.12.6-200.fc41.x86_64
efi
grub2
initramfs-0-rescue-12d7e45e4aa444d88c33a0bc2106e0d2.img
initramfs-6.11.4-301.fc41.x86_64.img
initramfs-6.12.6-200.fc41.x86_64.img
loader
symvers-6.11.4-301.fc41.x86_64.xz
symvers-6.12.6-200.fc41.x86_64.xz
System.map-6.11.4-301.fc41.x86_64
System.map-6.12.6-200.fc41.x86_64
vmlinuz-0-rescue-12d7e45e4aa444d88c33a0bc2106e0d2
vmlinuz-6.11.4-301.fc41.x86_64
vmlinuz-6.12.6-200.fc41.x86_64
[user@localhost ~]$
```

The *fsck* Command

The `fsck` (filesystem consistency check) command in Linux is a crucial tool for maintaining the integrity of filesystems. It is used to check and repair inconsistencies and errors in filesystems, ensuring that data remains accessible and the system operates smoothly. The `fsck` command can be run manually or automatically during system boot, depending on the configuration. Here's how you can use `fsck` to check and repair a filesystem:

1. First, identify the device name of the filesystem you want to check. You can use the `lsblk` or `df -h` command. Let's use the example of `/dev/sda1`.

```
[user@localhost ~]$ sudo lsblk
[sudo] password for user:
NAME            MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sr0              11:0    1  2.6G  0 rom
zram0           252:0    0  3.8G  0 disk [SWAP]
nvme0n1         259:0    0   40G  0 disk
|-nvme0n1p1     259:1    0    1M  0 part
|-nvme0n1p2     259:2    0    1G  0 part /boot
|-nvme0n1p3     259:3    0   39G  0 part
  |-fedora-root 253:0    0   15G  0 lvm  /
sdb             260:0    0    8G  0 disk
|-sdb1          260:1    0   39G  0 part /mnt/usb
[user@localhost ~]$
```

2. Before running the `fsck` command, ensure the filesystem is unmounted to prevent data corruption:

```
[user@localhost ~]$ sudo umount /dev/sda1
[sudo] password for user:
[user@localhost ~]$
```

3. Use the `fsck` command to check and repair the filesystem. You can add the `-y` option to automatically answer yes to all prompts:

```
[user@localhost ~]$ sudo fsck -y /dev/sda1
[sudo] password for user:
fsck from util-linux 2.40.2
e2fsck 1.46.5 (30-Dec-0040)
Files: clean, 1129/1200 files, 80340/162345 blocks
[user@localhost ~]$
```

4. The `fsck` command will display the results of the check and any repairs it performed. Review these results. The rule of thumb is that no errors is good news!
5. After the check and repair, remount the filesystem:

```
[user@localhost ~]$ sudo mount /dev/sdb1 /mnt/usb
[sudo] password for user:
[user@localhost ~]$
```

The Linux Boot Process

The Linux boot process, much like its Windows counterpart, is a sequence of events that transforms a powered-off machine into a fully operational system. While Windows relies on the BIOS/UEFI to load the Windows Boot Manager and subsequently the Windows OS loader, Linux employs a combination of EFI (Extensible Firmware Interface), GRUB (GRand Unified Bootloader), and the Linux kernel. Understanding these processes is essential for anyone looking to delve deeper into system administration or operating system internals. The following is a step-by-step breakdown of how these components work together to boot a Linux system:

1. When the computer is powered on, the EFI firmware initializes the hardware and performs a series of checks to ensure everything is functioning correctly. It then looks for the EFI bootloader on the EFI System Partition (ESP).
2. The EFI firmware loads the GRUB bootloader from the ESP. GRUB provides a user interface that allows you to select which operating system or kernel to boot if multiple options are available. It also loads its configuration file, typically `grub.cfg`, which contains the boot menu and options.
3. Once a selection is made, GRUB loads the chosen Linux kernel into memory. It also loads the initial RAM disk (`initramfs` or `initrd`), which contains essential drivers and tools needed to mount the root filesystem.

4. The kernel decompresses itself and initializes the hardware, including the CPU, memory, and peripheral devices. It then mounts the `initramfs` as a temporary root filesystem.
5. The kernel starts the `init` system, such as `systemd` or the traditional `init`. This system is responsible for starting and managing system services and user processes. It brings the system to a specific runlevel or target, defining the state of the system (e.g., multiuser mode, graphical mode).
6. Finally, the system reaches a state where it can present a login prompt or graphical login screen to the user. After login, the user session is started, and the system is ready for use.

While the Linux boot process involves unique components like EFI, GRUB, and the Linux kernel, it shares fundamental similarities with the Windows boot process. Both systems rely on firmware initialization, a bootloader, and a kernel to transition from power-on to a fully functional operating environment. Being familiar with these parallels helps you better understand the Linux operating system by unshrouding the mystery under the hood (so to speak).

Managing Processes

Every time the shell executes a command that you type, it starts a process. The shell itself is a process, as are any scripts or programs that the shell runs. The `ps` command will show a snapshot of the current processes running as the currently logged-in user. Use the `ps ax` command to see a list of processes for the entire operating system. When you type **`ps ax`**, Bash shows you the current set of processes. Here are a few lines of output from the command `ps ax --cols 80` (the `--cols 80` option is used to ensure that you see each command in its entirety):

```
[user@localhost ~]$ ps ax --cols 80
 PID TTY      STAT   TIME COMMAND
    1 ?        Ss     0:03 /usr/lib/systemd/systemd --system
    2 ?        S      0:00 [kthreadd]
    3 ?        S      0:00 [pool_workqueue_release]
    4 ?        I<     0:00 [kworker/R-rcu_gp]
    5 ?        I<     0:00 [kworker/R-sync_wq]
    6 ?        I<     0:00 [kworker/R-slub_flushwq]
    7 ?        I<     0:00 [kworker/R-netns]
   10 ?        I<     0:00 [kworker/0:0H-kblockd]
[Output Cut]
12303 ?        S      0:00 systemd-nsresourcework: waiting...
12304 ?        S      0:00 systemd-nsresourcework: waiting...
12319 ?        S      0:00 systemd-nsresourcework: waiting...
12321 ?        I      0:00 [kworker/0:2-events]
12341 ?        I      0:00 [kworker/1:0-events]
12342 ?        S      0:00 systemd-userwork: waiting...
12345 pts/0    R+     0:00 ps -ax --cols 80
[user@localhost ~]$
```

In this listing, the first column has the heading PID, and it shows a number for each process. PID stands for *process ID (identification)*, which is a sequential number assigned by the Linux kernel. If you look through the output of the ps ax command, you'll see that the systemd command is the first process and has a PID of 1. That's why systemd is referred to as the *mother of all processes*.

The COMMAND column shows the command that created each process, and the TIME column shows the cumulative CPU time used by the process. The STAT column shows the state of a process: S means that the process is sleeping, and R means that it's running. The symbols following the status letter have further meanings; for example, < indicates a high-priority process, and + means that the process is running in the foreground. The TTY column shows the terminal, if any, associated with the process.

The process ID, or process number, is useful when you have to stop an errant process forcibly. Look at the output of the ps ax command and note the PID of the offending process. Then use the kill command with that process number to stop the process. For example, to stop process number 8550, type the following command:

```
[user@localhost ~]$ kill 8550
[user@localhost ~]$
```

The ps command will allow you to see processes currently running at the time you initiate the command. However, if you want to see an interactive display of processes similar to the Windows Task Manager, you can use the top command, shown in Figure 4.23. The top command allows you to sort by columns and scroll through the various processes. You can even kill processes interactively.

As you can see, the top command is extremely similar to the Windows Task Manager. Some of the common uses for the top command are similar to its Windows counterpart:

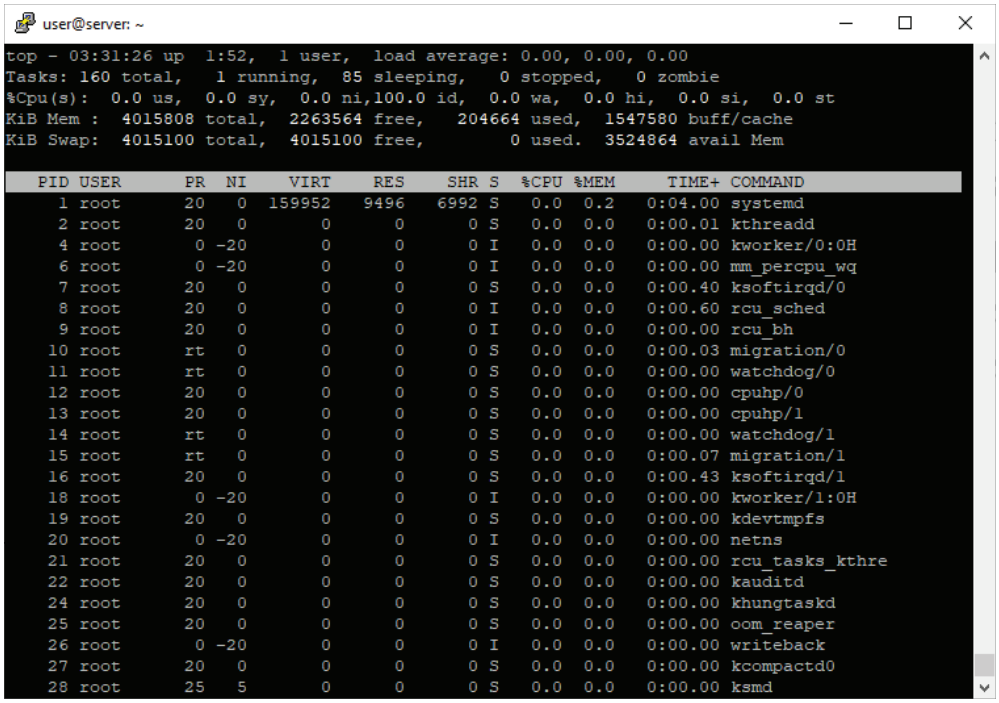
Monitoring System Performance The top command provides a real-time overview of system performance by displaying key metrics such as CPU and memory usage. This allows users to quickly identify any processes that are consuming excessive resources, which can help in diagnosing performance issues and ensuring that the system runs smoothly. By continuously updating the display, top offers an up-to-date snapshot of the system's health.

Process Management With top, users can view detailed information about each running process, including the process ID (PID), user, priority, and execution time. This information is crucial for managing processes, especially when troubleshooting issues or optimizing system performance. For instance, if a particular process is causing problems, users can easily identify it and take appropriate action, such as terminating or renicing the process.

Resource Allocation The top command helps users understand how system resources are being allocated by displaying resource usage for each process. This includes metrics like CPU and memory consumption, which are essential for effective resource management. By analyzing this data, users can make informed decisions about resource

allocation, such as adjusting the priority of processes or redistributing workloads to ensure optimal performance.

FIGURE 4.23 The `top` command



Interactive Commands One of the powerful features of `top` is its interactive interface, which allows users to perform actions directly from the command. For example, users can kill a process by pressing `k` and entering the PID, or change the priority of a process using the `r` command. These interactive commands provide a convenient way to manage processes without needing to exit the `top` interface, making it a highly efficient tool for system administration.

Customizable Display The `top` command offers a customizable display, allowing users to tailor the information shown to their specific needs. Users can sort processes by different criteria, such as memory usage or CPU time, and choose which columns to display. Additionally, these preferences can be saved for future sessions, ensuring a consistent and personalized experience. This flexibility makes `top` a versatile tool that can be adapted to various monitoring and management tasks.

System Uptime and Load Average In addition to process information, `top` displays system uptime and load averages, which provide insights into the system's overall health

and stability. Uptime indicates how long the system has been running without a reboot, while load averages show the average system load over the past 1, 5, and 15 minutes. These metrics are useful for assessing the long-term performance and reliability of the system, helping users identify trends and potential issues.

All of these uses for the `top` command should sound similar to its Windows counterpart. Remembering those similarities is the best way to learn Linux.

Directory Navigation

In Linux, when you log in as root, your home directory is `/root`. For other users, the home directory is usually in the `/home` directory. For example, the home directory for a user logging in as *testuser* is `/home/testuser`. This information is stored in the `/etc/passwd` file. By default, only you have permission to save files in your home directory, and only you can create subdirectories in your home directory to organize your files further.

Linux supports the concept of a *current directory*, which is the directory on which all file and directory commands operate. After you log in, for example, your current directory is the home directory. To see the current directory, type the **`pwd`** command.

```
[user@localhost ~]$ pwd
/home/user
[user@localhost ~]$
```

To change the current directory, use the `cd` command. To change the current directory to `/usr/lib`, type the following:

```
[user@localhost ~]$ cd /usr/lib
[user@localhost lib]$
```

Then to change the directory to the `modules` subdirectory in `/usr/lib`, type the following command:

```
[user@localhost lib]$ cd modules
[user@localhost modules]$
```

You should notice that your command prompt reflects a portion of your current directory or focus directory. This functionality will differ depending on the shell. However, the Bash shell typically shows the focus directory. If you need the full path, we can use the `pwd` command, which will output `/usr/lib/modules` as the current full working directory path:

```
[user@localhost modules]$ pwd
/usr/lib/modules
[user@localhost modules]$
```

These two examples show that you can refer to a directory's name in two ways: with an absolute pathname or a relative pathname. An example of an absolute pathname is `/usr/lib`, which is an exact directory in the directory tree. (Think of the absolute pathname as the

complete mailing address for a package that the postal service will deliver to your next-door neighbor.) An example of a relative pathname is `modules`, which represents the `modules` subdirectory of the current directory, whatever that may be. (Think of the relative directory name as giving the postal carrier directions from your house to the one next door so that the carrier can deliver the package.)

If you type **`cd modules`** in **`/usr/lib`**, the current directory changes to `/usr/lib/modules`. However, if you type the same command in `/home/testuser`, the shell tries to change the current directory to `/home/testuser/modules`. This directory of `/home/testuser/modules` probably doesn't exist, but I think you get the idea.

Use the `cd` command without any arguments to change the current directory back to your *home directory*. No matter where you are, typing **`cd`** at the shell prompt brings you back home. The tilde character (~) is an alias that refers to your home directory. Thus, you can also change the current directory to your home directory by using the command `cd ~`. You can refer to another user's home directory by appending that user's name to the tilde. Thus, `cd ~comptia` changes the current directory to the home directory of the CompTIA user. Of course, we need to elevate our privileges with the `su -` command first, since the one user account will not have the privileges to another user's home directory. However, the root user does!

```
[user@localhost modules]$ cd
[user@localhost ~]$ pwd
/home/user
[user@localhost ~]$ su -
Password:
[root@localhost comptia]# whoami
root
[root@localhost ~]# cd ~comptia
[root@localhost comptia]# pwd
/home/comptia
[root@localhost comptia]#
```

A single dot (`.`) and two dots (`..`), often referred to as *dot-dot*, also have special meanings. A single dot (`.`) indicates the current directory, whereas two dots (`..`) indicate the parent directory. For example, if the current directory is `/usr/share`, you go one level up to `/usr` by typing the following:

```
[root@localhost comptia]# pwd
/home/comptia
[root@localhost comptia]# cd ..
[root@localhost home]# pwd
/home
[root@localhost home]# ls -la
total 0
drwxr-xr-x.  4 root    root     33 Jan  1 13:44 .
```

```
dr-xr-xr-x. 18 root    root    235 Jan  1 12:11 ..
drwx------.  2 comptia comptia  62 Jan  1 13:44 comptia
drwx------.  3 user    user    114 Jan  1 13:22 user
[root@localhost home]#
```

After executing the `ls -la` command, you can see the directory structure you just navigated. You can also see the single dot and double dot referenced in the directory listing. In the next section, we'll examine directory listing more closely.

Directory Listings

You can get a directory listing by using the `ls` command. By default, the `ls` command, without any options, displays the contents of the current directory in a compact, multicolumn format. To tell the directories and files apart, use the `-F` option (`ls -F`). The output will show the directory names with a slash (/) appended to them. Plain filenames appear as is. The at sign (@) appended to a listing indicates that this file is a link to another file. (In other words, this filename simply refers to another file; it's a shortcut.) An asterisk (*) is appended to executable files. (The shell can run any executable file.)

```
[user@localhost /]$ ls -F
afs/  boot/  etc/   lib@    media/  opt/   root/  sbin@  sys/  usr/
bin@  dev/   home/  lib64@  mnt/    proc/  run/   srv/   tmp/  var/
[user@localhost /]$
```

You can see even more detailed information about the files and directories with the `-l` (long format) option. The rightmost column shows the name of the directory entry. The date and time before the name show when the last modifications to that file were made. To the left of the date and time is the size of the file in bytes. The file's group and owner appear to the left of the column that shows the file size. The next number to the left indicates the number of links to the file. (A *link* is like a shortcut in Windows.)

```
[user@localhost /]$ ls -l
total 20
dr-xr-xr-x.   2 root root    6 Jul 16 20:00 afs
lrwxrwxrwx.   1 root root    7 Jul 16 20:00 bin -> usr/bin
dr-xr-xr-x.   5 root root 4096 Jan  1 12:31 boot
drwxr-xr-x.  20 root root 3920 Jan  1 12:28 dev
drwxr-xr-x. 108 root root 8192 Jan  1 13:44 etc
drwxr-xr-x.   4 root root   33 Jan  1 13:44 home
lrwxrwxrwx.   1 root root    7 Jul 16 20:00 lib -> usr/lib
lrwxrwxrwx.   1 root root    9 Jul 16 20:00 lib64 -> usr/lib64
drwxr-xr-x.   2 root root    6 Jul 16 20:00 media
drwxr-xr-x.   3 root root   18 Jan  1 12:12 mnt
```

```
[Output Cut]

[user@localhost /]$
```

Linux Permissions

The leftmost column of a long style directory listing shows the file's permission settings, which determine who can read, write, or execute the file. This column shows a sequence of nine characters, which appear as `rwxrwxrwx` when each letter is present. Each letter indicates a specific permission. A hyphen (`-`) in place of a letter indicates no permission for a specific operation on the file. Think of these nine letters as three groups of three letters (`rwx`), interpreted as follows:

Leftmost Group Controls the read, write, and execute permissions of the file's owner. In other words, if you see `rwx` in this position, the file's owner can read (`r`), write (`w`), and execute (`x`) the file. A hyphen in the place of a letter indicates no permission. Thus, the string `rw-` means that the owner has read and write permissions but not execute permission. Although executable programs (including shell programs) typically have execute permission, directories treat execute permission as equivalent to *use* permission: a user must have execute permission on a directory before they can open and read the contents of the directory.

Middle Group Controls the read, write, and execute permissions of any user belonging to that file's group.

Rightmost Group Controls the read, write, and execute permissions of all other users (collectively thought of as *the world*).

Thus, a file with the permission setting `rwx------` is accessible only to the file's owner, whereas the permission setting `rwxr--r--` makes the file readable by the world.

An interesting feature of the `ls` command is that it doesn't list any file the name of which begins with a period. To see these files, you must use the `ls` command with the `-a` option, as follows:

```
[user@localhost ~]$ ls -a
.   .bash_history  .bash_profile  .config
..  .bash_logout   .bashrc        .lesshst
[user@localhost ~]$
```

Most Linux commands take single-character options, each with a hyphen as a prefix. When you want to use several options, type a hyphen and concatenate (string together) the option letters, one after another. Thus, `ls -al` is equivalent to `ls -a -l` as well as to `ls -l -a`.

```
[user@localhost ~]$ ls -al
```

```
total 20
drwx------. 3 user user 114 Jan  1 13:22 .
drwxr-xr-x. 4 root root  33 Jan  1 13:44 ..
-rw-------. 1 user user   7 Jan  1 12:16 .bash_history
-rw-r--r--. 1 user user  18 Aug 11 20:00 .bash_logout
-rw-r--r--. 1 user user 144 Aug 11 20:00 .bash_profile
-rw-r--r--. 1 user user 522 Aug 11 20:00 .bashrc
drwx------. 3 user user  20 Jan  1 12:47 .config
-rw-------. 1 user user  20 Jan  1 13:22 .lesshst
[user@localhost ~]$
```

Changing Permissions and Ownership

You may need to change a file's permission settings to protect it from others. Use the `chmod` command to change the permission settings of a file or a directory. To use `chmod` effectively, you have to specify the permission settings. A good way is to concatenate letters from the columns of Table 4.2 in the order shown (who/action/permission). You use only the single character from each column—the text in parentheses is for explanation only.

TABLE 4.2 Letter codes for file permissions

Who	Action	Permission
u (user)	+ (add)	r (read)
g (group)	– (remove)	w (write)
o (others)	= (assign)	x (execute)
a (all)	s (set user ID)	

For example, to give everyone read access to all the files in a directory, pick `a` (for *all*) from the first column, `+` (for *add*) from the second column, and `r` (for *read*) from the third column, to come up with the permission setting `a+r`. Then use the set of options with `chmod`, as follows:

```
[user@localhost folder]$ ls -l
total 0
-rw-r--r--. 1 user user   0 Jan  1 15:07 bin.exe
-rw-r--r--. 1 user user  34 Jan  1 15:05 file1.txt
-rw-r--r--. 1 user user  34 Jan  1 15:05 file2.txt
-rw-r--r--. 1 user user  36 Jan  1 15:05 file3.txt
```

```
[user@localhost folder]$ chmod a+rw *
[user@localhost folder]$ ls -l
total 0
-rw-rw-rw-. 1 user user   0 Jan  1 15:07 bin.exe
-rw-rw-rw-. 1 user user  34 Jan  1 15:05 file1.txt
-rw-rw-rw-. 1 user user  34 Jan  1 15:05 file2.txt
-rw-rw-rw-. 1 user user  36 Jan  1 15:05 file3.txt
[user@localhost folder]$
```

To permit everyone to execute one specific file, type the following:

```
[user@localhost folder]$ chmod a+x bin.exe
[user@localhost folder]$ ls -l
total 0
-rwxrwxrwx. 1 user user   0 Jan  1 15:07 bin.exe
-rw-rw-rw-. 1 user user  34 Jan  1 15:05 file1.txt
-rw-rw-rw-. 1 user user  34 Jan  1 15:05 file2.txt
-rw-rw-rw-. 1 user user  36 Jan  1 15:05 file3.txt
[user@localhost folder]$
```

Sometimes you have to change a file's user or group ownership in order for everything to work correctly. For example, suppose you're instructed to create a directory named *cups* and give it the ownership of user ID *lp* and group ID *sys*. You create the `cups` directory with the command `mkdir` as follows:

```
[user@localhost folder]$ mkdir cups
[user@localhost folder]$ ls -l
total 0
-rwxrwxrwx. 1 user user  0 Jan  1 15:07 bin.exe
drwxr-xr-x. 2 user user  6 Jan  1 15:13 cups
-rw-rw-rw-. 1 user user 34 Jan  1 15:05 file1.txt
-rw-rw-rw-. 1 user user 34 Jan  1 15:05 file2.txt
-rw-rw-rw-. 1 user user 36 Jan  1 15:05 file3.txt
[user@localhost folder]$
```

If you check the file's details with the `ls -l` command, you see that the user and group ownership are both assigned to user. To change the owner, use the `chown` command. For example, to change the ownership of the `cups` directory to user ID *lp* and group ID *sys*, type the following:

```
[user@localhost folder]$ sudo chown lp:sys cups
[sudo] password for user:
[user@localhost folder]$ ls -l
total 0
-rwxrwxrwx. 1 user user  0 Jan  1 15:07 bin.exe
```

```
drwxr-xr-x. 2 lp   sys   6 Jan  1 15:13 cups
-rw-rw-rw-. 1 user user 34 Jan  1 15:05 file1.txt
-rw-rw-rw-. 1 user user 34 Jan  1 15:05 file2.txt
-rw-rw-rw-. 1 user user 36 Jan  1 15:05 file3.txt
[user@localhost folder]$
```

Working with Files

Working with files in Linux is a fundamental skill for anyone using the Linux operating system. Whether you're a beginner or an experienced user, mastering file operations is essential for efficient system management and productivity. In the following, we'll explore key file management tasks such as copying files, moving files, and viewing file contents. Each of these operations plays a crucial role in organizing and manipulating data, ensuring that you can handle your files with ease and precision. Let's dive into the world of Linux file management and discover how to perform these tasks effectively.

Copying Files

To copy files from one directory to another, use the `cp` command. If you want to copy a file to the current directory but retain the original name, use a period (`.`) as the second argument of the `cp` command. Thus, the following command copies the `host.conf` file from the `/etc` directory to the current directory (denoted by a single period):

```
[user@localhost ~]$ cp /etc/host.conf .
[user@localhost ~]$ ls -l
total 4
drwxr-xr-x. 3 user user 84 Jan  1 15:13 folder
-rw-r--r--. 1 user user  9 Jan  1 15:22 host.conf
[user@localhost ~]$
```

The `cp` command makes a new copy of a file and leaves the original intact.

If you want to copy the entire contents of a directory—including all subdirectories and their contents—to another directory, use the command `cp -ar` *`sourcedir destdir`*. (This command copies everything in the *`sourcedir`* directory to the *`destdir`* directory.) For example, to copy all the files from the `/etc/ssl` directory to the current directory, type the following command:

```
[user@localhost ~]$ cp -ar /etc/ssl .
[user@localhost ~]$ ls -l
total 4
drwxr-xr-x. 3 user user 84 Jan  1 15:13 folder
-rw-r--r--. 1 user user  9 Jan  1 15:22 host.conf
drwxr-xr-x. 2 user user 91 Jan  1 12:11 ssl
[user@localhost ~]$
```

Moving Files

To move a file to a new location, use the `mv` command; the syntax is similar to the `cp` command, such as `mv source destination`. The original file or folder will be gone, and a new copy appears at the destination. You can use `mv` to rename a file. If you want to change the name of `host.conf` to `host.conf.bak`, use the `mv` command as follows:

```
[user@localhost ~]$ ls -l
total 4
drwxr-xr-x. 3 user user 84 Jan  1 15:29 folder
-rw-r--r--. 1 user user  9 Jan  1 15:22 host.conf
drwxr-xr-x. 2 user user 91 Jan  1 12:11 ssl
[user@localhost ~]$ mv host.conf host.conf.bak
[user@localhost ~]$ ls -l
total 4
drwxr-xr-x. 3 user user 84 Jan  1 15:29 folder
-rw-r--r--. 1 user user  9 Jan  1 15:22 host.conf.bak
drwxr-xr-x. 2 user user 91 Jan  1 12:11 ssl
[user@localhost ~]$
```

If you want to move the `host.conf.bak` file to a subdirectory named `folder`, use the following command:

```
[user@localhost ~]$ mv host.conf.bak folder/
[user@localhost ~]$ ls -l
total 0
drwxr-xr-x. 3 user user 105 Jan  1 15:31 folder
drwxr-xr-x. 2 user user  91 Jan  1 12:11 ssl
[user@localhost ~]$ ls -l folder/
total 4
-rwxrwxrwx. 1 user user  0 Jan  1 15:07 bin.exe
drwxr-xr-x. 2 lp   sys   6 Jan  1 15:13 cups
-rw-rw-rw-. 1 user user 34 Jan  1 15:05 file1.txt
-rw-rw-rw-. 1 user user 34 Jan  1 15:05 file2.txt
-rw-rw-rw-. 1 user user 36 Jan  1 15:05 file3.txt
-rw-r--r--. 1 user user  9 Jan  1 15:22 host.conf.bak
[user@localhost ~]$
```

An interesting feature of `mv` is that you can use it to move entire directories (with all their subdirectories and files) to a new location. If you have a directory named `data` that contains many files and subdirectories, you can move that entire directory structure to `old_data` by using the following command:

```
[user@localhost folder]$ ls -l
total 4
```

```
-rwxrwxrwx. 1 user user  0 Jan  1 15:07 bin.exe
drwxr-xr-x. 2 lp   sys   6 Jan  1 15:13 cups
drwxr-xr-x. 2 user user 57 Jan  1 15:34 files
-rw-r--r--. 1 user user  9 Jan  1 15:22 host.conf.bak
[user@localhost folder]$ ls -l files
total 0
-rw-rw-rw-. 1 user user 34 Jan  1 15:05 file1.txt
-rw-rw-rw-. 1 user user 34 Jan  1 15:05 file2.txt
-rw-rw-rw-. 1 user user 36 Jan  1 15:05 file3.txt
[user@localhost folder]$ mv files ~/
[user@localhost folder]$ cd ~
[user@localhost ~]$ ls -l
total 0
drwxr-xr-x. 2 user user 57 Jan  1 15:34 files
drwxr-xr-x. 3 user user 54 Jan  1 15:35 folder
drwxr-xr-x. 2 user user 91 Jan  1 12:11 ssl
[user@localhost ~]$
```

To delete files, use the `rm` command. For example, to delete a file named `file4.txt`, type the following command:

```
[user@localhost ~]$ ls -l
total 0
-rw-r--r--. 1 user user 34 Jan  1 15:36 file4.txt
drwxr-xr-x. 2 user user 57 Jan  1 15:34 files
drwxr-xr-x. 3 user user 54 Jan  1 15:35 folder
drwxr-xr-x. 2 user user 91 Jan  1 12:11 ssl
[user@localhost ~]$ rm file4.txt
[user@localhost ~]$ ls -l
total 0
drwxr-xr-x. 2 user user 57 Jan  1 15:34 files
drwxr-xr-x. 3 user user 54 Jan  1 15:35 folder
drwxr-xr-x. 2 user user 91 Jan  1 12:11 ssl
[user@localhost ~]$
```

To delete a folder, you need to add the `-rf` switch to the command line. The `-r` switch is for recursive operations and the `-f` is for force, so you don't get a bunch of message prompts asking if it is okay to delete a subfolder. The following displays what happens if you forget the `-rf` on the command line:

```
[user@localhost ~]$ ls -l
total 0
drwxr-xr-x. 2 user user 57 Jan  1 15:34 files
```

```
drwxr-xr-x. 3 user user 54 Jan  1 15:35 folder
drwxr-xr-x. 2 user user 91 Jan  1 12:11 ssl
[user@localhost ~]$ rm folder
rm: cannot remove 'folder': Is a directory
[user@localhost ~]$ rm -rf folder
[user@localhost ~]$ ls -l
total 0
drwxr-xr-x. 2 user user 57 Jan  1 15:34 files
drwxr-xr-x. 2 user user 91 Jan  1 12:11 ssl
[user@localhost ~]$
```

Be careful with the `rm` command, especially when you log in as root. You can inadvertently delete important files with `rm`.

Viewing Files

Sometime you just need to view the contents of a file. To view the contents of a file, use the `cat` command. The `cat` command will display the contents of a file, as shown in the following example:

```
[user@localhost ~]$ cat /etc/hosts
# Loopback entries; do not change.
# For historical reasons, localhost precedes localhost.localdomain:
127.0.0.1   localhost localhost.localdomain localhost4 localhost4.localdomain4
::1         localhost localhost.localdomain localhost6 localhost6.localdomain6
# See hosts(5) for proper format and other examples:
# 192.168.1.10 foo.example.org foo
# 192.168.1.13 bar.example.org bar
[user@localhost ~]$
```

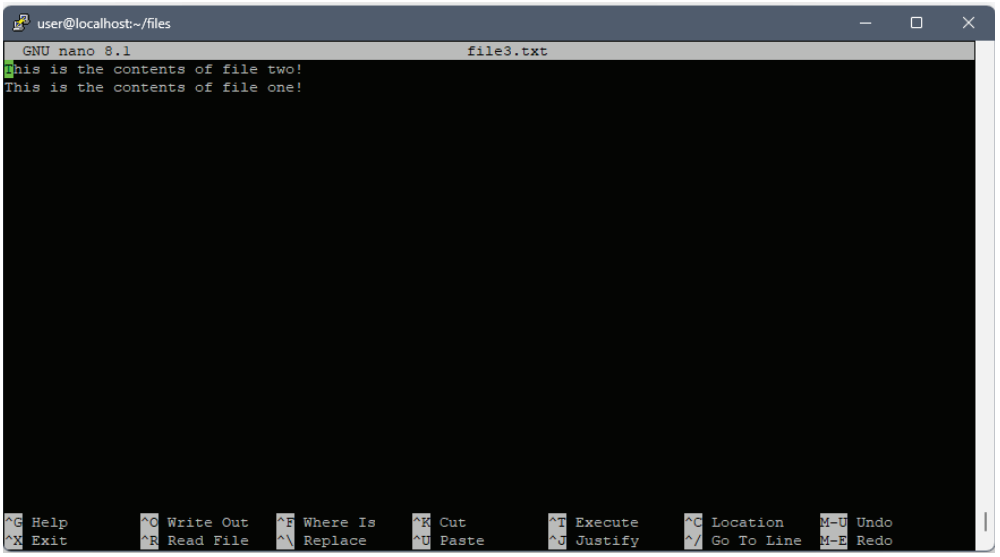
The `cat` file can also be used to concatenate multiple files. For example, you may have two files that you want to join together. By using the > redirector, you can redirect the output of two or more files into one file, as follows:

```
[user@localhost files]$ cat file1.txt
This is the contents of file one!
[user@localhost files]$ cat file2.txt
This is the contents of file two!
[user@localhost files]$ cat file2.txt file1.txt > file3.txt
[user@localhost files]$ cat file3.txt
This is the contents of file two!
This is the contents of file one!
[user@localhost files]$
```

Editing Files

The original text-based editor that came with Linux/UNIX was the `vi` editor, which was short for *visual editor*. The original `vi` editor was not end user–friendly, and it often required a cheat sheet to get anything done. Thankfully, the `nano` and `pico` editors were adopted by Linux, which made editing text-based config files much easier. You can launch the `nano` editor by typing the **nano** command followed by the file you want to edit, such as `nano file3`. This will launch the editor, as shown in Figure 4.24.

FIGURE 4.24 The nano editor



Once the editor is launched, it acts similar to the Windows Notepad utility. You can use the arrow keys to navigate text; find text; replace text; and cut, copy, and paste text. The editor is also very intuitive; by using the Control key sequences, you can perform all the functions and even get help.

Locating Files and Content

One of the most important functions you can perform on files is finding them. In Windows we can use the familiar Ctrl+F key sequence to search for files in a folder structure. In Linux the `find` command allows you to do the same. You can find a file in a folder structure, as shown here:

```
[user@localhost dev]$ find ttyS12
ttyS12
[user@localhost dev]$
```

The grep command in Linux is a powerful utility used for searching and filtering text within files. Whether you're a system administrator, developer, or just a casual user, mastering grep can significantly enhance your ability to find and manipulate data efficiently. The grep command includes a method of using regular expressions, and by combining grep with other commands, you'll get more advanced text processing.

The primary use of grep is to search for specific patterns within files. For example, to find all occurrences of the word Error in a log file, you can use the following command:

```
[root@localhost ~]# grep  "Error" /var/log/messages
Jan  1 12:16:26 localhost kernel: RAS: Correctable Errors collector
initialized.
Jan  1 12:28:33 localhost kernel: RAS: Correctable Errors collector
initialized.
[root@localhost ~]#
```

The grep command also supports regular expressions, allowing for more complex and flexible search patterns. For instance, to find lines that contain either error or info you can use the command grep -E "Error|Info" /var/log/messages. The -E option enables extended regular expressions, providing powerful pattern-matching capabilities.

```
[root@localhost ~]# grep -E "Error|Info" /var/log/messages
Jan  1 12:16:26 localhost kernel: RAS: Correctable Errors collector
initialized.
Jan  1 12:16:29 localhost systemd[1]: systemd-hibernate-clear.service - Clear
Stale Hibernate Storage Info was skipped because of an unmet condition check
(ConditionPathExists=/sys/firmware/efi/efivars/HibernateLocation-8cf2644b-4b0b
-428f-9387-6d876050dc67).
Jan  1 12:28:33 localhost kernel: RAS: Correctable Errors collector
initialized.
Jan  1 12:28:36 localhost systemd[1]: systemd-hibernate-clear.service - Clear
Stale Hibernate Storage Info was skipped because of an unmet condition check
(ConditionPathExists=/sys/firmware/efi/efivars/HibernateLocation-8cf2644b-4b0b
-428f-9387-6d876050dc67).
[root@localhost ~]#
```

Sometimes, you may need to perform case-insensitive searches. The -i option allows you to ignore case differences. The command line grep -i -E "Error|Info" /var/log/messages will search for error or info, as shown here:

```
[root@localhost ~]# grep -i -E "Error|Info" /var/log/messages
Jan  1 12:16:26 localhost kernel: ACPI: Using ACPI (MADT) for SMP
configuration information
Jan  1 12:16:26 localhost kernel: RAS: Correctable Errors collector
initialized.
Jan  1 12:16:29 localhost systemd[1]: systemd-hibernate-clear.service - Clear
Stale Hibernate Storage Info was skipped because of an unmet condition check
(ConditionPathExists=/sys/firmware/efi/efivars/HibernateLocation-8cf2644b-4b0b
-428f-9387-6d876050dc67).
```

```
Jan  1 12:16:30 localhost mcelog[934]: mcelog: ERROR: AMD Processor family 23:
mcelog does not support this processor.  Please use the edac_mce_amd module
instead.
Jan  1 12:16:30 localhost rsyslogd[936]: [origin software="rsyslogd"
swVersion="8.2312.0-5.fc41" x-pid="936" x-info="https://www.rsyslog
.com"] start
Jan  1 12:16:31 localhost chronyd[970]: Using leap second list /usr/share/
zoneinfo/leap-seconds.list
Jan  1 12:16:31 localhost NetworkManager[1057]: <info>  [1735751791.6809]
NetworkManager (version 1.50.0-1.fc41) is starting... (boot:c0e22ce0-c23f-45ee
-9801-0953d63c8548)
Jan  1 12:16:31 localhost NetworkManager[1057]: <info>  [1735751791.6820] Read
config: /etc/NetworkManager/NetworkManager.conf (lib: 22-wifi-mac-addr.conf,
99-nvme-nbft-no-ignore-carrier.conf)
[Output Cut]


[root@localhost ~]#
```

You can even search files recursively for a pattern within a directory and its subdirectories by using the `-r` option. The command `grep -r "sshd" /var/log/` will search for `sshd` in all files under the `/var/log/` directory, providing a search across multiple files, as shown here:

```
[root@localhost ~]# grep -r "sshd" /var/log/
/var/log/anaconda/journal.log:Jan 01 17:12:25 localhost.localdomain
groupadd[3823]: group added to /etc/group: name=sshd, GID=74
/var/log/anaconda/journal.log:Jan 01 17:12:25 localhost.localdomain
groupadd[3823]: group added to /etc/gshadow: name=sshd
/var/log/anaconda/journal.log:Jan 01 17:12:25 localhost.localdomain
groupadd[3823]: new group: name=sshd, GID=74
/var/log/boot.log:[  OK  ] Created slice system-sshd\x2dkeygen.slice - Slice /
system/sshd-keygen.
/var/log/boot.log:[  OK  ] Listening on sshd-UNIX-local.socket - OpenSSH
Server Socket (systemd-ssh-generator, AF_UNIX Local).
/var/log/boot.log:[  OK  ] Listening on sshd-vsock.socket - OpenSSH Server
Socket (systemd-ssh-generator, AF_VSOCK).
/var/log/boot.log:         Starting sshd-keygen@ecdsa.service - OpenSSH ecdsa
Server Key Generation...
/var/log/boot.log:         Starting sshd-keygen@ed25519.service - OpenSSH
ed25519 Server Key Generation...
/var/log/boot.log:         Starting sshd-keygen@rsa.service - OpenSSH rsa
Server Key Generation...
[Output Cut]


[root@localhost ~]#
```

The grep command can be combined with other commands using pipes to create powerful command-line workflows. For example, to list all running processes and filter for those containing ssh you can use the following:

```
[root@localhost ~]# ps aux | grep "ssh"
root        1215  0.0  0.2  15416 11076 ?        Ss   12:28   0:00 sshd-
session: user [priv]
user        1232  0.0  0.1  15680  7080 ?        S    12:28   0:01 sshd-
session: user@pts/0
root       11744  0.0  0.1   8160  6024 ?        Ss   12:31   0:00 sshd: /usr/
sbin/sshd -D [listener] 0 of 10-100 startups
root       14035  0.0  0.0   6500  2136 pts/0    S+   20:21   0:00 grep
--color=auto ssh
[root@localhost ~]#
```

By mastering these grep techniques, you can efficiently search and filter text in Linux, making grep an indispensable tool for managing and analyzing data.

In Exercise 4.3 you will work with some basic files on the Linux operating system to get some exposure to working with files.

EXERCISE 4.3

Working with Files

1. Log into a Linux or macOS operating system and open a terminal.
2. Type **cd ~** and press Enter. This will change the directory to your home folder.
3. Type **cat > file1** and press Enter. Begin typing **this is file1 contents**, then press Control+D.
4. Type **cat > file2** and press Enter. Begin typing **this is file2 contents**, then press Control+D.
5. Type **ls** and press Enter to examine the two files you've created.
6. Type **cat file2 file1 > file3** and press Enter. This will concatenate the two files.
7. Type **cat file3** and press Enter to examine the contents of file3.

In this example, we used the cat command to redirect console input to a file, such as file1 and file2. (We could have used the nano command if it was available to achieve the same outcome.) We then used the cat command to concatenate the two files.

Working with Directories

To organize files in your home directory, you have to create new directories. Use the `mkdir` command to create a directory. For example, to create a directory named `images` in the current directory, type the following:

```
[user@localhost ~]$ mkdir images
[user@localhost ~]$ ls -l
total 0
drwxr-xr-x. 2 user user 6 Jan  1 20:24 images
[user@localhost ~]$
```

After you create the directory, you can use the `cd images` command to change to that directory.

```
[user@localhost ~]$ cd images
[user@localhost images]$
```

You can create an entire directory tree by using the `-p` option with the `mkdir` command. For example, suppose you need to create a directory path of `files/src/jscript/examples`. To create this directory hierarchy, type the following command:

```
[user@localhost ~]$ mkdir -p files/src/jscript/examples
[user@localhost ~]$ ls -l
total 0
drwxr-xr-x. 3 user user 17 Jan  1 20:27 files
drwxr-xr-x. 2 user user  6 Jan  1 20:24 images
[user@localhost ~]$ cd files
[user@localhost files]$ ls -l
total 0
drwxr-xr-x. 3 user user 21 Jan  1 20:27 src
[user@localhost files]$ cd src/jscript/examples/
[user@localhost examples]$
```

When you no longer need a directory, use the `rmdir` command to delete it. You can delete a directory only when the directory is empty. To remove an empty directory tree, you can use the `-p` option, as follows:

```
[user@localhost examples]$ cd ~
[user@localhost ~]$ rmdir -p files/src/jscript/examples/
[user@localhost ~]$ ls -l
total 0
drwxr-xr-x. 2 user user 6 Jan  1 20:24 images
[user@localhost ~]$
```

This command removes the empty parent directories of applets. The command stops when it encounters a directory that's not empty.

Directory Usage

The du (disk usage) command in Linux is an invaluable tool for assessing and managing disk space usage. Understanding how to use du command can help you find out-of-control disk usage. In the following, we examine various applications of the du command, including checking directory sizes and summarizing disk usage.

The most common use of the du command is to check the size of directories. For example, to display the size of the /home directory, you can use the command `sudo du -sh /home`. The `-s` option provides a summary of the directory size, and the `-h` option makes the output human-readable, showing sizes in KB, MB, or GB. Of course, we need to use the sudo command, since our regular user account will not have permissions to everyone's home folder:

```
[user@localhost ~]$ sudo du -sh /home
[sudo] password for user:
32K     /home
[user@localhost ~]$
```

If we want to get a summary of disk usage for all directories within a specific path, you can add the asterisk to the end of the path. This command will display the size of each directory and file in the current directory, providing a quick overview of disk usage:

```
[user@localhost ~]$ sudo du -sh /home/*
12K     /home/comptia
20K     /home/user
[user@localhost ~]$
### Displaying Disk Usage for All Files
```

If you need to see the disk usage for all files within a directory, including subdirectories, use the command `sudo du -ah /home`. The `-a` option includes all files, and the `-h` option makes the output human-readable.

```
[user@localhost ~]$ sudo du -ah /home
4.0K    /home/user/.bash_logout
4.0K    /home/user/.bash_profile
4.0K    /home/user/.bashrc
4.0K    /home/user/.bash_history
0       /home/user/.config/procps
0       /home/user/.config
4.0K    /home/user/.lesshst
0       /home/user/images
20K     /home/user
4.0K    /home/comptia/.bash_logout
4.0K    /home/comptia/.bash_profile
4.0K    /home/comptia/.bashrc
```

```
12K     /home/comptia
32K     /home/
[user@localhost ~]$
```

By mastering these du techniques, you can efficiently manage disk space in Linux, ensuring that your filesystem remains organized and optimized.

Networking Utilities

Networking utilities in Linux are essential tools for managing and troubleshooting network connections. Understanding these utilities can greatly enhance your ability to diagnose and resolve network issues.

Managing Network Interfaces

Just as you can use the `ipconfig` command to see the status of IP configuration with Windows, the `ifconfig` command can be used in Linux. You can get information about the usage of the `ifconfig` command by using `ifconfig -help`. The following output provides an example of the basic `ifconfig` command run on a Linux system:

```
[user@localhost ~]$ ifconfig
ens160: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
    inet 172.16.1.13  netmask 255.240.0.0  broadcast 172.31.255.255
    inet6 fe80::20c:29ff:fe01:5345  prefixlen 64  scopeid 0x20<link>
    ether 00:0c:29:01:53:45  txqueuelen 1000  (Ethernet)
    RX packets 464139  bytes 624240392 (595.3 MiB)
    RX errors 0  dropped 32191  overruns 0  frame 0
    TX packets 79223  bytes 7326946 (6.9 MiB)
    TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
    inet 127.0.0.1  netmask 255.0.0.0
    inet6 ::1  prefixlen 128  scopeid 0x10<host>
    loop  txqueuelen 1000  (Local Loopback)
    RX packets 4  bytes 240 (240.0 B)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 4  bytes 240 (240.0 B)
    TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

[user@localhost ~]$
```

In addition to using `ifconfig`, Linux users can use the `iwconfig` command to view the state of their wireless network. By using `iwconfig`, you can view such important information as the link quality, access point (AP) MAC address, data rate, and encryption keys, which can be helpful in ensuring that the parameters in the network are consistent.

The ip utility has replaced the ifconfig utility on most modern Linux distributions. Red Hat Enterprise Linux has adopted the ip utility, and it's similar to the ifconfig utility. You can configure an address or show the configured IP addresses similar to the ifconfig utility. The output even looks very familiar to the ifconfig utility:

```
[root@localhost ~]# ip addr add 172.16.1.200/12 eth0
[root@localhost ~]# ip addr
eth0: <BROADCAST, MULTICAST, UP> mtu 1500 qlen 1000
 link/ether 00:0c:29:e9:08:92 brd ff:ff:ff:ff:ff:ff:ff:ff
 inet 172.16.1.200/12 scope global eth0
    valid_lft forever preferred_lft forever
 inet6 fe80::20c:29ff:fee9:892/64 scope link
    valid_lft forever preferred_lft forever
lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 noqueue
 link/loopback 00:00:00:00:00:00:00:00 brd 00:00:00:00:00:00:00:00
 inet 127.0.0.1/8 scope host lo
    valid_lft forever preferred_lft forever
 inet6 ::1/128 scope host
    valid_lft forever preferred_lft forever
[root@localhost ~]#
```

Resolvers

Resolvers in Linux are essential components that translate human-readable domain names into IP addresses, enabling communication over the Internet. This process, known as domain name resolution, is handled by the resolver library, which consists of functions that query DNS servers to obtain the corresponding IP addresses. When a user types a URL into their browser, the resolver takes over, converting the domain name into an IP address that the system can use to establish a connection. This seamless translation is crucial for accessing websites, sending emails, and performing other network-related tasks.

The /etc/resolv.conf file is how you configure the resolver for the operating system in Linux. This file contains information about DNS servers that the system should use for resolving domain names. Typically, it includes one or more nameserver entries, each specifying the IP address of a DNS server. For example, a basic /etc/resolv.conf file might look like this:

```
[user@localhost ~]$ cat /etc/resolv.conf

nameserver 8.8.8.8
nameserver 8.8.4.4
search sybex.com

[user@localhost ~]$
```

In this example, the resolver will first attempt to query the DNS server at `8.8.8.8`. If that server is unavailable, it will try the next one at `8.8.4.4`. This configuration ensures that the system can resolve domain names even if one of the DNS servers fails. Additionally, the `/etc/resolv.conf` file can include other options, such as search directives, which specify default search domains to append to queries. In the previous example, `sybex.com` will be appended to queries.

Modern Linux distributions often use tools like `systemd-resolved` to dynamically manage the contents of `/etc/resolv.conf`, ensuring that the resolver configuration adapts to changing network conditions. For instance, when the device connects to a new network or a VPN, `systemd-resolved` can dynamically adjust the DNS server entries to reflect the new network's DNS settings. This ensures that the system always uses the most appropriate DNS servers without requiring manual intervention. When `system-resolved` is employed, the configuration file will typically be `/etc/systemd/resolved.conf` and it has several different options, beyond the scope of this exam.

Resolving Domain Names

The Domain Information Groper (`dig`) tool is almost identical to the Windows `nslookup` tool and has become an adopted standard for name resolution testing on Linux/UNIX operating systems. The tool allows you to resolve any resource record for a given host and direct the query to a specific server.

The command does not offer an interactive mode like the `nslookup` command. The command by default queries A records for a given host, and the output has debugging turned on by default.

In the following example, you see a query being performed on the DNS server of 8.8.8.8 for an MX record of `sybex.com`. The debugging output shows that one query was given, two answers were retrieved, and nothing was authoritative (not the primary servers). The output also details the query made and the answers returned:

```
[root@localhost ~]# dig @8.8.8.8 mx sybex.com

; <<>> DiG 9.9.5-3ubuntu0.13-Ubuntu <<>> @8.8.8.8 mx sybex.com
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 49694
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;sybex.com.           IN   MX

;; ANSWER SECTION:
```

```
sybex.com.       899   IN   MX   10 cluster1.us.messagelabs.com.
sybex.com.       899   IN   MX   20 cluster1a.us.messagelabs.com.

;; Query time: 76 msec
;; SERVER: 8.8.8.8#53(8.8.8.8)
;; WHEN: Wed Nov 01 21:43:32 EDT 2017
;; MSG SIZE rcvd: 104

[root@localhost ~]#
```

The HOSTS File

The `/etc/hosts` file has a history that dates back to the early days of the Internet. Before the Domain Name System (DNS) was developed in the 1980s, the `HOSTS.TXT` file was used to map hostnames to IP addresses. This file was manually maintained and distributed among ARPANET members by the Stanford Research Institute. As the Internet grew, the manual maintenance of this file became impractical, leading to the creation of DNS. Despite the advent of DNS, the `/etc/hosts` file remains an integral part of modern operating systems, providing a simple and effective way to manage hostname resolution locally for the operating system.

Today the `/etc/hosts` file is primarily used for local hostname resolution. It allows systems to map IP addresses to hostnames without the need to query a DNS server. This can be particularly useful in development and testing environments, where developers need to simulate how websites will behave on a live server. By mapping domain names to local IP addresses, developers can test their applications in a controlled environment. Additionally, the `/etc/hosts` file can be used to override DNS settings for specific hostnames, providing a quick and easy way to redirect traffic or block access to certain websites by mapping them to non-routable IP addresses.

Administrators often edit the `/etc/hosts` file for various reasons. One common use is to troubleshoot network issues by temporarily overriding DNS settings to ensure that a hostname resolves to the correct IP address. This can help identify and resolve DNS-related problems without waiting for DNS propagation. Another use is during server migrations, where administrators can map old domain names to new IP addresses to ensure a smooth transition with minimal downtime. By updating the `/etc/hosts` file, administrators can test new server configurations and verify that everything is working correctly before making DNS changes.

Network Connectivity

Network connectivity is a fundamental aspect of Linux systems, enabling communication between devices and networks. This section will cover the tools that CompTIA has identified as exam objectives. However, there are many other tools depending on what you are trying to diagnose.

The *ping* Command

The `ping` command is a fundamental network utility used to test the reachability of a host on an IP network and measure the round-trip time for messages sent from the originating host to a destination computer. It operates by sending Internet Control Message Protocol (ICMP) Echo Request packets to the target host and waiting for an Echo Reply. This process helps in diagnosing network connectivity issues, determining network latency, and verifying that a specific IP address is active. For example, to check if a server with the IP address `8.8.8.8` is reachable, you can use the following command:

```
[user@localhost ~]$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=114 time=17.5 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=114 time=30.0 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=114 time=23.9 ms
^C
--- 8.8.8.8 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 17.505/23.794/29.973/5.090 ms
[user@localhost ~]$
```

This command will send a series of packets to the specified IP address and display the response time for each packet. The ping of packet will continue indefinitely. To stop the constant ping, you need to press Control+C to stop the command.

The *traceroute* Command

The `traceroute` command, similar to the `tracert` Windows command, is another valuable network diagnostic tool used to trace the path that packets take from your computer to a specified destination. By sending a series of ICMP or UDP packets with incrementally increasing Time-To-Live (TTL) values, `traceroute` reveals each hop along the route to the target host. This helps in identifying where delays or failures occur within the network. It also helps with troubleshooting connectivity issues and understanding network topology. For instance, to trace the route to `8.8.8.8`, you can use the following command:

```
[user@localhost ~]$ traceroute 8.8.8.8
traceroute to 8.8.8.8 (8.8.8.8), 30 hops max, 60 byte packets
 1  _gateway (172.16.1.1)  0.977 ms  0.867 ms  0.811 ms
 2  100.93.196.195 (100.93.196.195)  15.302 ms 100.93.196.194 (100.93.196.194)
15.351 ms  15.421 ms
 3  po-318-346-rur402.pittsburgh.pa.pitt.comcast.net (68.86.100.233)  15.303
ms  15.417 ms  15.349 ms
 4  po-400-xar02.pittsburgh.pa.pitt.comcast.net (68.85.75.65)  24.003 ms po-2-
rur402.pittsburgh.pa.pitt.comcast.net (96.108.91.26)  23.918 ms po-400-
xar02.pittsburgh.pa.pitt.comcast.net (68.85.75.65)  23.993 ms
```

```
 5  be-22-ar01.mckeesport.pa.pitt.comcast.net (162.151.65.5)  25.465
ms  25.520
 ms po-400-xar02.pittsburgh.pa.pitt.comcast.net (68.85.75.65)  23.944 ms
 6  be-31611-cs01.pittsburgh.pa.ibone.comcast.net (96.110.42.161)
25.676 ms be-
31621-cs02.pittsburgh.pa.ibone.comcast.net (96.110.42.165)  24.214 ms
be-31611-
cs01.pittsburgh.pa.ibone.comcast.net (96.110.42.161)  24.258 ms
 7  be-31621-cs02.pittsburgh.pa.ibone.comcast.net (96.110.42.165)
24.134 ms be-
1311-cr11.pittsburgh.pa.ibone.comcast.net (96.110.38.138)  17.068 ms
16.090 ms
 8  be-302-cr12.beaumeade.va.ibone.comcast.net (96.110.32.101)  21.190
ms be-301-cr12.beaumeade.va.ibone.comcast.net (96.110.39.165)  21.161 ms
21.115 ms
 9  be-1212-cs02.beaumeade.va.ibone.comcast.net (68.86.84.149)  33.702 ms
 24.445 ms be-3212-pe12.ashburn.va.ibone.comcast.net (96.110.34.118)
32.619 ms
10  be-3112-pe12.ashburn.va.ibone.comcast.net (96.110.34.114)  24.341 ms be-
3312-pe12.ashburn.va.ibone.comcast.net (96.110.34.122)  33.054 ms be-3213-
pe13.ashburn.va.ibone.comcast.net (68.86.166.214)  32.922 ms
11  * * *
12  * * *
13  dns.google (8.8.8.8)  21.258 ms  21.758 ms 142.251.252.71 (142.251.252.71)
22.597 ms
[user@localhost ~]$
```

This command will display each hop along the path to `8.8.8.8`, including the IP address and response time for each intermediate router, providing a detailed map of the network route and highlighting any potential points of failure or latency.

The *curl* Command

The `curl` command is a versatile and powerful tool used for transferring data to and from a server using various protocols, including HTTP, HTTPS, FTP, and more. It is commonly used for testing and interacting with web services, downloading files, and performing network requests. With `curl`, you can send `GET`, `POST`, and other types of HTTP requests, making it an essential utility for developers and system administrators. For example, to fetch the contents of a web page, you can use the following command:

```
[user@localhost ~]$ curl https://8.8.8.8
<HTML><HEAD><meta http-equiv="content-type" content="text/html;charset=utf-8">
<TITLE>302 Moved</TITLE></HEAD><BODY>
<H1>302 Moved</H1>
The document has moved
```

```
<A HREF="https://dns.google/">here</A>.
</BODY></HTML>
[user@localhost ~]$
```

This command will retrieve the HTML content of the specified URL and display it in the terminal, allowing you to inspect the response from the server. If you wanted to capture the file to a file, add the `-o` option and the filename of the file to capture to, as shown here:

```
[user@localhost ~]$ curl https://8.8.8.8 -o file.html
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100   216  100   216    0     0   2349      0 --:--:-- --:--:-- --:--:--  2373
[user@localhost ~]$ cat file.html
<HTML><HEAD><meta http-equiv="content-type" content="text/html;charset=utf-8">
<TITLE>302 Moved</TITLE></HEAD><BODY>
<H1>302 Moved</H1>
The document has moved
<A HREF="https://dns.google/">here</A>.
</BODY></HTML>
[user@localhost ~]$
```

Getting Help

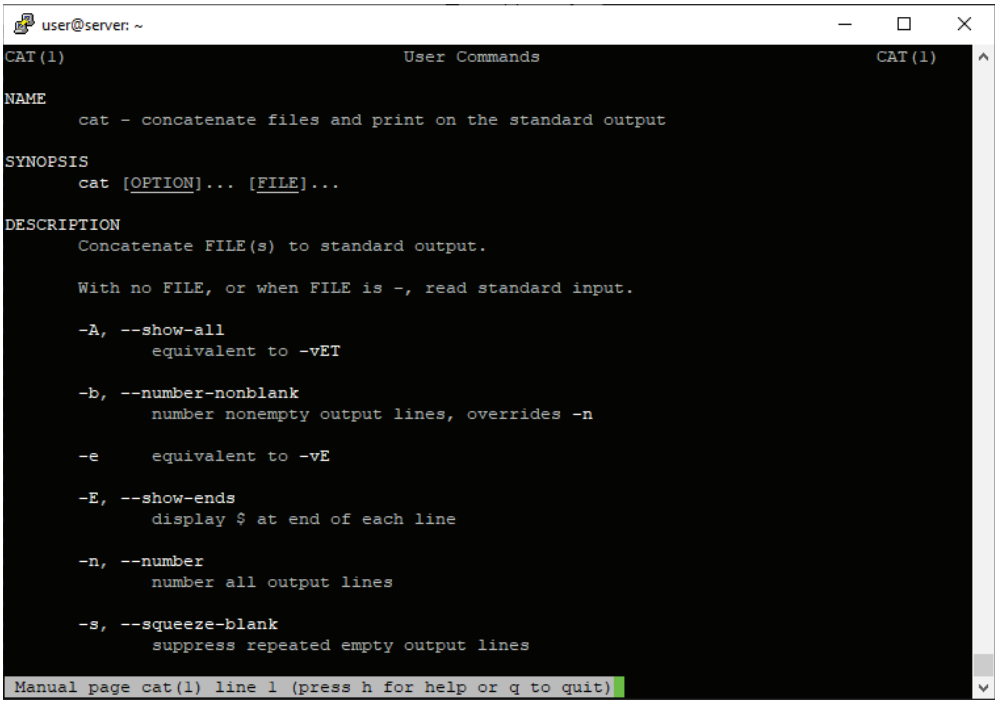
Linux is a great alternative to the Windows operating system, but the command line can be overwhelming. However, once you learn some of these basic commands, it can be very rewarding. From time to time you may need help, and this is where the `man` command comes in handy. The `man` command, also known as the manual command, allows you to quickly look up the command arguments or usage of a command, as shown in Figure 4.25. Enter the command **`man`**, followed by the command you need information on, to open the manual page. For example, you can see the manual page for the `cat` command by entering **`man cat`** at the command line.

The manual pages are installed with any package that is installed on the system. We will only be concerned with command-line usage, but for programmers, there are manual pages that explain system calls and APIs.

Summary

This chapter provided an overview of operating systems other than Microsoft Windows. In particular, we looked at macOS and Linux, including the features and various tools included with each that appear on the CompTIA A+ 220-1202 exam.

FIGURE 4.25 The man command



We covered the installation and uninstallation of applications in macOS. This included the various methods that can be used to download and install an application. We discussed the basic management of applications, such as creating shortcuts.

We also covered best practices for both macOS and Linux. There are best practices that technicians and administrators should follow regardless of which operating system(s) they are running, such as backup and antivirus.

The chapter concluded with some well-rounded knowledge of Linux administration. You learned a myriad of Linux commands that can be used in a variety of applications. You also learned how the Linux operating system works and how you can administer the OS. This chapter covers what you need to know for the 220-1202 exam. An entire book can be dedicated to the Linux operating system, and there is a CompTIA certification dedicated to Linux, called Linux+.

Exam Essentials

Know how applications are managed with common features and tools in macOS. Know that applications can be installed via the App Store or downloadable applications. Know

the various file types an application installation can be downloaded on. Know how applications are installed by dragging the application to the `Applications` folder, and be able to describe the uninstallation process of dragging the application to the Trash.

Be able to identify best practices. Best practices help keep systems and data usable. Among the best practices to follow, it is important to schedule backups, schedule disk maintenance, keep systems up to date, and make sure that antivirus/antimalware definition files are current.

Know the available System settings. You should be familiar with the various System settings. Know what each setting offers in the form of configuration for macOS.

Know how to use key macOS tools and features. A number of tools and features are included with macOS that make it an attractive operating system to use. Among them are Mission Control, Keychain, Spotlight, iCloud, Gestures, the Finder, Dock, Disk Utility, File Vault, Terminal, and Force Quit.

Know the basic Linux commands. Every version and distribution of Linux allows you to get to the command line in one way or another and interact with the shell. From there, you can give commands to navigate around (`cd`, `pwd`), to create and change file values (`chmod`, `chown`), to run commands (`su`, `sudo`), and to do many other tasks.

Know the common configuration files. The common configuration files you should know for the exam are `/etc/passwd`, `/etc/shadow`, `/etc/hosts`, `/etc/fstab`, and `/etc/resolv.conf`. Review the chapter for the various applications of each file.

Understand the various components of Linux. The various components of Linux that you should understand for the 220-1202 exam are the bootloader, kernel, and systemd. These components are all part of the boot process. The Linux boot process involves initializing the EFI firmware, loading the GRUB bootloader, and then loading and initializing the Linux kernel, which starts the systemd system.

Review Questions

The answers to the chapter review questions can be found in Appendix A.

1. Within a Linux terminal, you want to see all the files on your system in long format (using the `-l` option), including any hidden files (which requires the `-a` option). Which command should you use?
   - **A.** `ls -a | ls -l`
   - **B.** `ls -s; ls -l`
   - **C.** `ls -la`
   - **D.** `ls -a\ls -l`

2. Which of the following allows you to see all running programs in macOS?
   - **A.** Keychain
   - **B.** Mission Control
   - **C.** Finder
   - **D.** Force Quit

3. As part of your training program, you're trying to convince users to make backups on a regular basis. Which Apple app can be used to make backups of various types on a regular basis?
   - **A.** Time Machine
   - **B.** Finder
   - **C.** VSS
   - **D.** Keychain

4. Which of the following Linux commands/utilities can be used to edit a file?
   - **A.** `ps`
   - **B.** `nano`
   - **C.** `rm`
   - **D.** `ls`

5. Which of the following Linux commands/utilities can be used to edit an Ethernet connection's configuration settings?
   - **A.** `dd`
   - **B.** `apt`
   - **C.** `ip`
   - **D.** `pwd`

6. Which of the following is a macOS feature for password management in the latest version macOS 15.2 (Sequoia)?
   - **A.** Passwords
   - **B.** Keychain
   - **C.** Dock
   - **D.** Spotlight

7. The interpreter in Linux between the operating system and the user is known as the _____.
   - **A.** Shell
   - **B.** Translator
   - **C.** Login
   - **D.** GUI

8. What type of backup is kept on-site at the computer center for immediate recovery purposes?
   - **A.** On-path attack
   - **B.** Cloud copies
   - **C.** Journal copies
   - **D.** Working copies

9. Which of the following utilities can be used in Linux to download patches for installation on a workstation?
   - **A.** `update`
   - **B.** Shell/terminal
   - **C.** `apt`
   - **D.** `patch`

10. Which of the following commands can be used to change the owner of a file to a new owner in Linux?
    - **A.** `cd`
    - **B.** `chmod`
    - **C.** `chown`
    - **D.** `pwd`

11. Which Linux utility can be used to check and repair disks?
    - **A.** `fsck`
    - **B.** `chkdsk`
    - **C.** `du`
    - **D.** `dumgr`

12. Your iPad has an application that will not stop running. What feature/tool can you use to stop it?
    A. `kill`
    B. Force Quit
    C. Task Manager
    D. Close Quit

13. Which of the following is the most common shell used with Linux?
    A. Tcl/Tk
    B. Terminal
    C. Bash
    D. SSH

14. What is the name of the area at the bottom of a macOS screen where, by default, a bar of crucial icons appears?
    A. Footer
    B. Mission Control
    C. Taskbar
    D. Dock

15. Which key combination can you use to bring up Spotlight from within an app?
    A. Control+Shift
    B. Option+Tab
    C. Command+spacebar
    D. Alt+Home

16. Which Linux command can be used to let you run a single command as another user?
    A. `sudo`
    B. `su`
    C. `passwd`
    D. `ifconfig`

17. Which of the following Linux commands will show you a list of running processes?
    A. `ls`
    B. `cat`
    C. `ps`
    D. `su`

18. You are currently in a Linux terminal session and in the `/home/testuser/documents/mail` directory. Which command will take you to `/home/testuser/documents`?

A. `cd .`

B. `cd ..`

C. `cd ...`

D. `cd ~`

19. If the permissions for a file are `rwxrw-r--`, what permissions apply for a user who is a member of the group to which the owner belongs?

A. Read, write, and execute

B. Read and write

C. Read only

D. No access

20. What does the `-p` option with `mkdir` do?

A. Prompts the user before creating files

B. Prompts the user before creating subfolders

C. Creates subfolders as well as folders

D. None of the above

Performance-Based Question

You will encounter performance-based questions on the A+ exams. The questions on the exam require you to perform a specific task, and you will be graded on whether or not you were able to complete the task. The following requires you to think creatively in order to measure how well you understand this chapter's topics. You may or may not see similar questions on the actual A+ exams. To see how your answers compare to the authors', refer to Appendix B.

By default, not all files and folders in a Linux directory are shown when you do an `ls` listing. Entries that start with a period (.) are considered "hidden" and not shown. Try this command in your home directory, and then compare the result with what you see when you don't use the `-a` option:

1. Type **`cd /`** to change to the root directory.
2. Type **`ls -F`** to see the files and directories in the root directory.
3. Type **`ls -aF`** to see everything, including hidden files.
4. Type **`cd ~`** to change to your home directory.
5. Type **`ls -l`** to see the files and directories in your home directory.
6. Type **`ls -al`** to see everything, including hidden files.

Chapter

5

Security Concepts

THE FOLLOWING COMPTIA A+ 220-1202 EXAM OBJECTIVES ARE COVERED IN THIS CHAPTER:

✓ **2.0 Security**

✓ **2.1 Summarize various security measures and their purposes.**

- Physical security
  - Bollards
  - Access control vestibule
  - Badge reader
  - Video surveillance
  - Alarm systems
  - Motion sensors
  - Door locks
  - Equipment locks
  - Security Guards
  - Fences
- Physical access security
  - Key fobs
  - Smart cards
  - Mobile digital key
  - Keys
  - Biometrics
    - Retina scanner
    - Fingerprint scanner
    - Palm print scanner
    - Facial recognition technology (FRT)
    - Voice recognition technology

- Lighting
- Magnetometers

- Logical Security
  - Principle of least privilege
  - Zero Trust model
  - Access control lists (ACLs)
  - Multifactor authentication (MFA)
    - Email
    - Hardware token
    - Authenticator application
    - Short message service (SMS)
    - Voice call
    - Time-based one-time pass word (TOTP)
    - One-time password/passcode (OTP)
  - Security Assertions Markup Language (SAML)
  - Single sign-on (SSO)
  - Just-in-time access
    - Privileged access management (PAM)
  - Mobile device management (MDM)
  - Data loss prevention (DLP)
  - Identity access management (IAM)
  - Directory services

✓ **2.2 Given a scenario, configure and apply basic Microsoft Windows OS security settings.**

- Active Directory
  - Joining domain
  - Assigning log-in script
  - Moving objects within organizational units
  - Assigning home folders
  - Applying Group Policy

- Selecting security groups
- Configuring folder redirection

✓ **2.4 Summarize types of malware and tools/methods for detection, removal, and prevention.**

- Malware
  - Trojan
  - Rootkit
  - Virus
  - Spyware
  - Ransomware
  - Keylogger
  - Boot sector virus
  - Cryptominer
  - Stalkerware
  - Fileless
- Adware
  - Potentially unwanted program (PUP)
- Tools and methods
  - Recovery console
  - Endpoint detection and response (EDR)
  - Managed detection and response (MDR)
  - Extended detection and response (XDR)
  - Antivirus
  - Anti-malware
  - Email security gateway
  - Software firewalls
  - User education regarding common threats
    - Anti-phishing training
  - OS reinstallation

✓ **2.5 Compare and contrast common social engineering attacks, threats, and vulnerabilities.**

- Social engineering
  - Phishing
    - Vishing
    - Smishing
    - QR code phishing
    - Spear phishing
    - Whaling
  - Shoulder surfing
  - Tailgating
  - Impersonation
  - Dumpster diving
- Threats
  - Denial of service (DoS)
  - Distributed denial of service (DDoS)
  - Evil twin
  - Zero-day attack
  - Spoofing
  - On-path attack
  - Brute-force attack
  - Dictionary attack
  - Insider threat
  - Structured Query Language (SQL) injection
  - Cross-site scripting (XSS)
  - Business email compromise (BEC)
  - Supply chain/pipeline attack
- Vulnerabilities
  - Non-compliant systems
  - Unpatched systems

- Unprotected systems (missing antivirus/missing firewall)
- EOL
- Bring your own device (BYOD)

✓ **2.7 Given a scenario, apply workstation security options and hardening techniques.**

- Data-at-rest encryption
- Password considerations
  - Length
  - Character types
  - Uniqueness
  - Complexity
  - Expiration
- Basic input/output system (BIOS)/Unified Extensible Firmware Interface (UEFI) passwords
- End-user best practices
  - Use screensaver locks
  - Log off when not in use
  - Secure/protect critical hardware (e.g., laptops)
  - Secure personally identifiable information (PII) and passwords
  - Use password managers
- Account management
  - Restrict user permissions
  - Restrict log-in times
  - Disable guest account
  - Use failed attempts lockout
  - Use timeout/screen lock
  - Apply account expiration dates
- Change default administrator's user account/password
- Disable AutoRun
- Disable unused services

✓ **2.9 Compare and contrast common data destruction and disposal methods.**

- Physical destruction of hard drives
  - Drilling
  - Shredding
  - Degaussing
  - Incineration
- Recycling or repurposing best practices
  - Erasing/wiping
  - Low-level formatting
  - Standard formatting
- Outsourcing concepts
  - Third-party vendor
  - Certification of destruction/recycling
- Regulatory and environmental requirements

Think of how much simpler an administrator's life was in the days before every user had to be able to access the Internet, and how much simpler it must have been when you only had to maintain a number of dumb terminals connected to a mini-tower or mainframe computer. Much of what has created headaches for an administrator since then is the inherent security risk that comes about as the network expands. As our world—and our networks—have become more connected, the need to secure data and keep it away from the eyes of those who can do harm has increased exponentially.

Realizing this, CompTIA added the Security domain to the A+ exams a number of years back. Security is now a topic that every administrator and technician must not only be aware of and concerned about, but also be actively involved in implementing methods to enforce and monitor. In the world of production, quality may be job one, but in the IT world, it is security.

This chapter, one of two chapters that focus primarily on security, will cover myriad security concepts. First, we will explore the physical aspects of security, and then we will dive deeper into the logical aspects of security. We will then look at how external forces, such as malware, social engineering, and vulnerabilities, can impact security. We will finish this chapter by looking at some common ways that you can safeguard yourself from security breaches. We will cover the proper destruction and disposal methods as well as security measures you can employ in network installations.

A+ is not the only IT certification that CompTIA offers. Security+ is one of the more popular choices. The topics found in this chapter are a subset of what you need to know for that certification.

Many of the security issues that plague networks today can be solved through the implementation of basic security elements. Some of those elements are physical (e.g., locked doors), and others are digital (e.g., antivirus/antimalware software), but all share in common the goal of keeping problems out. The following six topic areas are key:

- Physical security
- Digital security
- User education
- The principle of least privilege
- Email security
- Virtual private networks (VPNs)

As you study for the exam, know the types of physical security elements that you can add to an environment to secure it. Know, as well, what types of digital security you should implement to keep malware at bay. Understand that the first line of defense is the user. You need to educate users to understand why security is important, and you must impose the principle of least privilege to prevent them from inadvertently causing harm.

Physical Security Concepts

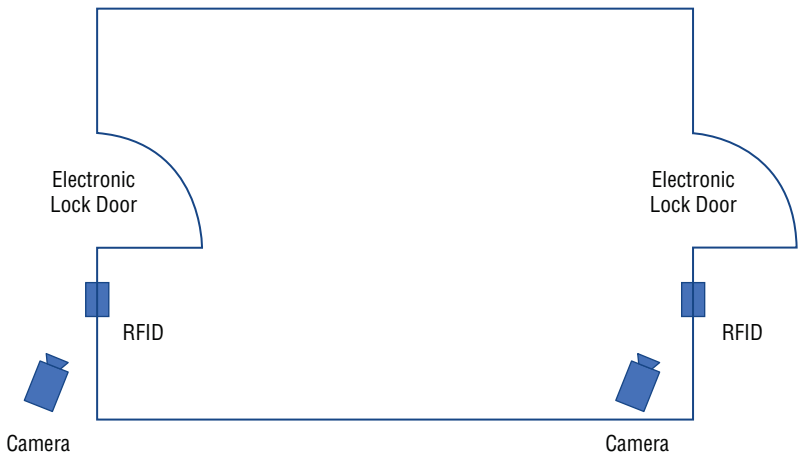
Physical security is the most overlooked element of security in a network. A simple lock can keep out the most curious prying eyes from a network closet or server room. A more layered approach can be implemented for higher security installations. However, the simple fact is that not a lot of time is spent on physically securing the network. In this section, we will cover the CompTIA objectives related to physical security of networks.

Exam Objective 2.1, "Summarize various security measures and their purposes," requires you to know the various security measures for both physical and logical access.

Access Control Vestibule

The use of an access control vestibule helps to prevent nonauthorized users from tailgating. An *access control vestibule* is a small room that has two controlled doors, as shown in Figure 5.1. When a person enters the first door, they are trapped in the room until they have been authorized to enter the second controlled door. The close proximity between people in this confined space makes it uncomfortable both for the authorized user and for the nonauthorized user attempting to tailgate. In this example, the doors are controlled by radio frequency identification (RFID) readers, which we will cover later in this section.

FIGURE 5.1 A common access control vestibule setup



Badge Reader

Identification (ID) badges are used to provide proof of access. Badges can be any form of identification intended to differentiate the holder from everyone else. This can be as simple as a name badge or a photo ID. When the badge contains a photo ID, it is considered the authentication factor of something that you are.

Many ID badges also contain a magnetic strip or RFID provision so that the badge can be used in conjunction with a *badge reader*. When the information is read by the badge reader, it is sent to an access control system for authorization through the controlled door. A benefit of implementing badge readers is that it creates an electronic audit of all access to an area.

Video Surveillance

Video surveillance is the backbone of physical security. It is the only detection method that allows an investigator to identify what happened, when it happened, and, most important, who made it happen. Two types of cameras can be deployed: fixed and *pan-tilt-zoom (PTZ)*. Fixed cameras are the best choice when recording for surveillance activities. Pan-tilt-zoom (PTZ) cameras allow for 360-degree operations and zooming in on an area. PTZs are most commonly used for intervention, such as covering an area outside during an accident or medical emergency. PTZ cameras are usually deployed for the wrong reasons, mainly because they are cool! PTZs are often put into patrol mode to cover a larger area than a fixed camera can. However, when an incident occurs, they are never pointed in the area you need them! It is always best to use a fixed camera or multiple fixed cameras, unless you need a PTZ for a really good reason. They are usually more expensive and require more maintenance than fixed cameras.

Video surveillance can be deployed using two common media types: coaxial cable and Ethernet. Coaxial cable is used typically in areas where preexisting coaxial lines are in place or distances are too far for typical Ethernet. These systems are called closed-circuit television (CCTV). Coaxial camera systems generally use appliance-like devices for recording of video. These CCTV recorders generally have a finite number of ports for cameras and a finite amount of storage in the form of direct-attached storage (DAS).

Most video installations for CCTV are coaxial cable and Ethernet, as previously described. However, wireless is popular for consumer applications, such as doorbells and home surveillance cameras. These devices generally use cloud storage and require an Internet connection.

Ethernet (otherwise known as IP) surveillance is becoming the standard for new installations. Anywhere an Ethernet connection can be installed, a camera can be mounted. Power over Ethernet (PoE) allows power to be supplied to the camera, so the additional power supplies used with coaxial cameras are not needed. Ethernet also provides the flexibility of virtual local area networks (VLANs) for added security so that the camera network is isolated from operational traffic. IP surveillance uses network video recorder (NVR) software to record cameras. Because NVRs are server applications, you can use traditional storage such as network area storage (NAS) or storage area network (SAN) storage. This allows you to store and treat the video recordings like traditional data.

Coaxial camera networks can be converted to IP surveillance networks with the use of a device called a *media converter*. These devices look similar to a CCTV recorder. They have a limited number of ports for the coaxial cameras and are generally smaller than the CCTV recorder. This is because they do not have any DAS. The sole purpose of the media converter is to convert the coaxial camera to an Ethernet feed to the NVR.

The use of IP video surveillance allows for a number of higher-end features such as camera-based motion detection, license plate recognition (LPR), and motion fencing. Advanced NVR software allows cameras to send video only when motion is detected at the camera; this saves on storage for periods of nonactivity. LPR is a method of detecting and capturing license plates in which the software converts the plate to a searchable attribute for the event. With motion fencing, an electronic fence can be drawn on the image so that any activity within this region will trigger an alert. Among the many other features are facial recognition and object recognition.

Motion Sensors

There are several different motion sensor types that you can use to detect unauthorized access. Passive infrared (PIR) is the most common motion detection used today, mainly because of price. PIR sensors operate by monitoring the measurement of infrared radiation from several zones. In Figure 5.2, you can see the reflective panel that divides the infrared zones. A PIR sensor will always have this grid pattern on the sensor's face.

FIGURE 5.2 A typical PIR sensor

Microwave detectors also look like PIR sensors, but they do not have a reflective panel. Microwave detectors are common in areas where wide coverage is needed. They operate by sending pulses of microwaves out and measuring the microwaves received. These detectors are more expensive than PIR sensors and are susceptible to external interference, but they have a wider area of coverage.

Vibration sensors are another type of sensor used for motion detection. Although you may have seen them in the latest over-the-top heist movie, vibration sensors are really used in physical security systems. They are most often implemented as seismic sensors. They help protect from natural disasters and accidental drilling, or the occasional over-the-top heist.

Alarm System

An alarm system is another type of physical security system. It provides a method to alert security personnel in the event of unauthorized access or a break in. An alarm system can be configured to trigger in the event of an access control system logging unauthorized access to a controlled door. However, it is more common to find alarm systems installed for break-in detection and response.

An alarm system can be configured to use motion sensors, video surveillance, magnetic contacts, and a multitude of other sensors. Each sensor will be installed in a different logical zone. The perimeter of the building might be zone 1, the server room might be zone 2, and so on. There could be many different zones configured for each sensor. The main purpose of the zone is to communicate the location of the sensor being tripped so that law enforcement agents can respond to that location.

A monitoring company is typically contracted with the installation. The purpose of the monitoring company is to act as a buffer between the zone being tripped and a law enforcement agency. The alarm panel will dial out to a monitoring station and it will transmit the account number and the zone that is tripped. The monitoring station can then process its call-down list. The call-down list will typically consist of the phone number of the supervisor of the area, the in-house security personnel, and ultimately a law enforcement agency. The monitoring company can also monitor the health of the alarm panel, depending on the model of alarm panel and its features.

Door Locks

The most common physical prevention tactic is the use of locks on doors and equipment. This might mean the installation of a tumbler-style lock or an elaborate electronic combination lock for the switching closet. If a tumbler-style lock is installed, then the appropriate authorized individuals who require access will need a physical key. Using physical keys can become a problem, because you may not have the key with you when you need it the most, or you can lose the key. The key can also be copied and used by unauthorized individuals. Combination locks, also called cipher locks, can be reprogrammed and do not require physical keys, as shown in Figure 5.3. Combination locks for doors can be purchased as mechanical or electronic.

FIGURE 5.3 A typical combination door lock

When physical locks use keys, the factor of authentication is considered something that you have—because you must have the key. When physical locks use ciphers, the authentication is considered something you know—because you must know the cipher.

Equipment Locks

There are many different types of *equipment locks* that can secure the information and the device that holds the information. Simply thwarting the theft of equipment containing data and restricting the use of USB thumb drives can secure information. In the following sections, we will cover several topics that are directly related to the physical aspects of information security.

Cable Locks

Cable locks are used to secure laptops and any device with a Universal Security Slot (USS), as shown in Figure 5.4. A cable lock is just that—a cable with a lock at one end. The lock can be a tumbler or a combination, as shown in Figure 5.5. The basic principle is that the end of the lock fits into the USS. When the cable is locked, the protruding slot of metal turns into a cross that cannot be removed. This provides security to expensive equipment that can be stolen due to its portability or size.

FIGURE 5.4 A Universal Security Slot

FIGURE 5.5 A standard cable lock

Server Locks

Most servers come with a latch-style lock that prevents someone from opening the server, but the tumbler-style lock is trivial to open. Anyone with a paperclip can open these locks if they have forgotten the keys. Other types of server locks are holes for padlocks that latch through the top cover and the body of the server. However, over the past 10 years, a

declining number of servers come with this feature. This is mainly due to the fact that servers can be better secured behind a locked rack-mounted enclosure. Rack-mounted enclosures generally come with a tumbler-style lock that can protect all the servers and network equipment installed in the cabinet, while still providing airflow.

USB Locks

Universal Serial Bus (USB) locks can be put into place to physically lock out USB ports on a workstation or server from use. These devices are extremely rare to find, because most equipment and operating systems allow for the USB ports to be deactivated. USB locks work by inserting a small plastic spacer into the USB port. Once inserted, the spacer latches to the USB detent with plastic teeth. A tool is required to remove the USB spacer.

Security Guards

Physical security begins with personnel—specifically, security-focused personnel, such as security guards. *Security guards* should be responsible for limiting access from the outer perimeter of your installation. Security guards typically use photo IDs, also known as ID badges, to allow access to the installation. Exceptions to this are people on the entry control roster; in some secured buildings, only people on the entry control roster are allowed to enter. In this type of scenario, the ID badge is used only to provide ID. This is common in government and sensitive installations.

Fences

Fences are a physical security barrier to keep unauthorized persons out of a secure area. Exterior fences can be arranged so that they create a choke point where a guard can inspect credentials to allow authorized personnel into the area. Guards can also be replaced with electronic locks and RFID readers to limit access. When installed in conjunction with a video camera system to surveil the area around the entry point, a fence creates a very secure outer layer for your facility. Fences should be considered the outermost security layer of a multiple-barrier system.

In addition to an exterior fence, the building that houses the data should have RFID readers and electronic door locks. The innermost area surrounding the equipment can also be segmented with a fence and additional access controls, such as standard keyed locks or electronic access control. When fences are used in the interior of the data center, air quality can be maintained while preventing unauthorized access, as shown in Figure 5.6.

Using multiple barriers as described allows contractors for HVAC systems to maintain their systems, while preventing direct physical access to the servers and equipment.

FIGURE 5.6 Interior data center fences

Bollards

A *bollard* is an architectural structure that acts as a visual indicator for a perimeter. They are also very sturdy, since their second function is to act as a barrier for the perimeter and protect the area. They are commonly found around areas where a truck or other vehicle can cause damage. In Figure 5.7, the bollard is protecting a fiber-optic vault from accidental damage by a vehicle. Bollards can also be found in the interior of a building if there is potential for damage to a protected area from a vehicle, such as a forklift or equipment cart.

FIGURE 5.7 A typical bollard

Physical Security for Staff

Organizations should authorize and audit staff access for sensitive areas inside a facility. Implementing physical security for staff is one way you can control access to the physical equipment and the data that is stored on the equipment. By limiting access to the equipment and the underlying data, you can prevent service disruptions or loss of data. This section will focus on ways that you can implement physical security for staff.

Key Fobs

Key fobs are named after the chains that used to hold pocket watches to clothes. *Key fobs* are embedded *radio frequency identification (RFID)* circuits that fit on a set of keys and are used with physical access control systems, as shown in Figure 5.8. They are often used for access to external and internal doors for buildings. Key fobs are close-proximity devices that authorize the user for entry; an electronic lock is actuated when the device is presented, and the door can be opened. This is an authentication factor of something that you have.

FIGURE 5.8 A key fob

Smartcards and RFID Badges

A *smartcard* is the size of a credit card with an integrated circuit embedded into the card (also called an integrated circuit chip [ICC]). The chip is exposed on the face of the card with surface contacts, as shown in Figure 5.9. Smartcards are used for physical authentication to electronic systems and access control systems and require a PIN or password. A smartcard is considered a multifactor authentication method because it is something you have (card) and something you know (PIN or password). The U.S. military uses smartcards called Common Access Cards (CACs) for access to computer systems and physical access controls.

FIGURE 5.9 A typical smartcard

An RFID badge is a wireless, no-contact technology used with RFID transponders. RFID badges typically work on the 125 kHz or 13.56 MHz radio frequency and are passively powered by the RFID transponder. When an RFID badge is placed in close proximity to the RFID transponder, the radio frequency (RF) energy emitted by the transponder powers a chip in the RFID badge. The RFID chip then varies the frequency back to the transponder in the effort to transmit its electronic signature (number). This type of authentication is considered something you have.

Keys

Physical keys are extremely hard to control and do not allow for the auditing of their usage. A physical key can be lent to someone, copied, stolen, or used by an unauthorized person. Because of the problems surrounding physical keys, their use should largely be avoided.

If keys are absolutely necessary, then a two-person system should be considered. A two-person system requires that two people must use their keys to open one lock, although nothing stops the keys from being lent to the same person to open a door.

Another option is to use an electronic lock box for management of the keys. When a technician needs a particular key, they will log into the key box and check out the key needed. This system allows for auditing controls, but it does not prevent copying of keys.

Mobile Digital Key

The history of mobile digital keys displays the rapid technological advancements and evolving security needs, beginning in the early 2010s with the hospitality industry. InterContinental Hotels Group (IHG) first experimented with mobile keys in 2010, followed by Starwood Hotels' SPG Keyless in 2014, which allowed guests to unlock rooms with smartphones. Major hotel chains like Hilton and Marriott quickly adopted and refined this technology, enhancing guest experiences and setting new security standards. By 2019, Hilton had implemented digital keys in over 4,000 hotels, while Marriott developed a user-friendly system leveraging Starwood's legacy technology.

Today, the use of mobile digital keys extends far beyond hotels. They are now commonly used in residential buildings, offices, and even vehicles. Digital keys utilize technologies like Bluetooth, NFC (near-field communication), and Wi-Fi to communicate securely with locks. This wireless communication allows users to unlock doors with their smartphones, providing a seamless and convenient access experience.

Mobile device keys offer superior convenience and security compared to traditional keys. They eliminate the need to carry physical keys, reduce the risk of lockouts, and provide seamless access through smartphones. Enhanced security features like encryption and two-factor authentication protect against unauthorized access, while the ability to remotely deactivate keys if a device is lost or stolen adds an extra layer of protection. This combination of convenience and robust security makes mobile device keys a better choice for modern access control.

Biometrics

Biometrics refers to the technology that identifies individuals based on their unique physical or behavioral traits. This method of identification is increasingly popular due to its high level of security and convenience. This factor of authentication is considered "something that you are," and we will discuss other factors later in this chapter.

In more recent years, most mobile phones have implemented biometrics in the access control of the mobile device. Several manufacturers have adopted fingerprint access control, and some have even adopted facial recognition via the forward-pointing camera.

Biometric systems typically follow three main steps: enrollment, where the system captures and stores a biometric sample; storage, where the sample is converted into a digital format and stored in a database; and comparison, where the system compares a new sample with the stored data to verify identity. These systems are designed to be difficult to forge or

steal, making them a reliable form of authentication. The following are the various types of biometric systems outlined for the 220-1202 exam:

Retina Scanner Retina scanners work by using low-energy infrared light to scan the unique pattern of blood vessels in the retina, located at the back of the eye. The light traces a path on the retina, and the differences in light absorption by the blood vessels create a unique pattern that is digitized and stored. Retinal patterns are highly stable and unique, making them difficult to replicate. To safeguard against false positives, retina scanners use sophisticated algorithms to ensure the scanned pattern matches the stored data with high precision.

Fingerprint Scanner Fingerprint scanners capture the unique patterns of ridges and valleys on a person's fingertip. There are several types of fingerprint scanners, including optical, capacitive, and ultrasonic. Optical scanners use light to capture an image of the fingerprint, while capacitive scanners use electrical currents to map the ridges and valleys. Ultrasonic scanners use sound waves to create a detailed image. These scanners compare the captured fingerprint with stored data to verify identity. Figure 5.10 shows a typical biometric device.

FIGURE 5.10 A fingerprint biometric lock

Palm Print Scanner Palm print scanners work similarly to fingerprint scanners but capture the larger and more complex patterns found on the palm. These patterns include ridges, lines, and creases that are unique to each individual. Palm print scanners use

imaging sensors to capture these details and convert them into a digital format for storage and comparison.

Facial Recognition Technology (FRT) Facial recognition technology identifies individuals by analyzing the unique features of their faces. This includes the distance between the eyes, the shape of the cheekbones, and the contour of the lips, among other features. The system captures an image of the face, converts it into a digital format, and compares it with stored data.

Voice Recognition Technology Voice recognition technology identifies individuals based on their unique vocal characteristics, such as pitch, tone, and cadence. The system captures a sample of the person's voice, converts it into a digital format, and analyzes it for unique patterns. To safeguard against false positives, voice recognition systems use advanced algorithms that can distinguish between similar-sounding voices and account for variations in speech due to illness or emotional state.

With the passing of time, the definition of biometrics is expanding from simply identifying physical attributes about a person to being able to describe patterns in their behavior. Recent advances have been made in the ability to authenticate someone based on the key pattern that they use when entering their password (how long they pause between keys, the amount of time each key is held down, and so forth). An organization adopting biometric technologies needs to consider the controversy they may face. Some authentication methods are considered more intrusive than others. The error rate also needs to be considered, along with an acceptance of the fact that errors can include both false positives, where the reader allows access falsely, and false negatives, where the reader denies access erroneously. Therefore, biometrics are often used with another factor of authentication, such as a PIN number. This approach provides multifactor authentication.

Lighting

Most security cameras work on the principle of collecting light to record a picture. As light levels decrease, the quality of the picture decreases significantly. Therefore, areas in which you have video cameras should have sufficient levels of lighting. In reality, any area that is sensitive should have a level of lighting, since threat agents often hide in darker areas to avoid being seen.

Lighting the area doesn't always require a visible light source. Most cameras sensors built in the last 10 years allow for the collection of light from infrared (IR) light sources. This means that even in the dark you can surveil and record an area, although a visible light source does deter unauthorized access.

Magnetometers

The *magnetometer*, also known as a metal detector, uses an electromagnetic field to detect metallic objects. We have all seen these devices at a choke point in the airport or government building. When a metal detector is used for people entering a facility, you can detect

weapons, such as guns or knives. When a metal detector is deployed in this fashion, it protects your staff from threat agents with malicious intent.

A metal detector can also be used to monitor people leaving a facility. The metal detector can monitor staff leaving with equipment. When it is used in this way, it will protect against data loss and theft. However, unless your organization is regulated as a highly classified facility, this approach will be hard to enforce and could infringe on your employees' privacy.

Logical Security

Whereas the topic of physical security concepts, from CompTIA's standpoint, focuses on keeping individuals out, logical security focuses on keeping harmful data and malware out as well as on authorization and permissions. This logical security includes devices and methods that protect the environment logically, such as firewalls, antivirus software, and directory permissions, just to name a few. The areas of focus are antivirus software, firewalls, antimalware, user authentication/strong passwords, and directory permissions. Each of these topics is addressed in this section.

Principle of Least Privilege

The *principle of least privilege* is a common security concept that states a user should be restricted to the fewest number of privileges that they need to do their job. By leveraging the principle of least privilege, you can limit internal and external threats. For example, if a frontline worker has administrative access on their computer, they have the ability to circumvent security; this is an example of an internal threat. Along the same lines, if a worker has administrative access on their computer and receives a malicious email, a bad actor could now have administrative access to the computer; this is an example of an external threat. Therefore, only the permissions required to perform their tasks should be granted to users, thus providing least privilege.

Security is not the only benefit to following the principle of least privilege, although it does reduce the surface area of attack because users have less access to sensitive data that can be leaked. When you limit workers to the least privilege they need on their computers or the network, fewer intentional or accidental misconfigurations will happen that can lead to downtime or help desk calls. Some regulatory standards require following the principle of least privilege. By following the principle of least privilege, an organization can improve on compliance audits by regulatory bodies.

Zero Trust

In a traditional trust model, everything within the perimeter of the network is trusted and everything outside is not. When an external user needs to access an internal resource, they connect to the internal network via a virtual private network (VPN). Connecting to the

network with a VPN makes the device they are connecting with trusted because the user is authenticated with their credentials.

The traditional model works well when the expectation is that you drive to work and log on to a trusted computer to access sensitive information. However, with today's mixture of work from home and mobile device access, the traditional model doesn't always work. The traditional model also doesn't address cloud resource access. This is where the Zero Trust model excels. With a Zero Trust model, nothing is trusted regardless of the location and everything is authenticated. If a user logs on to a cloud resource, the device is authenticated and the user is authenticated before they can connect to the data or application. This prevents a bad actor from using a user's compromised device to access sensitive data. Every vendor has a different implementation of Zero Trust, but the concept remains the same; nothing is trusted and everything and everyone must be authenticated.

Access Control Lists

Access control lists (ACLs) are used to control traffic and applications on a network. Every network vendor supports a type of ACL method; for the remainder of this section, I will focus on Cisco ACLs.

An ACL method consists of multiple access control entries (ACEs) that are condition actions. Each entry is used to specify the traffic to be controlled. Every vendor will have a different type of control logic. However, understanding the control logic of the ACL system allows you to apply it to any vendor and be able to effectively configure an ACL. The control logic is defined with these simple questions:

- How are the conditions of an ACL evaluated?
- What is the default action if a condition is not met?
- How is the ACL applied to traffic?
- How are conditions edited for an ACL?

Let's explore the control logic for a typical Cisco layer 3 switch or router. The conditions of the ACL are evaluated from top to bottom. If a specific condition is not met for the ACL, the default action is to deny the traffic. Only one ACL can be configured per interface, per protocol, and per direction. When you are editing a traditional standard or extended ACL, the entire ACL must be negated and reentered with the new entry. With traditional ACLs, there is no way to edit a specific ACL on the fly. When editing a named access list, each condition is given a line number that can be referenced so that the specific entry can be edited. For the remainder of this section, we will use named access lists to illustrate an applied access list for controlling traffic.

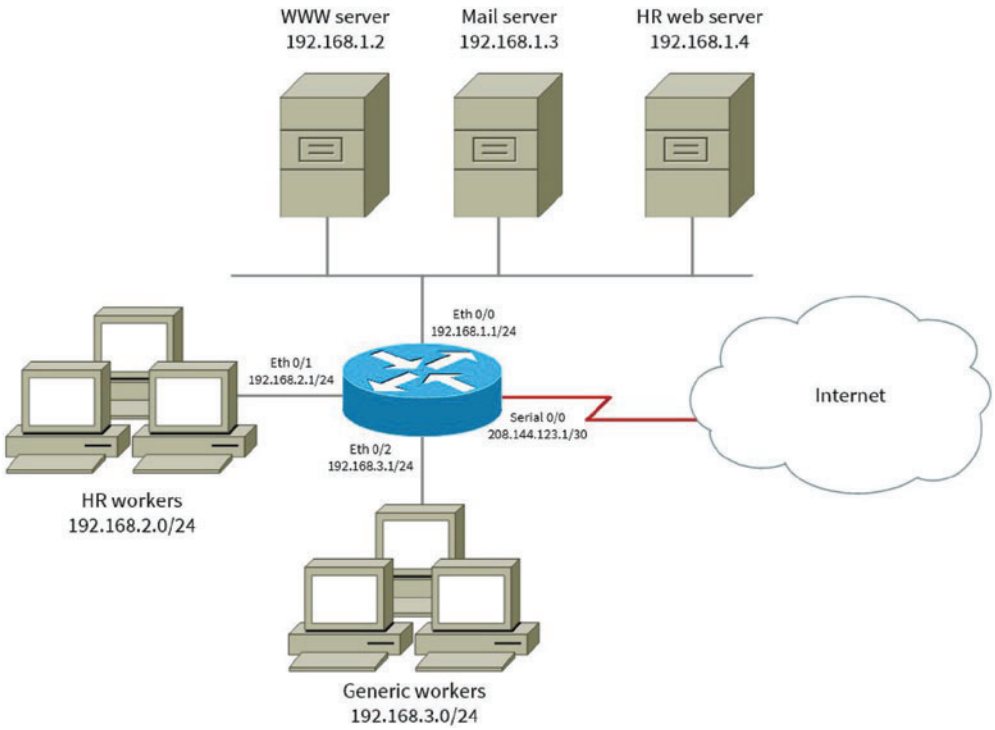
In Figure 5.11 you can see a typical corporate network. There are two different types of workers: HR workers and generic workers. We want to protect the HR web server from access by generic workers.

We can protect the HR server by applying an ACL to outgoing traffic for Eth 0/0 and describing the source traffic and destination to be denied. We can also apply an ACL to the incoming interface of Eth 0/2 describing the destination traffic to be denied. For this

example, we will build an access list for incoming traffic to Eth 0/2, blocking the destination of the HR server.

```
Router(config)# ip access-list extended block-hrserver
Router(config-ext-nacl)# deny ip any host 192.168.1.4
Router(config-ext-nacl)# permit ip any any
Router(config-ext-nacl)# exit
Router(config)# interface ethernet 0/2
Router(config-if)# ip access-group block-hrserver in
```

FIGURE 5.11 A typical corporate network



This ACL, called `block-hrserver`, contains two condition action statements. The first denies any source address to the specific destination address of 192.168.1.4. The second allows any source address to any destination address. We then enter the interface of Eth 0/2 and apply the ACL to the inbound direction of the router interface. The rule will protect the HR server from generic worker access while allowing the generic workers to access all other resources and the Internet.

It is important to note that the focus of this section is to understand how ACLs are used to protect resources. It is not important to understand how to build specific ACLs, since commands will be different from vendor system to vendor system.

Access control lists can also describe the logical controls for file access. These types of ACLs control who has access with which files. We will cover file-based ACLs in Volume 2 Chapter 6, "Securing Operating Systems."

Authentication Factors

This section discusses various components of physical security that control access. When components are used to control access, they do so based on the authentication of people. Authentication can happen in several different ways:

Something You Know This is commonly a username and password or *personal identification number (PIN)*. You can make passwords more complex by requiring uppercase, lowercase, numeric, and symbol combinations.

Something You Have This is based on something you physically have. This chapter covers several types of devices, including key fobs, keys, and smartcards, just to name a few.

Something You Are This authentication factor is based on something that makes you physically unique. Biometrics such as fingerprints, palm, face, your voice, and retina scans should be the first things that come to mind.

Somewhere You Are A relatively newer factor of authentication is based on somewhere you are. With the proliferation of global positioning system (GPS) chips, your current location can authenticate you for a system. This is performed by authentication rules based on the user's location.

Something You Do Another relatively newer factor of authentication for network systems is based on something you do. Although it has been used for hundreds of years for documents and contracts, a signature is something that you do and don't even think about how you do it. It is unique to you and only you, because there is a specific way you sign your name. Typing your name into the computer is something you do and don't think about, but there is a slight hesitation that you make without knowing it.

Multifactor Authentication

All authentication is based on something that you know, have, are, or do, or a location you are in. A common factor of authentication is a password, but passwords can be guessed, stolen, or cracked. A fingerprint can be lifted with tape, a key can be stolen, or a location spoofed. No one factor is secure by itself because it can be compromised easily.

When more than one item (factor) is used to authenticate a user, this is known as *multifactor authentication (MFA)*. It may take two, three, or four factors to authenticate, but as long as it is more than one, as the name implies, it is known as multifactor. One of the most common examples where this is used in everyday life is at an ATM. In order to withdraw money, a user must provide a card (one factor) and a PIN number (a second factor). If you know the PIN number but do not have the card, you cannot get money from the machine. If you have the card but do not have the PIN number, you cannot get money from the machine.

In this section we will cover the most common two-factor (2FA)/multifactor authentication methods used by protected applications. The following methods are generally used in conjunction with a traditional user and password combination. It should be assumed that when we talk about 2FA it provides the same functionality as MFA.

Email

Some applications use email as a 2FA method. However, using email as a 2FA option is probably the least secure method. This is mainly due to the fact that people reuse passwords. If your banking website username and password is compromised (something you know) and you reuse the same credentials on email, it provides no protection. Hopefully the email account is protected with 2FA in a way that it requires something you have.

Email is useful as a notification method when someone logs into a secure login. However, keep in mind the threat agents know this as well. If your email account is compromised, a threat agent will often create a rule in your email box to dump these notifications directly to the trash.

Short Message Service (SMS)

Some applications will allow the use of short message service (SMS) text messages as the 2FA method. When this method is used, a simple text message is sent to the user's phone number. The message will contain a random 5- to 8-digit code that the user will use to satisfy the 2FA requirement. When you first set up this 2FA method, the protected application will request the code before turning on 2FA. This is done to verify that the phone number is correct and that you can receive text messages.

Voice Call

Some applications that are protected by 2FA will allow voice calls to be initiated to the end user. This is usually done if the person does not have a phone that accepts text messages. The voice call will recite a 5- to 8-digit code that the user will use to satisfy the 2FA requirement. This process is similar to SMS, with the difference being it is an automated voice call.

Hardware and Software Tokens

A hard token is a physical device, often resembling a key fob or a USB thumb drive. All of these types of devices serve as a "something you have" authentication factor, providing an additional layer of security. SecurID from RSA is one of the best-known examples of a physical hardware token in the form of a key fob, as shown in Figure 5.12.

FIGURE 5.12 An RSA security key fob

The various hardware tokens operate differently based upon the type of hardware token being used and the implementation. The following are the various token systems you might encounter.

Time-based One-Time Password (TOTP) Time-based One-Time Password (TOTP) is a dynamic authentication method that generates a unique, time-sensitive code based on a shared secret key and the current time. When a user attempts to log in, the TOTP algorithm combines the secret key with the current timestamp to produce a one-time password that is valid for a short period, typically 30 to 60 seconds. This code is then entered by the user to complete the authentication process. The time-sensitive nature of TOTPs enhances security by ensuring that even if a code is intercepted, it quickly becomes invalid. This method is widely used in two-factor authentication (2FA) systems, providing an additional layer of security by requiring both something the user knows (a password) and something the user has (the TOTP code).

One-Time Password/Passcode (OTP) One-Time Password (OTP) is a static code that is generated for a single login session or transaction. Unlike TOTPs, OTPs are not time-dependent and can be delivered through various channels such as SMS, email, or hardware tokens. When a user initiates a login or transaction, an OTP is generated and sent to the user, who then enters the code to complete the process. Because OTPs are not time-sensitive, they can be vulnerable if intercepted and used before they expire. Despite this, OTPs remain a popular choice for enhancing security in various applications, particularly where time-based constraints are less critical.

USB Tokens USB tokens, such as Fast Identity Online (FIDO) keys, are another popular form of hardware token. These devices plug directly into a computer's USB port and use cryptographic techniques to authenticate the user. When a user attempts to log in, they simply insert the USB token and press a button on the device to complete the authentication process. This method is highly secure as it requires physical possession of the token and cannot be easily replicated.

A new type of hardware token is becoming the new standard, and it is really considered a software token or soft token. It operates the same as a hardware token, but it is an application on your cell phone that provides the code. Google Authenticator is one example of

these types of applications. Microsoft also has an authenticator application called Microsoft Authenticator that is similar to Google Authenticator.

When configuring 2FA on an application, you have two ways of adding an account to the authenticator application. You can take a picture of a quick response (QR) code, or you can enter a security code into the authenticator application. If you choose to use a QR code, then the application turning the 2FA on will present a QR code that can be scanned by the authenticator application. If you choose to use a setup key, the application turning on the 2FA will provide a key. There is generally a second step before the application is protected by 2FA, where you will be required to enter a code from the authenticator application to the protected application. A lengthy one-time-use backup key is also generated, in case you need to turn 2FA off because your device is lost or stolen.

Single Sign-On (SSO)

Single sign-on (SSO) is an authentication method that allows users to access multiple applications and services with a single set of login credentials. This approach simplifies the user experience by eliminating the need to remember and manage multiple passwords, reducing the likelihood of password fatigue and the associated security risks. SSO is becoming increasingly popular due to its ability to streamline access management, enhance security, and improve productivity. By centralizing authentication, organizations can enforce stronger security policies and reduce the burden on IT support teams, as users are less likely to forget their credentials or require password resets.

SSO operates based on a trust relationship between an identity provider (IdP) and the service provider (SP). The IdP might be Windows Active Directory or Microsoft Entra ID which are two popular IdPs today. The SP is typically the cloud-based application that requires the authentication. When a user attempts to access the application, the service provider redirects the user to the identity provider for authentication. The identity provider verifies the user's credentials and, upon successful authentication, generates a token that is sent back to the service provider. This token contains information about the user's identity and is used to grant access to the requested application. This process allows users to log in once and gain access to multiple applications without needing to re-enter their credentials. The token is often stored in the user's web browser in the form of a web browser cookie.

When an account in the IdP is deactivated, access to all associated applications and services are also deactivated. This centralized control ensures that once a user's account is disabled, they can no longer access any of the linked applications, thereby enhancing security and preventing unauthorized access. This feature is particularly useful in scenarios such as employee offboarding, where it is crucial to promptly revoke access to sensitive information and systems.

Some applications will constantly check the validity of the token, and other applications will just check to see if the web browser cookie is still valid. It is always advisable to revoke the token during an employee offboarding so the token is invalidated, even if it still lives in someone's web browser cache.

Security Assertion Markup Language (SAML)

Security Assertion Markup Language (SAML) is a standard protocol used to facilitate SSO by enabling the exchange of authentication and authorization data between the identity provider and service providers. SAML works by passing XML-based authentication information between the IdP and SPs. When a user tries to access a service, the service provider sends a SAML request to the identity provider. The IdP authenticates the user and responds with a SAML assertion, which includes the user's identity and authorization information. The service provider then uses this assertion to grant the user access to the application. SAML enhances security by ensuring that authentication data is transmitted securely and that users need to authenticate only once to access multiple services.

Conditional Access

Conditional access is a security framework that enhances the protection provided by SSO by enforcing specific access controls based on predefined conditions. These conditions can include factors such as user location, device compliance, and real-time risk detection. By evaluating the context of access requests, conditional access ensures that only authenticated and authorized users can access resources. For example, an organization might require MFA for users attempting to log in from unfamiliar locations or block access from devices that are noncompliant with security policies. This adaptive approach to security helps mitigate risks associated with lost or stolen devices and unauthorized access attempts, further strengthening the overall security posture.

Just-in-Time Access

Just-in-time access is a security strategy designed to minimize the risk of unauthorized access by granting users elevated privileges only for the duration necessary to complete a specific task. This approach aligns with the principle of least privilege, ensuring that users have the minimum level of access required to perform their duties. By reducing the window of time during which elevated privileges are active, just-in-time access helps to mitigate potential security threats and prevent misuse of sensitive information.

Microsoft Entra ID Governance plays a crucial role in supporting just-in-time access by automating and streamlining identity and access management processes. With Entra ID Governance, organizations can define and enforce policies that ensure the right individuals have the appropriate access at the right times. This includes setting up approval workflows, access reviews, and automated expiration of access rights. As an example, imagine an employee needs temporary access to a sensitive financial system to complete a quarterly report. They submit a request through the Entra ID Governance portal, detailing the reason and duration needed. The request goes to their supervisor for approval. Their supervisor reviews and approves it, granting the employee access for 48 hours. After 48 hours, the employee's access is automatically revoked. The supervisor or an administrator can later review access logs to ensure compliance. This process ensures the employee gets the right access at the right time while maintaining security and compliance. The approval process is not bound only to a supervisor's approval; it can be automated based on a number of conditions.

Privileged Access Management (PAM)

Privileged access management (PAM) is a comprehensive security strategy designed to control, monitor, and protect privileged accounts within an organization. These accounts have elevated permissions that allow users to perform critical administrative tasks, access sensitive information, and make system changes. PAM solutions work by enforcing strict controls over who can access these accounts, monitoring their use, and ensuring that only authorized individuals can perform high-level administrative tasks. This approach helps to mitigate the risks of credential theft and privilege misuse, which can lead to significant security breaches.

Microsoft Entra ID Privileged Identity Management (PIM) is a robust tool that enhances PAM by providing granular control over privileged access within Microsoft environments. PIM allows organizations to manage, control, and monitor access to critical resources in Microsoft Entra ID, Azure, and other Microsoft Online Services such as Microsoft 365 and Microsoft Intune. Key features of PIM include just-in-time access, which grants temporary permissions to users only when needed, and time-bound access, which limits the duration of access to sensitive resources. Additionally, PIM enforces multifactor authentication for role activation, ensuring that only verified users can gain elevated access.

Mobile Device Management (MDM)

The traditional workforce has become a mobile workforce, with employees working from home, on the go, and in the office. Mobile devices such as laptops, tablets, and smartphones are used by employees to connect to the organization's cloud resources. *Bring your own device (BYOD)* has been embraced as a strategy by organizations to alleviate the capital expense of equipment by allowing employees to use devices they already own.

Because employees are supplying their own devices, a formal document called the BYOD policy should be drafted. The BYOD policy defines a set of minimum requirements for the devices, such as size and type, operating system, connectivity, antivirus/antimalware solutions, patches, and many other requirements the organization will deem necessary.

Many organizations use *mobile device management (MDM)* software that dictates the requirements for the BYOD policy. MDM software helps organizations protect their data on devices that are personally owned by the employees. When employees are terminated or a device is lost, the MDM software allows a secure remote wipe of the company's data on the device. The MDM software can also set policies requiring passwords on the device. All of these requirements should be defined in the organization's BYOD policy.

Data Loss Prevention

When an organization suffers data loss, the information is not lost—it is just out of the control of the organization. Data loss prevention (DLP) is a system used to ensure that end users do not send sensitive and/or critical information outside the organization. An example of sensitive information is a Social Security number or the secret sauce recipe for a hamburger chain. The user may not even know they are doing anything wrong in many instances.

DLP systems ensure that data loss will not occur because the damaging effects can be long term. A common DLP system is an outbound email filter. The email transport filters can sanitize or restrict information such as credit card numbers, Social Security numbers, or any type of data they are programmed to detect. Microsoft Exchange Server allows administrators to create custom DLP policies or use prebuilt DLP policies that ship with Microsoft Exchange. Other systems can classify data based on their sensitivity and restrict users from emailing the information to users outside the company or uploading the information to public cloud services.

DLP systems are efficient at identifying potential data loss and they provide a hard control for administrators, but it is best practice to implement soft controls as well by working with the organization's HR department to develop policies and effective training for end users.

Identity and Access Management

Identity and access management (IAM) is a security framework used for the authentication and authorization of users. The IAM model has been adopted by service providers as a means of managing users with their application. The IAM framework allows a service provider to define the authentication of users by means of a username and password, certificate keypair, or even single sign-on (SSO), just to name a few methods.

The framework also defines the authorization of user access for a resource within the service provider. Many service providers have adopted a model of security in which the object and accompanying permission is defined. This allows a granular control of what can be accessed by the user and what can be done with the service or object by the user. For example, if a database is created on a service provider, the specific database can be secured so that a particular user has access only to read objects within the database. However, administrators might have a higher level of permissions to alter information.

Directory Services

Logical security and directory services work together to enhance the overall security and management of network resources. Logical security protects data and resources through software-based measures, such as authentication, authorization, encryption, and access controls. Directory services, on the other hand, provide a centralized repository for storing and managing information about network resources, including users, computers, and services.

Authentication and Authorization Directory services like Microsoft Active Directory (AD) and Lightweight Directory Access Protocol (LDAP) play a crucial role in authentication and authorization processes. When a user attempts to access a network resource, the directory service verifies their identity (authentication) and checks their permissions (authorization) to ensure they have the appropriate access rights. This helps prevent unauthorized access and ensures that only legitimate users can access sensitive information.

Access Control Directory services enable administrators to define and enforce access control policies. By organizing users into groups and assigning permissions based on roles, directory services streamline the management of access rights. This ensures that users have access only to the resources they need, reducing the risk of data breaches and unauthorized access.

Policy Enforcement Directory services allow administrators to implement and enforce security policies across the network. For example, in an Active Directory environment, Group Policy Objects (GPOs) can be used to enforce password policies, software installation rules, and other security settings. This centralized management simplifies the enforcement of security policies and ensures consistency across the network.

Audit and Monitoring Directory services often include auditing and monitoring capabilities that help track user activities and access patterns. By logging authentication attempts, access to resources, and changes to directory information, administrators can detect and respond to suspicious activities, enhancing the overall security posture of the network.

By integrating logical security measures with directory services, organizations can create a robust security framework that protects their network resources, ensures compliance with security policies, and provides a seamless user experience.

Exam Objective 2.2, "Given a scenario, configure and apply basic Microsoft Windows OS security settings," requires you to know the basics of administration for Active Directory (AD). You should understand the various Active Directory components and their features.

Active Directory

Microsoft originally released *Active Directory (AD)* with Windows 2000 Server to compete with Novell Directory Services (NDS). Active Directory is a highly scalable directory service that can contain many different objects, including users, computers, and printers, just to name a few. Active Directory uses the LDAP protocol to quickly look up objects. It's important to understand that Active Directory is *not* the authentication mechanism; it is only the directory for storing and for the lookup of objects. Active Directory works in conjunction with Kerberos, which is the protocol that performed the authentication of users.

Active Directory uses a directory partition called the *schema partition* to describe classes of objects and the attributes that define each object. Each attribute is defined in the schema based on the value it will contain. For example, a user class can have a first name, last name, middle initial, description, and a number of other attributes. A specific user account is then created and configured in the GUI Microsoft Management Console (MMC) called Active Directory Users and Computers. There are also several attributes that don't show up in the

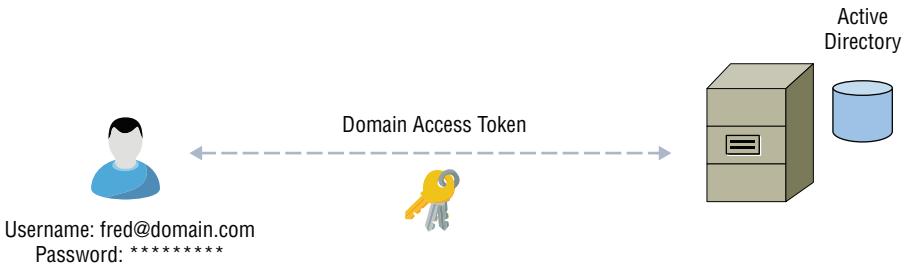
management tool, such as when an object was last changed or replicated, as well as many other attributes used by Active Directory for management.

Domain

A domain is a hierarchical collection of security objects, such as users, computers, and policies, among other components. Active Directory domains are named with a Domain Name System (DNS) name. For example, `sybex.com` would be the root domain; if you wanted to add a new domain, you would append the namespace to the left, as follows: `east.sybex.com`. Using a DNS namespace is one of the ways that Active Directory is scalable and hierarchical. Many organizations never need anything more than one domain to contain all their security objects.

When a user authenticates against an Active Directory domain, a domain access token is issued, as shown in Figure 5.13. You can consider these to be keys for the various locks (ACLs) on resources. If the user is a member of a particular security group, it will be in their security token. When the user encounters a file that is secured with an ACL, the security token is presented. If there is a matching credential, then the user is granted the associated file permission on the ACL.

FIGURE 5.13 Active Directory security tokens



Joining Active Directory Domain

Joining an Active Directory (AD) domain offers numerous benefits, particularly for organizations seeking centralized management and enhanced security. When a computer joins an AD domain, it becomes part of a network where administrators can manage user accounts, enforce security policies, and deploy software updates from a single point of control. This centralized approach simplifies IT management, reduces administrative overhead, and ensures consistent policy enforcement across all devices. Additionally, users benefit from single sign-on (SSO) capabilities, allowing them to access various network resources with one set of credentials.

When a computer joins an AD domain, it establishes a trust relationship with the domain controller, which authenticates the computer and creates a computer account in the AD database. This trust allows the computer to securely communicate with the domain controller and other domain resources. When a domain user logs on, the computer forwards their

credentials to the domain controller for verification. If valid, the domain controller generates an access token, enabling the user to access network resources without re-entering credentials.

To learn how to join Windows 11 to an Active Directory domain, refer to the "Networking Models" section in Volume 2 Chapter 3, "Windows Administration." You will find step-by-step instructions on joining a domain in this section.

Organizational Units

A domain can hold security objects, but you need to have some organization to the many different objects that you will create in your domain. Organizational units (OUs) enable you to group objects together so that you can apply a set of policies to the objects. OUs should be designed to group objects by the following criteria:

Object Class Objects can be organized based on their class or type. You can organize all the computers and users into their respective OUs.

Geographic Location Objects can be organized based on their location. It is common practice to use airport codes when the organization is spread across a large distance. However, town names can also be used when the organization branches are in relatively close proximity.

Function Objects can be organized based on their function in the organization, such as servers, workstations, and users. Objects can also be grouped by their job function, such as Sales, Marketing, and HR.

Hybrid Nothing says that you have to stick to one organization of objects. Depending on how you are planning your domain, you could use a combination. It is not uncommon to see airport codes to represent the various offices, with functions under each airport code, as shown in Figure 5.14.

Moving Objects

Moving objects between OUs is a straightforward process that allows administrators to adapt to organizational changes or refine their directory structure. When an object, such as a user or computer, needs to be relocated to a different OU, it can be done through directory management tools like the Active Directory Users and Computers (ADUC) MMC console.

When you move an object from one OU to another, several things happen behind the scenes to ensure the object is properly integrated into its new location:

Policy Application The object will inherit the Group Policy Objects (GPOs) associated with the new OU. This means any policies applied to the new OU, such as security settings, software installations, or login scripts, will now apply to the moved object.

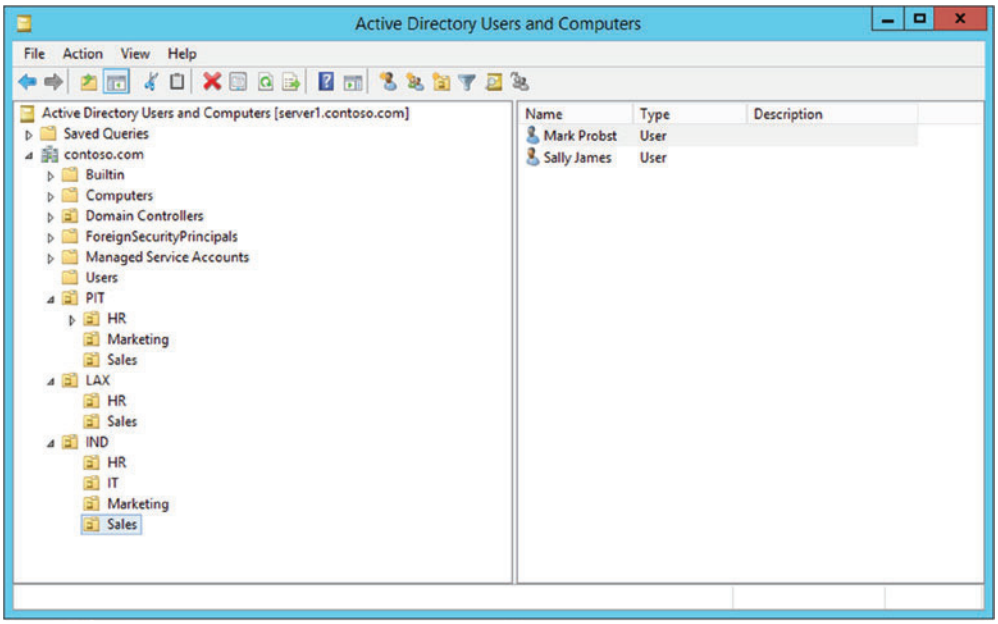
Permissions and Access Control The object will also inherit the permissions and access controls defined for the new OU. This can affect the object's access to resources and its ability to perform certain actions within the network.

Replication In a multidomain controller environment, the change will be replicated across all domain controllers to ensure consistency. This ensures that the object's new location is recognized throughout the entire network.

Audit and Logging The move will be logged in the directory service's audit logs. This helps administrators track changes and maintain a record of object movements for security and troubleshooting purposes.

Script and Automation Adjustments If there are any scripts or automated processes that reference the object's original location, they may need to be updated to reflect the new OU. This ensures that automated tasks continue to function correctly.

FIGURE 5.14 A hybrid OU structure

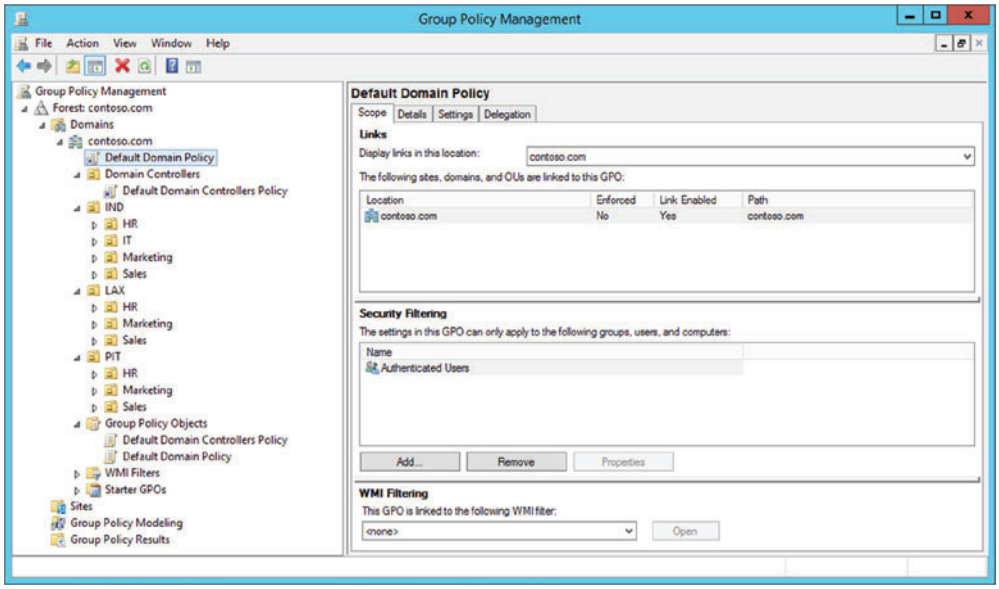


Group Policy

Group Policy is a feature of Active Directory that enables you to apply policies to control users and computers. Typically, you do not apply policies to individual users or computers but instead to groups of users or computers. A Group Policy Object (GPO) is a type of object in Active Directory that allows you to apply a set of policies against an organizational

unit. Group Policy Objects are created, linked, and edited in the Group Policy Management Console (GPMC), as shown in Figure 5.15.

FIGURE 5.15 The Group Policy Management Console



You can control thousands of settings for both the user and computer objects, as shown in Figure 5.16. Policies are hard controls that you can force on an object. Policies are refreshed in the background every 90 minutes. So, if a setting that has a policy applied changes, it will be set back during the refresh cycle. Most of the time, however, settings are grayed out when they are being managed by GPO and cannot be changed at all. Preferences allow for files, Registry, environment variable, and Control Panel items to be modified. Preferences set an initial setting and are applied only during first login, so these settings are a preference, not a policy, and the user can change these settings afterward.

Login Scripts

Login scripts are one of the configurable attributes for a user account. As covered in Volume 2 Chapter 8, "Scripting, Remote Access, and Artificial Intelligence," you can use VBScript or Windows batch scripts as login scripting languages. Login scripts are useful on an Active Directory network for connecting network-mapped drives and printers, among other administrative tasks. Login scripts also provide uniformity across an enterprise by running the same commands for each user configured with the script. The location of the setting is found on the Profile tab of the user account, as shown in Figure 5.17.

FIGURE 5.16 Group Policy Object settings

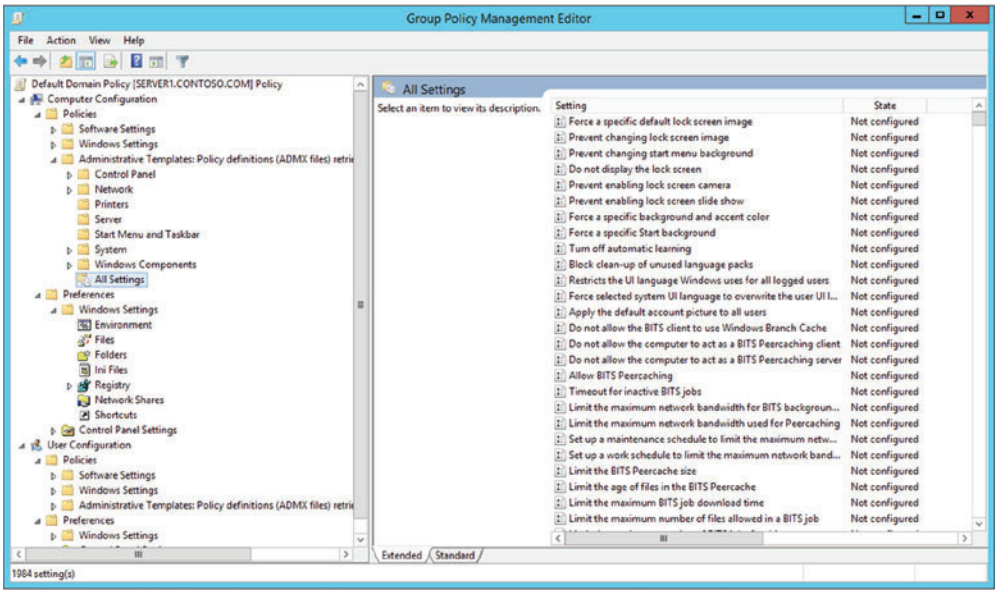
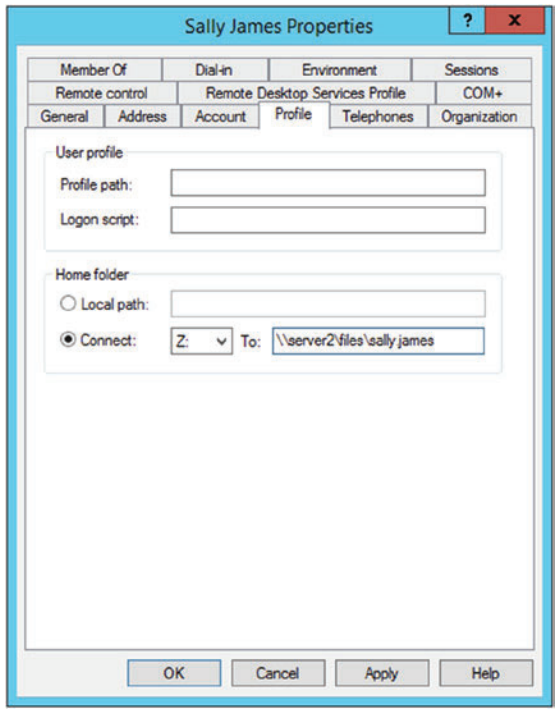


FIGURE 5.17 Profile settings for a user account



Home Folders

A *home folder* is a private network location in which the user can store their personal files. The home folder is an attribute that can be set for a user account in the Active Directory Users and Computers MMC on the Profile tab, as shown in Figure 5.17. The location can be a local path, if the user will use the same computer, and the files should be stored locally for the user. However, it is most useful when you connect a network drive to a remote file server. This allows for centralized file storage, and you can then perform backups on the data.

Folder Redirection

Normally, when a user logs into the network and a roaming profile exists for the user, the profile is completely downloaded to the computer the user is working on. During logout, all data is written back to the roaming profile location on the network file server. Profiles can become extremely large in size, sometimes even gigabytes, and slow down the login and logout processes.

Folder redirection is a Group Policy setting that allows the redirection of portions of users' profile folders to a network location. When folder redirection is used, the roaming profile is still downloaded. However, the redirected folders are not downloaded; they are simply redirected to the network location. This speeds login and logout times because the entire profile is no longer downloaded (login) and uploaded (logout).

Security Groups

The use of Active Directory domains allows you to adopt a centralized administration model for user accounts. You also have the ability to secure files, printers, and other resources on the domain with these centralized credentials. However, if you secure the resources solely with user accounts, very quickly it will become a daunting task, as more and more people need access to resources. This is a common pitfall of new administrators, because most resources only need to be accessed by a few people initially.

All security should be done using *security groups* for a few reasons. The first reason is a simple one: You want to administer groups of users and not individual users. It is easier to apply permissions to a group of users than individual users. As the resource needs to be shared by more people, you never have to revisit the resource to apply the new permissions if a group is used. All you need to do is add the new users to the group and they will have access.

One additional benefit that accompanies using security groups is the centralized auditing of permissions. If you want to know who has access to a resource, all you have to do is look at the membership of the group associated with the resource. If you need to check if someone specific has permission to the resource, you just have to look at their group membership.

When securing resources, you should always create a new group that is associated with the resource and never reuse a group for multiple resources. For example, if you were securing a main office printer, you should create a group that explains the resource and the level of access, such as creating a group called `perm_print_mainprinter`. Just by looking

at the group you can identify that it is a permissions group, that it allows printing, and that the resource is the main printer.

Exam Objective 2.4, "Summarize types of malware and tools/methods for detection, removal, and prevention," requires you to know the types of malware. You should know the difference between a Trojan, rootkit, virus, spyware, ransomware, keylogger, boot sector virus, cryptominer, stalkerware, and fileless malware. In addition, you should be familiar with the tools and methods to detect, remove, and prevent the various malware.

Malware

Malware is a broad term describing any software with malicious intent. Although we use the terms *malware* and *virus* interchangeably, distinct differences exist between them. The lines have blurred because the delivery mechanisms of malware and viruses are sometimes indistinguishable.

A virus is a specific type of malware, the purpose of which is to multiply, infect, and do harm. A virus distinguishes itself from other malware because it is self-replicating code that often injects its payload into documents and executables. This is done in an attempt to infect more users and systems. Viruses are so efficient in replicating that their code is often programmed to deactivate after a period of time, or they are programmed to only be active in a certain region of the world.

Malware can be found in a variety of other forms, such as covert cryptomining, web search redirection, adware, spyware, and even ransomware, and these are just a few. Today the largest threat of malware is ransomware because it's lucrative for criminals.

Ransomware

Ransomware is a type of malware that is becoming popular because of anonymous currency, such as Bitcoin. Ransomware is software that is often delivered through an unsuspecting random download, phishing email, malicious attachment, compromised website, or a targeted attack by a threat actor. Once on a computer or in the network, it takes control of systems and demands that a third party be paid. The "control" can be accomplished by encrypting the hard drive, by changing user password information, or via any of a number of other creative ways. Users are usually assured that by paying the extortion amount (the ransom), they will be given the code needed to revert their systems back to normal operations. CryptoLocker was a popular ransomware that made headlines across the world (see Figure 5.18), but there have been many others since its discovery. You can protect yourself from ransomware by having antivirus/anti-malware software with up-to-date definitions and by keeping current on patches.

FIGURE 5.18 CryptoLocker

Trojans

Trojan horses are programs that enter a system or network under the guise of another program. A Trojan horse may be included as an attachment or as part of an installation program. The Trojan horse can create a backdoor or replace a valid program during installation. It then accomplishes its mission under the guise of another program. Trojan horses can be used to compromise the security of your system, and they can exist on a system for years before they're detected.

The best preventive measure for Trojan horses is to not allow them entry into your system. Immediately before and after you install a new software program or operating system, back it up! If you suspect a Trojan horse, you can reinstall the original program(s), which should delete the Trojan horse. A port scan may also reveal a Trojan horse on your system. If an application opens a TCP or UDP port that isn't supported in your network, you can track it down and determine which port is being used.

Keyloggers

A *keylogger* is normally a piece of software that records an unsuspecting victim's keystrokes. Keyloggers can stay loaded in memory and wait until you log into a website or other authentication system. They will then capture and relay the information to an awaiting host on the Internet.

Keyloggers don't always have to be in the form of software. Some keyloggers are hardware dongles that sit between the keyboard and computer. These must be retrieved and the data must be downloaded manually, so they are not very common.

Rootkits

Rootkits are software programs that have the ability to hide certain things from the operating system. They do so by obtaining (and retaining) administrative-level access. With a rootkit, there may be a number of processes running on a system that don't show up in Task Manager, or connections that don't appear in a Netstat display of active network connections that may be established or available. The rootkit masks the presence of these items by manipulating function calls to the operating system and filtering out information that would normally appear.

Unfortunately, many rootkits are written to get around antivirus and antispyware programs that aren't kept up-to-date. The best defense you have is to monitor what your system is doing and catch the rootkit in the process of installation.

Spyware

Spyware differs from other malware in that it works—often actively—on behalf of a third party. Rather than self-replicating, like viruses and worms, spyware is spread to machines by users who inadvertently ask for it. The users often don't know they have asked for it but have done so by downloading other programs, visiting infected sites, and so on.

The spyware program monitors the user's activity and responds by offering unsolicited pop-up advertisements (sometimes known as *adware*), gathers information about the user to pass on to marketers, or intercepts personal data, such as credit card numbers.

Cryptominers

With the rise of bitcoin, so came the rise of cryptominers. A cryptominer is typically a purpose-built device that grinds out cryptographic computations. When the computation is balanced, a cryptocoin is created and equates to real money, such as Bitcoin, Ethereum, and Dogecoin, just to name a few. A cryptominer does not always have to be a dedicated purpose-built device—it can also be a distributed group of computers called a cryptopool.

Malware in the form of cryptominers became very popular, because it is a very lucrative way for threat agents to make money. The problem is that the threat agents use your computer to grind out the computations. The most common way a threat agent will run a cryptominer remotely is with JavaScript embedded on a malicious web page. Threat agents have also been known to create viruses in which the payload (cryptominer) uses your video card to grind out the computations. However, the JavaScript variant is more common to find in the wild.

Fileless

Fileless malware is a sophisticated type of cyberthreat that operates without leaving a trace on the victim's hard drive, making it exceptionally difficult to detect and remove. Unlike

traditional malware, which relies on executable files to infect a system, fileless malware exploits existing software, applications, and protocols to execute malicious activities directly in the system's memory. This stealthy approach allows it to bypass conventional antivirus solutions and persist in the system undetected. Cybercriminals often use fileless malware to gain unauthorized access, steal sensitive information, or disrupt operations, posing a significant challenge for cybersecurity professionals. As technology evolves, so do the tactics of these malicious actors, necessitating advanced and adaptive security measures to combat such elusive threats.

Stalkerware

Stalkerware is a particularly insidious form of spyware designed to monitor and track an individual's activities without their knowledge or consent. Often disguised as legitimate applications, stalkerware can be secretly installed on a victim's device, granting the perpetrator access to personal data such as messages, call logs, location, and even real-time audio and video feeds. This invasive software is frequently used in abusive relationships to exert control and surveillance over the victim, leading to severe privacy violations and emotional distress. The covert nature of stalkerware makes it challenging to detect and remove, highlighting the need for robust cybersecurity measures and increased awareness to protect individuals from such intrusive threats. As technology continues to advance, it is crucial to stay vigilant and informed about the potential risks posed by stalkerware and other forms of digital surveillance.

Viruses

Viruses can be classified as polymorphic, stealth, retrovirus, multipartite, armored, companion, phage, and macro viruses. Each type of virus has a different attack strategy and different consequences.

The following sections introduce the symptoms of a virus infection, explain how a virus works, and describe the types of viruses you can expect to encounter and how they generally behave. We'll also discuss how a virus is transmitted through a network and look at a few hoaxes.

Estimates for losses due to viruses are in the billions of dollars. These losses include financial loss as well as lost productivity.

Symptoms of a Virus/Malware Infection

Many viruses will announce that you're infected as soon as they gain access to your system. They may take control of your system and flash annoying messages on your screen or destroy your hard disk. When this occurs, you'll know that you're a victim. Other viruses will cause your system to slow down, cause files to disappear from your computer, or take over your disk space.

Because viruses are the most common type of malware, the term *virus* is used in this section.

You should look for some of the following symptoms when determining if a virus infection has occurred:

- The programs on your system start to load more slowly. This happens because the virus is spreading to other files in your system or is taking over system resources.
- Unusual files appear on your hard drive, or files start to disappear from your system. Many viruses delete key files in your system to render it inoperable.
- Program sizes change from the installed versions. This occurs because the virus is attaching itself to these programs on your disk.
- Your browser, word processing application, or other software begins to exhibit unusual operating characteristics. Screens or menus may change.
- The system mysteriously shuts itself down or starts itself up and does a great deal of unanticipated disk activity.
- You mysteriously lose access to a disk drive or other system resources. The virus has changed the settings on a device to make it unusable.
- Your system suddenly doesn't reboot or gives unexpected error messages during startup.

This list is by no means comprehensive. What is an absolute, however, is the fact that you should immediately quarantine the infected system. It is imperative that you do all you can to contain the virus and keep it from spreading to other systems within your network, or beyond.

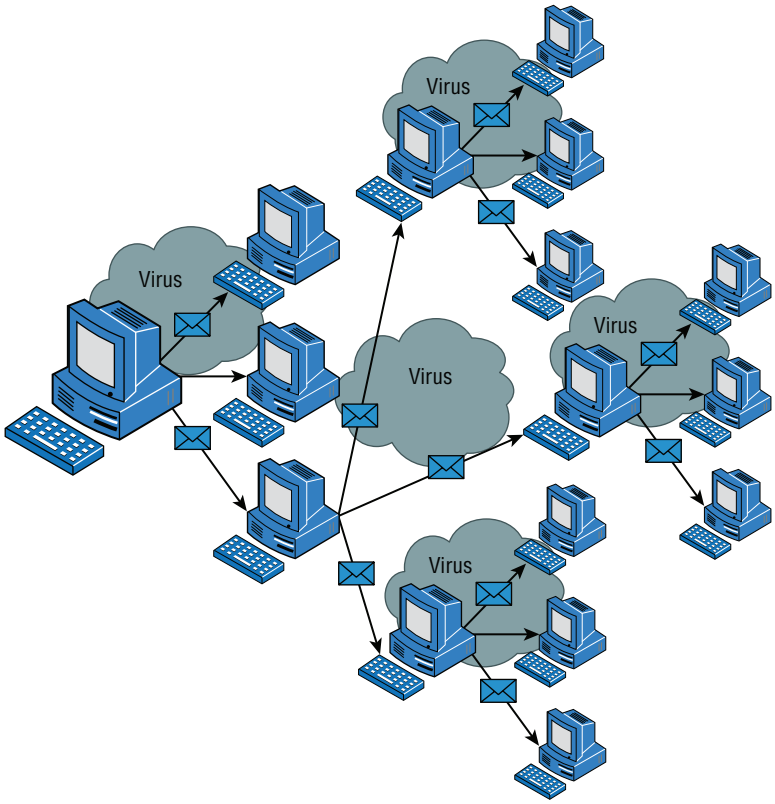
How Viruses Work

A virus, in most cases, tries to accomplish one of two things: render your system inoperable or spread to other systems. Many viruses will spread to other systems given the chance and then render your system unusable. This is common with many of the newer viruses.

If your system is infected, the virus may try to attach itself to every file in your system and spread each time you send a file or document to other users. Some viruses spread by infecting files that are either transmitted through a network or by removable media, such as backup tapes, USB thumb drives, among others. When you give removable media to another user or put it into another system, you then infect that system with the virus.

Many viruses today are spread using email. The infected system attaches a file to any email that you send to another user. The recipient opens this file, thinking it's something that you legitimately sent them. When they open the file, the virus infects the target system. The virus might then attach itself to all the emails that the newly infected system sends, which in turn infects the computers of the recipients of the emails. Figure 5.19 shows how a virus can spread from a single user to literally thousands of users in a very short period of time using email.

FIGURE 5.19 A virus spreading from an infected system using email



Types of Viruses

Viruses take many different forms. The following list briefly introduces these forms and explains how they work.

The best defense against a virus attack is up-to-date antivirus/antimalware software installed and running. The software should be on all workstations as well as the server.

These are the most common types of viruses, but this isn't a comprehensive list:

Armored Virus An armored virus is designed to make itself difficult to detect or analyze. Armored viruses cover themselves with protective code that stops debuggers or disassemblers from examining critical elements of the virus. The virus may be written in such a way that some aspects of the programming act as a decoy to distract analysis while the actual code hides in other areas in the program.

From the perspective of the creator, the more time that it takes to deconstruct the virus, the longer it can live. The longer it can live, the more time it has to replicate and spread to as many machines as possible. The key to stopping most viruses is to identify them quickly and educate administrators about them—the very things that the armor makes difficult to accomplish.

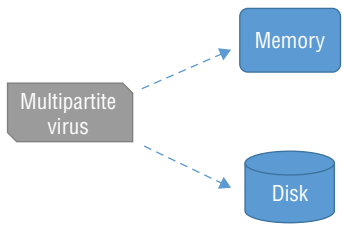
Boot Sector Virus This virus infects the Master Boot Record (MBR) of a hard disk. This type of virus loads when the computer boots and can re-infect an operating system. Secure Boot is used to verify the entire boot process with digital signatures and identify any part of the boot process that has been modified.

Companion Virus A companion virus attaches itself to legitimate programs and then creates a program with a different filename extension. This file may reside in your system's temporary directory. When a user types the name of the legitimate program, the companion virus executes instead of the real program. This effectively hides the virus from the user. Many of the viruses that are used to attack Windows systems make changes to program pointers in the Windows Registry so that they point to the infected program. The infected program may perform its dirty deed and then start the real program.

Macro Virus A macro virus exploits the enhancements made to many application programs. Programmers can expand the capability of applications such as Microsoft Word and Excel. Word, for example, supports a mini-BASIC programming language that allows files to be manipulated automatically. These programs in the document are called *macros*. For example, a macro can tell your word processor to spell-check your document automatically when it opens. Macro viruses can infect all the documents on your system and spread to other systems via email or other methods. Macro viruses are one of the fastest-growing forms of exploitation today.

Multipartite Virus A multipartite virus attacks your system in multiple ways. It may attempt to infect your boot sector, infect all your executable files, and destroy your application files. The hope here is that you won't be able to correct all the problems and will allow the infestation to continue. The multipartite virus depicted in Figure 5.20 attacks a system's boot sector, infects application files, and attacks Word documents.

FIGURE 5.20 A multipartite virus commencing an attack on a system



Phage Virus A phage virus alters programs and databases, and the only way to remove this virus is to reinstall the programs that are infected. If you miss even a single instance

of this virus on the victim system, the process will start again and infect the system once more.

Polymorphic Virus Polymorphic viruses change form to avoid detection. These types of viruses attack your system, display a message on your computer, and delete files on your system. The virus will attempt to hide from your antivirus software. Frequently, the virus will encrypt parts of itself to avoid detection. When the virus does this, it's referred to as mutation. The mutation process makes it hard for antivirus software to detect common characteristics of the virus.

A *signature* is an algorithm or other element of a virus that uniquely identifies it. Because some viruses have the ability to alter their signature, it is crucial that you keep signature files current, whether you choose to download them manually or configure the antivirus engine to do so automatically.

Retrovirus A *retrovirus attack* bypasses the antivirus software installed on a computer. You can consider a retrovirus to be an anti-antivirus. Retroviruses can directly attack your antivirus software and potentially destroy the virus definition database file. When this information is destroyed without your knowledge, you are left with a false sense of security. The virus may also directly attack an antivirus program to create bypasses for itself.

Stealth Virus A *stealth virus* attempts to avoid detection by masking itself from applications. It may attach itself to the boot sector of the hard drive. When a system utility or program runs, the stealth virus redirects commands around itself to avoid detection. An infected file may report a file size different from what is actually present. Stealth viruses may also move themselves from file A to file B during a virus scan for the same reason.

Virus Transmission in a Network

Upon infection, some viruses destroy the target system immediately. The saving grace is that the infection can be detected and corrected. Some viruses won't destroy or otherwise tamper with a system; instead, they use the victim system as a carrier. The victim system then infects servers, fileshares, and other resources with the virus. The carrier then infects the target system again. Until the carrier is identified and cleaned, the virus continues to harass systems in this network and spread.

Present Virus Activity

New viruses and threats are released on a regular basis to join the cadre of those already in existence. From an exam perspective, you need only be familiar with the world as it existed at the time the questions were written. From an administrative standpoint, however, you need to know what is happening today.

To find information on alerts and threats, visit the Cybersecurity and Infrastructure Security Agency (CISA) Cybersecurity Alerts & Advisories web page at `www.cisa.gov/news-events/cybersecurity-advisories`. You'll find a detailed description of the most current threats as well as links to pages on older threats and advisories of vulnerable systems.

Botnets

A *botnet* is a group of zombies, which sounds like a ridiculous beginning to a horror movie. When malware infects a computer, its purpose is often to lie dormant and await a command from a command-and-control server. When this happens, the computer is considered a zombie. When enough infected computers (zombies) check in, the *threat agent* will send a command to the command-and-control server, and the botnet of zombies will work on the task. Often the task is to launch a malicious DDoS attack or to send spam. DDoS will be covered later in this chapter, in the "Common Security Threats" section.

Worms

A *worm* is different from a virus in that it can reproduce itself, it's self-contained, and it doesn't need a host application to be transported. Many of the so-called viruses that make the news are actually worms. However, it's possible for a worm to contain or deliver a virus to a target system.

By their nature and origin, worms are supposed to propagate, and they use whatever services they're capable of using to do that. Early worms filled up memory and bred inside the RAM of the target computer. Worms can use TCP/IP, email, Internet services, or any number of possibilities to reach their target.

Adware

Adware is a type of software that automatically delivers advertisements to a user's device, often without their consent. These ads can appear in various forms, such as pop-ups, banners, or in-browser ads, and are typically designed to generate revenue for the software's creator. While adware itself is not inherently malicious, it can significantly degrade the user experience by slowing down the device, consuming bandwidth, and creating a cluttered and intrusive environment. Additionally, some adware may track user behavior and collect personal information, raising privacy concerns. To mitigate the impact of adware, users should be cautious when downloading free software, as adware is often bundled with these applications, and should use reputable security tools to detect and remove unwanted programs.

Potentially Unwanted Programs (PUPs)

Potentially unwanted programs (PUPs) are software applications that a user may not have intended to install on their device. These programs often come bundled with legitimate

software and can include adware, toolbars, or other unwanted utilities. While PUPs are not always harmful, they can negatively affect system performance, compromise privacy, and create security vulnerabilities. PUPs can be particularly deceptive, as they often masquerade as useful tools or enhancements, tricking users into installing them. To avoid PUPs, users should carefully read installation prompts, opt for custom installation options to deselect unwanted components, and regularly scan their devices with trusted antimalware software to identify and remove any PUPs that may have been inadvertently installed.

Mitigating Software Threats

Now that you understand some of the common software threats, let's look at how you can protect yourself from them. This section discusses practical tools and methods that you can use to safeguard yourself from common software-based threats. We will also discuss tactics you can use to mitigate risk as well as ways to recover from security mishaps.

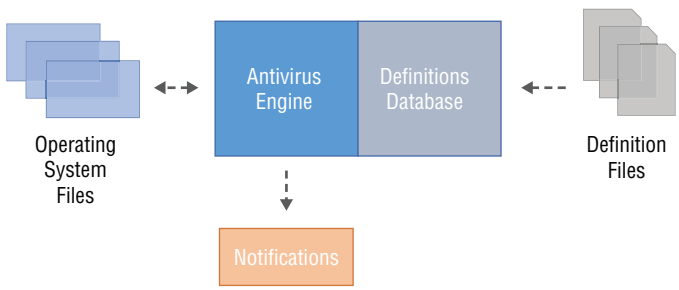
Antivirus

Most malware can be simply prevented with the use of antivirus software. Back when Windows XP came out, the running joke was that you would get a virus before you could get a chance to install antivirus software. To some extent this was true, if you had to get online to retrieve the software.

Microsoft introduced Microsoft Security Essentials as a download for Windows XP, and the Windows Vista operating system started to ship with it installed. Today, Windows comes preinstalled with Windows Virus & Threat Protection, so if you don't purchase antivirus software you are still protected. As a result of these tactics, Microsoft has made the Windows operating system safer than it used to be.

Although Microsoft's antivirus program will work fine for most computing needs, there are some advantages to purchasing antivirus products from third-party vendors. To understand some of the differences, you need to be familiar with the components of antivirus software. Antivirus software comprises two main components: the antivirus engine and the definitions database, as shown in Figure 5.21.

FIGURE 5.21 Antivirus software components



Antivirus Engine The antivirus engine is responsible for the real-time scanning of operating system files and the notifications to the user. Various antivirus engines will scan operating system files differently. For example, an antivirus engine might recognize when a game is being played, and it will temporarily stop scanning files so that the game loads faster. Notifications can also automatically submit files to the cloud for more extensive scanning.

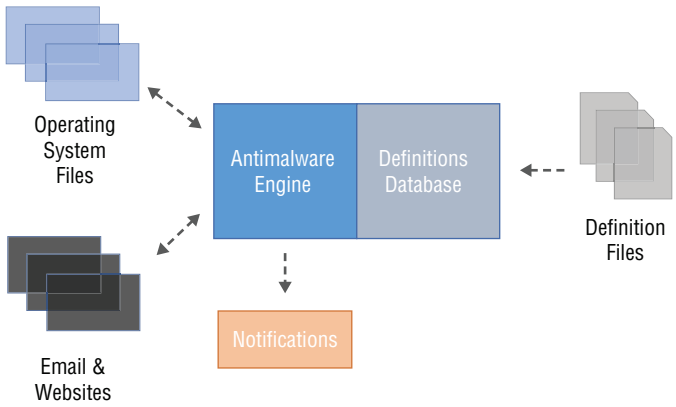
Definitions Database The definitions database is why you would select one antivirus product over another. A definitions database's effectiveness is based on two attributes: the frequency of updates and the comprehensiveness of the database signatures. Antivirus definitions, also known as antivirus signatures, are discovered daily and added to the antivirus database. Some antivirus products will download deltas to the initial database every hour, and some will download them once a day.

Antimalware

Malware is a broad term and covers many different software threats that we learned about in this chapter. Antimalware and antivirus are extremely similar in their functionality, and sometimes vendors have a hard time differentiating their products. This is because many antivirus products now check more than just files.

An antimalware software package will not only check the filesystem for threats, like rootkits and trojans, but will also watch incoming email for phishing scams and malicious websites, as shown in Figure 5.22. When these threats are detected, the user gets a notification, and the threat is usually mitigated or avoided completely.

FIGURE 5.22 Antimalware components



In Exercise 5.1 you will test your antimalware protection with a harmless file called an Eicar file.

EXERCISE 5.1

Testing Your Antimalware

1. Navigate to the Eicar antimalware test file site at `www.eicar.org/download-anti-malware-testfile`.
2. Scroll down to the download section.
3. Download a few of the Eicar test files and notice how your antivirus detects the malware.
4. Examine the alerts as your antimalware software reports the malware.

The Eicar website contains a totally benign piece of malware that triggers your antimalware engine. Any search for Eicar will produce similar results and the contents are benign.

Detection and Response

The increasing sophistication and frequency of cyberthreats necessitate advanced detection and response mechanisms beyond traditional antivirus (AV) or antimalware solutions. While AV and antimalware software are essential for identifying and blocking known threats, they often fall short in detecting and responding to more complex and evolving attacks. This is where endpoint detection and response (EDR), managed detection and response (MDR), and extended detection and response (XDR) come into play. These advanced solutions offer distinct capabilities and advantages, such as continuous monitoring, threat hunting, and comprehensive visibility across multiple security layers, making them crucial for addressing specific security challenges and requirements in today's dynamic threat landscape. The following explains how each of these detection and response packages differentiate themselves from each other:

Endpoint Detection and Response (EDR) Endpoint detection and response (EDR) is a cybersecurity solution designed to monitor and protect endpoints, such as computers, mobile devices, and servers, from malicious activities. EDR solutions continuously collect and analyze data from endpoints to detect suspicious behavior, enabling rapid identification and response to potential threats. By providing detailed visibility into endpoint activities, EDR assists security teams in quickly investigating and mitigating incidents, thereby reducing the risk of data breaches and minimizing the impact of cyberattacks.

Managed Detection and Response (MDR) Managed detection and response (MDR) is a comprehensive cybersecurity service that combines advanced technology with human expertise to detect, analyze, and respond to threats. MDR providers offer 24/7 monitoring and threat hunting, leveraging sophisticated tools and techniques to identify and neutralize threats before they can cause significant harm. By outsourcing these critical functions to a team of skilled security professionals, organizations can enhance their

security posture without the need for extensive in-house resources. MDR services are particularly beneficial for small to medium-sized businesses that may lack the expertise or budget to maintain a full-fledged security operations center (SOC).

Extended Detection and Response (XDR) Extended detection and response (XDR) is an integrated cybersecurity solution that extends beyond traditional EDR by correlating data across multiple security layers, including endpoints, networks, servers, and cloud environments. XDR provides a holistic view of an organization's security landscape, enabling more effective threat detection and response. By breaking down silos and integrating various security tools and data sources, XDR enhances the ability to detect complex, multivector attacks and streamline incident response processes. This comprehensive approach helps organizations improve their overall security posture and reduce the time and effort required to manage and respond to threats.

Each detection and response platform plays a vital role in protecting an organization from compromise. EDR focuses on endpoint protection, providing detailed visibility and rapid response capabilities. MDR offers a managed service approach, combining technology and expertise to deliver continuous threat monitoring and response. XDR takes a broader view, integrating data from multiple sources to provide a comprehensive security solution. By understanding and leveraging these products, organizations can better protect themselves against cyberthreats, and ensure the safety and integrity of their data.

Email Security Gateway Systems

Email security gateway systems are a critical component of an organization's cybersecurity defenses. They are designed to protect against a number of email threats such as phishing, malware, ransomware, and of course spam. These systems function as a protective layer, inspecting incoming and outgoing email traffic to filter out malicious content and prevent it from reaching end users. Email security gateways can be deployed in two primary forms: on-premises and cloud-based systems.

On-Premises Email Security Gateways On-premises email security gateways are installed and managed within the organization's own data centers. This deployment model provides organizations with direct control over their security policies, configurations, and data. On-premises solutions are often favored by organizations with stringent data privacy requirements, such as those in the healthcare or financial sectors, or those that need to comply with specific regulatory standards like GDPR or HIPAA. However, on-premises solutions require significant investment in hardware, software, and ongoing maintenance, as well as dedicated IT staff to manage and update the system.

Cloud-Based Email Security Gateways Cloud-based email security gateways, on the other hand, are hosted by third-party providers and accessed over the Internet. This deployment model offers several advantages, including scalability, flexibility, and reduced maintenance overhead. Cloud-based solutions can easily scale to accommodate the growing needs of an organization, making them ideal for businesses of all sizes.

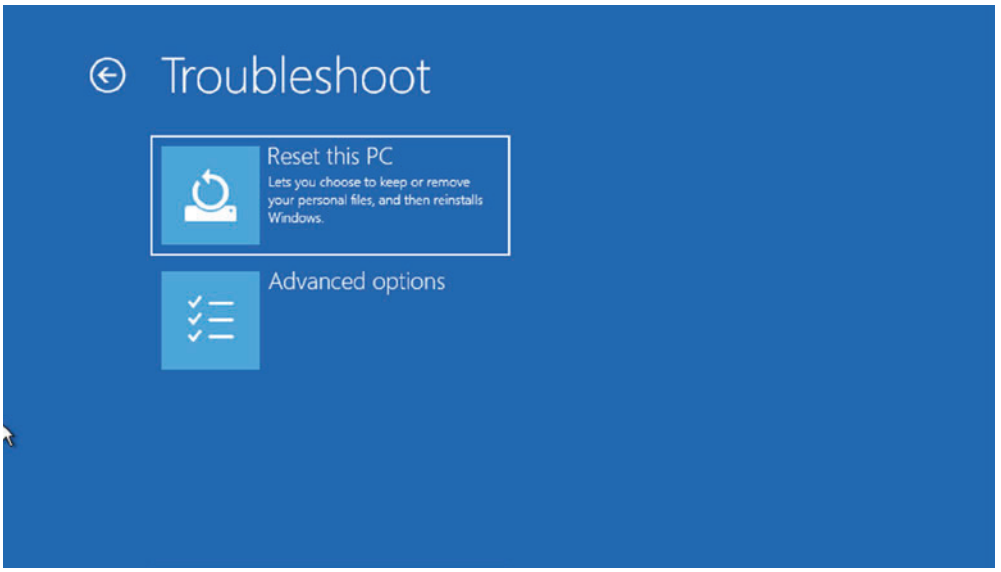
They also benefit from the provider's advanced threat intelligence, which is gathered from a global network of users and continuously updated to protect against the latest threats. Additionally, cloud-based gateways often come with built-in redundancy and disaster recovery capabilities, ensuring continuous protection even in the event of a system failure. Organizations with limited IT resources or those looking to quickly adapt to evolving threat landscapes may find cloud-based solutions particularly beneficial. However, reliance on a third-party provider means that organizations must trust the provider's security measures and data handling practices.

Both on-premises and cloud-based email security gateways play a crucial role in safeguarding an organization from email compromise. They filter out malicious content and prevent data breaches, thus protecting against financial loss, and ensure compliance with regulatory requirements.

Recovery Console

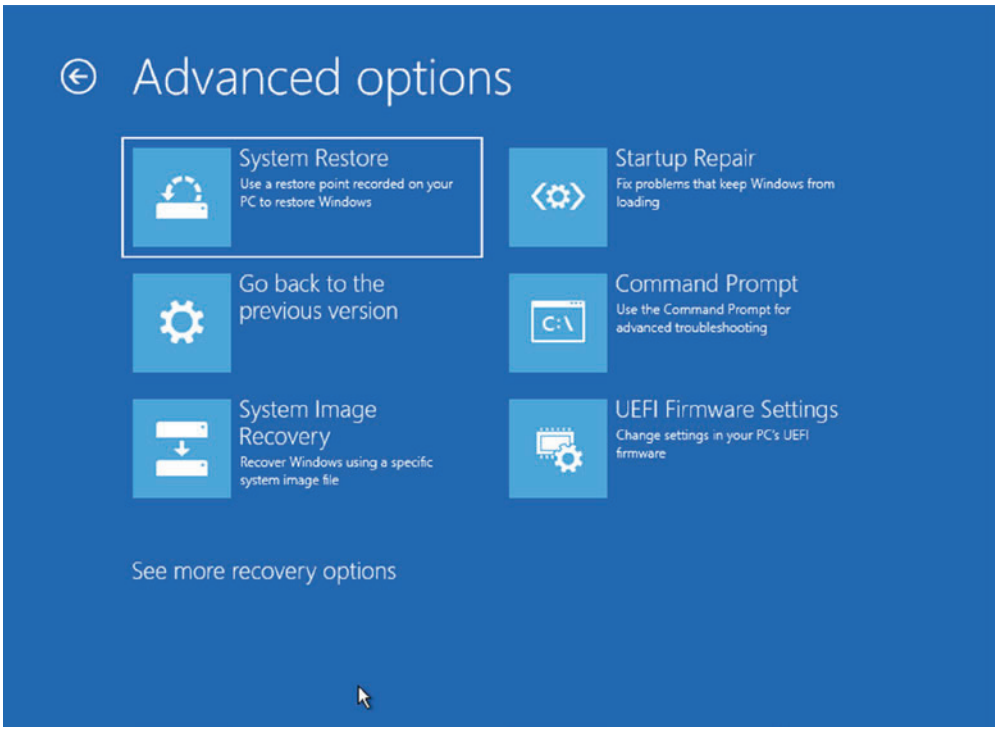
A recovery console can perform a number of useful functions for recovery from a security threat. The Windows Recovery Environment (WinRE) is a recovery console that can perform a number of useful functions, as we'll cover in this section. The most useful function is the Reset This PC option, which allows you to refresh the operating system while keeping your data files or remove everything and start from scratch, as shown in Figure 5.23. The latter of the two options assumes you have backups of your data files.

FIGURE 5.23 Windows Recovery Environment



The Windows Recovery Environment also allows you to perform a system restore, whereby you can restore the operating system back to a specific point in time. If a system recovery image exists, you can also recover with the System Image Recovery option. This option will reset the operating system back to the point in the recovery image, which is usually just like the day you turned it on. Figure 5.24 shows the Advanced Options menu.

FIGURE 5.24 Windows Advanced Options



To access Windows Recovery Advanced Options, go to Start ➢ Settings ➢ System ➢ Recovery, then under Advanced Startup, select Restart Now; this will boot you into the Windows Recovery Environment where you can choose Troubleshoot and then Advanced Options to access further recovery tools.

End-User Education

By far the best prevention of security threats is the education of your end users regarding common threats. For example, the most effective method of preventing viruses, malware, spyware, and harm to data is to teach your users not to open suspicious files and to open only those files that they're reasonably sure are virus/malware free. End users should also

be educated on how to identify Trojans and phishing emails scams. The end-user education should also identify guidelines for physical destruction of data, in particular any paperwork that has sensitive information on it. End users should also be educated on the various social engineering threats and how to identify them. An end user who has foresight and who exercises vigilance is more powerful than any antivirus or antimalware product on the market.

End-user education in an organization is normally part of the employee onboarding process for new hires. However, it should not stop there, because threats change every day. Many organizations revisit the training for their employees once a year in ideal circumstances. This training can be performed in a formal classroom setting or through an online service. Some online services offer educational videos that have interactive questions to verify that the employee has learned the objectives of the video.

Because phishing is such a widespread problem for organizations, special antiphishing training is often mandatory for employees a few times a year. Often organizations will phish their employees with specially crafted emails in an attempt to see how well their training is working. When an employee spots the phishing attempt, they can earn rewards, like a gift card. However, if they get phished, then they must retake the antiphishing training or they may be targeted in the future. A popular month for these tactics is October, because it is the cybersecurity awareness month.

Software Firewalls

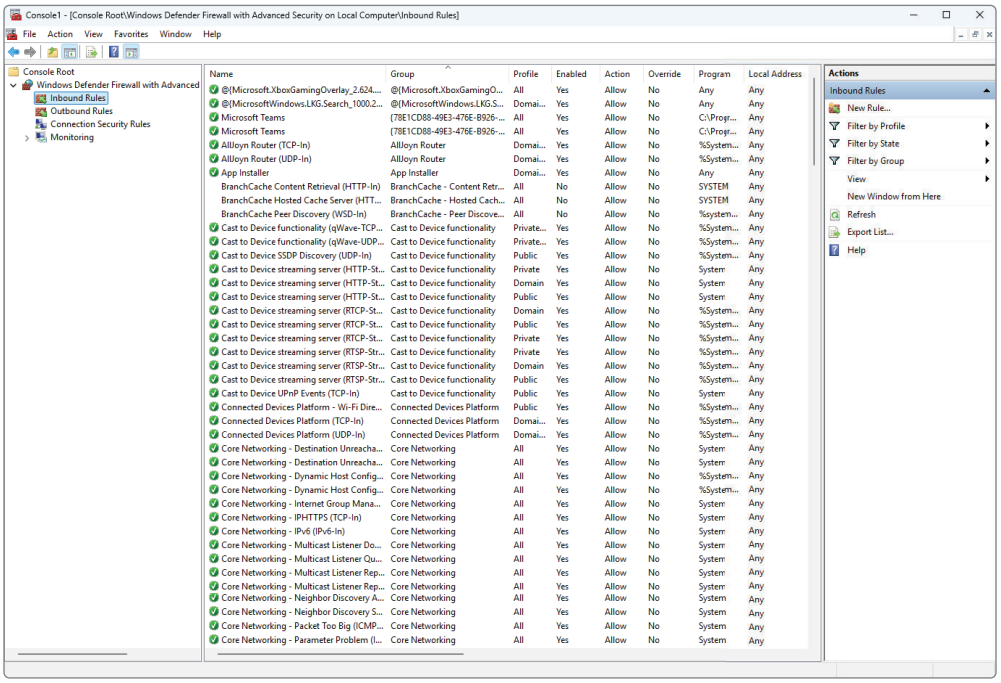
"Software firewalls" is a misnomer for this section, since all firewalls are software-based in some way. Sure, you might purchase a piece of equipment that is classified as a hardware firewall, but there is software running on the firewall to protect your network. However, when we discuss firewalls in respect to operating systems, we call them software firewalls, because they are part of the operating system and thus considered software.

Let's look at where we came from and where we are now. A major gamechanger in the history of Microsoft was the release of Windows XP Service Pack 2, which switched on the built-in firewall by default. It was long overdue for the operating system at the time. It was also not well received, because administrators had to learn firewall rules when they installed a new software package that required incoming network traffic. As a result, the firewall was the first thing that got shut off when there was a problem with connectivity.

Throughout the years of new Windows versions, the firewall has received new features and became a polished product. Windows 10 renamed the product Windows Defender Firewall and Windows Defender Firewall with Advanced Security, as shown in Figure 5.25. Both of these configure the same firewall service, with the latter of the two allowing for much more granular control. The inner workings are almost identical to those of the original firewall that shipped with Windows XP SP2.

The Windows firewall has achieved its original purpose of protecting the operating system from malicious worms and malicious inbound network connections. By default, the outbound network traffic is allowed and inbound network traffic is blocked, unless a rule exists, as shown in Figure 5.26.

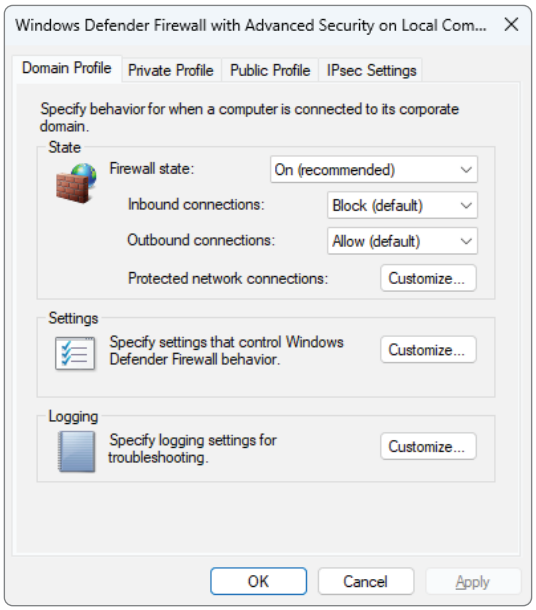
FIGURE 5.25 Windows Defender Firewall with Advanced Security on Windows 11



You will also notice that there are three different profiles: Domain, Private, and Public. When the network service starts up, it contacts the default gateway (router) and configures itself to a profile. This allows the operating system to be location-aware and protect itself differently based on your location. In prior versions of Windows, if the router has never been seen before, then you'll get prompted with a dialog box asking you to choose if you want to allow your PC to be discoverable. If you answer Yes, then the firewall profile will be configured as Private, and any rules associated with the Private Profile will be active. If you answer No, then any rules associated with the Public Profile will be active. In Windows 11, the location profile will automatically default to the Public Profile. If you want to change the profile, you can navigate to Start menu ➤ Settings ➤ Network & Internet ➤ Properties of the current connection, then Public Network or Private Network. The Domain Profile is automatically selected if the network is the corporate network and the operating system is joined to the domain.

Firewalls are also built into other operating systems, including Linux and macOS. Depending on the distribution of Linux or macOS, the firewall included will vary, as well as the way you would configure it. However, most distributions of Linux, such as Ubuntu and Debian, come with the iptables firewall installed. Fedora Linux comes with `firewalld`, which also supports location-based firewall rules.

FIGURE 5.26 Windows Defender Firewall with Advanced Security defaults



Regardless of which type of firewall an operating system comes preinstalled with, a third-party firewall can be installed. These firewalls can offer intrusion-detection capabilities that alert you when someone is attacking. In almost all cases, the firewall that comes with the operating system is more than adequate, but you have to keep it on at all times to prevent unwanted connections.

Reinstalling the OS

When you are compromised by a virus or other type of malware, the only way to be sure you have removed it completely is to reinstall the operating system. This may seem like an extreme measure, but virus researchers do not always know what the threat agent embeds in the operating system. The threat agent's mission is to gain access to your operating system and to keep a persistent connection. If that means opening a few other backdoors, then that is what they will embed in their malware.

Fortunately, the Windows operating system makes it easy to reinstall the operating system. The Windows operating system allows you to reset the PC with the use of the recovery console or from the Settings app. Many devices also have a recovery tool embedded in the browser to factory-reset/reimage the device.

Exam Objective 2.5, "Compare and contrast common social engineering attacks, threats, and vulnerabilities," requires you to know the types of social engineering. You should be familiar with all of the different social engineering types, threats, vulnerabilities, and the tactics a threat actor will take to compromise your organization with social engineering.

Social Engineering Attacks, Threats, and Vulnerabilities

Social engineering is a process in which an attacker attempts to acquire information about your network and system by social means, such as talking to people in the organization. A social engineering attack may occur over the phone, by email, or in person. The intent is to acquire access information, such as user IDs and passwords. When the attempt is made through email or instant messaging, it is known as *phishing* (discussed later), and it's often made to look as if a message is coming from sites where users are likely to have accounts. (Banks, bills, and credit cards are popular.)

These are relatively low-tech attacks and are more akin to con jobs. Take the following example: Your help desk gets a call at 4:00 a.m. from someone purporting to be the vice president of your company. They tell the help desk personnel that they are out of town to attend a meeting, their computer just failed, and they are sitting in a FedEx office trying to get a file from their desktop computer back at the office. They can't seem to remember their password and user ID. They tell the help desk representative that they need access to the information right away or the company could lose millions of dollars. Your help desk rep knows how important this meeting is and gives the user ID and password over the phone. At this point, the attacker has just successfully socially engineered an ID and password that can be used for an attack by impersonating a high-profile person.

Another common approach is initiated by a phone call or email from someone who pretends to be your software vendor, telling you that they have a critical fix that must be installed on your computer system. It may state that if this patch isn't installed right away, your system will crash and you'll lose all your data. For some reason, you've changed your maintenance account password, and they can't log in. Your system operator gives the password to the person. You've been hit again.

In Exercise 5.2, you'll test your users to determine the likelihood of a social engineering attack. The steps are suggestions for tests; you may need to modify them slightly to be appropriate at your workplace. Before proceeding, make certain that your manager knows that you're conducting such a test and approves of it.

EXERCISE 5.2

Testing Social Engineering

1. Call the receptionist from an outside line when the sales manager is at lunch. Tell the receptionist that you're a new salesperson, that you didn't write down the username and password the sales manager gave you last week, and that you need to get a file from the email system for a presentation tomorrow. Does the receptionist direct you to the appropriate person or attempt to help you retrieve the file?

2. Call the human resources department from an outside line. Don't give your real name but instead say that you're a vendor who has been working with this company for years. You'd like a copy of the employee phone list to be emailed to you, if possible. Do they agree to send you the list, which would contain information that could be used to try to guess usernames and passwords?
3. Pick a user at random. Call them and identify yourself as someone who works with the company. Tell them that you're supposed to have some new software ready for them by next week and that you need to know their password to finish configuring it. Do they do the right thing?

The best defense against any social engineering attack is education. Make certain that the employees of your company know how to react to the requests presented here. Social engineering works on the premise that people try to help when they are vested in your efforts, such as a co-worker or if you are trying to help them.

Phishing

Phishing is a form of social engineering in which you ask someone for a piece of information that you are missing by making it look as if it is a legitimate request. An email might look as if it is from a bank and contain some basic information, such as the user's name. These types of messages often state that there is a problem with the person's account or access privileges. The person will be told to click a link to correct the problem. After they click the link, which goes to a site other than the bank's, they are asked for their username, password, account information, and so on. The person instigating the phishing attack can then use this information to access the legitimate account.

One of the best countermeasures to phishing is to mouse over the Click Here link and read the URL. Almost every time, the URL is an adaptation of the legitimate URL, as opposed to a link to the real thing.

The only preventive measure in dealing with social engineering attacks is to educate your users and staff never to give out passwords and user IDs over the phone or via email or to anyone who isn't positively verified as being who they say they are.

Vishing

When phishing is combined with *Voice Over IP (VoIP)*, it becomes known as *vishing*, which is just an elevated form of social engineering. While crank calls have existed since the invention of the telephone, the rise in VoIP now makes it possible for someone to call you from almost anywhere in the world, without the worry of tracing, caller ID, and other features of landlines, and pretend to be someone they are not in order to get data from you.

Smishing

Smishing is the same thing as vishing and phishing, except they text it to your phone instead with a Short Message Service (SMS) code. The message typically has an emotional appeal, urgency, and call to action. I currently get about 10 texts a day that range from saying my packages can't be delivered and I must call, my online services are being shut off for nonpayment, or the government has decided to give me money! In fact, I looked at my phone while writing this and found this message:

Netflix: We couldn't process your monthly payment, so your account is temporarily on hold. Please update your billing information here: `https://Netflix-Canada-update.com`.

Typically the threat actors will want iTunes gift cards and these can then be turned into money. That or the threat actors really dig music, but I think the motivation is money.

Quishing

QR code phishing, also known as *quishing*, is a sophisticated social engineering tactic where cybercriminals use QR codes to lure victims into scanning them with their mobile devices. Once scanned, these QR codes redirect users to malicious websites designed to steal sensitive information, such as login credentials or financial data. Attackers often embed these QR codes in seemingly legitimate emails or PDF documents, leveraging brand impersonation and urgency to trick users into scanning them. This method is particularly effective because traditional email filters struggle to detect QR codes, and the use of multiple devices, such as scanning with a personal phone can bypass corporate security measures. As a result, quishing poses a unique challenge for both individuals and organizations in maintaining cybersecurity. Don't scan random QR codes on gas station pumps when you are pumping gas; they might be malicious in nature.

Spear Phishing

With spear phishing, the attacker uses information that the target would be less likely to question because it appears to be coming from a trusted source. Suppose, for example, that you receive a message that appears to be from your spouse that says to click here to see that video of your children from last Christmas. Because it appears far more likely to be a legitimate message, it cuts through your standard defenses like a spear, and the likelihood that you would click this link is higher. Generating the attack requires much more work on the part of the attacker, and it often involves using information from contact lists, friend lists from social media sites, and so on.

Trust is a key issue with security. There are trusted software sources that you know and work with all the time (such as Microsoft), and there are untrusted sources—and you should differentiate between them. Likewise, there are trusted sites and untrusted sites, and you should similarly differentiate between them. Let common sense be your guide.

Whaling

Whaling is nothing more than phishing, or spear phishing, for so-called "big" users—thus, the reference to the ocean's largest creatures. Instead of sending out a To Whom It May Concern message to thousands of users, the whaler identifies one person from whom they can gain all the data that they want—usually a manager or business owner—and targets the phishing campaign at them.

Shoulder Surfing

Another form of social engineering is known as *shoulder surfing*. It involves nothing more than watching someone when they enter their sensitive data. They can see you entering a password, typing in a credit card number, or entering any other pertinent information. A privacy filter can be used to block people from looking at your screen from an angle. However, privacy filters do not protect you as you are entering a password, since a shoulder surfer will watch your keystrokes. The best defense against this type of attack is to survey your environment before entering personal data. It is also proper etiquette to look away when someone is entering their password.

Tailgating

Tailgating is another form of social engineering, and it works because we want to be helpful. Tailgating is the act of entering a building that requires a swipe card or other authentication factor by using the person in front of you.

You may be walking toward an entry that requires some authentication, when someone walking the same way introduces themselves as new to the company and shares some stories about their first day. By the time you get to the door, you may hold it open for them and wish them luck. It can even happen without you knowing it, if the door barely closes and they grasp it. Several different tactics, such as access control vestibules and guards, can be used to mitigate this threat. The best prevention is education of your staff to make sure that it does not happen.

Impersonation

Impersonation is prevalent in many different social engineering attacks. The threat agent portrays (impersonates) another employee for many of the attacks to work. Most employees want to help another fellow employee. The threat agent might be impersonating Bill Jones from accounting asking IT to reset his password. Or the threat agent might impersonate the IT department calling Bill Jones and instructing him to type some well-crafted commands into his computer. Many phishing emails also impersonate your bank, an online store, or some other reputable source in an attempt to steal your credentials. The best method for combatting impersonation is end-user training. Training should help users identify suspicious email or phone calls.

Dumpster Diving

Dumpster diving is the act of a person rifling through the trash with the expectation to find information. A strong policy to prevent dumpster diving is the physical destruction of any sensitive data. Destruction can be performed with the use of a mechanical shredder on-site or a service that destroys materials off-site on behalf of the organization. If an off-site shredding service is performed, always request a signed certificate of destruction to prove that sensitive material was destroyed.

Common Security Threats

A *threat* is a potential danger to the network or the assets of the organization. The potential danger to a network or organization is the attack that a threat agent can carry out. All attacks upon an organization are either technology based or physically based. A technology-based attack is one in which the network and operating systems are used against the organization in a negative way. Physically based attacks use human interaction or physical access, which we previously covered as social engineering attacks. We will now cover several different types of technology-based attacks that are commonly used against networks and organizations.

Denial-of-Service Attacks

A *denial-of-service (DoS)* is an attack launched to disrupt the service or services a company receives or provides via the Internet. A DoS attack is executed with an extremely large number of false requests; because of the attack, the servers will not be able to fulfill valid requests for clients and employees. There are several different types of DoS attacks:

Reflective A reflective DoS is not a direct attack; it requires a third party that will inadvertently execute the DoS. The attacker will send a request to a third-party server and forge the source address of the packet with the victim's IP address. When the third party responds, it responds back to the victim. There are two victims in this type of DoS attack: the first is the victim the attack is aimed at, and the second is the third-party server used to carry out the attack. Figure 5.27 shows an ICMP-based smurf attack, which is a reflective attack.

Amplified An amplified DoS is a variant of a reflective DoS attack. It is carried out by making a small request to the third-party server that yields a larger response to the victim. The most common third-party servers used to carry out the attack are DNS and NTP. For example, an attacker will request a DNS query for a single hostname that contains 20 aliases while forging the source IP address. The victim is then barraged with the 20 answers from the query, as shown in Figure 5.28.

FIGURE 5.27 An ICMP-based smurf attack

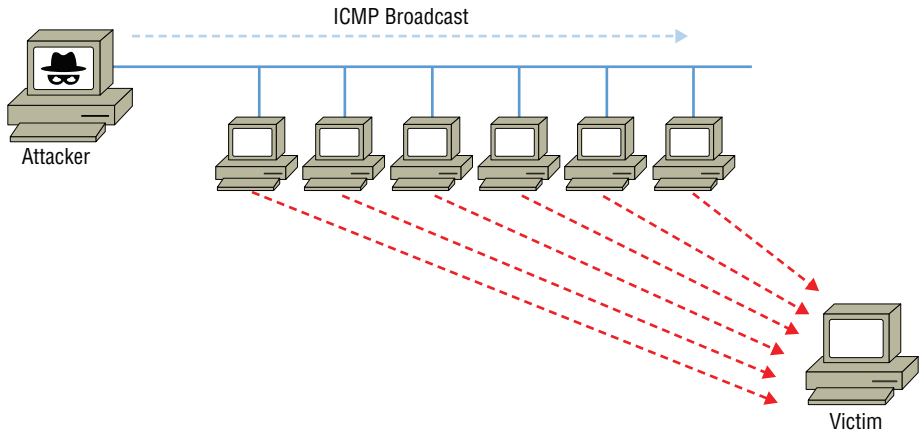
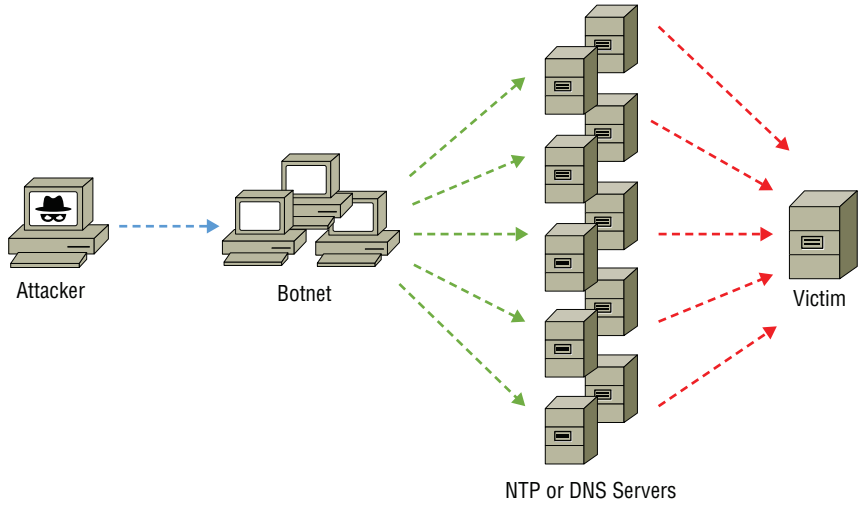
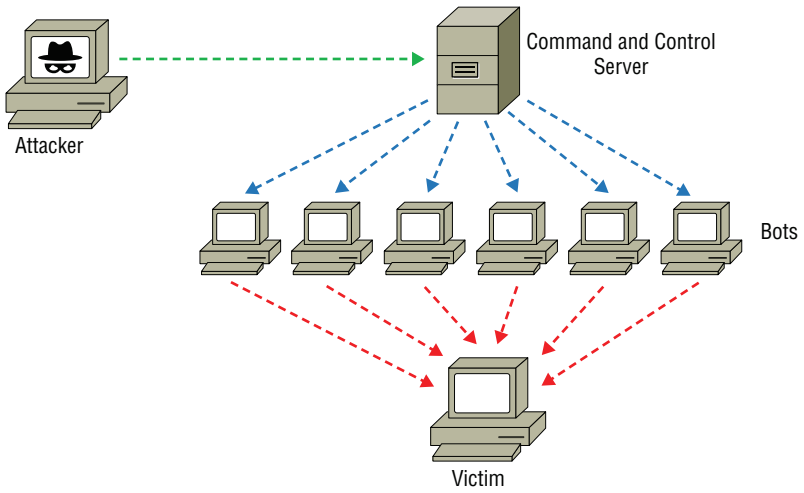


FIGURE 5.28 An amplified attack



Distributed A distributed denial-of-service (DDoS) has become the most common type of DoS, because the source of the DoS is varied. It is common for botnets to launch DDoS attacks on organizations. When a single host is used to create a DoS, it can simply be blocked. However, when traffic is coming from millions of different hosts, it is impossible to isolate the DoS and firewall the source. Figure 5.29 shows a DDoS attack.

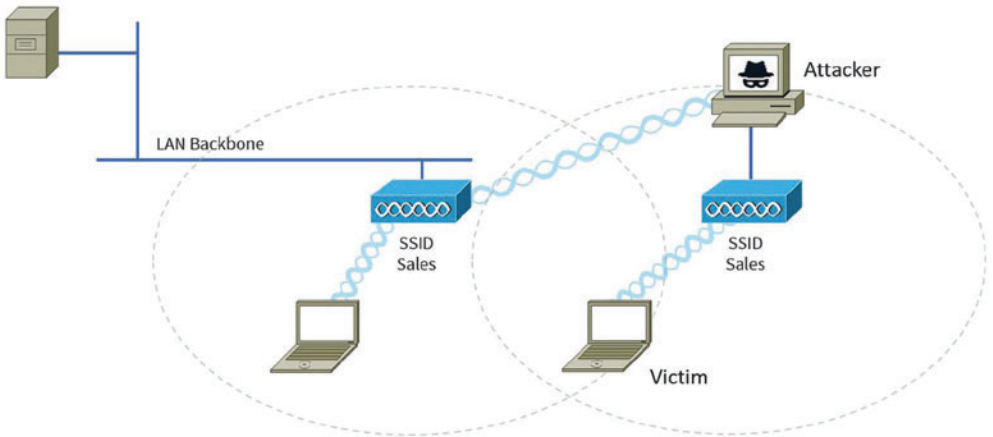
FIGURE 5.29 A DDoS attack



Evil Twin

An *evil twin* attack is a wireless phishing attack in which the attacker sets up a wireless access point to mimic the organization's wireless access points. When a user connects to the evil twin, it allows the attacker to listen in on the user's traffic. Evil twin access points often report a stronger signal to entice the user to connect to the specific access point, as shown in Figure 5.30. The attacker will then create a connection back to the wireless network and passively sniff network traffic as it routes the traffic to the original destination. The best way to mitigate against evil twin attacks is to perform wireless site surveys on a regular basis to ensure that only valid access points are being used.

FIGURE 5.30 Evil twin attack



Zero-Day Attacks

When a hole (vulnerability) is found in a web browser or other software, and attackers begin exploiting it the very day it is discovered by the developer (bypassing the one-to-two-day response time that many software providers need to put out a patch once the hole has been found), it is known as a *zero-day* attack (or exploit). It is very difficult to respond to a zero-day exploit. If attackers learn of the weakness the same day as the developer, then they have the ability to exploit it until a patch is released. Often, the only thing that you as a security administrator can do, between the discovery of the exploit and the release of the patch, is to turn off the service. You can do this by isolating or disconnecting the system(s) from the network until a patch is released. Although this can be a costly undertaking in terms of productivity, it is the only way to keep the network safe.

Several years ago, Stuxnet was found to be using a total of four zero-day vulnerabilities to spread from host to host. You can learn more about Stuxnet here:

`www.kaspersky.com/resource-center/definitions/what-is-stuxnet`

Spoofing Attacks

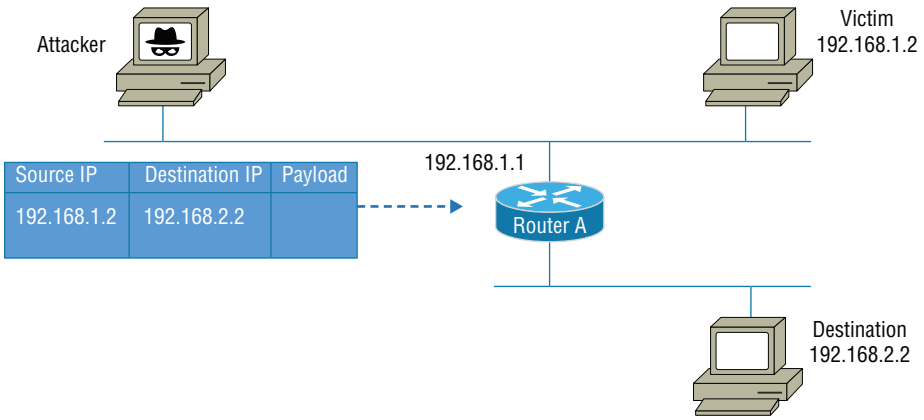
A *spoofing attack* is an attempt by someone or something to masquerade as someone else. This type of attack is usually considered an access attack. A common spoofing attack that was popular for many years on early UNIX and other timesharing systems involved a programmer writing a fake login program. It would prompt the user for a user ID and password. No matter what the user typed, the program would indicate an invalid login attempt and then transfer control to the real login program. The spoofing program would write the login and password into a disk file, which was retrieved later.

The most popular spoofing attacks today are IP spoofing, ARP spoofing, and DNS spoofing. With IP spoofing, the goal is to make the data look as though it came from a trusted host when it didn't (thus spoofing the IP address of the sending host), as shown in Figure 5.31. The threat agent will forge their packet with the victim's source address.

With *ARP spoofing* (also known as *ARP poisoning*), the media access control (MAC) address of the data is faked. By faking this value, it is possible to make it look as though the data came from a networked device that it did not come from. This can be used to gain access to the network, to fool the router into sending to the device data that was intended for another host, or to launch a DoS attack. In all cases, the address being faked is an address of a legitimate user, making it possible to get around such measures as allow/deny lists.

With *DNS spoofing*, the DNS server is given information about a name server that it thinks is legitimate when it isn't. This can send users to a website other than the one to which they wanted to go, reroute mail, or do any other type of redirection for which data from a DNS server is used to determine a destination. Another name for this is *DNS poisoning*.

FIGURE 5.31 IP address spoofing attack



Always think of spoofing as fooling. Attackers are trying to fool the user, system, and/or host into believing that they're something they are not. Because the word *spoof* can describe any false information at any level, spoofing can occur at any level of a network.

The important point to remember is that a spoofing attack tricks something or someone into thinking that something legitimate is occurring.

On-Path Attack

Many of the attacks we're discussing can be used in conjunction with an *on-path attack*, which was previously known as a man-in-the-middle (MitM) attack. For example, the evil twin attack mentioned earlier allows the attacker to position themselves between the compromised user and the destination server. The attacker can then eavesdrop on a conversation and possibly change information contained in the conversation. Conventional on-path attacks allow the attacker to impersonate both parties involved in a network conversation. This allows the attacker to eavesdrop and manipulate the conversation without either party knowing. The attacker can then relay requests to the server as the originating host attempts to communicate on the intended path, as shown in Figure 5.32.

Password Attacks

Password attacks occur when an account is attacked repeatedly. This is accomplished by using applications known as password crackers, which send possible passwords to the account in a systematic manner. The attacks are initially carried out to gain passwords for an access or modification attack. There are several types of password attacks:

Brute-Force Attacks A *brute-force* attack is an attempt to guess passwords until a successful guess occurs. As an example of this type of attack, imagine starting to guess with

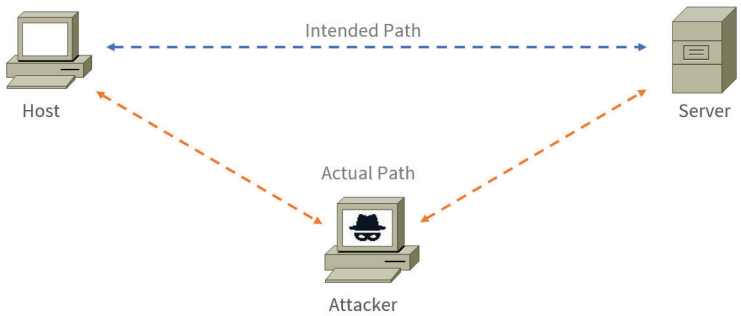
A and then going through *z*; when no match is found, the next guess series goes from *AA* to *zz*, and then it adds a third value (*AAA* to *zzz*). Because of the nature of this routine, this type of attack usually occurs over a long period of time. To make passwords more difficult to guess, they should be much longer than a few characters. It is recommended that you make the password at least 12 digits and complex. You should also have password lockout policies.

Dictionary Attacks A *dictionary attack* uses a dictionary of common words to attempt to find the user's password. Dictionary attacks can be automated, and several tools exist in the public domain to execute them. As an example of this type of attack, imagine guessing words and word combinations found in a standard English-language dictionary.

Rainbow Tables When passwords are stored in the operating system, they are stored in their hashed format. A hash is a one-way cryptographic algorithm. If you have access to the password hash, a rainbow table can be used to find the password. Rainbow tables are databases that have had every permutation run through a hashing algorithm, so the hash can be looked up and cross-referenced back to the password that created it. These databases can be really large. It is not uncommon for a rainbow table to ship on a portable hard drive.

Hybrid Attacks A *hybrid attack* typically uses a combination of dictionary entries and brute force. For example, if you know that there is a good likelihood that the employees of a particular company are using derivatives of the company name in their passwords, then you can seed those values into the values attempted.

FIGURE 5.32 On-path attack



Insider Threat

Insider threats are threats that originate from within your organization. Employees know the organization and can navigate the organization to get the information they need. A disgruntled employee can carry out an attack on the organization by leaking information or

selling it. When information is sold to a competitor for profit, it is considered corporate espionage. The inside threat does not always need to be criminal in intent. It can also be as simple as an employee plugging an unauthorized wireless access point into the corporate network.

SQL Injection

A *Structured Query Language (SQL) injection* attack occurs when a threat agent enters a series of escape codes along with a well-crafted SQL statement into a URL. The seamlessly harmless page on the backend that is awaiting the request runs the SQL query along with its normal query. For example, a normal post URL might look like this:

```
http://www.wiley.com/phone.php?name=jones
```

The threat agent will add their SQL injection after the normal post query string, such as the following:

```
http://www.wiley.com/phone.php?name=jones; DROP TABLE Users
```

This would generate the following SQL query on the backend and send the malicious query to the SQL database:

```
SELECT FullName, PhoneNum
From Phones
Where FullName Like '%jones%'; DROP TABLE USERS
```

The first two and half lines to the semicolon are generated by the page the query is posted to. The line basically tells the SQL database to return the full name and phone number for anything that contains jones. However, the threat agent appended `DROP TABLE Users` to the query with a semicolon. This will delete the Users table and cause disruption. Technically the SQL injection causes a DoS attack. Other malicious queries that are not so obvious and disruptive can be submitted to discover information like table structure and consequently steal data. Many retailers, banks, and online stores, just to name a few, have fallen prey to SQL injection attacks and made front-page news.

The best way to combat this attack is by building input validation into the rendered page on the backend; this is also known as *sanitization*. This mitigation tactic is well outside the scope of the exam, but understanding the attack is the key takeaway.

Cross-Site Scripting (XSS)

Cross-site scripting (XSS) is a tactic a threat agent uses to deliver a malicious script to the victim by embedding it into a legitimate web page. Common delivery methods for XSS are message boards, forums, or any page that allows comments to be posted. The threat agent will submit a post to these types of pages with their malicious script, such as JavaScript. When the victim browses the page, the threat agent's script will execute.

JavaScript and other scripting languages are controlled tightly by the browser, so direct access to the operating system is usually not permitted. However, the script will have access

to the web page you are browsing or the cookies the actual page stores. This type of attack is common in hijacking web pages and trying to force the user into installing a piece of malware.

Business Email Compromise (BEC)

Business email compromise (BEC) is a form of cybercrime that targets organizations by exploiting them through email. Threat actors either use compromised legitimate business email accounts, or create convincing spoofed accounts to deceive employees, partners, or clients into transferring funds or sensitive information. This type of attack relies heavily on social engineering tactics and detailed research to make fraudulent requests appear authentic. The components of a BEC attack are as follows:

Research and Targeting Threat actors begin by gathering information about the target organization. They use publicly available data from company websites, social media, and other online sources to identify key individuals, such as executives or finance employees.

Email Compromise or Spoofing The threat actor either compromises the email accounts of these key employees through phishing or other means. If the threat actor cannot compromise a legitimate account, they will create spoofed email addresses that closely resemble a legitimate account.

Crafting Deceptive Emails Using the compromised or spoofed accounts, attackers send carefully crafted emails that appear to come from trusted sources. These emails often contain urgent requests for wire transfers, sensitive data, or changes to payment details.

Exploiting Trust and Urgency The emails are designed to exploit the recipient's trust in the sender and create a sense of urgency. This makes it difficult for the recipient to recognize the fraud.

The consequences of BEC attacks can be severe financially for the organization. Financial losses can sometimes amount to millions of dollars, depending on the extent of the compromise. Additionally, the breach of trust can damage business relationships and the organization's reputation.

Supply Chain/Pipeline Attacks

A supply chain attack, also known as a pipeline attack, targets the tight integration of today's modern business operations. These attacks exploit vulnerabilities in third-party vendors or suppliers to gain access to a primary target's network. By compromising a trusted partner, the threat actor can bypass any security measures that the primary target might have in place.

One of the most infamous examples is the SolarWinds attack in 2020. Attackers inserted malicious code into the software updates for a product called Orion, which was used for IT management. When customers installed the updates, they unknowingly introduced a

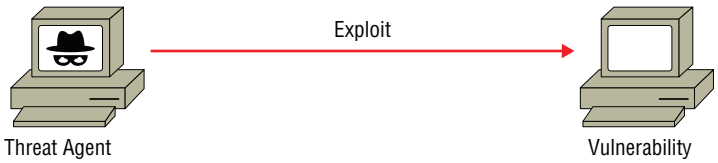
backdoor into their systems. This allowed the attackers to conduct espionage and data theft on a massive scale. Many customers of the SolarWinds software were government agencies and Fortune 500 companies.

Supply chain/pipeline attacks are effective for threat actors because they can leverage the trust and access granted to third-party vendors. This makes detection and mitigation challenging, since the malicious activity often appears legitimate. To combat these threats, organizations must implement stringent security measures, such as thorough vetting of suppliers, continuous monitoring of third-party activities, and adopting a zero-trust approach to network access.

Exploits and Vulnerabilities

Exploits and vulnerabilities both have the same effect of compromising systems. Vulnerabilities are weaknesses in security for an operating system or network product. Vulnerabilities are the reason we need to constantly patch network systems. Exploits are scripts, code, applications, or techniques used in exploiting the vulnerabilities by a threat agent, as shown in Figure 5.33. In the following section we will cover the most common vulnerabilities as they pertain to the CompTIA exam.

FIGURE 5.33 Threat agents, exploits, and vulnerabilities



Noncompliant Systems

One of the easiest ways to make your systems vulnerable and expose them to threats is to fail to keep them compliant. As an administrator, you should always follow security regulatory standards as well as compliance standards.

One product that can keep your operating systems compliant is Microsoft Endpoint Configuration Manager (MECM). MECM allows for the publishing of a baseline for the Windows operating system. It will then monitor the baseline against the operating systems in your organization and will remediate them if they fall out of compliance.

MECM is just one of many tools that can be used for compliance, several of which are third-party tools. Third-party compliance solutions provide other unique benefits, such as the compliance of third-party applications in addition to Windows.

Patching and Updates

When operating systems are installed, they are usually point-in-time snapshots of the current build of the operating system. From the time of the build to the time of installation, several vulnerabilities can be published for the operating system. When an OS is installed, you should patch it before placing it into service. Patches remediate the vulnerabilities found in the OS and fixed by the vendor. Updates add new features not included with the current build. However, some vendors may include vulnerability patches in updates. Network devices also have patches and updates that should be installed prior to placing them into service.

After the initial installation of the device or operating system and the initial patches and updates are installed, you are not done! Vendors continually release patches and updates to improve security and functionality, usually every month and sometimes outside of the normal release cycle. When patches are released outside of the normal release cycle, they are called out-of-band patches and are often in response to a critical vulnerability.

Microsoft products are patched and updated through the Windows Update functionality of the operating system. However, when an administrator is required to patch and update an entire network, Windows Server Update Services (WSUS) can be implemented. A WSUS server enables the administrator to centrally manage patches and updates. The administrator can also report on which systems still need to be patched or updated.

Microsoft had officially announced the Windows Server Update Services (WSUS) is now deprecated. No new features will be developed for the product, but the feature will still function as designed. In the future this will leave a void for administrators that need internal patch management. Ultimately third-party patching products will become popular again.

When endpoints are Azure joined, Intune can provide control over update policies, allowing administrators to schedule updates, enforce compliance, and monitor the update status of all managed devices. By integrating Microsoft Intune with Windows Update for Business, organizations can take advantage of features like Delivery Optimization and Windows Update rings, which can optimize the update process and minimize disruption to end users.

Operating Systems' Life Cycle

All operating systems have a life cycle of release, support, and eventually end-of-life (EOL). When most operating systems reach their EOL, the vendor stops supplying security patches. The void of the most current patches creates a giant vulnerability for the organization, since the operating system is no longer protected from the latest vulnerabilities. To combat these vulnerabilities, keep the operating system current. This is accomplished with continual upgrades to the operating system as new versions are released.

Unprotected Systems

Obviously, an unpatched system presents a huge vulnerability in your network, but an unprotected system is just as vulnerable. A workstation or server without antivirus protection or firewall protection poses a significant risk. A workstation or server without antivirus software can contract malware and potentially infect other computers and ultimately leak data. A missing or misconfigured firewall is equally as vulnerable. The firewall is used to prevent unauthorized connections that could exploit a vulnerability in the operating system. Firewalls are also often used in place of an undesirable patch that causes other issues. For example, turning on a firewall rule that prevents connecting to the Print Spooler service on Windows is a good protection method. It can shield you from print spooler vulnerabilities if the workstation or server is not providing print services. However, if the operating system is not protected by the firewall and an exploit is released, it is consequentially vulnerable.

BYOD

Security is the biggest concern as it applies to BYOD devices. The biggest reason is that the organization has less control over BYOD devices than over devices it issues and owns. BYOD devices come with two inherent risks: data leakage and data portability. Data leakage happens when a device is lost or compromised in some way. There are tactics to mitigate this, such as full device encryption. However, the user's device is then forcefully encrypted by the organization and there could be legal ramifications. Another common tactic is to use mobile device management (MDM) software that creates a partition for company data. This would allow the company to encrypt their data and not affect user data.

Data portability means that the user can cart away organizational data when they leave. Although most of the time this is not a risk, an unscrupulous salesperson may be a big risk to the organization. A line-of-business (LOB) application should be selected that displays only the data on a mobile device and does not allow data storage. Another tactic is to employ MDM software that allows remote wiping of the organization's data. When an employee leaves, the wipe is executed and the organization's data is gone. This type of functionality is also useful if a device is lost, so it also mitigates the risk of data leakage.

Exam Objective 2.7, "Given a scenario, apply workstation security options and hardening techniques," requires you to know data encryption, password considerations, and end-user best practices and account management.

Security Best Practices

A best practice is a technique or methodology that has largely been adopted by the professional community. A number of security best practices are techniques that can keep the

organization safe and secure. Many of the best practices will mitigate some of the risk from attacks that I previously discussed. The benefits of other best practices might not be immediately apparent, but they will keep an organization safe in the long run.

Data Encryption

To prevent the loss of data, data encryption should be considered—not that the data is really ever lost, but it's no longer within your control. Consider an example of a laptop with sensitive patient record information stored on it. If the laptop were to be stolen, there are a number of utilities that could provide unauthorized access. However, with encryption (such as BitLocker) enabled, both the operating system and the data would remain encrypted and inaccessible.

Three concepts are associated with data encryption: data in use, data in transit, and data at rest, as shown in Figure 5.34. *Data in use* is the concept of data that is in an inconsistent state and/or currently resident in memory. Most of the time, you don't need to be too concerned with data in memory, since that is normally a function of the operating system. However, when data is written to a temporary location, it is considered data in use and therefore should be encrypted. *Data in transit* is information traversing the network and should always be encrypted so that it is not intercepted. Over the past decade, just about every website and application has adopted some form of encryption, so there is no reason not to use encryption in transit.

FIGURE 5.34 Data and encryption

Data at rest is a point of contention because it is believed that once the data hits the server, it's safe. However, the data is more vulnerable because it's in one spot. If a drive needed to be replaced because it went bad, outside of physical destruction there is no way to assure the data is inaccessible. When used for backup tapes, it is not only a good idea but should be a requirement.

Setting Strong Passwords

One of the most effective ways to keep a system safe is to employ strong passwords and educate your users about their best practices. Many password-based systems use a one-way hashing approach. You can't reverse the hash value in order to guess the password. This makes it impossible to reverse the hashes if the database of stored passwords is lifted (stolen) from the

operating system. Because the hash is sent over the network in lieu of the actual password, the password is harder to crack.

Passwords should be as long and complex as possible. The current recommendation by the National Institute of Standards and Technology (NIST) is a password should be at least 15 characters. Keep in mind, you should use 20 or more characters if security is a real concern. If you use only the lowercase letters of the alphabet, you have 26 characters with which to work. If you add the numeric values 0–9, you get another 10 characters. Adding uppercase letters, you gain an additional 26 characters. If you go one step further by using symbol characters (such as `!"#$%&'()*+,-./:;<=>?@[\]^_Ë‹{|}` and ~, including a blank space), you have an additional 33 more characters. You then have a pallet of 95 characters for each position in your password. A typical example of a complex password using all these elements might be `%s4@7dFs#D2$`. If you have a hard time coming up with a strong password on your own, you can always use an online password generator, such as `https://passwords generator.net`.

When the *Password Complexity* policy in Group Policy is enabled for the Windows operating system, three of the four categories—lowercase, uppercase, numbers, and symbols—must be used in your password. Windows Server and/or Active Directory can also require a minimum password size, which guarantees a secure password when coupled with complexity.

Most vendors recommend that you use symbols in your password, and some go so far as to require it. Some symbols, however, do not work for web-based applications. Therefore, you may find that a vendor accepts a reduced set of characters. Although this reduces the complexity, using characters in addition to letters and numbers makes the password more secure.

Let's look further at password complexity. If you used a 4-character password, this would be 95 × 95 × 95 × 95, or approximately 81.5 million possibilities. A 5-character password would give you 95^5, or approximately 7.7 billion possibilities. And a 10-character password would give you 95^{10}, or 5.9×10^{18} (a *very* big number) possibilities. As you can see, these numbers increase exponentially with each position added to the password. A 4-digit password could probably be broken in a fraction of a day, whereas a 10-digit password would take considerably longer and much more processing power.

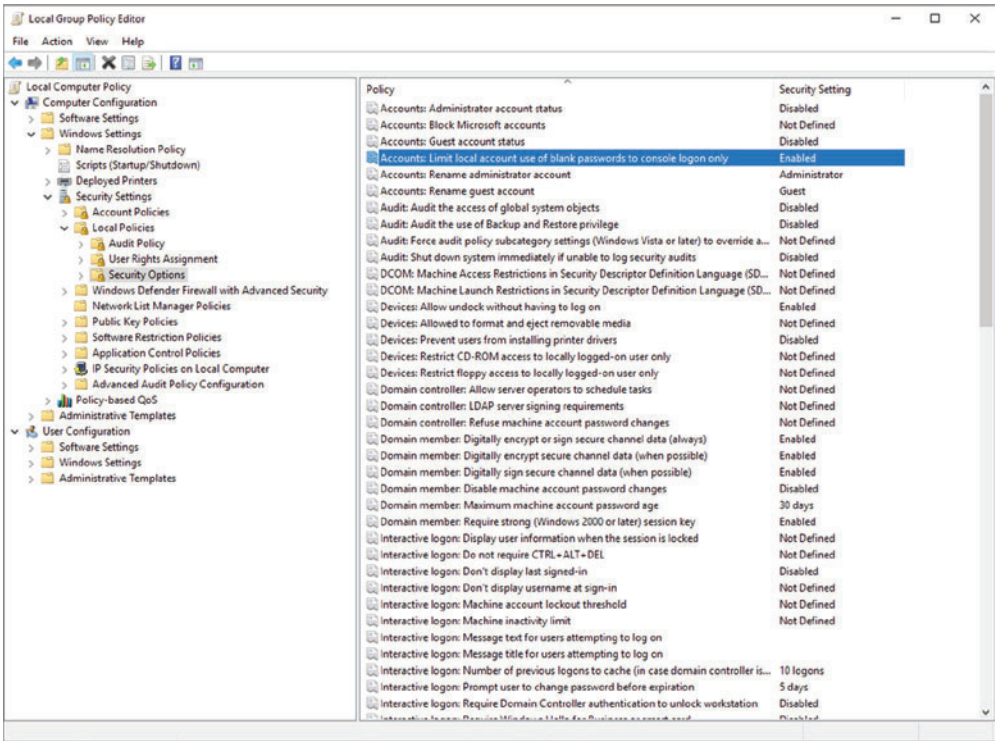
If your password consisted of only the 26 lowercase letters, the 4-digit password would have 26^4, or 456,000 combinations. A 5-character password would have 26^5, or over 11 million combinations, and a 10-character password would have 26^{10}, or 1.4×10^{14} combinations. The number of combinations is still a big number, but it would take considerably less time to break it compared to a longer password. This is all based on the notion that a brute-force password attack is being performed. If a dictionary attack were being performed, a 4- or 5-digit lowercase password could take less than 5 minutes to crack.

Mathematical methods of encryption are primarily used in conjunction with other encryption methods as part of authenticity verification. The message and the hashed value of the message can be encrypted using other processes. In this way, you know that the message is secure and hasn't been altered.

Requiring Passwords

Make absolutely certain that you require passwords for all accounts. It's such a simple thing to overlook in a small network, but it's not something a malicious user will overlook. By default, Windows will not allow an account to connect over the network if it has a blank password. It will, however, allow a person to log in locally with a blank password. There is a security option in the local Group Policy that specifies this behavior, as shown in Figure 5.35.

FIGURE 5.35 Windows Group Policy security options



The operating system is not the only place where you should use a password for security. You should also use passwords on the *basic input/output system (BIOS)* and *Unified Extensible Firmware Interface (UEFI)* firmware. If a malicious user has access, they could possibly circumvent your security by booting a live operating system.

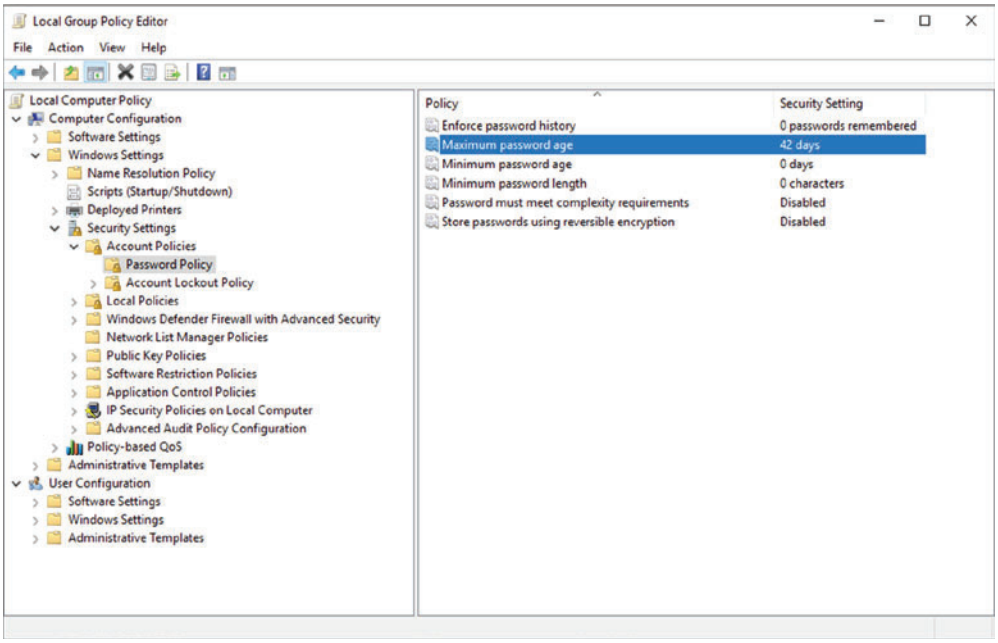
You should also change the *default passwords* on system accounts. There are dedicated sites on the web that document the default username and password for various vendor devices. A common hacker can easily pull up these sites and find the default username and password for your wireless access point, camera system, or any other system on which you've neglected to change the default password.

Password Expiration

Password expiration should be a consideration because passwords can be compromised as time goes on. Whether passwords are compromised by shoulder surfing or keylogging, or intercepted via the network when a user is logging in, the fact remains they are only one factor of authentication. Therefore, passwords should be set to expire on a monthly, bi-monthly, quarterly, semi-annual, or annual basis. The more sensitive an account is, the more frequently the password should be changed.

Windows has a default password expiration of 42 days, as shown in Figure 5.36. You should put in place a system to expire passwords on a periodic basis, as stated previously. You would then communicate this to your users via the onboarding process when they are hired.

FIGURE 5.36 Password expiration



End-User Best Practices

In addition to administrator best practices, there are several different end-user best practices that you should advocate to your users. In the following, we will cover the top end-user best practices covered by the CompTIA exam. However, when it comes to end-user best practices and training, these are just the tip of the iceberg.

Locking Screens

When a user walks away from their computer and leaves themself logged in, anyone who walks up to the computer has the same level of access as the owner of the account. This type of attack requires that the threat agent be physically present. However, leaving a computer logged in also invites insider threats, unauthorized access to information, or even data loss.

Training users to lock their screen when they walk away is the best way to prevent unauthorized access. By simply pressing the Windows key and L, a user can lock their screen as they walk away.

Alternately, the administrator can require a user to use a screen saver lock. For example, the screen saver lock can be set to 15 minutes. After 15 minutes of idle time, the screen saver will turn on. The user will not be able to access the desktop until they enter their password. This setting provides two benefits: First, it provides a visual deterrent to potential threat agents, and second, it prevents threat agents from carrying out an attack.

Logging Off

When users are not utilizing a system, they should be encouraged to log off the system. When users remain logged in, the programs that they were running stay running as well. If there is malware on the system, it will stay running as well, potentially allowing threat agents to carry out attacks.

When a user logs off the operating system, any malware running will terminate and hopefully not launch on next login. Malware that launches on the next login is considered to be persistent. Outside of malware, if the system has a resource that is shared, then having users log off will free the resource for the next person.

The administrator has control at their disposal that allows them to police the user. After a period of time in which the system is idle, the administrator can forcibly log off the user automatically. This is usually performed on shared systems, such as a terminal server that serves applications or virtual desktop infrastructure that serves desktops.

Securing Equipment and Information

It is our job as administrators to protect information, such as personally identifiable information (PII), as well as usernames and passwords. However, we also bestow this responsibility onto our users, since many times they have direct access to information. Users should be trained to identify PII and methods to protect such information. Examples of end-user measures to protect sensitive information can be as simple as controlling printouts, using discretion when viewing information with others around, and destroying sensitive trash, just to name a few.

End users also have portables devices that can contain sensitive data, and these devices should be secured when not in use. Many an organization has made front-page headlines with the loss of a simple laptop containing PII. Locks can be used to secure devices, and most portable devices contain a security slot for a lock. A laptop with sensitive data is not the only device that can be lost or stolen—portable hard drives can also contain PII and thus should be physically controlled, or their use should be prevented.

Password Managers

Password managers offer a solution for managing the thousands of passwords required in today's online environment. These applications securely store and encrypt credentials in a vault. The vault can only be accessed with a master password or biometric authentication. This ensures that even if the device is compromised, the stored passwords remain protected. Most password managers also have the ability to upload the vault entries to the cloud. This allows you the ability to sync vault on any computer you are using.

One of the primary benefits of using a password manager is the ability to generate strong, unique passwords for each account. This practice significantly reduces the risk of credential stuffing attacks, where threat actors use stolen passwords from one site to gain access to other accounts. By having unique passwords for each account, the impact of a single data breach is minimized.

Password managers also provide user convenience by automatically filling in login credentials on websites and applications. This not only saves time, but also reduces the likelihood of phishing attacks. Since the password manager will only autofill credentials on legitimate sites.

Another common feature of password managers is their ability to store other sensitive information, such as credit card details, secure notes, and personal identification numbers (PINs). This provides a single, secure location for all critical data.

The macOS Keychain Access app, and its successor, the Passwords app, was covered in Volume 2 Chapter 4, "Working with macOS and Linux." Although the Windows Credential Manager is not a true password manager, it stores web credentials for applications running on the Windows operating system.

Account Management

Given a security-related scenario, account management can take into consideration such settings as restricting user permissions, setting login time restrictions, disabling the Guest account, locking an account after a certain number of failed attempts, and configuring a screen lock when the system times out after a specified length of inactivity.

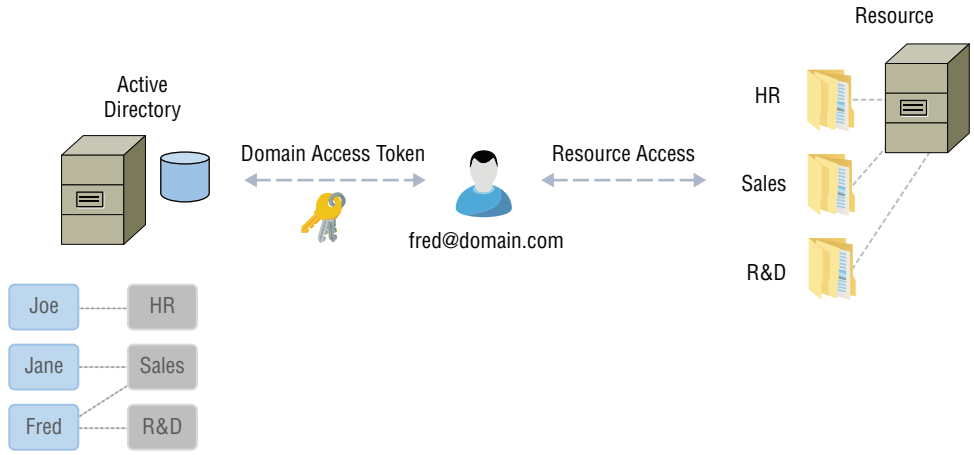
Restricting User Permissions

When assigning user permissions, follow the *principle of least privilege*: give users only the bare minimum that they need to do their job. Assign permissions to groups rather than to users, and make users members of groups (or remove them from groups) as they change roles or positions.

The use of groups is crucial to account management, because when you apply permission on NTFS for the user, you need to visit the resource to identify the permissions granted. When you use a group and apply the NTFS permission to that specific group, you can

then look at either the membership of the group or the membership of the user in Active Directory. This allows you to see who has access to the resource without having to visit the resource. Figure 5.37 shows an oversimplified example. A user, Fred, is a member of both the Sales and R&D groups; therefore, he has access to the Sales and R&D folders. In a real-world application, the group would be more descriptive, such as Perm_RW_Sales_Server1. This naming would describe what the group is used for (permissions), what level of permissions (RW), what resource (Sales share), and on what server (Server1).

FIGURE 5.37 Users, groups, and resources



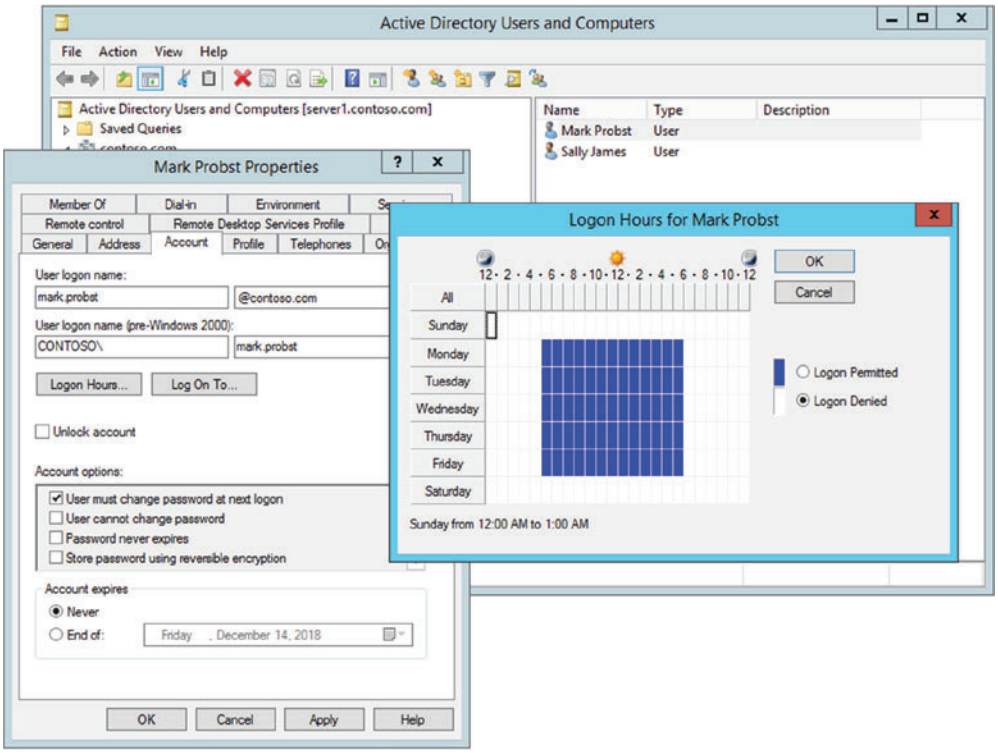
Setting Time Restrictions

Configure user accounts so that logins can occur only during times that the user can be expected to be working. Preventing logins at 2:00 a.m. can be an effective method of keeping hackers from your systems. This can be performed in Active Directory by clicking the user's account, selecting the Account tab, and then clicking Logon Hours, as shown in Figure 5.38. From this interface, you can configure the permitted hours for logins and denied hours for logins.

Account Expiration

You can also set account expiration on the Properties tab of the user. By default, the account is set to never expire. However, you can add a date at which time the account will expire. This is best used on contractor accounts, where the terms of use can be defined. Because contractor accounts often don't go through typical human resource processes, the accounts can be forgotten about. By adding an account expiration, you can be assured that at the end of the contract the account will be disabled (expired).

FIGURE 5.38 Account restrictions



Limiting the Number of Failed Login Attempts

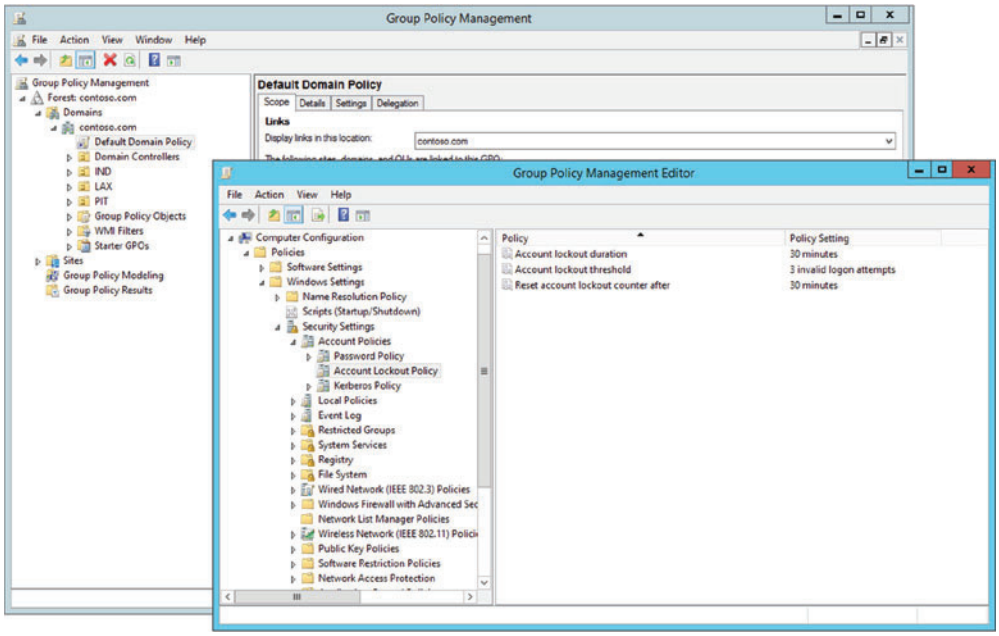
Configure user account settings to limit the number of login attempts before the account is locked for a period of time. Legitimate users who need to get in before the block expires can contact the administrator and explain why they weren't able to give the right password three times in a row, and illegitimate users will go away in search of another system to try to enter.

When choosing the number of failed attempts, you need to consider the number of calls you get to the help desk versus the security in having few failed attempts before lockout. You'll find that when you set it to three failed attempts, the help desk will get more calls than necessary, but it allows for better security. Setting the number of failed login attempts to five may be better for users, because many users realize after the third failed attempt that their Caps Lock key was on, but it's less secure than three failed attempts. This setting needs to be evaluated against your security requirement and help desk volume.

You should also consider the length of the lockout. If it's a Monday morning and a person enters their password wrong X number of times and gets locked out, 5 minutes

might be appropriate. The time it takes to get a cup of coffee and unlock might be just enough time on a Monday morning to allow the user to wake up. You can specify these settings for an entire domain, as shown in Figure 5.39. As shown here, the user will be locked out for 30 minutes after three failed attempts. By default, there is no account lockout policy set for a domain.

FIGURE 5.39 Account lockout policy settings



If a password attack is being executed, no protection could easily allow millions of password attempts. By setting a failed login attempt counter and a lockout duration, you slow the progress of the attack.

Changing Default Usernames

Default accounts represent a huge weakness because everyone knows they exist. When an operating system is installed—whether on a workstation or a server—certain default accounts are created. Knowing the names of those accounts simplifies the process of potential attackers accessing them because they only have to supply the password.

The default username should be changed whenever possible. Several websites are dedicated to documenting the default username and password for routers, switches, and other network equipment. These sites are useful, especially when you lose the documentation for the device and have to *factory-reset* it. A login after you have successfully just factory-reset the device is

really the only time that it's acceptable to use the default username and password for security. You don't need a website to guess the common administrative accounts on equipment or an operating system. They are usually admin, administrator, root, or sysadmin. Changing the default username makes it more challenging for someone to try to guess the credentials.

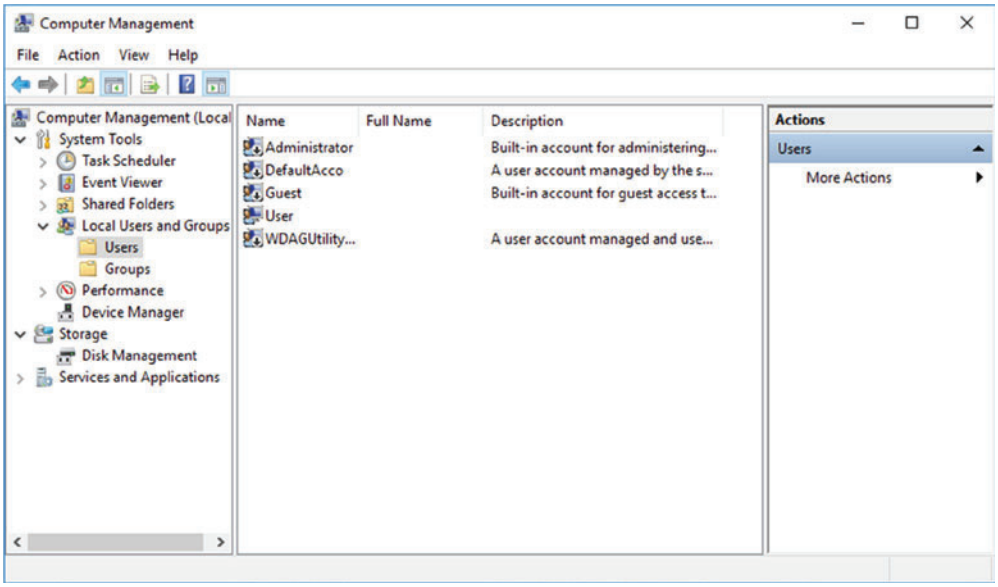
Changing the default password to a complex password is also a good practice in hardening the device. However, changing the username will also ensure that a brute-force attack cannot be performed against the default username. There are many different websites dedicated to listing the default credentials for network devices, so it doesn't take tremendous skill to obtain the default username and password of a device.

Disabling the Guest Account

When Windows is installed, one of the default accounts it creates is Guest. This represents a weakness that can be exploited by an attacker. While the account cannot do much, it can provide initial access to a system, which the attacker can use to find another account or acquire sensitive information about the system.

You should disable all accounts that are not needed, especially the Guest account. Starting with Windows 10, the operating system disables the Guest account by default, as shown in Figure 5.40. After you disable accounts that are not needed, rename the accounts, if you can. (Microsoft won't allow you to rename some.) Finally, change the passwords from the defaults and add them to the list of passwords that routinely get changed.

FIGURE 5.40 Disabled Guest account

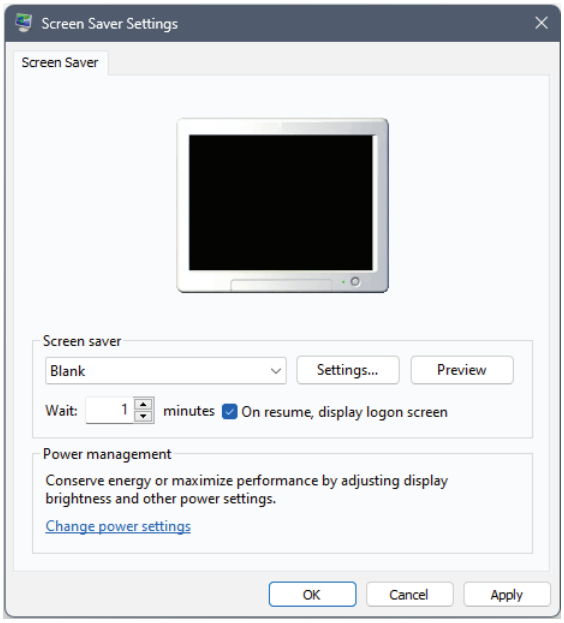


Requiring Screen Saver Passwords

You can require lengthy, complex passwords for all your users as well as lock down the operating system with passwords, but if a user goes to the restroom or on break without locking their workstation, any person wandering by can access everything the user has privileges to.

A *screen saver* should automatically start after a short period of idle time, and a password should be required before the user can begin the session again. This method of locking the workstation adds one more level of security. A Group Policy can be put in place to turn on password-protected screen savers. Adding a password-protected screen saver can ensure that if a workstation is left unattended, it will lock and require a password to resume access. You can access this setting on Windows 11 by right-clicking an empty portion of the Desktop and selecting Personalize ➢ Lock Screen ➢ Screen Saver. Then, in the Screen Saver Settings dialog box, to manually require a password after a screen saver has activated, select the On Resume, Display Logon Screen check box, as shown in Figure 5.41.

FIGURE 5.41 The Screen Saver Settings dialog box



Disable AutoRun

It is never a good idea to put any media in a workstation if you do not know where it came from or what it is. The simple reason is that said media (CD, DVD, USB) could contain malware. This attack is commonly referred to as a *drop attack*. Compounding matters, that malware could be referenced in the `autorun.inf` file, causing it to be summoned when the

media is inserted in the machine and requiring no other action. `autorun.inf` can be used to start an executable, access a website, or do any of a large number of different tasks. The best way to prevent a user from falling victim to such a ploy is to disable the *AutoRun* feature on the workstation.

Microsoft has changed the function on Windows so that it no longer acts as it previously did (prior to Windows 7). The feature is now disabled by default. The reason Microsoft changed the default action can be summed up in a single word: security. That text-based `autorun.inf` file can not only take your browser to a web page, it can also call any executable file, pass along variable information about the user, or do just about anything else imaginable. Simply put, it is never a good idea to plug any media into your system if you have no idea where it came from or what it holds. Such an action opens up the user's system—and the network—to any number of possible risks. An entire business's data could be jeopardized by someone with elevated privileges inadvertently putting a harmful USB thumb drive or CD into a computer at work.

AutoPlay

The AutoRun feature is disabled by default, so that malicious software does not start automatically. However, the ability to automatically start a video, music, or open a folder when removable media is inserted into the computer is really useful. The functionality of automatically performing an action or asking what should be done when media is inserted is still enabled by default on Windows, via a feature called *AutoPlay*. AutoPlay distinguishes itself from AutoRun, since it does not look at the `autorun.inf` file and does not start an executable unless the user specifically clicks an option that they want to do so.

Exam Objective 2.9, "Compare and contrast common data destruction and disposal methods," requires you to know the various methods for destroying hard drives. In addition, you should know the recycling and repurposing best practices, the outsourcing of disposal, as well as regulatory and environmental requirements.

Destruction and Disposal Methods

Think of all the sensitive data written to a hard drive. The drive can contain information about students, clients, users—about anyone and anything. The hard drive can be in a desktop PC, in a laptop, or even in a printer. Many laser printers above consumer grade offer the ability to add a hard drive to store print jobs. If the drive falls into the wrong hands, you can not only lose valuable data but also risk a lawsuit for not properly protecting privacy. An appropriate data destruction/disposal plan should be in place to avoid any potential problems.

Since data on media holds great value and liability, that media should never simply be tossed away for prying eyes to stumble on. For the purpose of this objective, the media in question is hard drives, and there are three key concepts to understand with regard to

them: formatting, sanitation, and destruction. Formatting prepares the drive to hold new information (which can include copying over data already there). Sanitation involves wiping the data off the drive, whereas destruction renders the drive no longer usable.

Although this objective is heavily focused on hard drives, it is also possible to have data stored on portable flash drives, backup tapes, CDs, or DVDs. In the interest of security, it is recommended that you destroy them before disposing of them.

Recycling or Repurposing Best Practices

For exam purposes, the best practices for recycling or repurposing fall into the categories of low-level formats (as opposed to standard formatting), overwrites, and drive wipes.

Low-Level Format vs. Standard Format

There are multiple levels of formatting that can be done on a drive. A standard format, accomplished using the operating system's `format` utility (or similar), can mark space occupied by files as available for new files without truly deleting what was there. Such erasing—if you want to call it that—doesn't guarantee that the information isn't still on the disk and recoverable.

A low-level format (typically accomplished only in the factory) can be performed on the system, or a utility can be used to completely wipe the disk clean. This process helps to ensure that information doesn't fall into the wrong hands.

The manufacturer performs a low-level format on hard drives. Low-level formatting must be performed even before a drive can be partitioned. In low-level formatting, the drive controller chip and the drive meet for the very first time and learn to work together. Because controllers are integrated into SATA and IDE drives, low-level formatting is a factory process. Low-level formatting is not operating system–dependent.

Never perform a low-level format on SATA, or SCSI drives! These drives are formatted at the factory, and you may cause problems by using low-level utilities on them.

The main thing to remember for the exam is that most forms of formatting included with the operating system do not actually erase the data completely. Formatting the drive and then disposing of it has caused many companies problems when individuals who never should have seen it retrieve the data using applications that are commercially available.

Hard Drive Sanitation and Sanitation Methods

A number of vendors offer hard drives with *Advanced Encryption Standard (AES)* cryptography built in. However, it's still better to keep these secure hard drives completely out of the

hands of others than to trust their internal security mechanisms once their usable life span has passed for the client. Some vendors include freeware utilities to erase the hard drive. If it is a Serial ATA (SATA) drive, you can always run HDDErase, but you are still taking your chances.

Sanitation Utilities

HDDErase is a freeware utility that is included with many different boot images. HDDErase can be downloaded with Hiren's BootCD from `www.hirensbootcd.org`.

If you were to perform a web search, you would find several other sanitization utilities like HDDErase, but it is important to recognize and acknowledge that many of these do not meet military or GSA specifications. Those specifications should be considered as guidelines to which you must also adhere when dealing with your own, or a client's, data. The only surefire method of rendering the hard drive contents completely eradicated is physical destruction.

Solid-state drives (SSDs) pose a greater problem since the media is flash memory and not mechanical, like conventional hard disk drives (HDDs). Low-level formats can be performed, as mentioned in the preceding section, but the 1s and 0s will still be technically on the flash memory. Therefore, many vendors have a sanitization utility for scrubbing information from SSDs. It is best to check with the vendor, as these tools are specific to the vendor and model of SSD.

Overwrite

Overwriting the drive entails copying over the data with new data. A common practice is to replace the data with 0s. A number of applications allow you to recover what was there prior to the last write operation, and for that reason, most overwrite software will write the same sequence and save it multiple times.

DBAN is a utility that comes with its own boot disk from `https://dban.org`. You can find a number of other software "shredders" by doing a quick web search.

Drive Wipe

If it's possible to verify beyond a reasonable doubt that a piece of hardware that's no longer being used doesn't contain any data of a sensitive or proprietary nature, then that hardware can be recycled (sold to employees, sold to a third party, donated to a school, and so on). That level of assurance can come from wiping a hard drive or using specialized utilities.

If you can't be assured that the hardware in question doesn't contain important data, then the hardware should be destroyed. You cannot, and should not, take a risk that the data your company depends on could fall into the wrong hands.

Physical Destruction

Physically destroying the drive involves rendering it no longer usable. While the focus is on hard drives, you can also physically destroy other forms of media, such as flash drives and optical media.

Shredder

Many commercial paper shredders are also capable of destroying DVDs and CDs. Paper shredders, however, are not able to handle hard drives; you need a shredder created for just such a purpose. A low-volume hard drive shredder that will destroy eight drives per minute can carry a suggested list price of around $20,000.

Drill/Hammer

If you don't have the budget for a hard drive shredder, you can accomplish similar results in a much more time-consuming way with a power drill. The goal is to physically destroy the platters in the drive. Start the process by removing the cover from the drive—this is normally done with a Torx driver. (Although #8 does not work with all drives, it is a good one to try first.) You can remove the arm with a slotted screwdriver and then the cover over the platters using a Torx driver. Don't worry about damaging or scratching anything—nothing is intended to be saved. Everything but the platters can be tossed away.

As an optional step, you can completely remove the tracks using a belt sander, grinder, or palm sander. The goal is to turn the shiny surface into fine powder. Again, this step is optional, but it adds one more layer of assurance that nothing usable remains. Always wear eye protection and be careful not to breathe in any of the fine particles that are generated during the grinding/destruction process.

Following this, use the power drill to create as small a set of particles as possible. A drill press works much better for this task than trying to hold the drive and drill it with a handheld model.

Do You Really Want to Do It Yourself?

Even with practice, you will find that manually destroying a hard drive is time consuming. There are companies that specialize in this and can do it efficiently. One such company is Shred-it, which will pick up your hard drive and provide a chain-of-custody assurance and a certificate of destruction on completion. You can find out more about what they offer at `www.shredit.com`.

Electromagnet (Degaussing)

A large electromagnet can be used to destroy any magnetic media, such as a hard drive or backup tape set. The most common of these is the degaussing tool. *Degaussing* involves

applying a strong magnetic field to initialize the media. (This is also referred to as *disk wiping*.) This process helps ensure that information doesn't fall into the wrong hands.

Degaussing involves using a specifically designed electromagnet to eliminate all data on the drive, including the factory-prerecorded servo tracks. You can find wand model degaussers priced at just over $500 or desktop units that sell for up to $30,000.

Incineration

A form of destruction not to be overlooked is incineration. It is possible to destroy most devices by burning them up, using an accelerant such as gasoline or lighter fluid to aid the process. However, burning hard drives at extremely high temperatures, typically between 870°C and 1,200°C causes unrecoverable data. The intense heat causes the materials inside the hard drive, such as aluminum or glass platters, to melt and deform, ensuring that the data cannot be recovered.

This method is typically used for securely disposing of sensitive information. However, there are environmental considerations and regulations that must be followed. Many professional data destruction services offer incineration as a secure disposal option.

Be careful with any fire, particularly those in which accelerants are used. Be sure that you are not burning anything capable of releasing toxic fumes and that you have the fire controlled and contained at all times.

Environmental and Regulatory Concerns

The proper disposal of e-waste is crucial for both environmental and regulatory reasons. Environmentally, the improper disposal of computer equipment can lead to the release of hazardous materials like lead, mercury, and cadmium. These materials can contaminate soil and water, posing significant health risks to humans and wildlife. Additionally, e-waste contains valuable materials that can be recycled and reused, reducing the need for new raw materials. From a regulatory perspective, many governments have strict laws governing the disposal of e-waste to prevent environmental damage. Compliance with these regulations not only helps protect the environment but also shields organizations from potential legal penalties and fines.

Certificate of Destruction

A *certificate of destruction* (or *certificate of recycling*) may be required for audit purposes. Such a certificate, usually issued by the organization carrying out the destruction, is intended to verify that the asset was properly destroyed and usually includes serial numbers, type of destruction done, and so on.

Securing Physical Documents/Passwords/Shredding

The type and amount of information that can be gleaned from physical documents is amazing, even in the age when there is such a push to go paperless. *Dumpster diving* is a common problem that puts systems at risk. Companies normally generate a huge amount of paper,

most of which eventually winds up in dumpsters or recycle bins. Dumpsters can contain highly sensitive information (such as a password a user has written on a piece of paper because they haven't memorized it yet).

In high-security and government environments, sensitive papers should either be shredded or burned. Most businesses don't do this. In addition, the advent of "green" companies has created an increase in the amount of recycled paper, which can often contain all kinds of juicy information about a company and its individual employees.

Real World Scenario

Document-Shredding Services

Many small companies purchase document-shredding equipment and perform this task themselves. Larger companies with a higher volume can contract a shredding service. These services place document receptacles in the offices for workers to dispose of sensitive documents. The service will then come in and dispose of the documents in a controlled fashion, sometimes shredding the documents on the premises with special equipment. These services can sometimes furnish certifications of disposal, depending on the type of destruction you require.

Summary

In this chapter, you learned about the various issues related to security that appear on the A+ 220–1202 exam. Security is a popular topic in computing, and the ways in which a troublemaker can cause harm increase regularly. CompTIA expects everyone who is A+ certified to understand the basic principles of security and be familiar with solutions that exist.

You also learned of security problem areas and issues that can be easily identified. Problem areas include viruses, Trojans, worms, and malware. Security solutions include implementing encryption technology, using authentication, implementing firewalls, and incorporating security at many levels.

Exam Essentials

Know the various security measures and their purpose. There are several different physical security measures that can be applied to the public and your staff. In addition, there are several different logical security measures, such as access control lists (ACLs), multifactor

authentication (MFA), and several others discussed in this chapter. Be familiar with their practical application.

Know the various physical security devices. You should be familiar with the various physical security devices, such as the use of an access control vestibule, badge readers, video surveillance, alarm systems, motion sensors, door locks, equipment locks, guards, bollards, and fences. In addition, you should be able to describe security devices as they relate to staff, such as the use of key fobs, smartcards, keys, biometrics, lighting, and magnetometers.

Know the various logical security methods. You should be familiar with logical methods of security, such as the implementation and use of access control lists (ACLs), implementing principle of least privilege, and multifactor authentication (MFA), including email, hard tokens, soft tokens, Short Message Service (SMS), voice calls, and authenticator applications as factors of authentication.

Be able to describe why antivirus/antimalware software is needed. Antivirus/antimalware software looks at a virus and takes action to neutralize it based on a virus definition database. Virus definition database files are regularly made available on vendor sites.

Understand the need for user education. Users are the first line of defense against most threats, whether physical or digital. They should be trained on the importance of security and how to help enforce it.

Know the types of malware and mitigation. You should be familiar with the various types of malware commonly found, such as Trojan, rootkit, virus, spyware, ransomware, keylogger, boot sector virus, and cryptominers. In addition, you should be familiar with the various ways to mitigate malware with the use of the recovery console, antivirus, anti-malware, software firewalls, antiphishing training, user education, and OS reinstallation.

Know the various best practices for security. You should know the best security practices, such as implementing data-at-rest encryption, password complexity, and password expiration, as well as protecting BIOS or UEFI with passwords. In addition, you should know the various end-user best practices, such as using screen saver locks, logging off when a computer or application is not in use, and implementing security and protection of critical hardware as well as security of personally identifiable information (PII) and passwords.

Know how to compare and contrast social engineering, threats, and vulnerabilities. Social engineering variants include shoulder surfing (watching someone work) and phishing (tricking someone into believing they are communicating with a party other than the one with whom they are communicating). Variations on phishing include vishing, shoulder surfing, whaling, tailgating, impersonation, dumpster diving, and evil twin.

Know the various threats to a network. You should be familiar with the various threats to a network, such as distributed denial-of-service (DDoS), denial-of-service (DoS), zero-day attack, spoofing, on-path attacks, brute-force password attacks, dictionary password attacks, insider threats, Structured Query Language (SQL) injection, cross-site scripting (XSS), Business email compromise (BEC), and supply chain/pipeline attack.

Understand the need for good passwords. Passwords are the first line of defense for protecting an account. A password should be required for every account, and strong passwords should be enforced. Users need to understand the basics of password security and work to keep their accounts protected by following company policies regarding passwords.

Understand the difference between standard and low-level formatting. Standard formatting uses operating system tools and marks the drive as available for holding data without truly removing what was on the drive (thus, the data can be recovered). A low-level format is operating system–independent and destroys any data that was on the drive.

Understand how to implement appropriate data destruction and disposal methods. A hard drive can be destroyed by tossing it into a shredder designed for such a purpose, or it can be destroyed with an electromagnet in a process known as degaussing. You can also disassemble the drive and destroy the platters with a drill or other tool that renders the data irretrievable. Third party vendors can also be used and they will supply certification of destruction/recycling. Proper disposal is required by regulatory and environmental requirements.

Review Questions

The answers to the chapter review questions can be found in Appendix A.

1. Which component of physical security addresses outer-level access control?
   - **A.** Fences
   - **B.** Access control vestibule
   - **C.** Multifactor authentication
   - **D.** Strong passwords
2. Which type of device can detect weapons on a person entering a facility?
   - **A.** Biometrics
   - **B.** Magnetometer
   - **C.** Motion sensor
   - **D.** Badge reader
3. As part of your training program, you're trying to educate users on the importance of security. You explain to them that not every attack depends on implementing advanced technological methods. Some attacks, you explain, take advantage of human shortcomings to gain access that should otherwise be denied. Which term do you use to describe attacks of this type?
   - **A.** Social engineering
   - **B.** IDS
   - **C.** Perimeter security
   - **D.** Biometrics
4. You're in the process of securing the IT infrastructure by adding fingerprint scanners to your existing authentication methods. This type of security is an example of which of the following?
   - **A.** Access control
   - **B.** Physical barriers
   - **C.** Biometrics
   - **D.** Softening
5. Which type of attack denies authorized users access to network resources?
   - **A.** DoS
   - **B.** Worm
   - **C.** Trojan
   - **D.** Social engineering

6. As the security administrator for your organization, you must be aware of all types of attacks that can occur and plan for them. Which type of attack uses more than one computer to attack the victim?
   - **A.** DoS
   - **B.** DDoS
   - **C.** Worm
   - **D.** Rootkit

7. A vice president of your company calls a meeting with the IT department after a recent trip to competitors' sites. She reports that many of the companies she visited granted access to the operating system or applications after an employee presented a number that rotated. Of the following, which technology relies on a rotating number for users for authentication?
   - **A.** Smartcard
   - **B.** Biometrics
   - **C.** Geofencing
   - **D.** Token

8. You've discovered that credentials to a specific application have been stolen. The application is accessed from only one computer on the network. Which type of attack is this most likely to be?
   - **A.** On-path attack
   - **B.** Zero-day
   - **C.** Denial-of-service (DoS)
   - **D.** Smurf

9. A junior administrator comes to you in a panic. After looking at the log files, he has become convinced that an attacker is attempting to use a legitimate IP address to disrupt access elsewhere on the network. Which type of attack is this?
   - **A.** Spoofing
   - **B.** Social engineering
   - **C.** Worm
   - **D.** Password

10. Which of the following is different from a virus in that it can reproduce itself, is self-contained, and doesn't need a host application to be transported?
    - **A.** Worm
    - **B.** Smurf
    - **C.** Phish
    - **D.** Trojan

11. A reflective attack attempts to use a broadcast ping on a network. The return address of the ping may be that of a valid system in your network. Which protocol does the reflective attack use to conduct the attack?

A. TCP

B. IP

C. UDP

D. ICMP

12. Which type of attack involves passing a database query with a web request?

A. Insider threat

B. Evil twin

C. SQL injection

D. Tailgating

13. Which is an example of an authentication method in which you have something?

A. Password

B. Key fob

C. Fingerprint

D. Place

14. You need to protect your users from potentially being phished via email. Which of the following should you use to protect them?

A. Antivirus software

B. End-user education

C. SecureDNS

D. The principle of least privilege

15. Your help desk has informed you that they received an urgent call from the vice president last night requesting his login ID and password. When you talk with the VP today, he says he never made that call. What type of attack is this?

A. Spoofing

B. Replay

C. Social engineering

D. Trojan horse

16. Internal users suspect there have been repeated attempts to infect their systems, as reported to them by pop-up messages from their antivirus software. According to the messages, the virus seems to be the same in every case. What is the most likely culprit?

A. A server is acting as a carrier for a virus.

B. A password attack is being carried out.

C. Your antivirus software has malfunctioned.

D. A DoS attack is under way.

17. You're working late one night and notice that the hard drive on your new computer is very active even though you aren't doing anything on the computer and it isn't connected to the Internet. What is the most likely suspect?
    - **A.** A spear phishing attack is being performed.
    - **B.** A virus is spreading in your system.
    - **C.** Your system is under a DoS attack.
    - **D.** TCP/IP hijacking is being attempted.

18. You're the administrator for a large bottling company. At the end of each month, you routinely view all logs and look for discrepancies. This month, your email system error log reports a large number of unsuccessful attempts to log in. It's apparent that the email server is being targeted. Which type of attack is most likely occurring?
    - **A.** Brute-force
    - **B.** Backdoor
    - **C.** Worm
    - **D.** TCP/IP hijacking

19. Your boss needs you to present to upper management the need for a firewall for the network. What is the thesis of your presentation?
    - **A.** The isolation of one network from another
    - **B.** The scanning of all packets for viruses
    - **C.** Preventing password attacks
    - **D.** The hardening of physical security

20. Which Active Directory component maps printers and drives during login?
    - **A.** Home folders
    - **B.** Organizational unit
    - **C.** Login script
    - **D.** Microsoft Management Console (MMC)